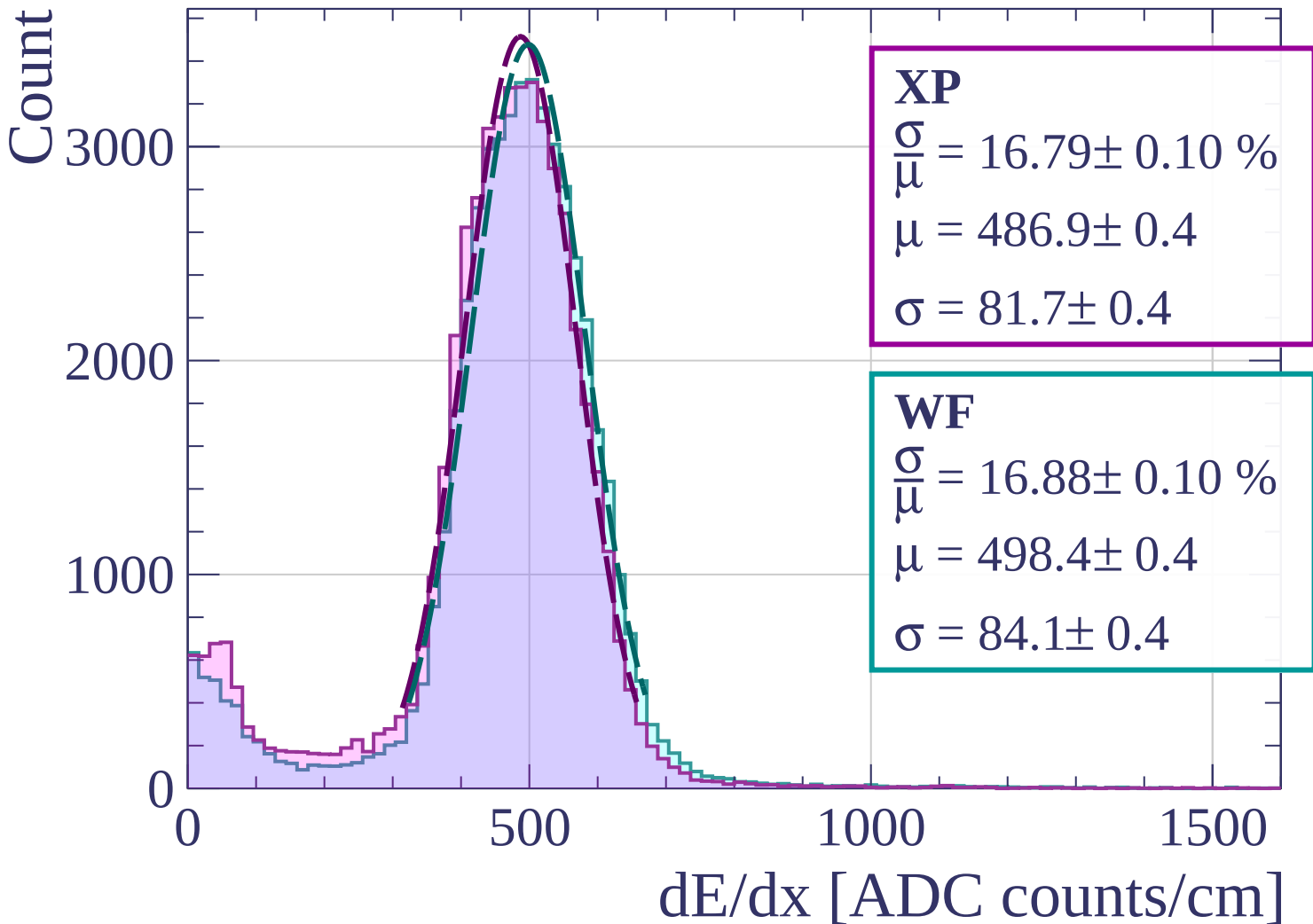
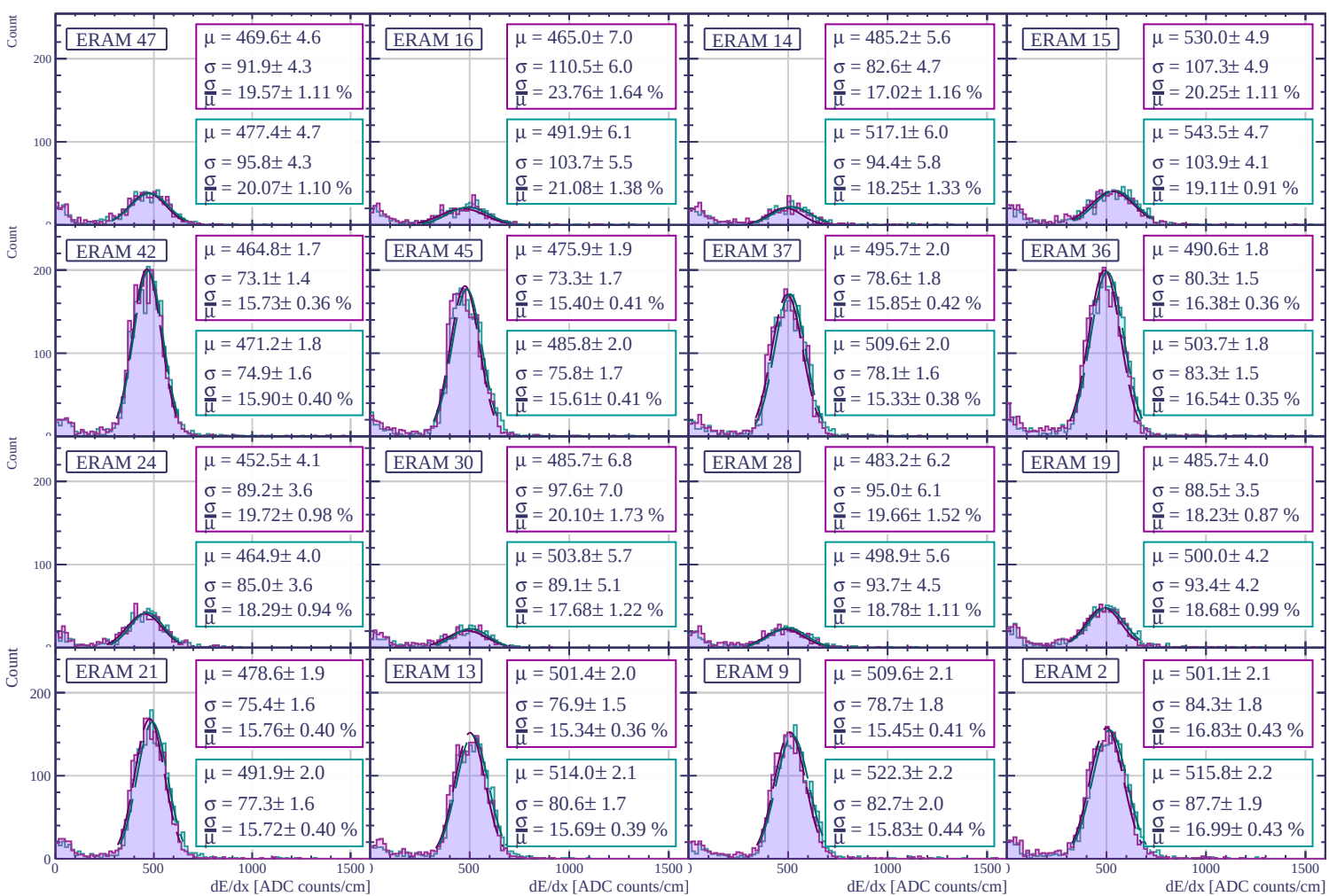
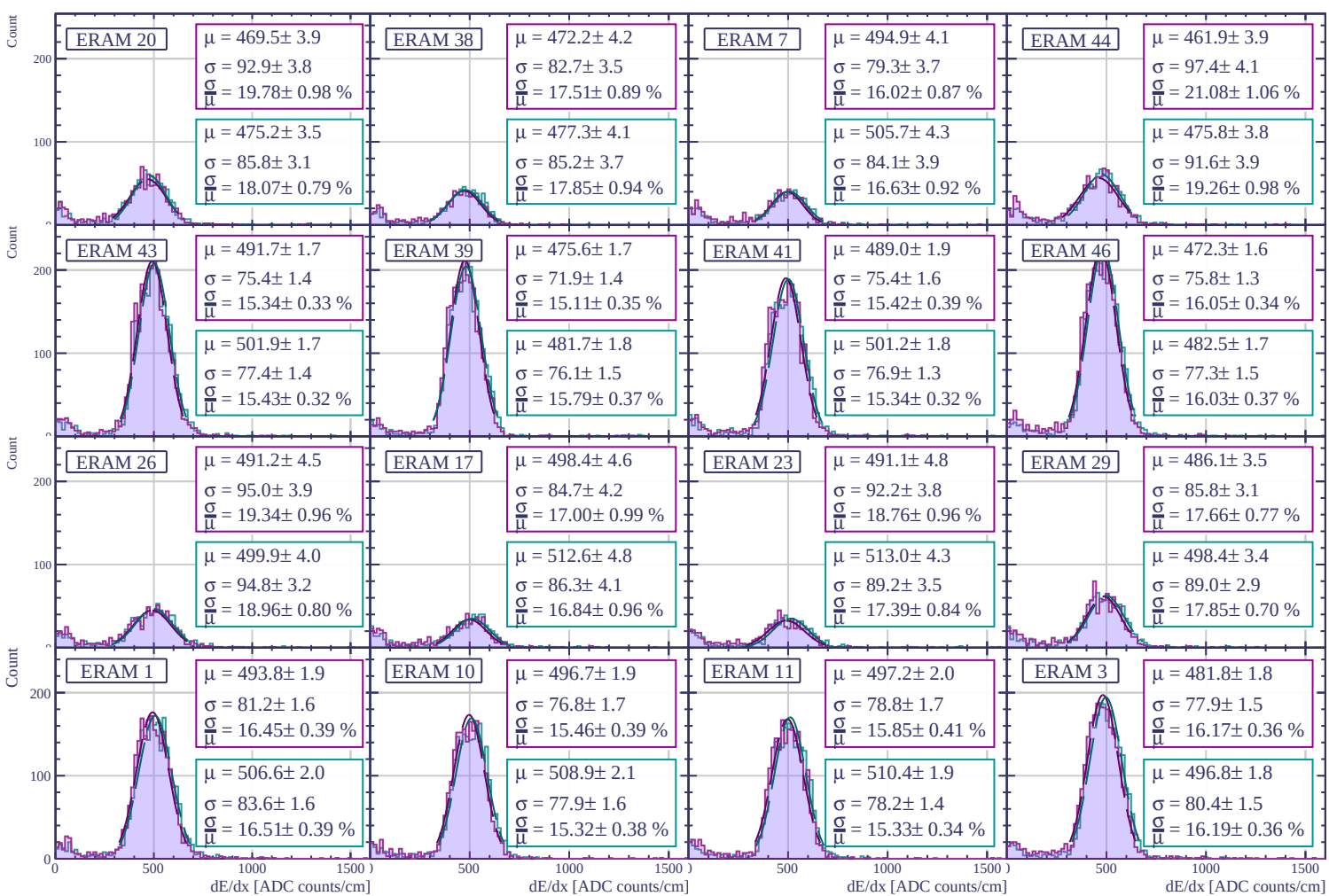
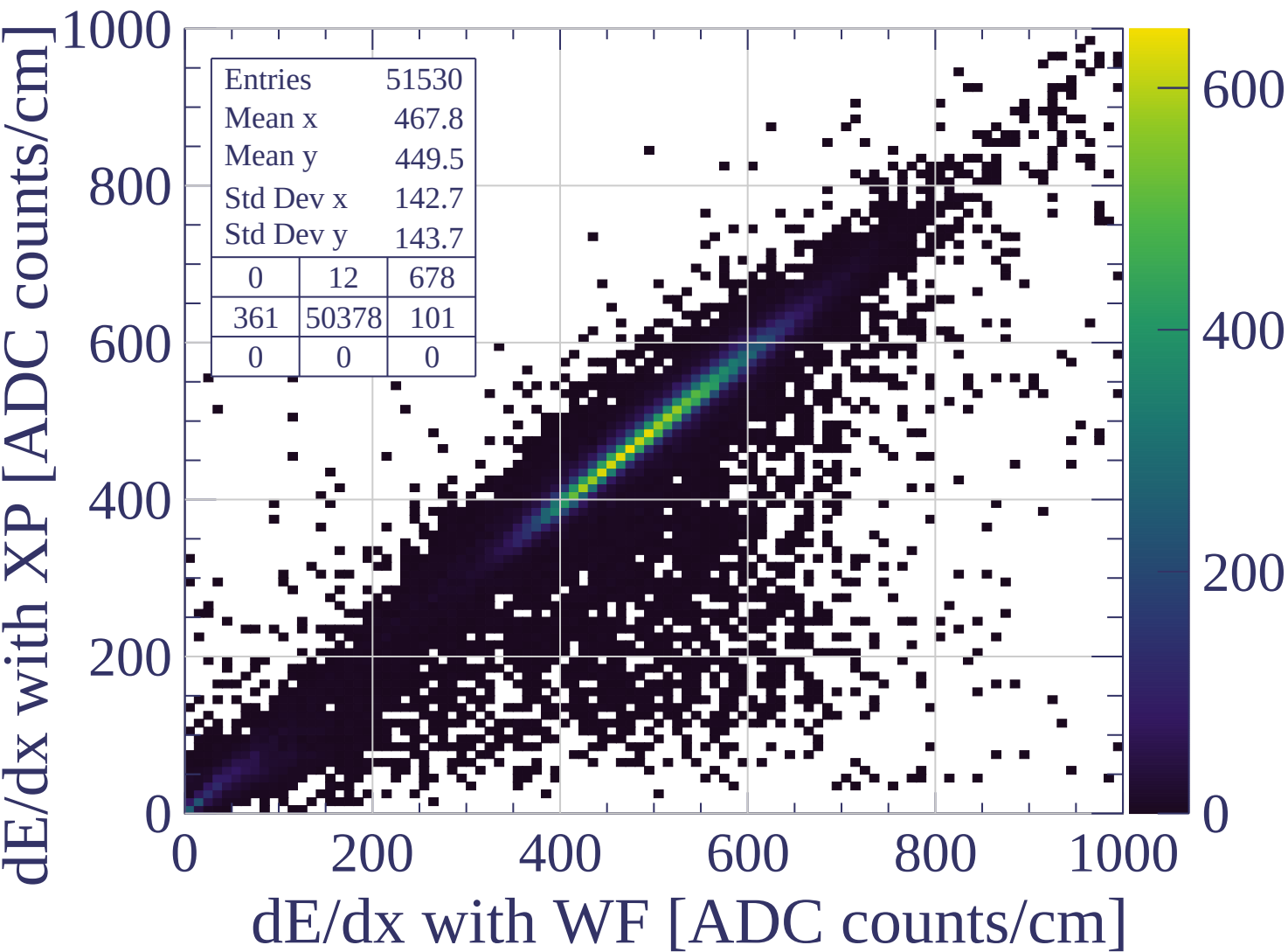
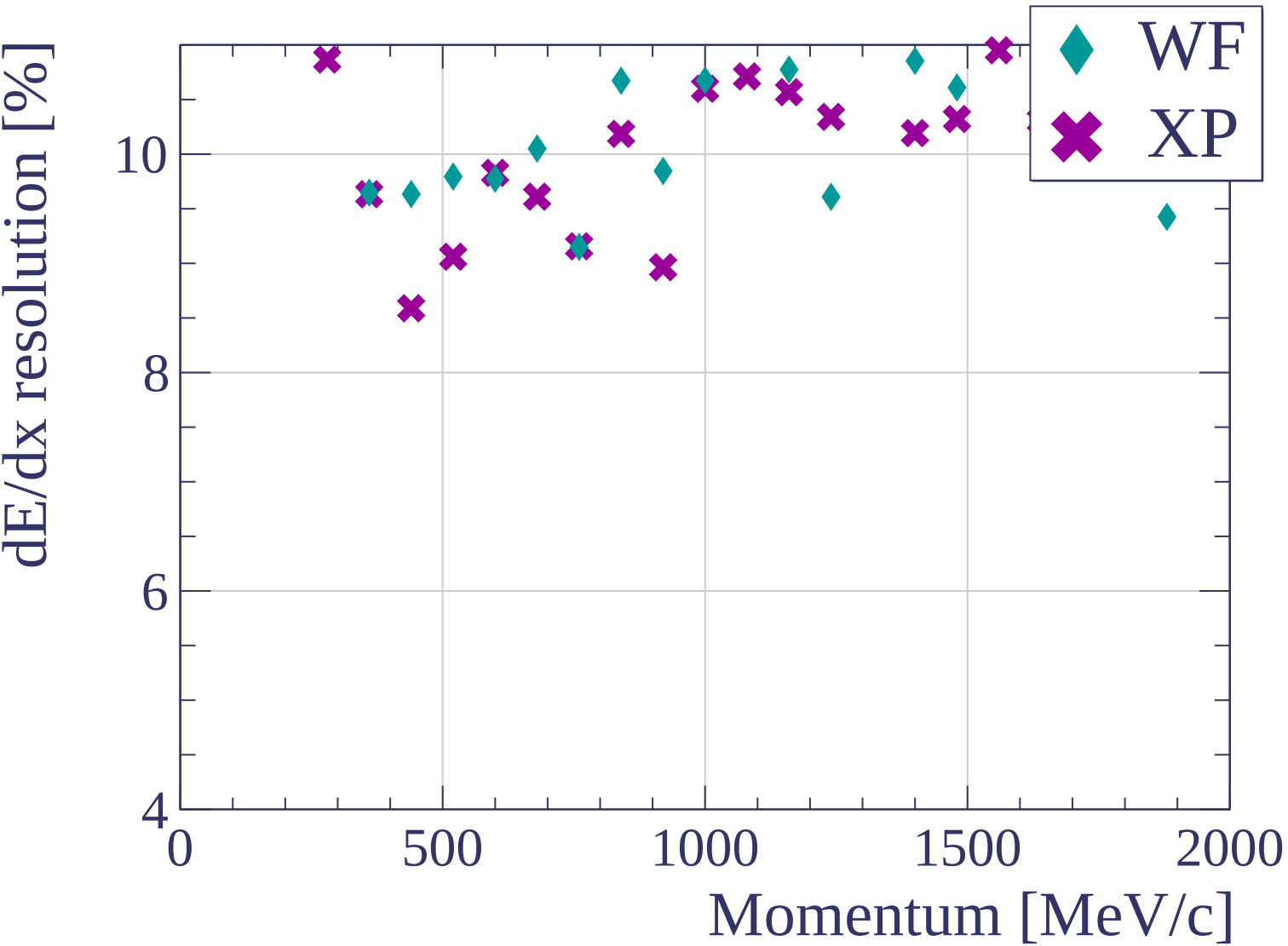


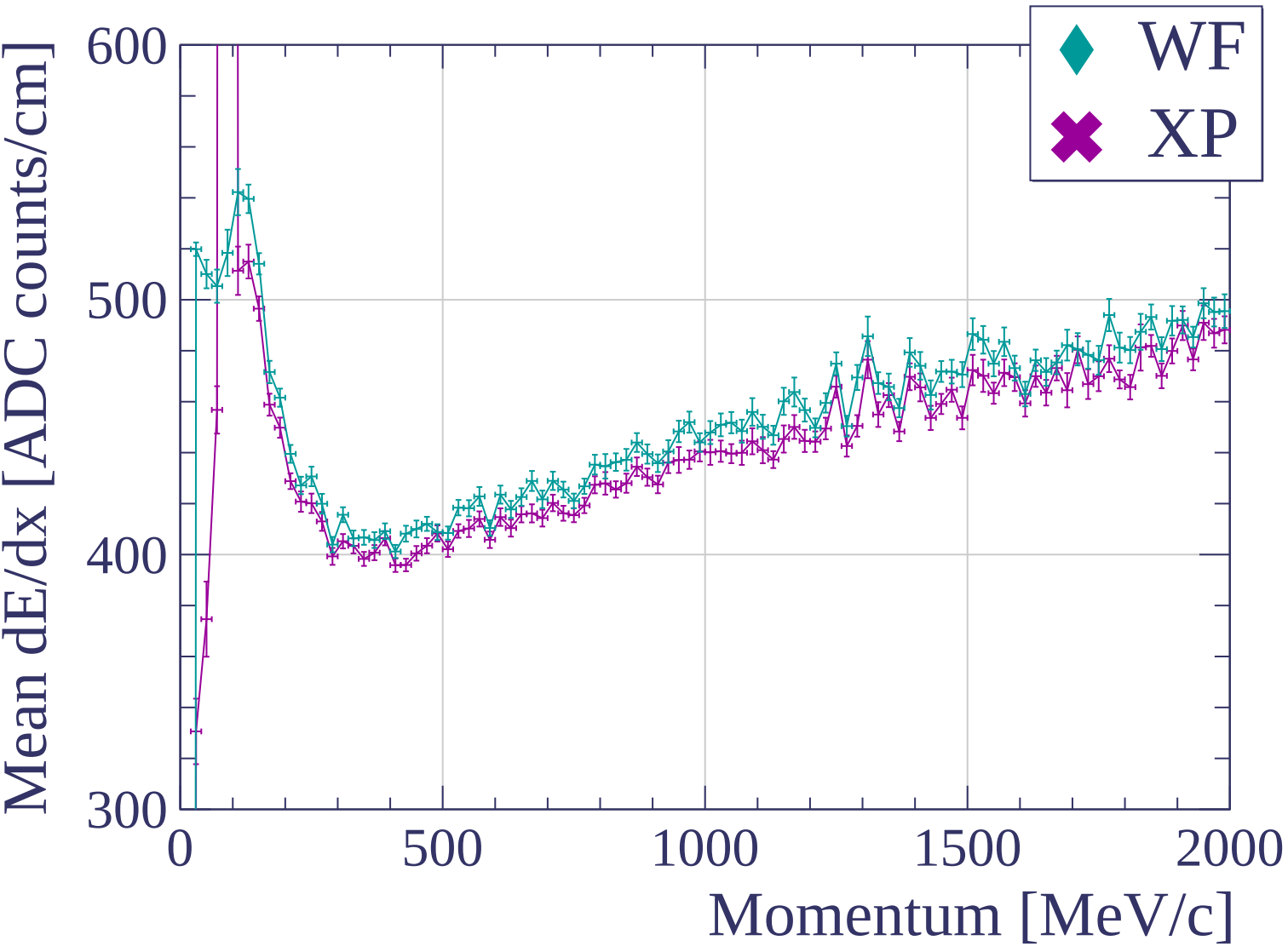


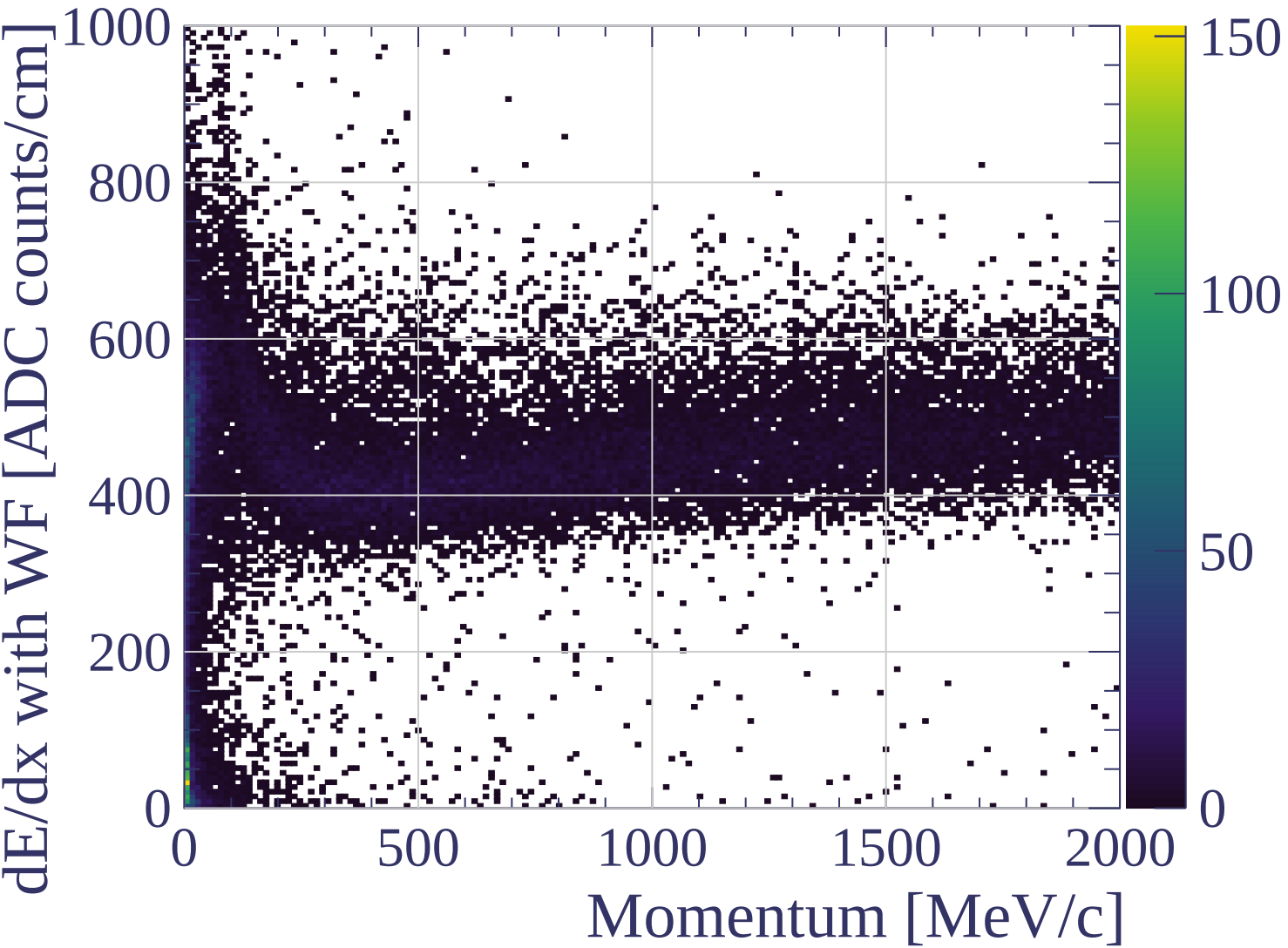


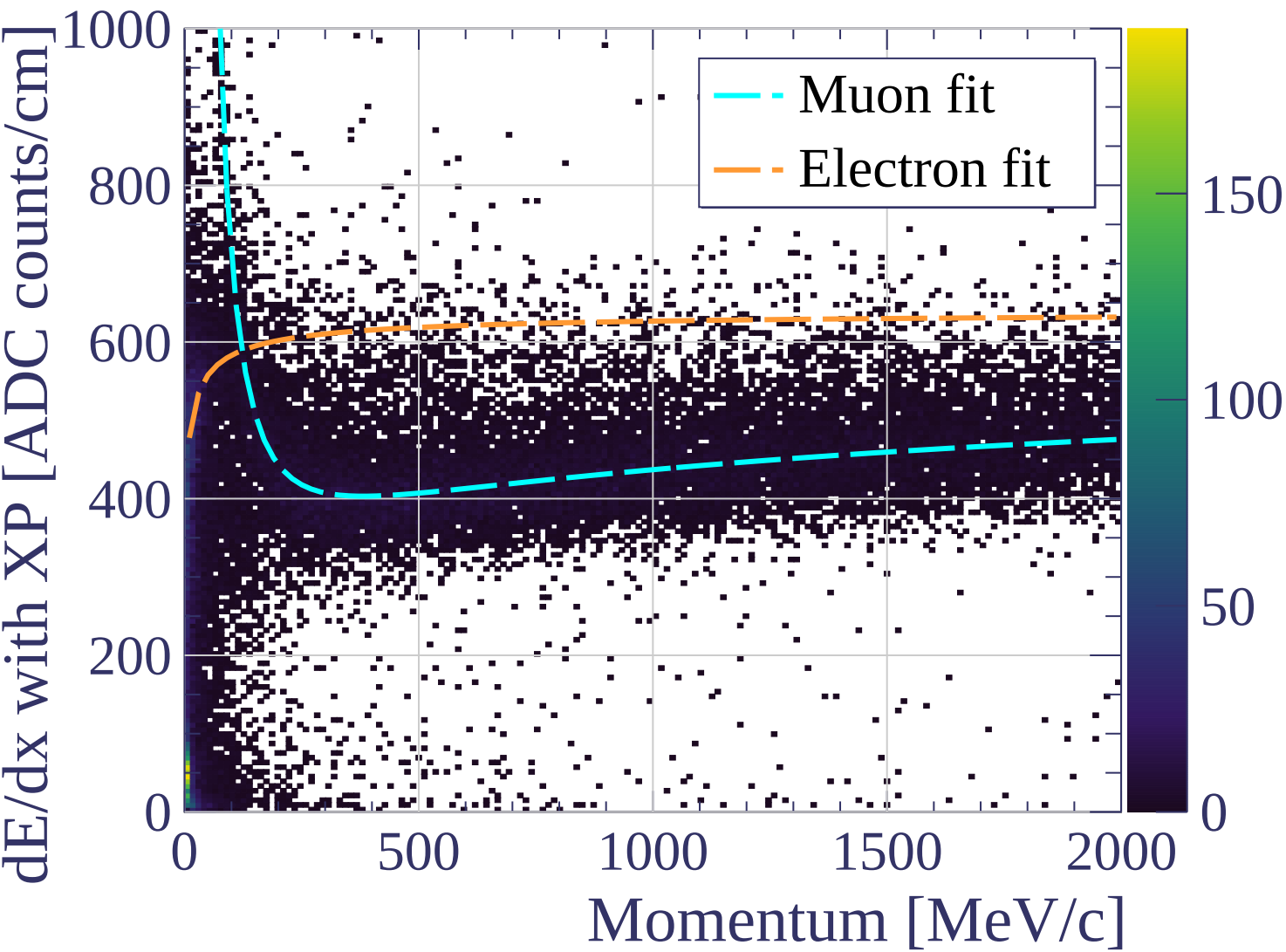




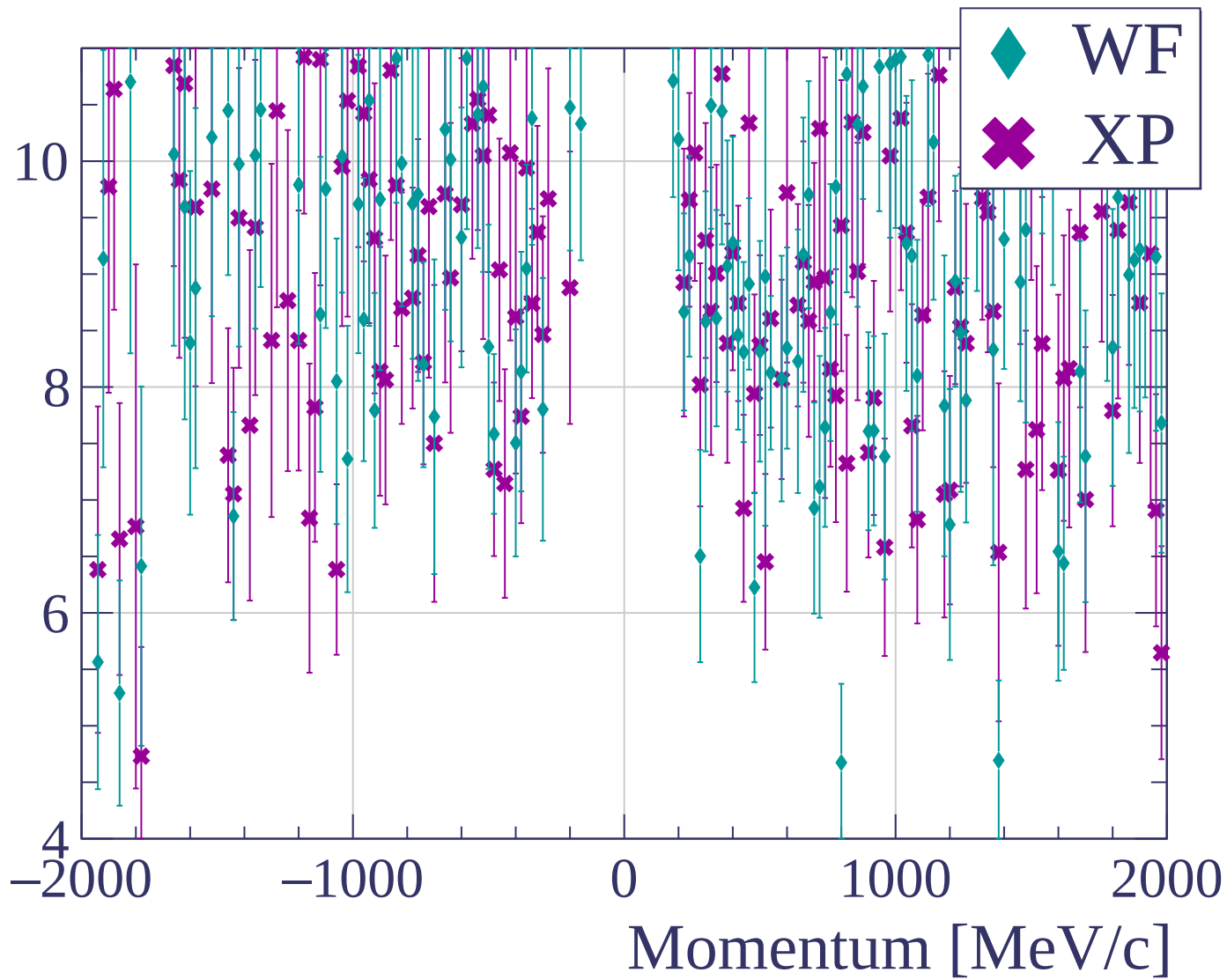


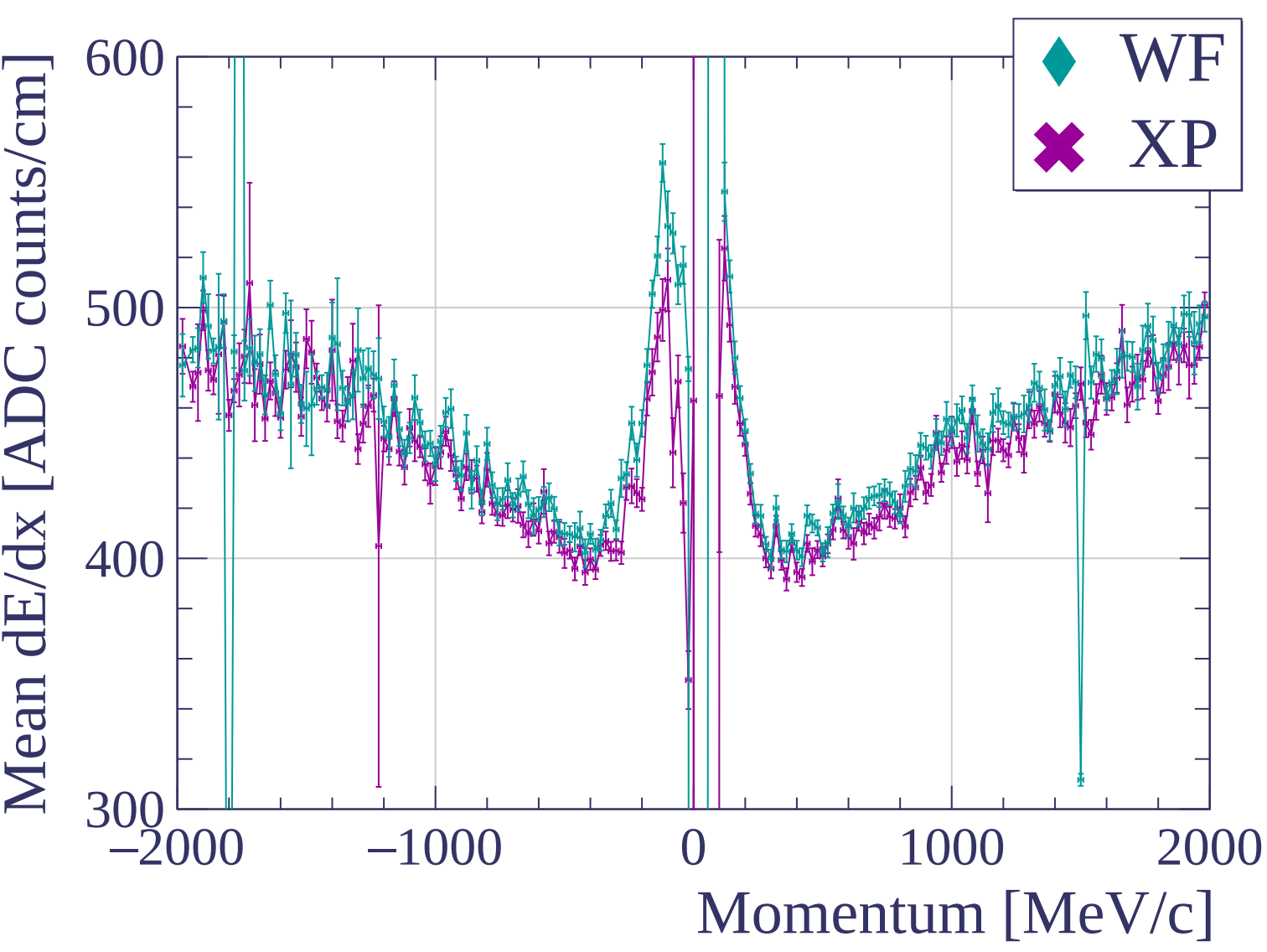


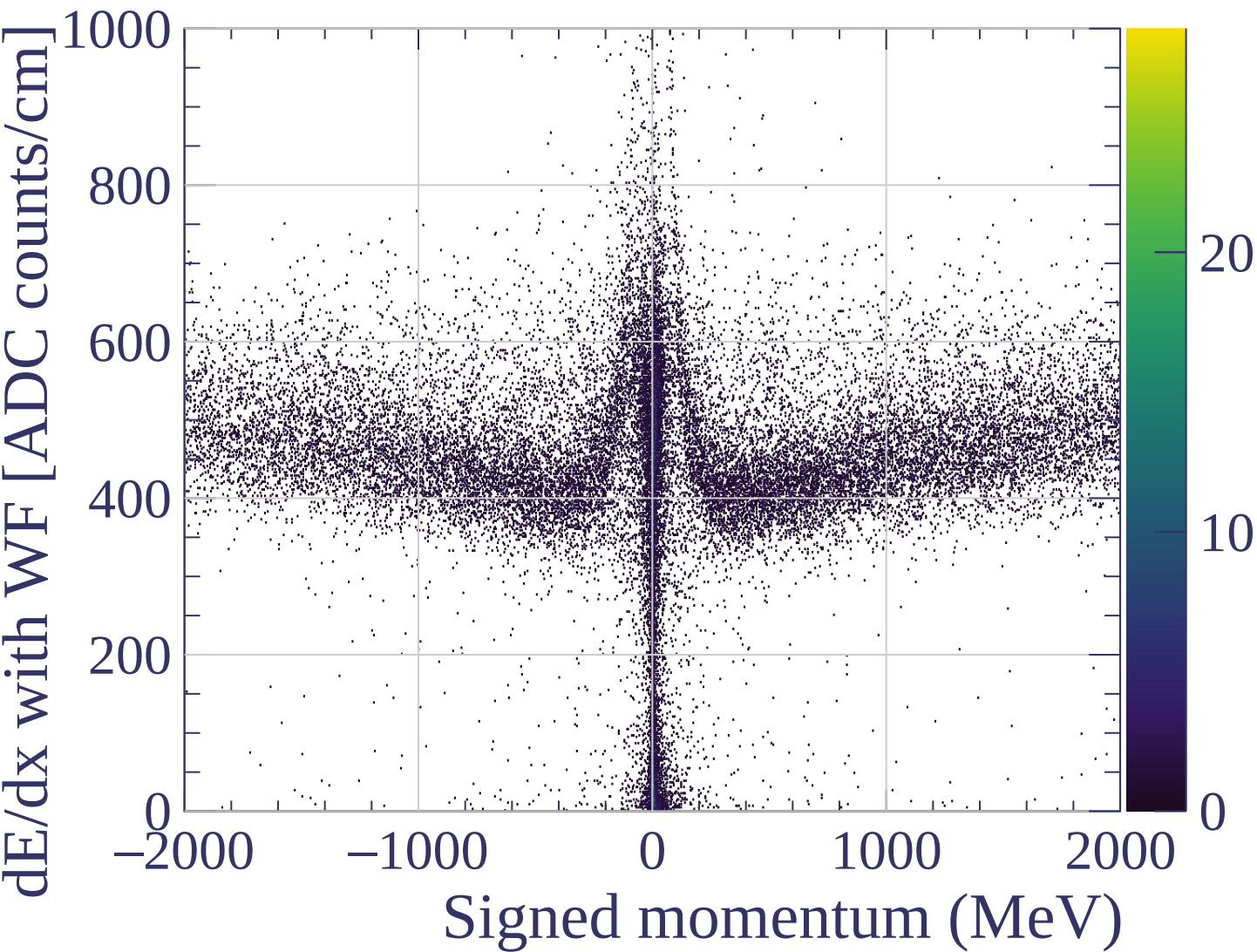


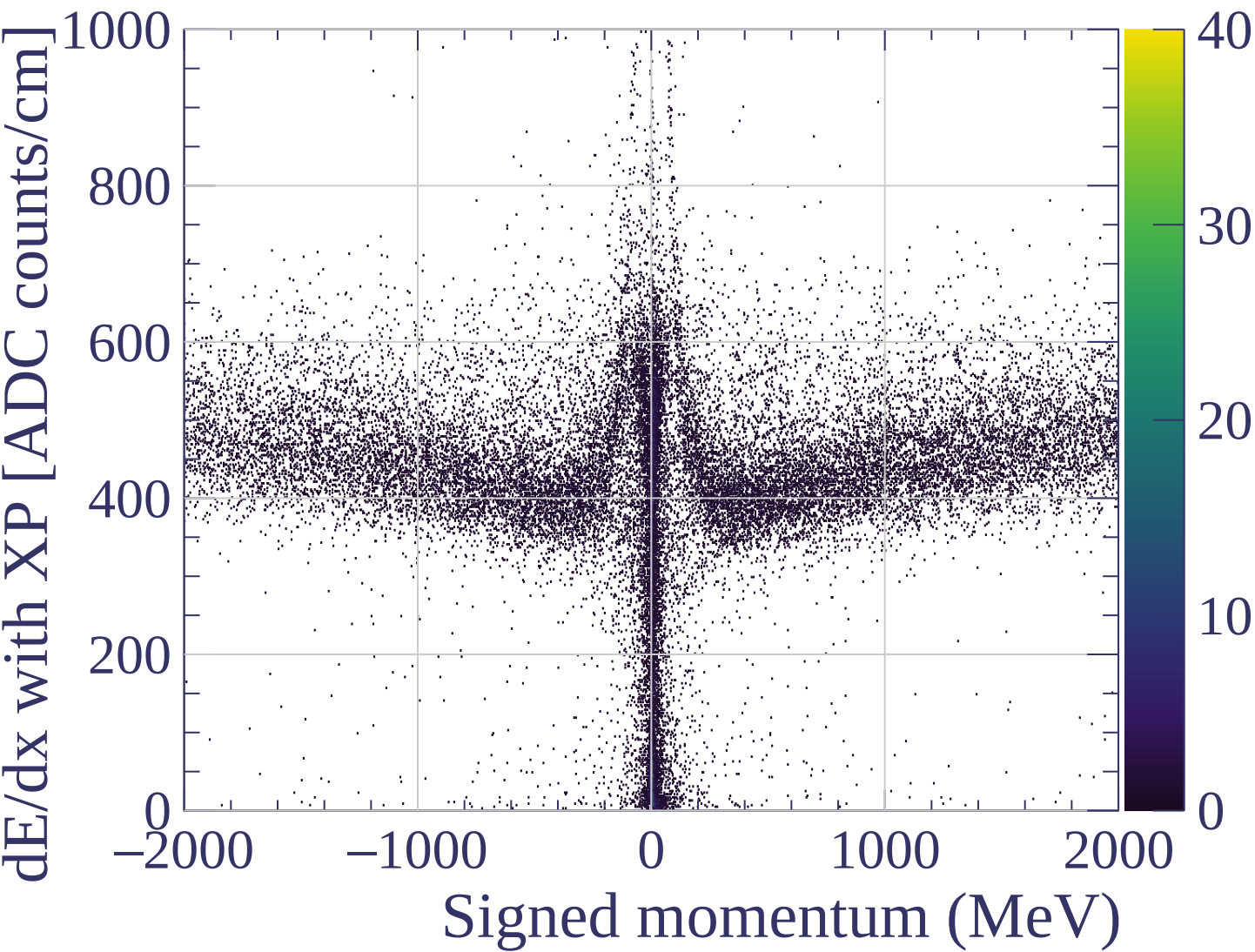


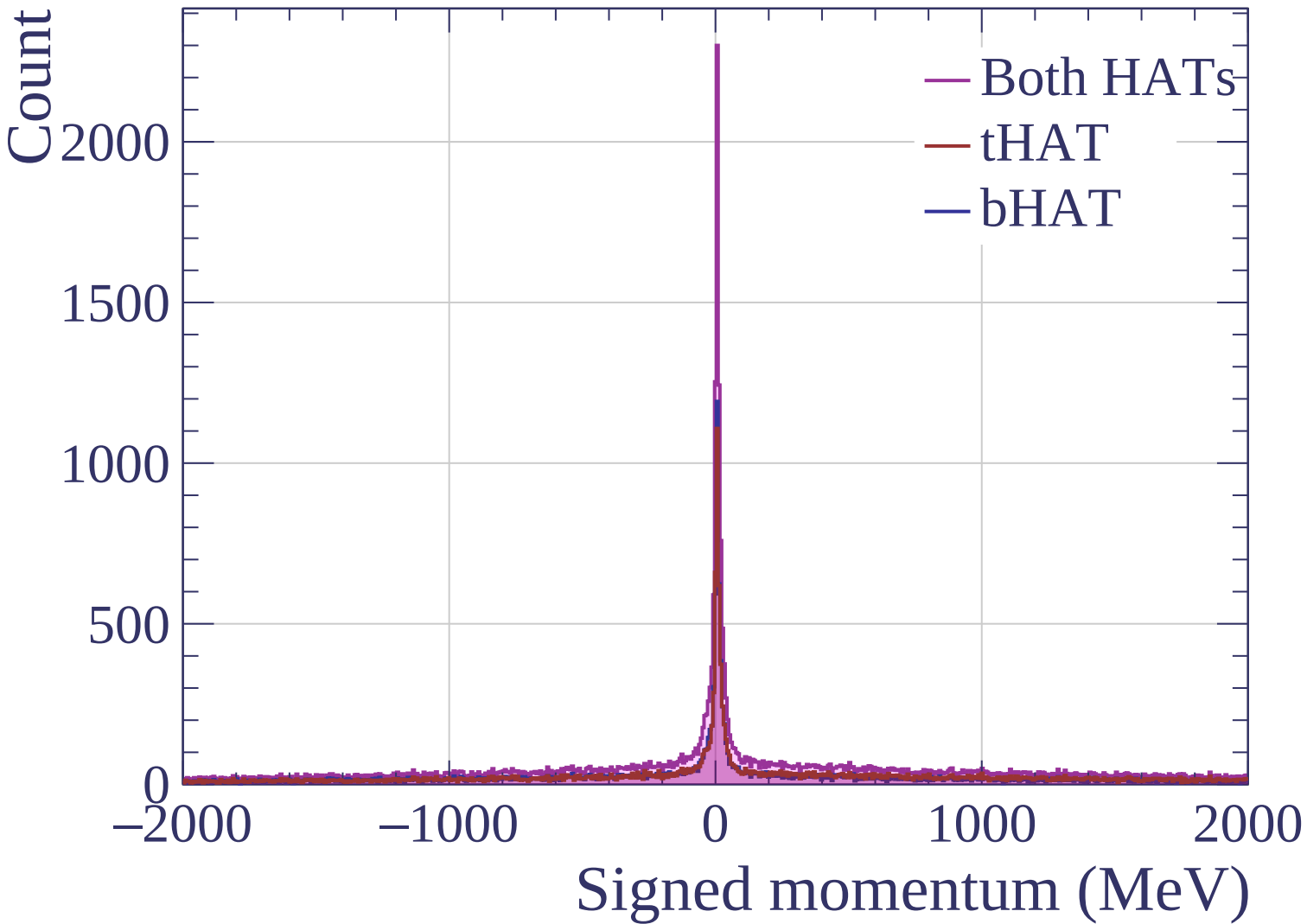
dE/dx resolution [%]

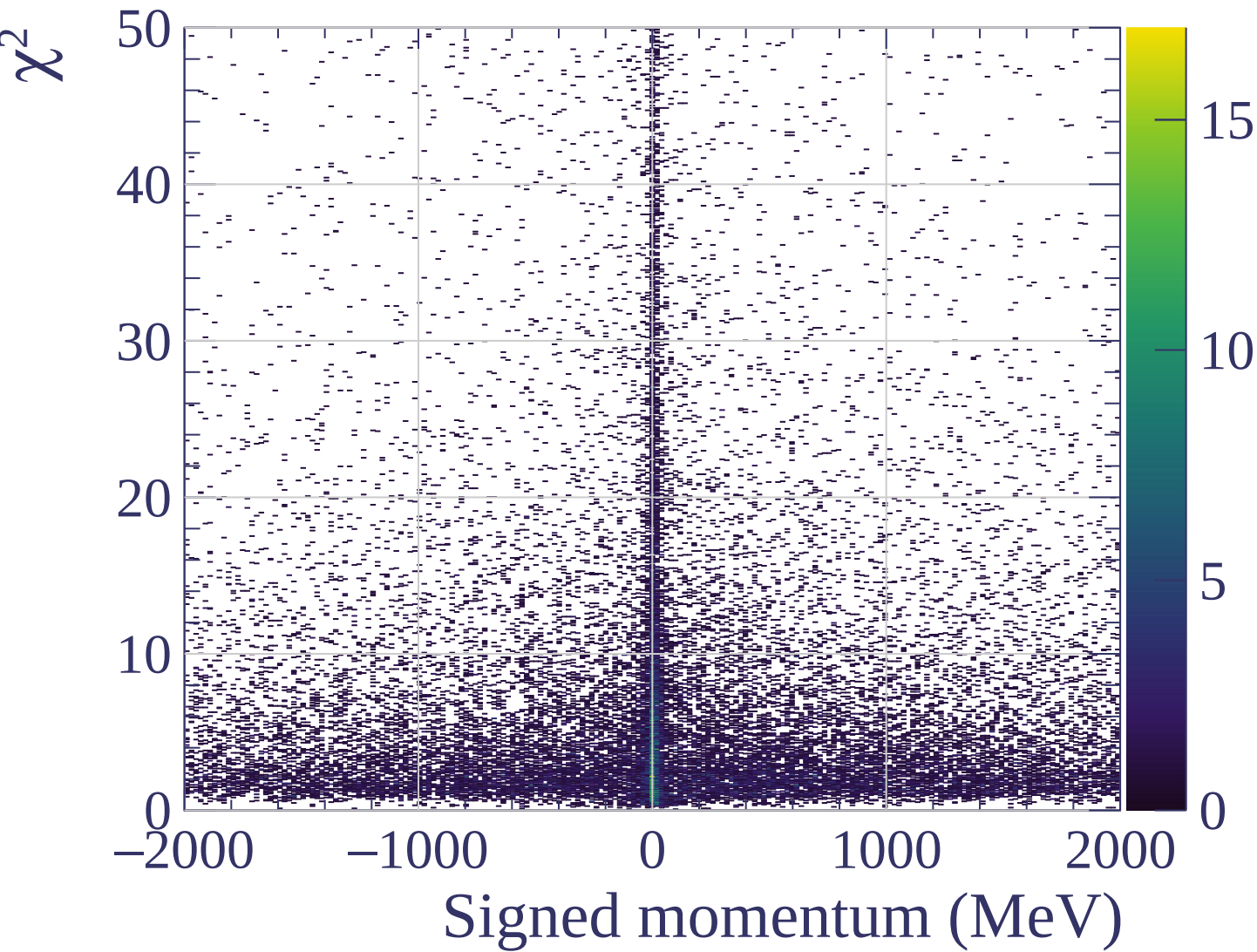


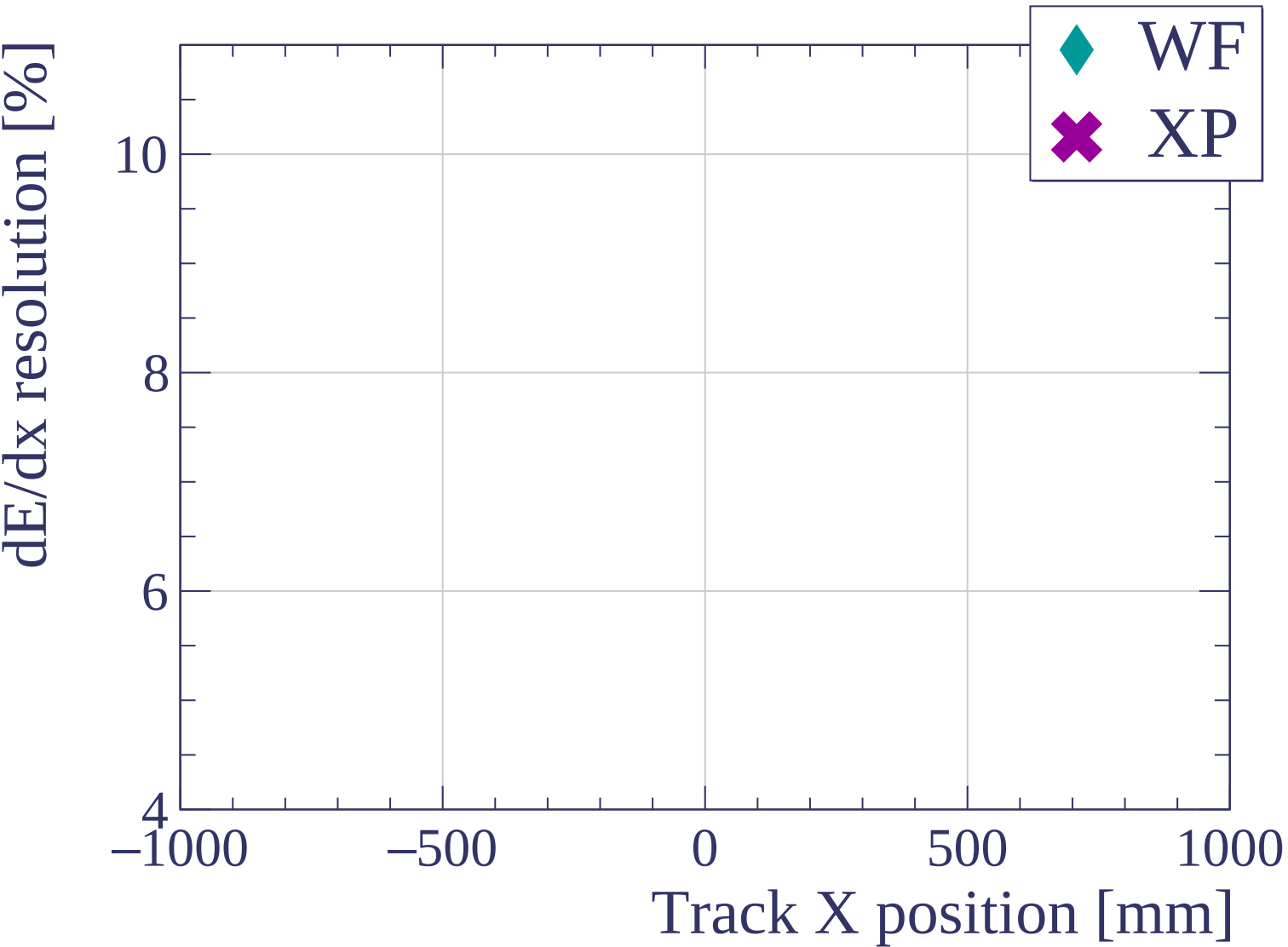


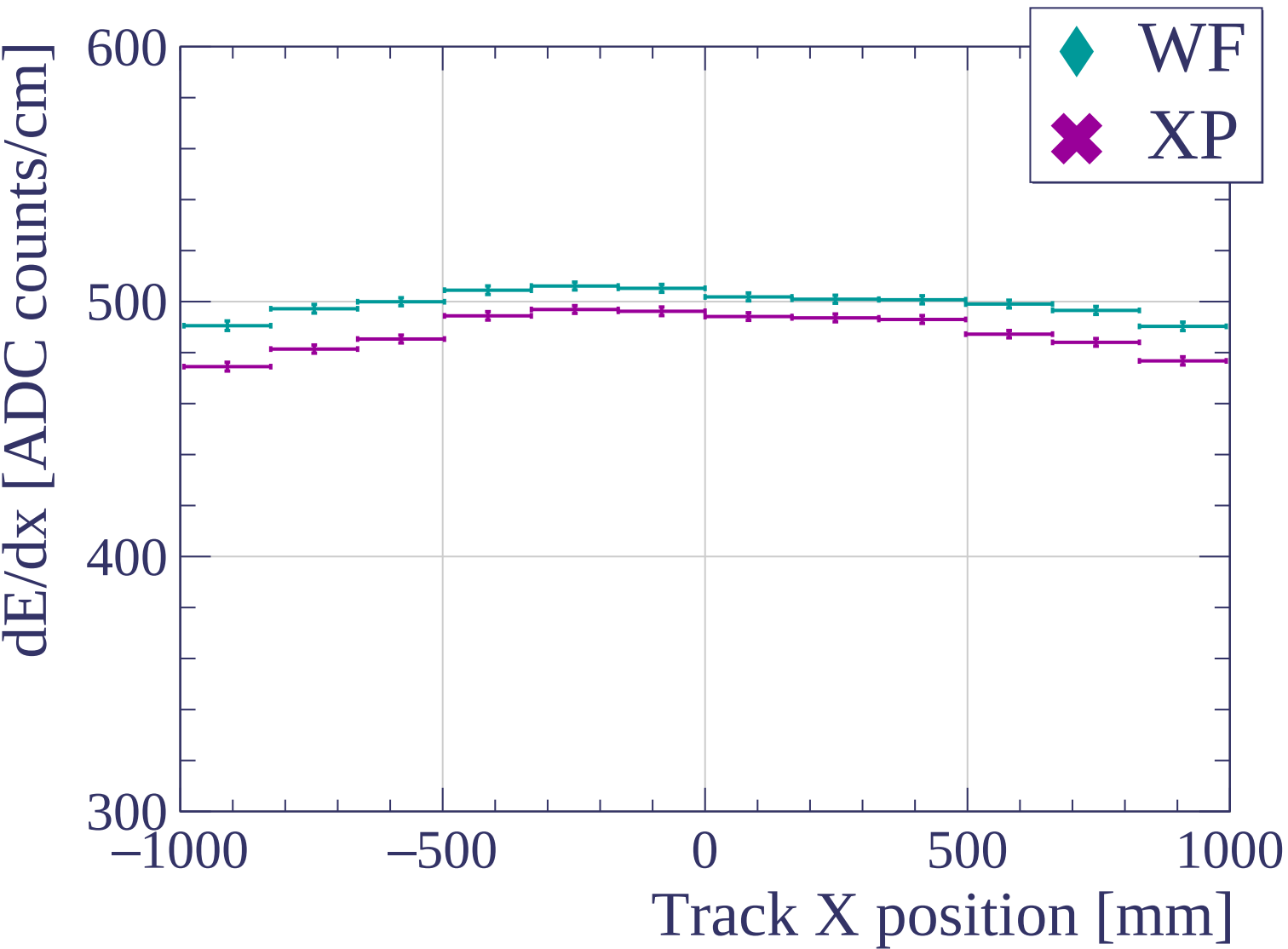


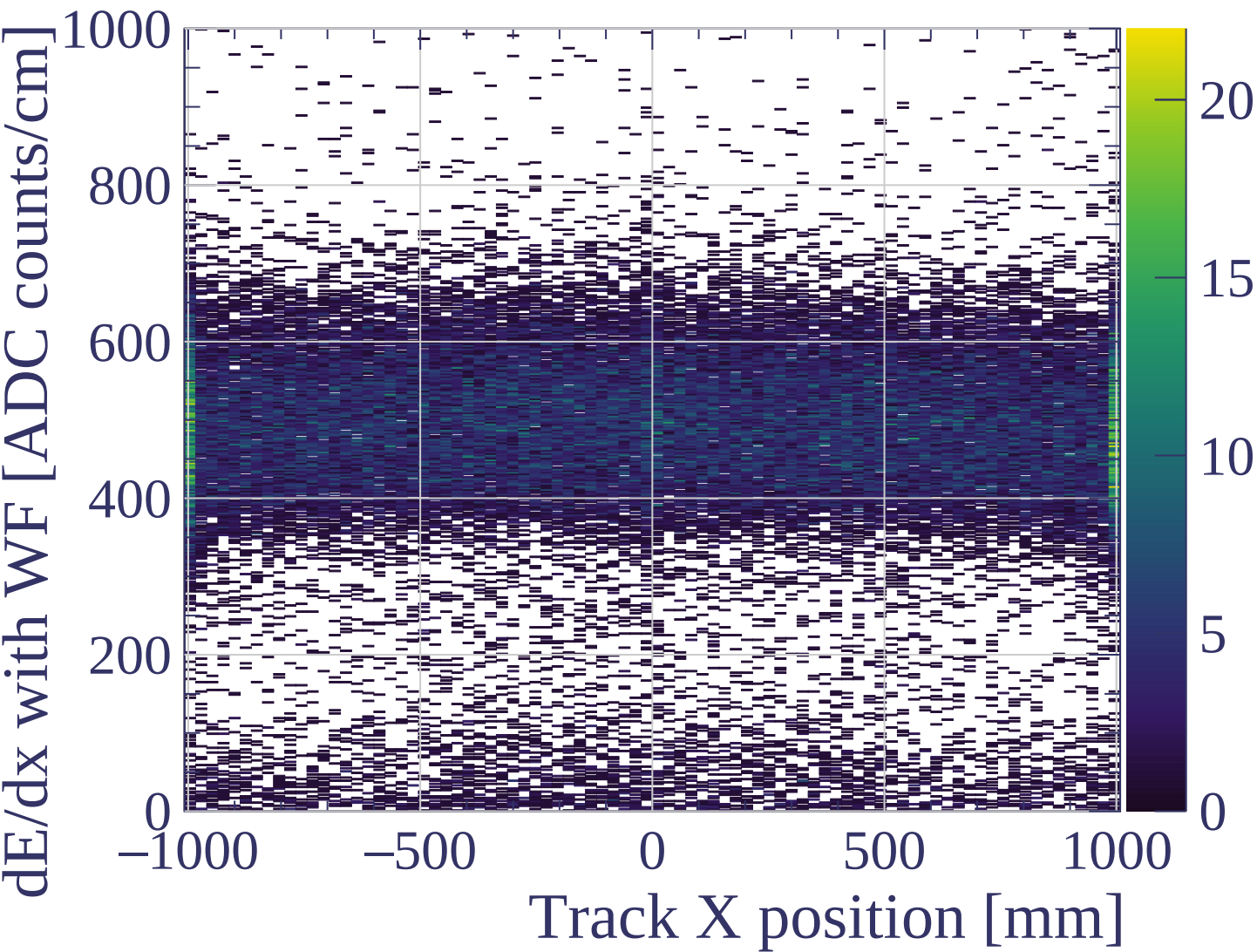


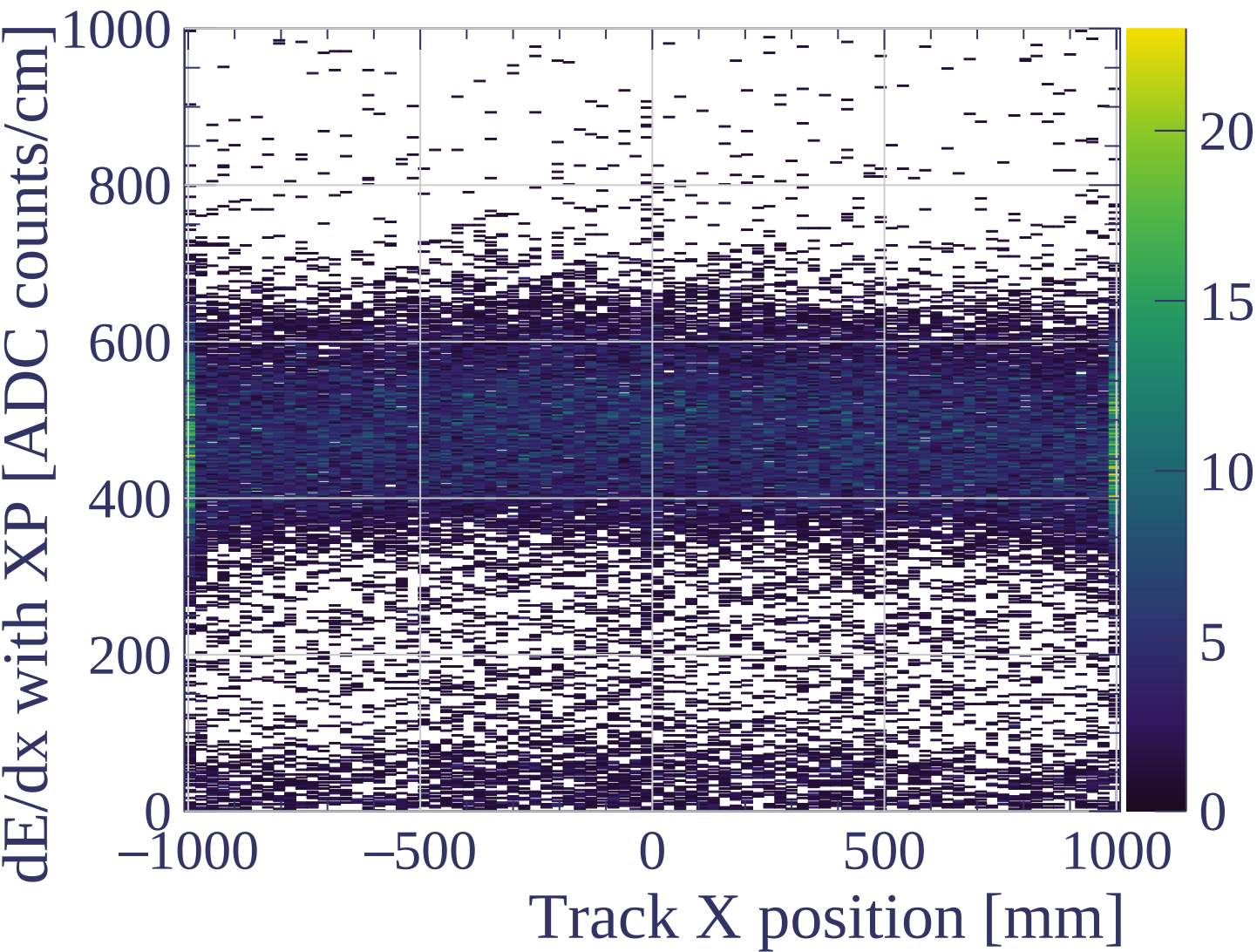


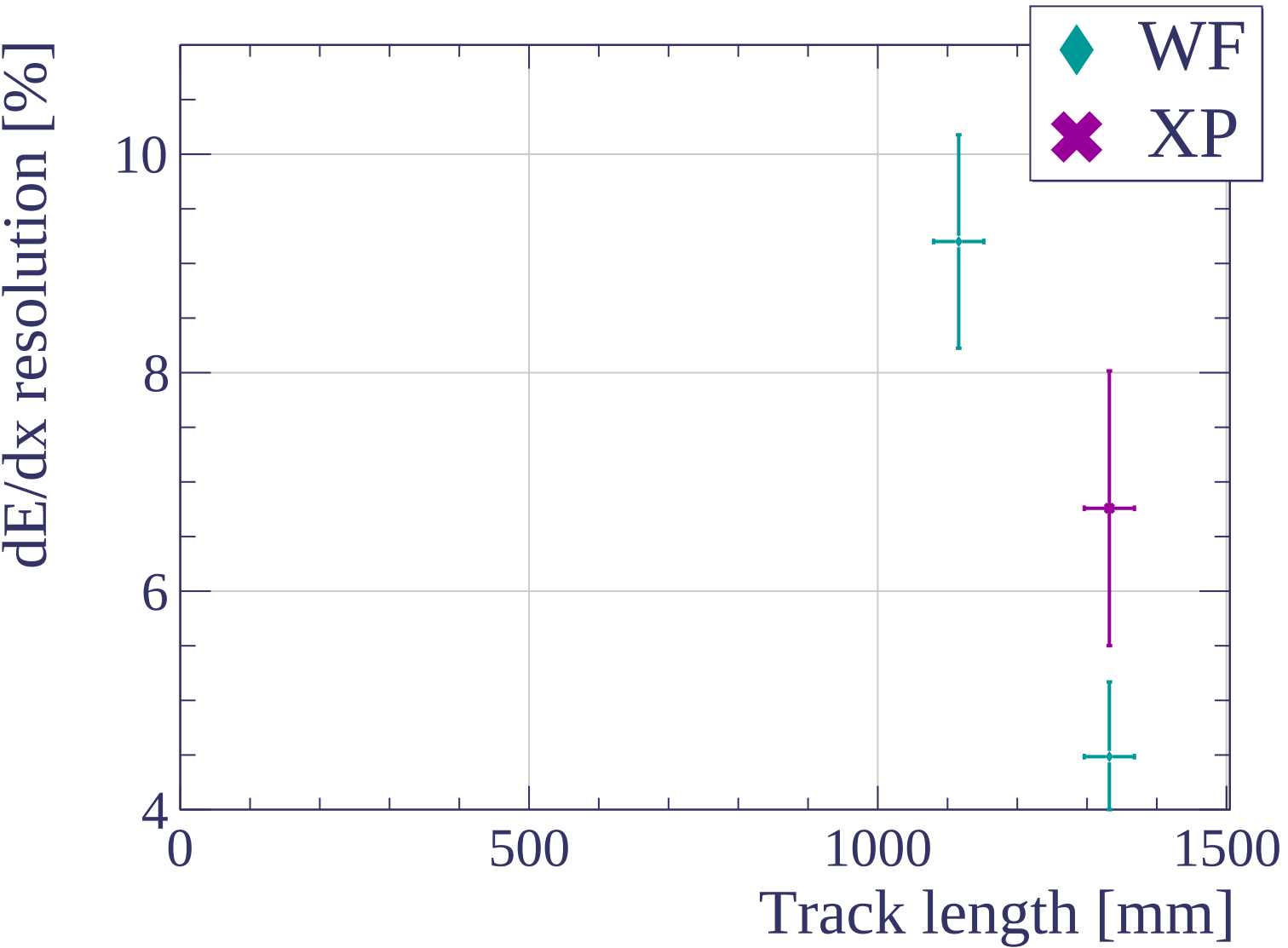


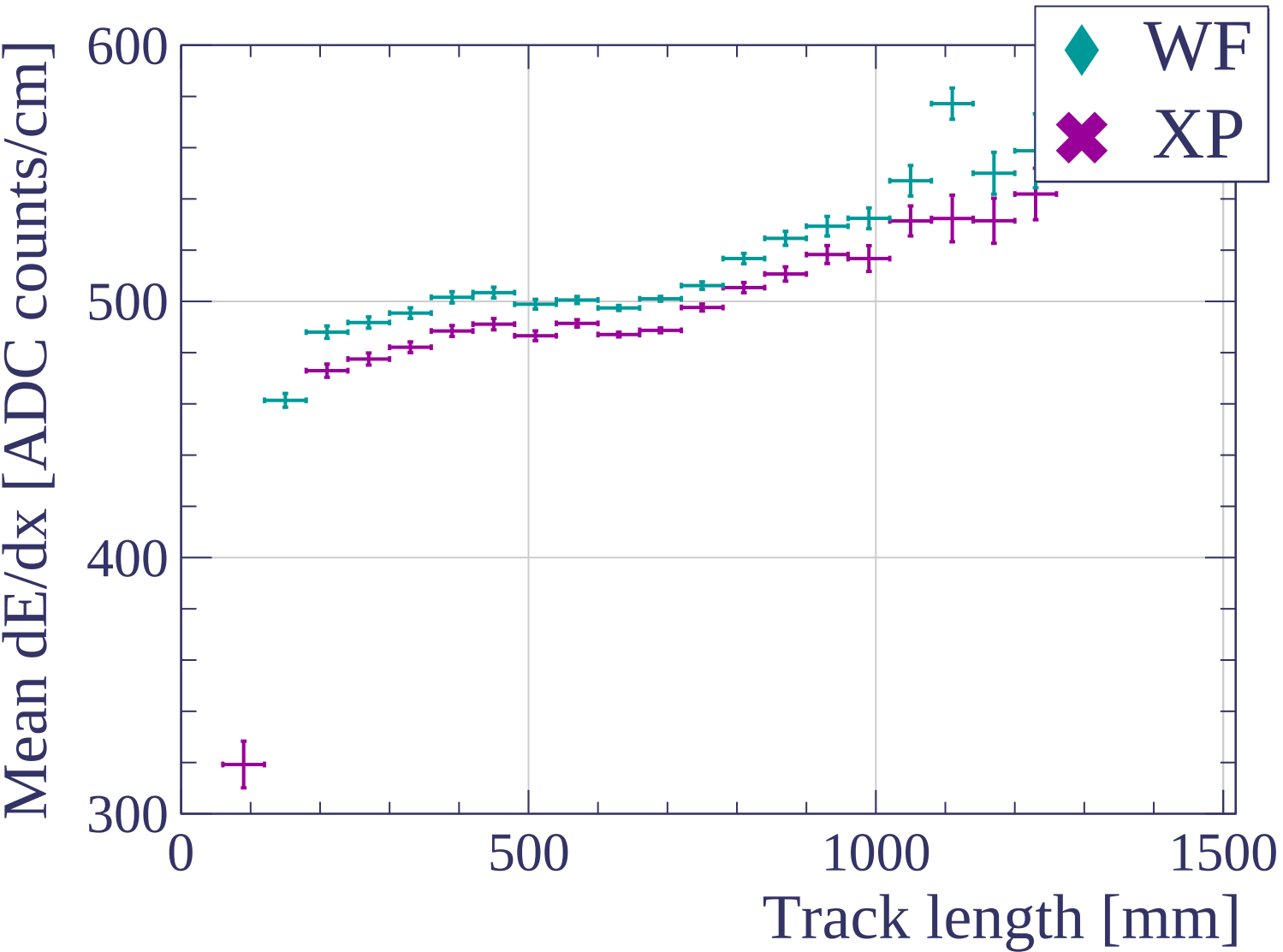


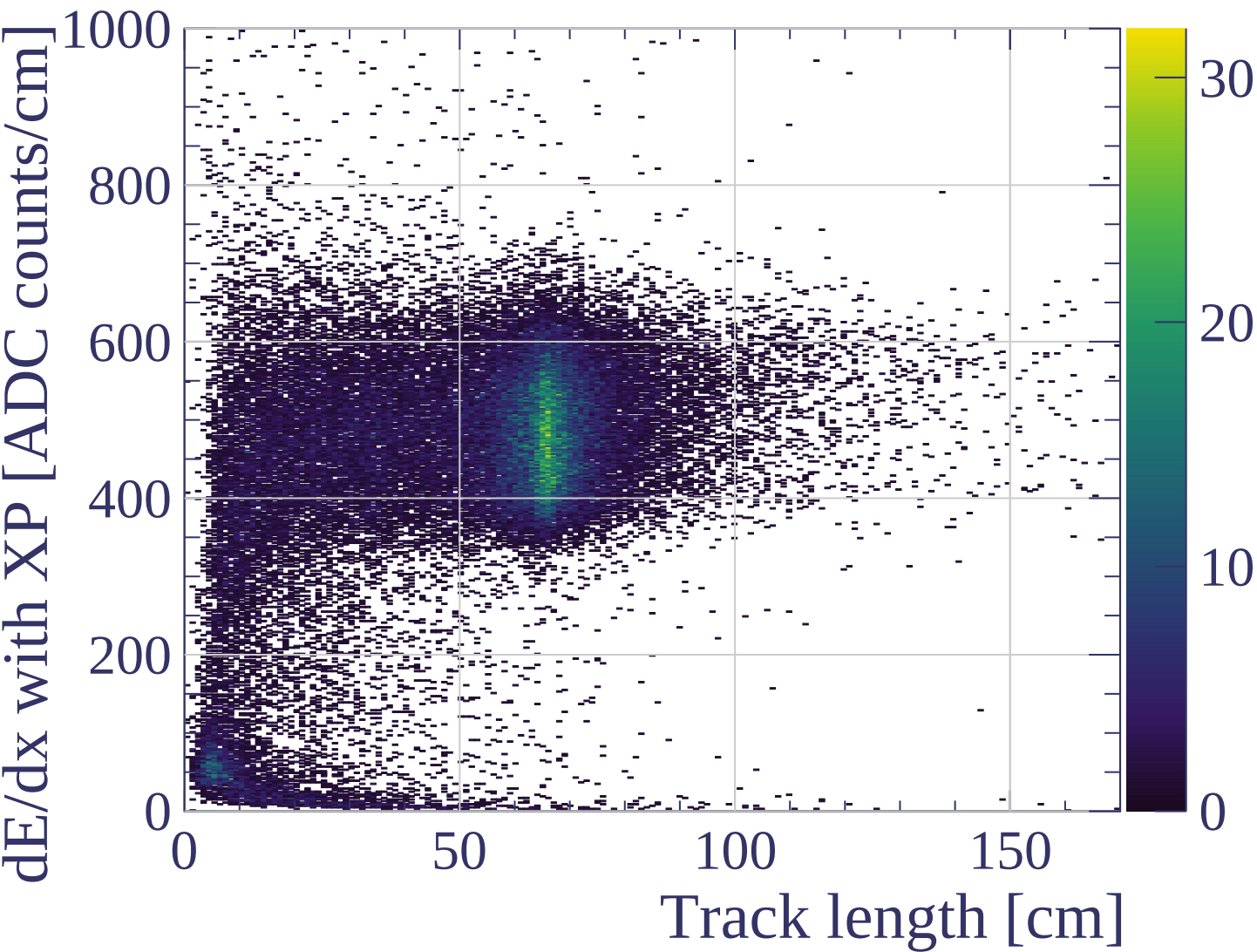


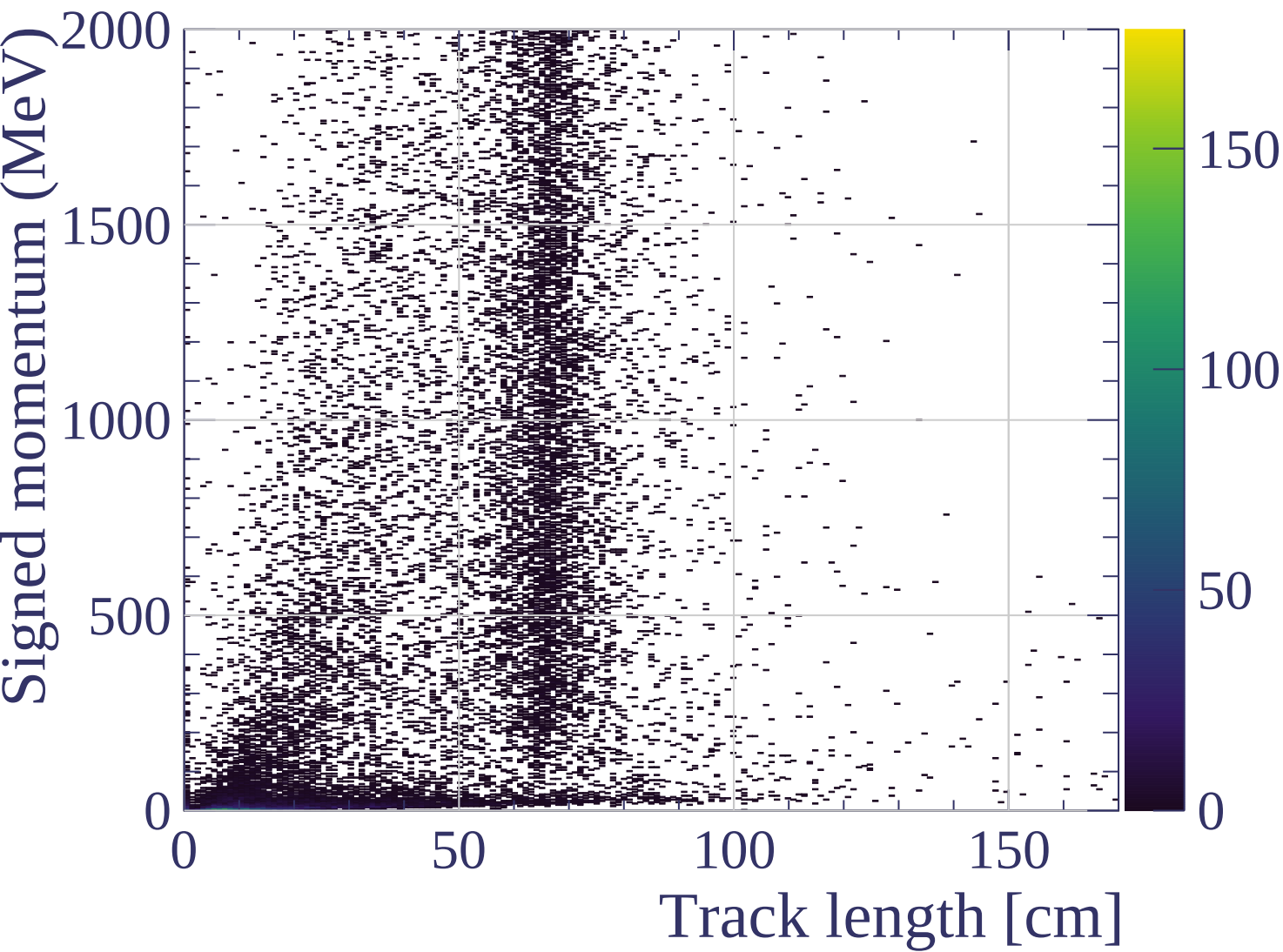


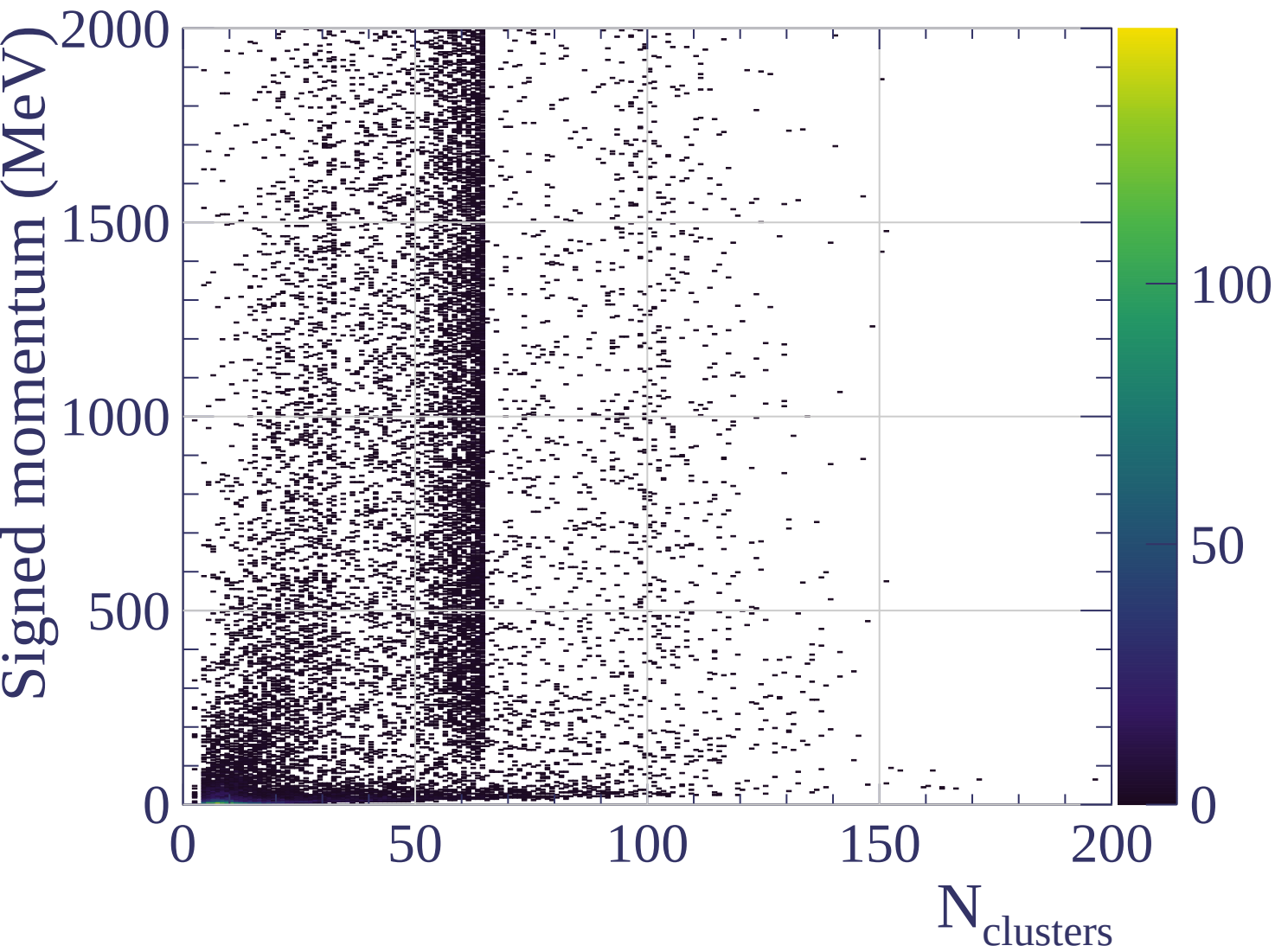


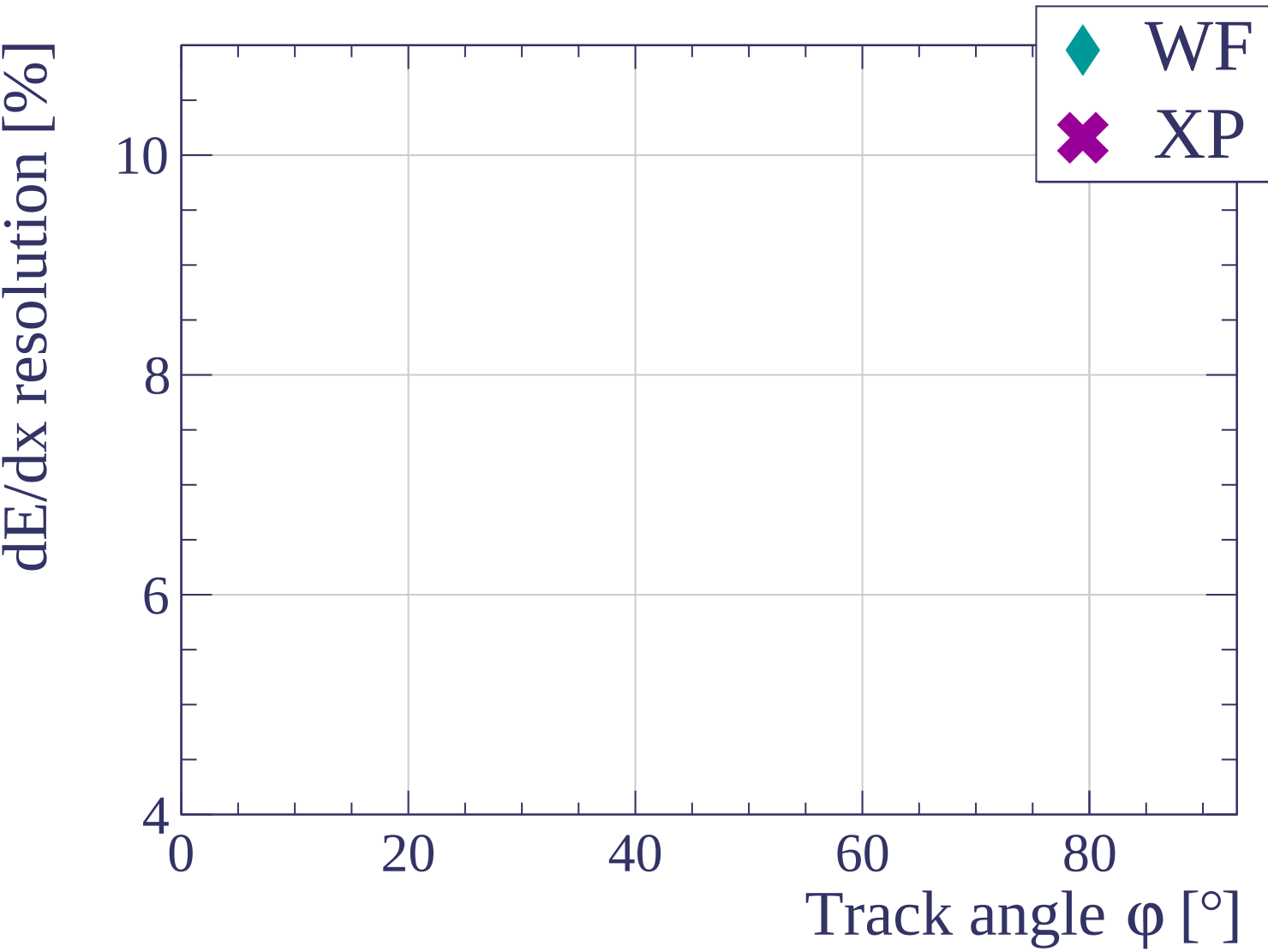


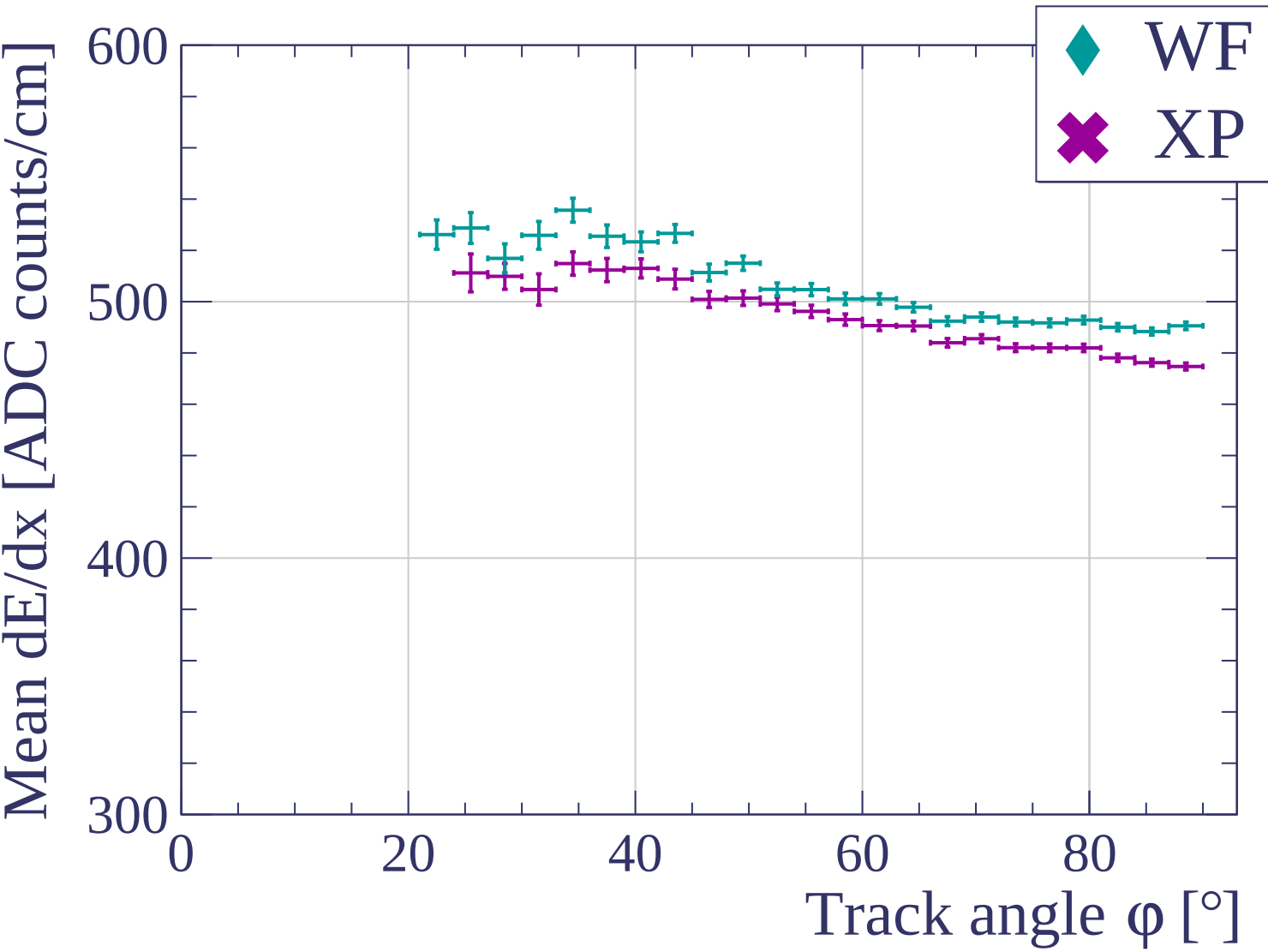


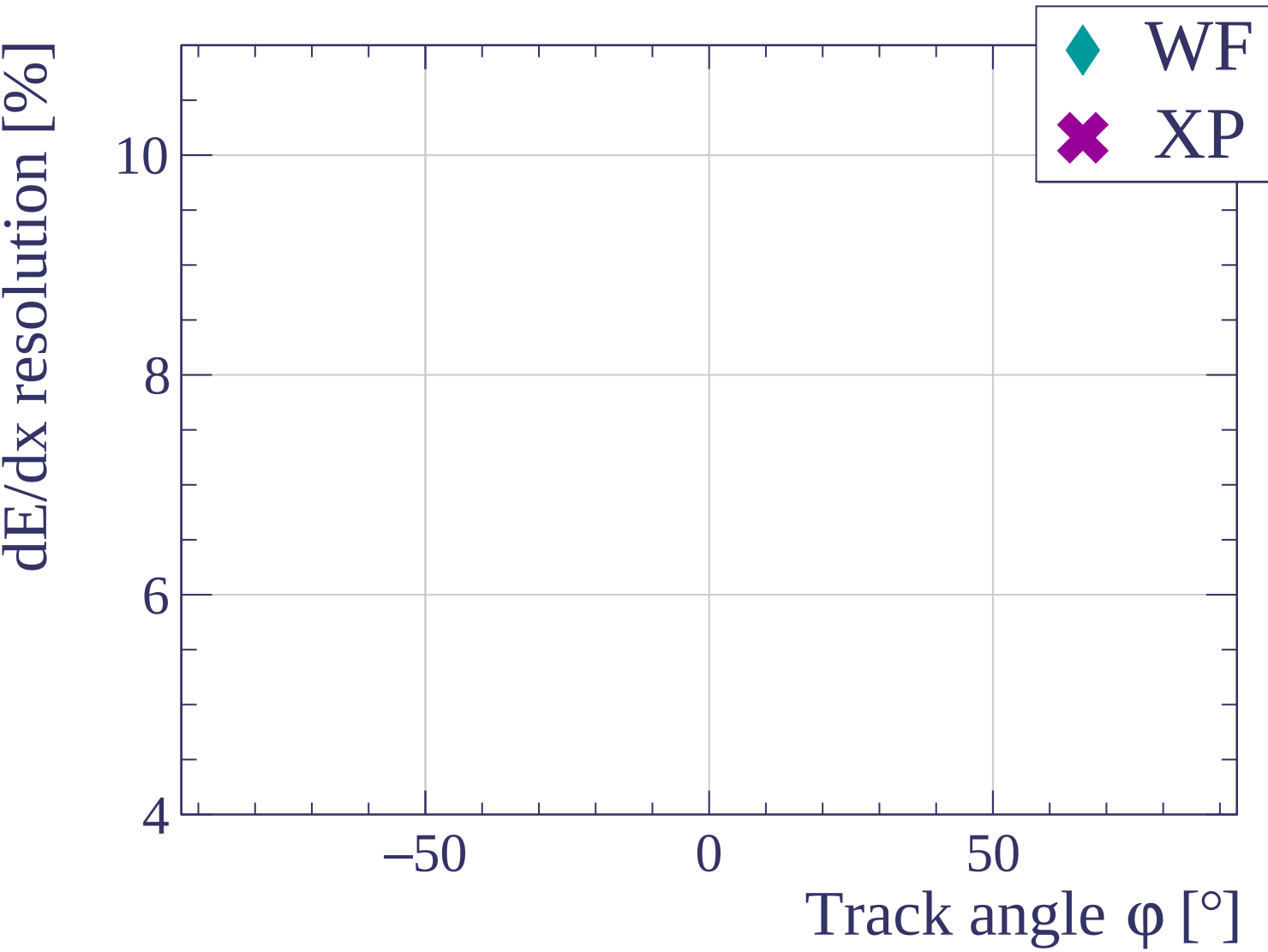


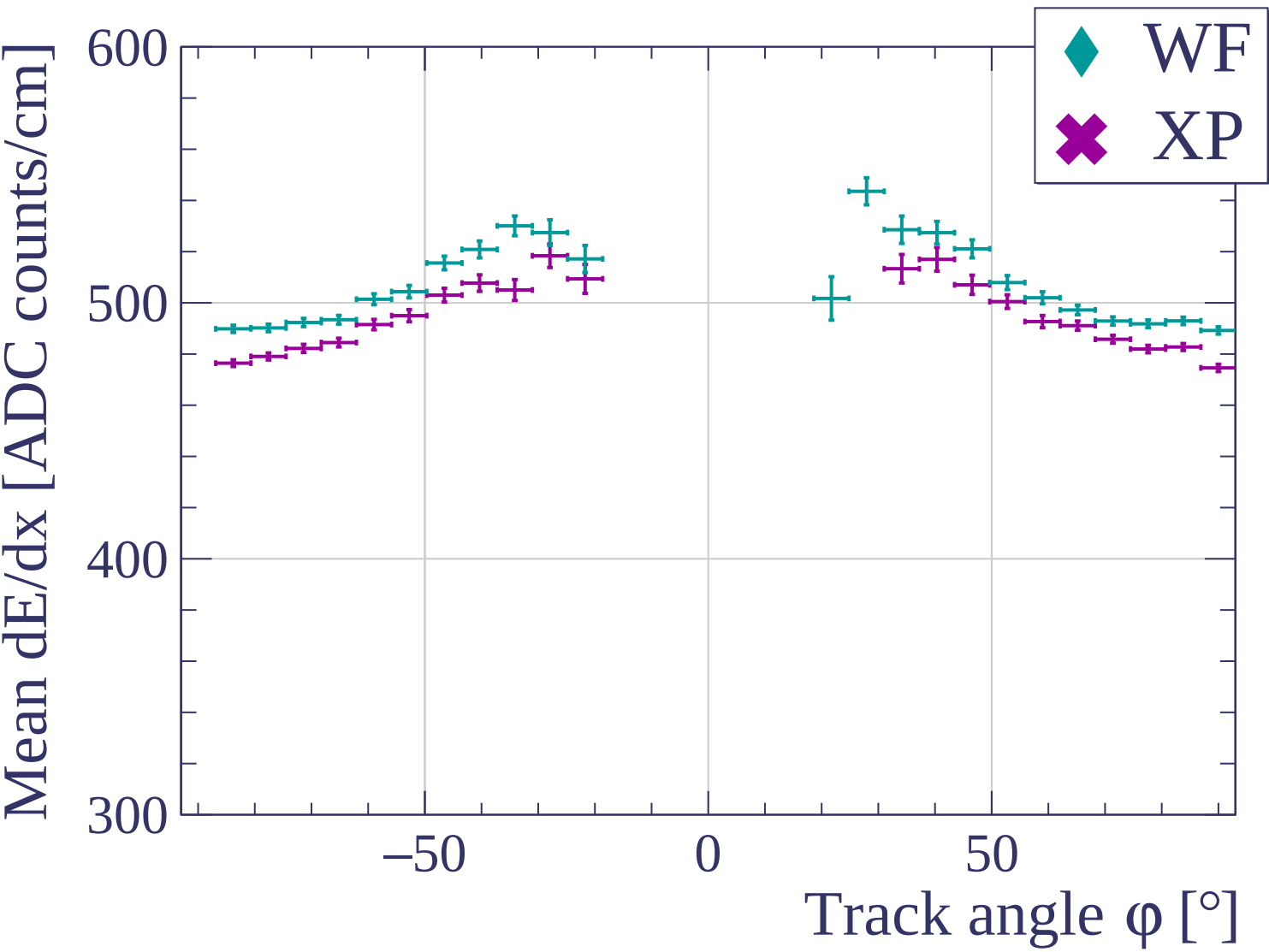


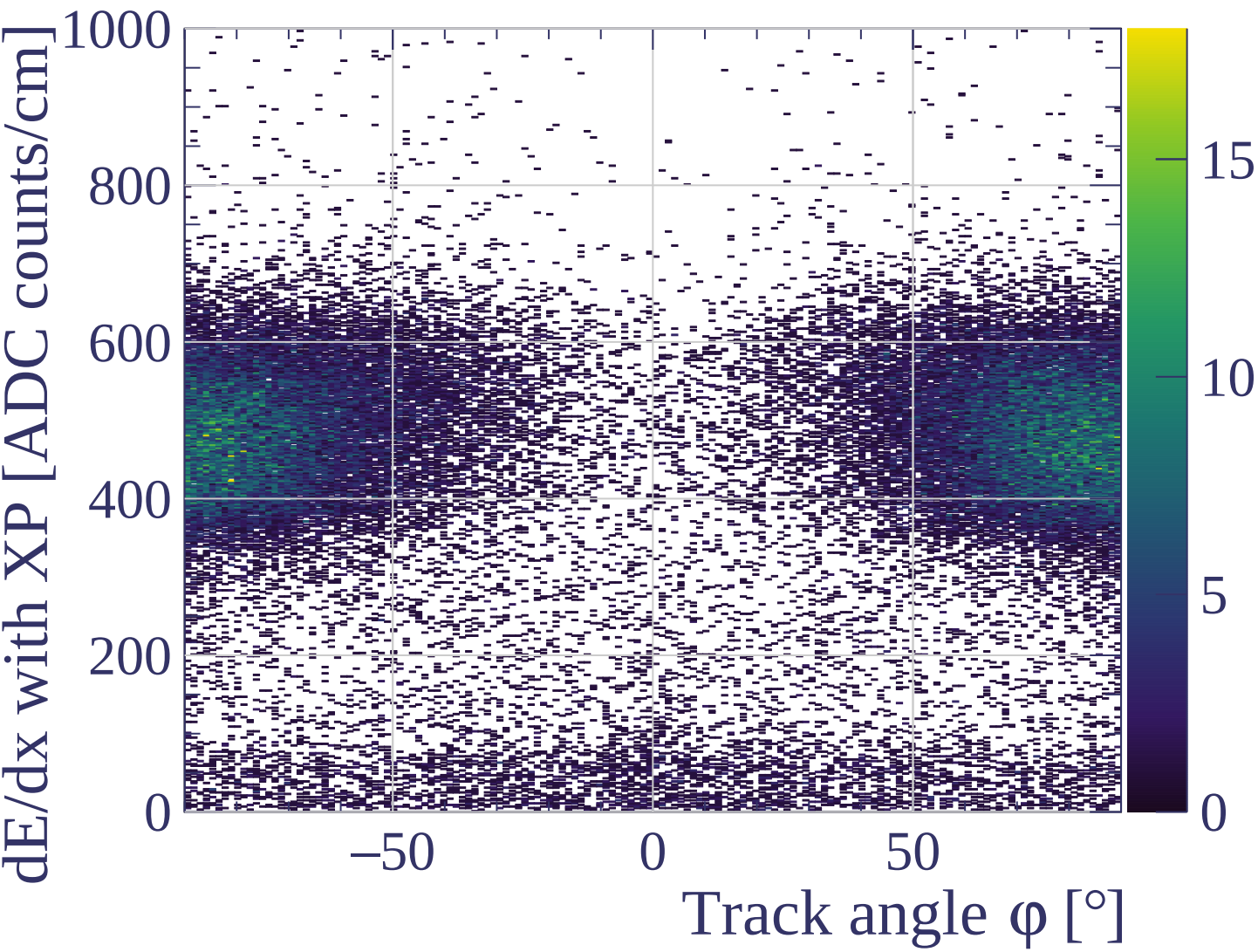


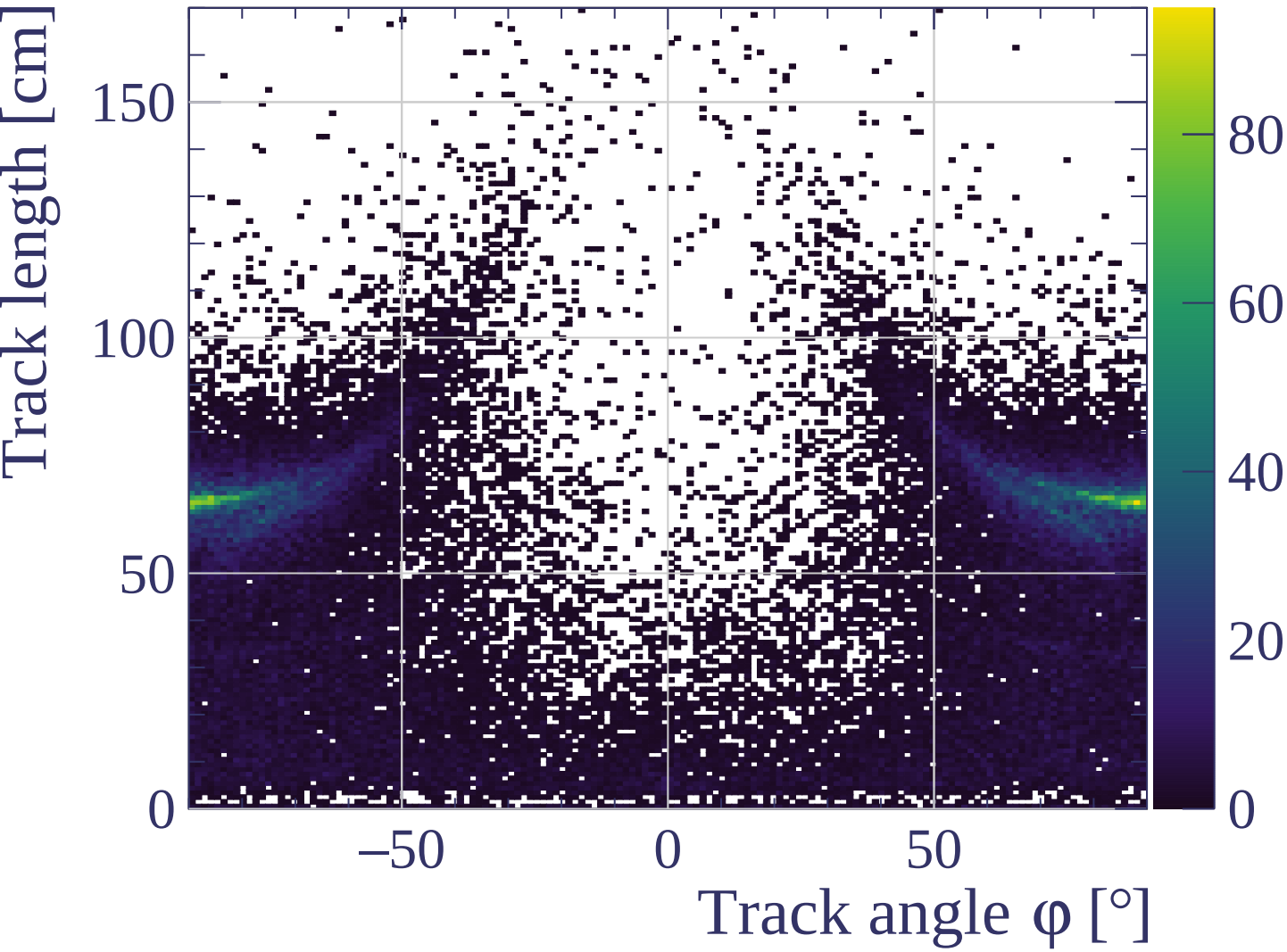


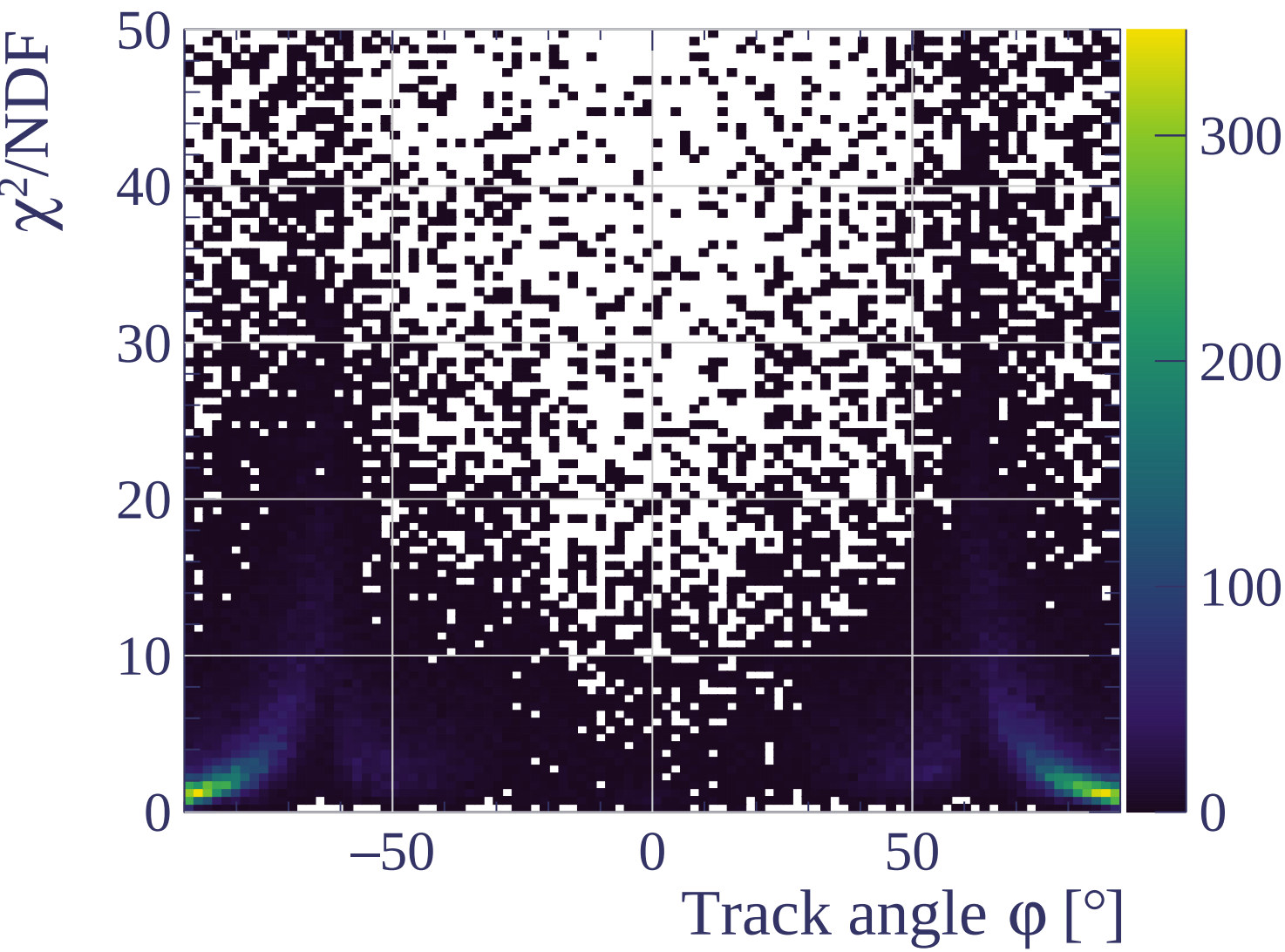


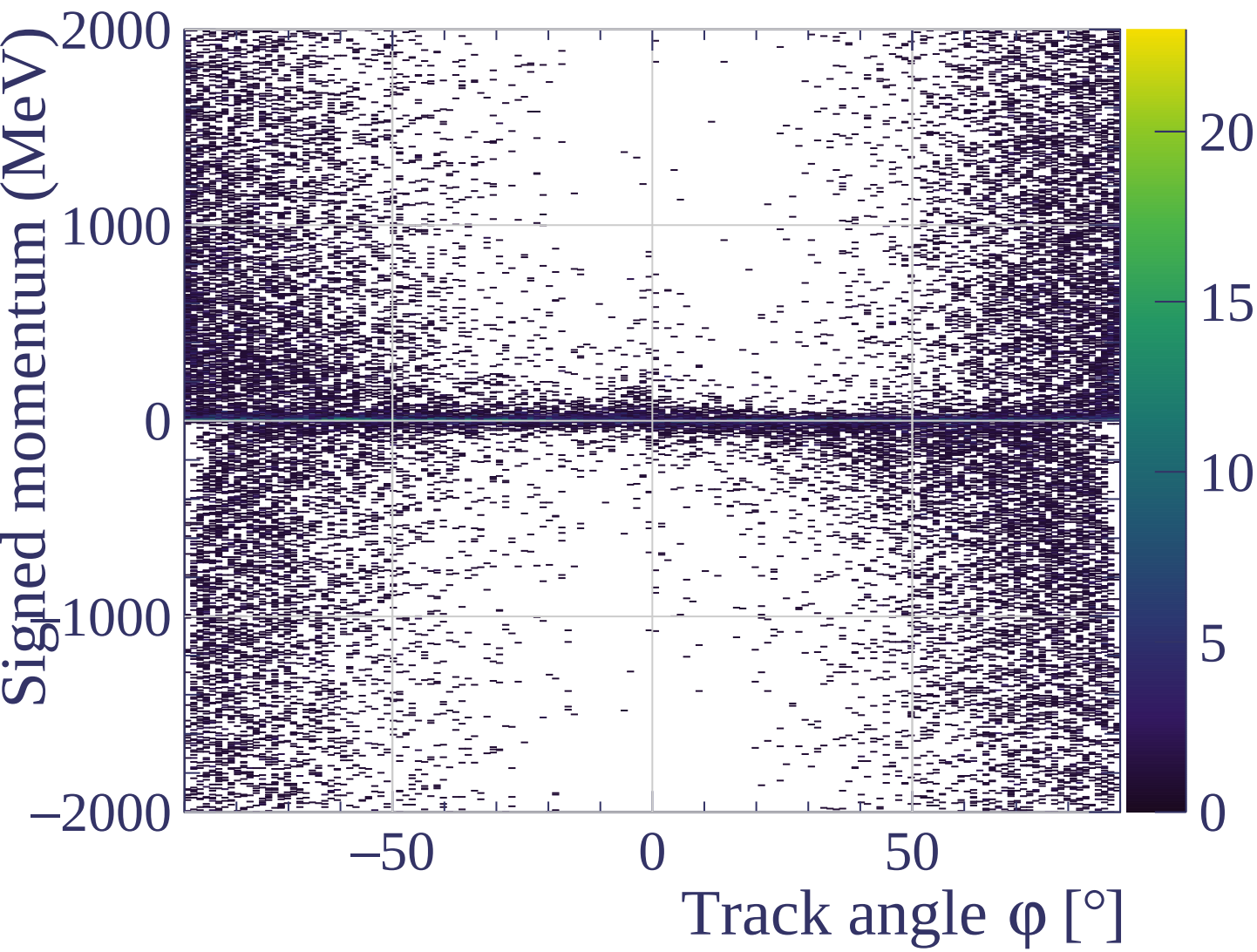




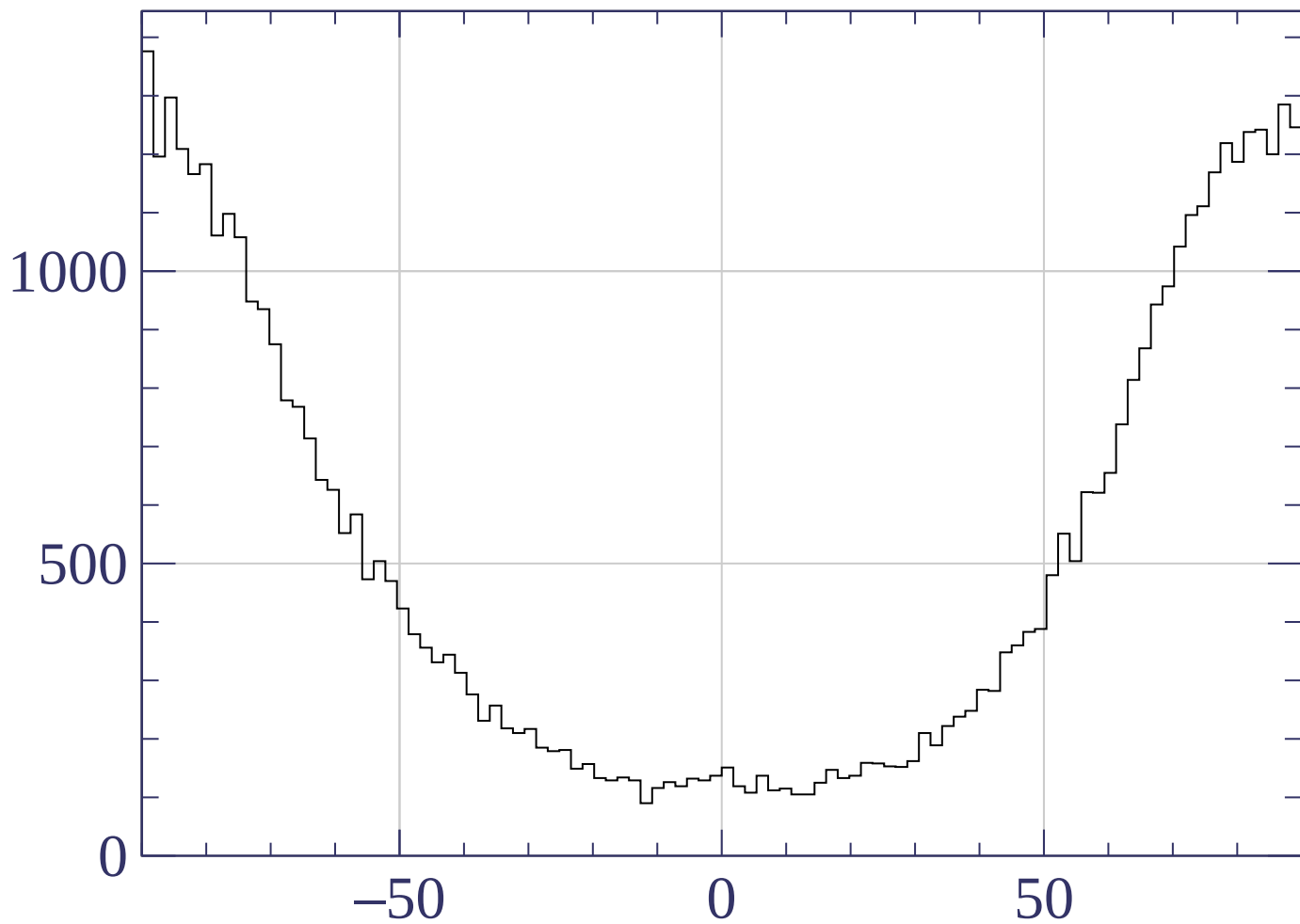




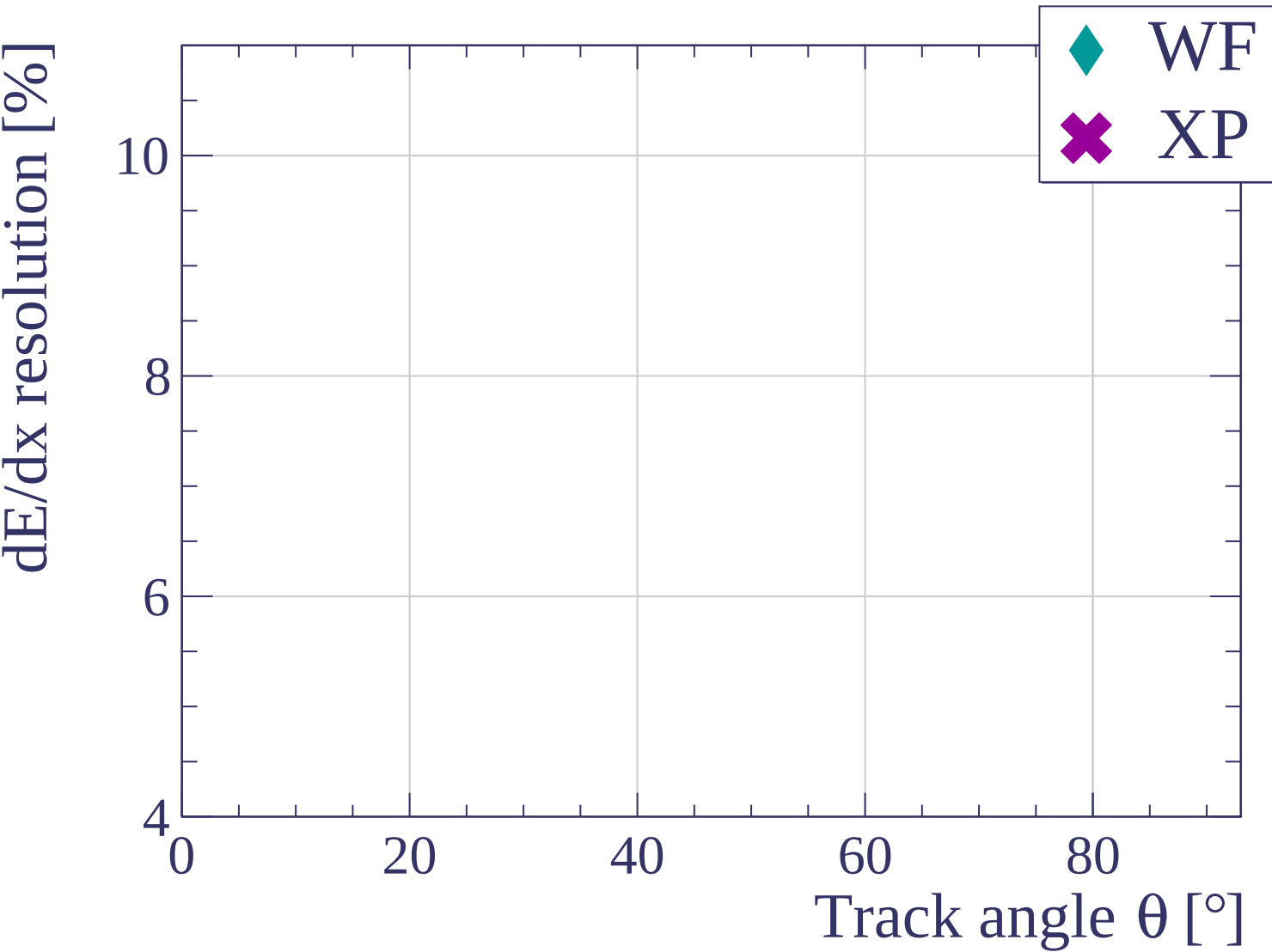


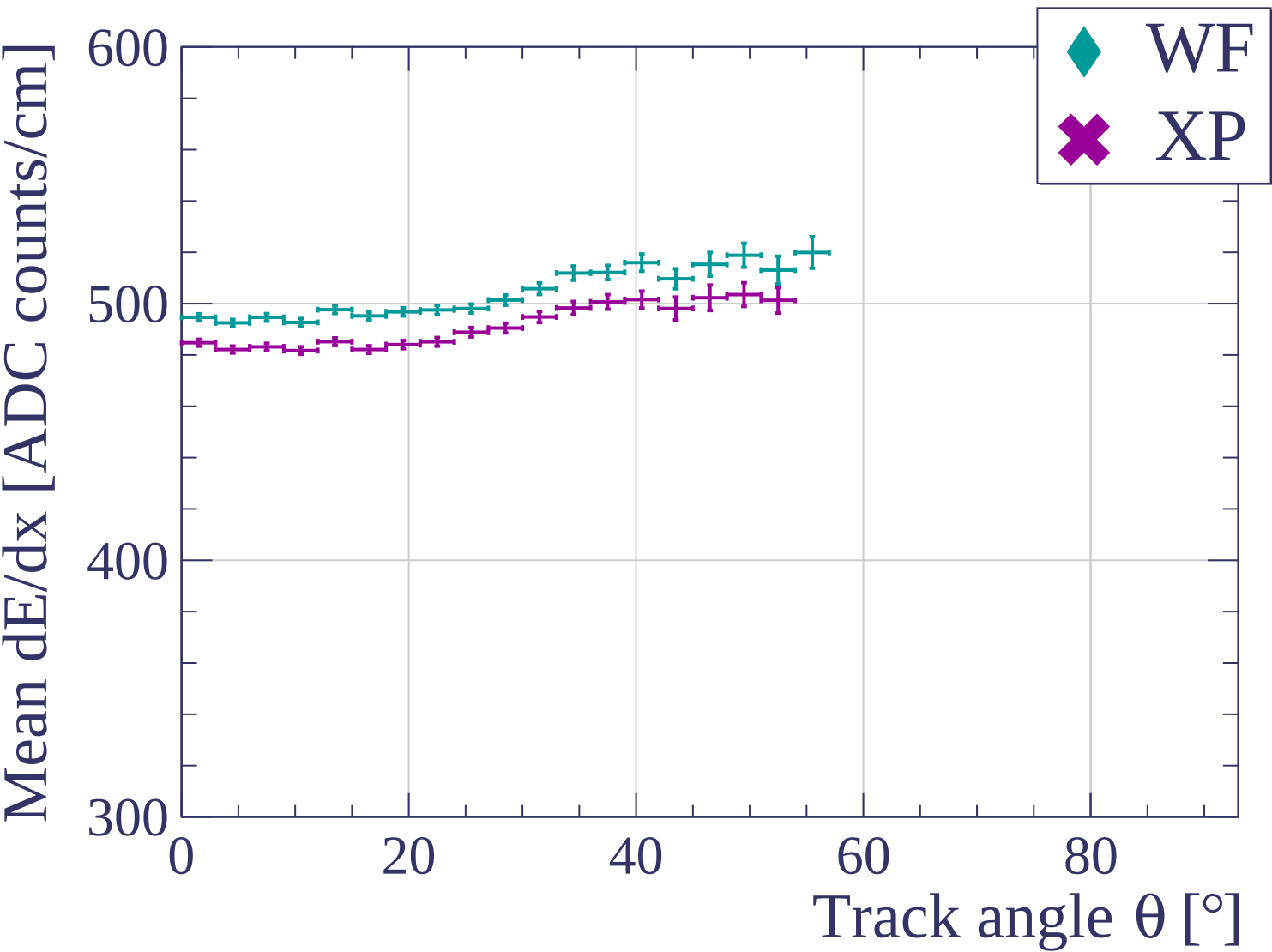


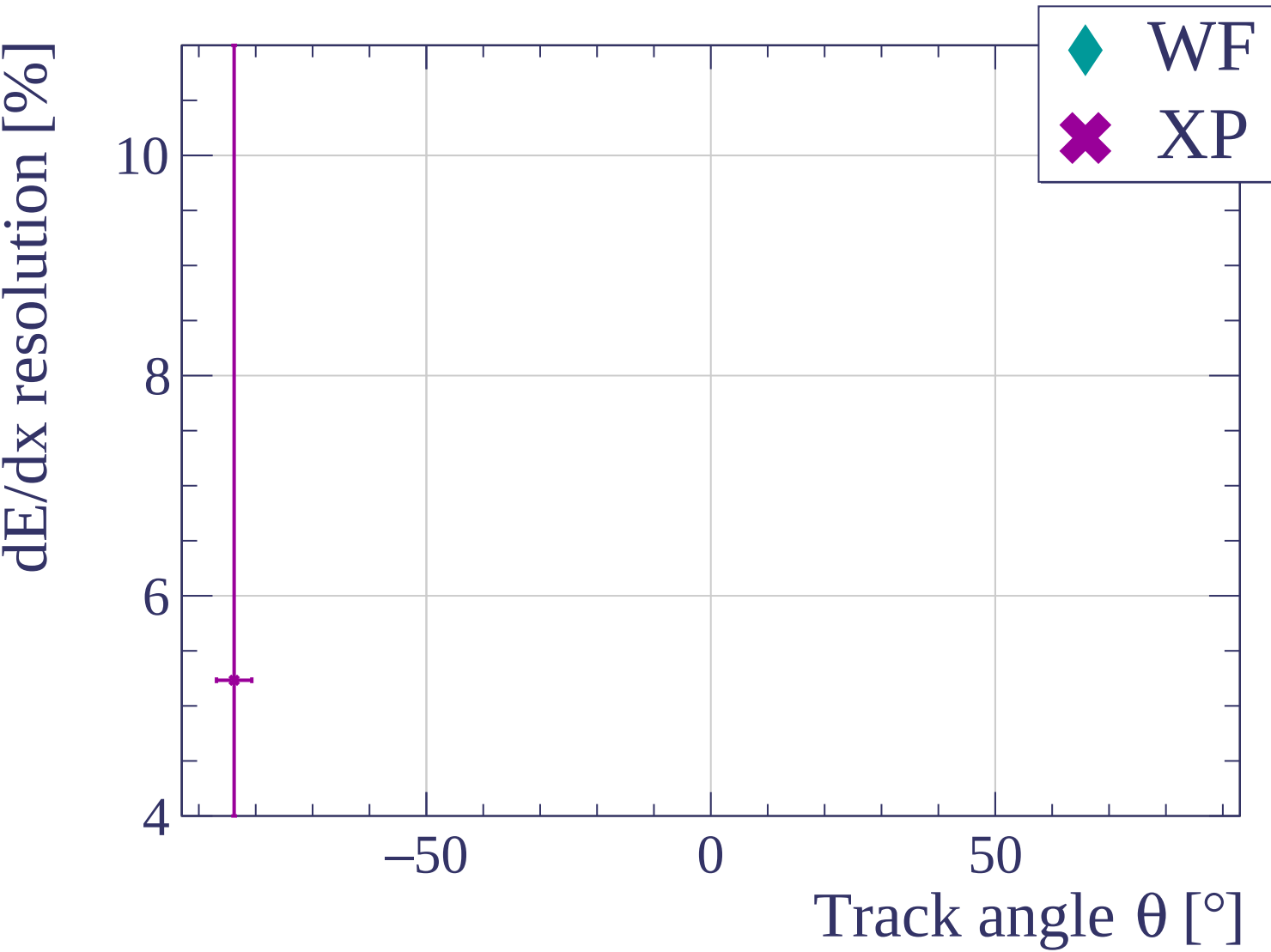
Count

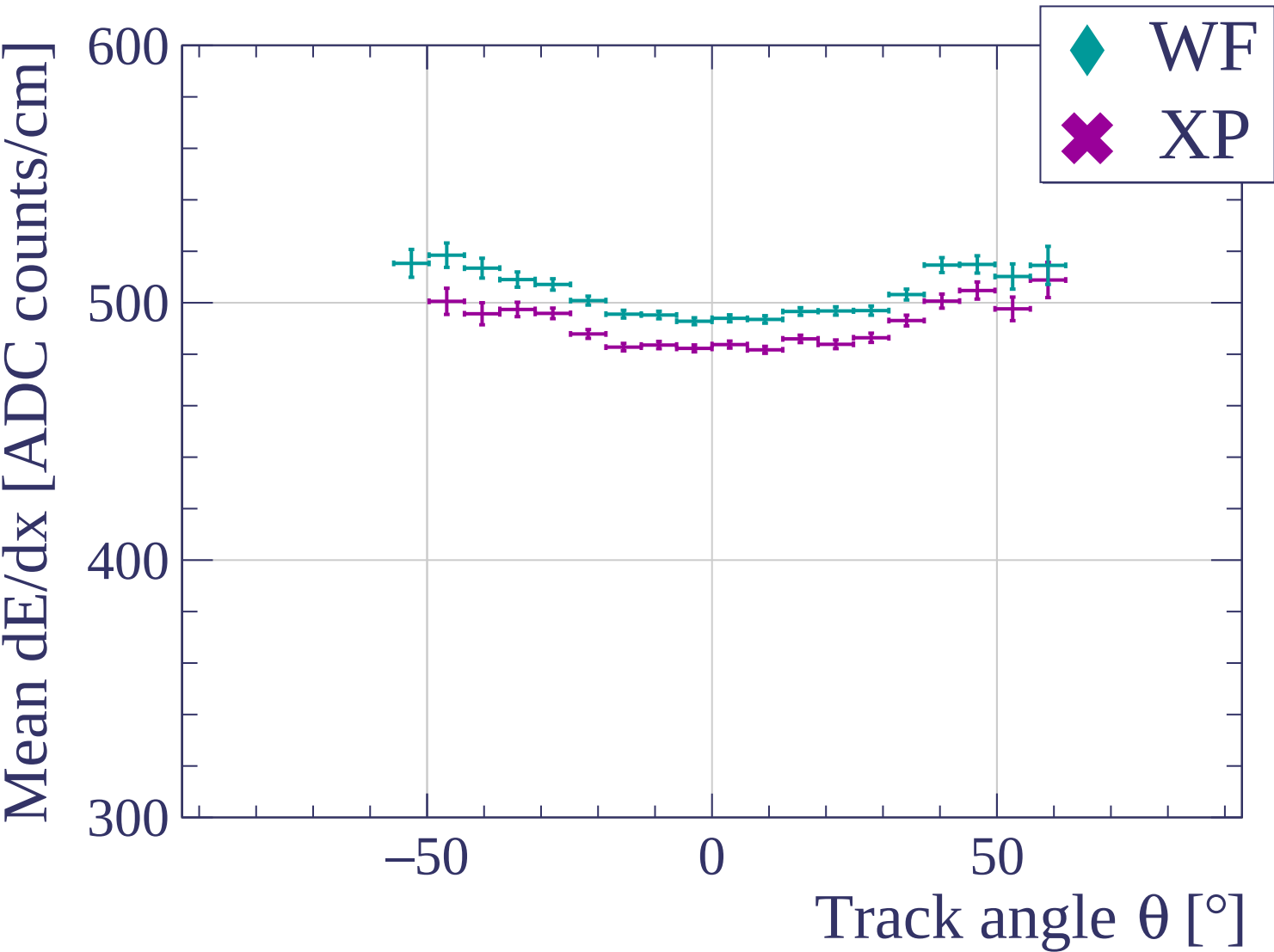


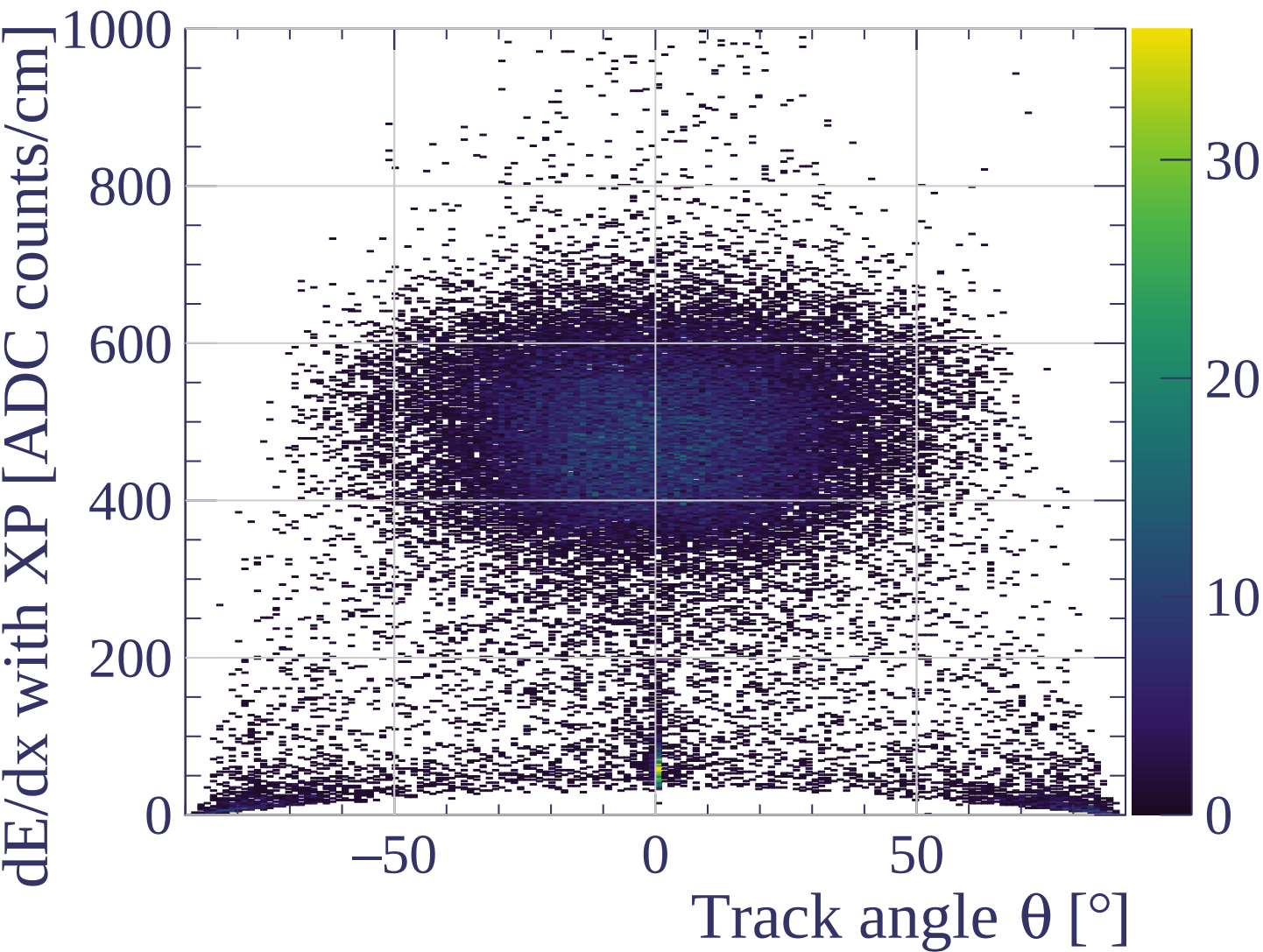
Track angle ϕ [°] angle

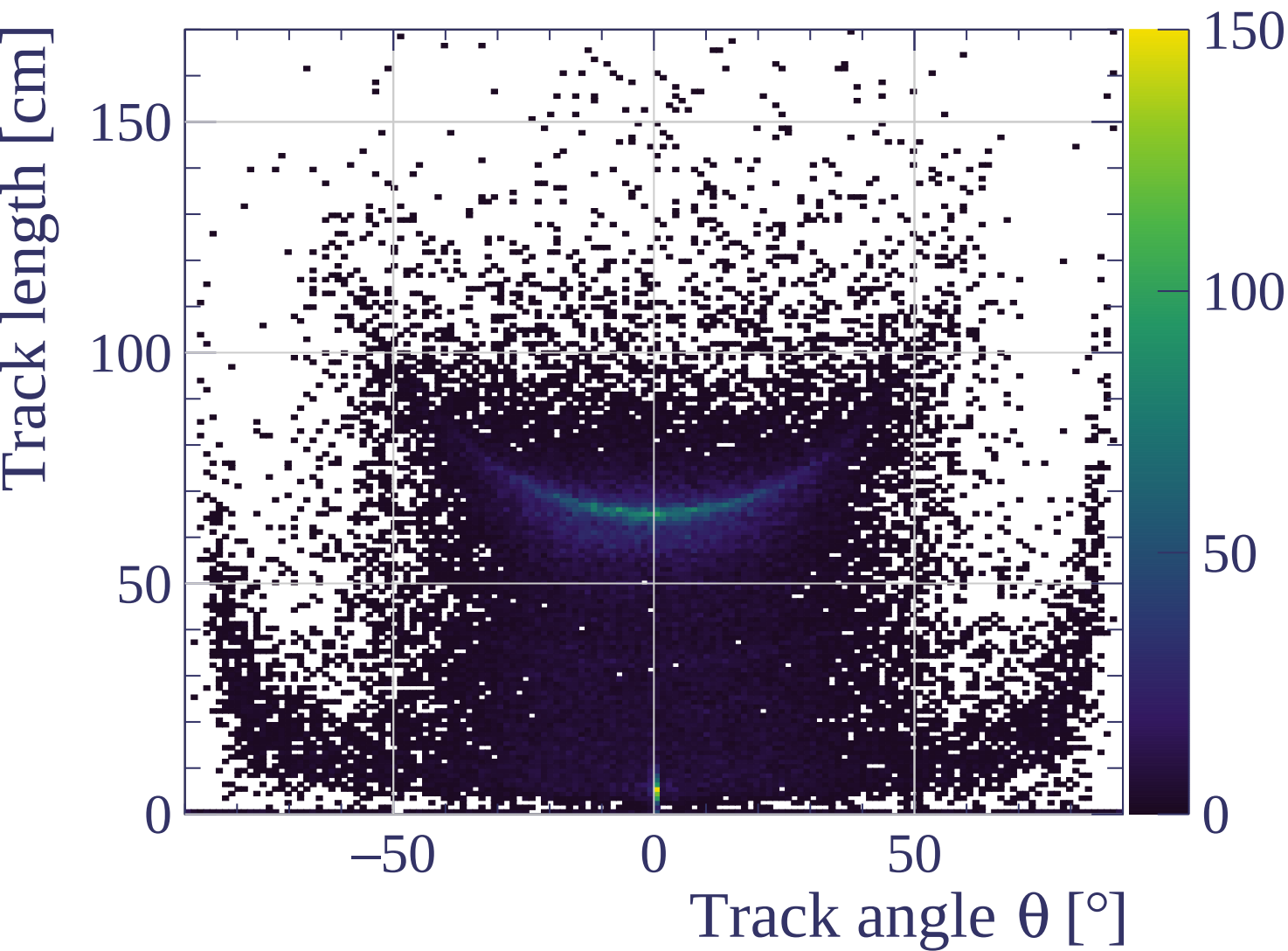


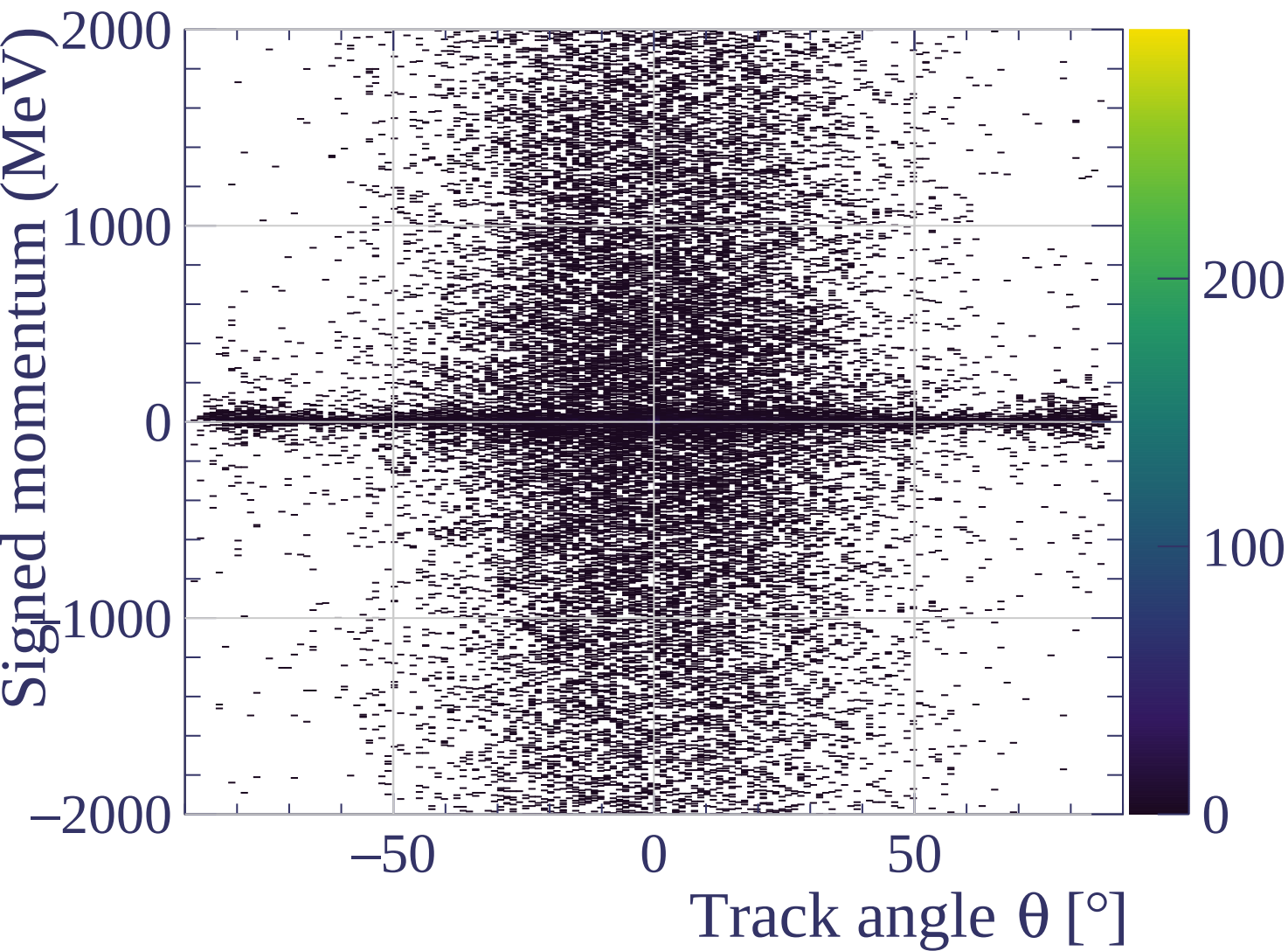


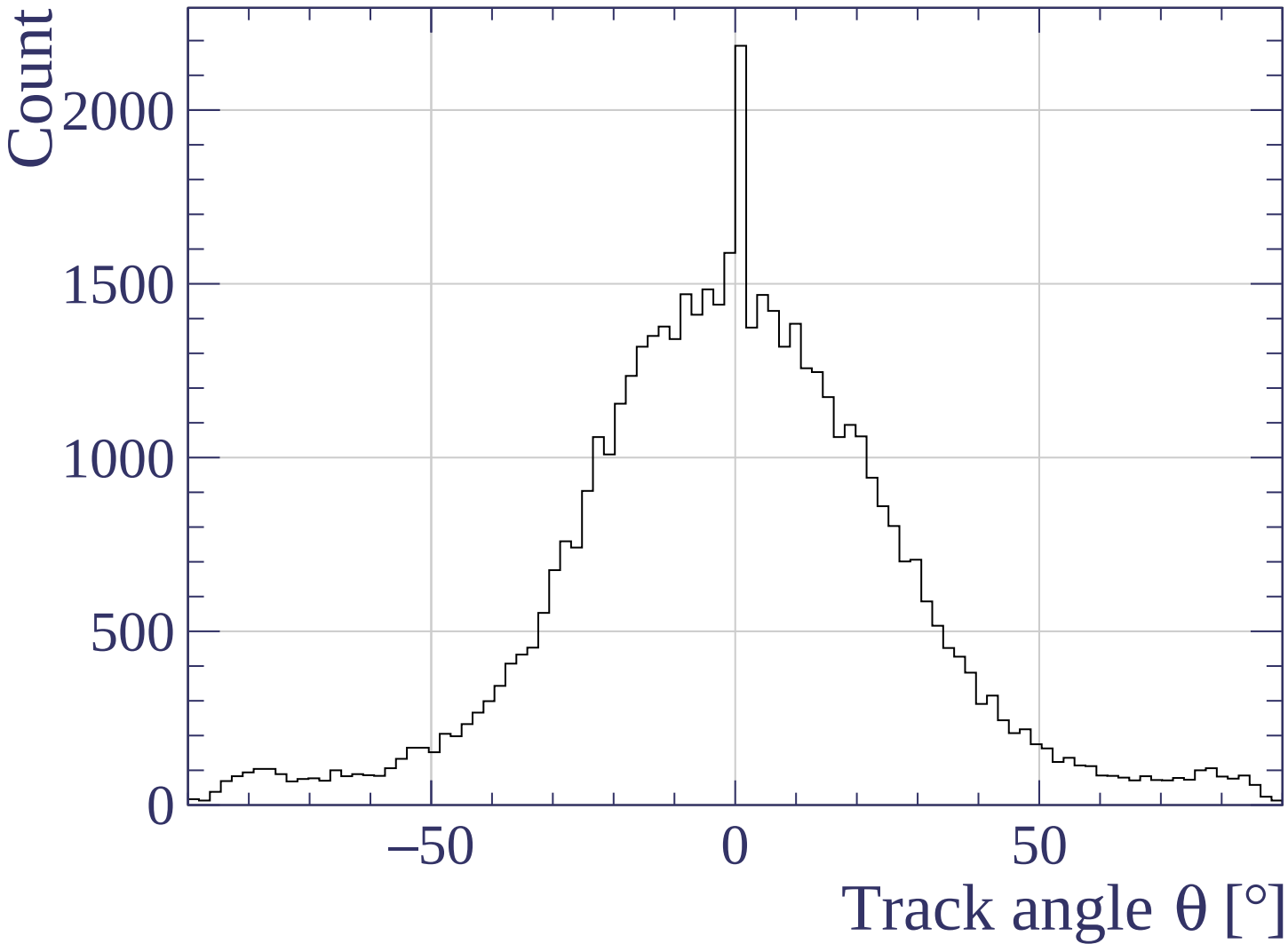




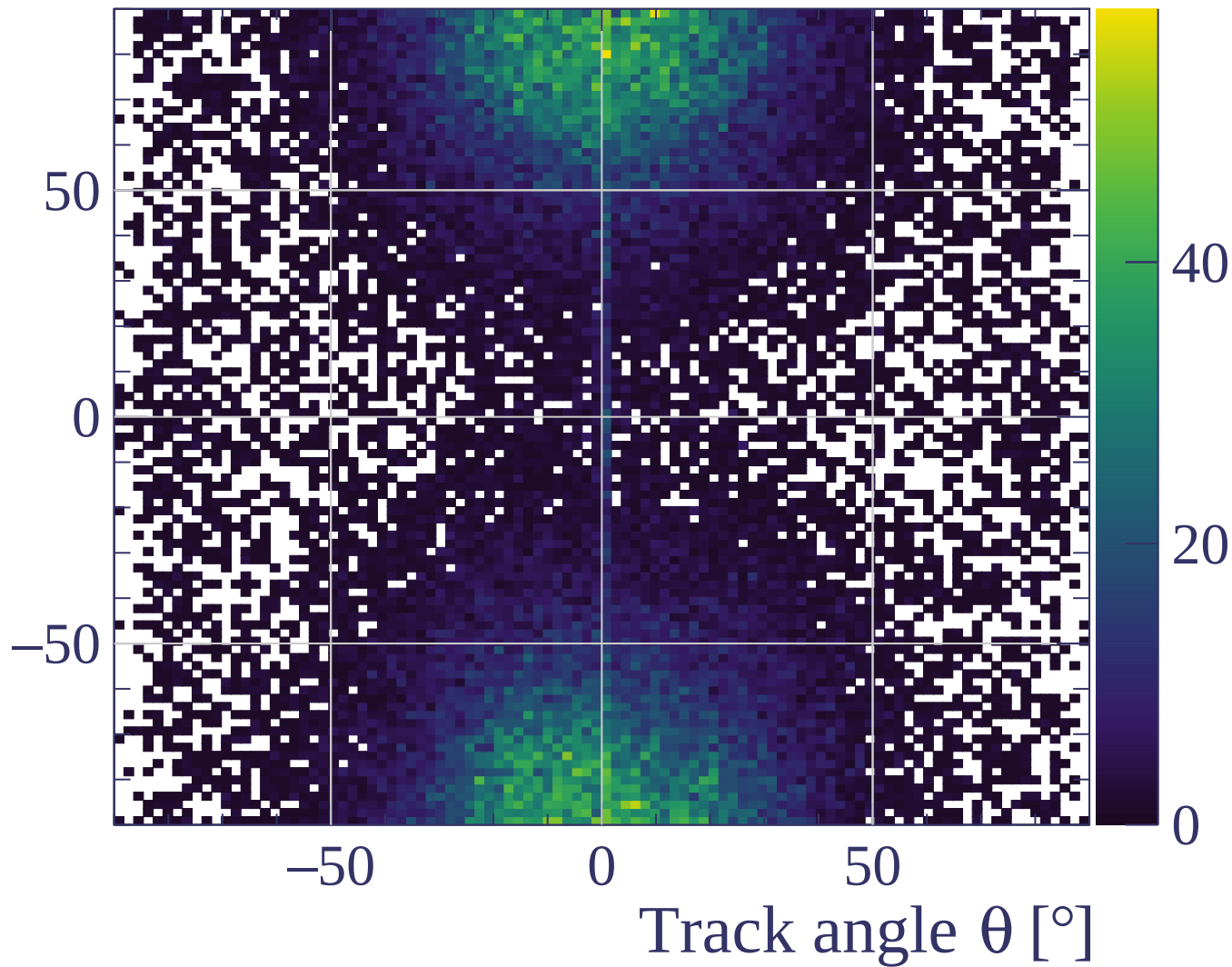




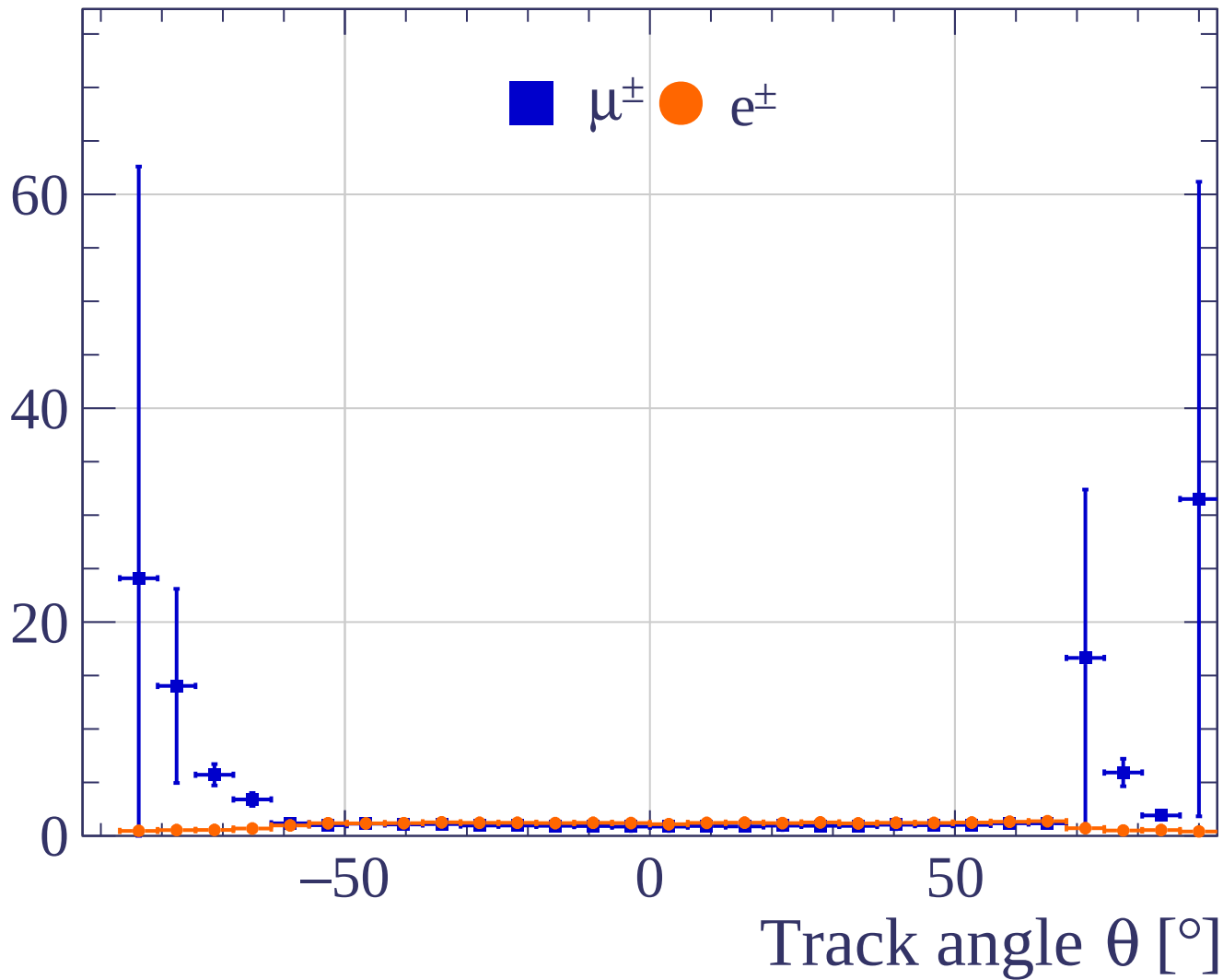




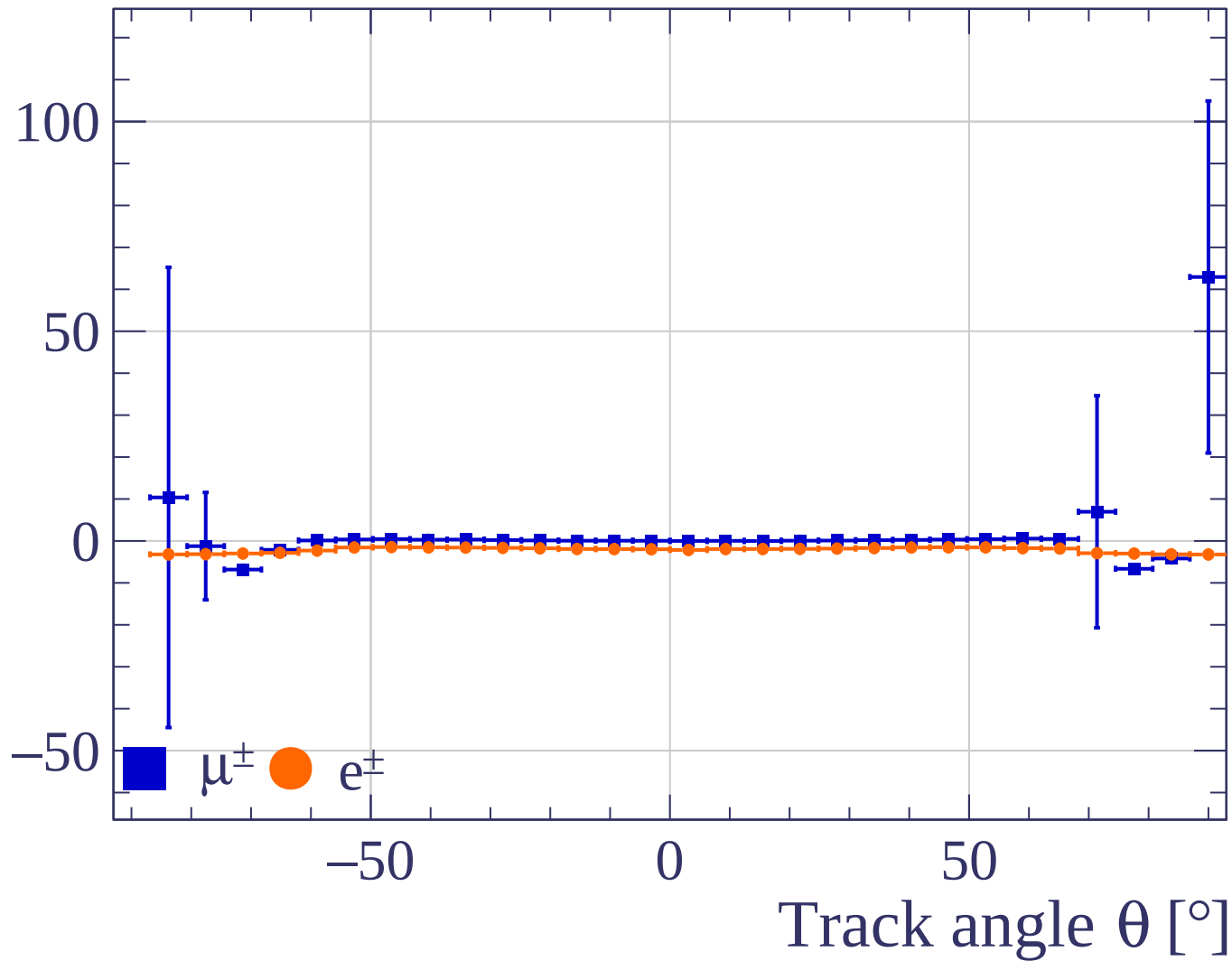
Track angle ϕ [°]



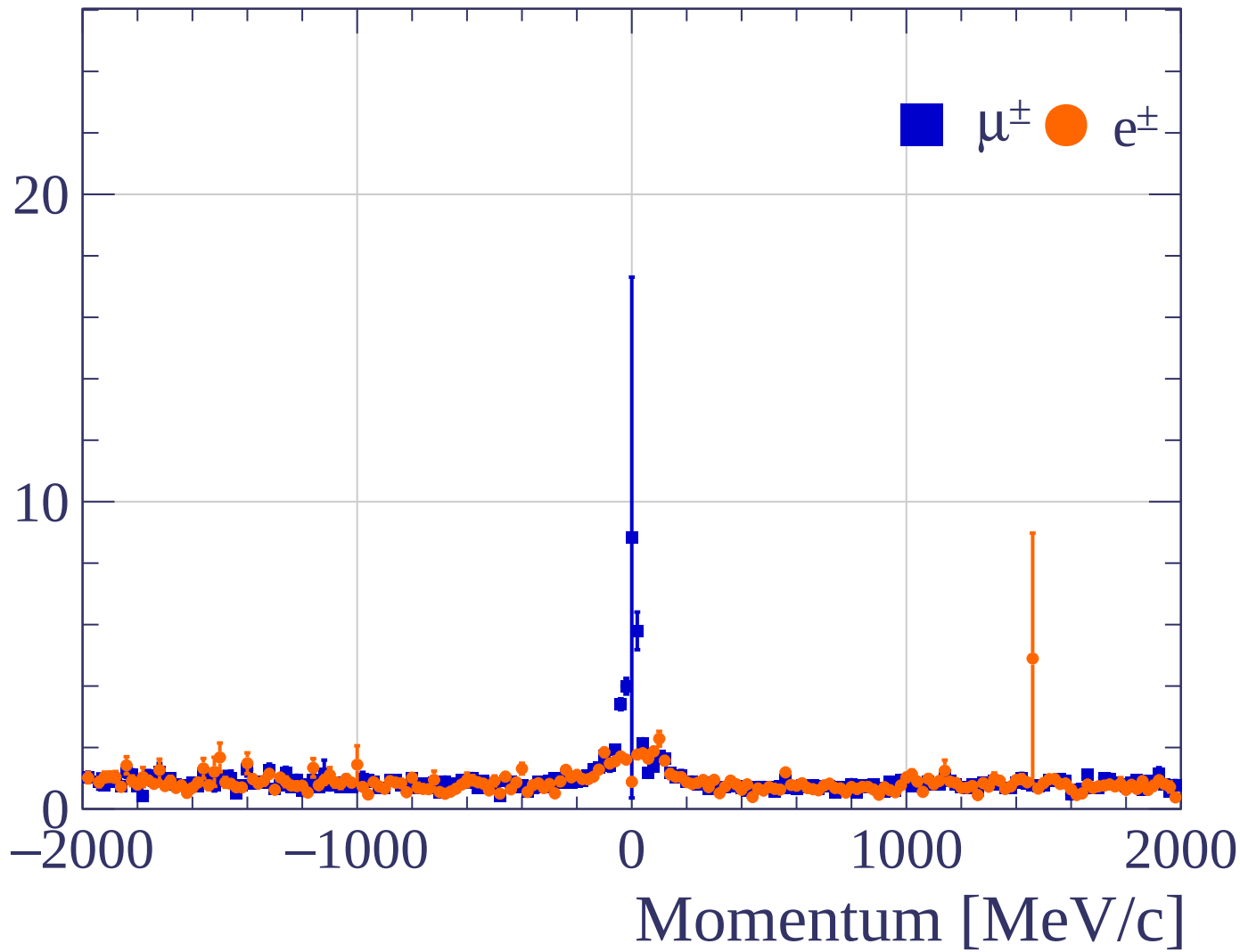
Pulls standard deviation



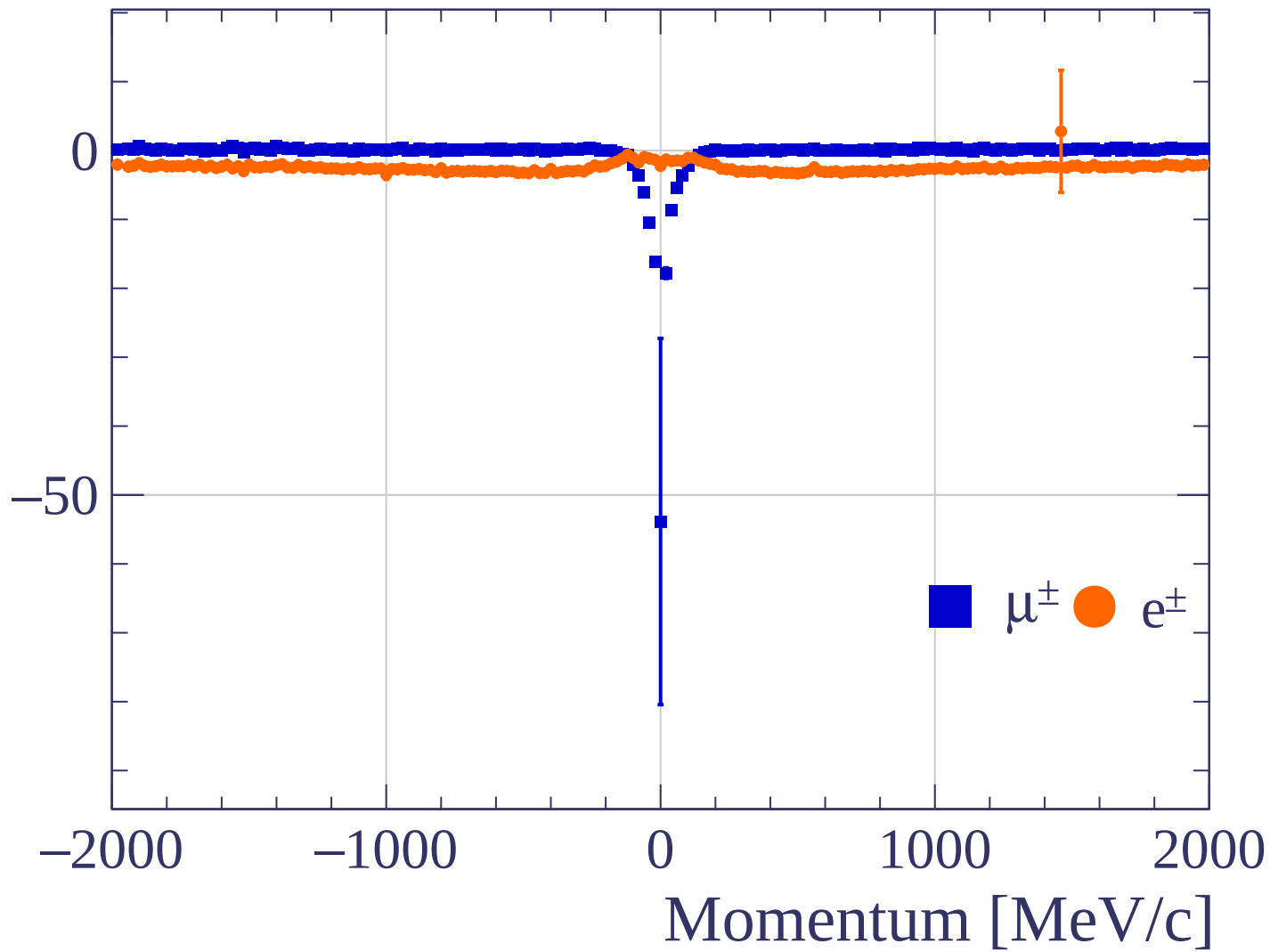
Pulls mean



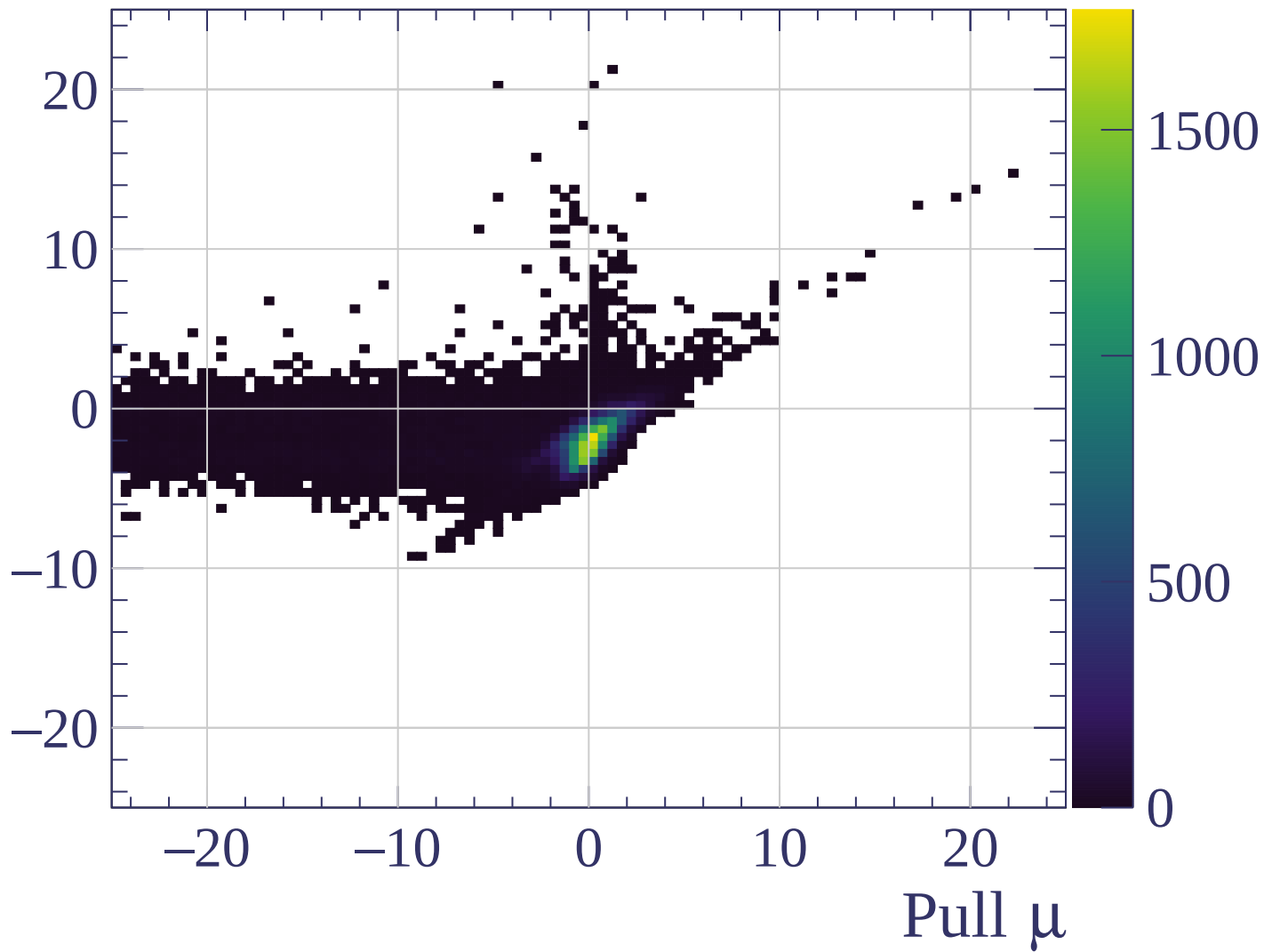
Pulls standard deviation

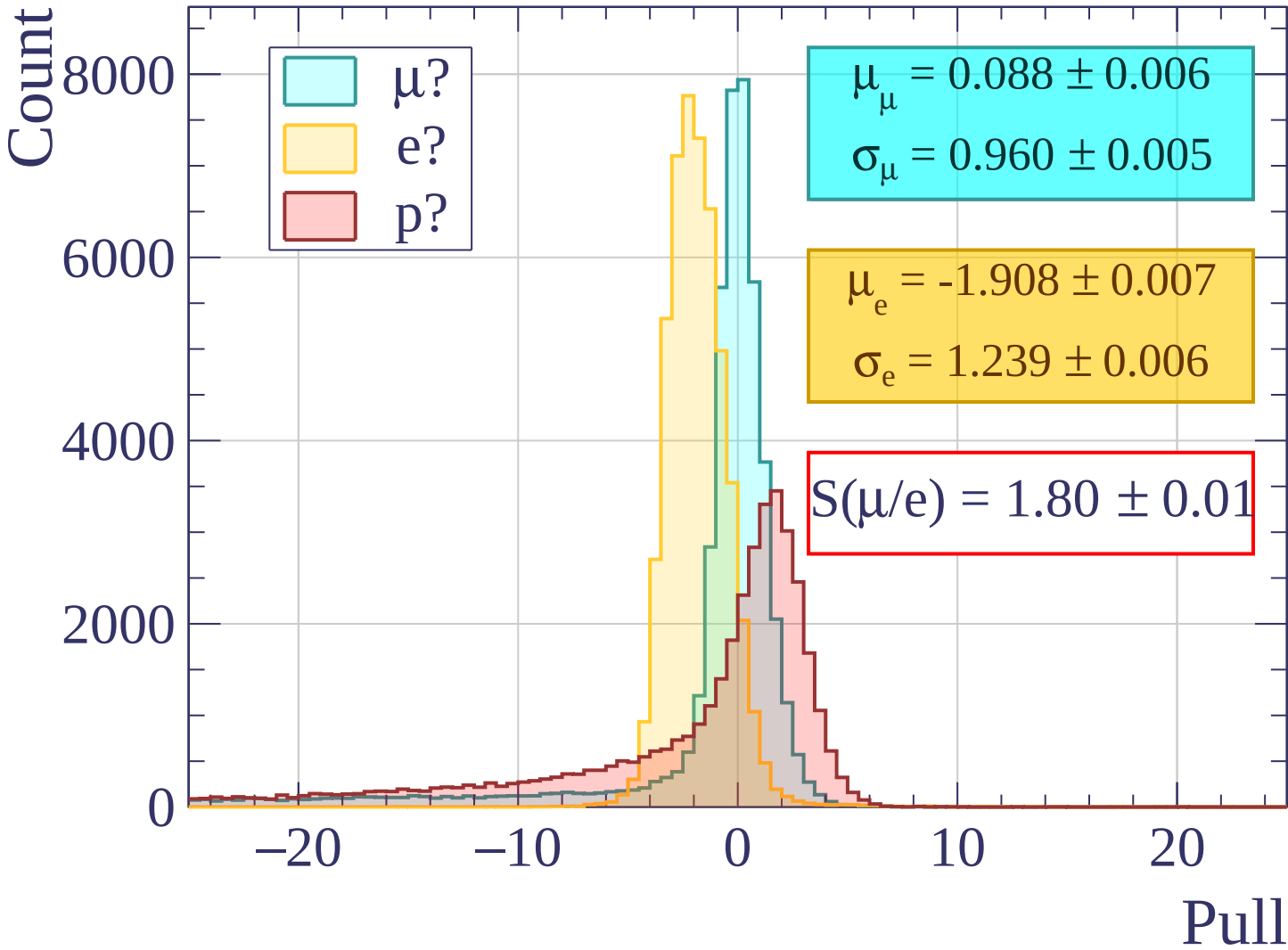


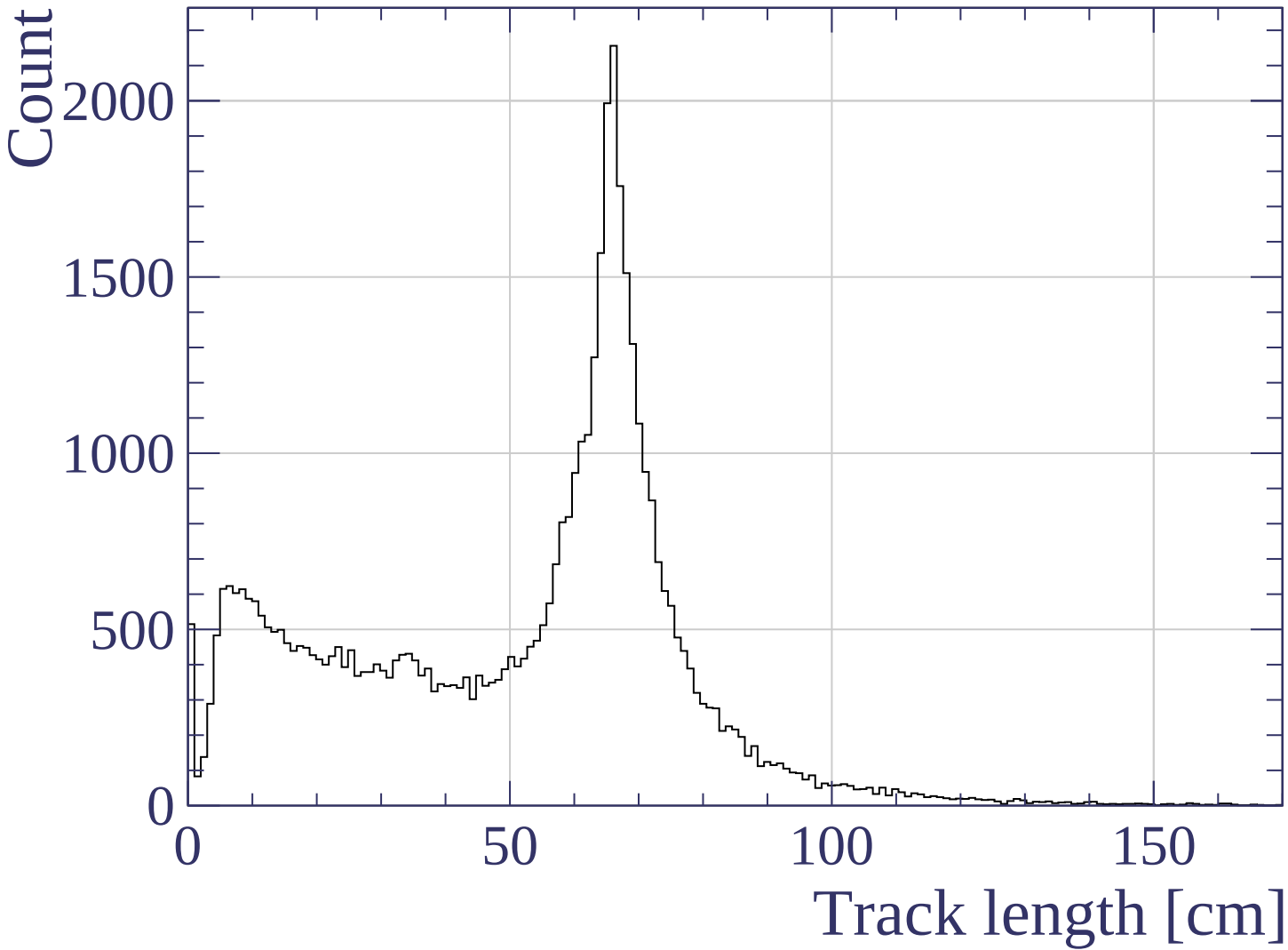
Pulls mean



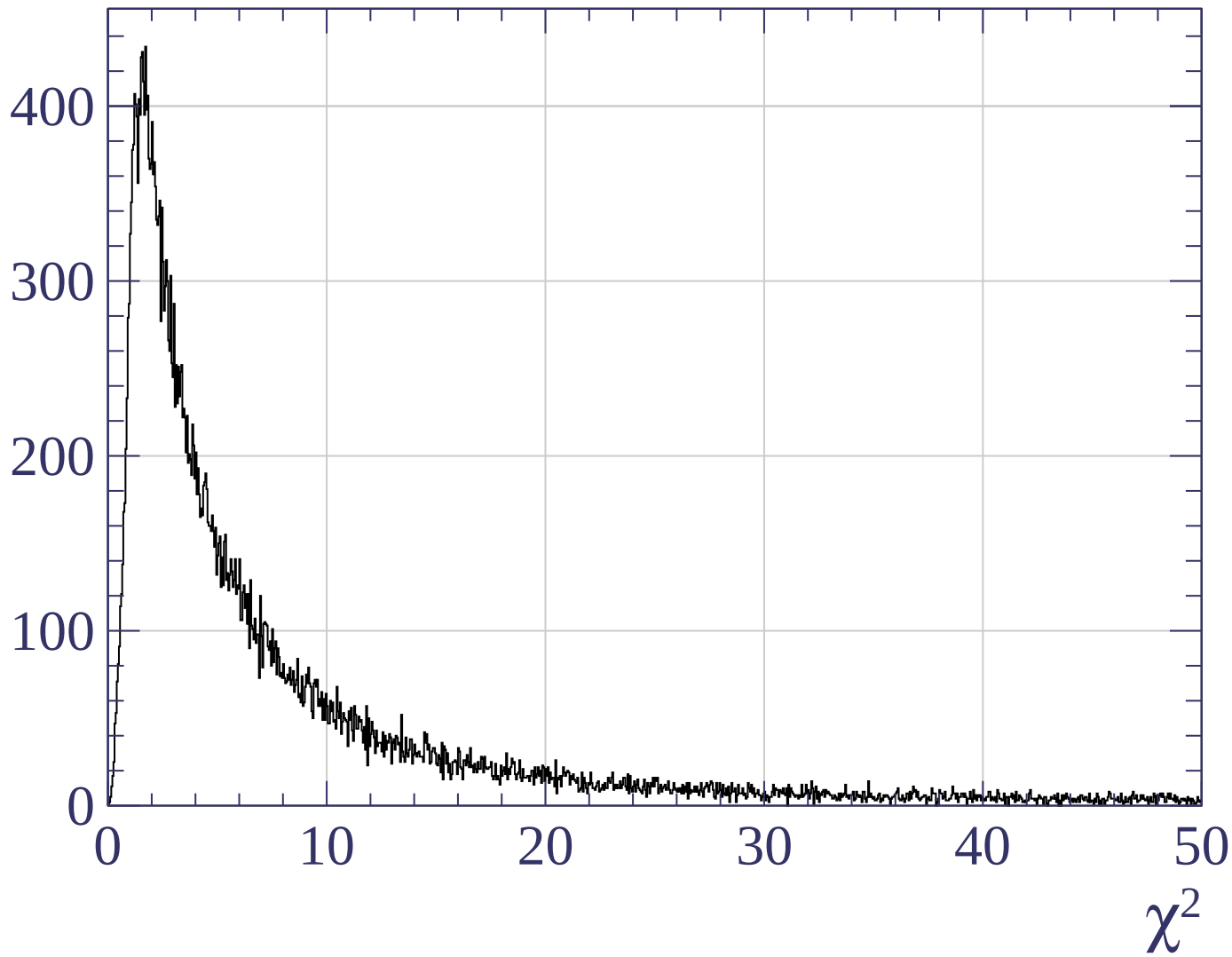
Pull e



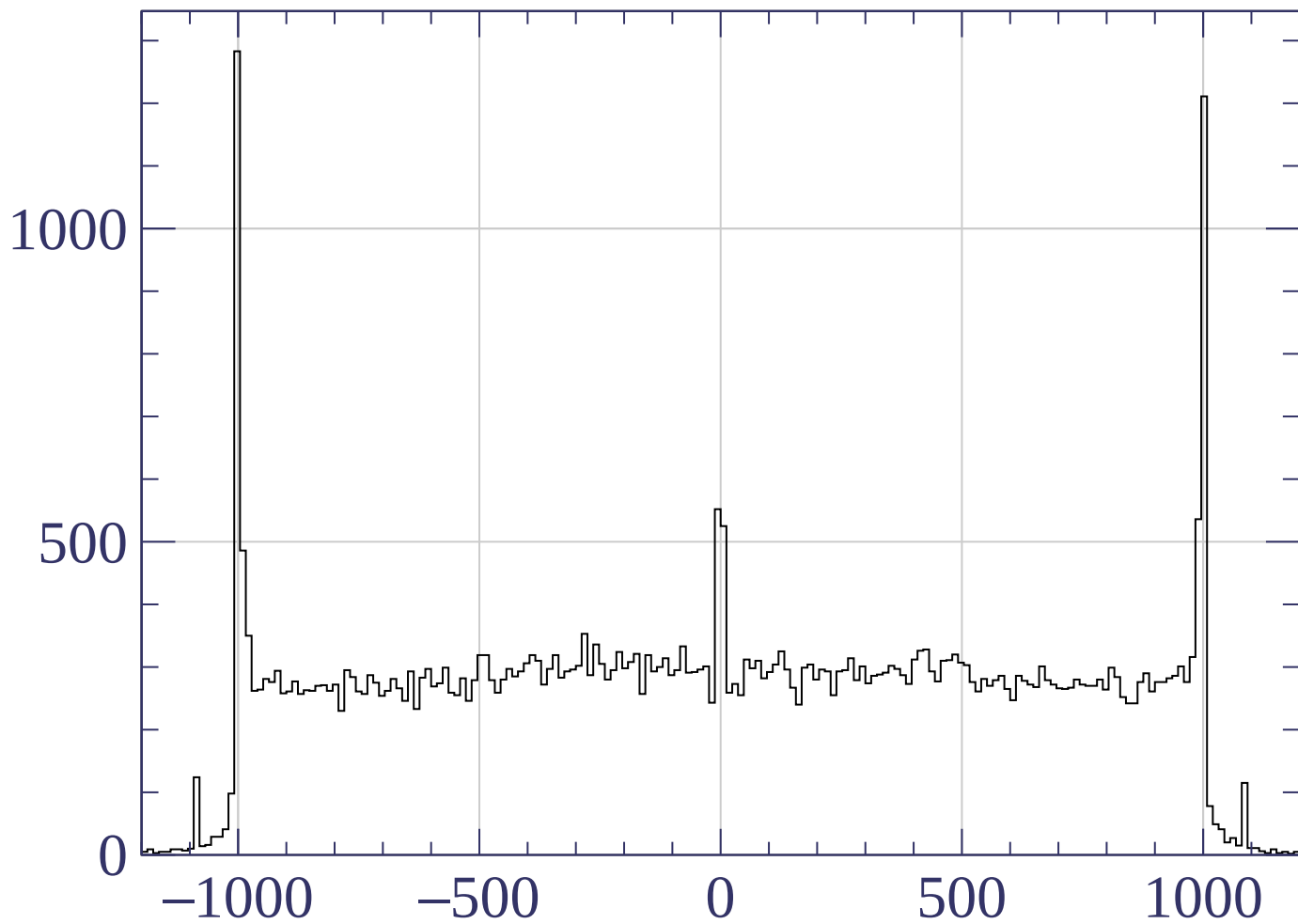




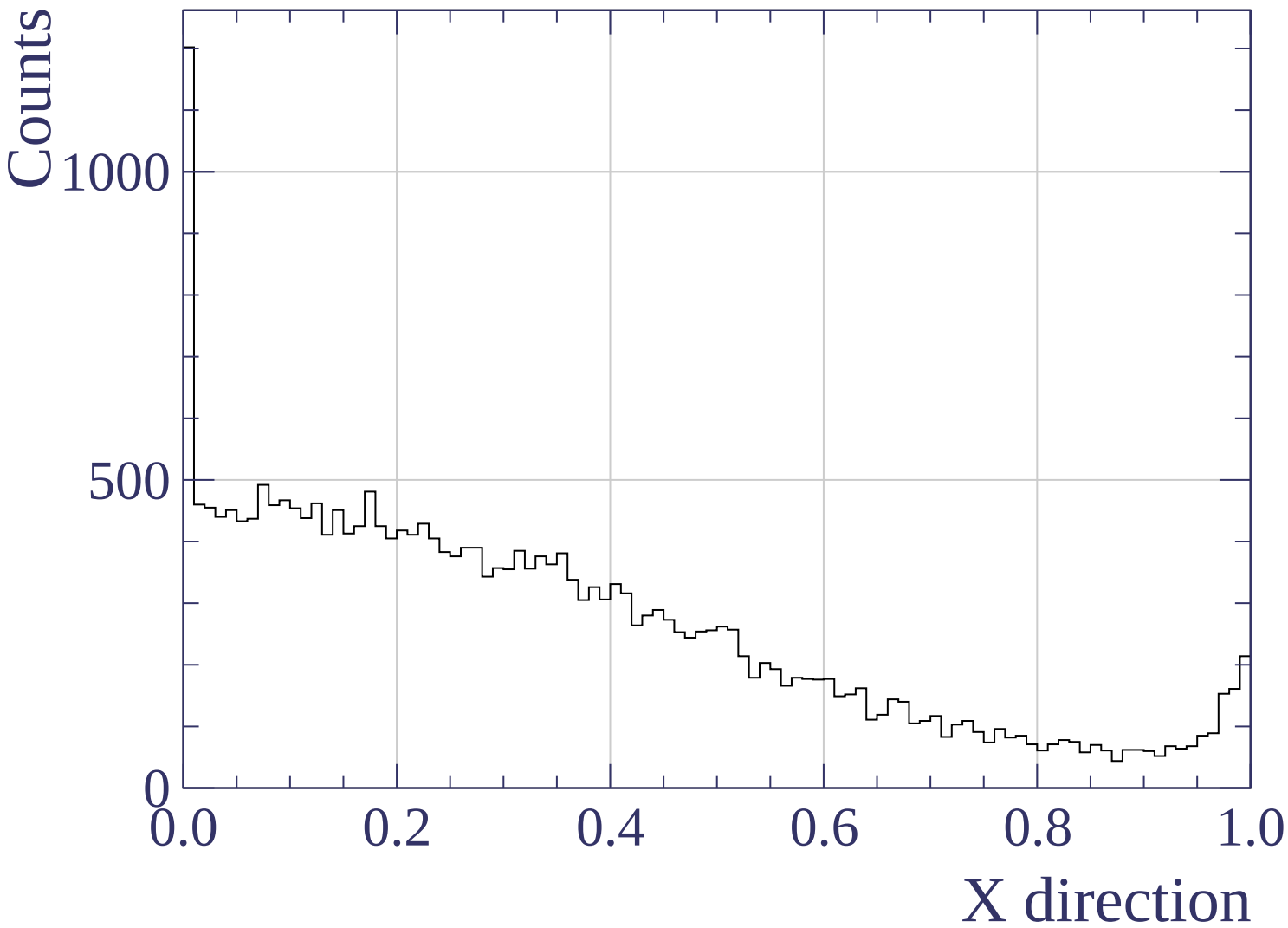
Count

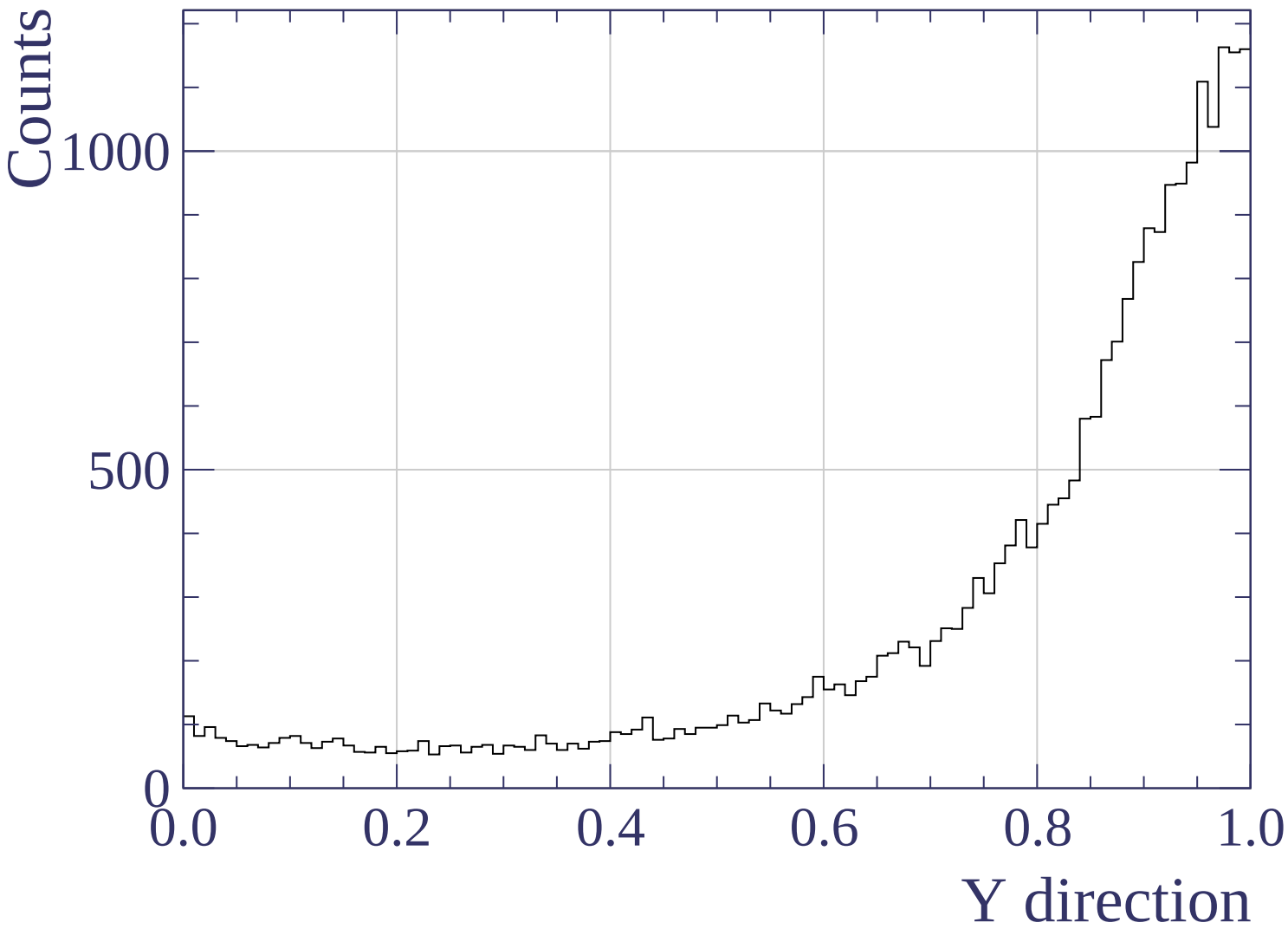


Count

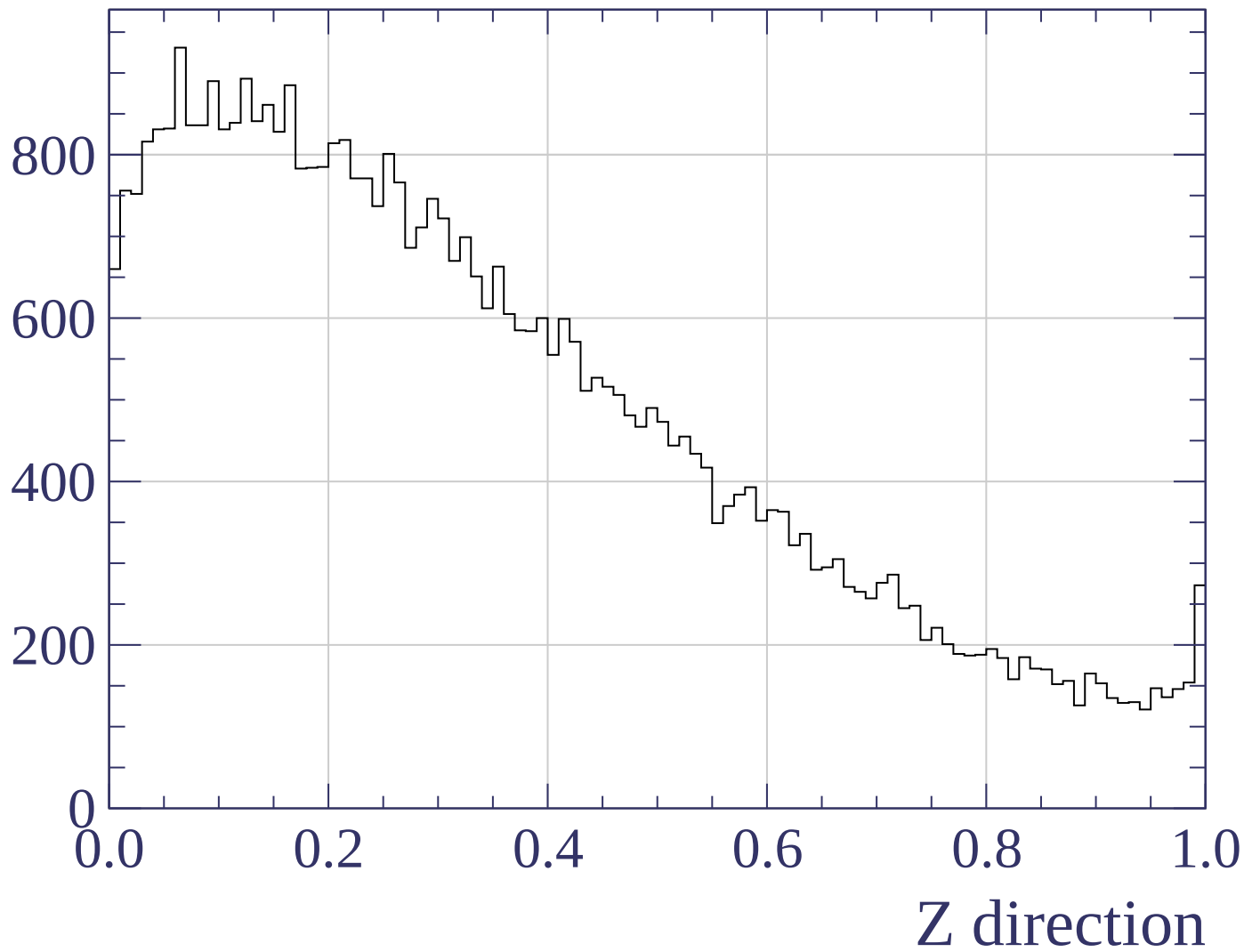


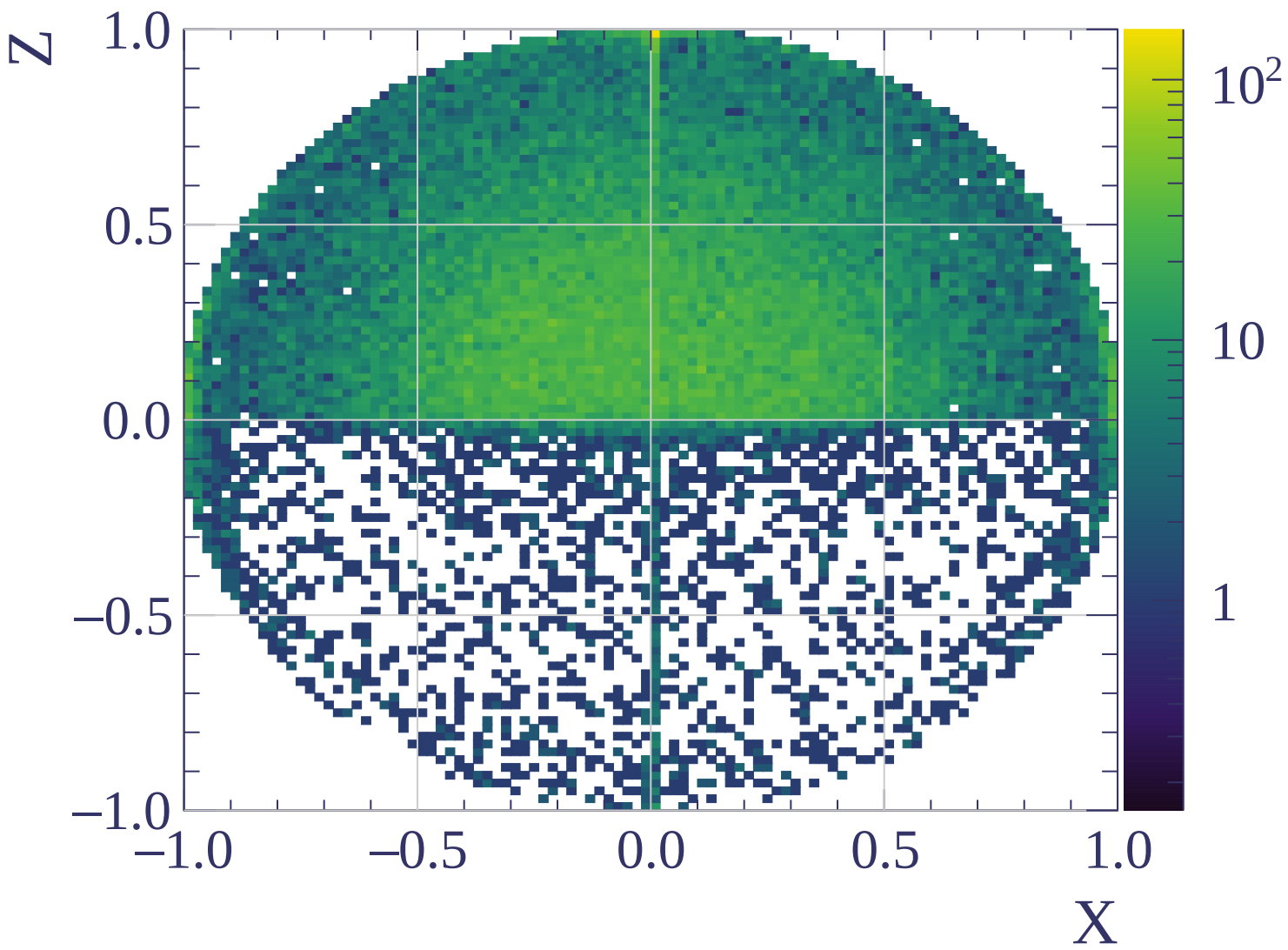
Track X position [mm]

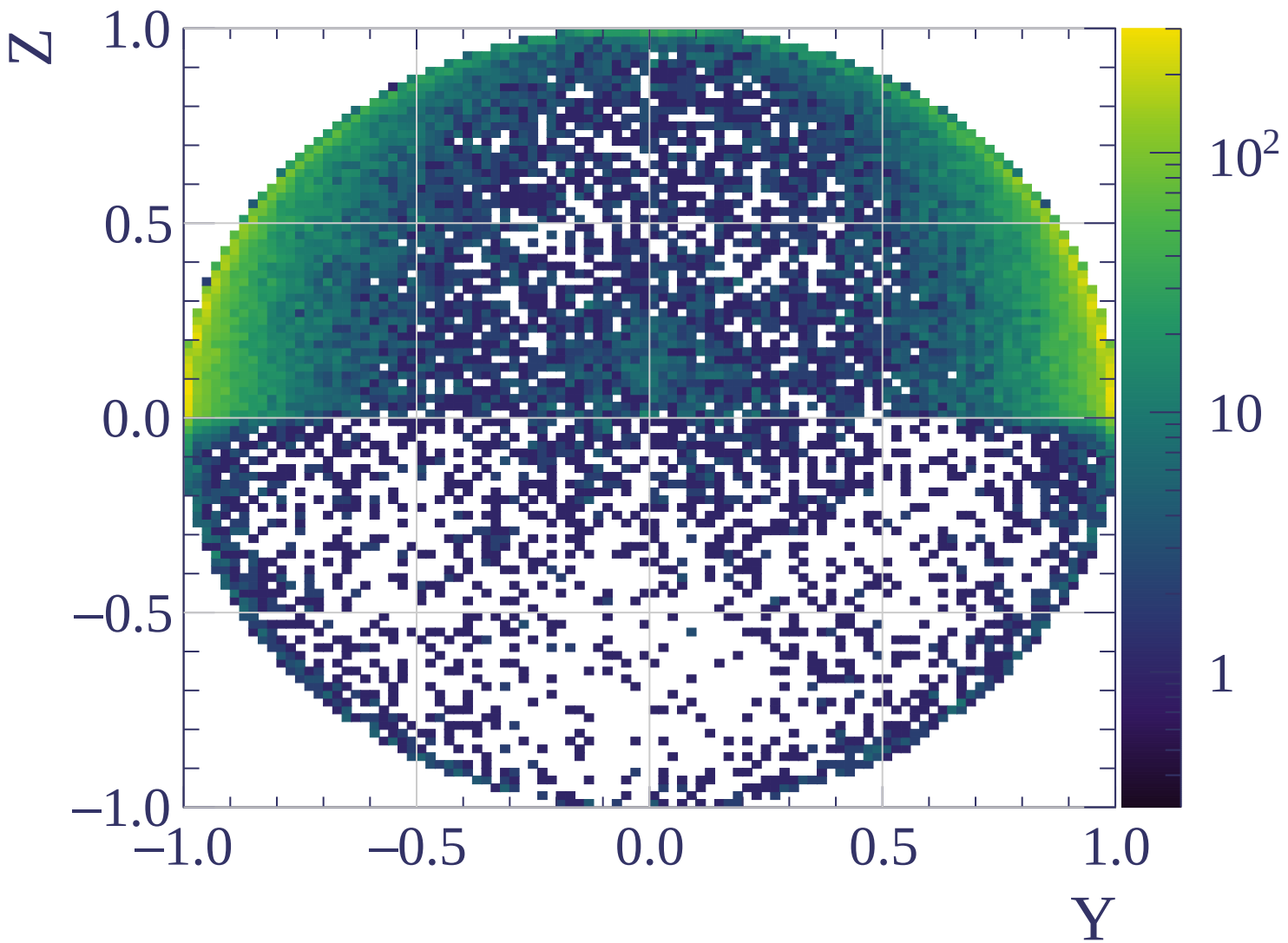


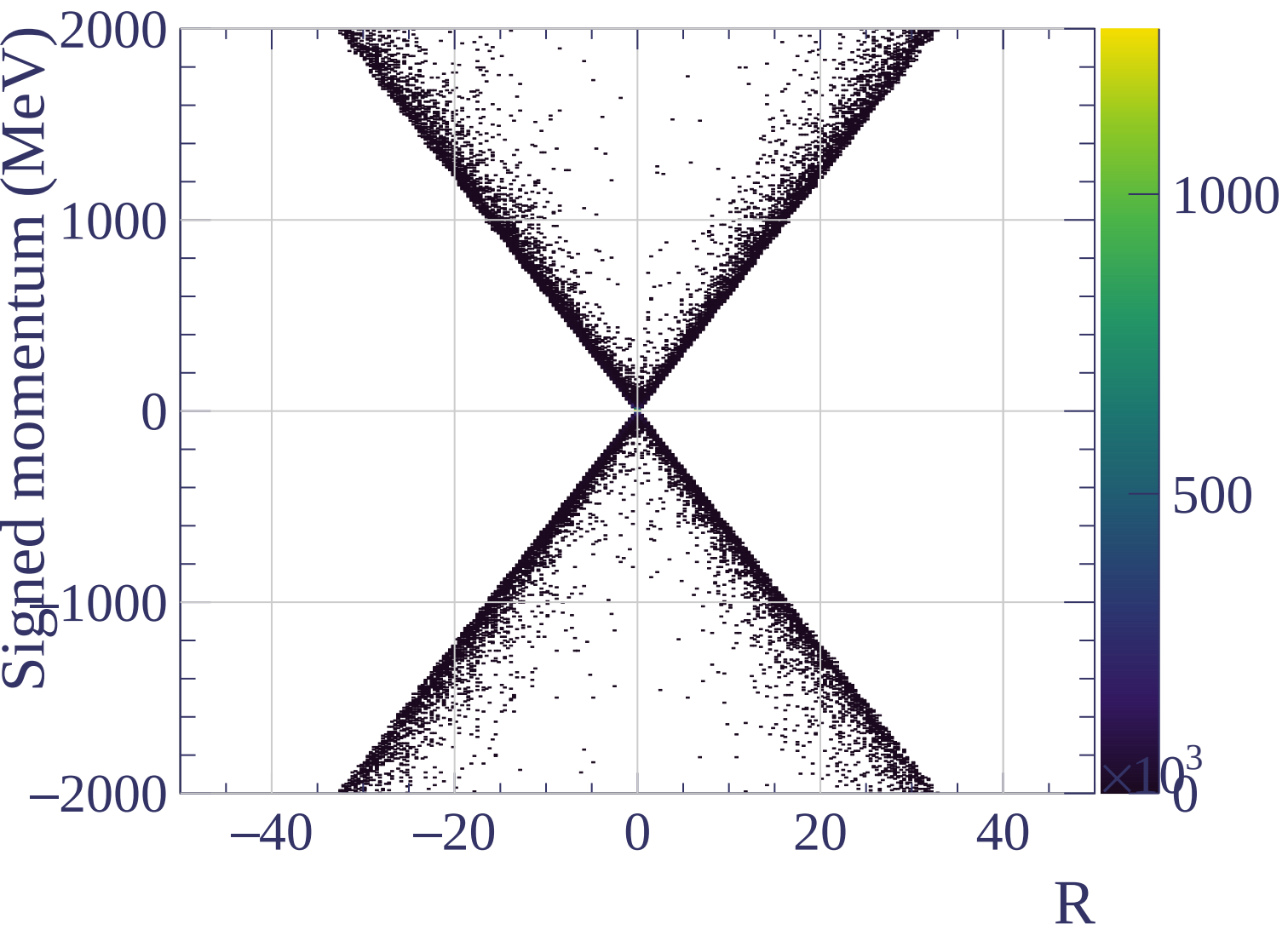


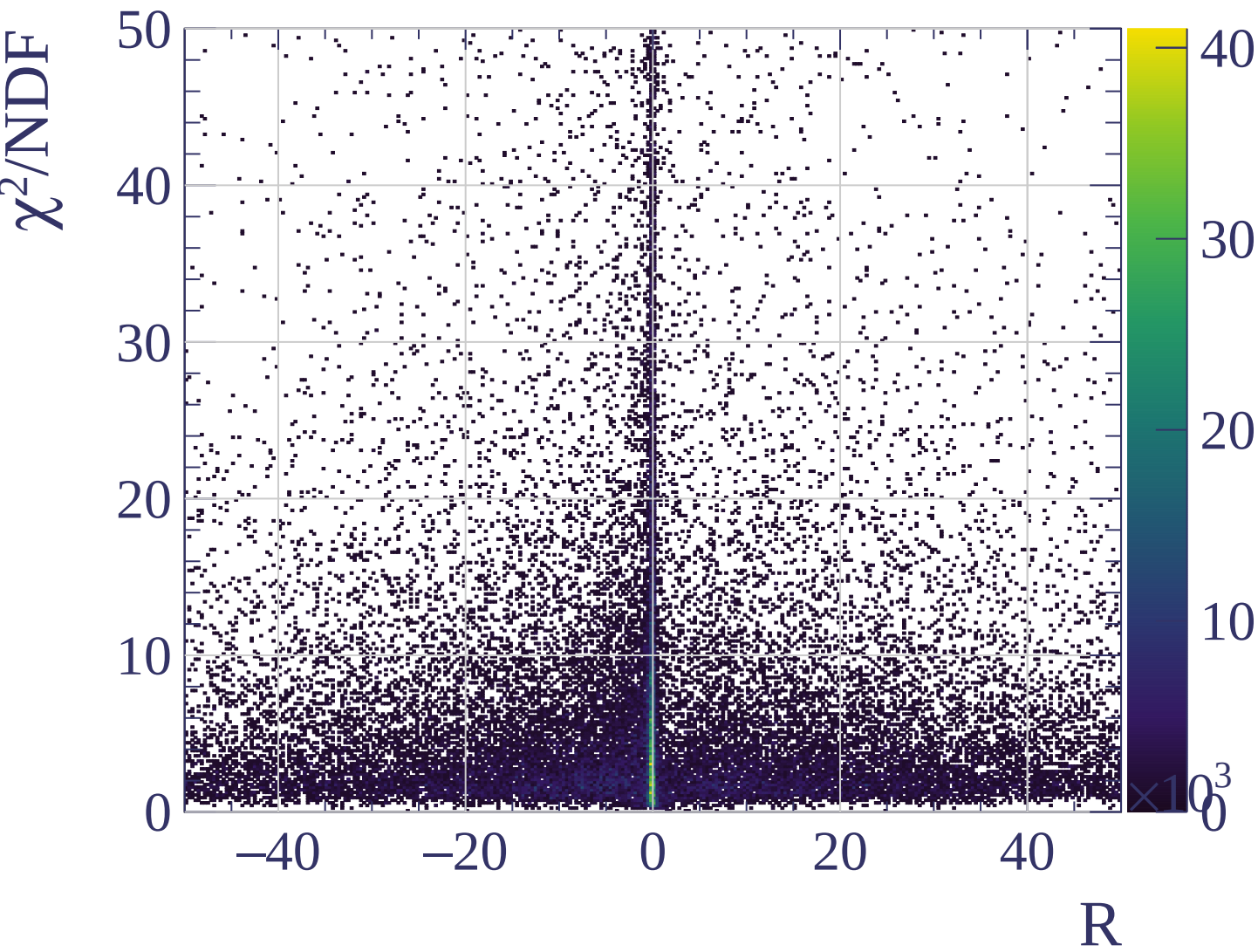
Counts



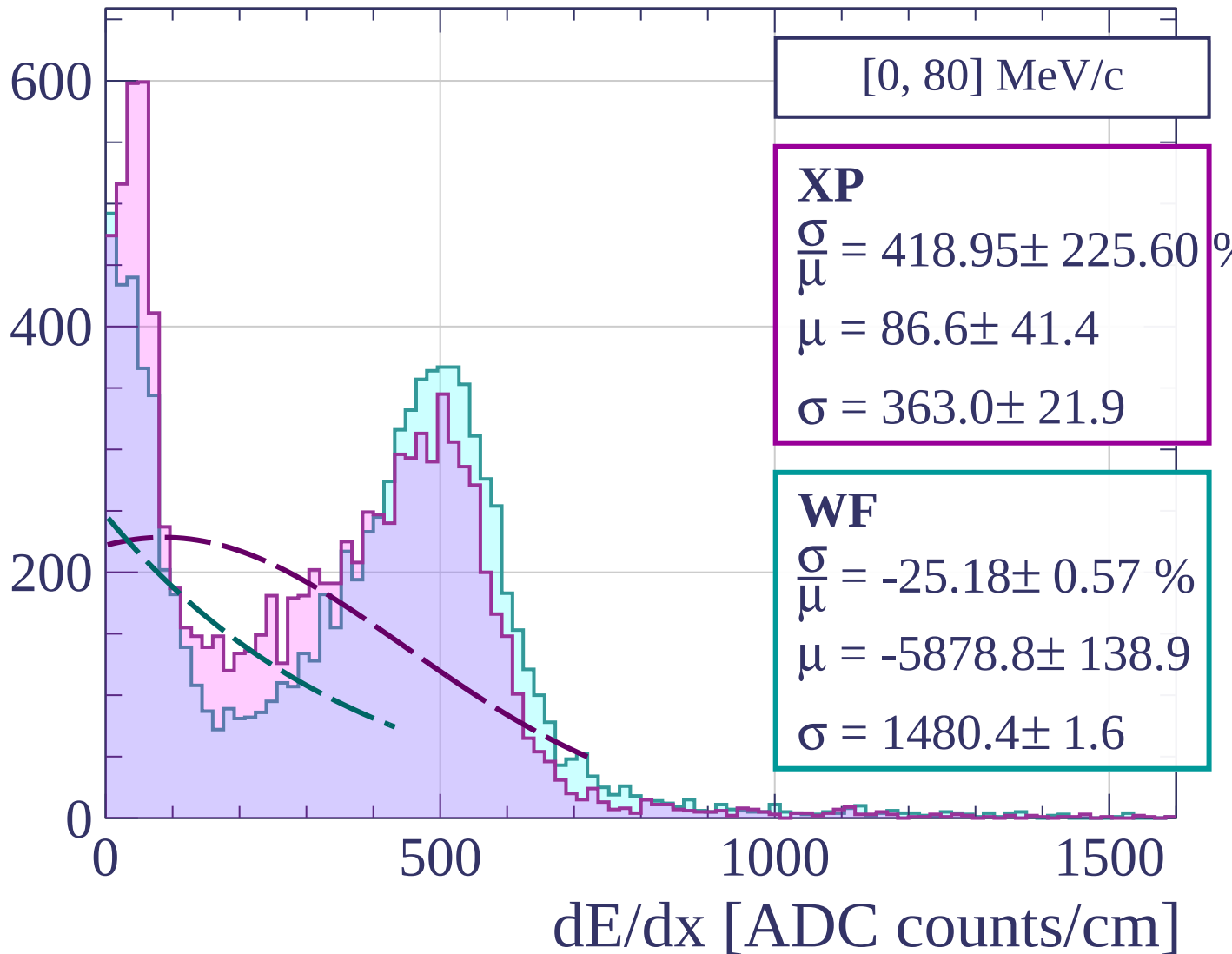


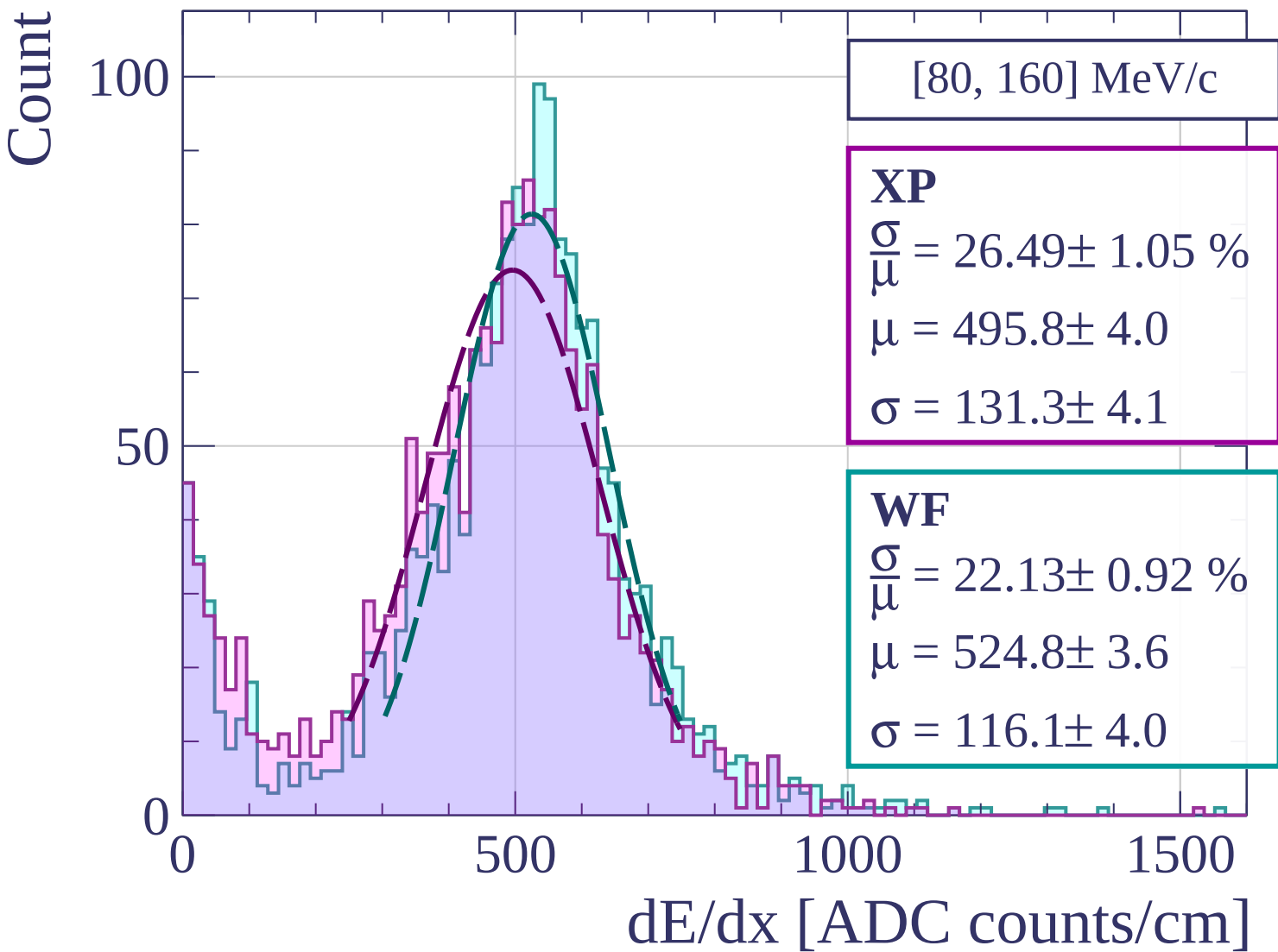




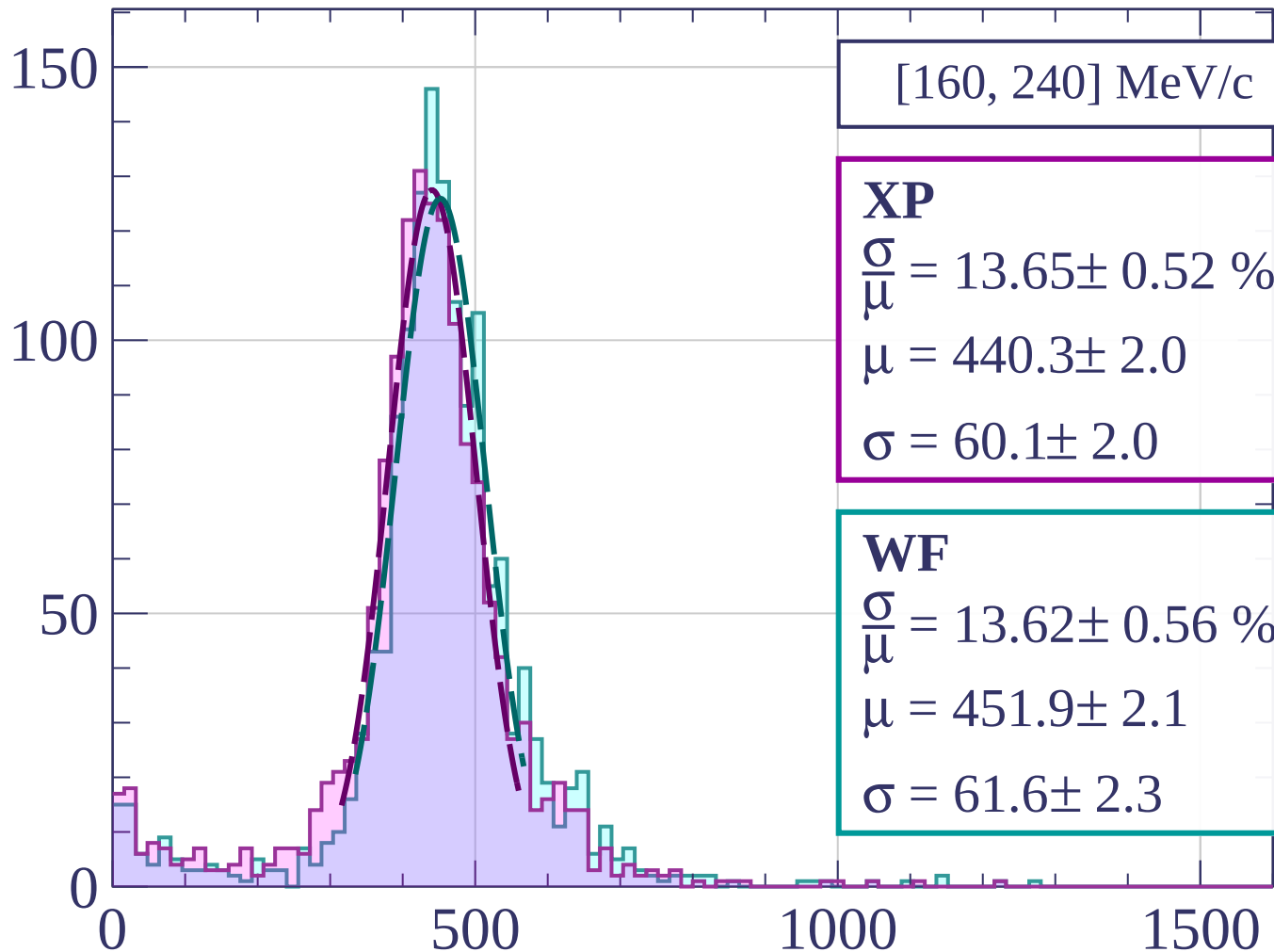


Count





Count



[160, 240] MeV/c

XP

$$\frac{\sigma}{\mu} = 13.65 \pm 0.52 \%$$

$$\mu = 440.3 \pm 2.0$$

$$\sigma = 60.1 \pm 2.0$$

WF

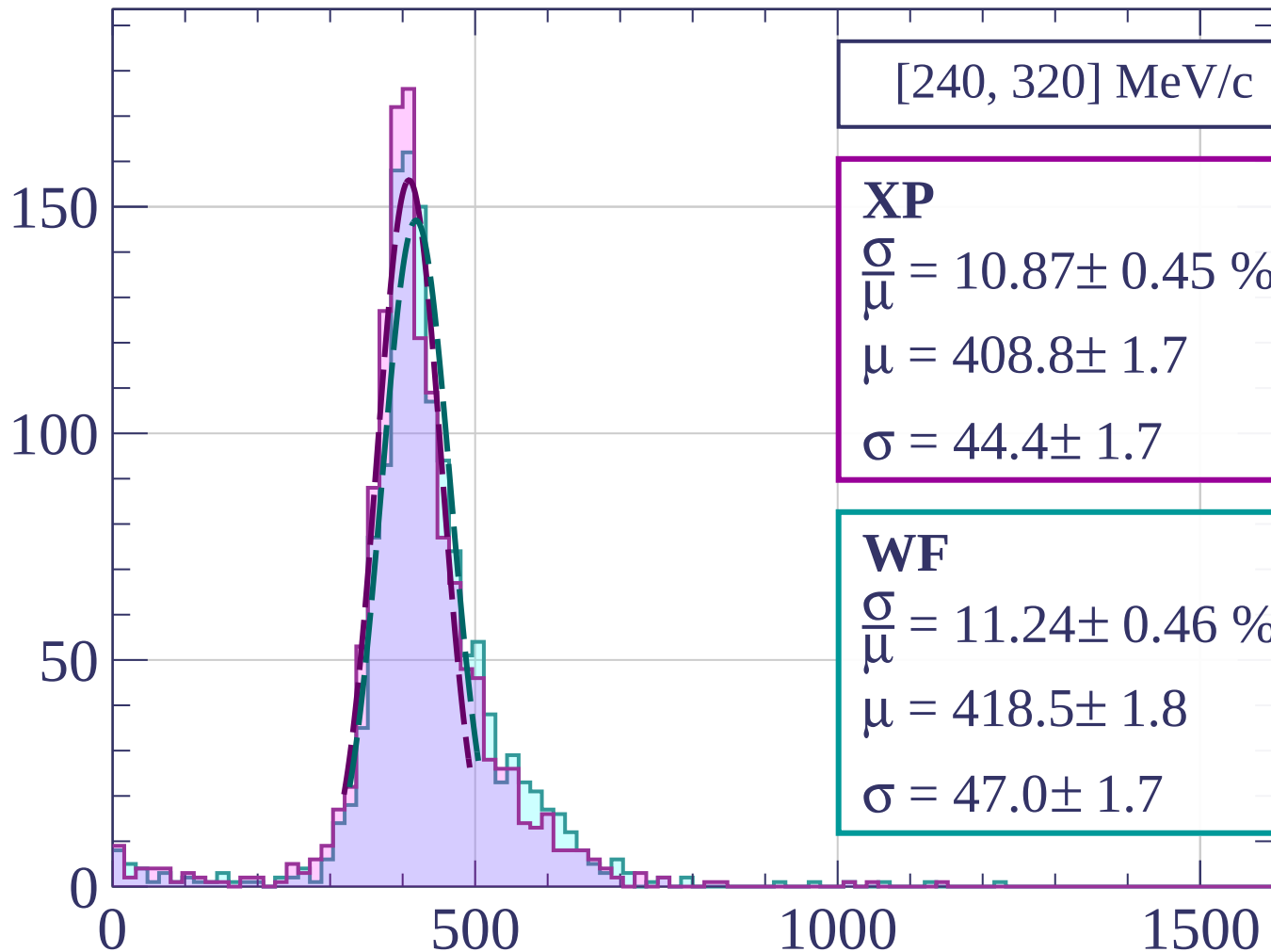
$$\frac{\sigma}{\mu} = 13.62 \pm 0.56 \%$$

$$\mu = 451.9 \pm 2.1$$

$$\sigma = 61.6 \pm 2.3$$

dE/dx [ADC counts/cm]

Count



[240, 320] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.87 \pm 0.45 \%$$

$$\mu = 408.8 \pm 1.7$$

$$\sigma = 44.4 \pm 1.7$$

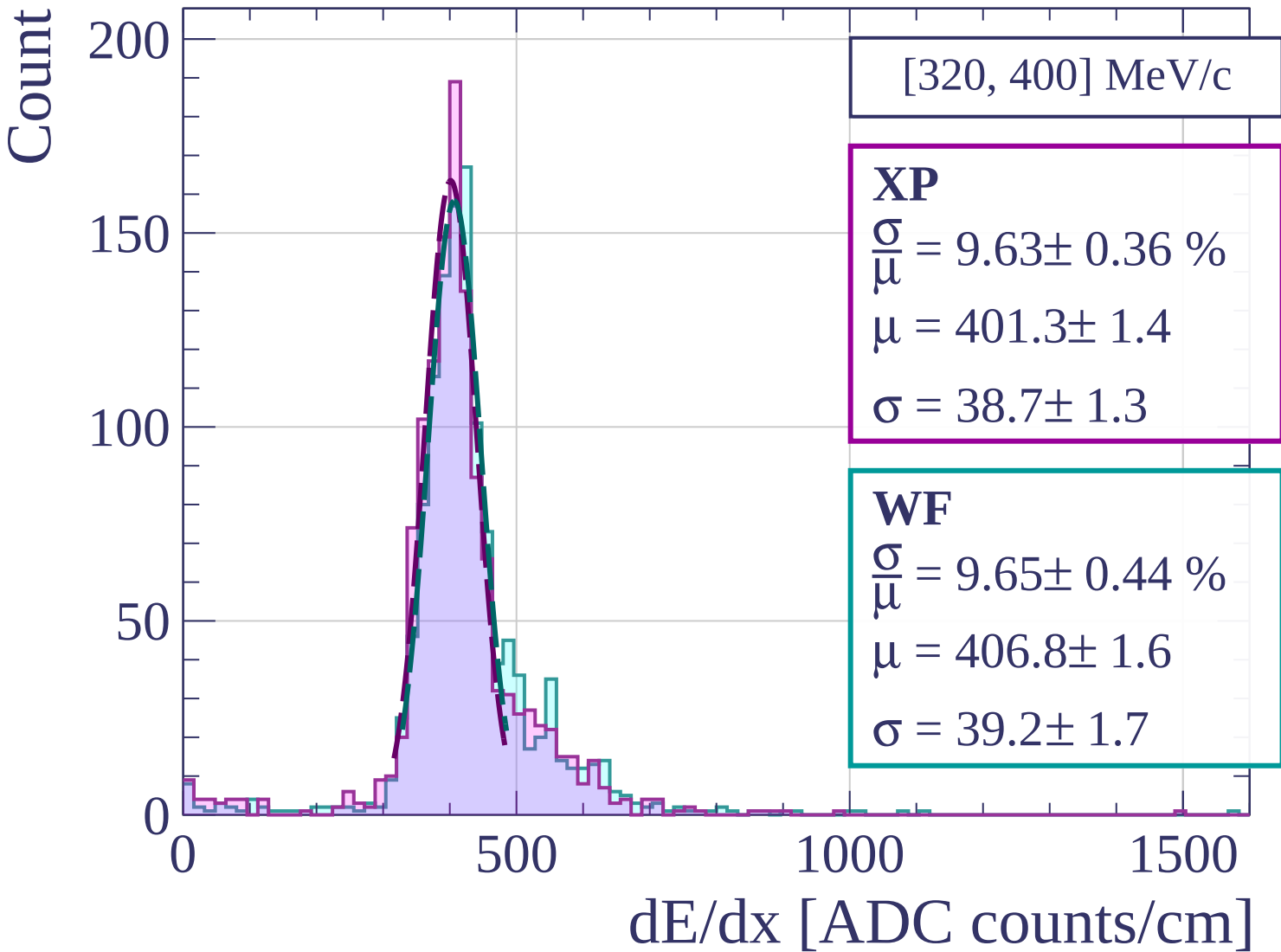
WF

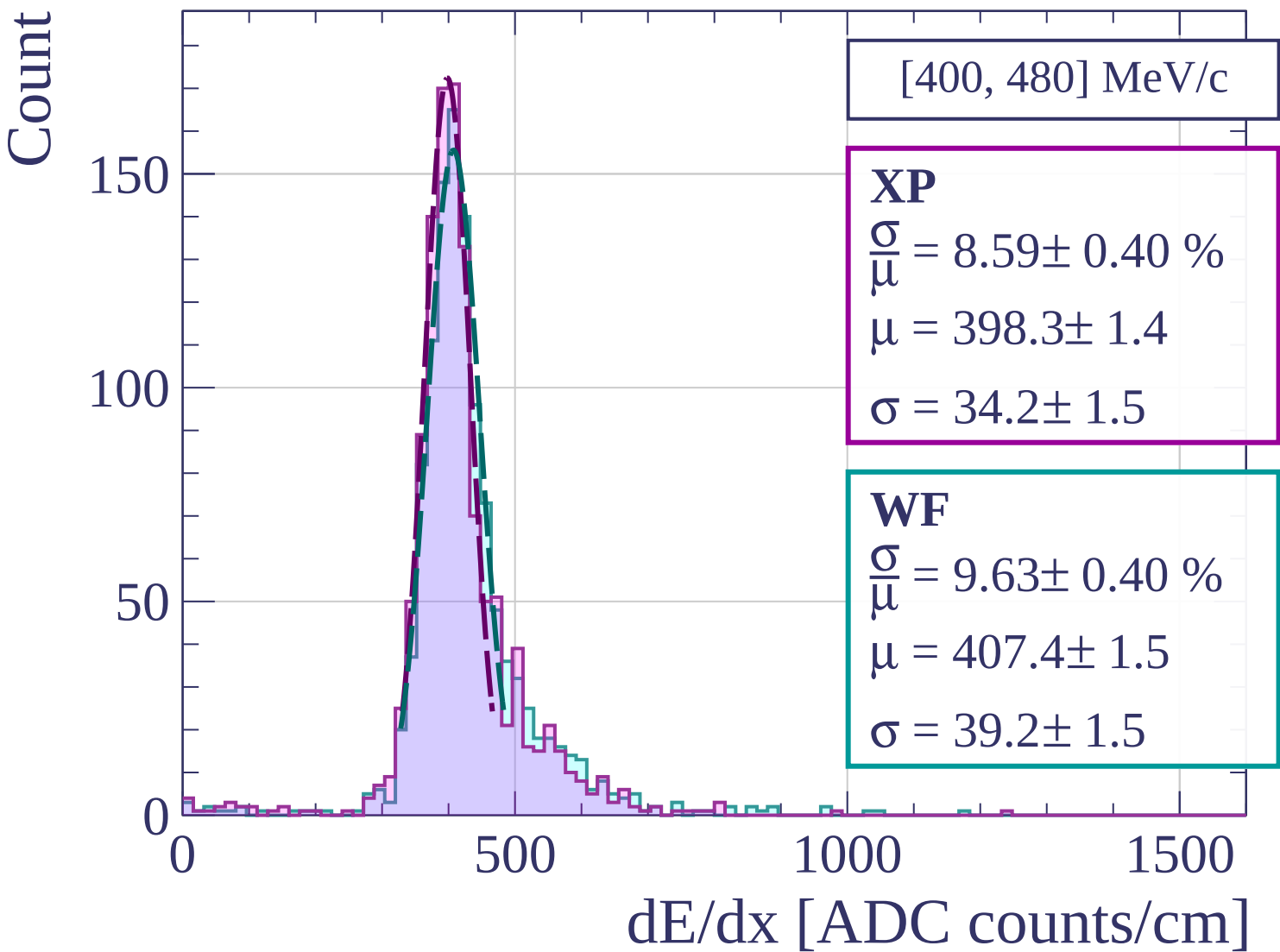
$$\frac{\sigma}{\mu} = 11.24 \pm 0.46 \%$$

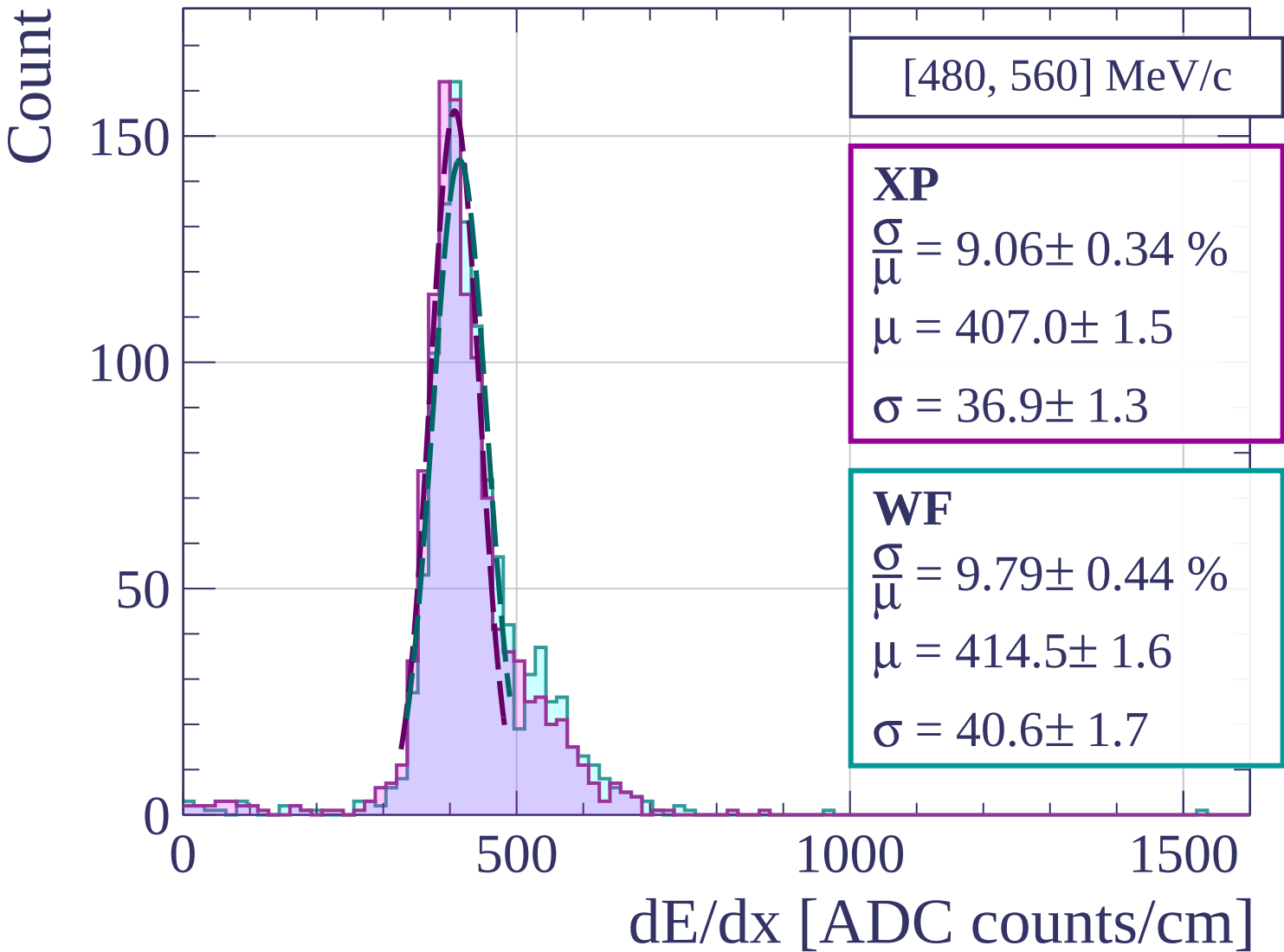
$$\mu = 418.5 \pm 1.8$$

$$\sigma = 47.0 \pm 1.7$$

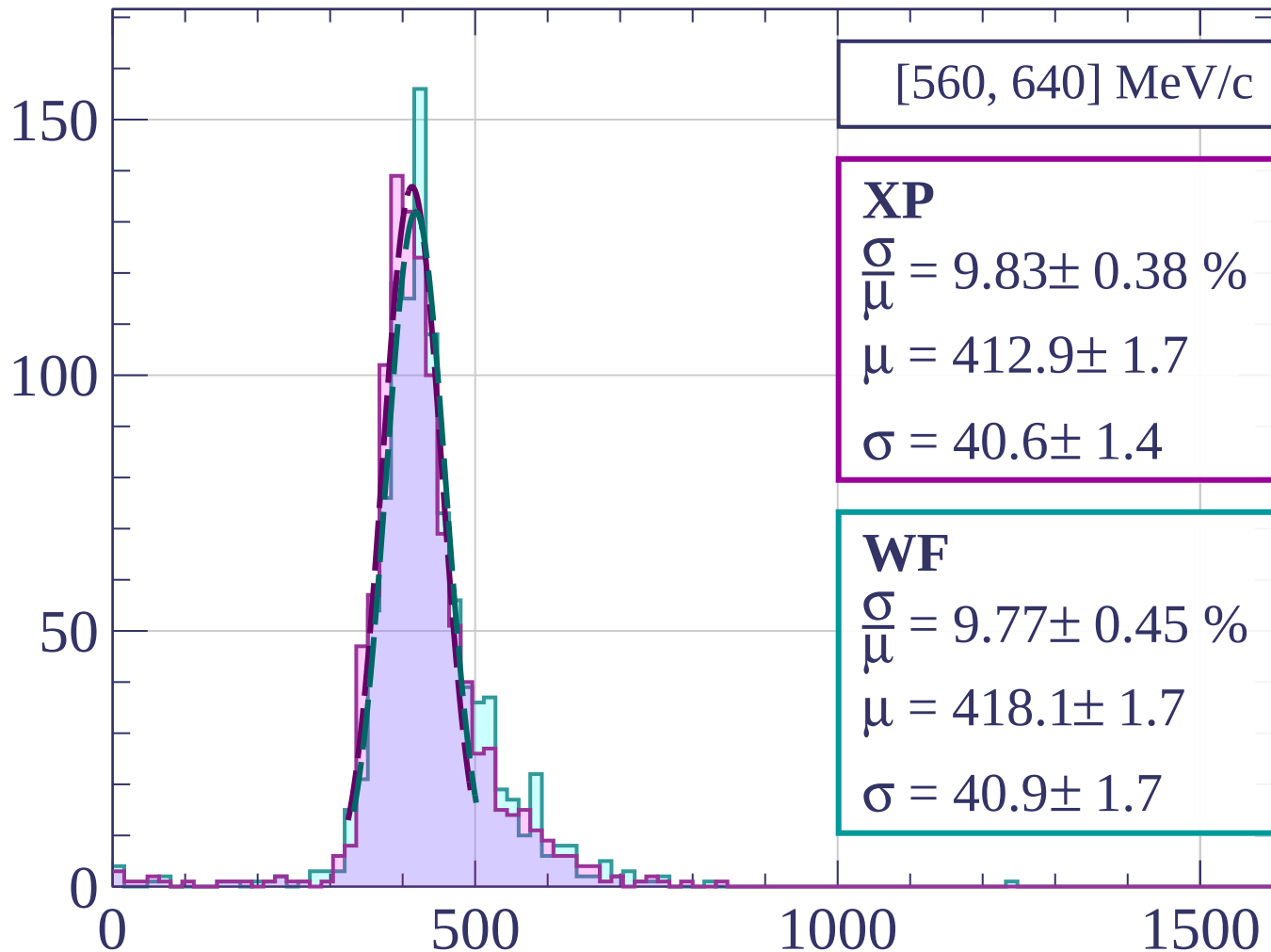
dE/dx [ADC counts/cm]



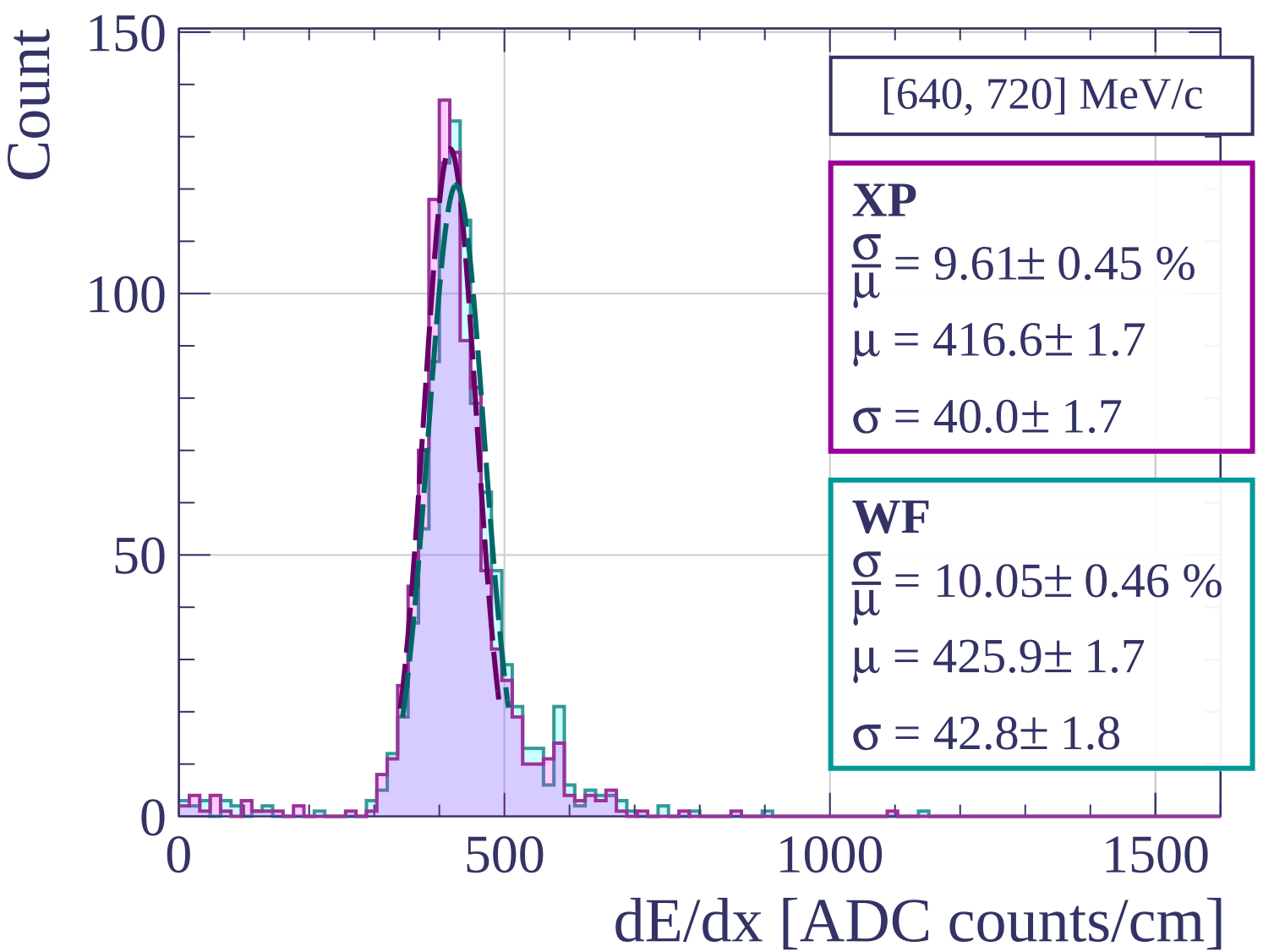


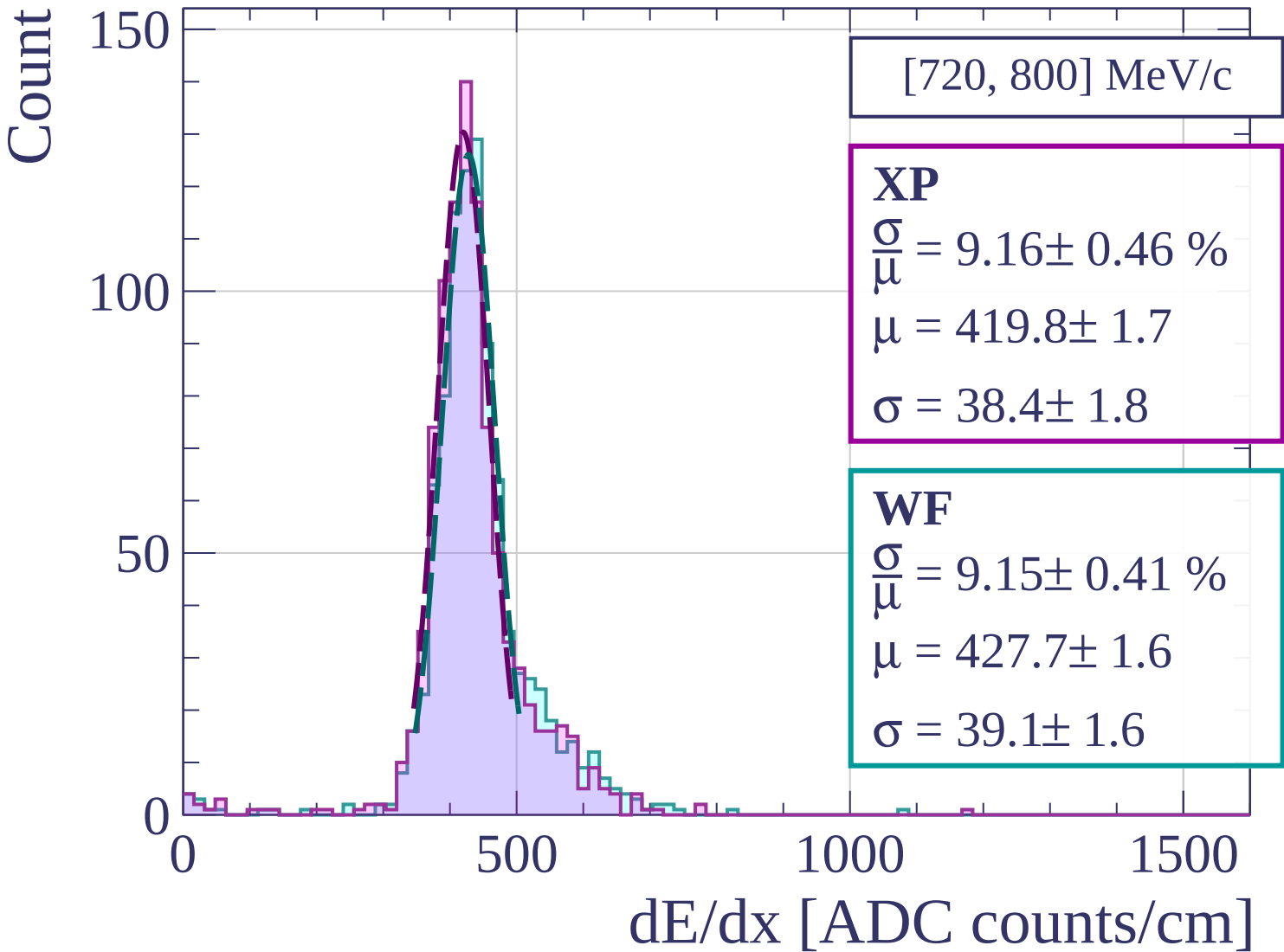


Count

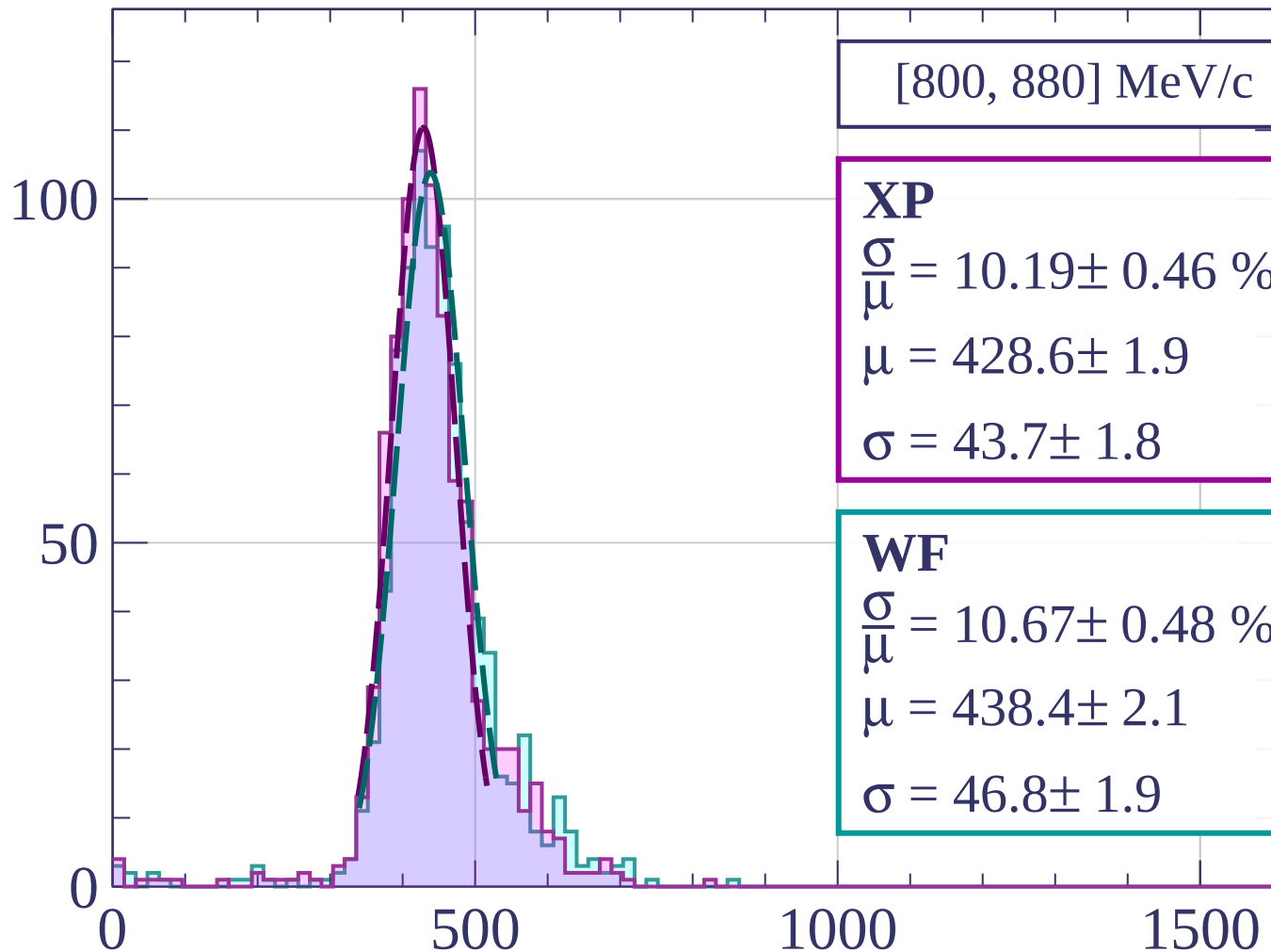


dE/dx [ADC counts/cm]



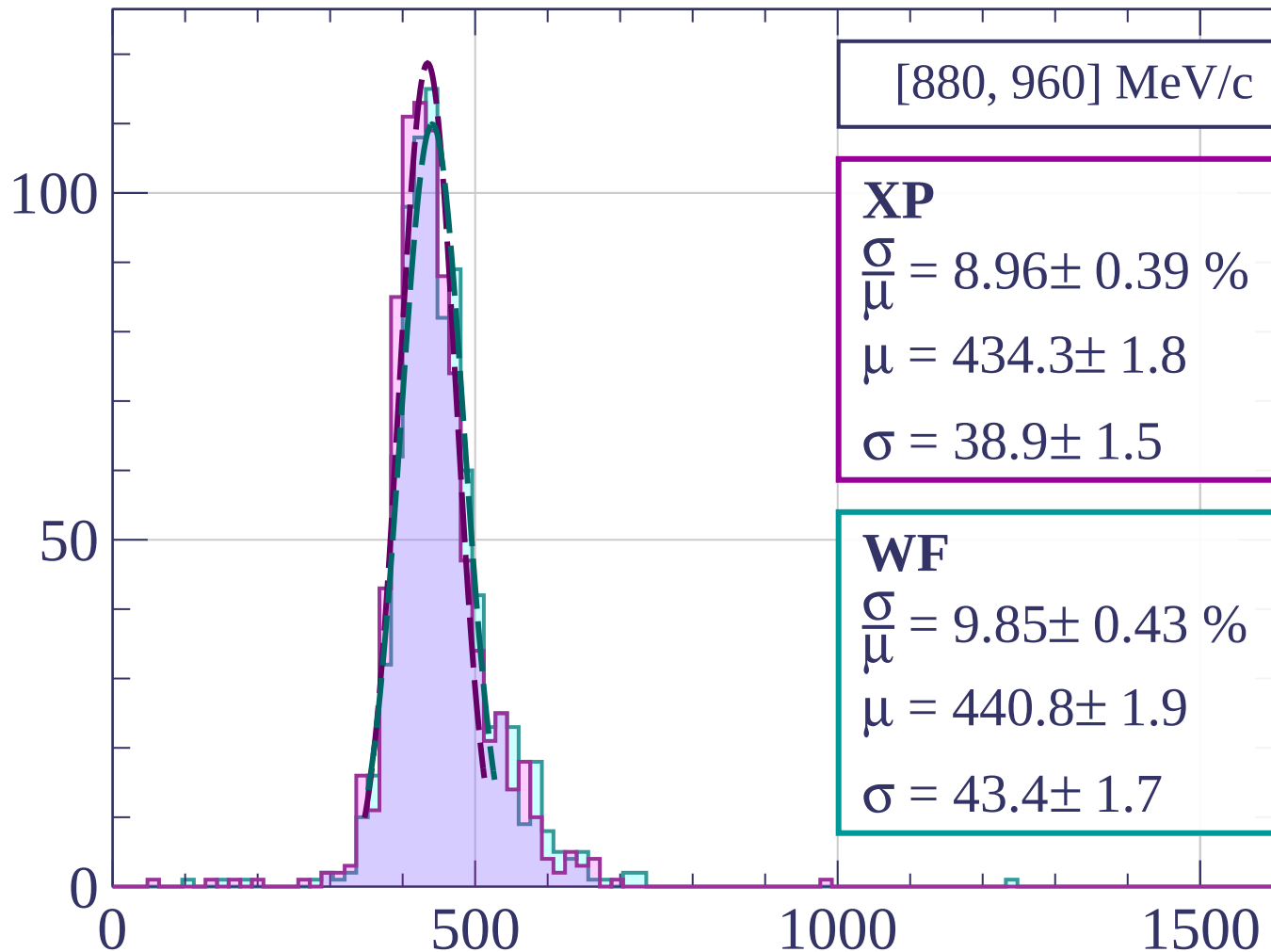


Count



dE/dx [ADC counts/cm]

Count



[880, 960] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.96 \pm 0.39 \%$$

$$\mu = 434.3 \pm 1.8$$

$$\sigma = 38.9 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.85 \pm 0.43 \%$$

$$\mu = 440.8 \pm 1.9$$

$$\sigma = 43.4 \pm 1.7$$

dE/dx [ADC counts/cm]

Count

[960, 1040] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.60 \pm 0.54 \%$$

$$\mu = 441.2 \pm 2.1$$

$$\sigma = 46.8 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 10.67 \pm 0.52 \%$$

$$\mu = 449.4 \pm 2.1$$

$$\sigma = 48.0 \pm 2.1$$

0

500

1000

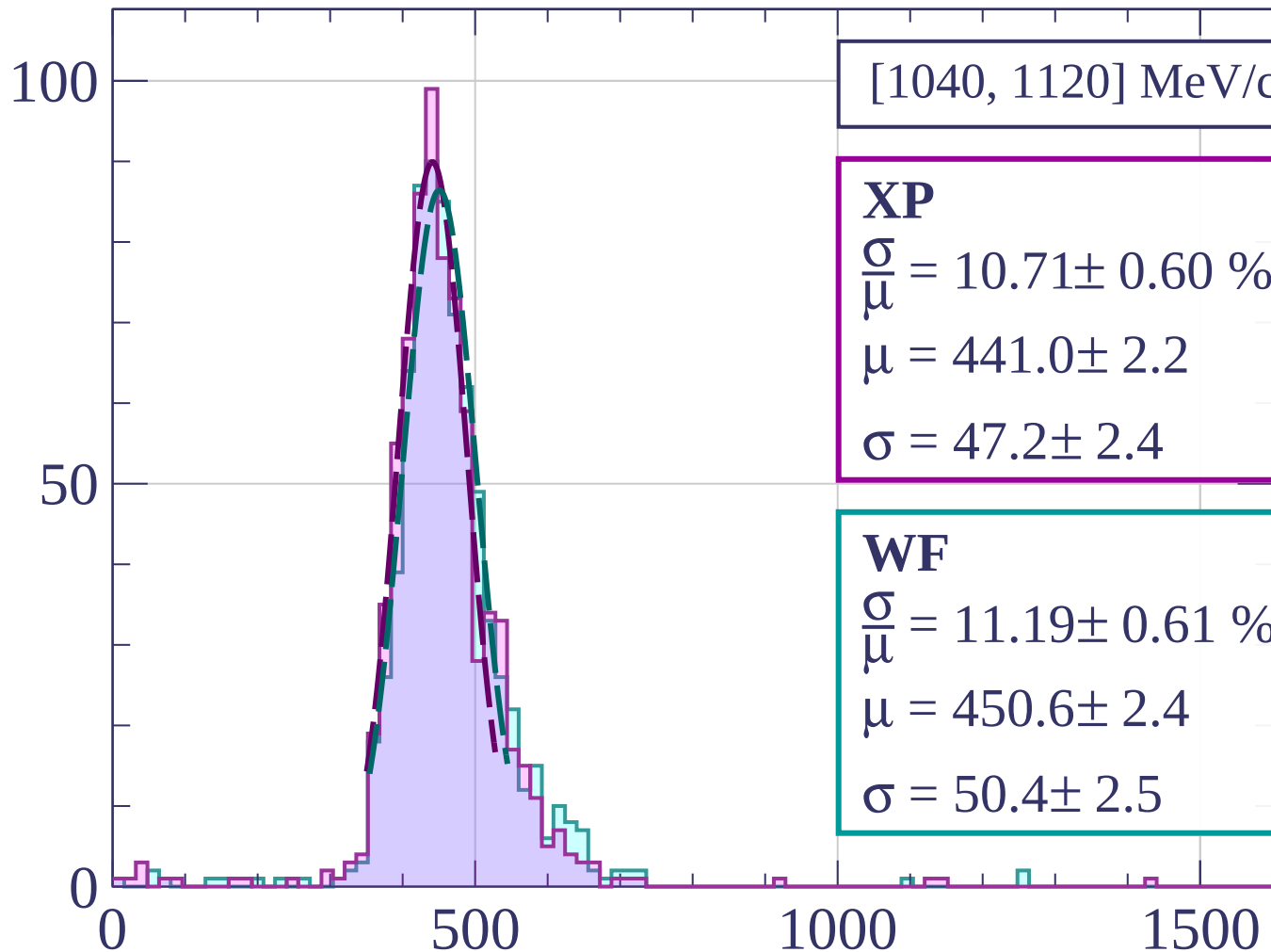
1500

dE/dx [ADC counts/cm]

100

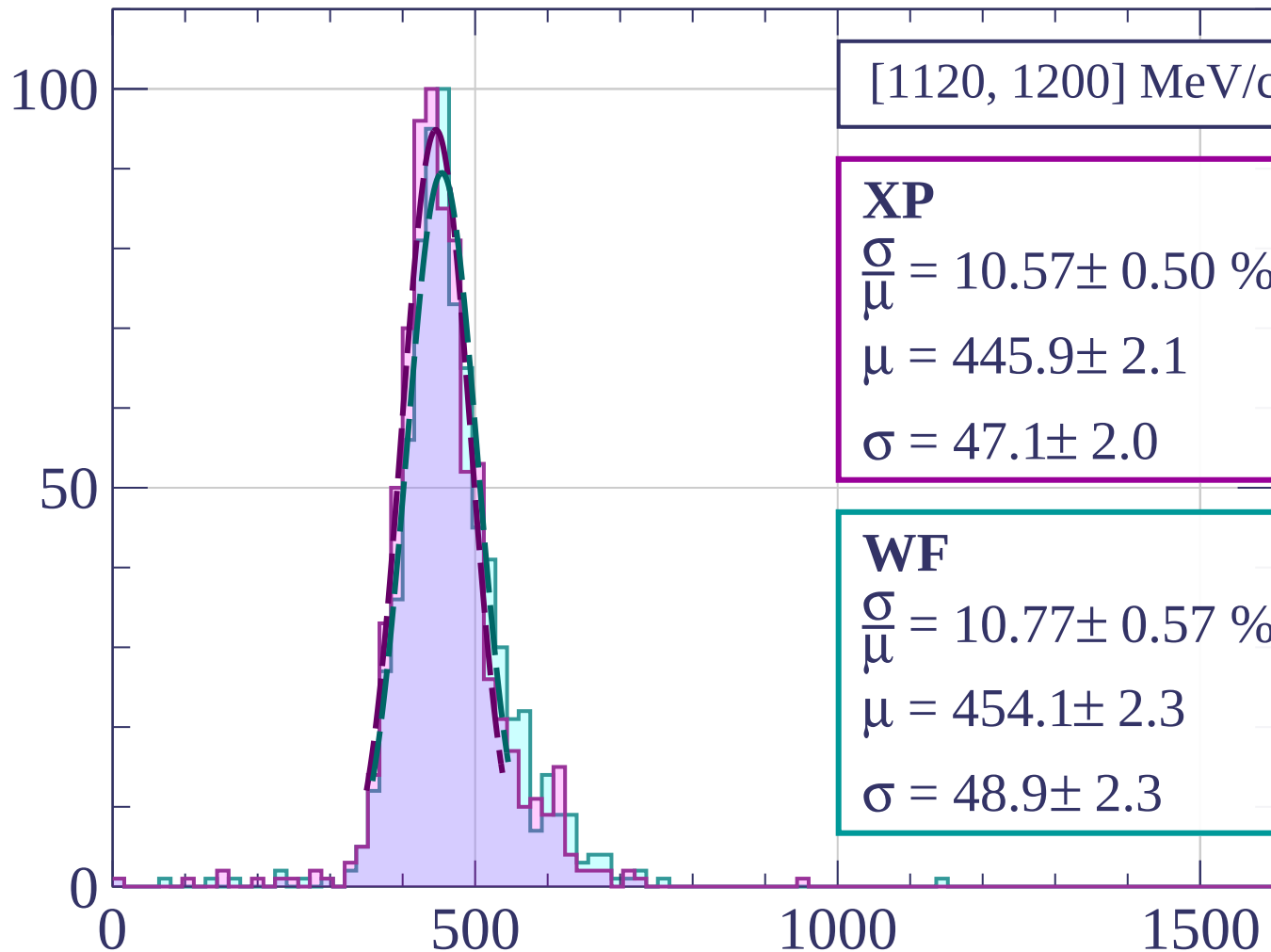
50

Count



dE/dx [ADC counts/cm]

Count



[1120, 1200] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.57 \pm 0.50 \%$$

$$\mu = 445.9 \pm 2.1$$

$$\sigma = 47.1 \pm 2.0$$

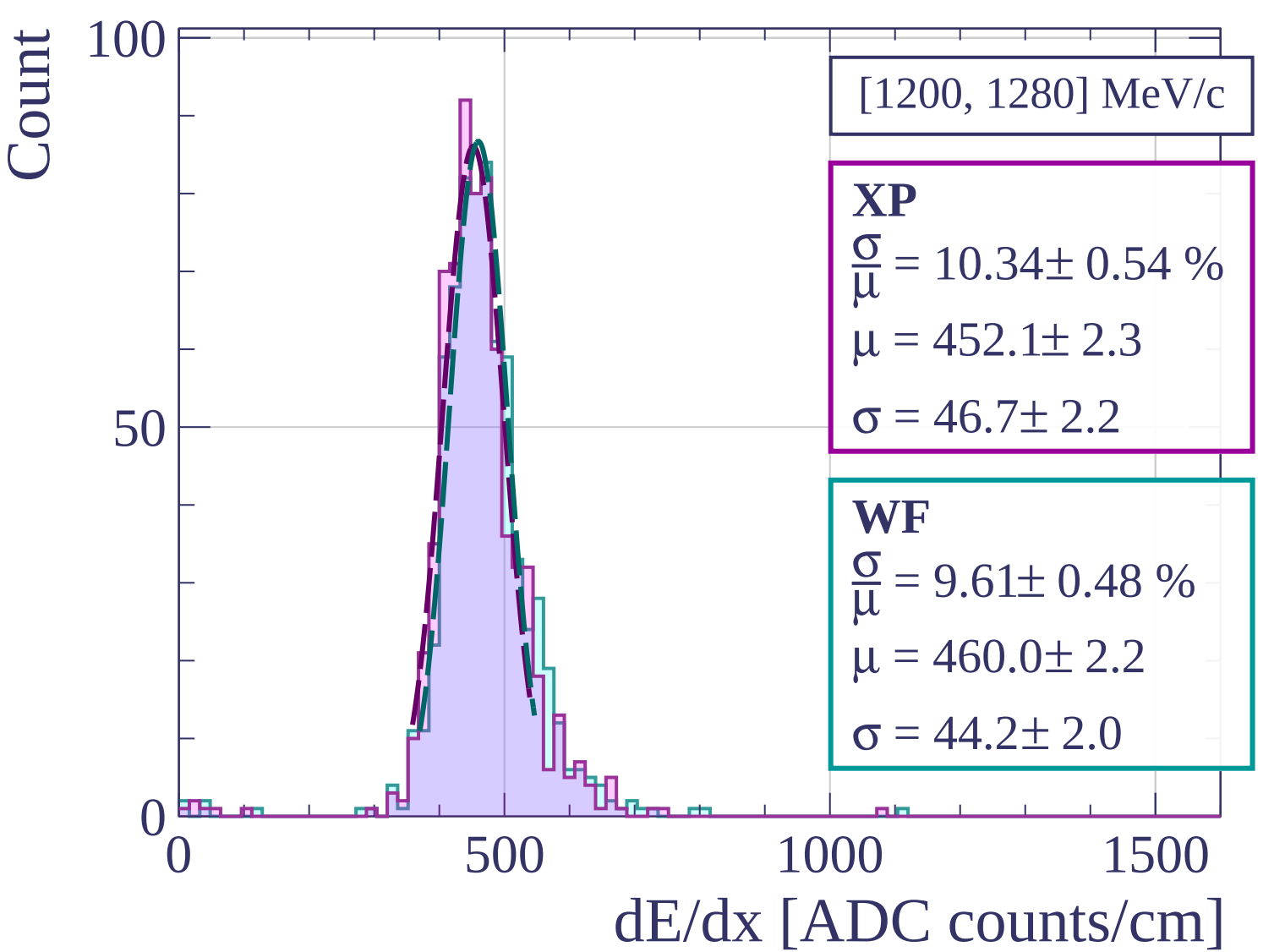
WF

$$\frac{\sigma}{\mu} = 10.77 \pm 0.57 \%$$

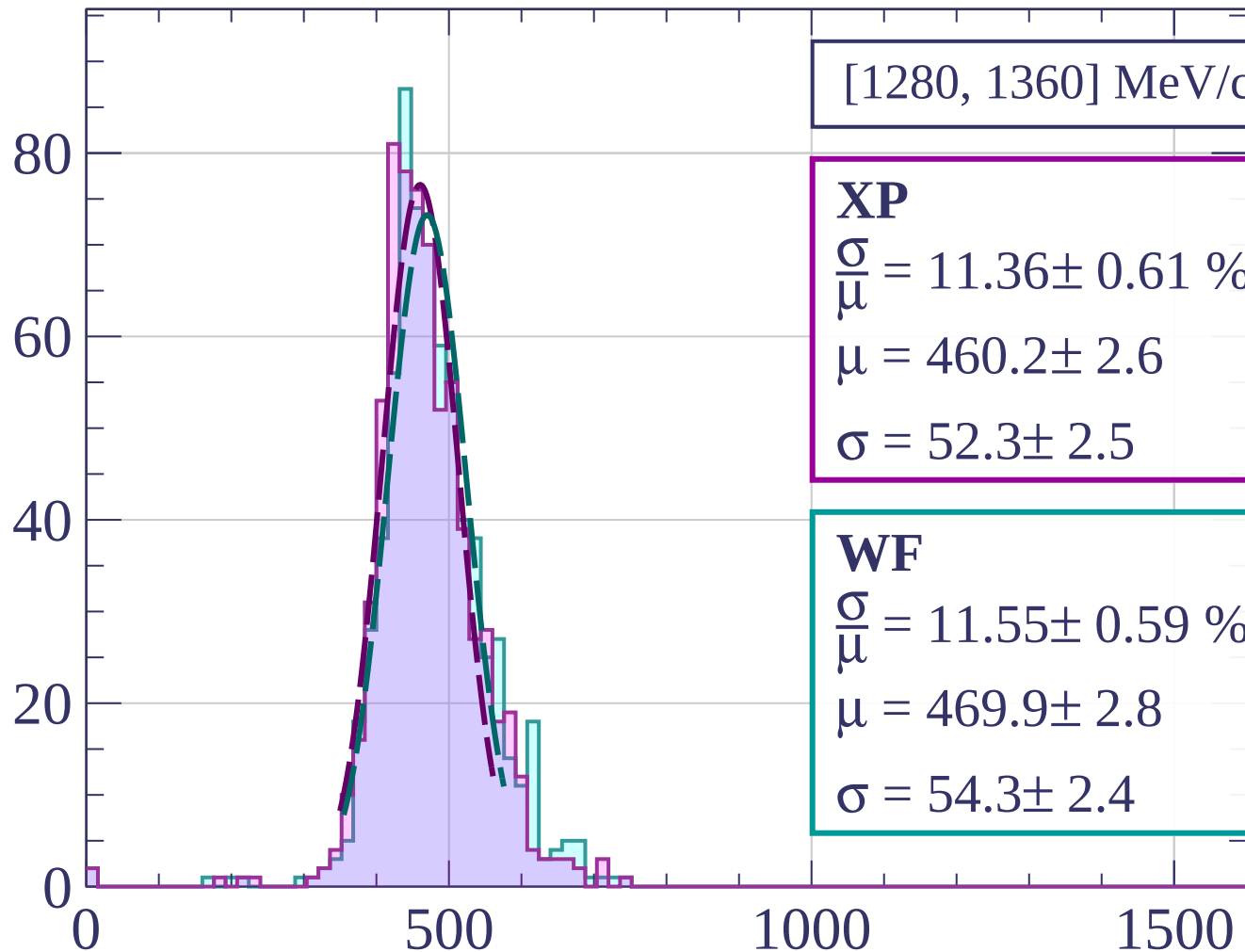
$$\mu = 454.1 \pm 2.3$$

$$\sigma = 48.9 \pm 2.3$$

dE/dx [ADC counts/cm]



Count



[1280, 1360] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.36 \pm 0.61 \%$$

$$\mu = 460.2 \pm 2.6$$

$$\sigma = 52.3 \pm 2.5$$

WF

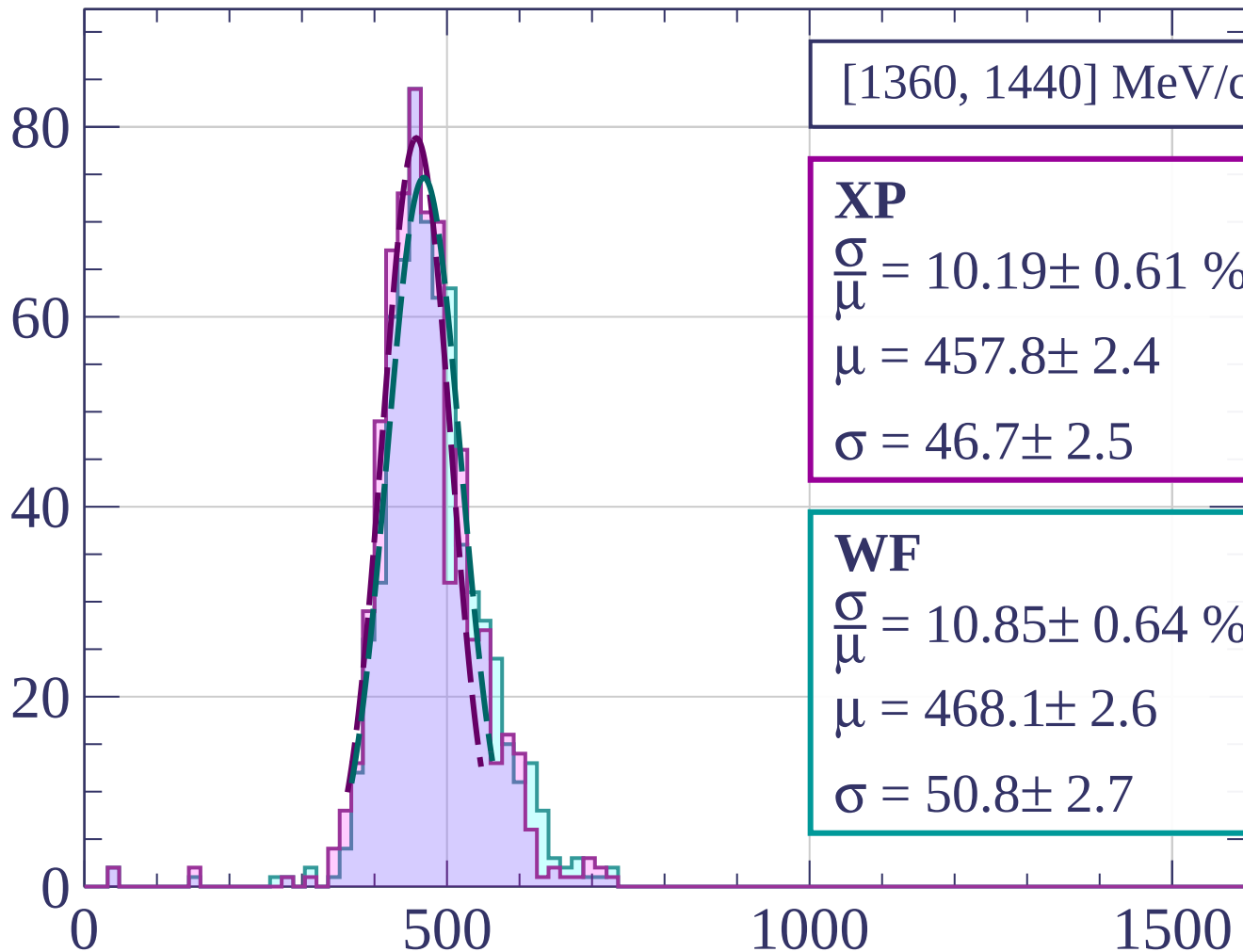
$$\frac{\sigma}{\mu} = 11.55 \pm 0.59 \%$$

$$\mu = 469.9 \pm 2.8$$

$$\sigma = 54.3 \pm 2.4$$

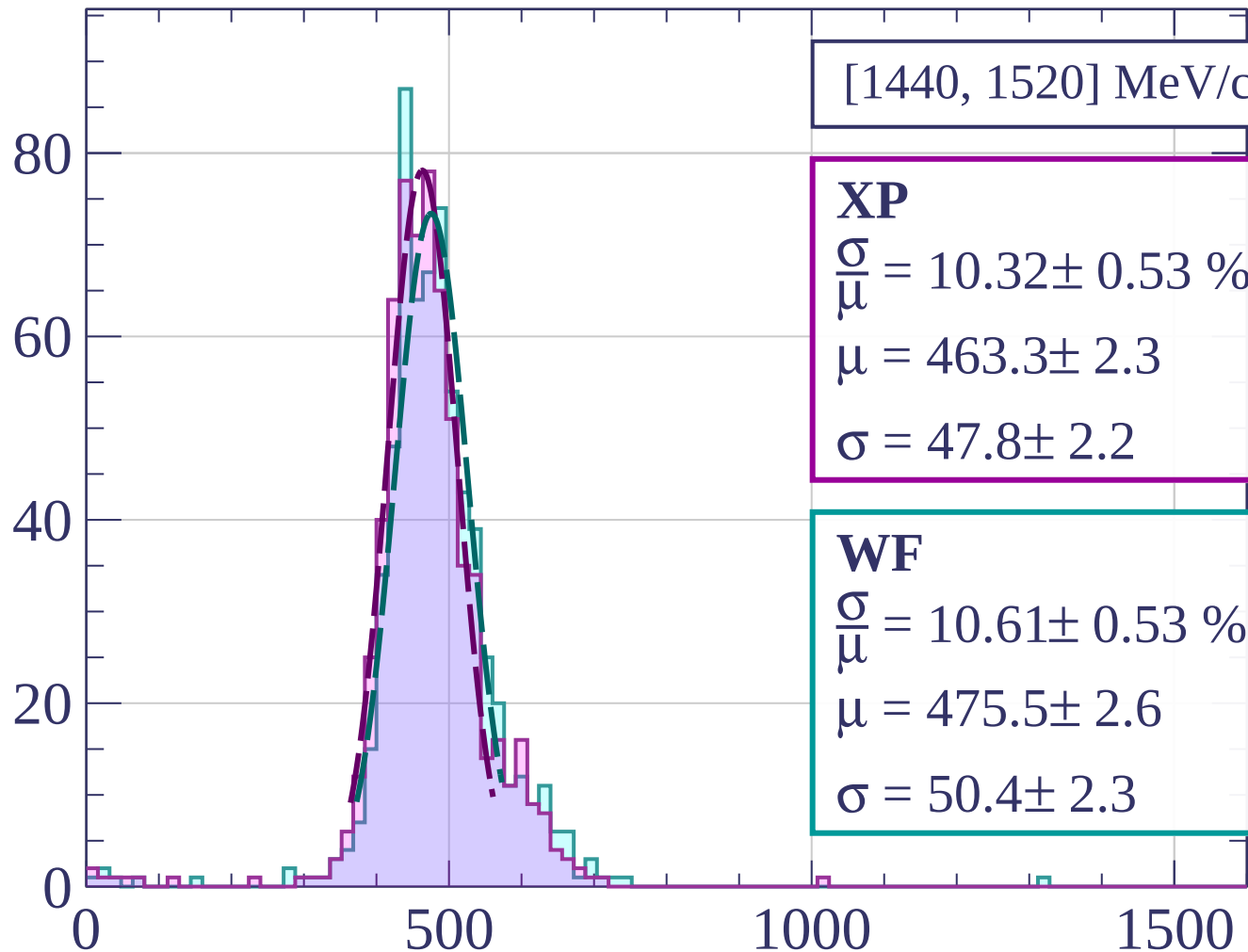
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[1440, 1520] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.32 \pm 0.53 \%$$

$$\mu = 463.3 \pm 2.3$$

$$\sigma = 47.8 \pm 2.2$$

WF

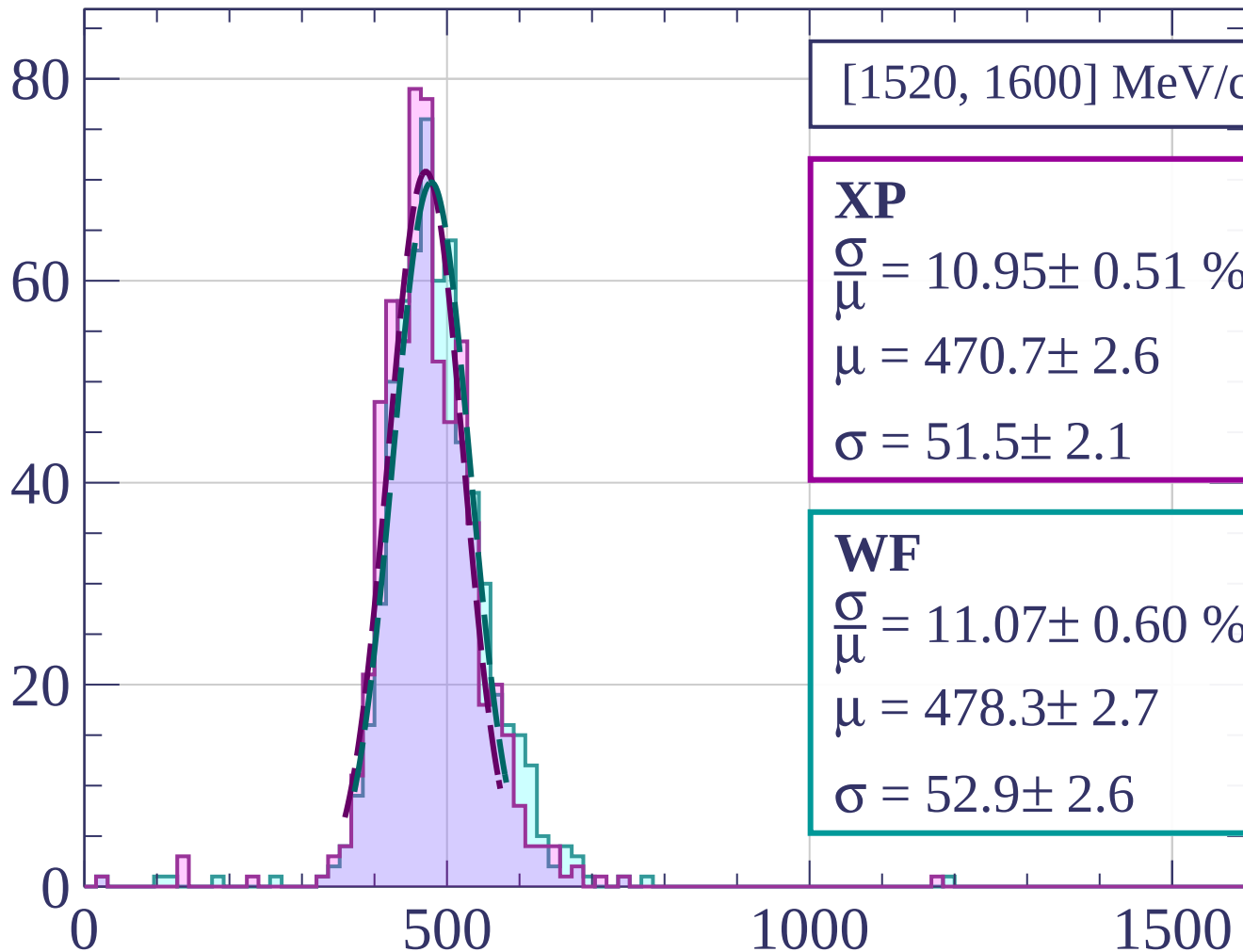
$$\frac{\sigma}{\mu} = 10.61 \pm 0.53 \%$$

$$\mu = 475.5 \pm 2.6$$

$$\sigma = 50.4 \pm 2.3$$

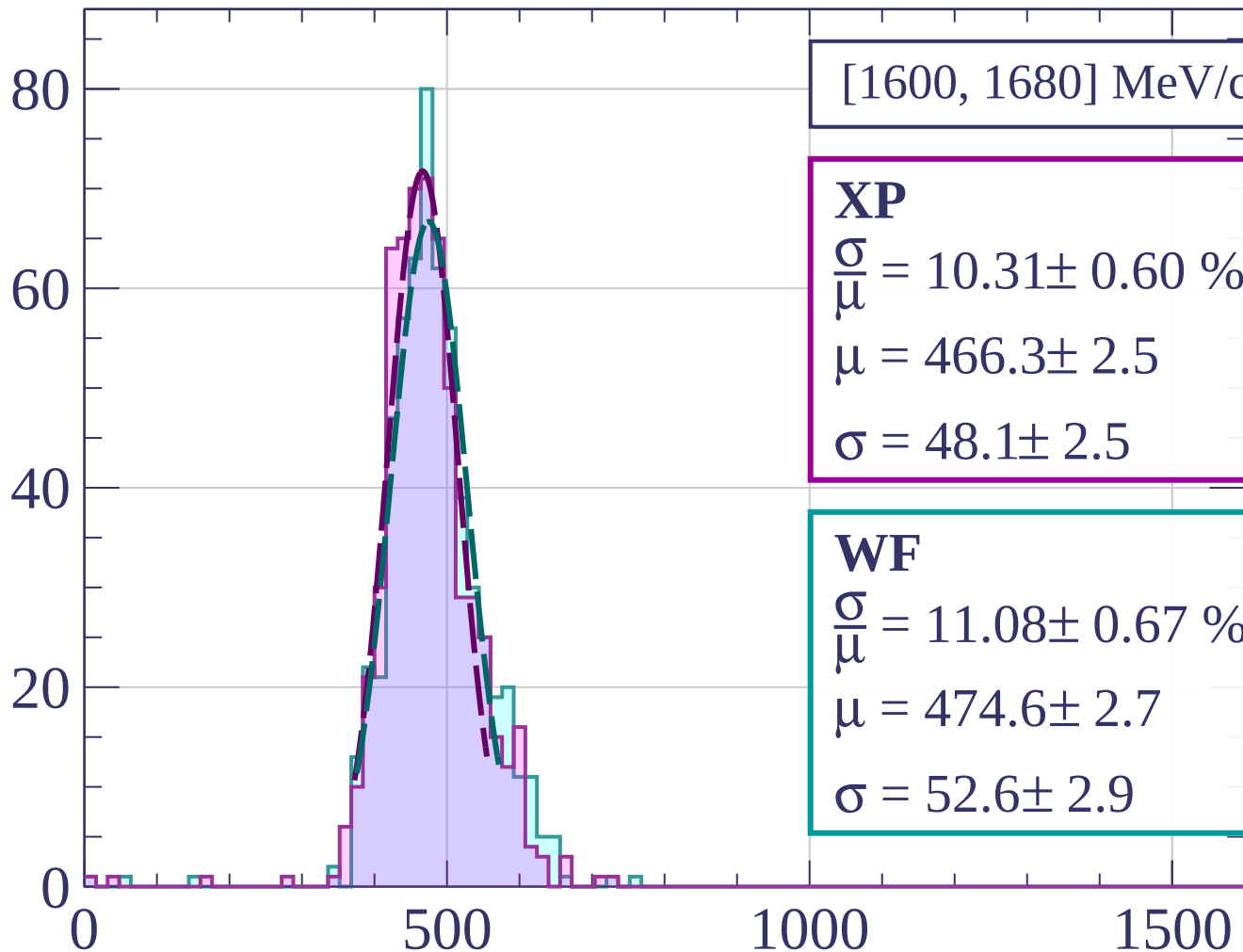
dE/dx [ADC counts/cm]

Count



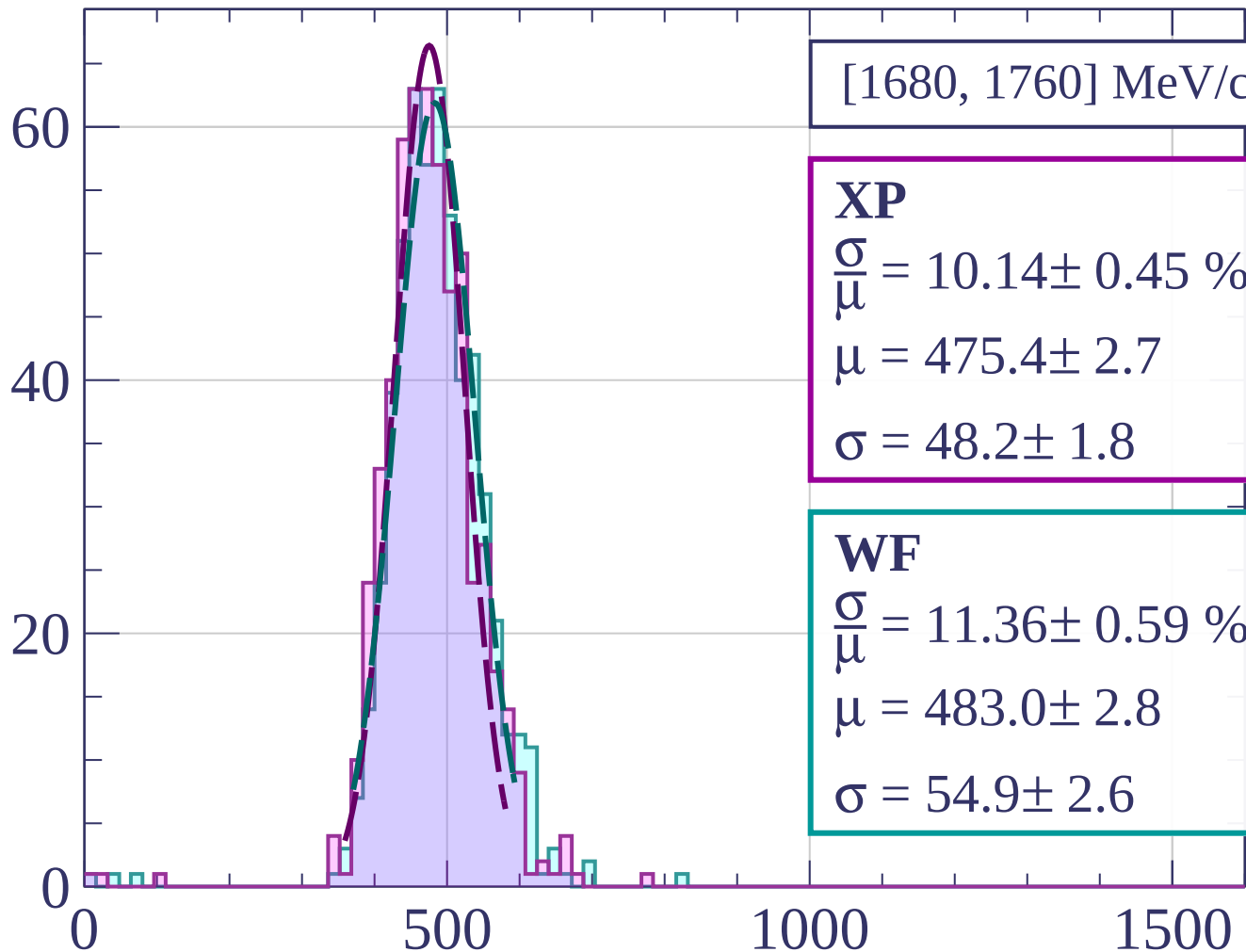
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[1680, 1760] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.14 \pm 0.45 \%$$

$$\mu = 475.4 \pm 2.7$$

$$\sigma = 48.2 \pm 1.8$$

WF

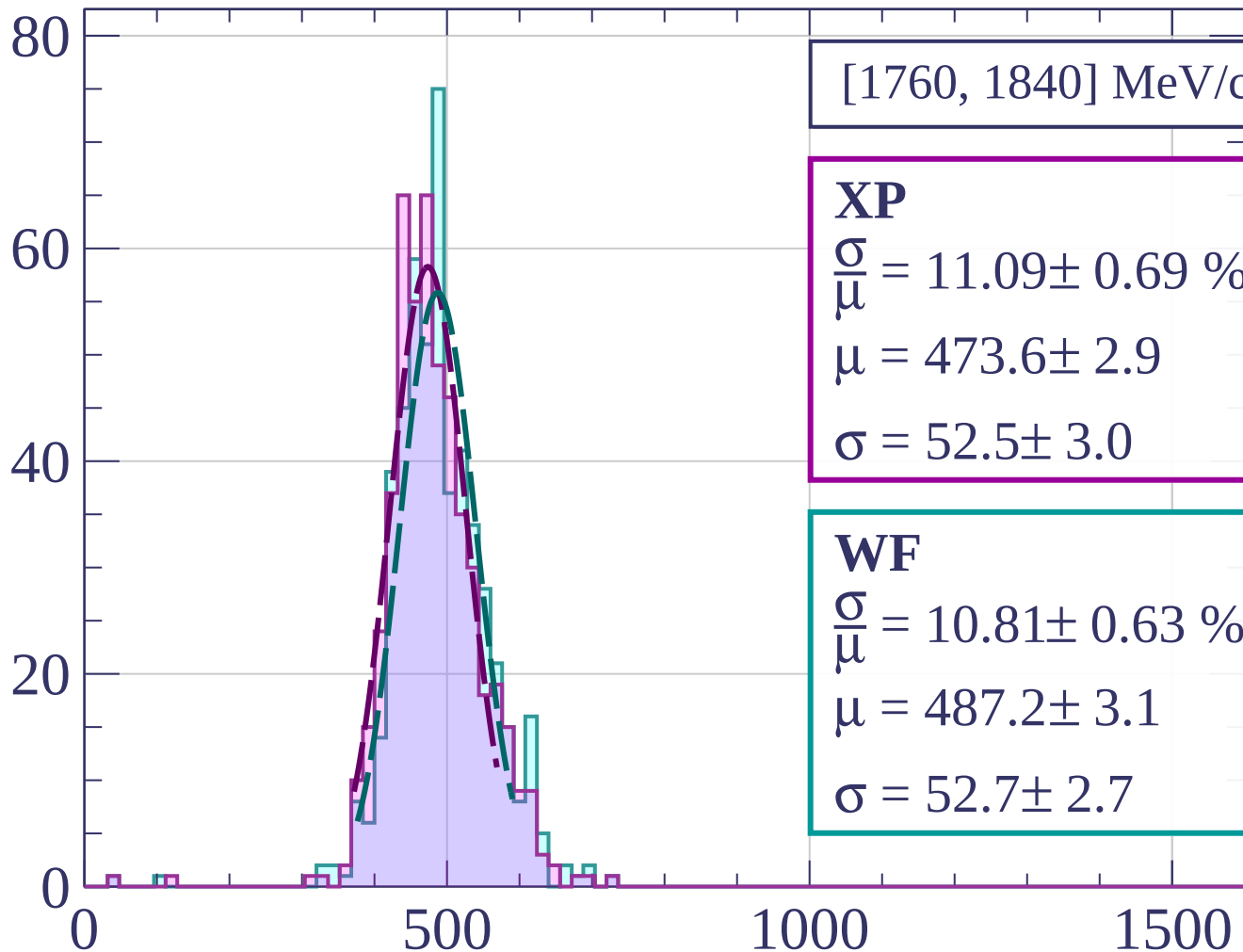
$$\frac{\sigma}{\mu} = 11.36 \pm 0.59 \%$$

$$\mu = 483.0 \pm 2.8$$

$$\sigma = 54.9 \pm 2.6$$

dE/dx [ADC counts/cm]

Count



[1760, 1840] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.09 \pm 0.69 \%$$

$$\mu = 473.6 \pm 2.9$$

$$\sigma = 52.5 \pm 3.0$$

WF

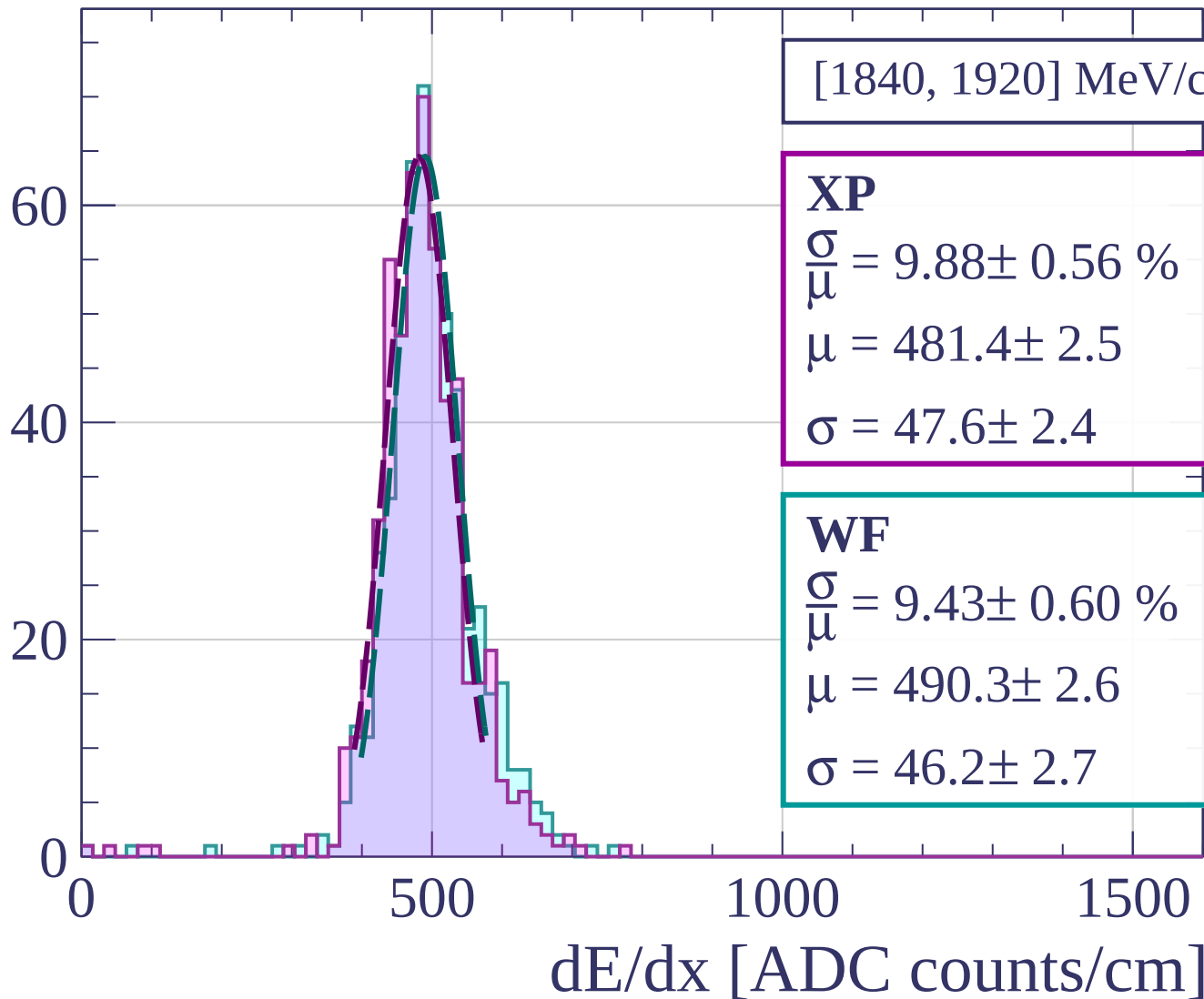
$$\frac{\sigma}{\mu} = 10.81 \pm 0.63 \%$$

$$\mu = 487.2 \pm 3.1$$

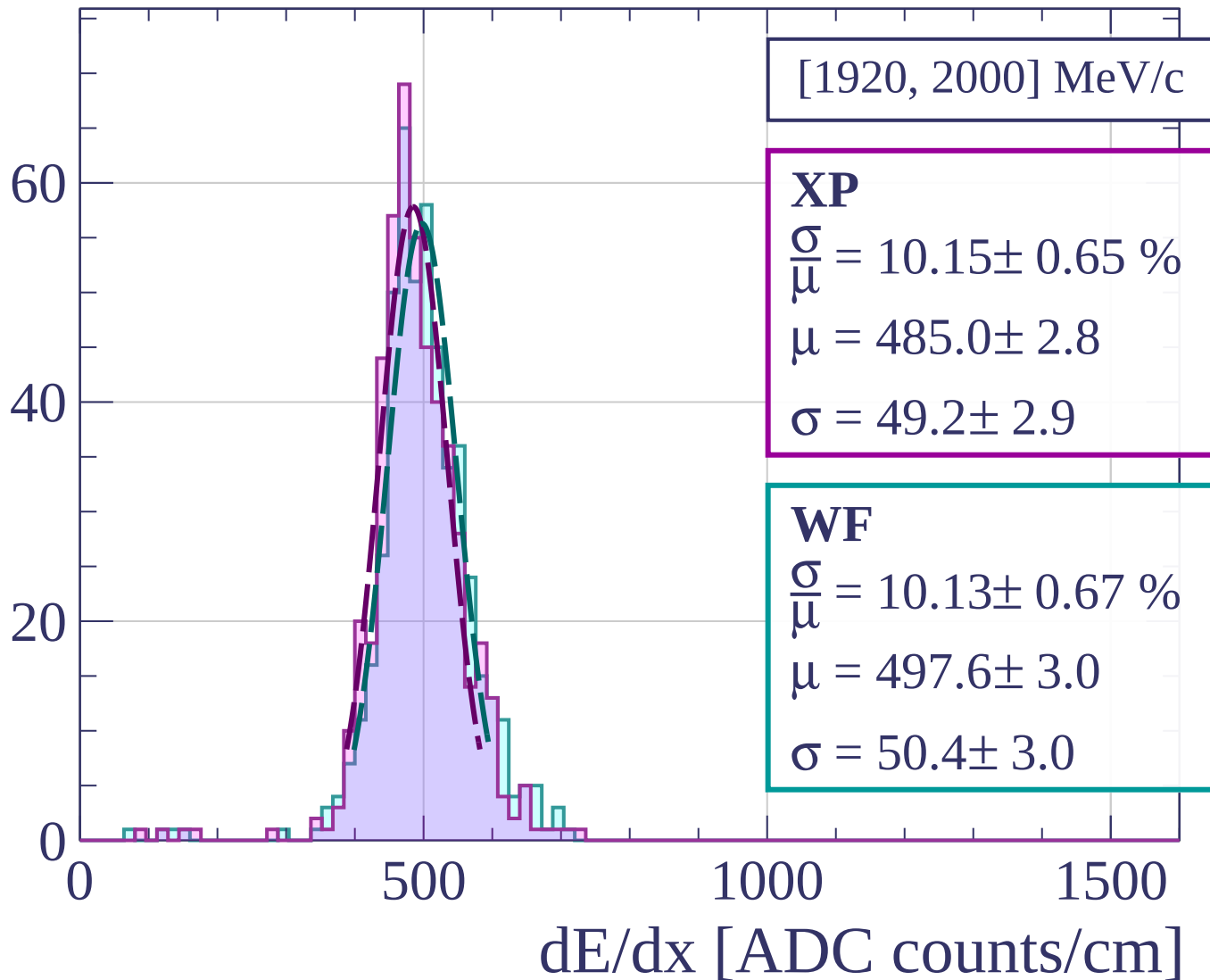
$$\sigma = 52.7 \pm 2.7$$

dE/dx [ADC counts/cm]

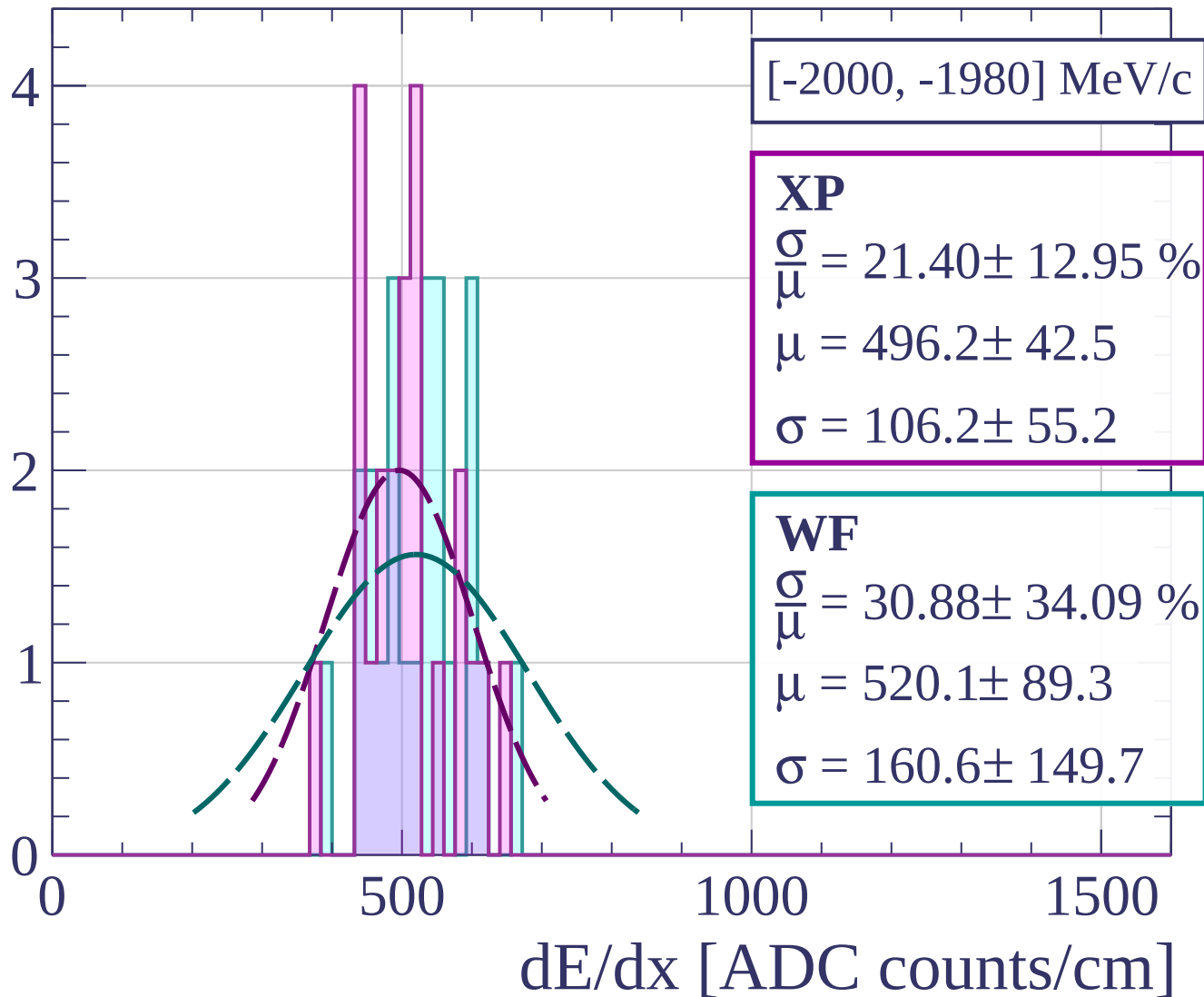
Count



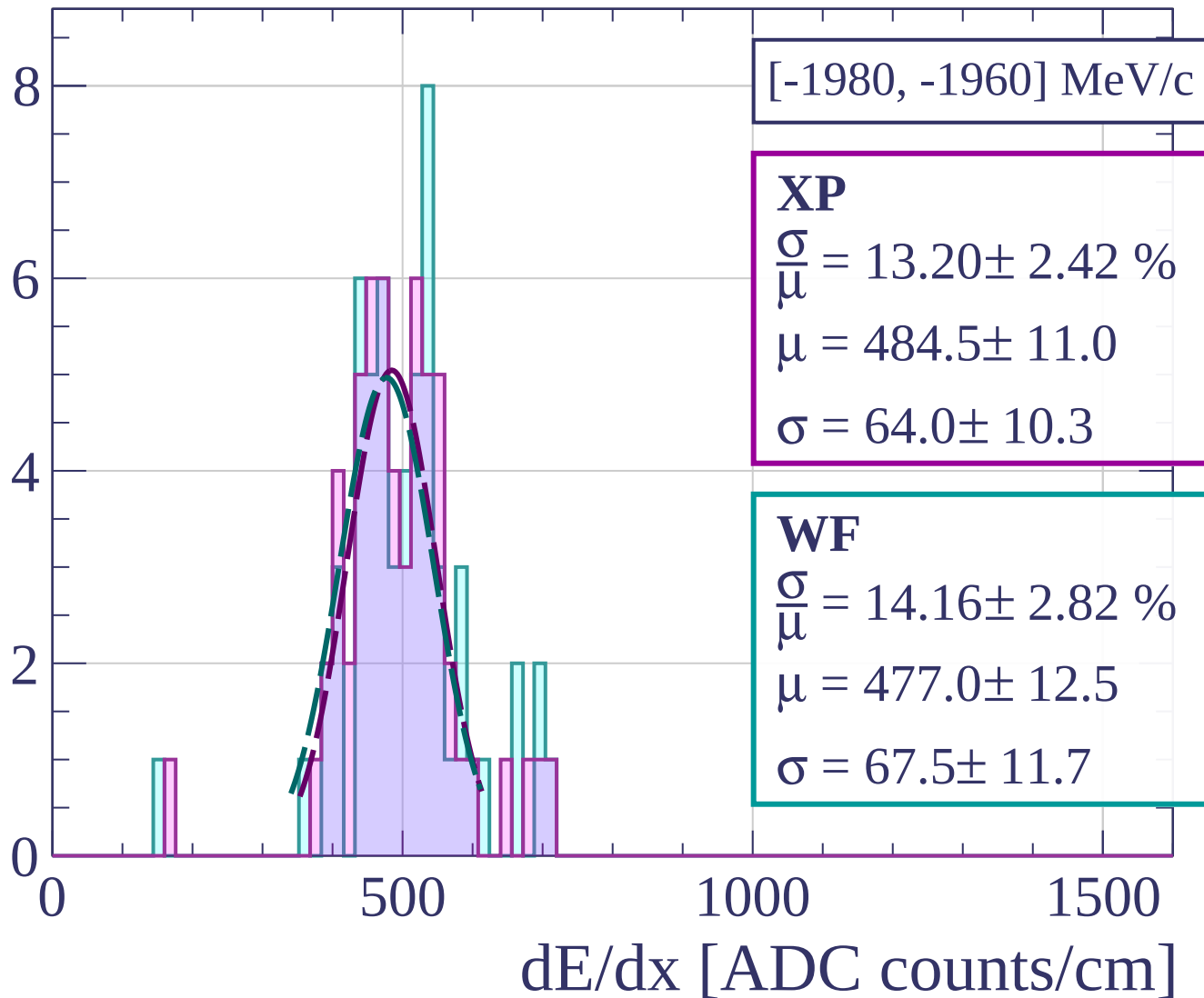
Count



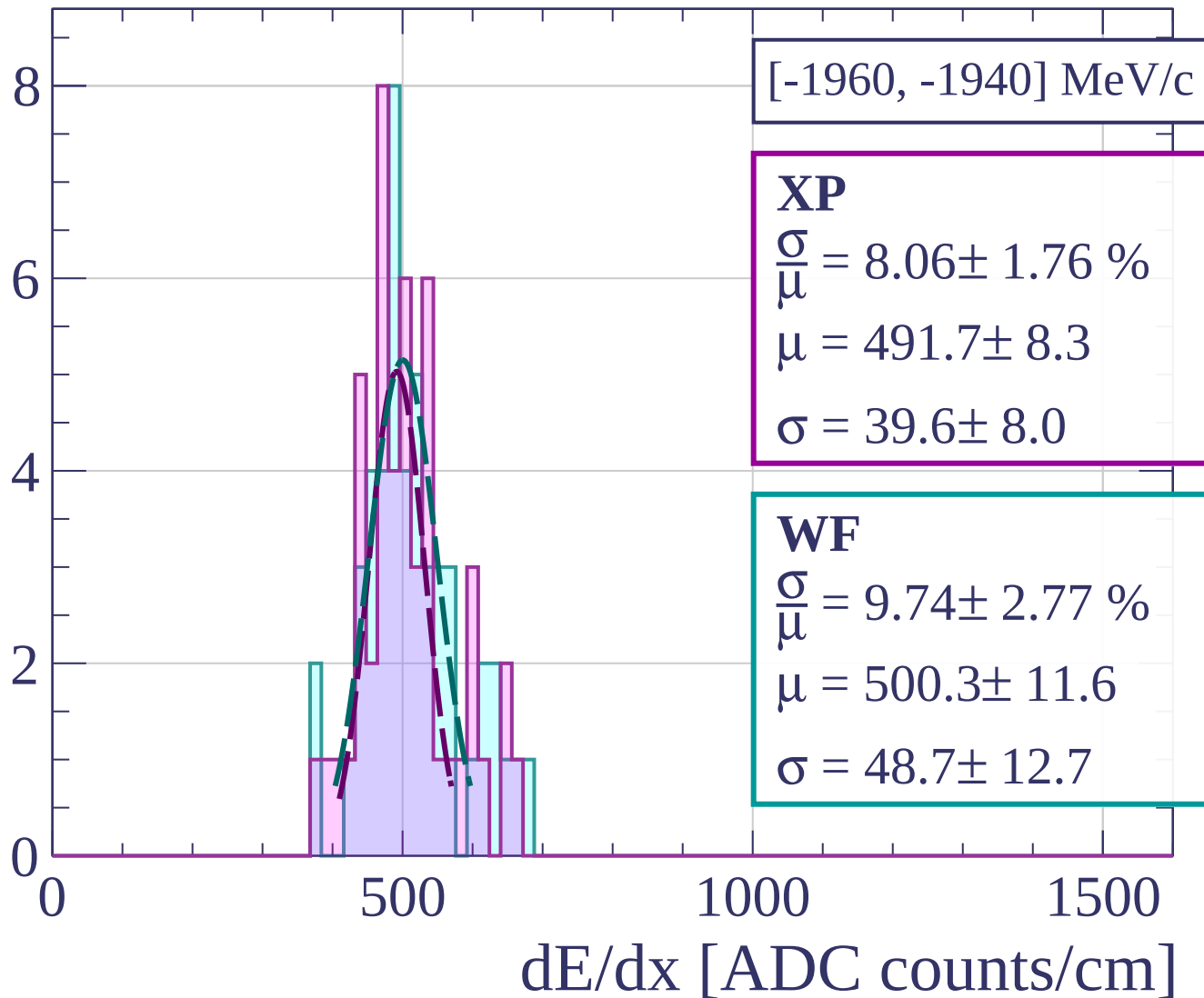
Count



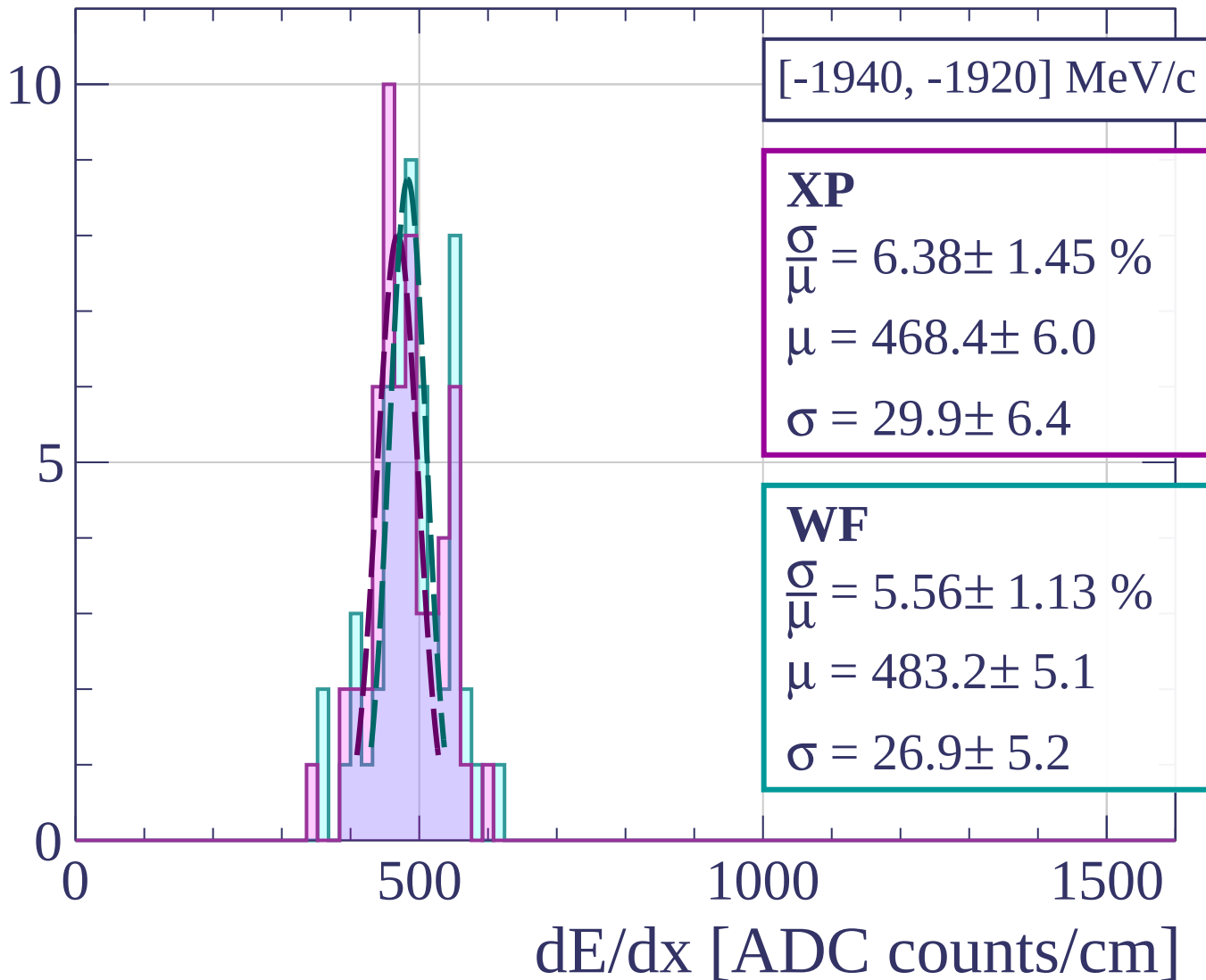
Count



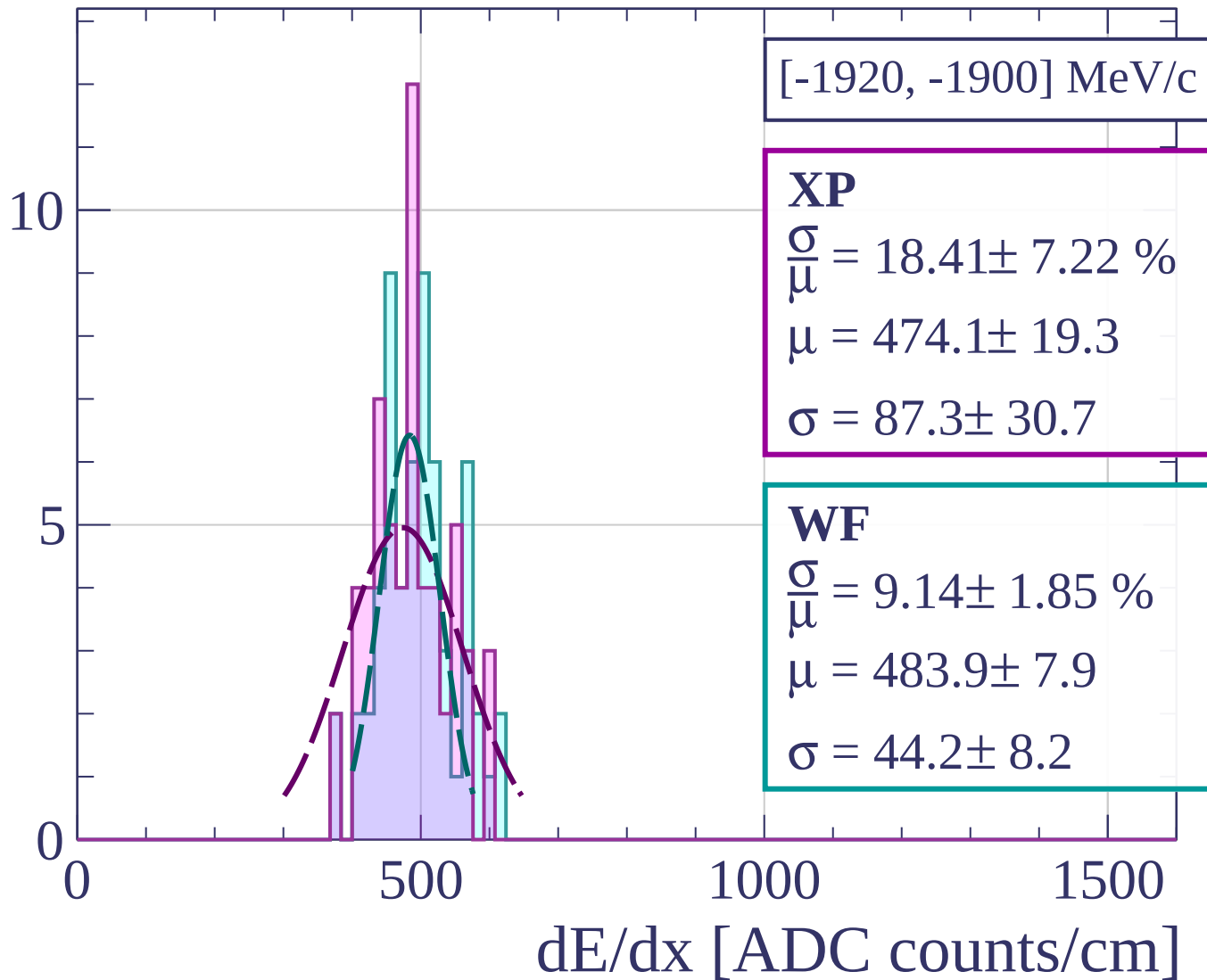
Count



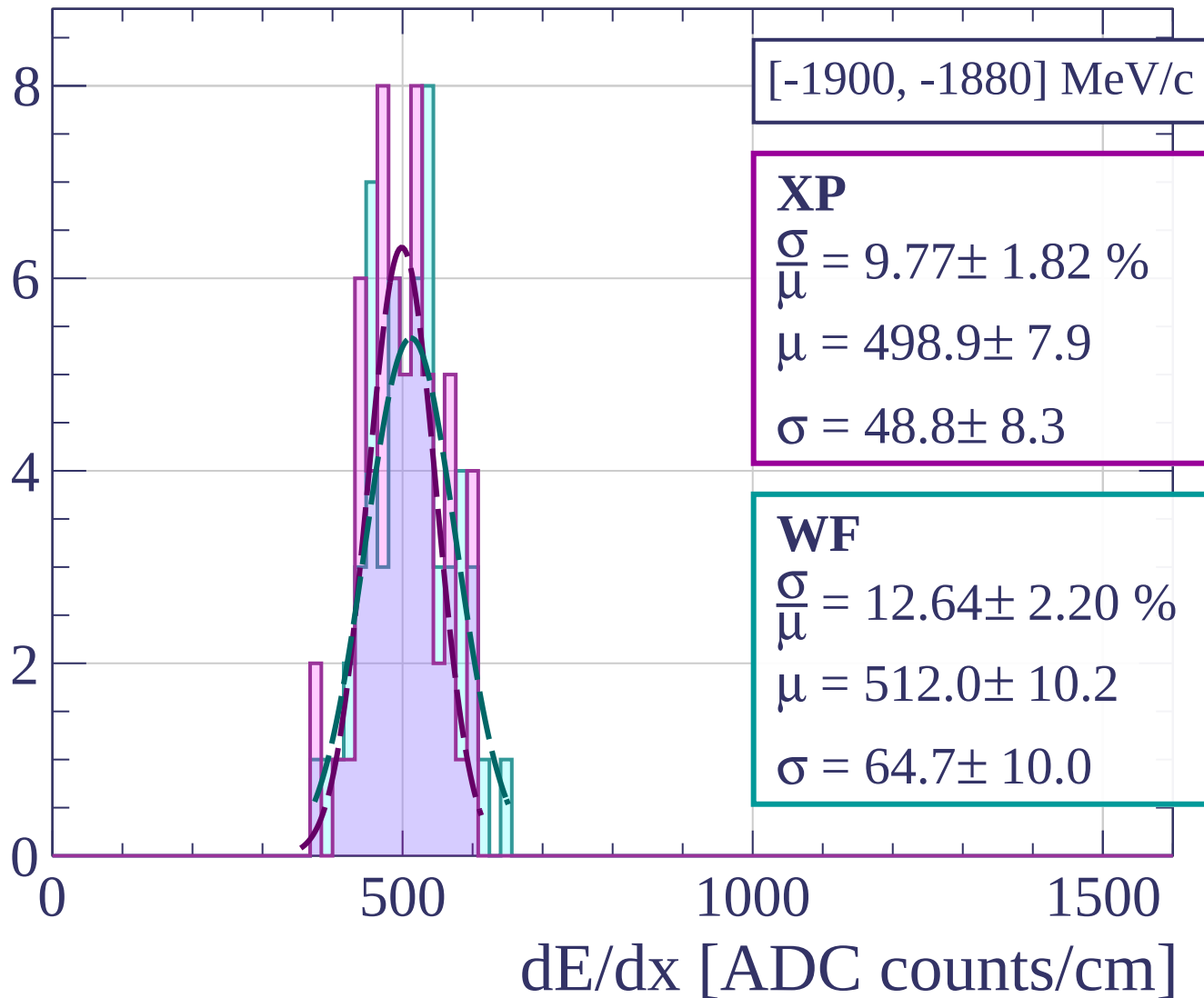
Count



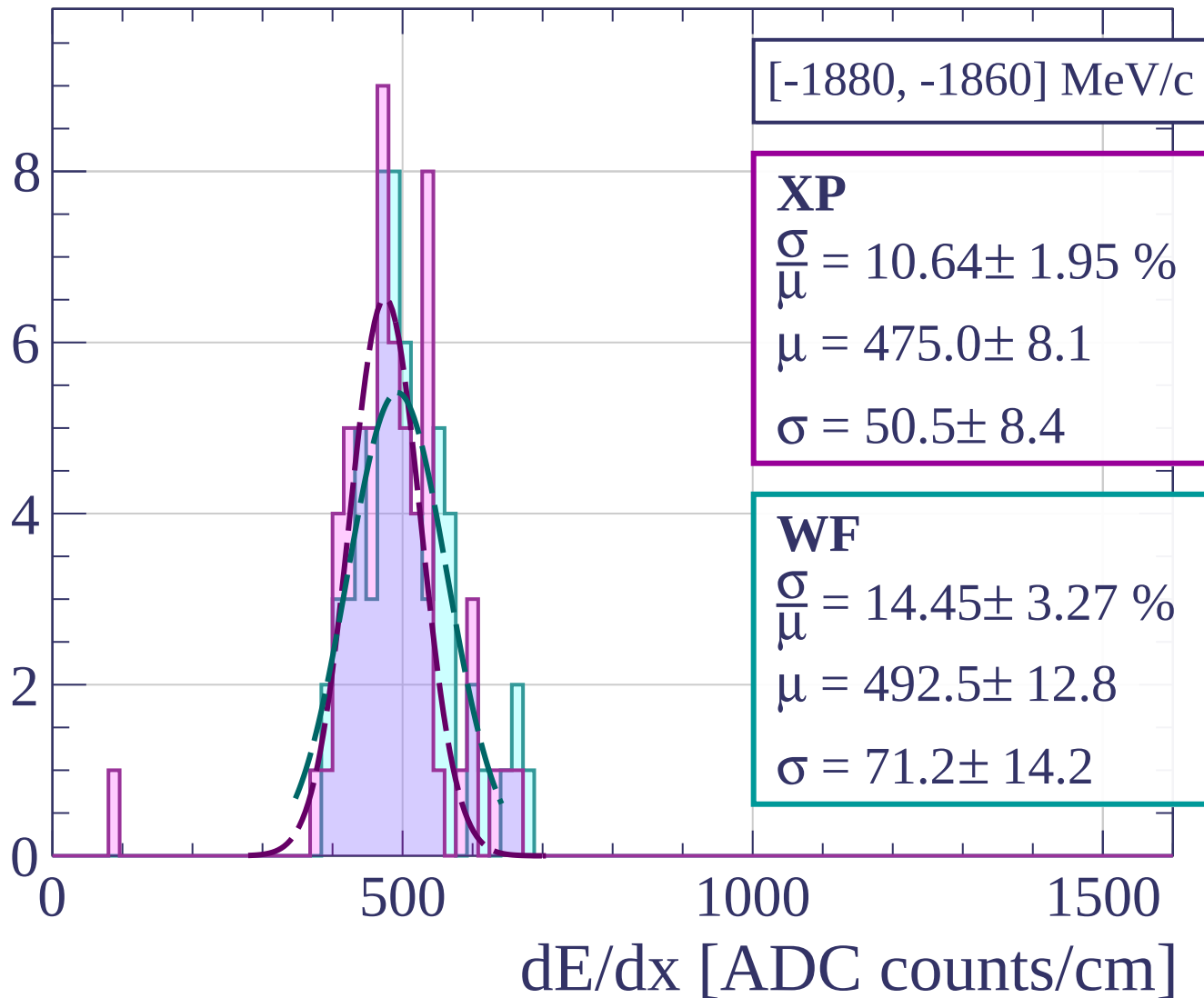
Count



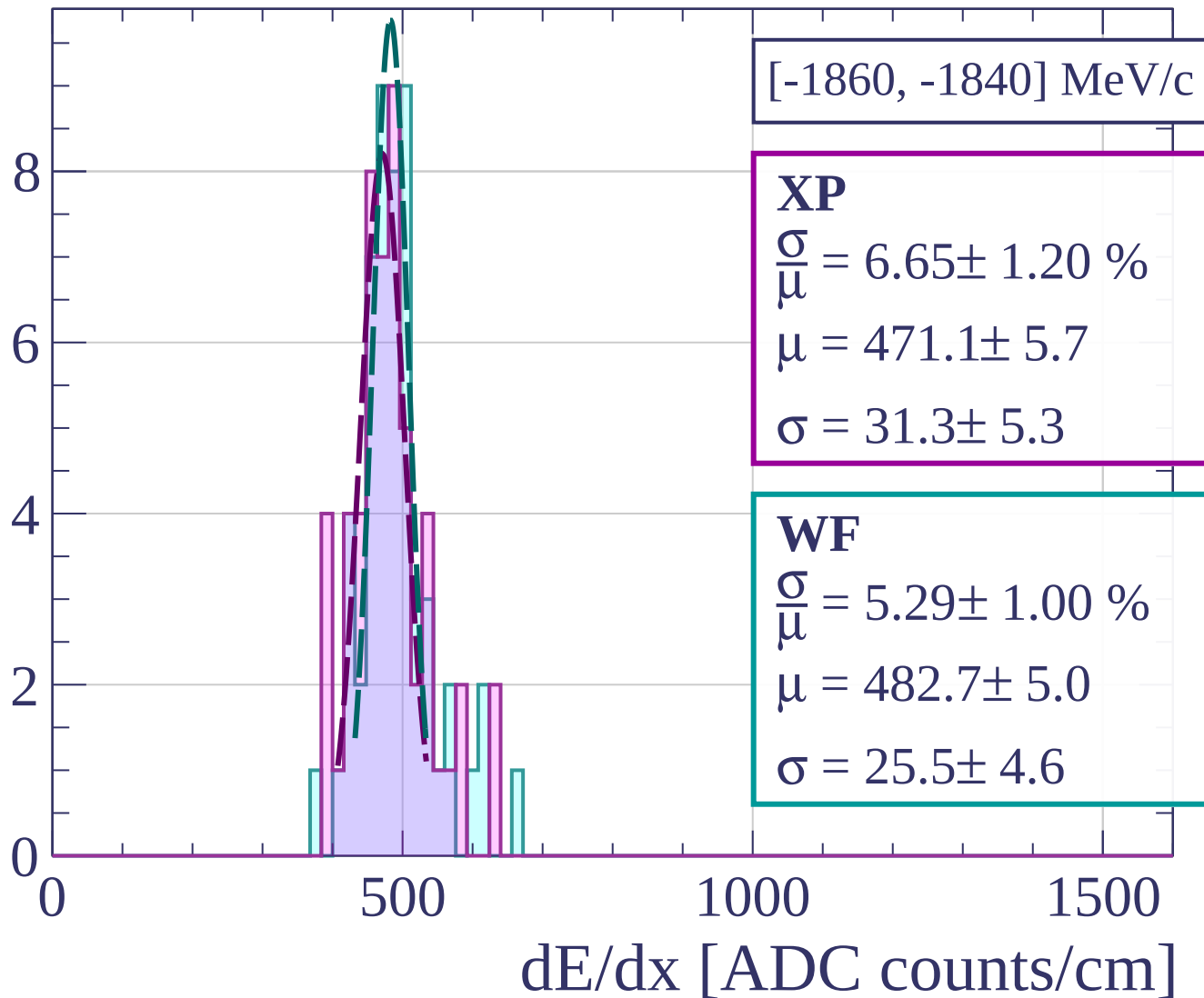
Count



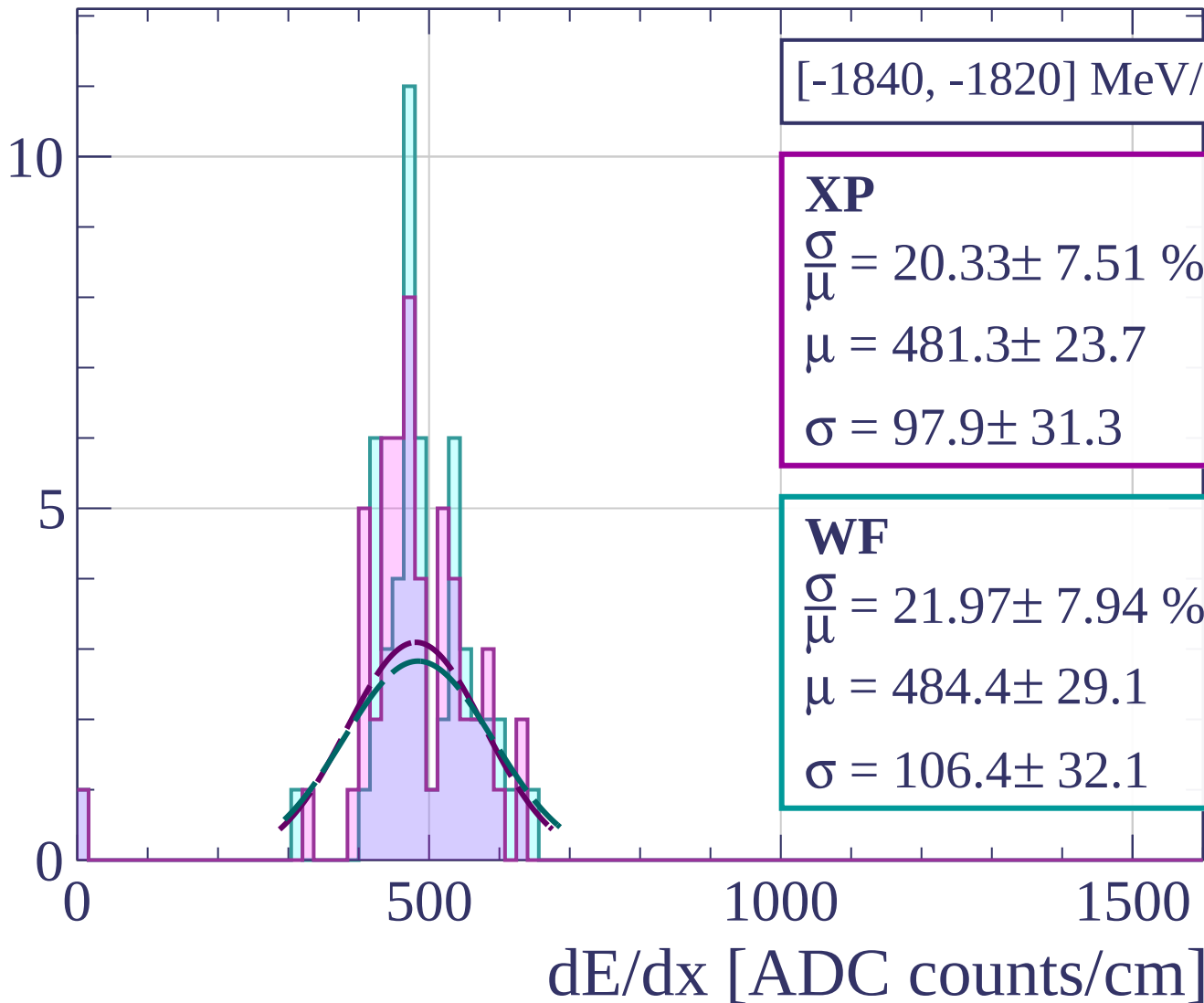
Count



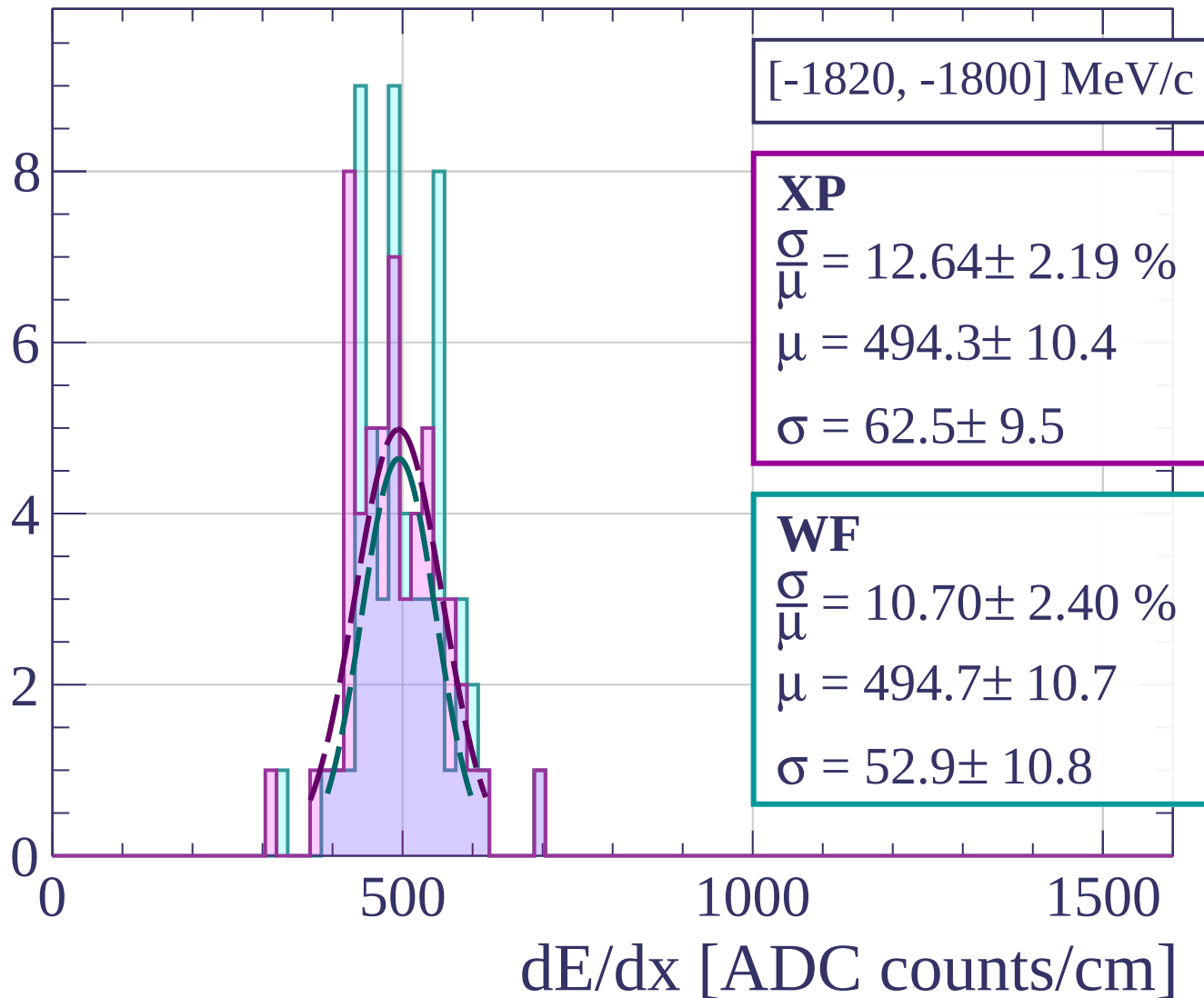
Count



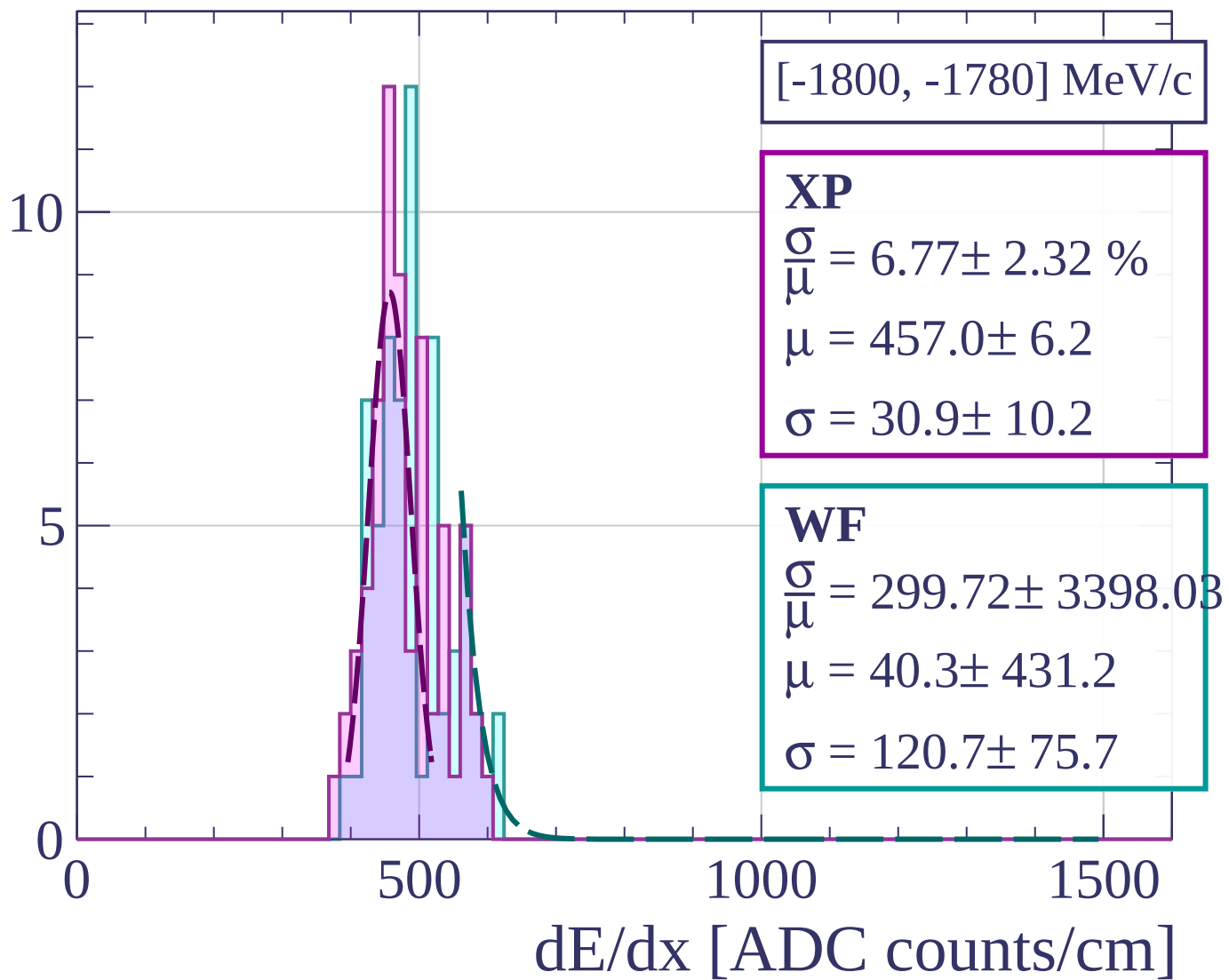
Count



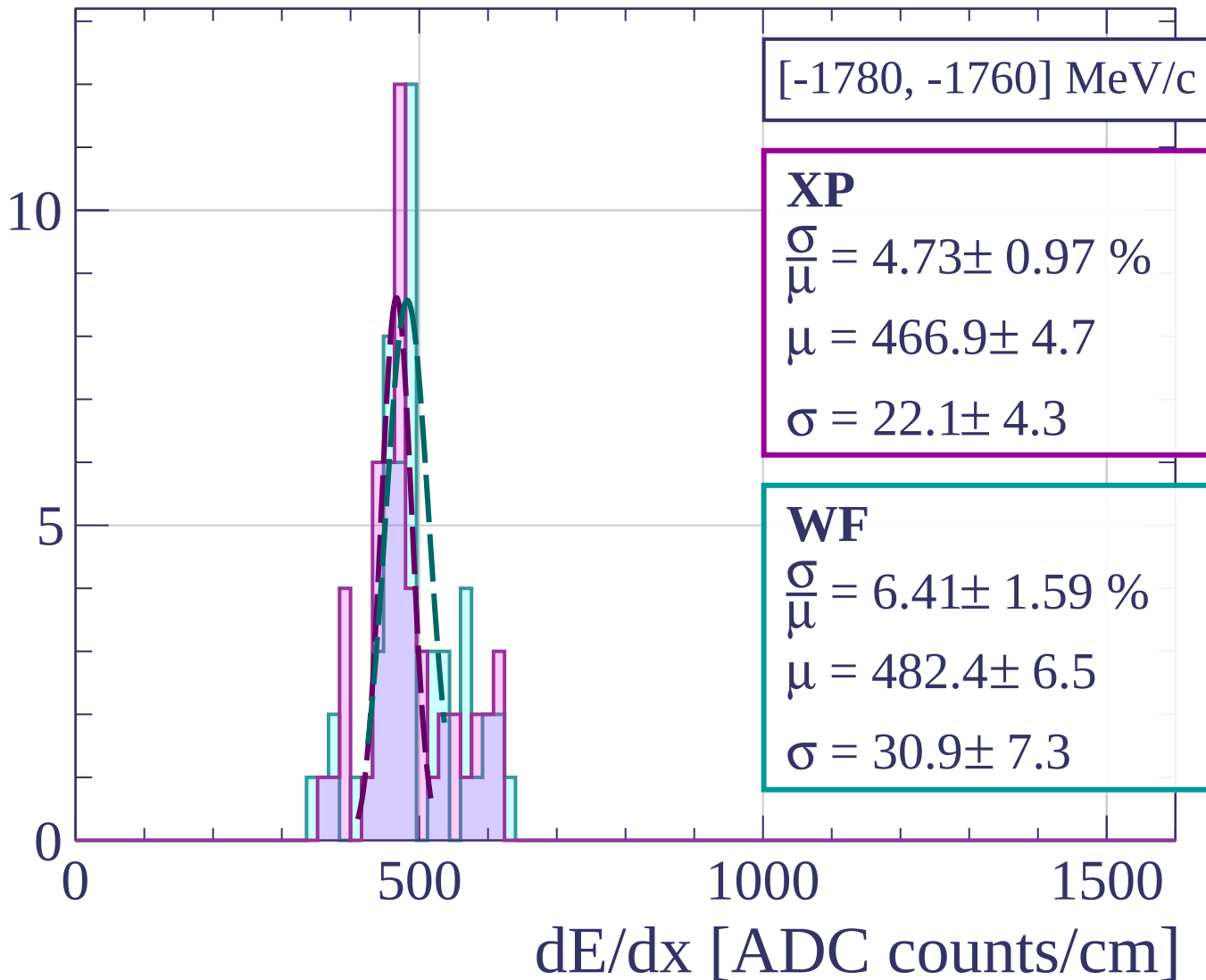
Count



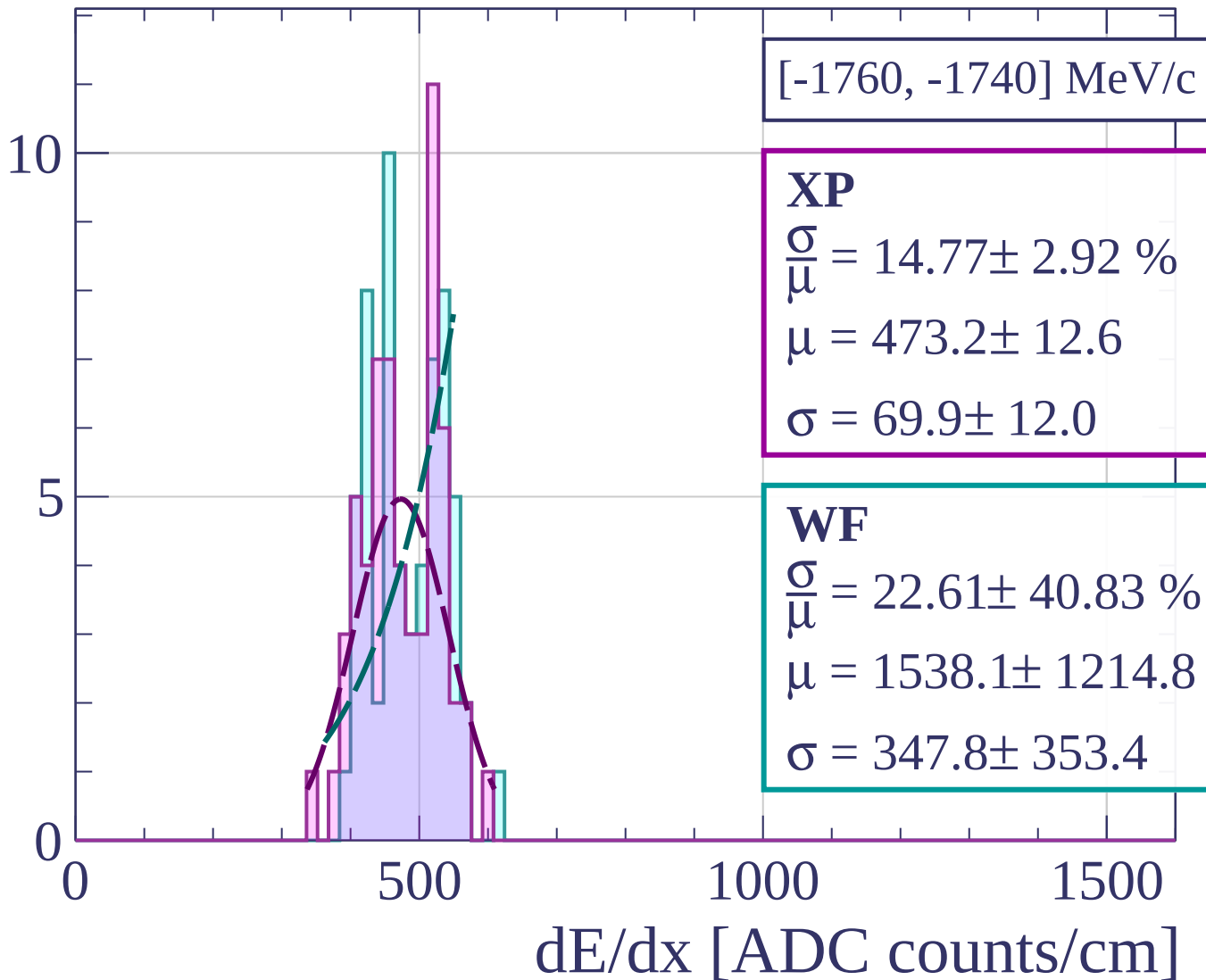
Count



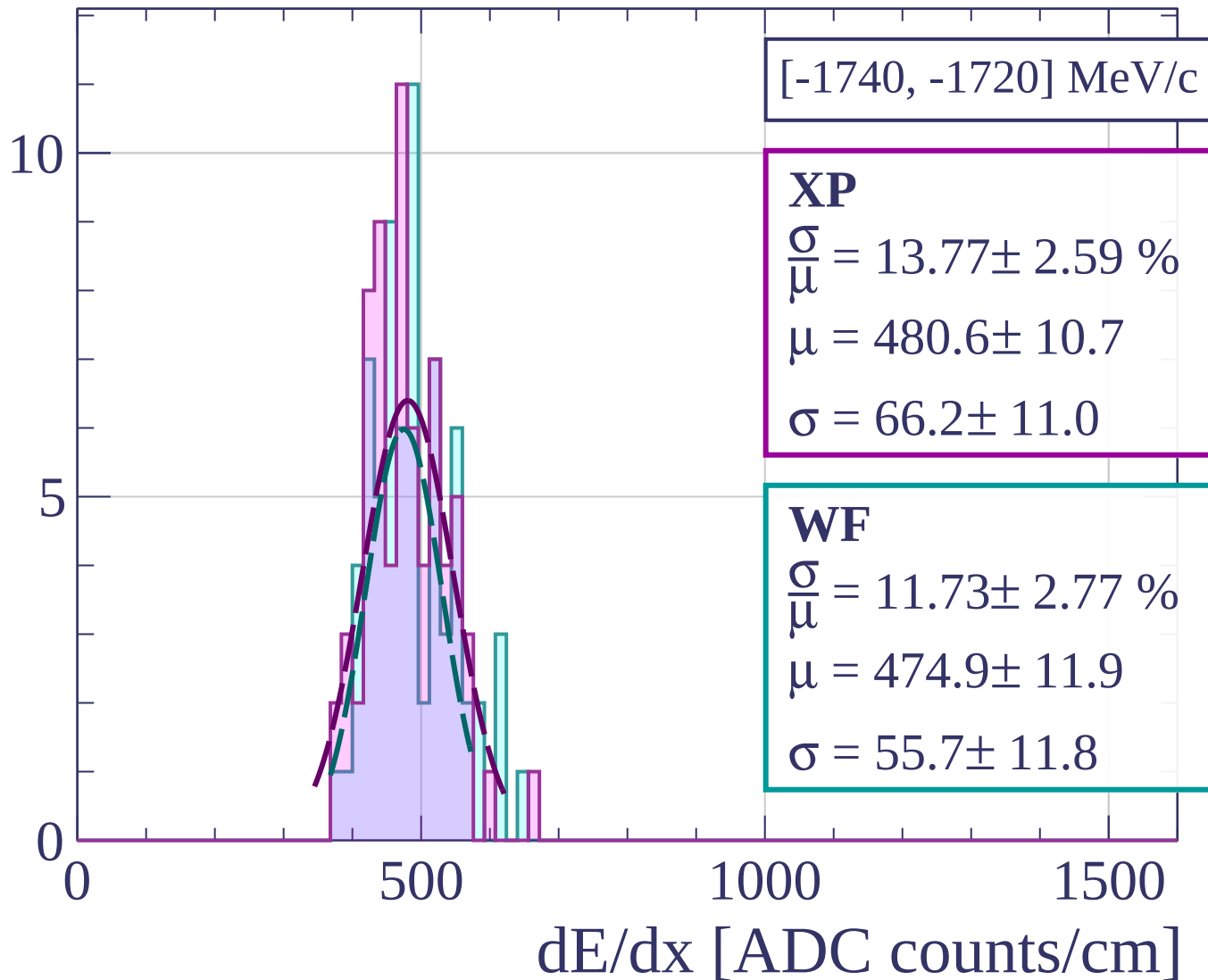
Count



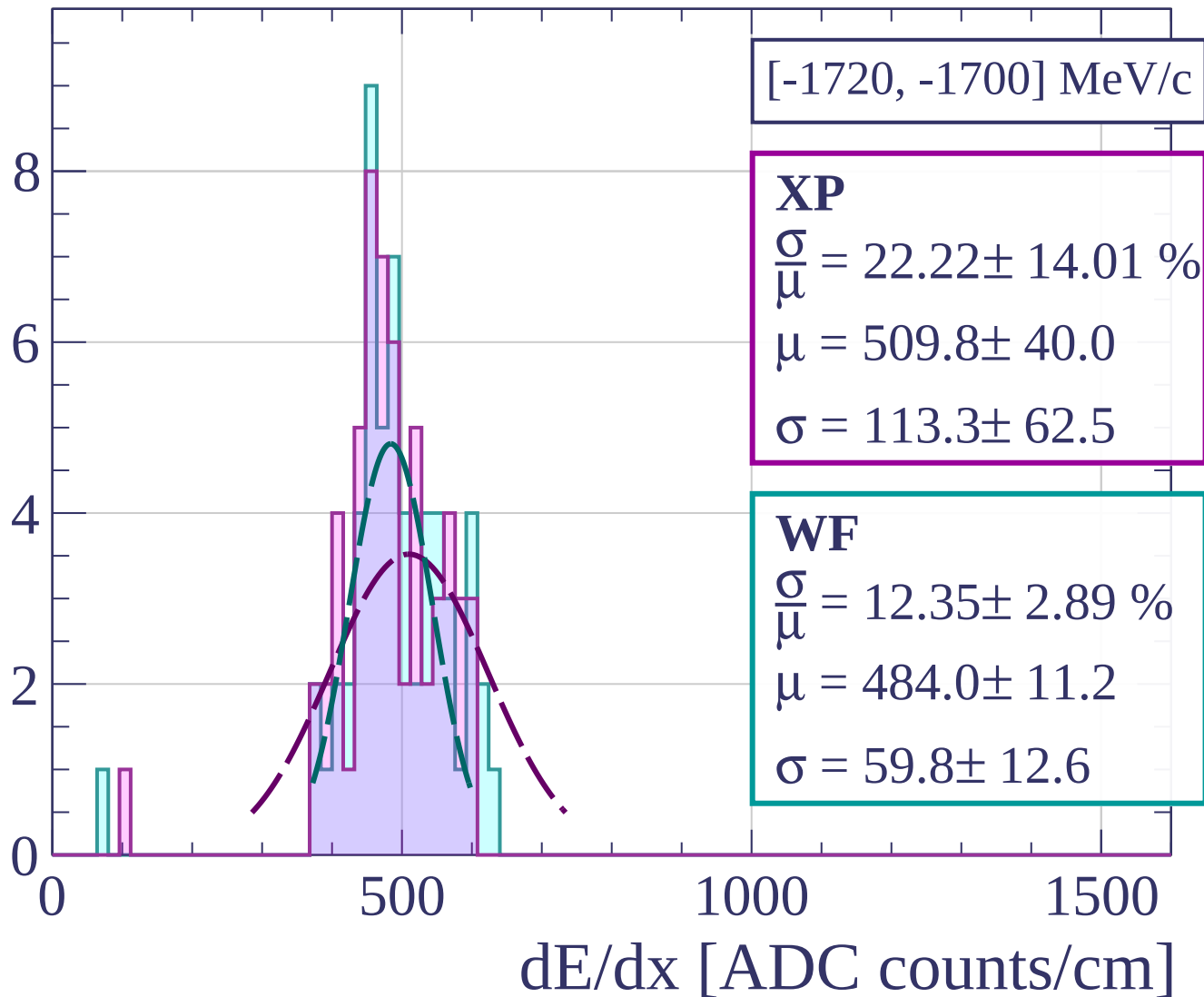
Count



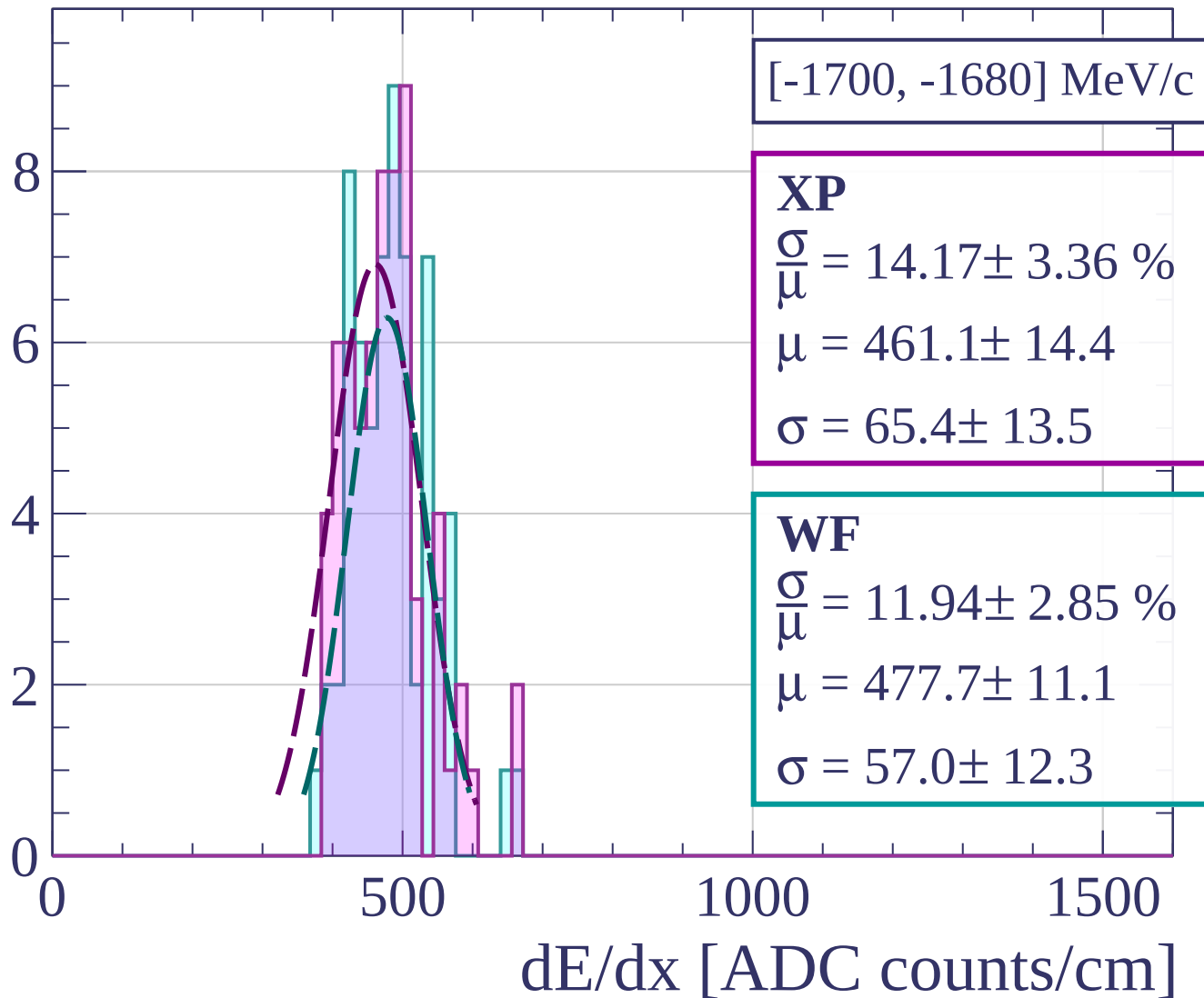
Count



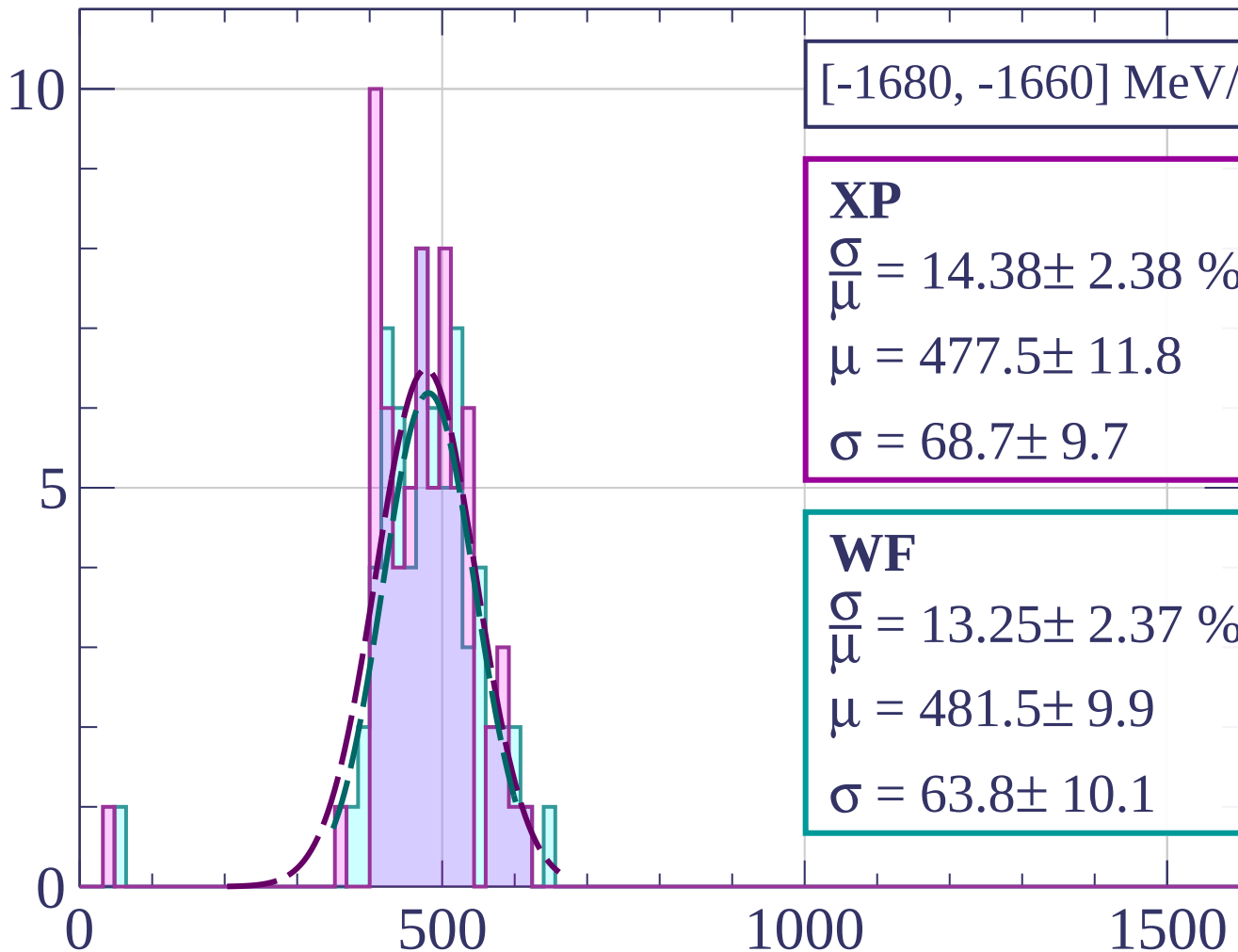
Count



Count



Count



[-1680, -1660] MeV/c

XP

$$\frac{\sigma}{\mu} = 14.38 \pm 2.38 \%$$

$$\mu = 477.5 \pm 11.8$$

$$\sigma = 68.7 \pm 9.7$$

WF

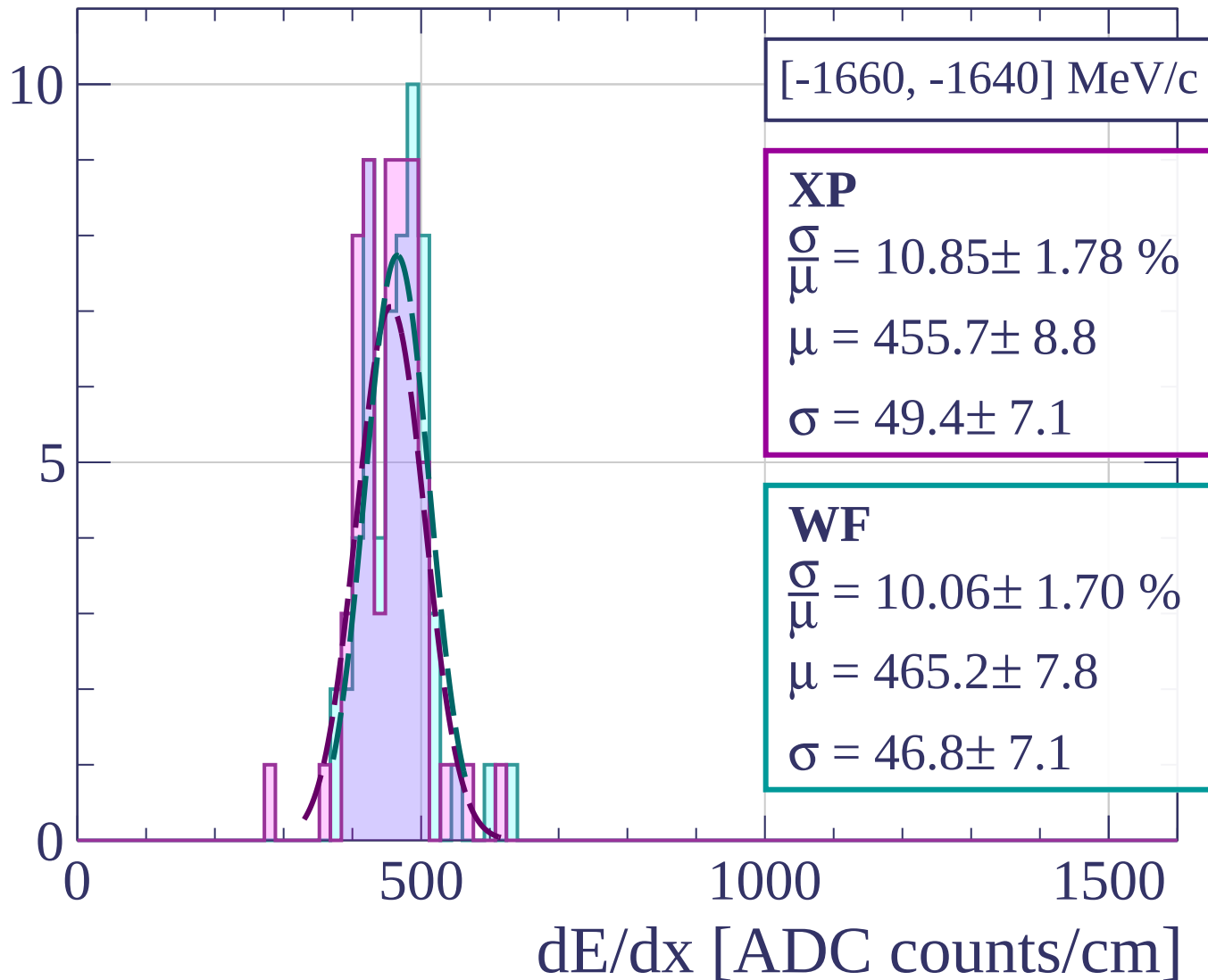
$$\frac{\sigma}{\mu} = 13.25 \pm 2.37 \%$$

$$\mu = 481.5 \pm 9.9$$

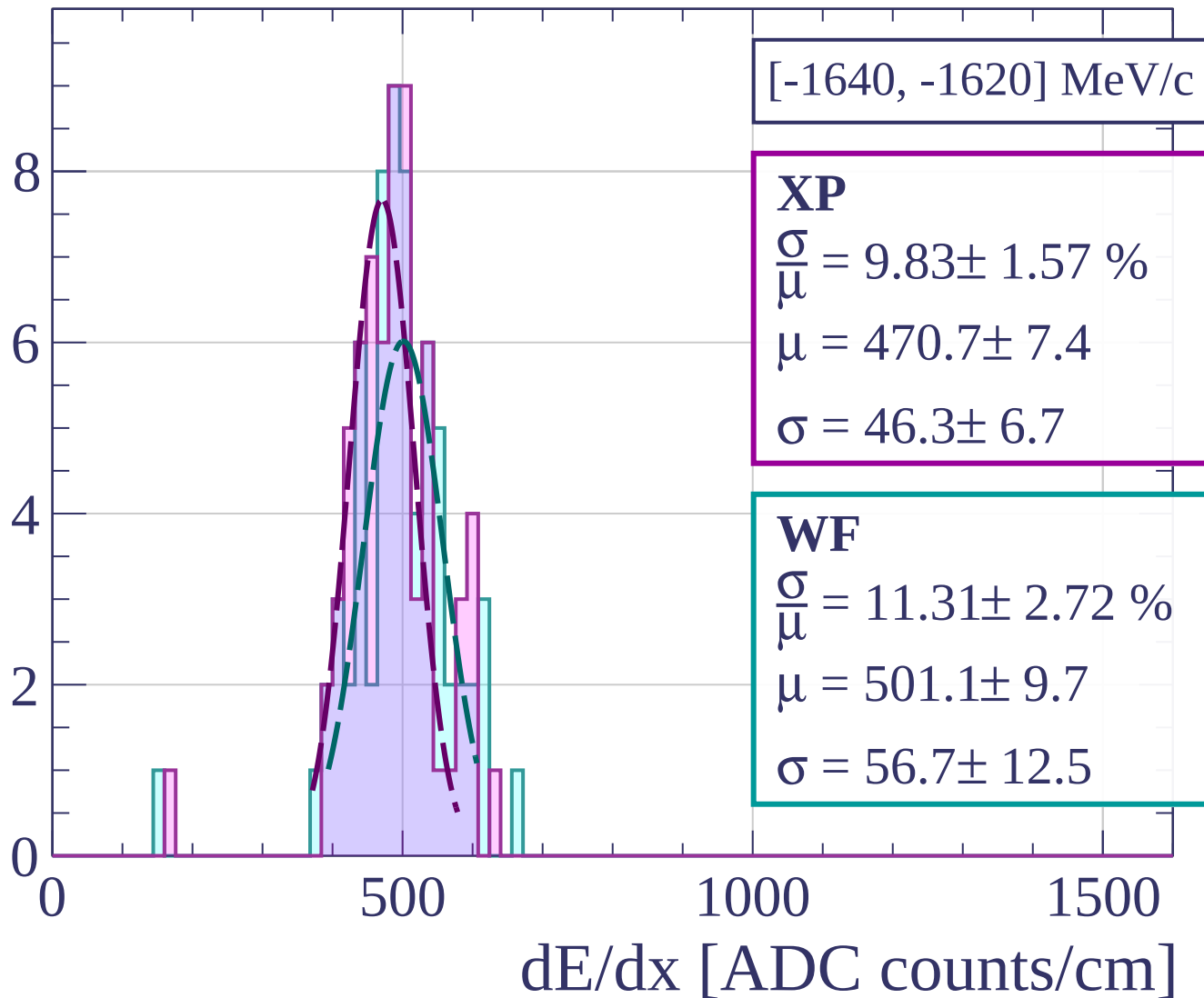
$$\sigma = 63.8 \pm 10.1$$

dE/dx [ADC counts/cm]

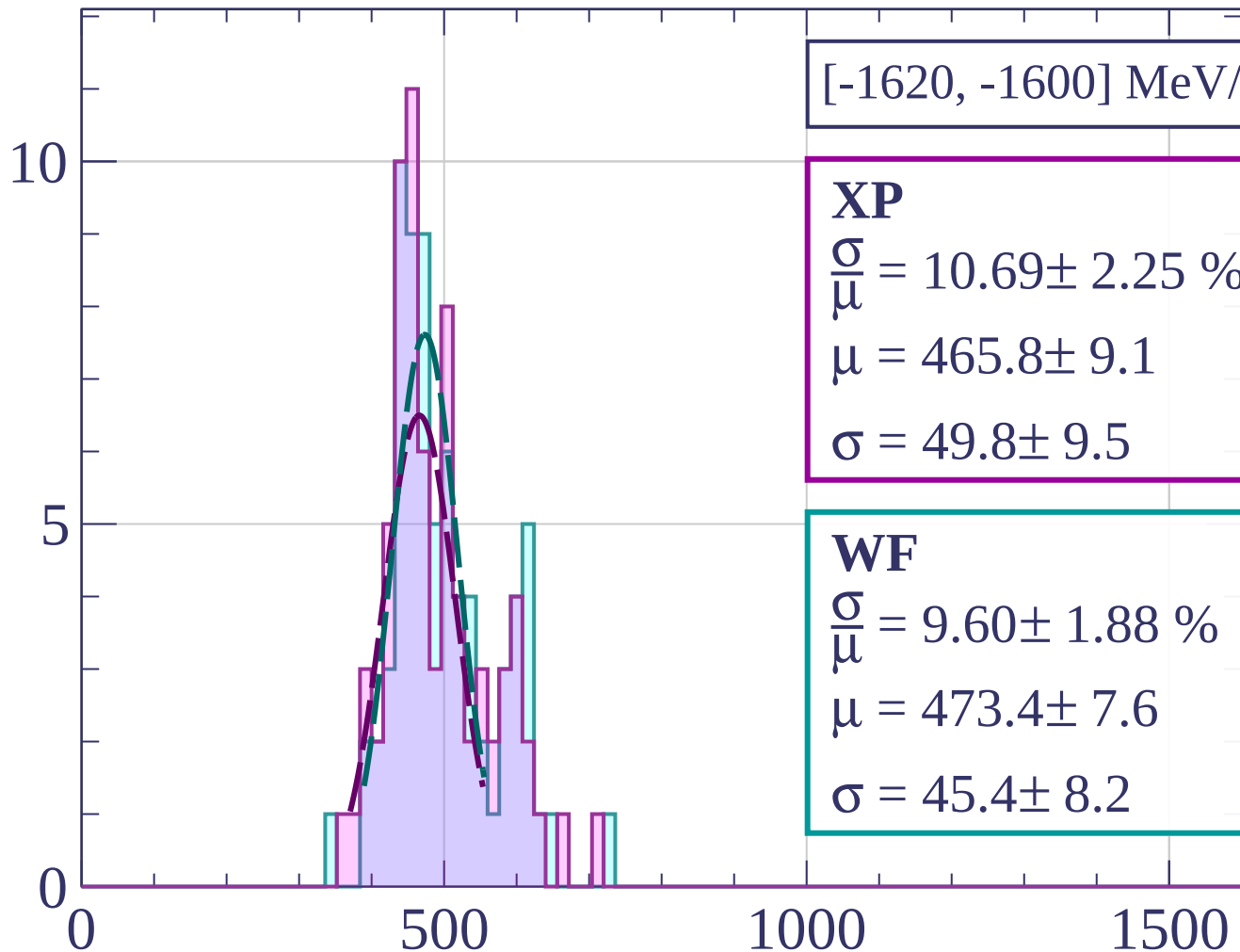
Count



Count

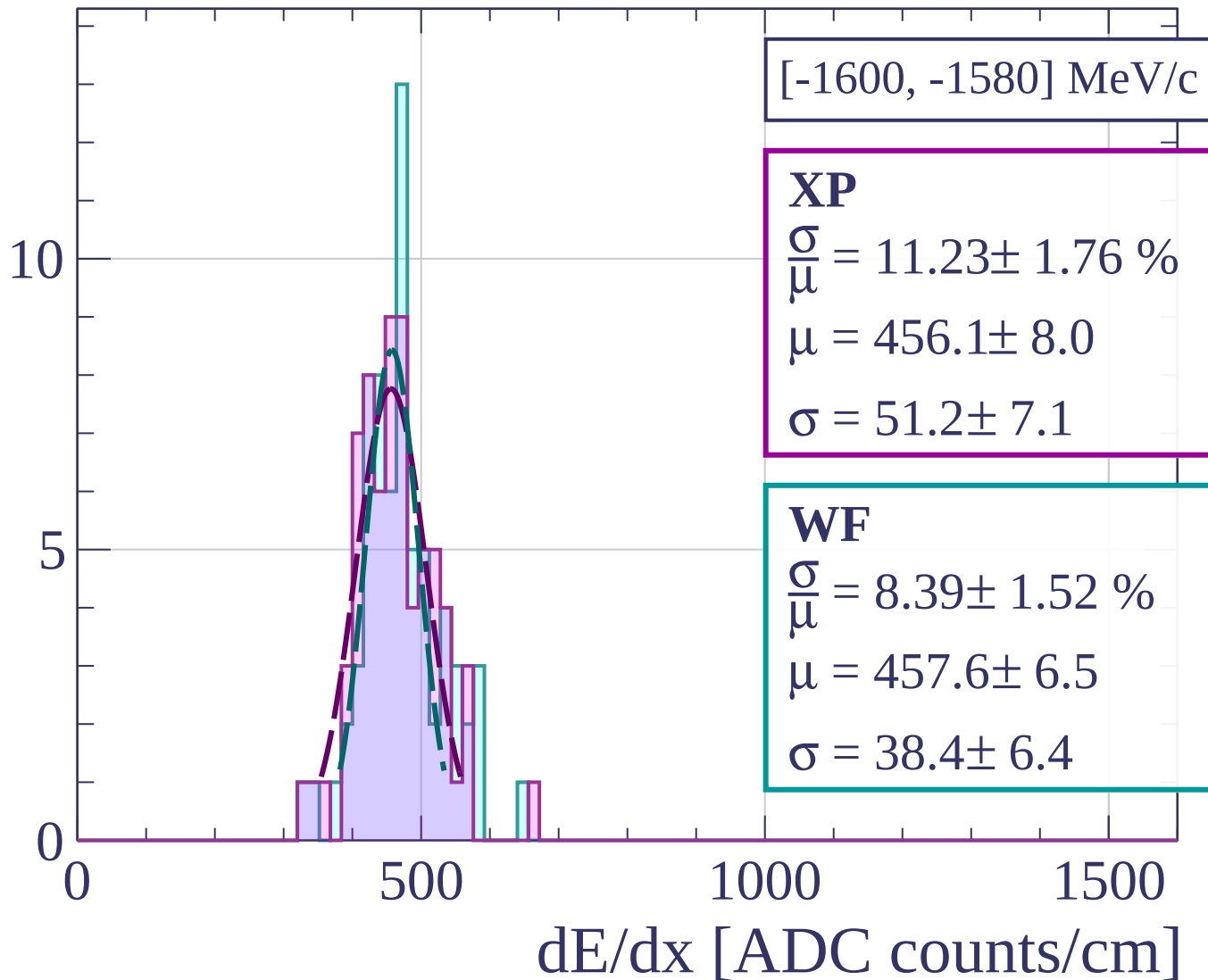


Count

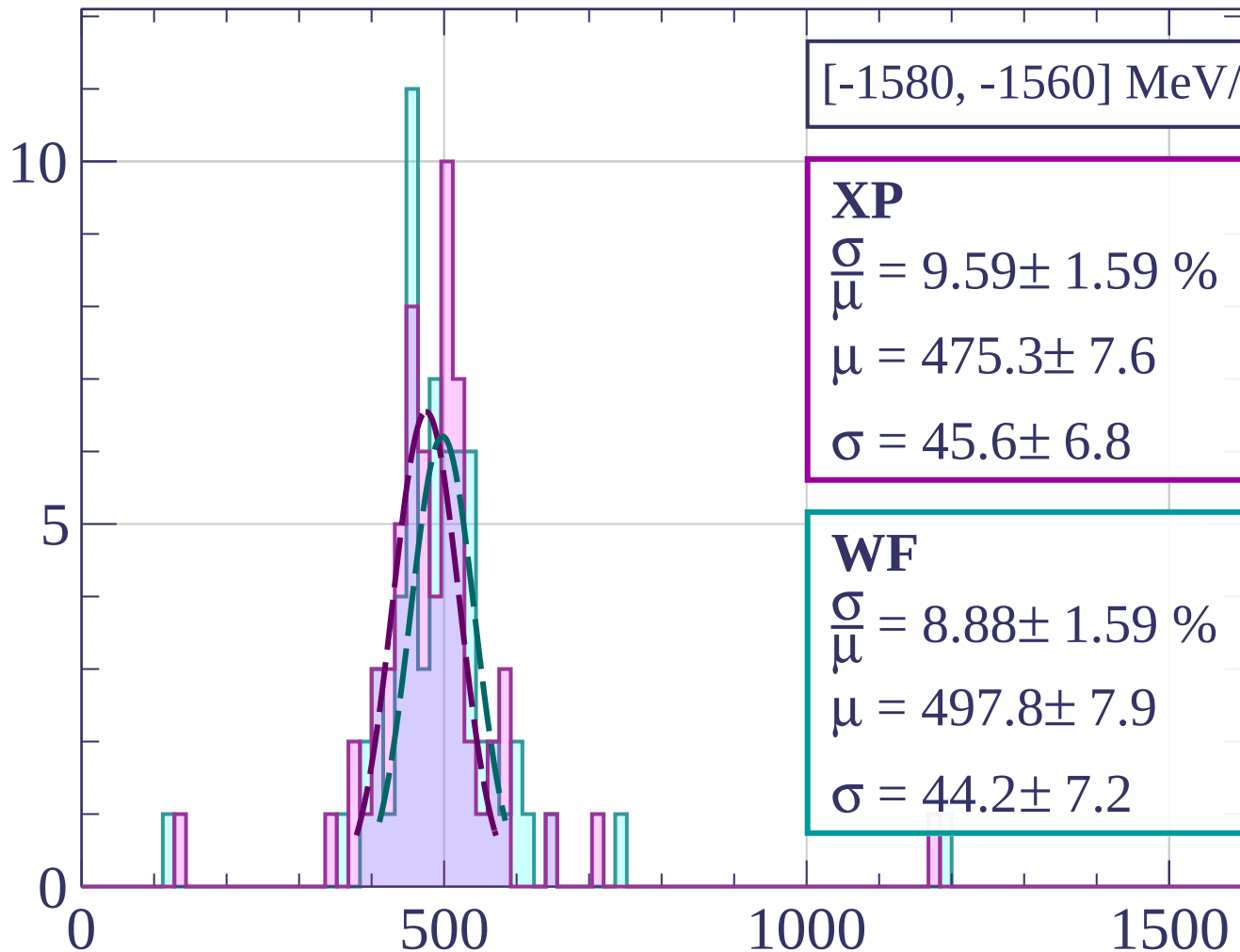


dE/dx [ADC counts/cm]

Count



Count



[-1580, -1560] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.59 \pm 1.59 \%$$

$$\mu = 475.3 \pm 7.6$$

$$\sigma = 45.6 \pm 6.8$$

WF

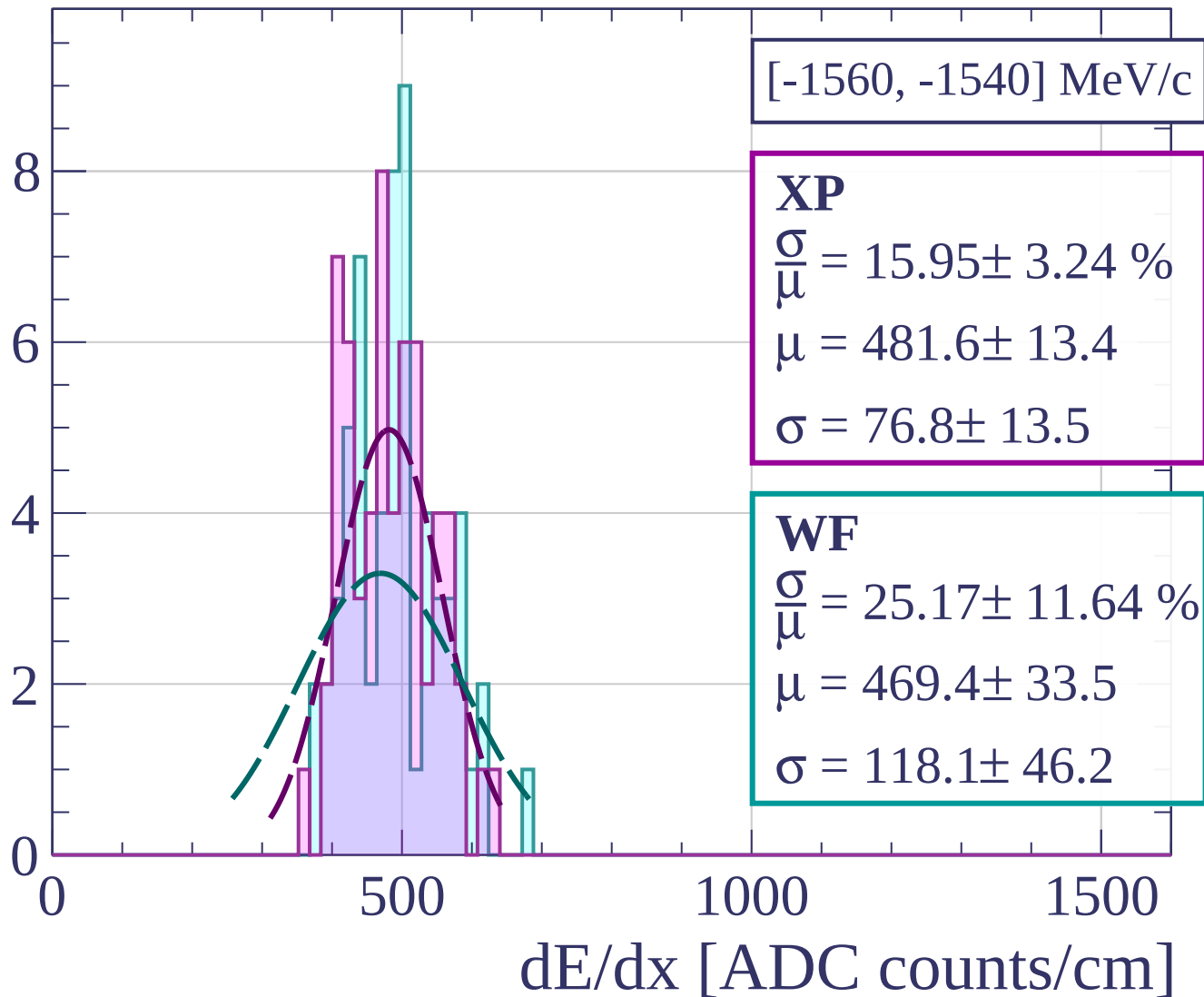
$$\frac{\sigma}{\mu} = 8.88 \pm 1.59 \%$$

$$\mu = 497.8 \pm 7.9$$

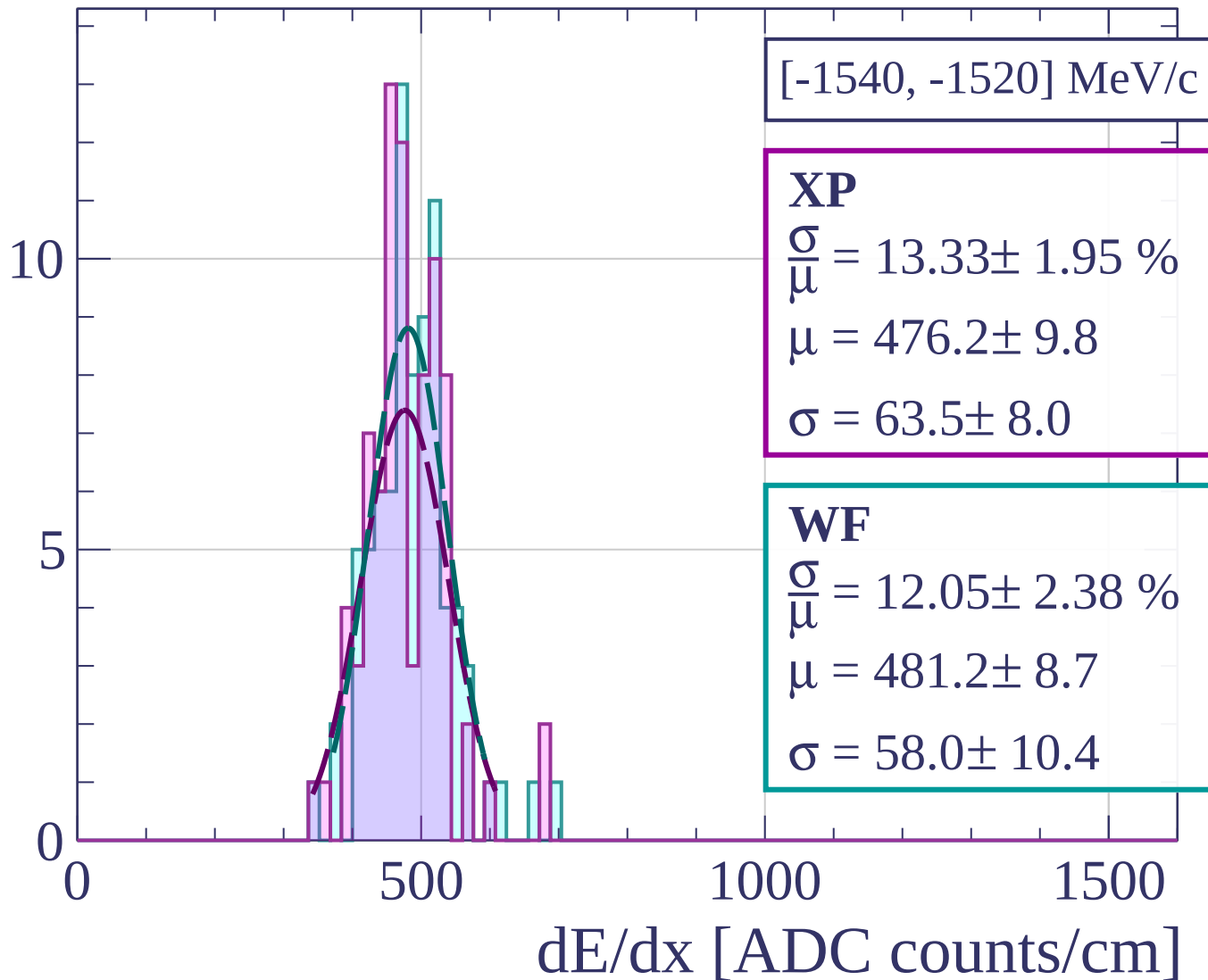
$$\sigma = 44.2 \pm 7.2$$

dE/dx [ADC counts/cm]

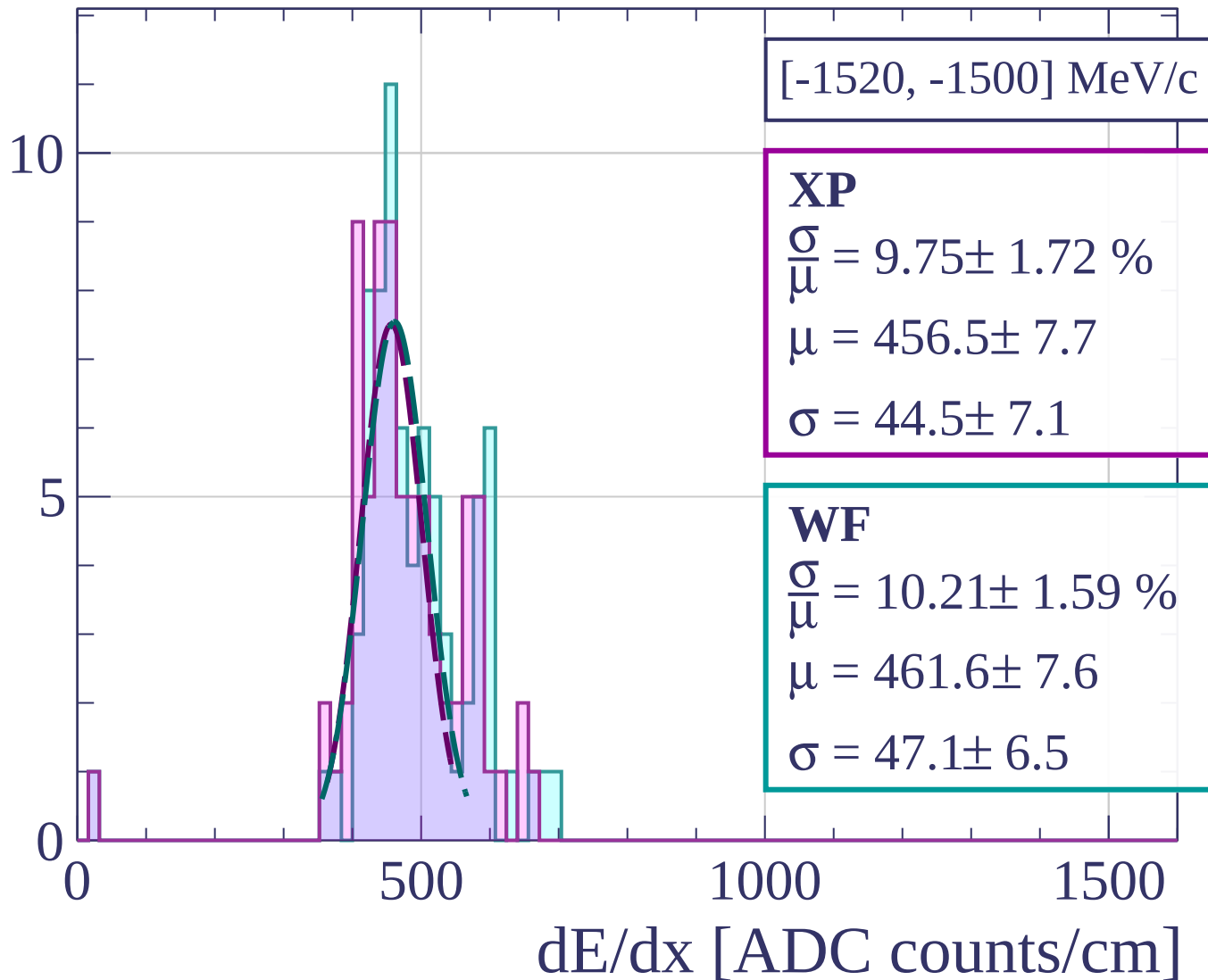
Count



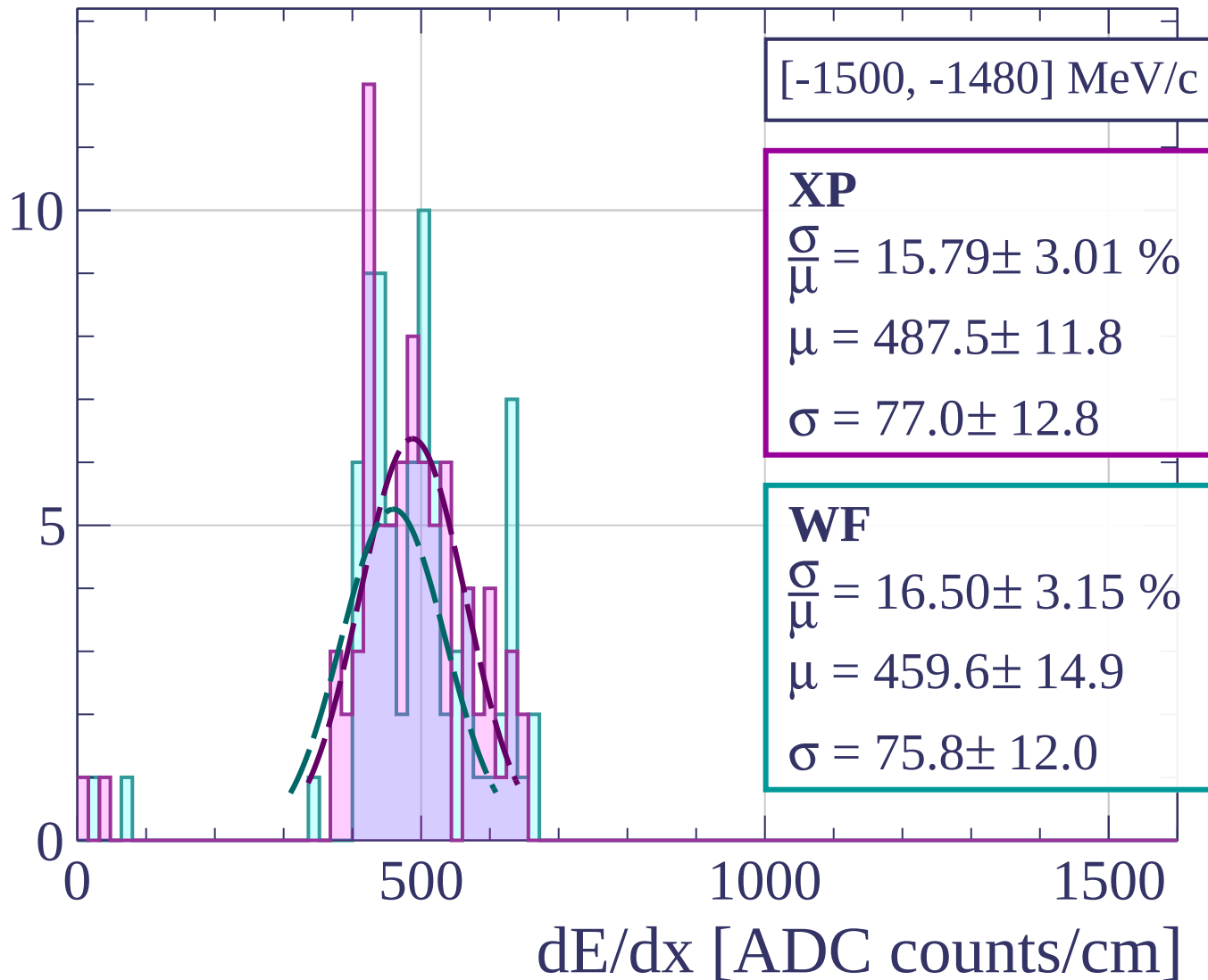
Count



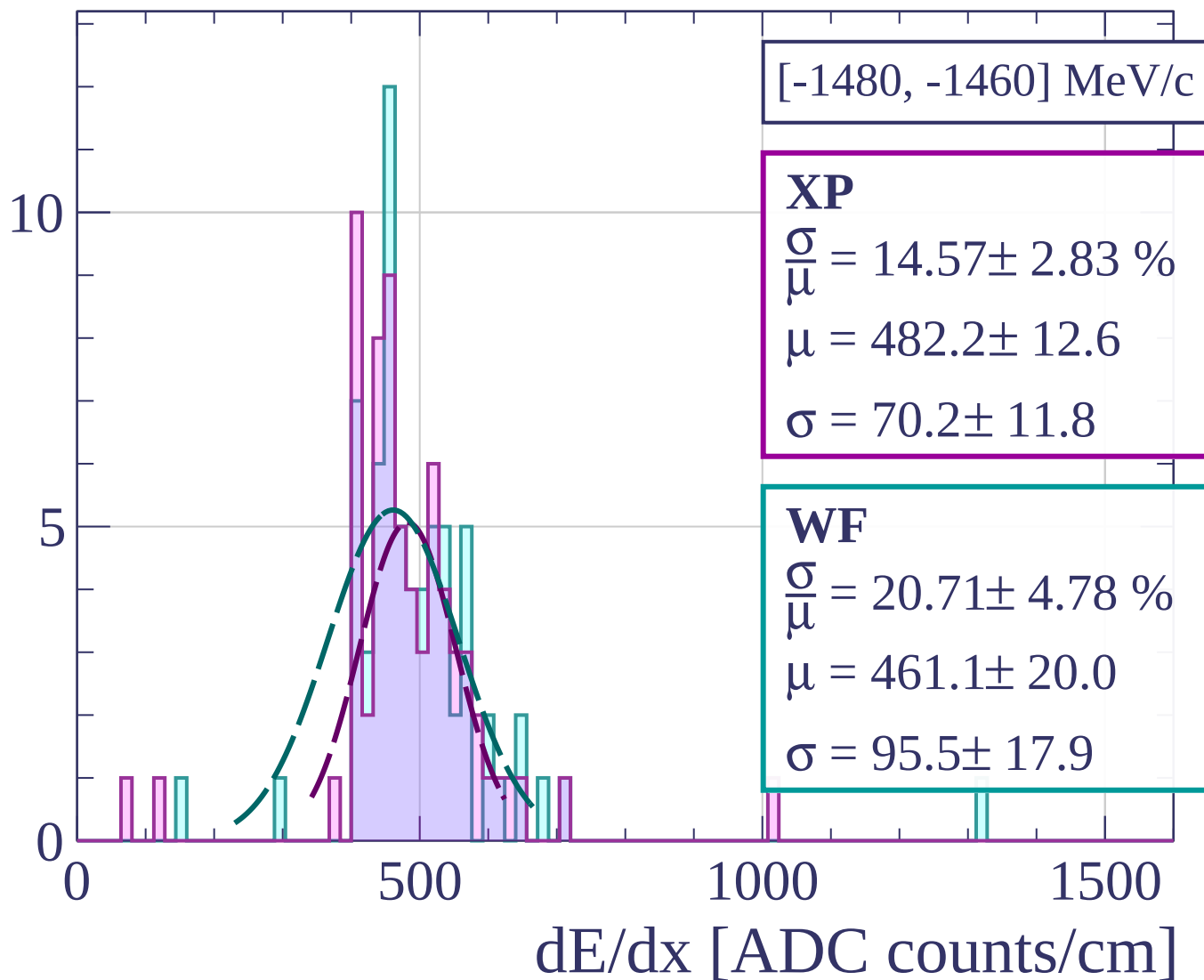
Count



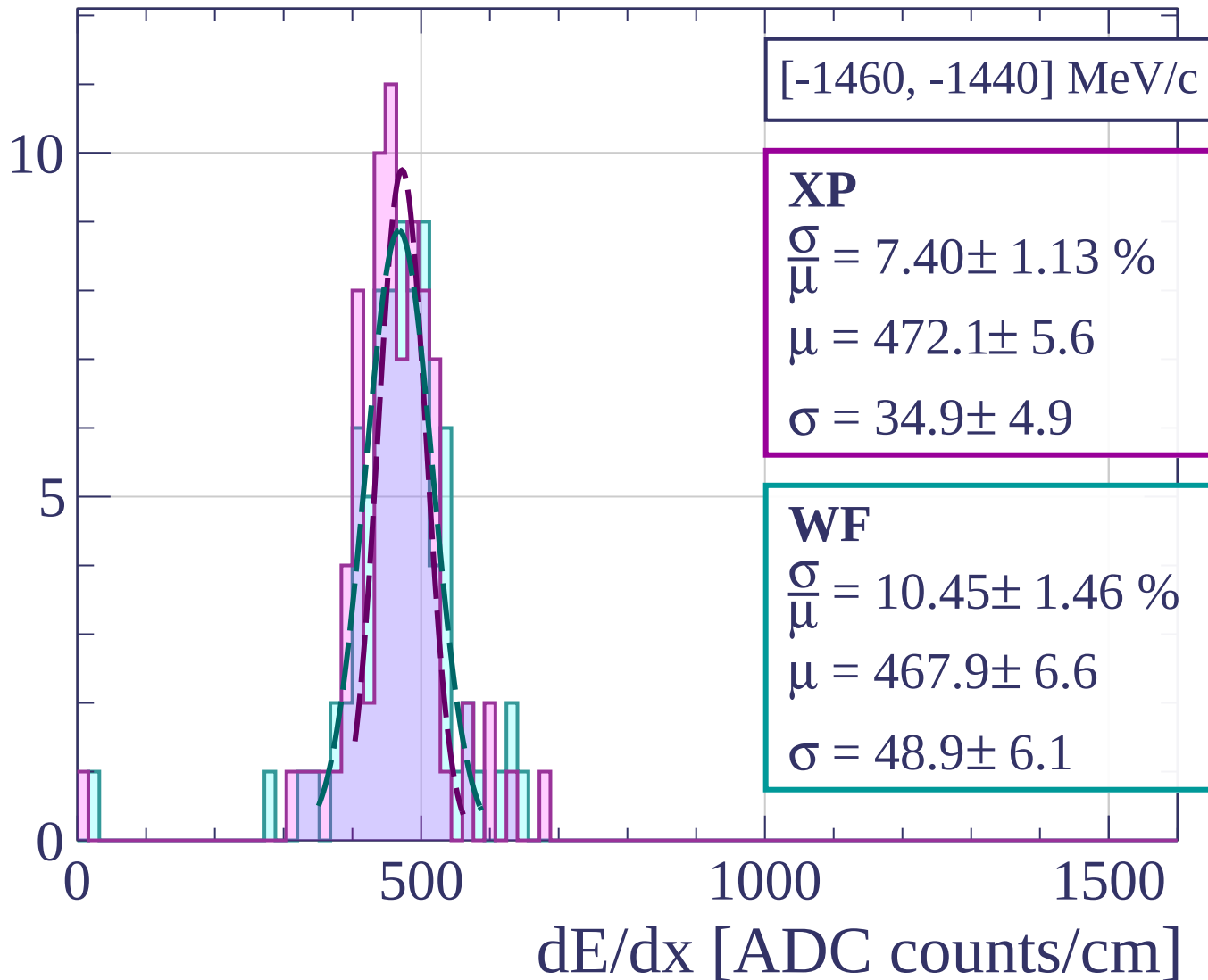
Count



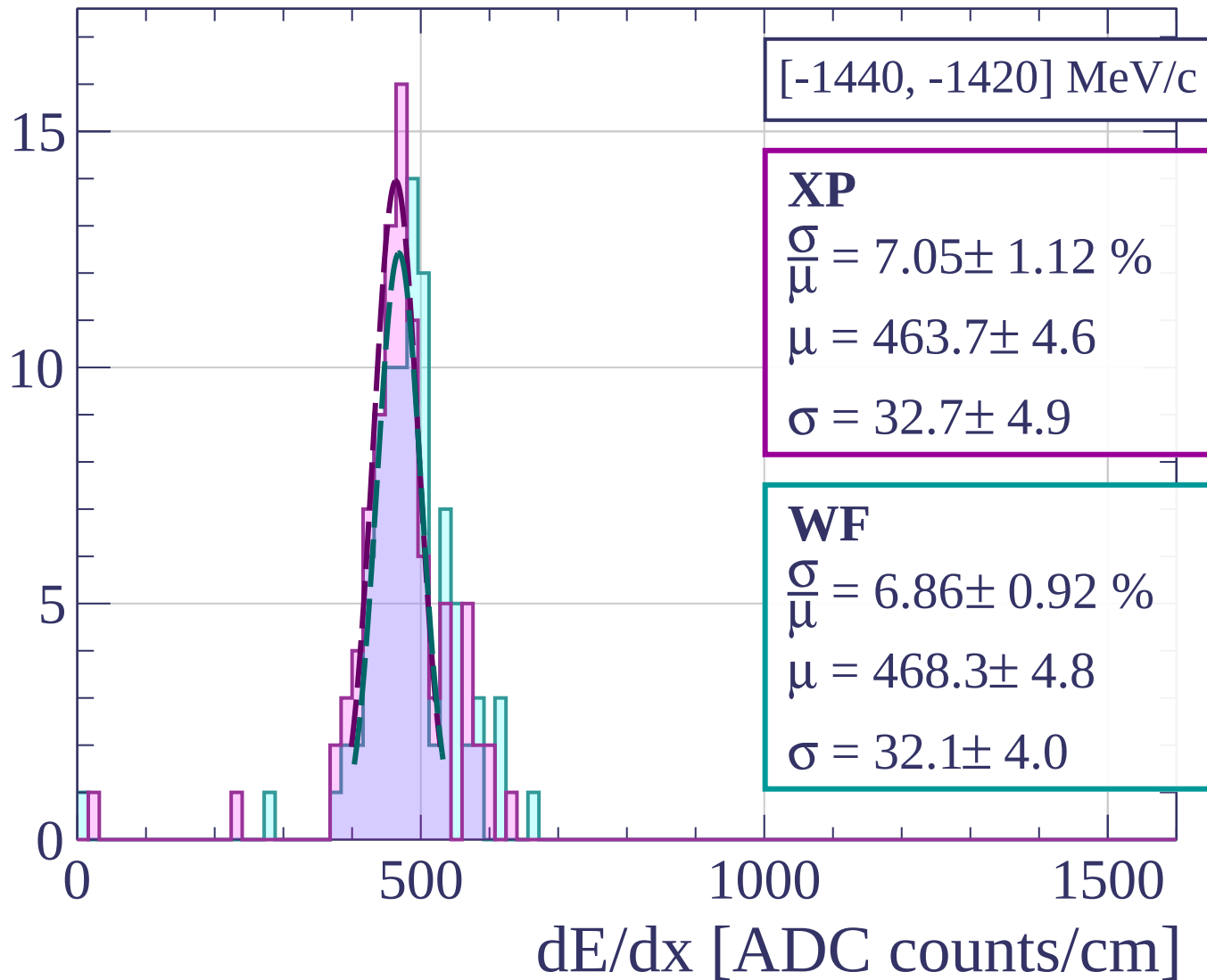
Count



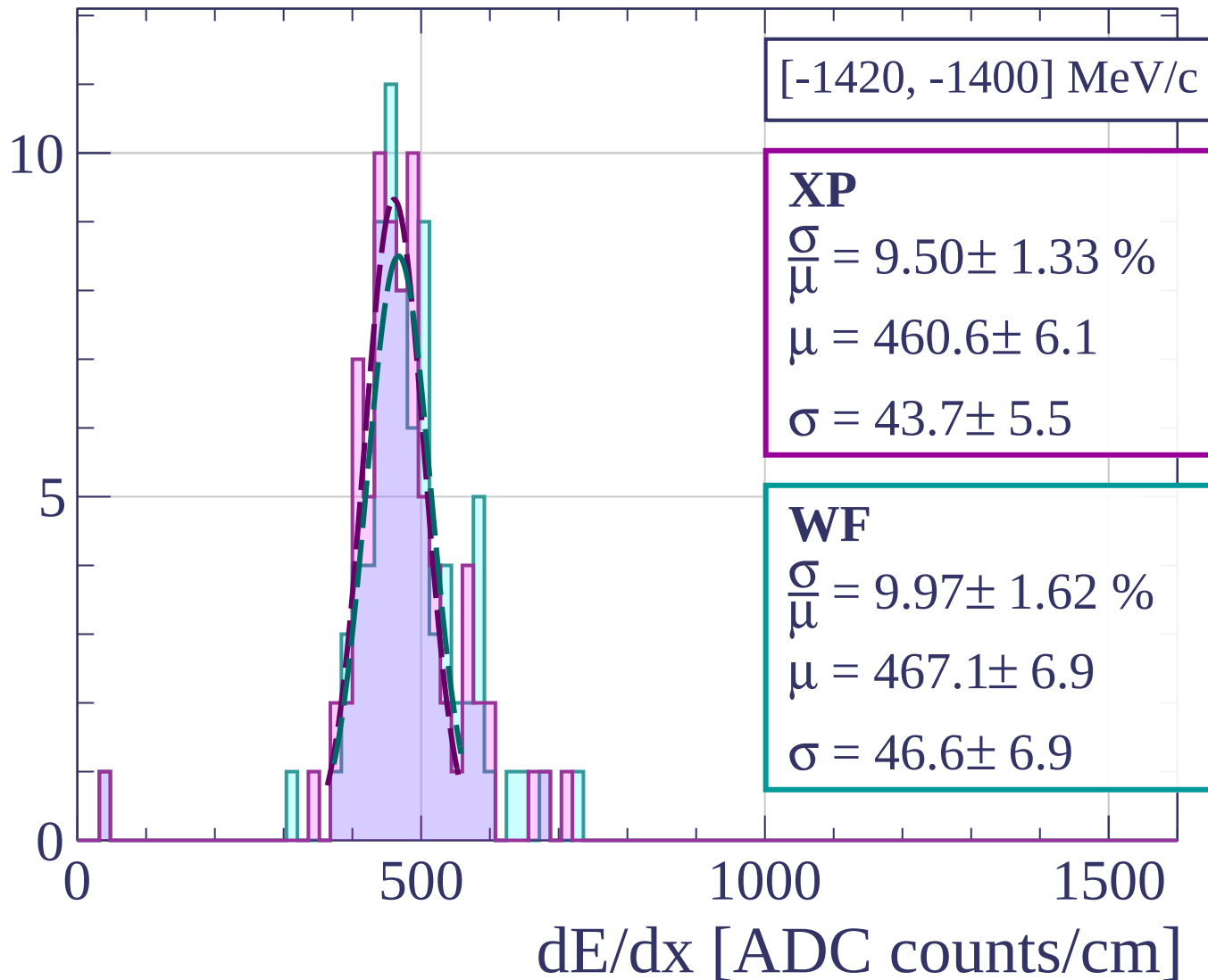
Count



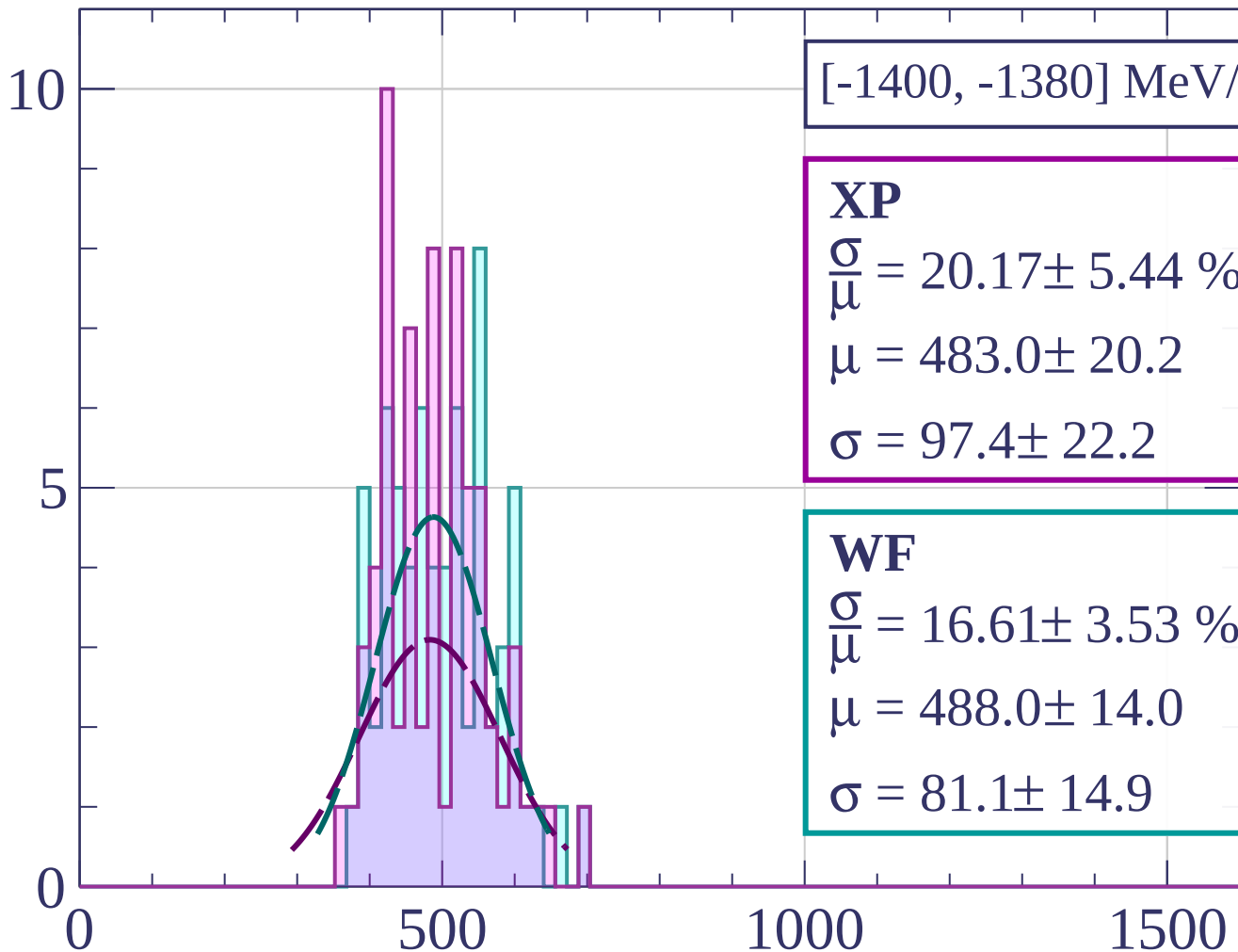
Count



Count



Count



[-1400, -1380] MeV/c

XP

$$\frac{\sigma}{\mu} = 20.17 \pm 5.44 \%$$

$$\mu = 483.0 \pm 20.2$$

$$\sigma = 97.4 \pm 22.2$$

WF

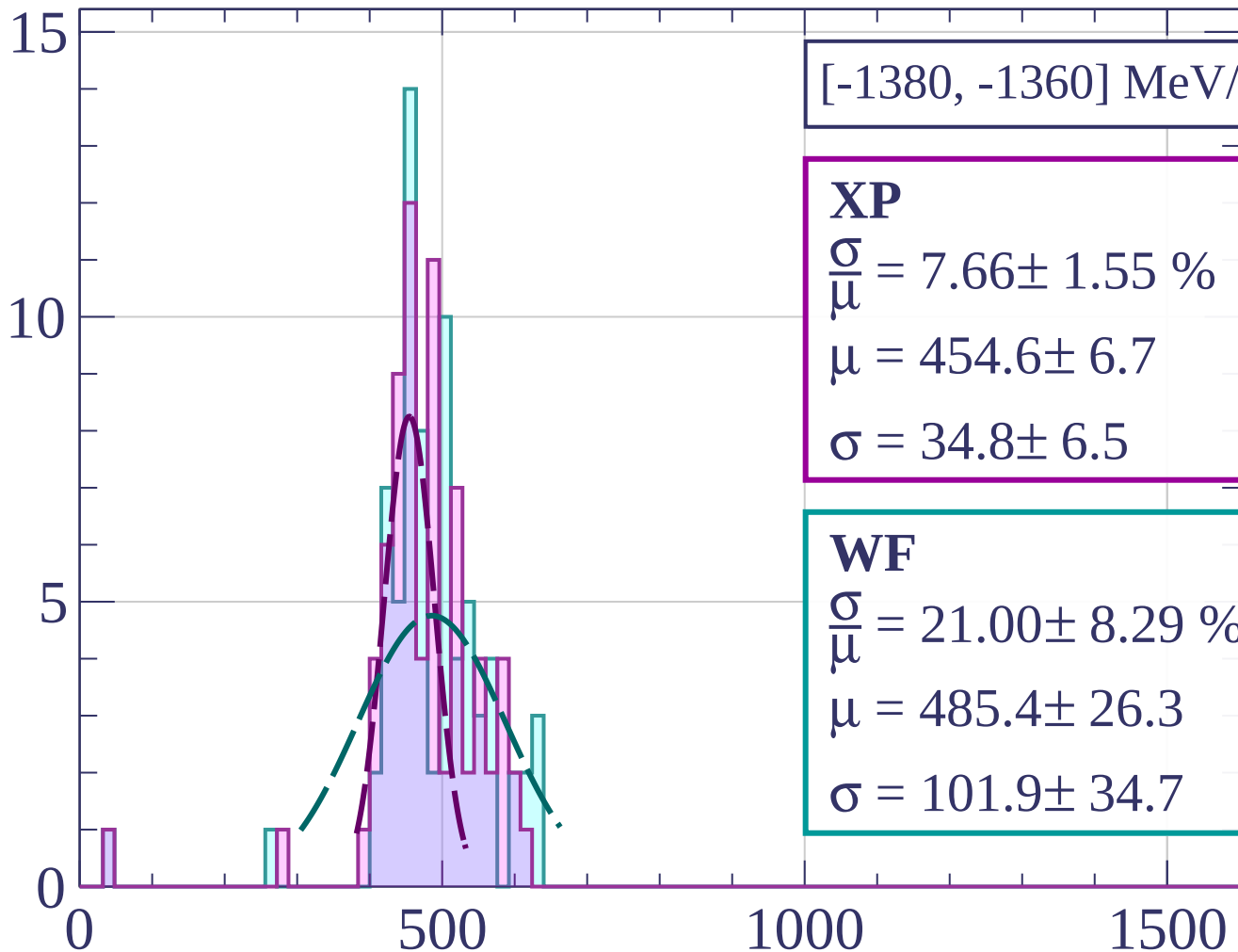
$$\frac{\sigma}{\mu} = 16.61 \pm 3.53 \%$$

$$\mu = 488.0 \pm 14.0$$

$$\sigma = 81.1 \pm 14.9$$

dE/dx [ADC counts/cm]

Count



[-1380, -1360] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.66 \pm 1.55 \%$$

$$\mu = 454.6 \pm 6.7$$

$$\sigma = 34.8 \pm 6.5$$

WF

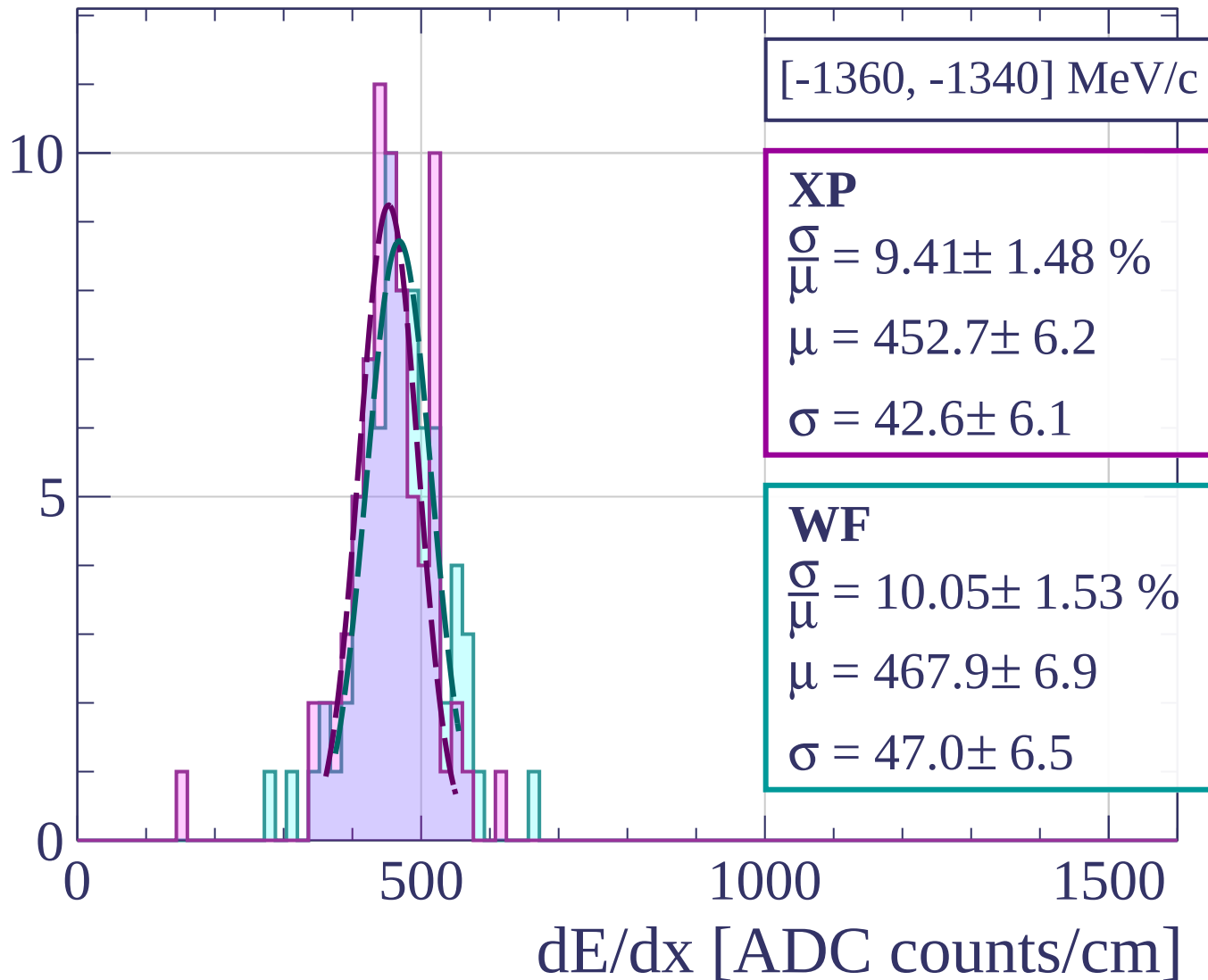
$$\frac{\sigma}{\mu} = 21.00 \pm 8.29 \%$$

$$\mu = 485.4 \pm 26.3$$

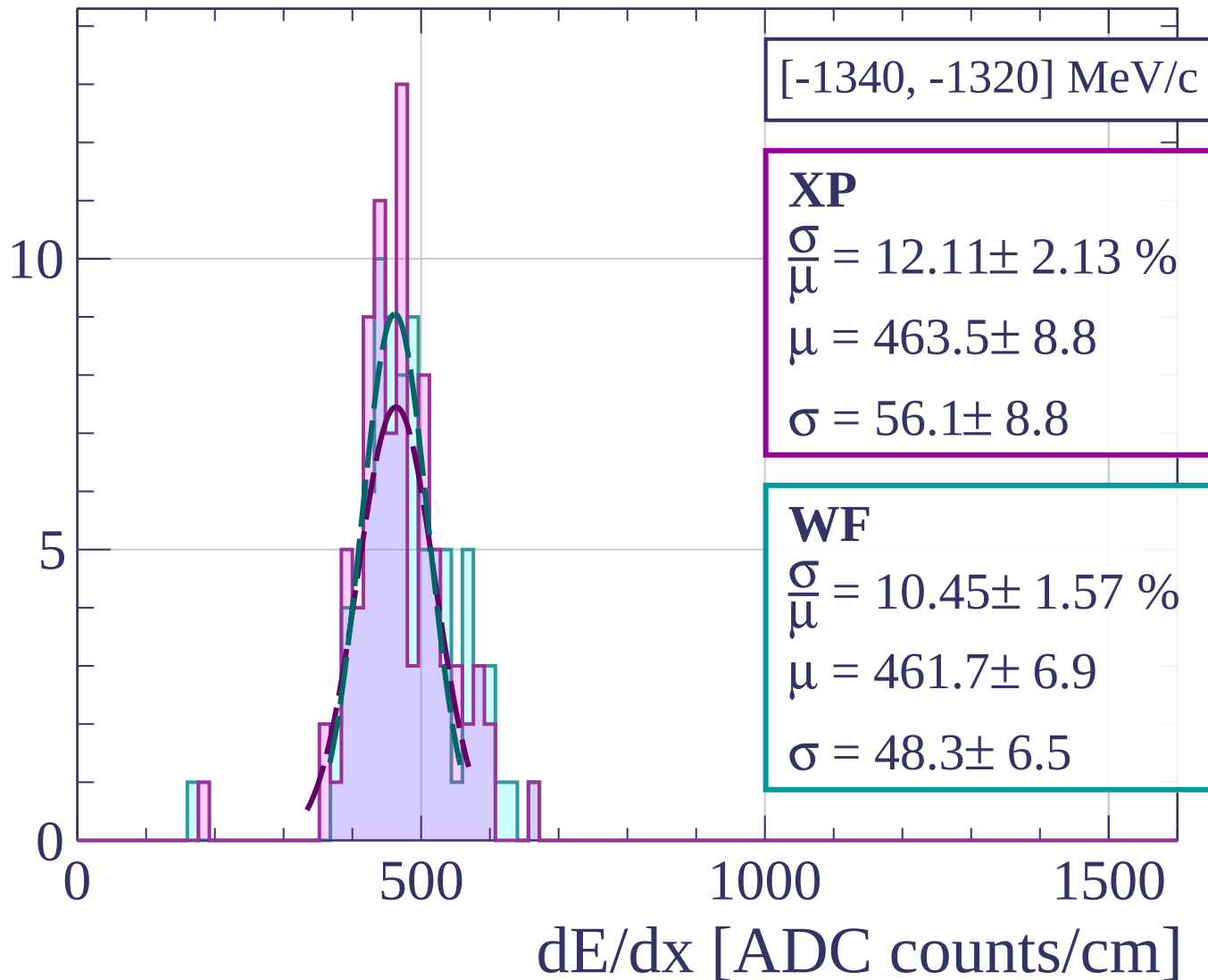
$$\sigma = 101.9 \pm 34.7$$

dE/dx [ADC counts/cm]

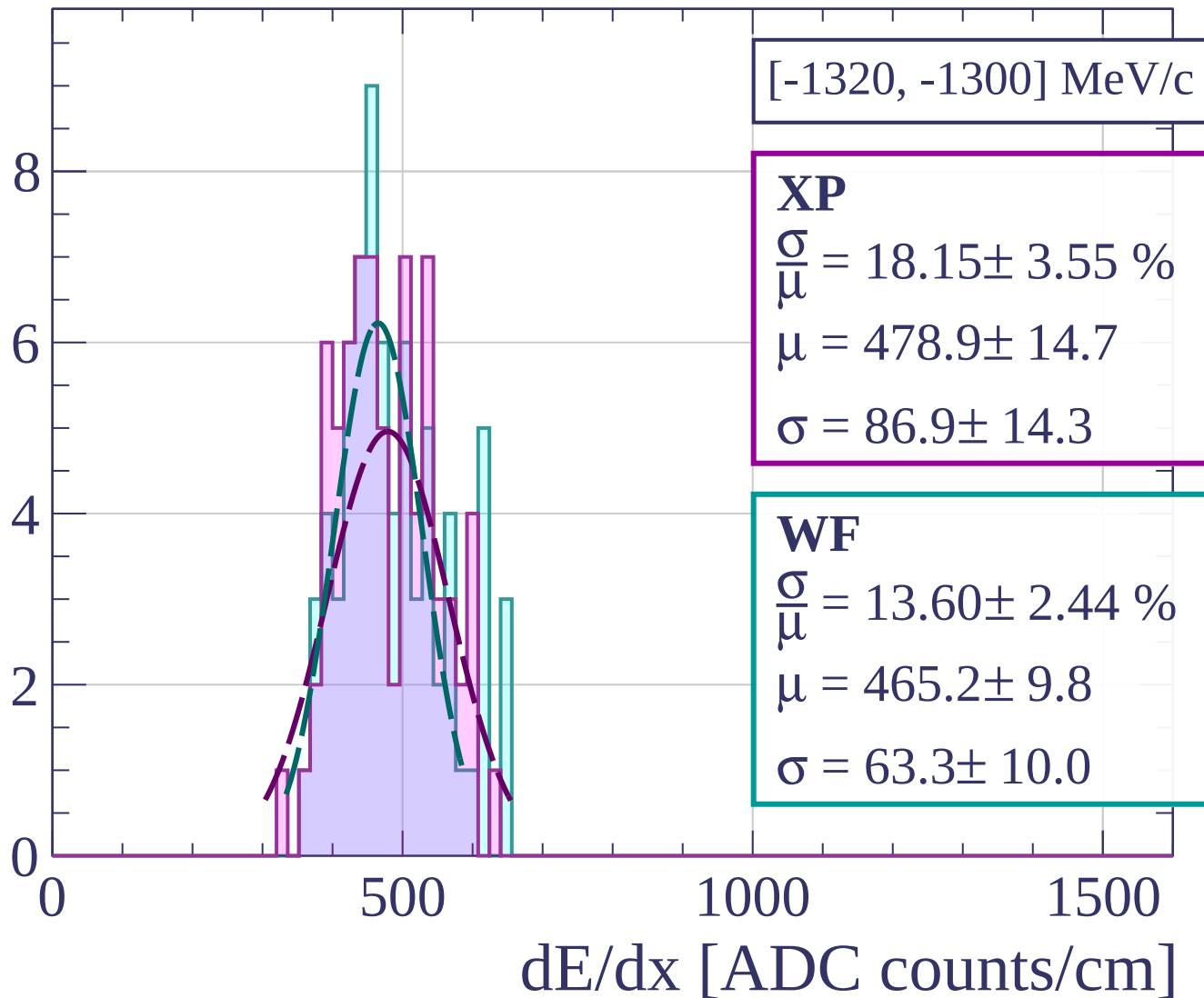
Count



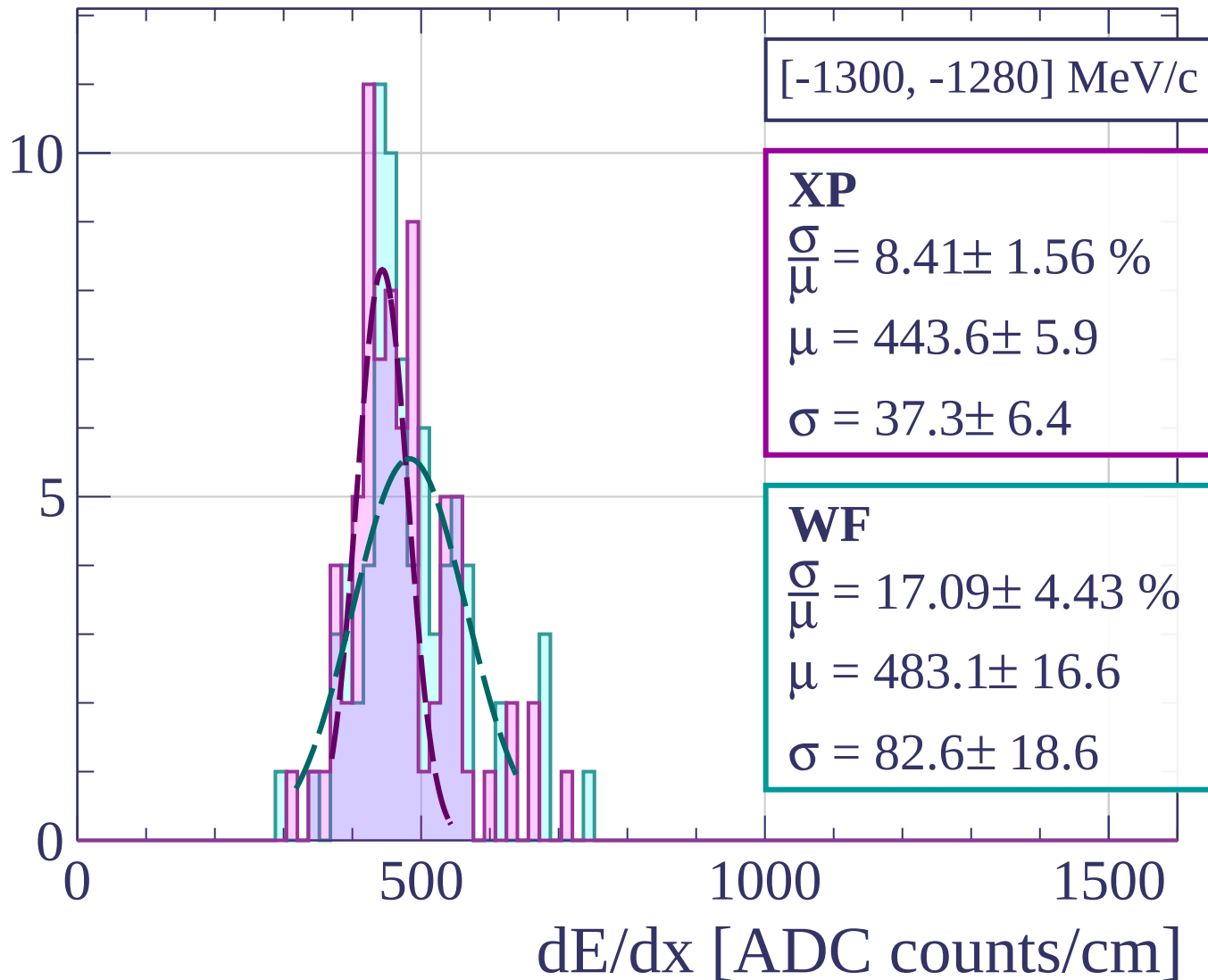
Count



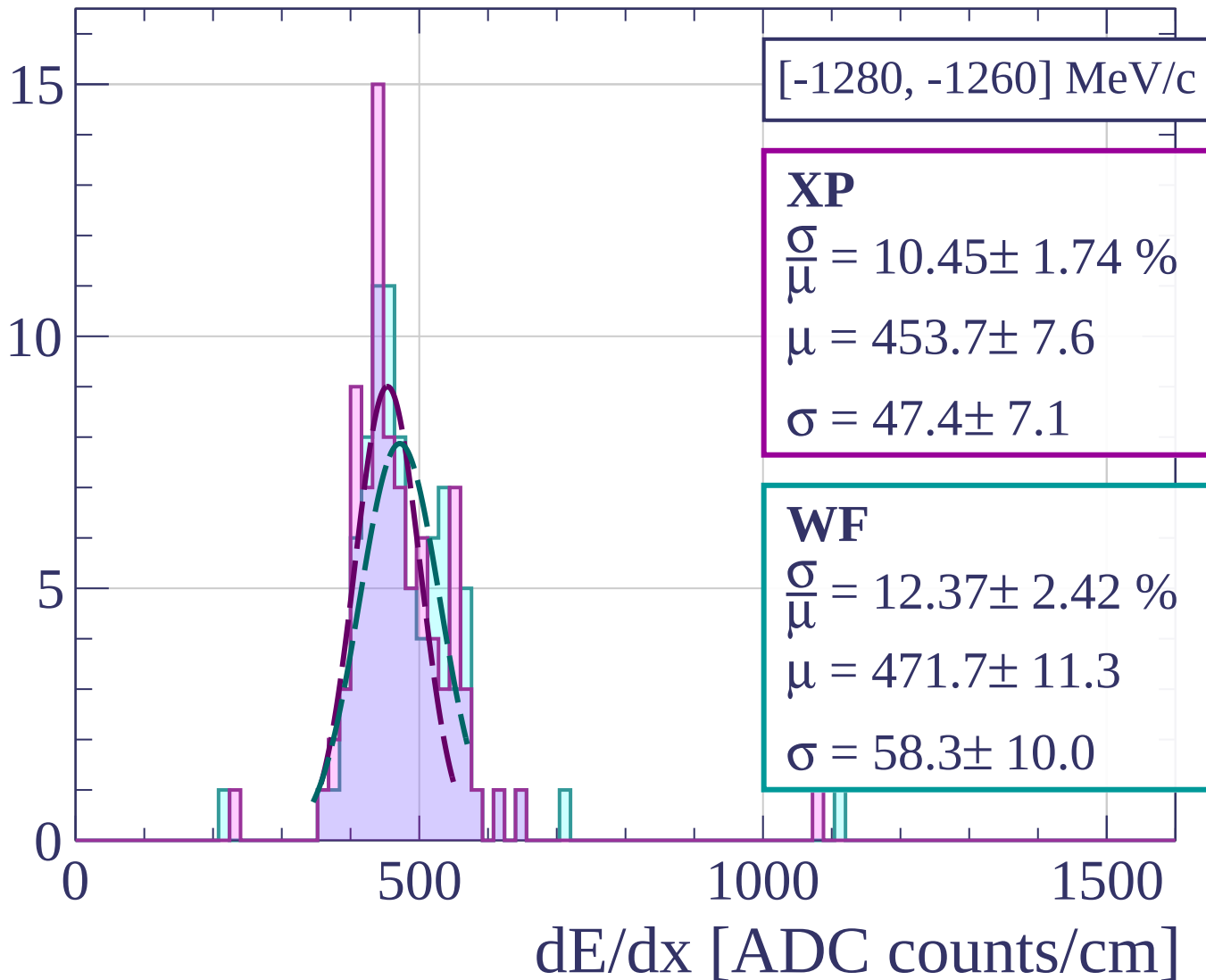
Count



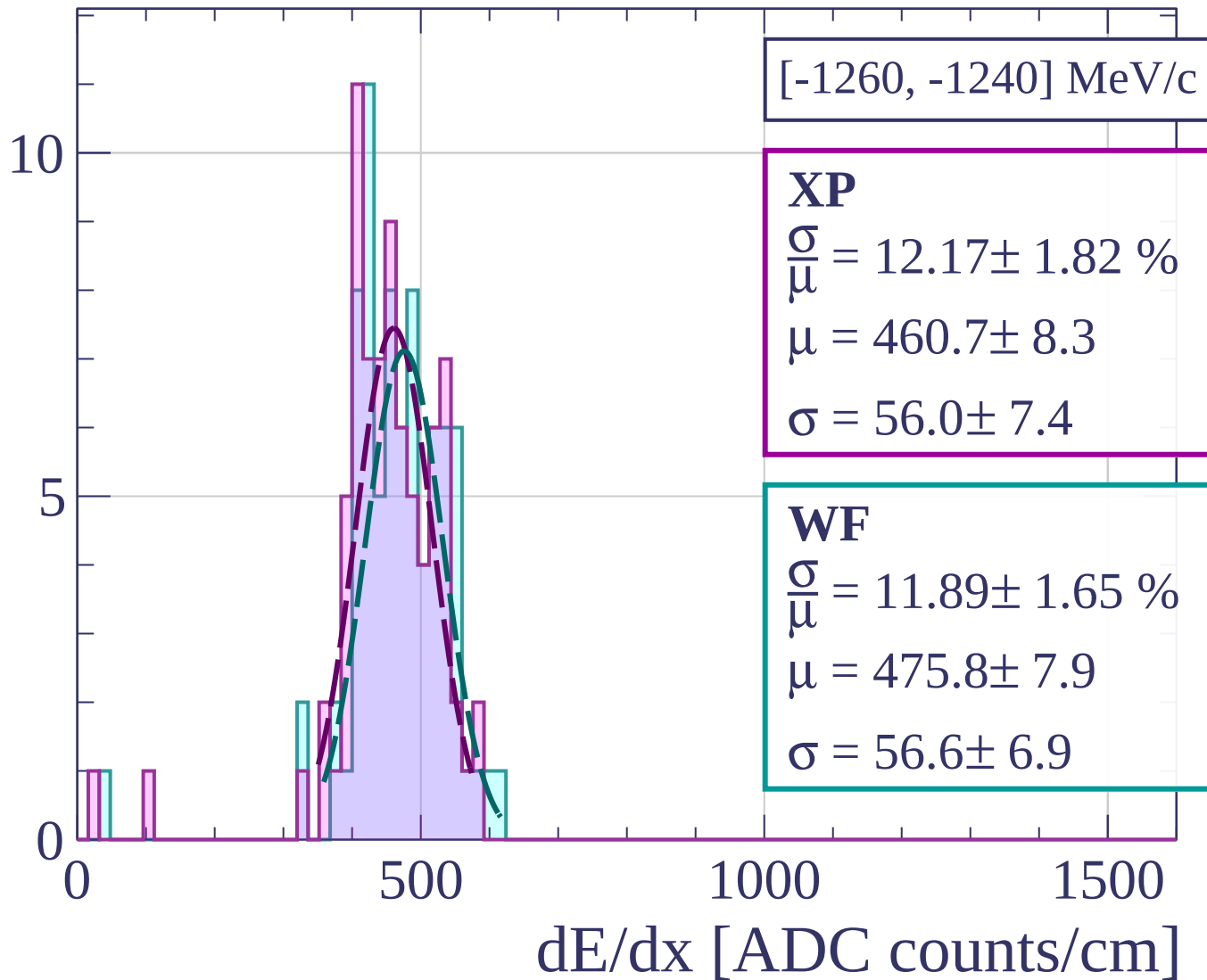
Count



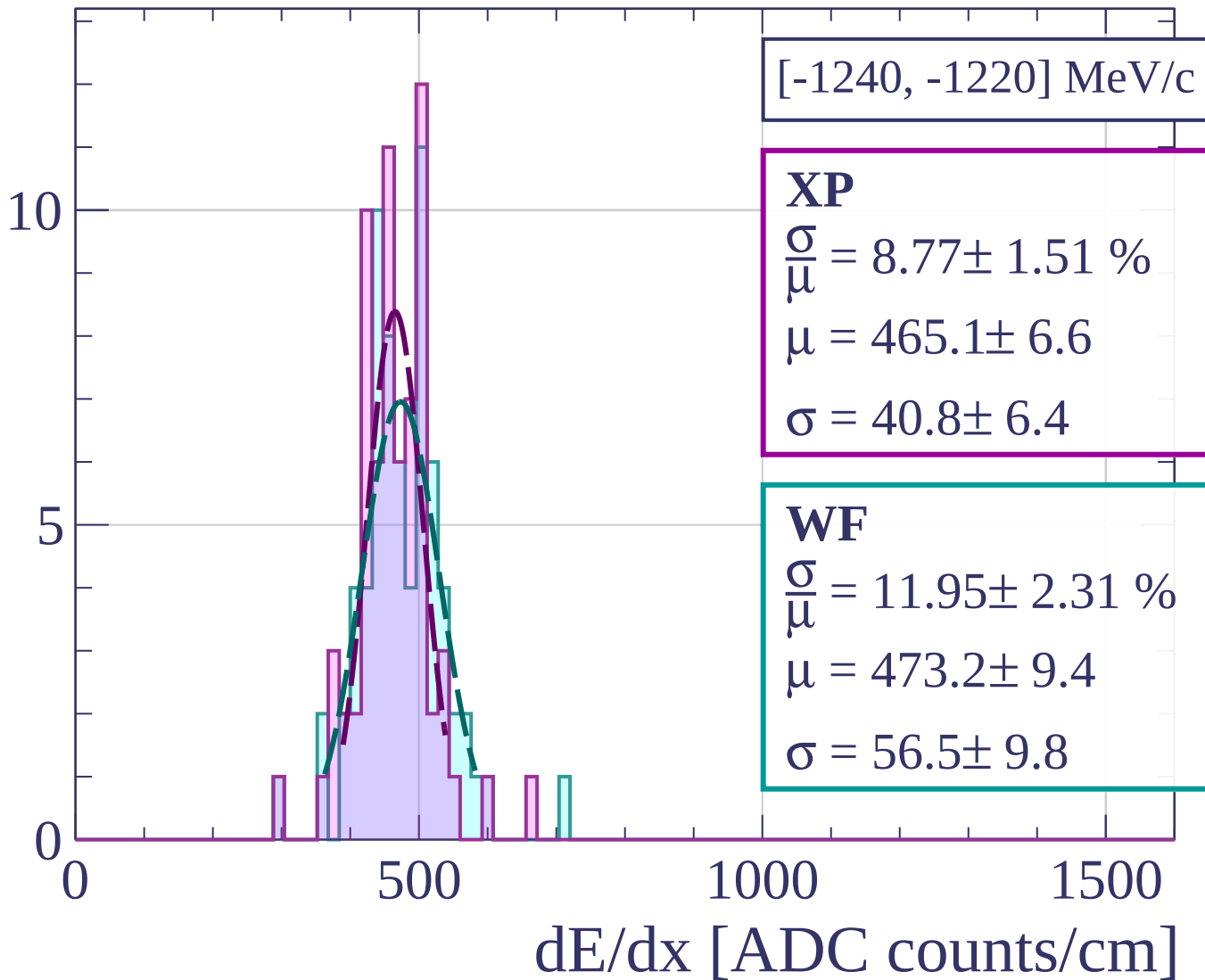
Count



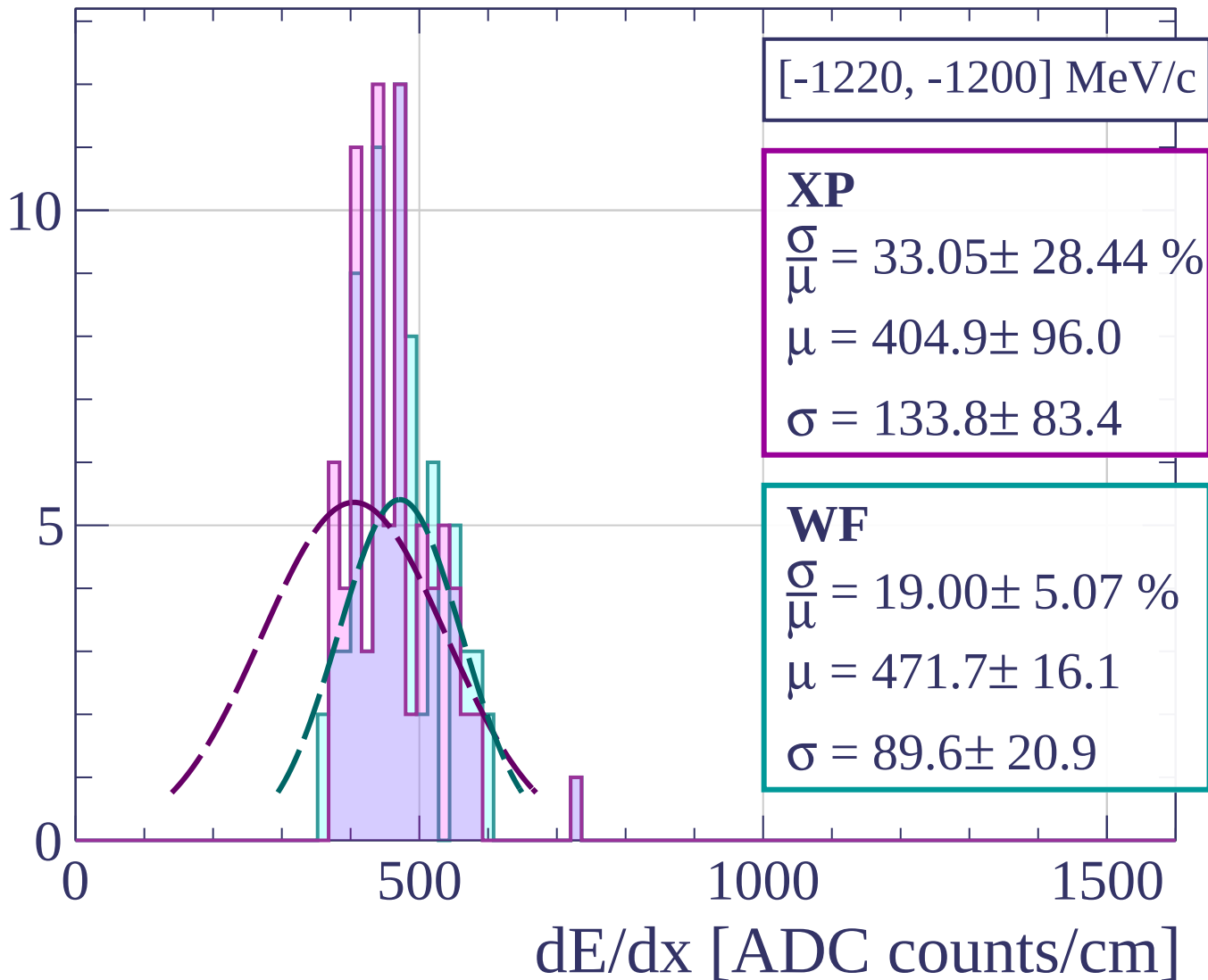
Count



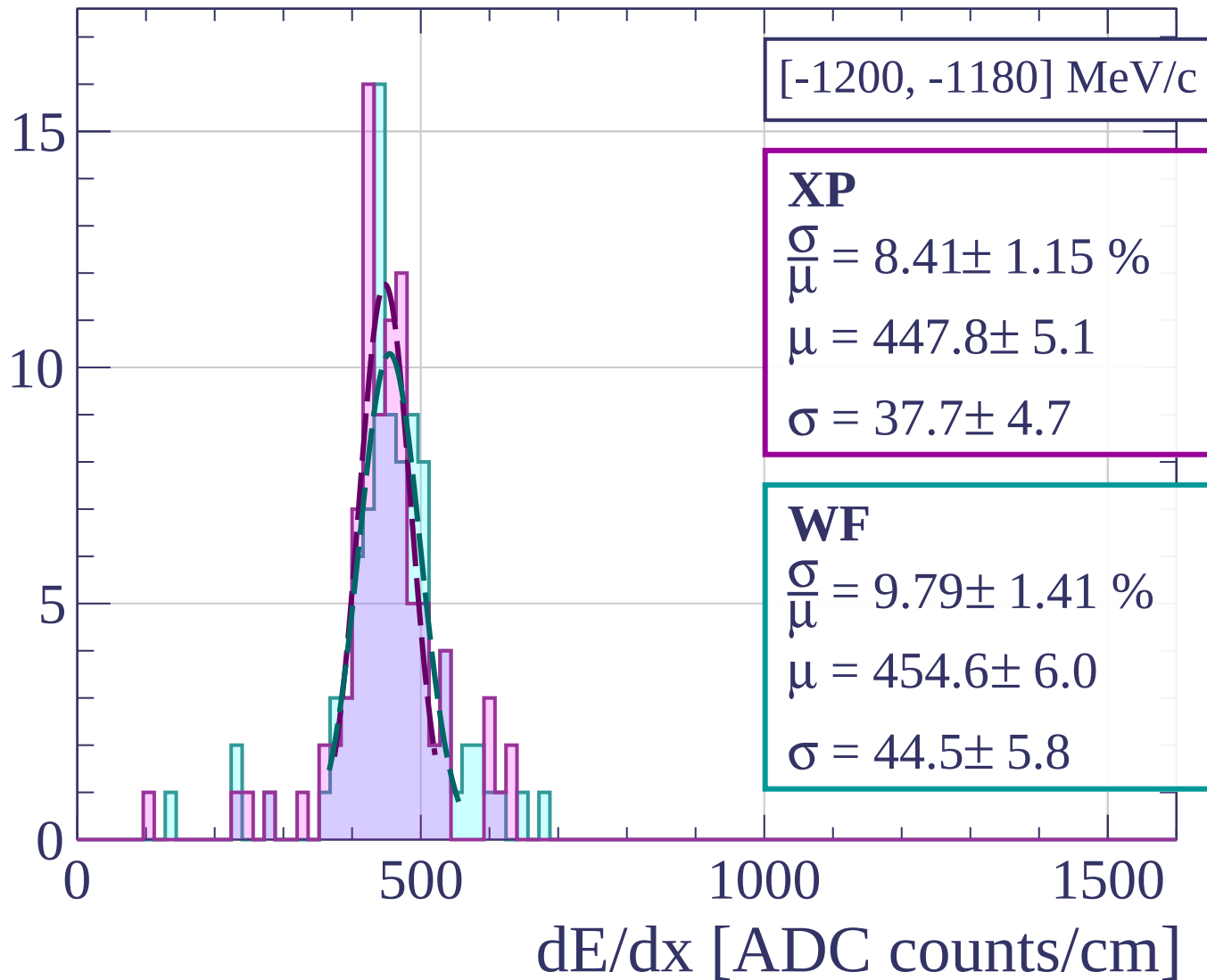
Count



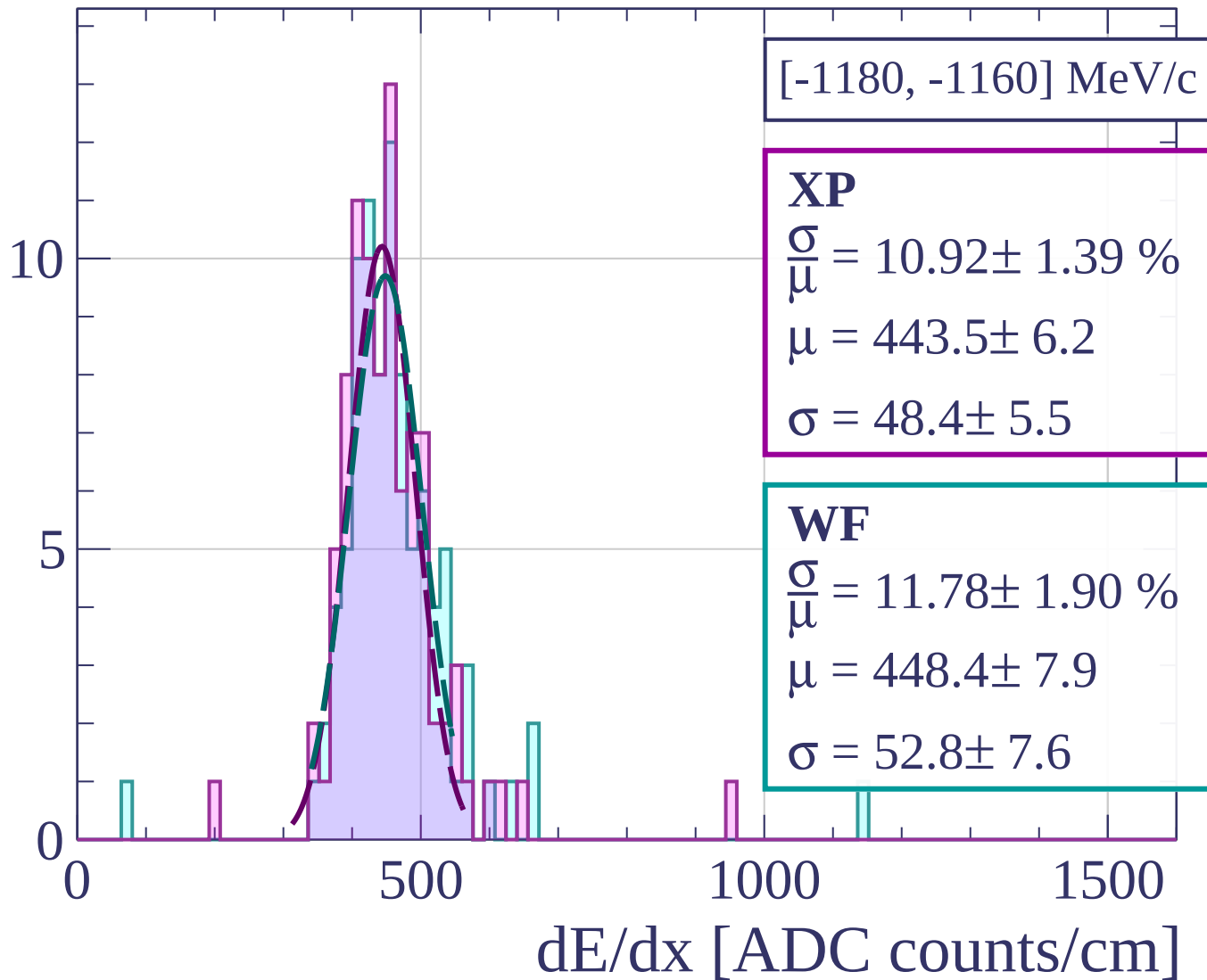
Count



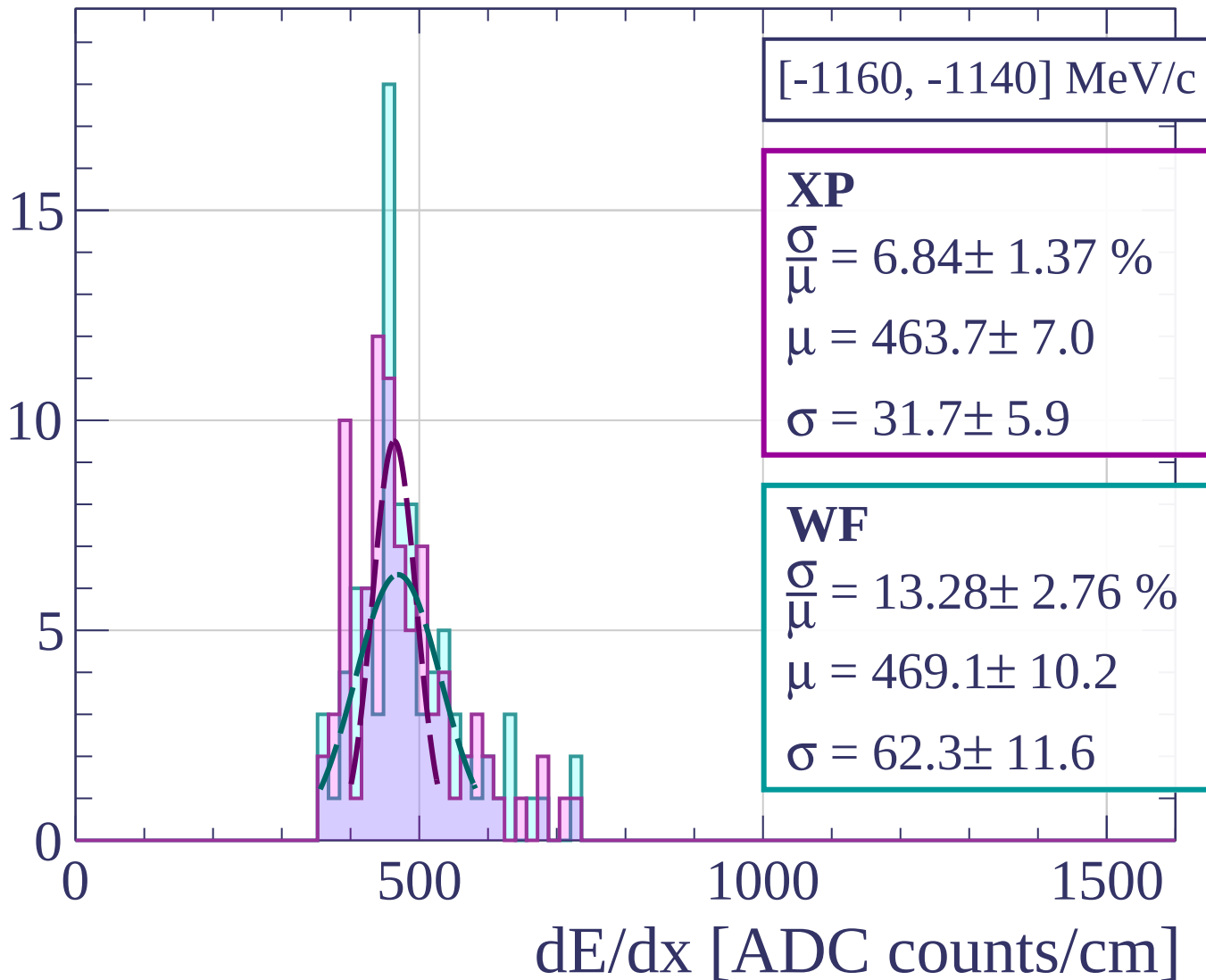
Count



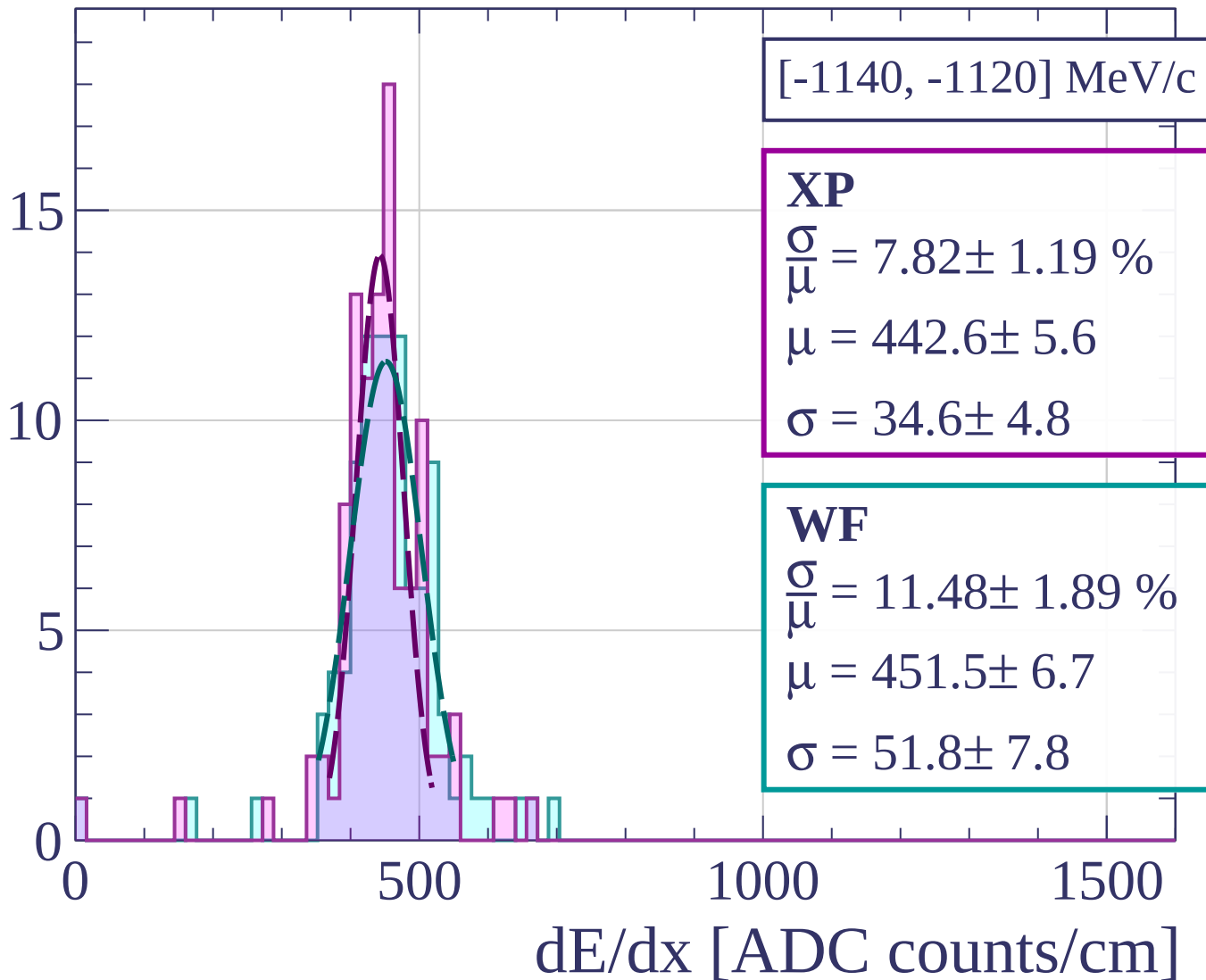
Count



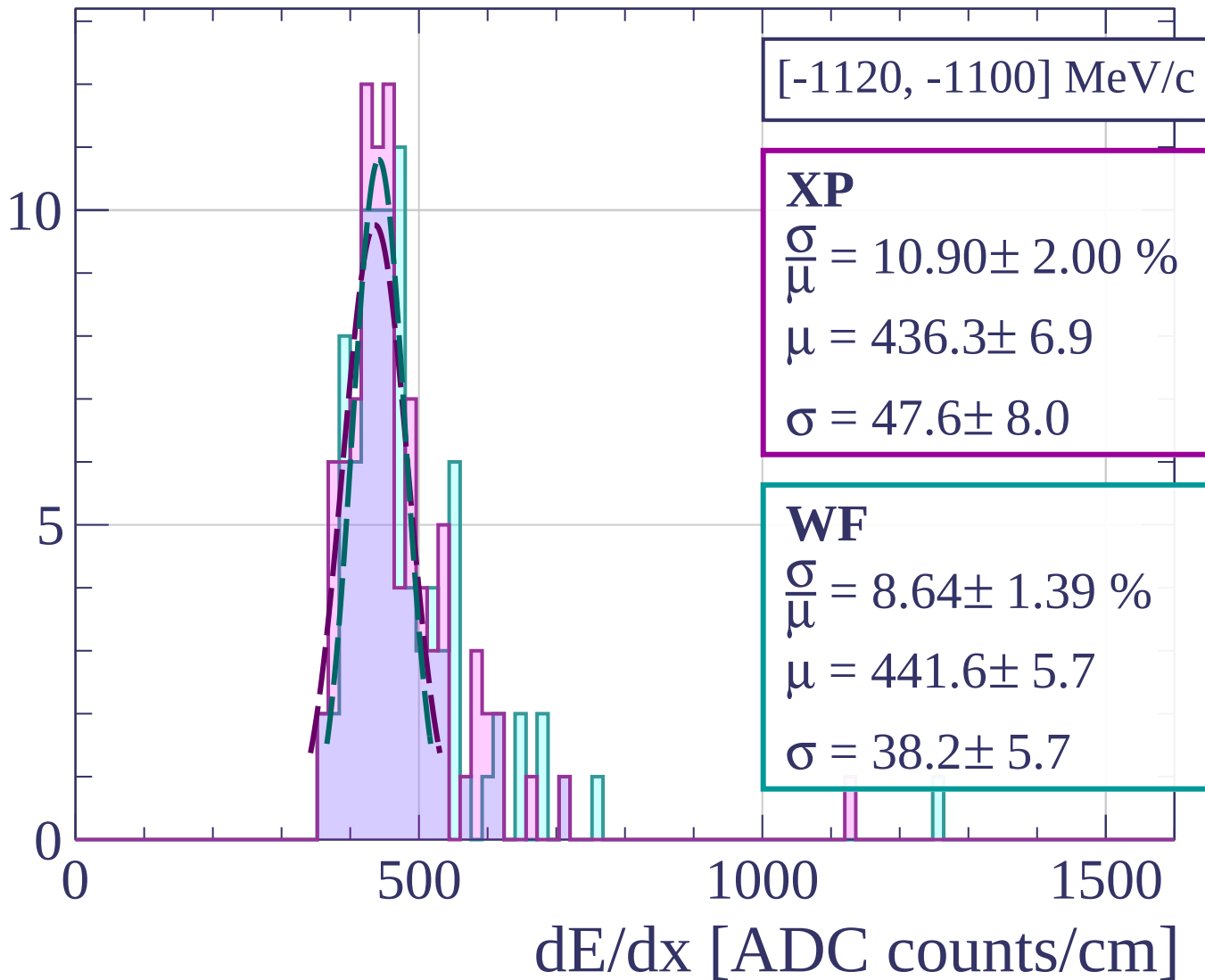
Count



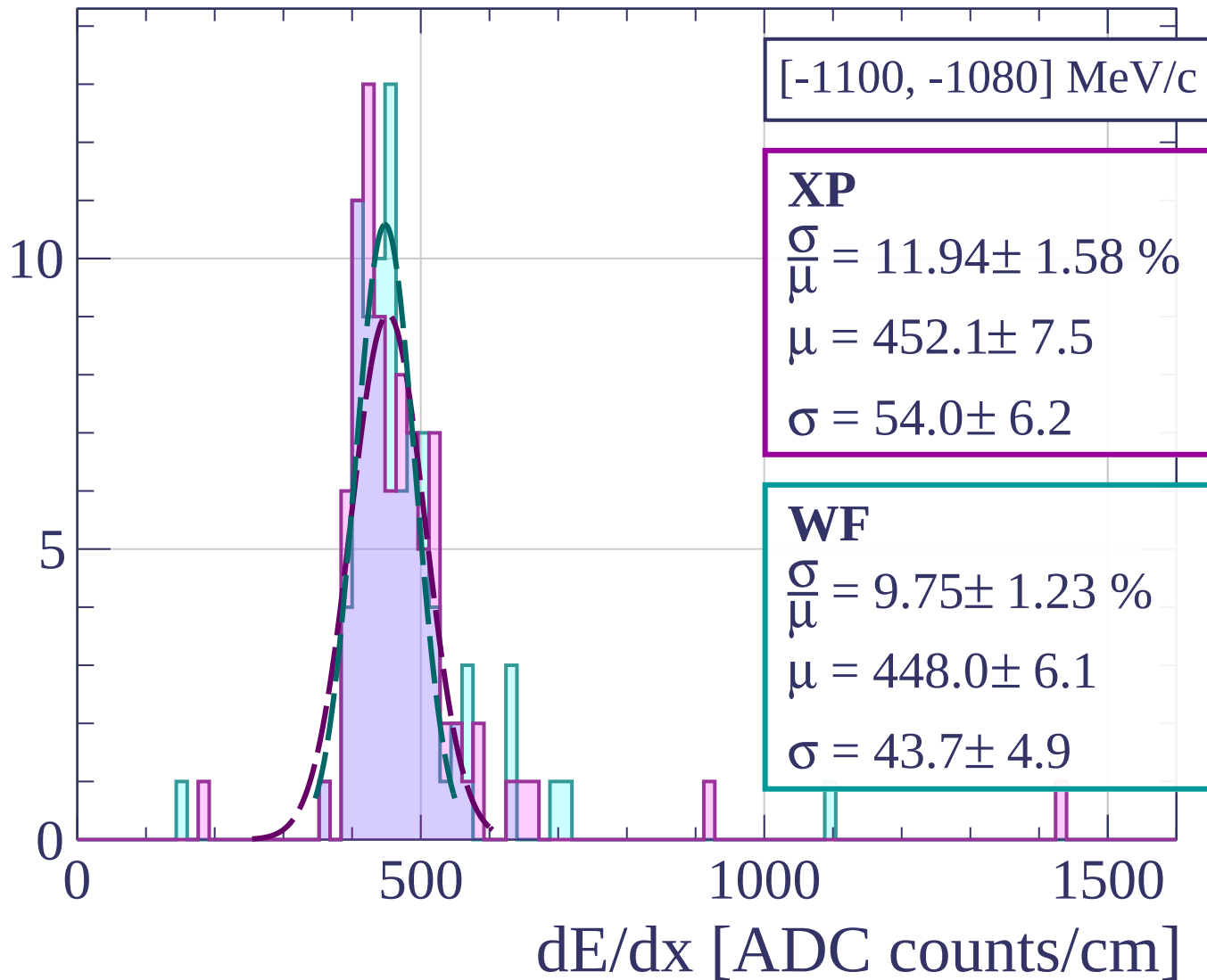
Count



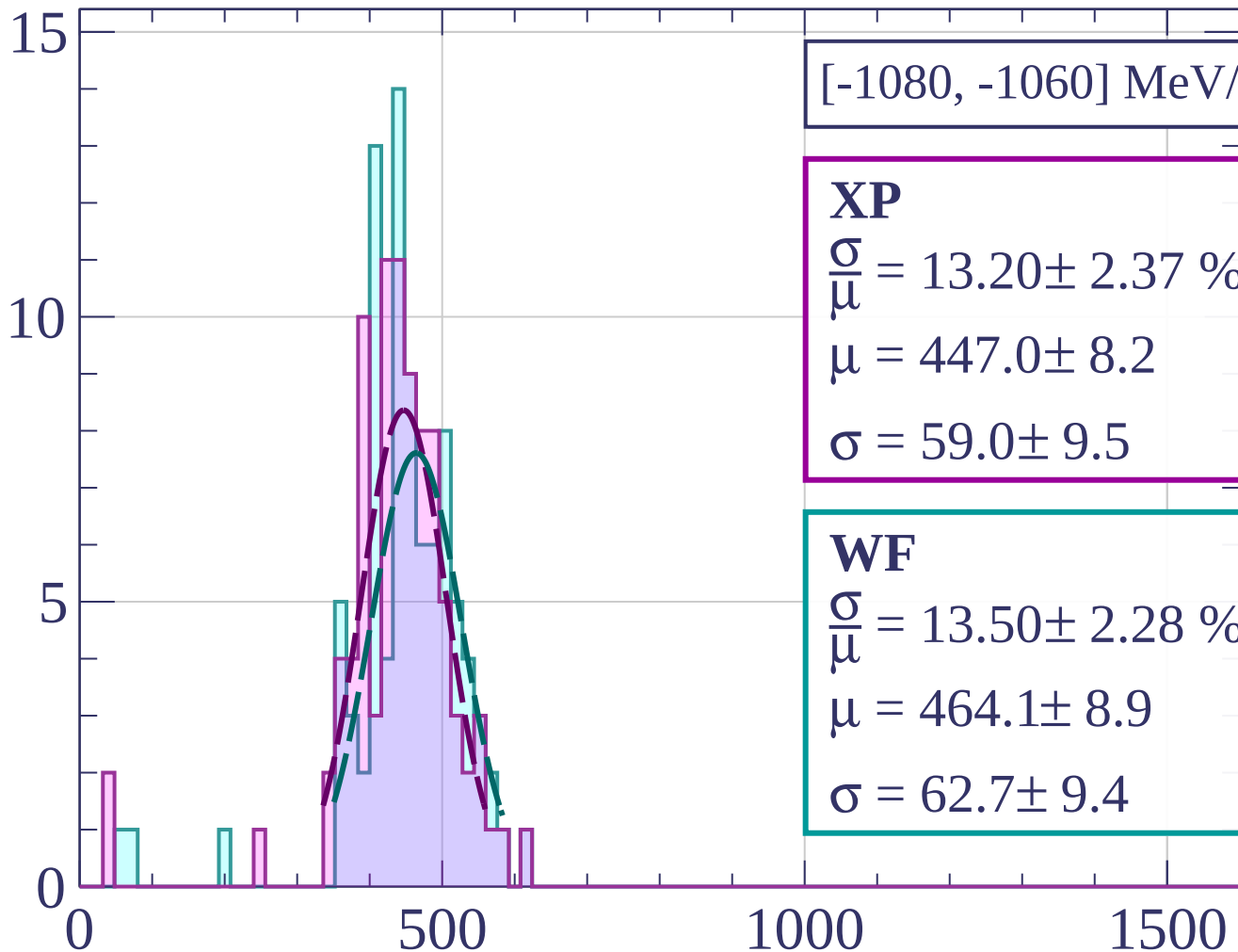
Count



Count



Count



[-1080, -1060] MeV/c

XP

$$\frac{\sigma}{\mu} = 13.20 \pm 2.37 \%$$

$$\mu = 447.0 \pm 8.2$$

$$\sigma = 59.0 \pm 9.5$$

WF

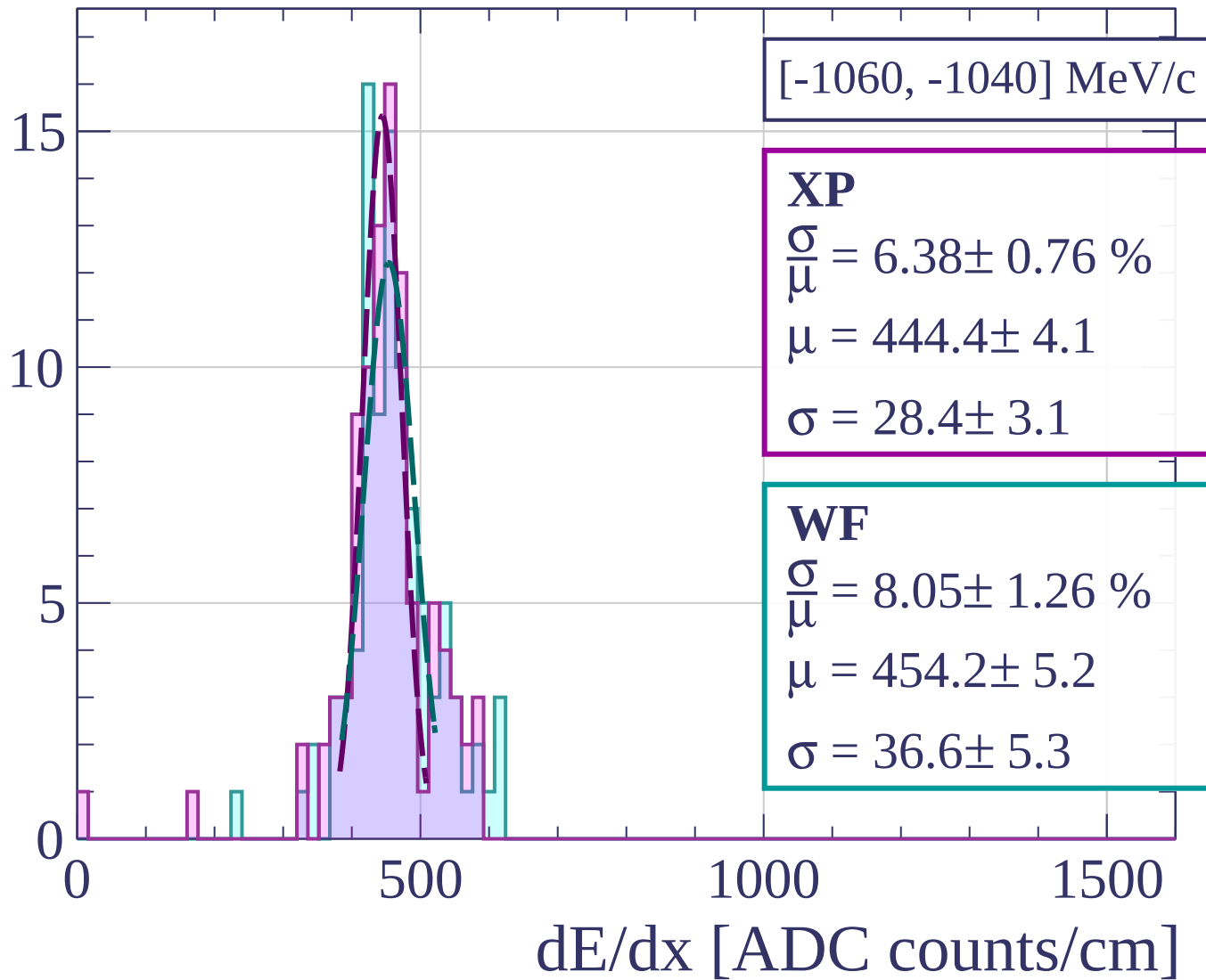
$$\frac{\sigma}{\mu} = 13.50 \pm 2.28 \%$$

$$\mu = 464.1 \pm 8.9$$

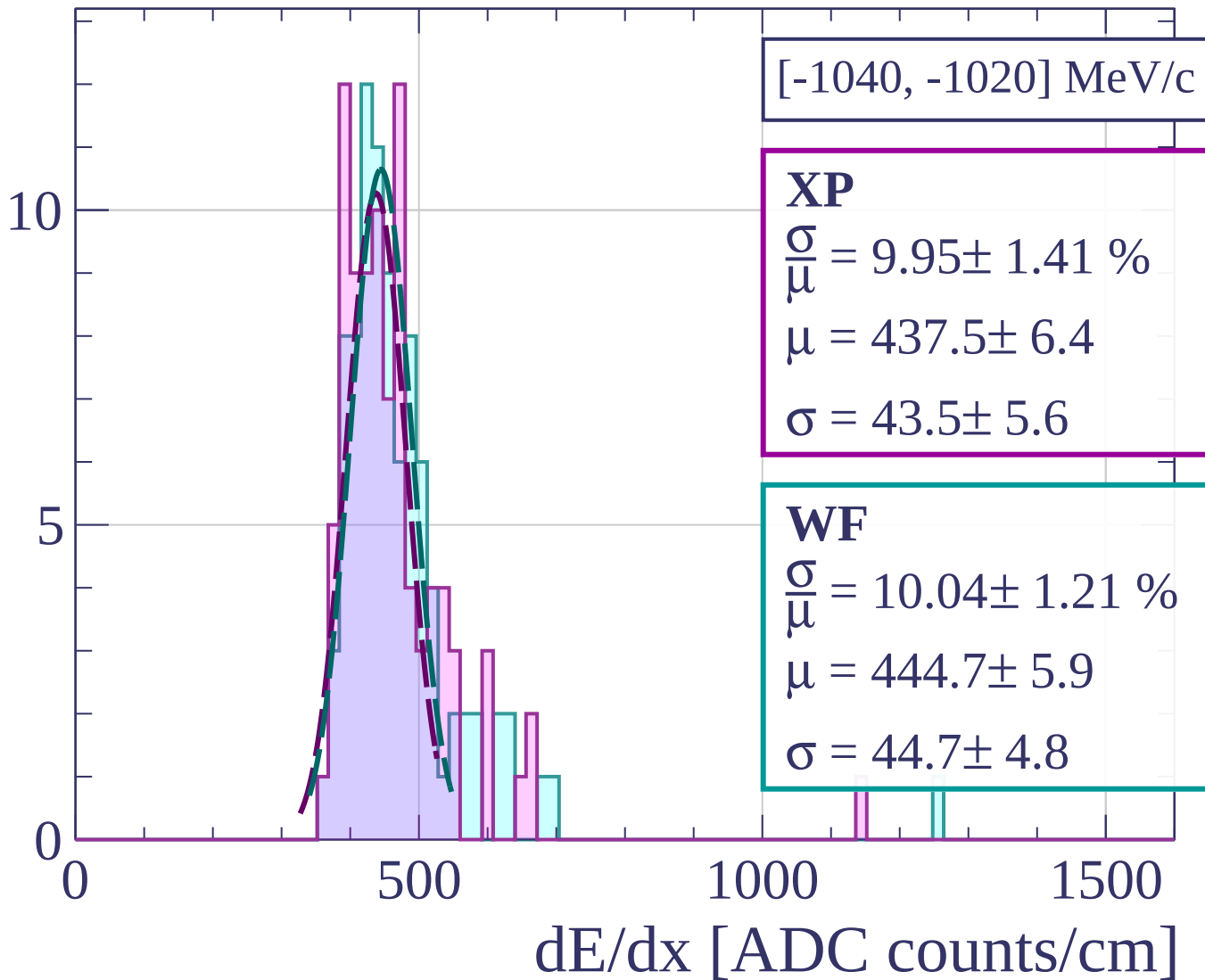
$$\sigma = 62.7 \pm 9.4$$

dE/dx [ADC counts/cm]

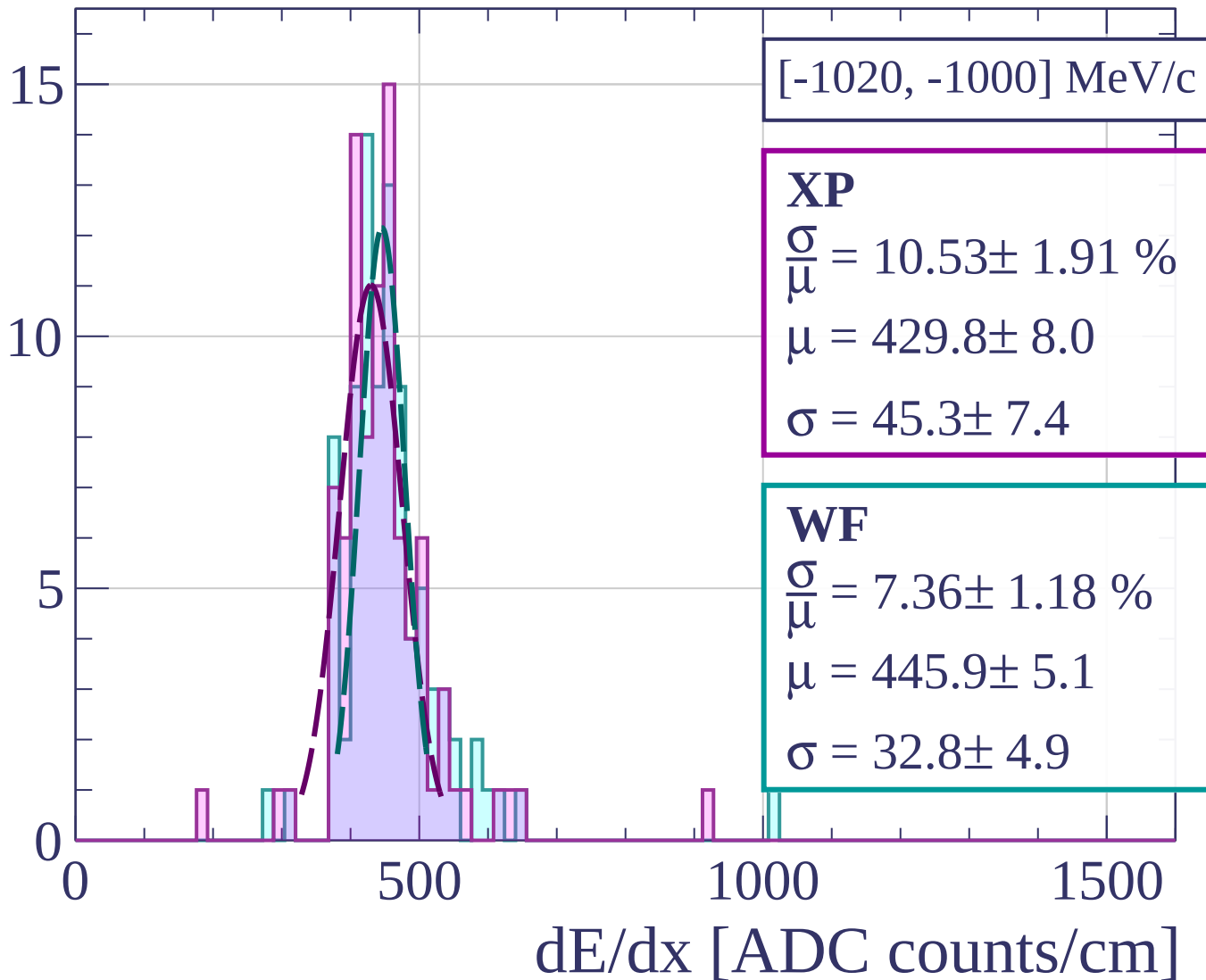
Count



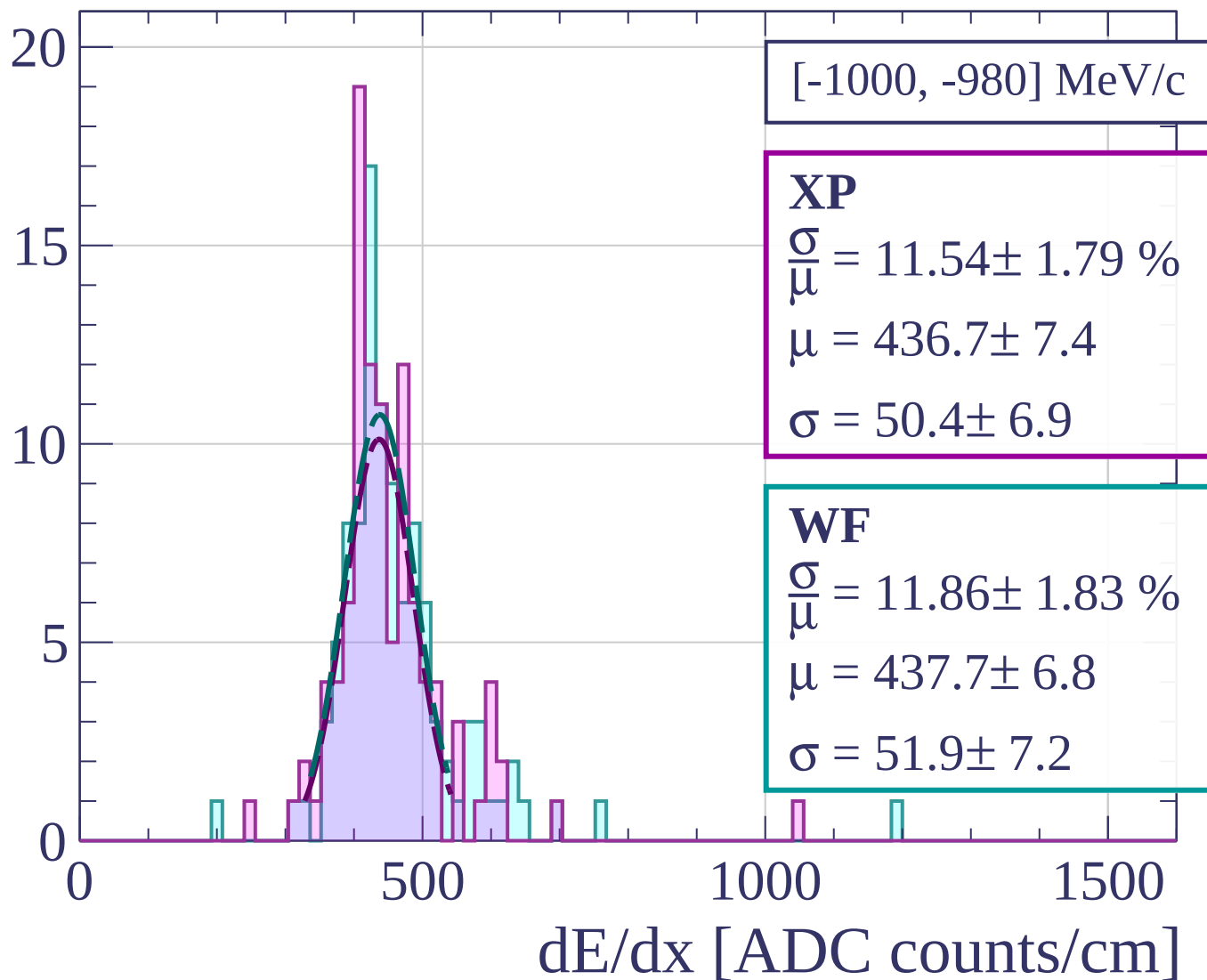
Count



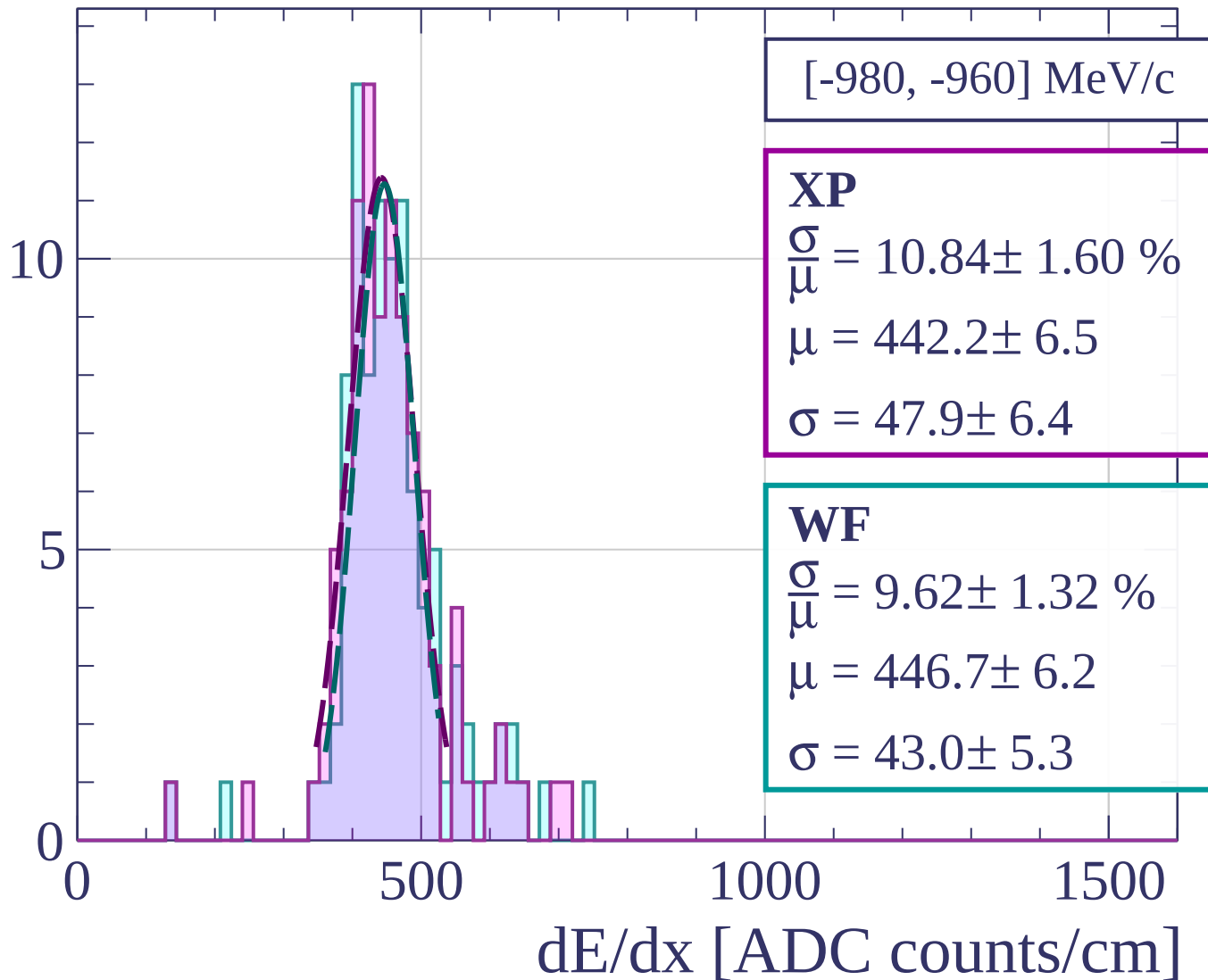
Count



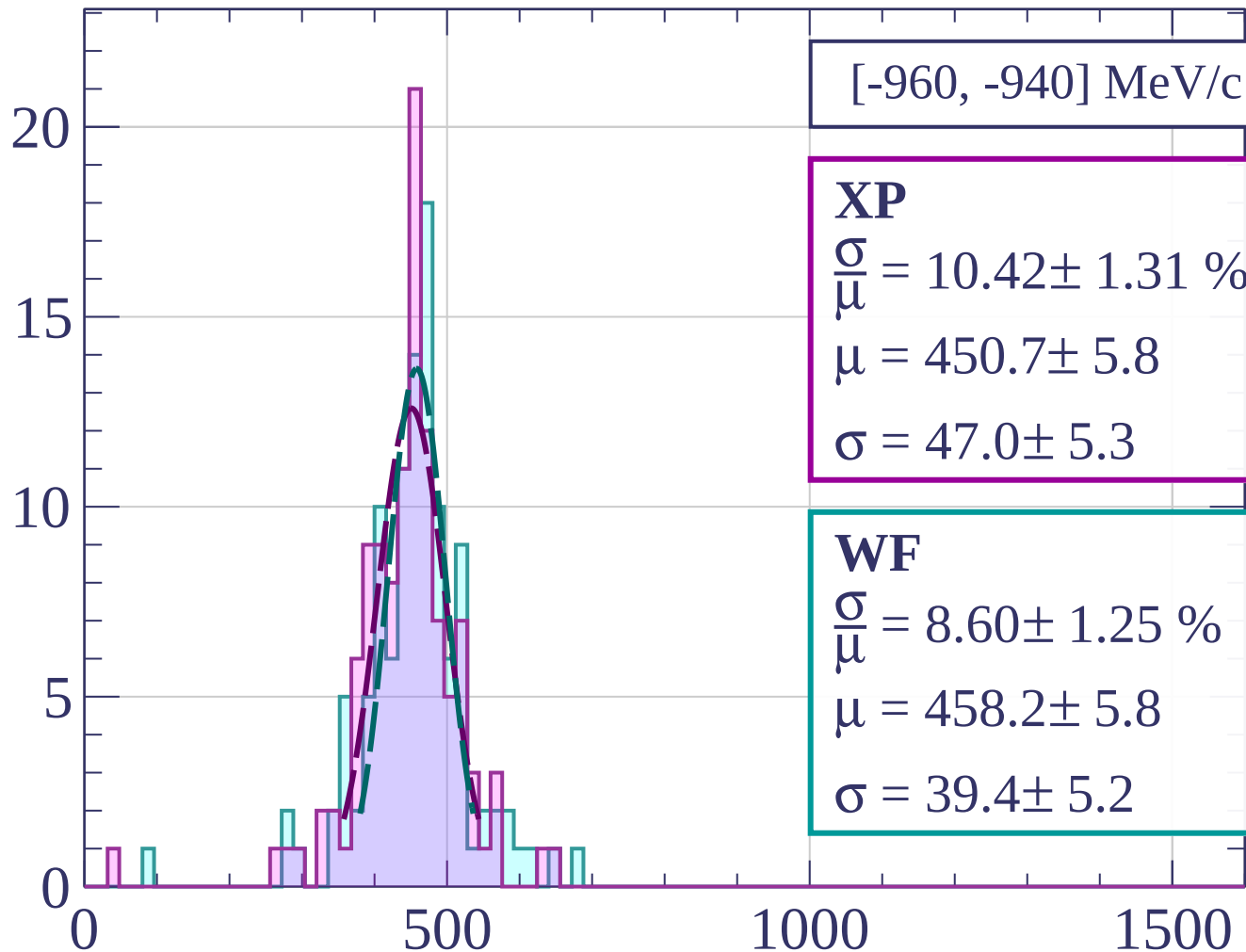
Count



Count



Count



[-960, -940] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.42 \pm 1.31 \%$$

$$\mu = 450.7 \pm 5.8$$

$$\sigma = 47.0 \pm 5.3$$

WF

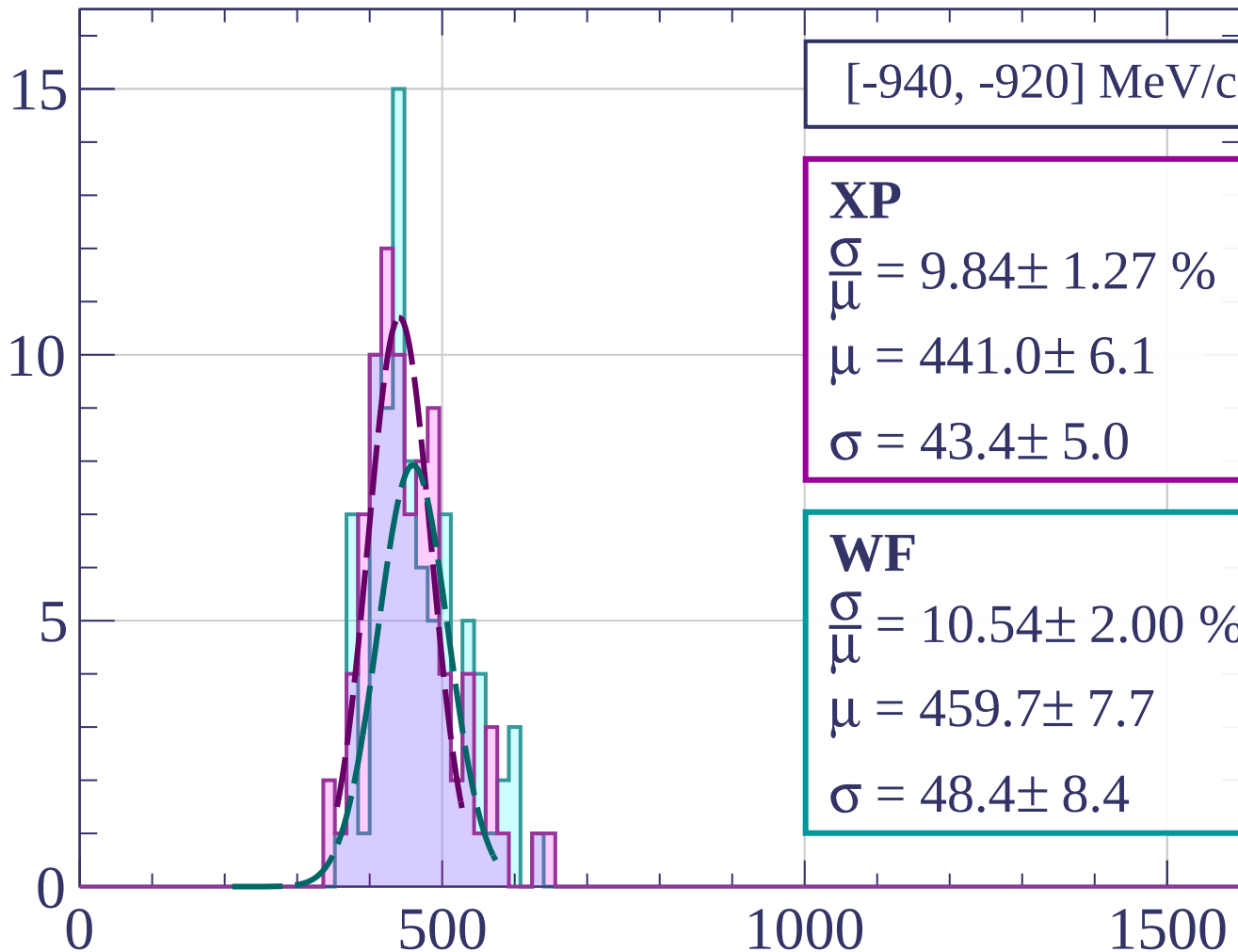
$$\frac{\sigma}{\mu} = 8.60 \pm 1.25 \%$$

$$\mu = 458.2 \pm 5.8$$

$$\sigma = 39.4 \pm 5.2$$

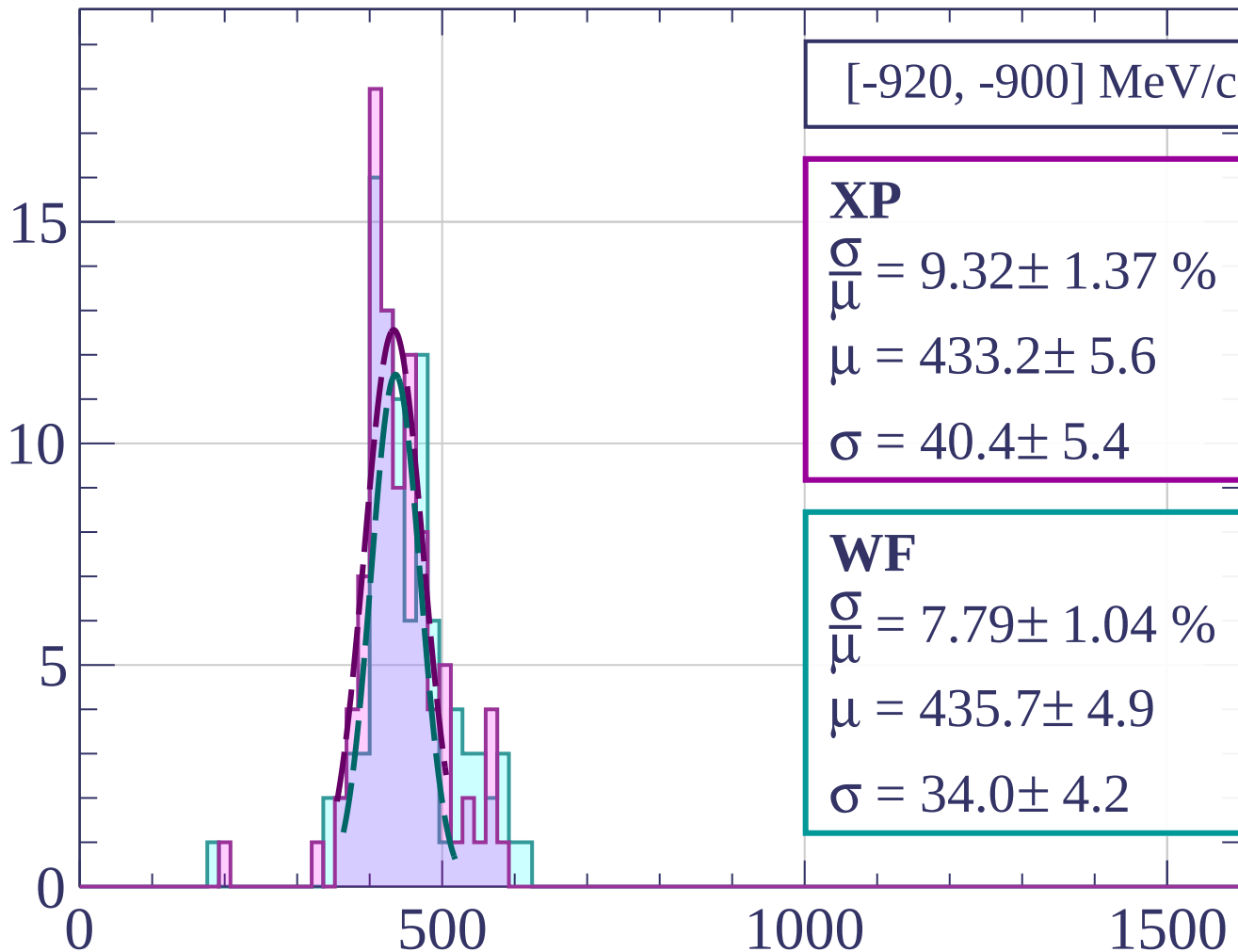
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[-920, -900] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.32 \pm 1.37 \%$$

$$\mu = 433.2 \pm 5.6$$

$$\sigma = 40.4 \pm 5.4$$

WF

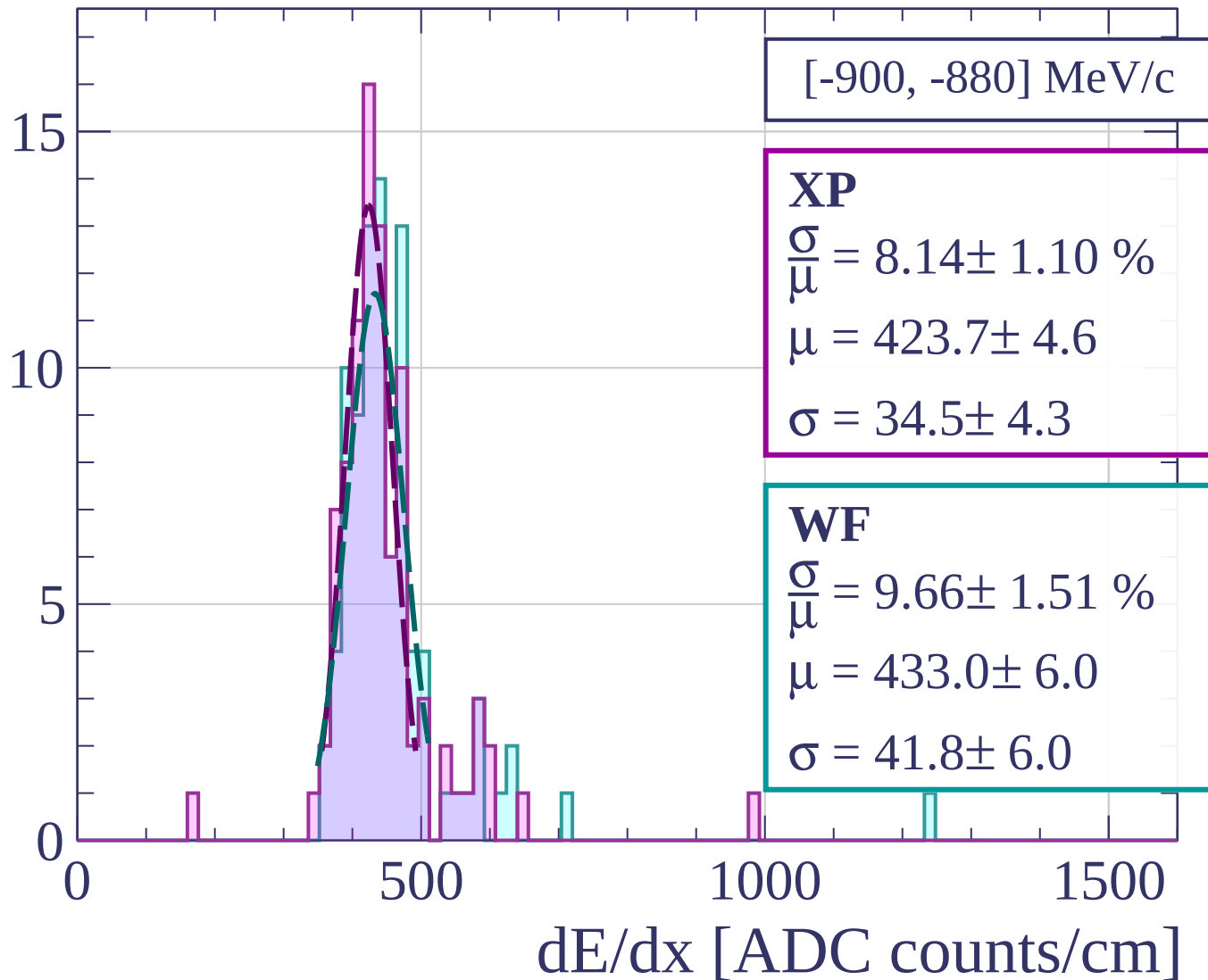
$$\frac{\sigma}{\mu} = 7.79 \pm 1.04 \%$$

$$\mu = 435.7 \pm 4.9$$

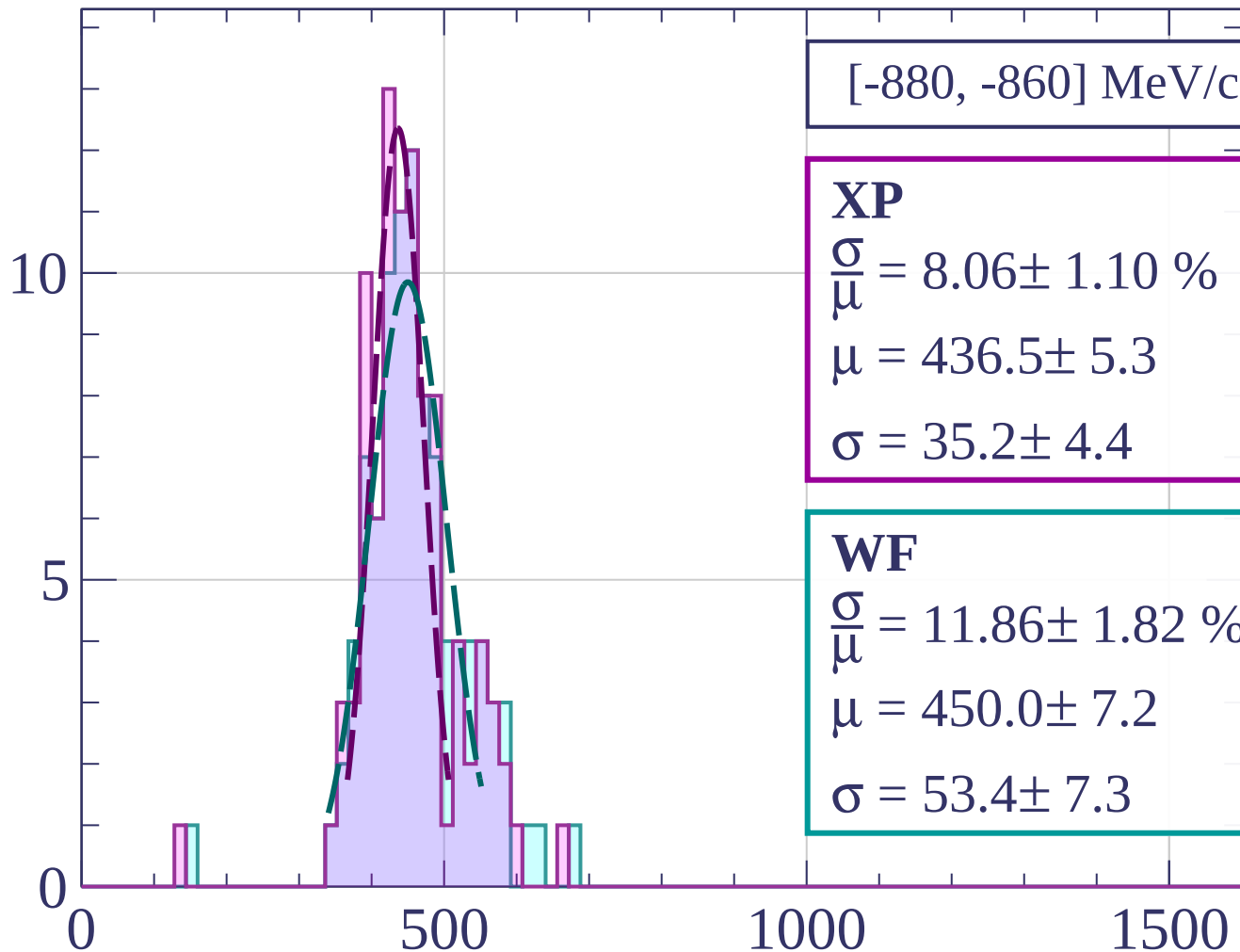
$$\sigma = 34.0 \pm 4.2$$

dE/dx [ADC counts/cm]

Count



Count



[-880, -860] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.06 \pm 1.10 \%$$

$$\mu = 436.5 \pm 5.3$$

$$\sigma = 35.2 \pm 4.4$$

WF

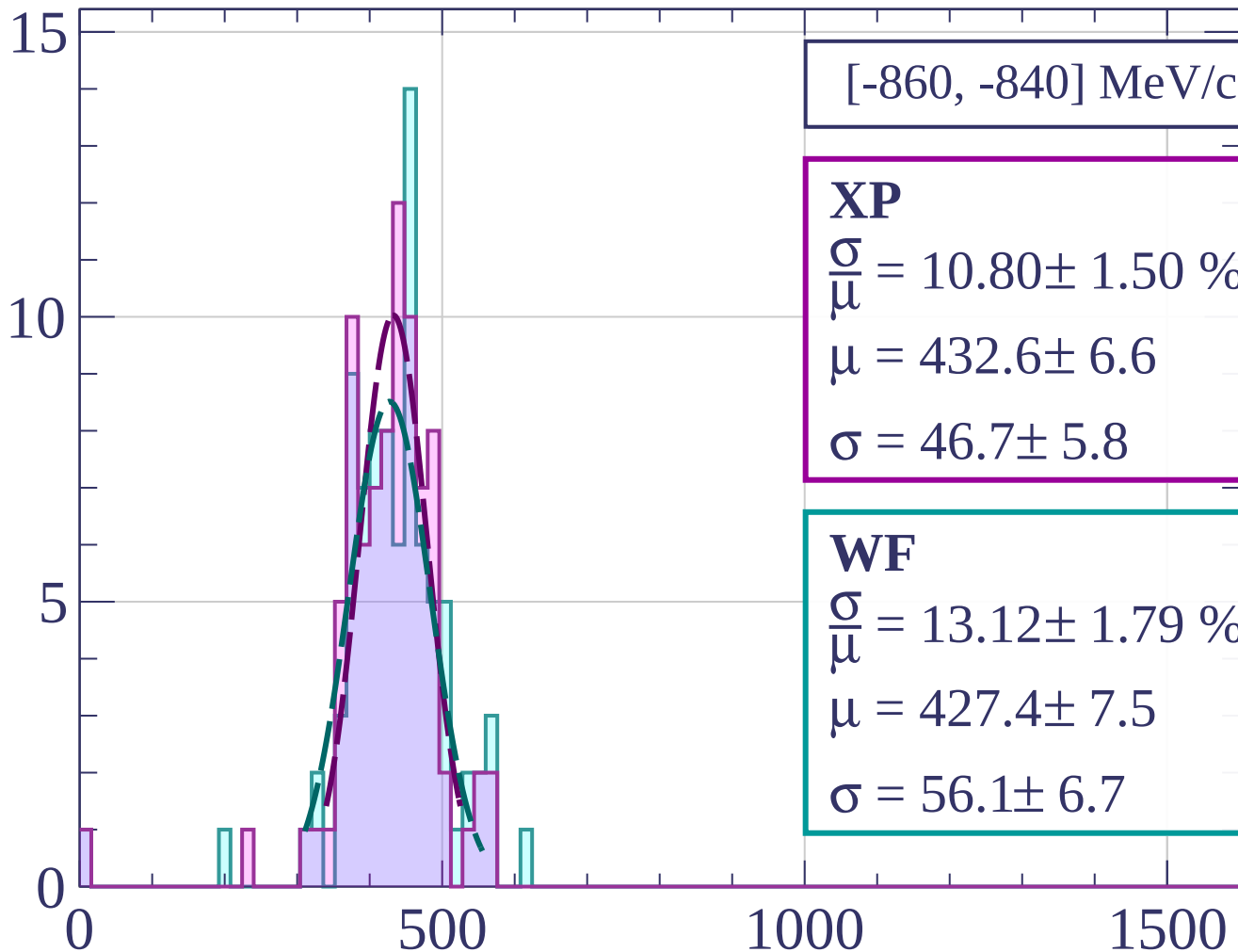
$$\frac{\sigma}{\mu} = 11.86 \pm 1.82 \%$$

$$\mu = 450.0 \pm 7.2$$

$$\sigma = 53.4 \pm 7.3$$

dE/dx [ADC counts/cm]

Count



[-860, -840] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.80 \pm 1.50 \%$$

$$\mu = 432.6 \pm 6.6$$

$$\sigma = 46.7 \pm 5.8$$

WF

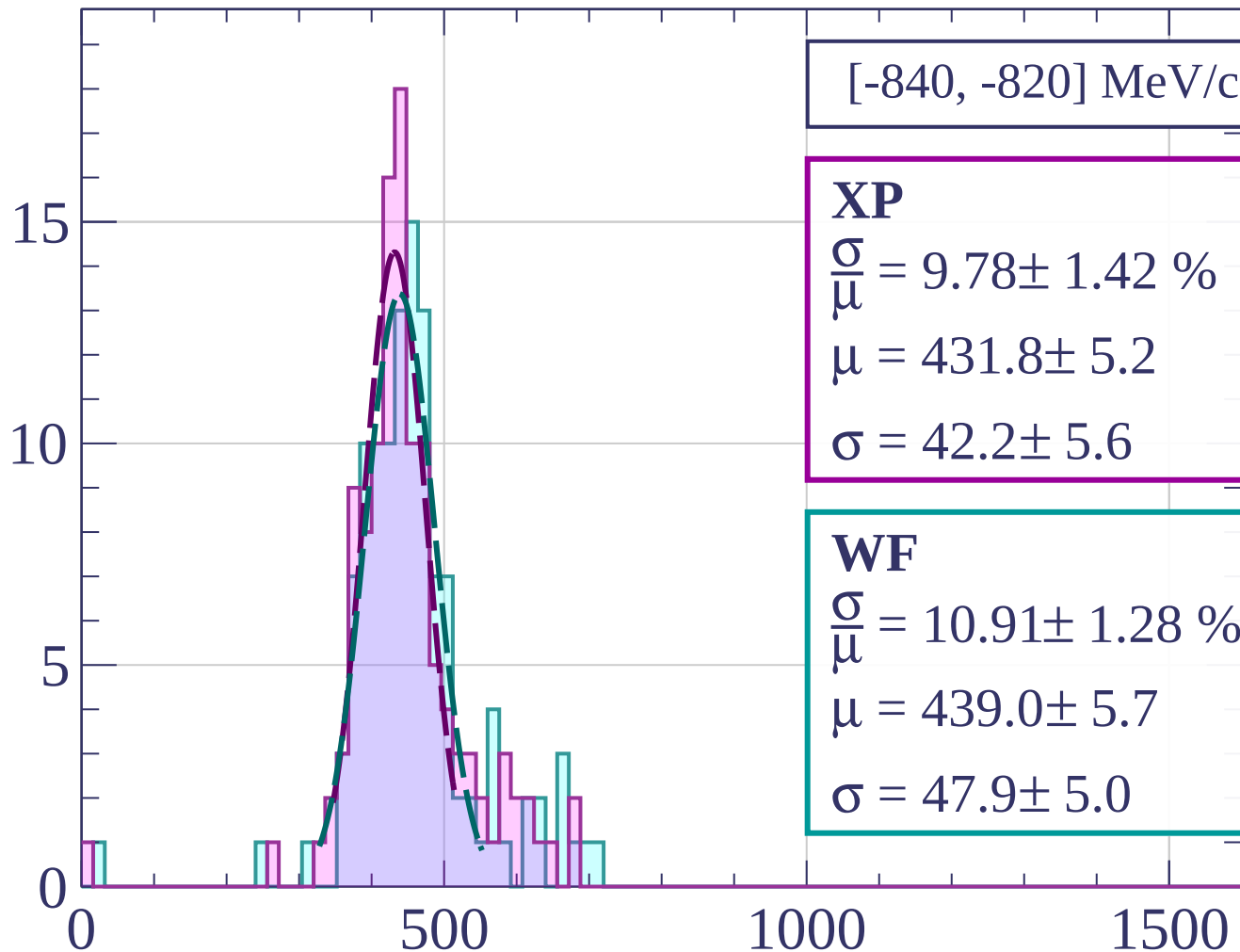
$$\frac{\sigma}{\mu} = 13.12 \pm 1.79 \%$$

$$\mu = 427.4 \pm 7.5$$

$$\sigma = 56.1 \pm 6.7$$

dE/dx [ADC counts/cm]

Count



[-840, -820] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.78 \pm 1.42 \%$$

$$\mu = 431.8 \pm 5.2$$

$$\sigma = 42.2 \pm 5.6$$

WF

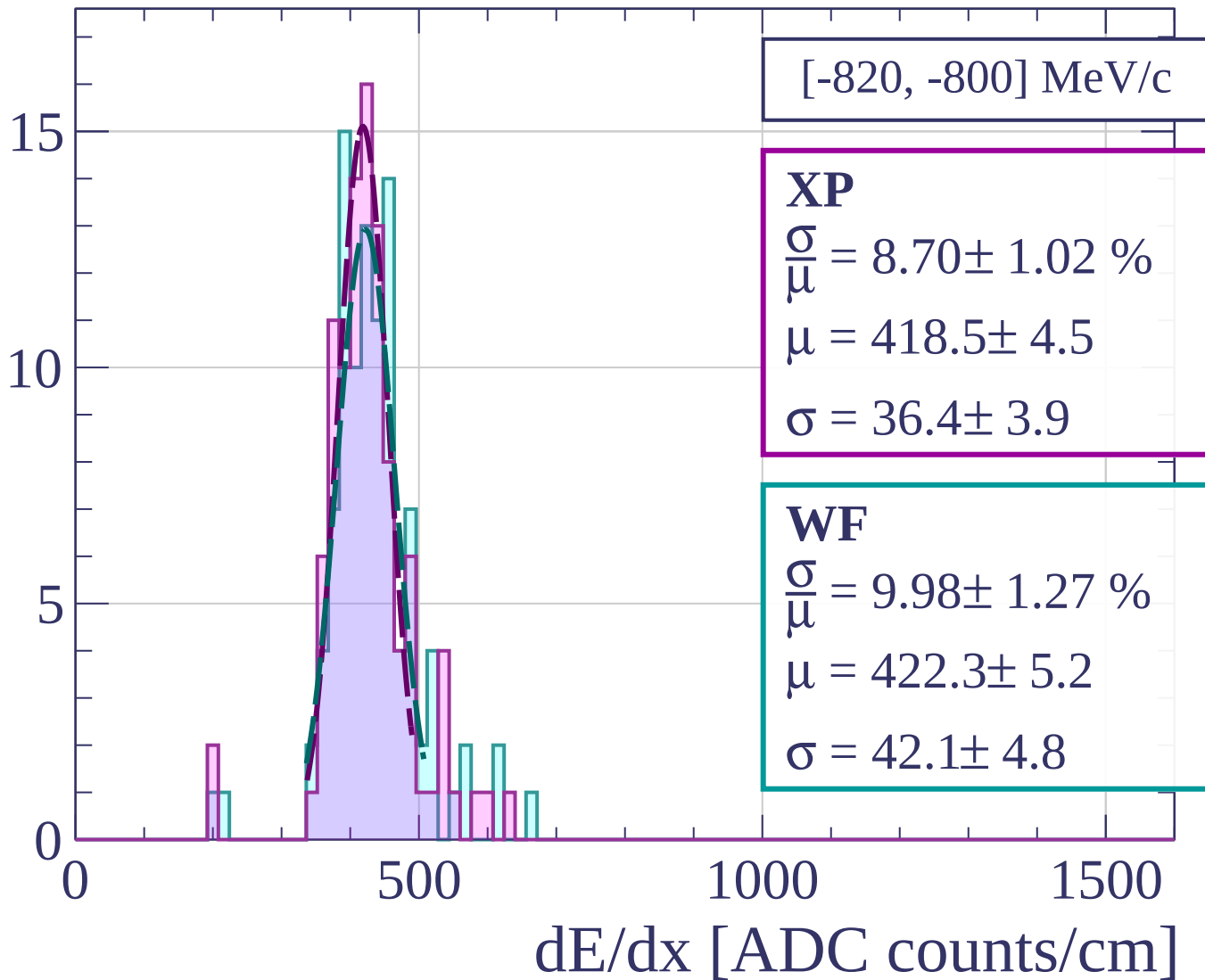
$$\frac{\sigma}{\mu} = 10.91 \pm 1.28 \%$$

$$\mu = 439.0 \pm 5.7$$

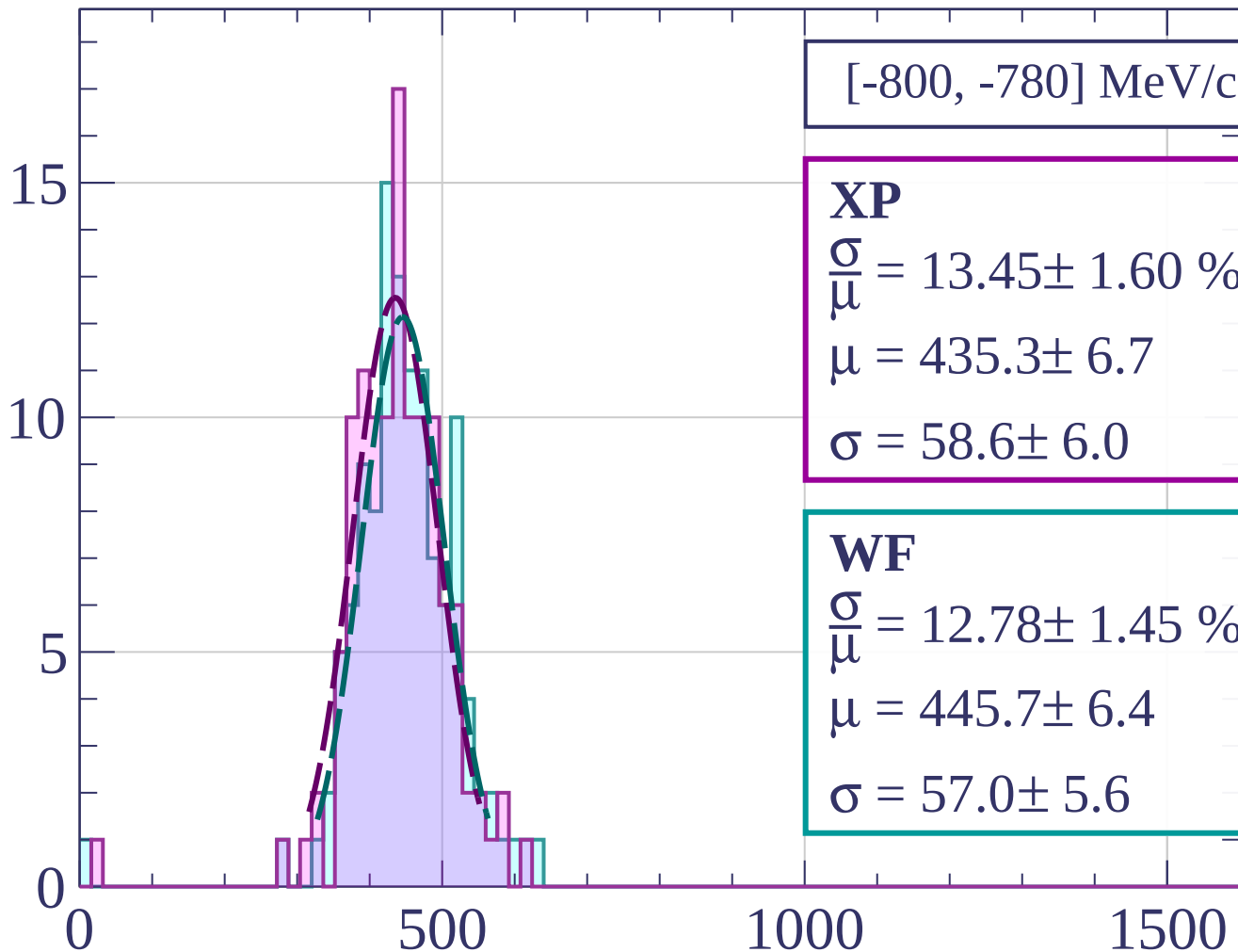
$$\sigma = 47.9 \pm 5.0$$

dE/dx [ADC counts/cm]

Count



Count



[-800, -780] MeV/c

XP

$$\frac{\sigma}{\mu} = 13.45 \pm 1.60 \%$$

$$\mu = 435.3 \pm 6.7$$

$$\sigma = 58.6 \pm 6.0$$

WF

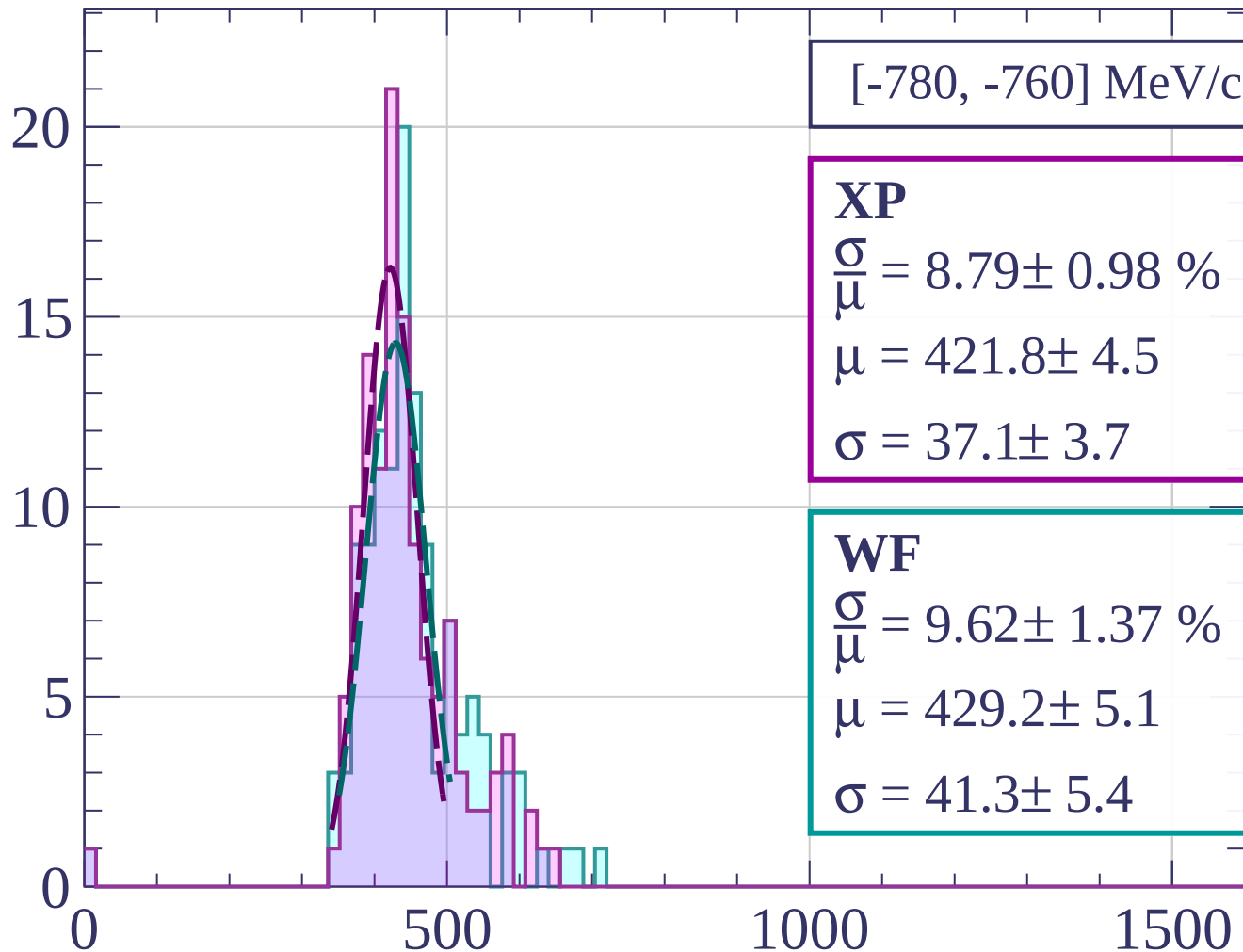
$$\frac{\sigma}{\mu} = 12.78 \pm 1.45 \%$$

$$\mu = 445.7 \pm 6.4$$

$$\sigma = 57.0 \pm 5.6$$

dE/dx [ADC counts/cm]

Count



[-780, -760] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.79 \pm 0.98 \%$$

$$\mu = 421.8 \pm 4.5$$

$$\sigma = 37.1 \pm 3.7$$

WF

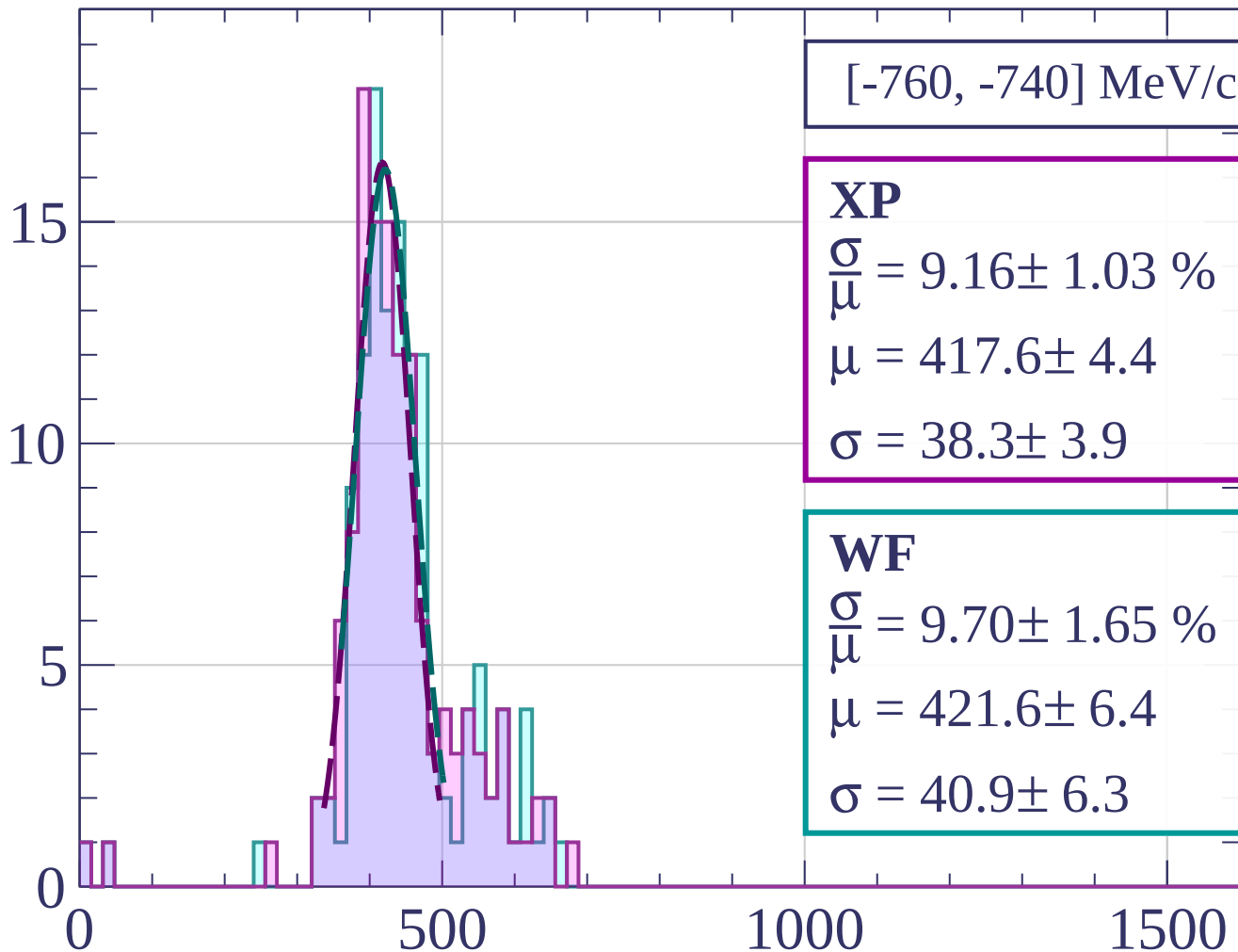
$$\frac{\sigma}{\mu} = 9.62 \pm 1.37 \%$$

$$\mu = 429.2 \pm 5.1$$

$$\sigma = 41.3 \pm 5.4$$

dE/dx [ADC counts/cm]

Count



[-760, -740] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.16 \pm 1.03 \%$$

$$\mu = 417.6 \pm 4.4$$

$$\sigma = 38.3 \pm 3.9$$

WF

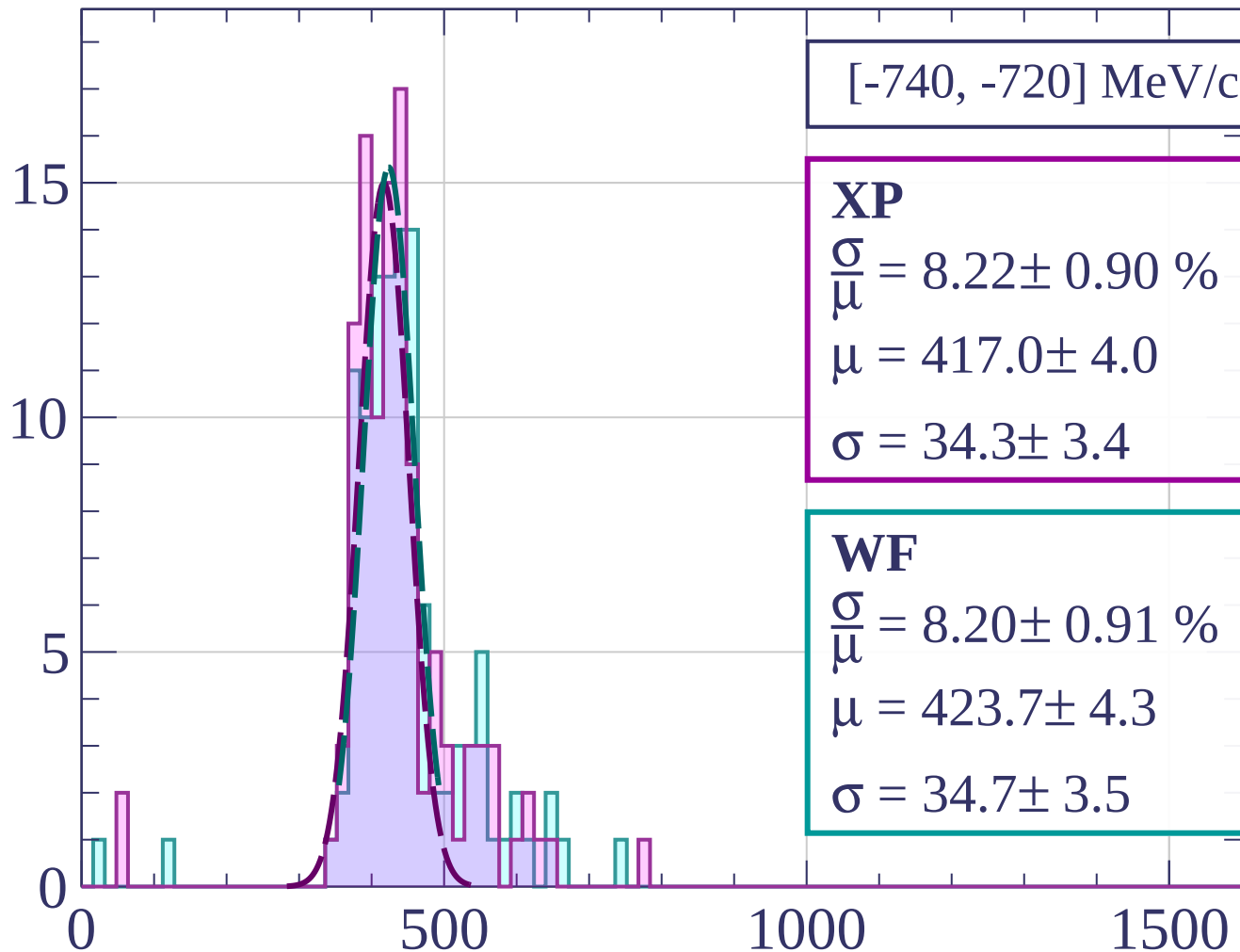
$$\frac{\sigma}{\mu} = 9.70 \pm 1.65 \%$$

$$\mu = 421.6 \pm 6.4$$

$$\sigma = 40.9 \pm 6.3$$

dE/dx [ADC counts/cm]

Count



[-740, -720] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.22 \pm 0.90 \%$$

$$\mu = 417.0 \pm 4.0$$

$$\sigma = 34.3 \pm 3.4$$

WF

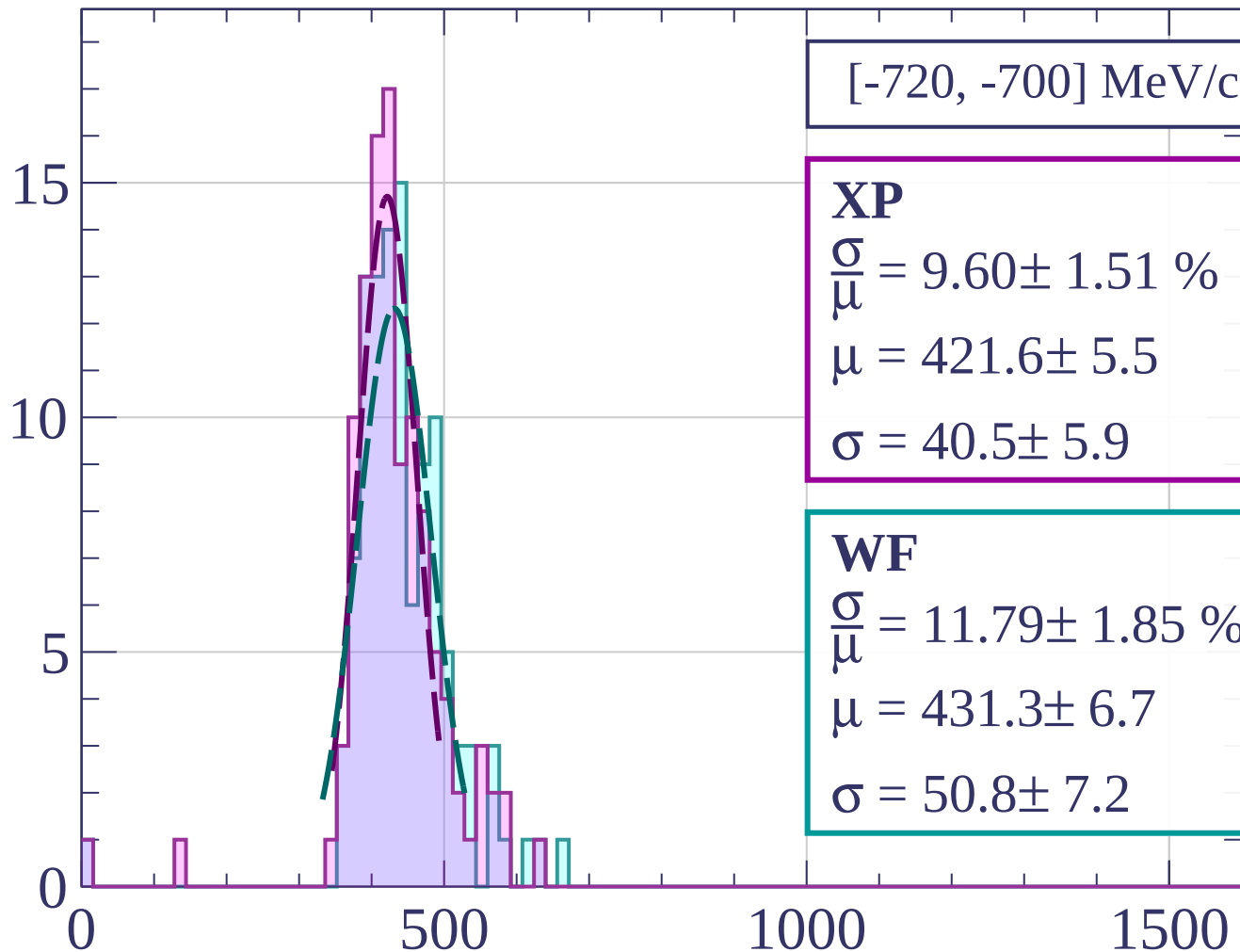
$$\frac{\sigma}{\mu} = 8.20 \pm 0.91 \%$$

$$\mu = 423.7 \pm 4.3$$

$$\sigma = 34.7 \pm 3.5$$

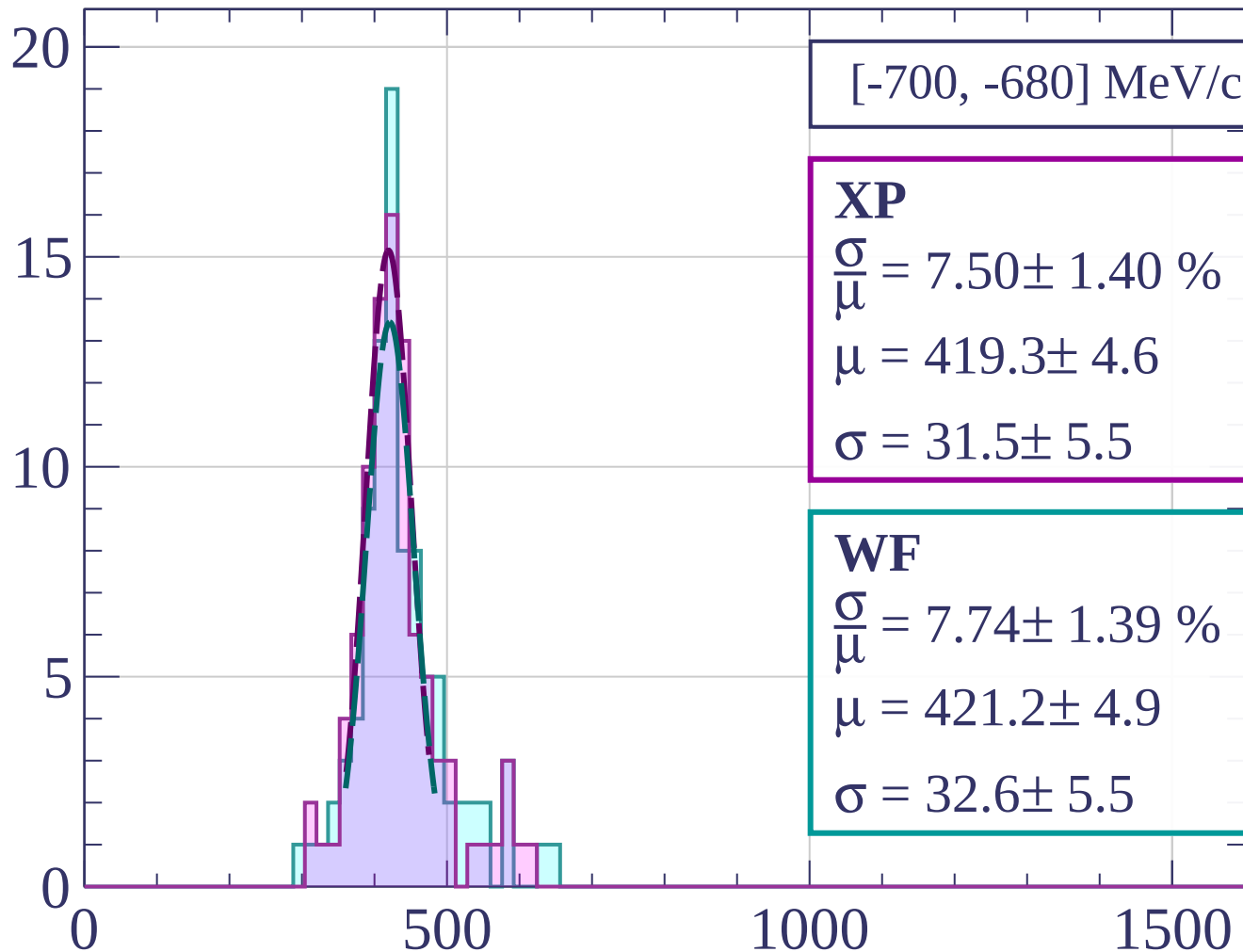
dE/dx [ADC counts/cm]

Count



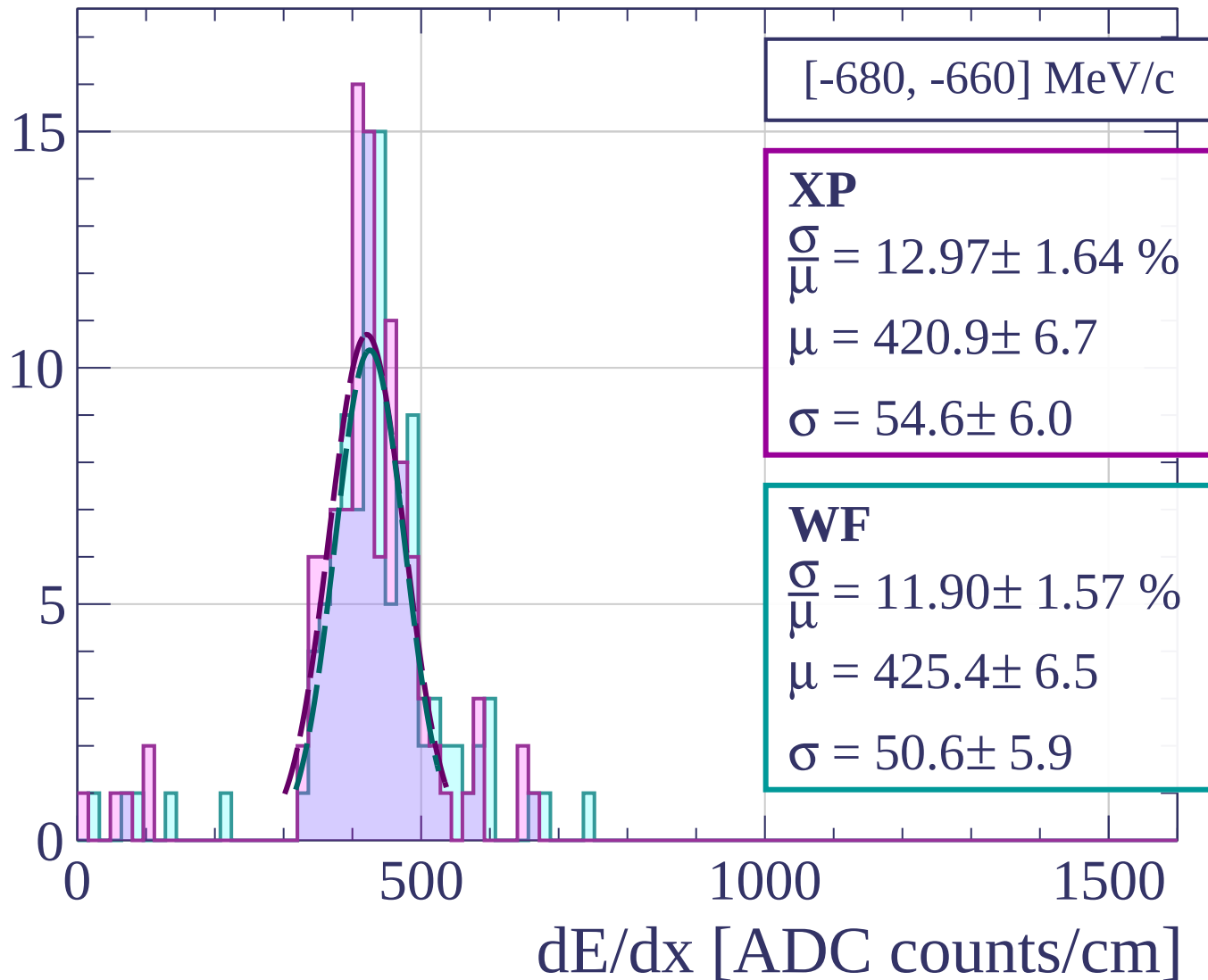
dE/dx [ADC counts/cm]

Count

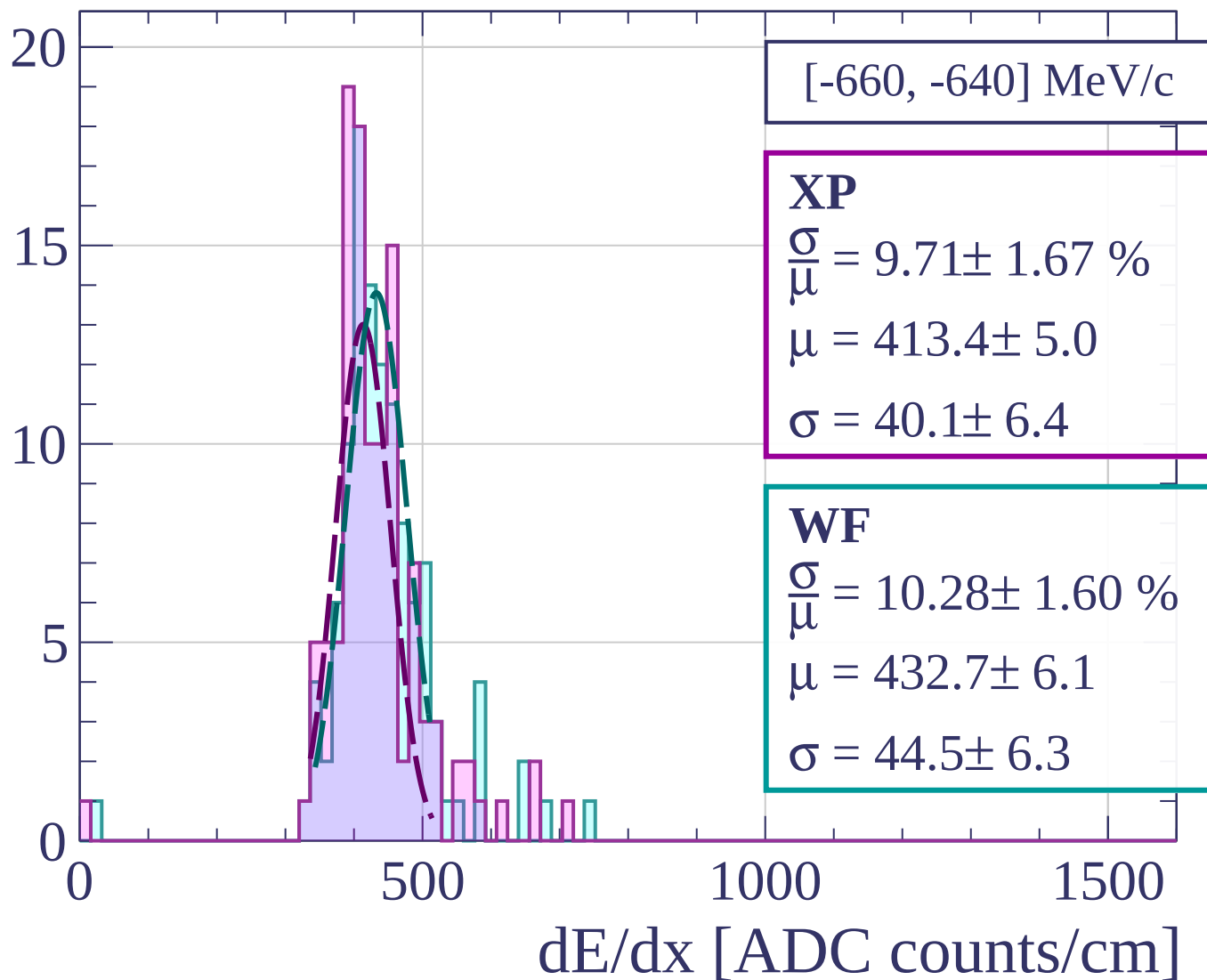


dE/dx [ADC counts/cm]

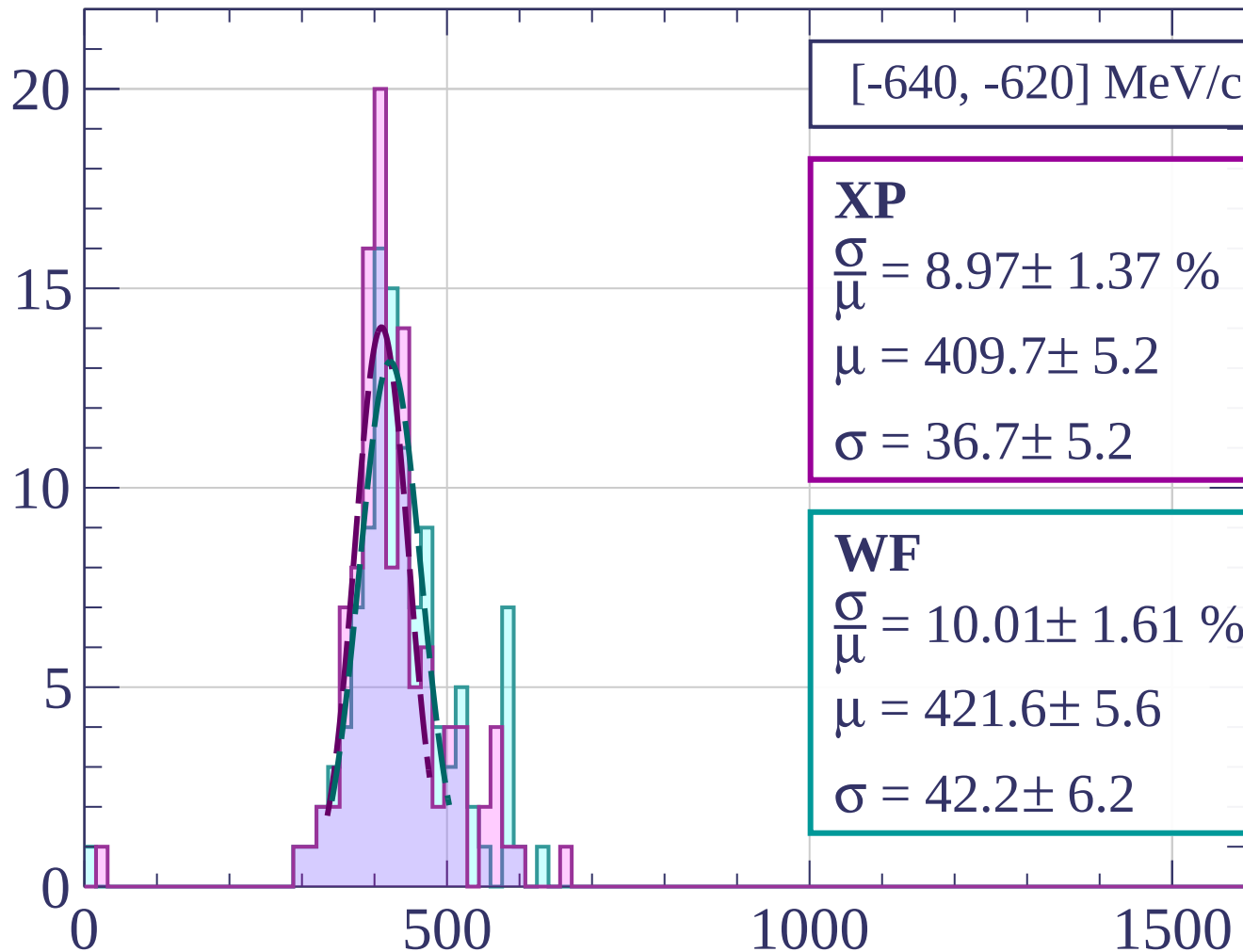
Count



Count



Count



[-640, -620] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.97 \pm 1.37 \%$$

$$\mu = 409.7 \pm 5.2$$

$$\sigma = 36.7 \pm 5.2$$

WF

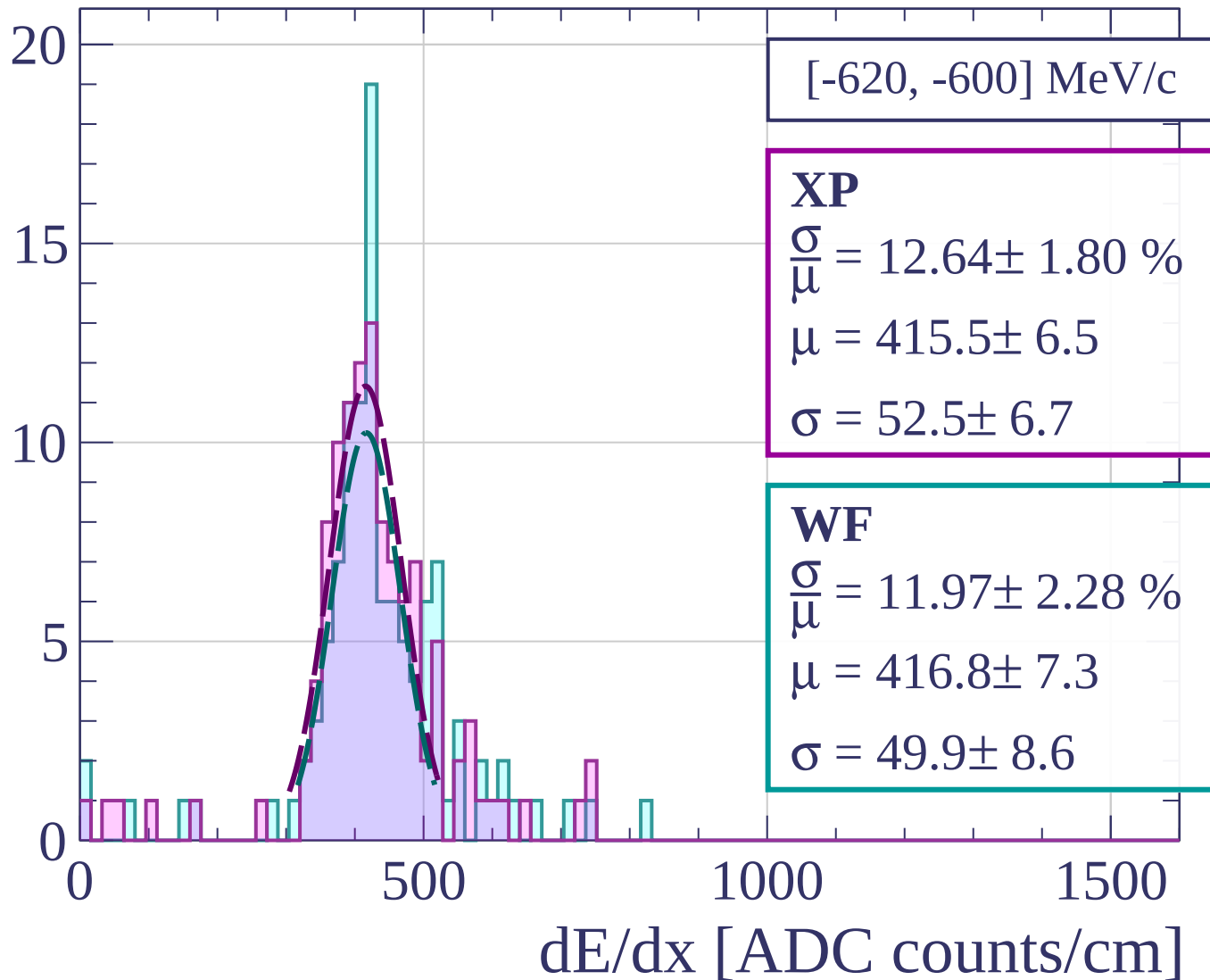
$$\frac{\sigma}{\mu} = 10.01 \pm 1.61 \%$$

$$\mu = 421.6 \pm 5.6$$

$$\sigma = 42.2 \pm 6.2$$

dE/dx [ADC counts/cm]

Count



[-620, -600] MeV/c

XP

$$\frac{\sigma}{\mu} = 12.64 \pm 1.80 \%$$

$$\mu = 415.5 \pm 6.5$$

$$\sigma = 52.5 \pm 6.7$$

WF

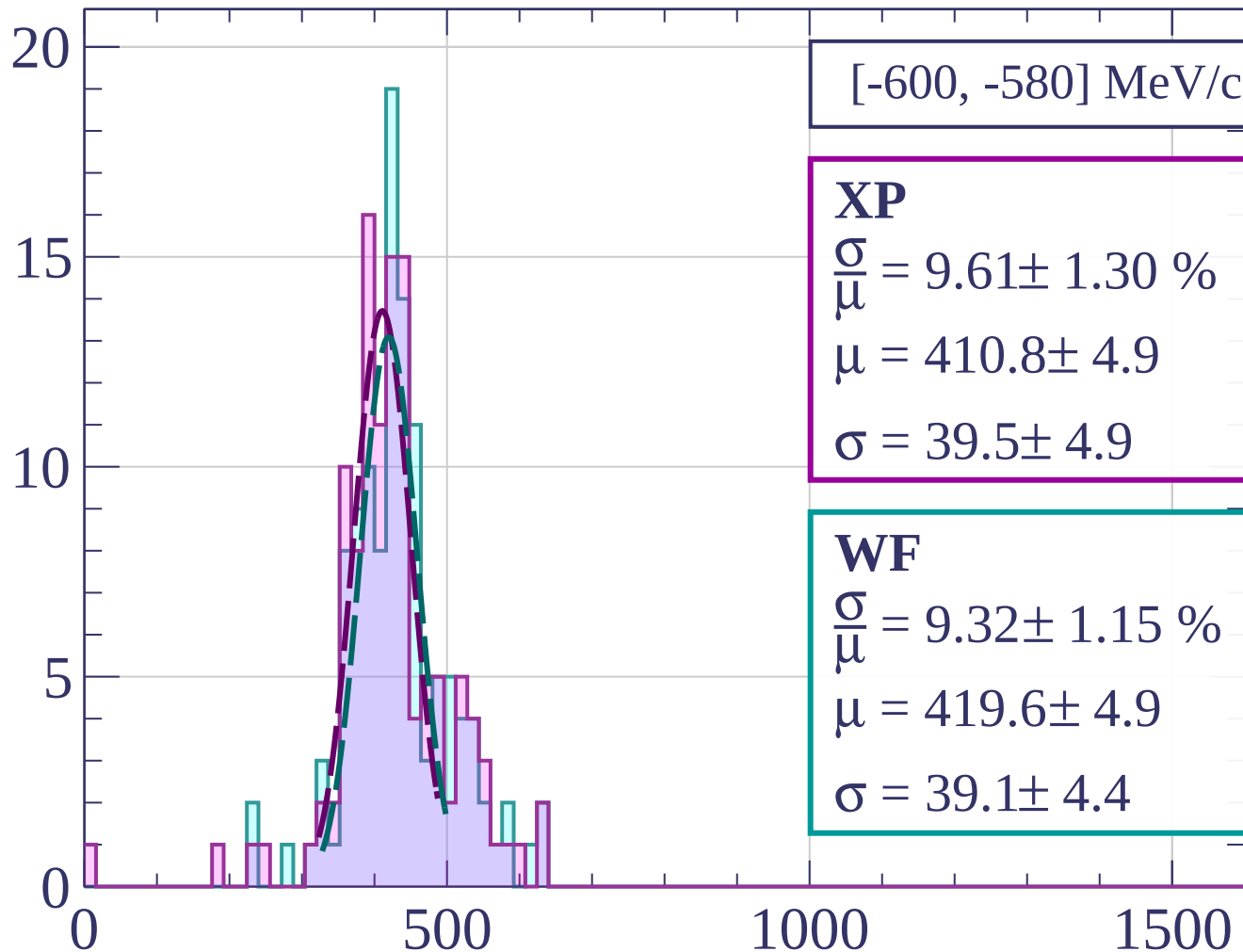
$$\frac{\sigma}{\mu} = 11.97 \pm 2.28 \%$$

$$\mu = 416.8 \pm 7.3$$

$$\sigma = 49.9 \pm 8.6$$

dE/dx [ADC counts/cm]

Count



[-600, -580] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 9.61 \pm 1.30 \%$$

$$\mu = 410.8 \pm 4.9$$

$$\sigma = 39.5 \pm 4.9$$

WF

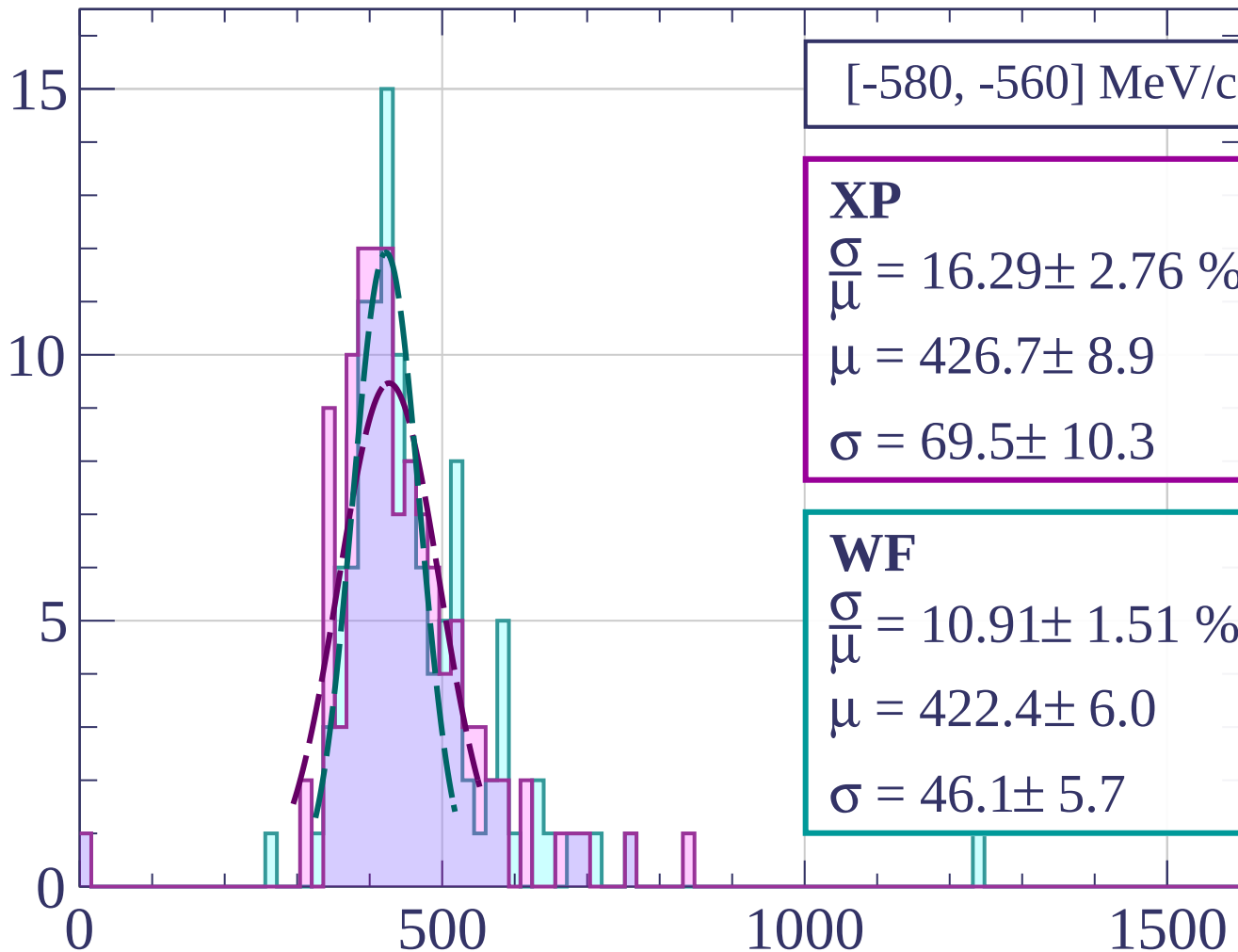
$$\frac{\sigma}{\bar{\mu}} = 9.32 \pm 1.15 \%$$

$$\mu = 419.6 \pm 4.9$$

$$\sigma = 39.1 \pm 4.4$$

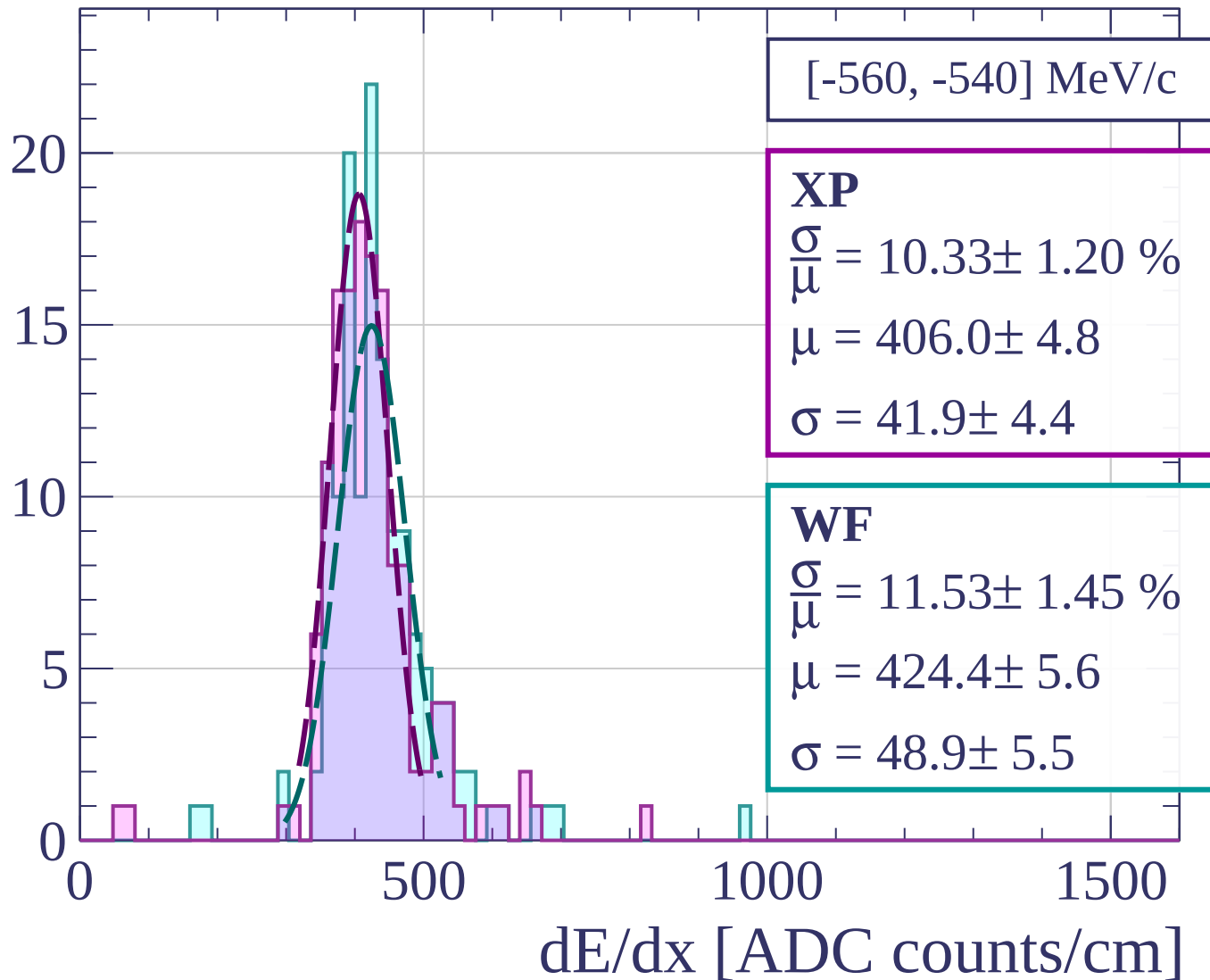
dE/dx [ADC counts/cm]

Count

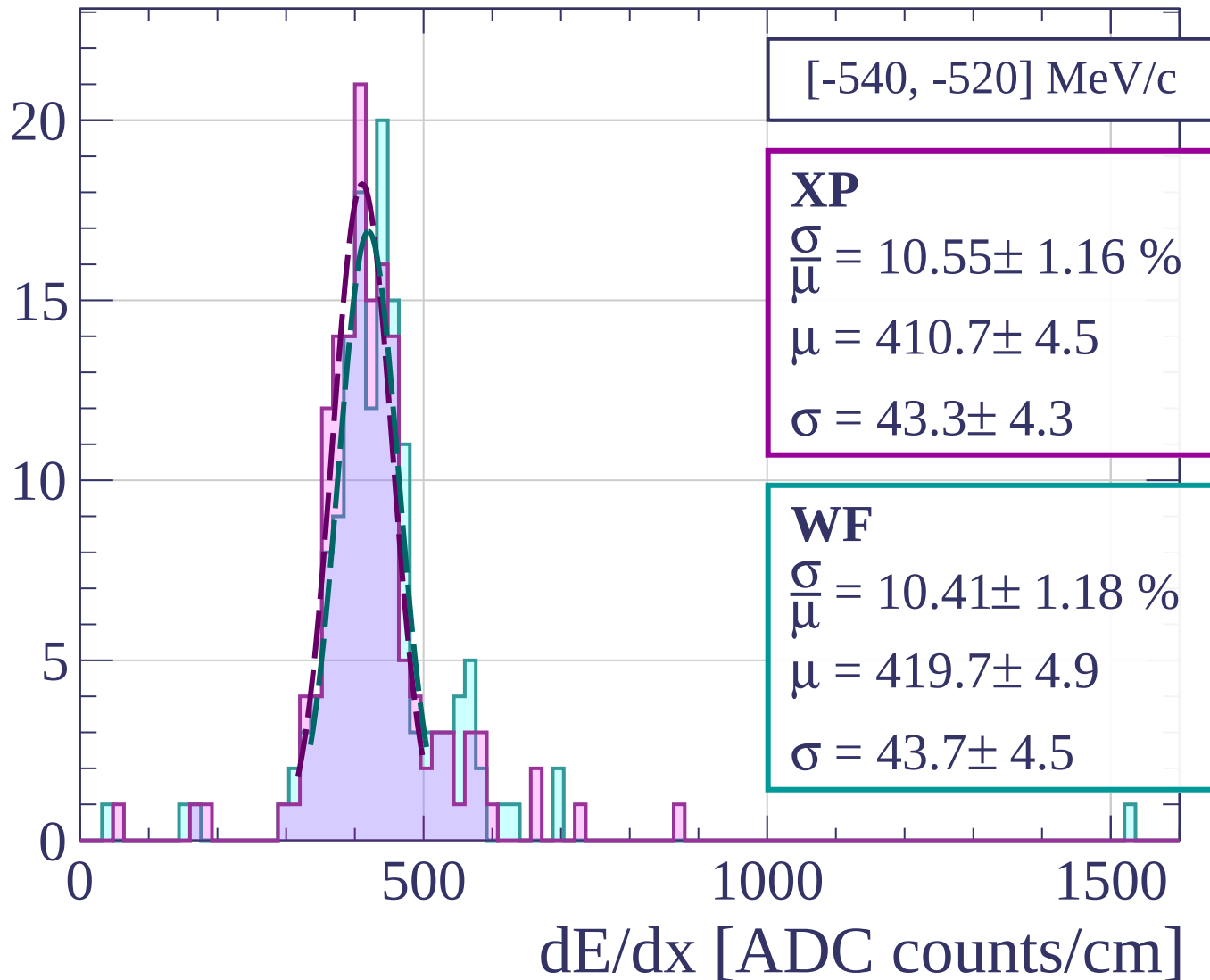


dE/dx [ADC counts/cm]

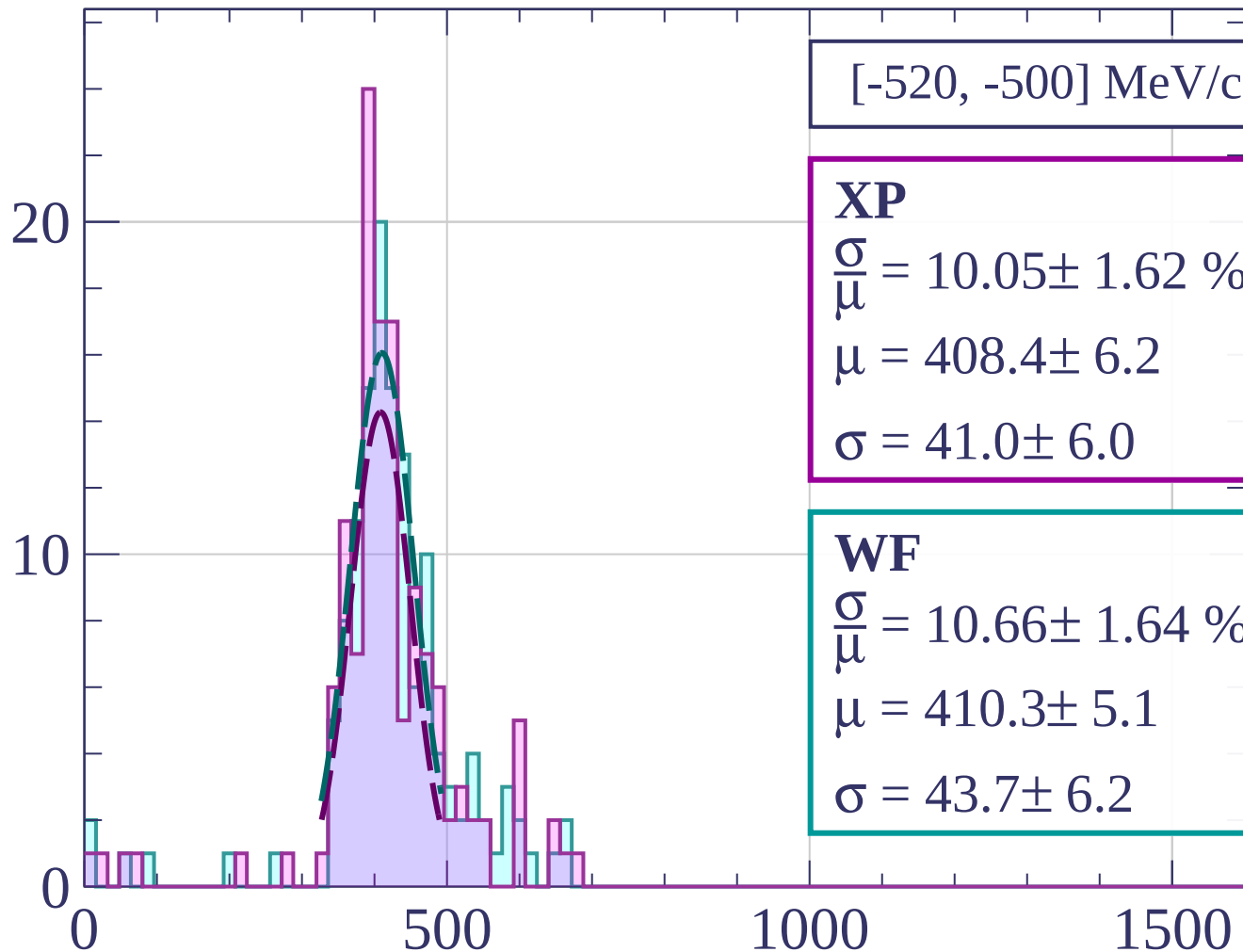
Count



Count



Count



[-520, -500] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.05 \pm 1.62 \%$$

$$\mu = 408.4 \pm 6.2$$

$$\sigma = 41.0 \pm 6.0$$

WF

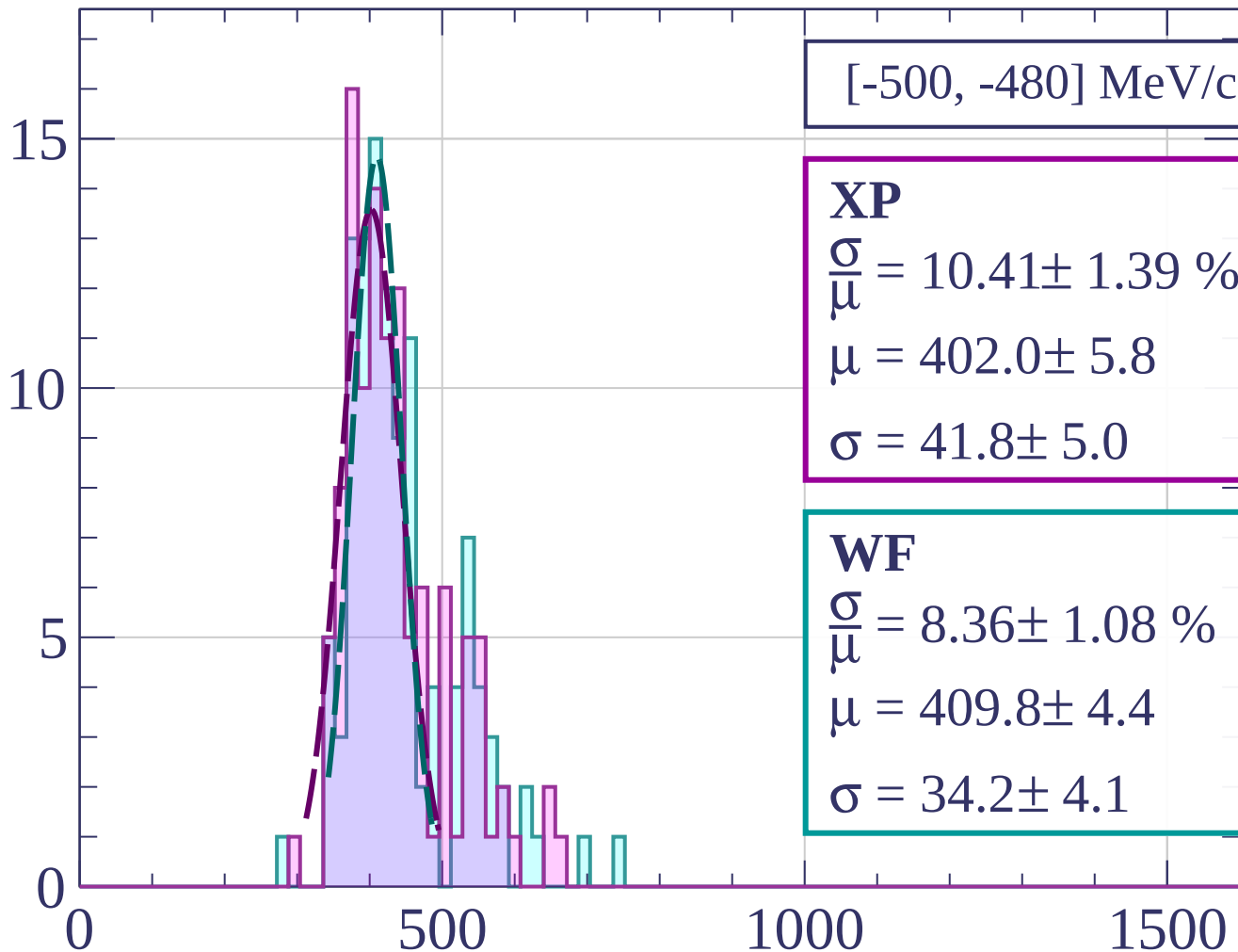
$$\frac{\sigma}{\mu} = 10.66 \pm 1.64 \%$$

$$\mu = 410.3 \pm 5.1$$

$$\sigma = 43.7 \pm 6.2$$

dE/dx [ADC counts/cm]

Count



[-500, -480] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.41 \pm 1.39 \%$$

$$\mu = 402.0 \pm 5.8$$

$$\sigma = 41.8 \pm 5.0$$

WF

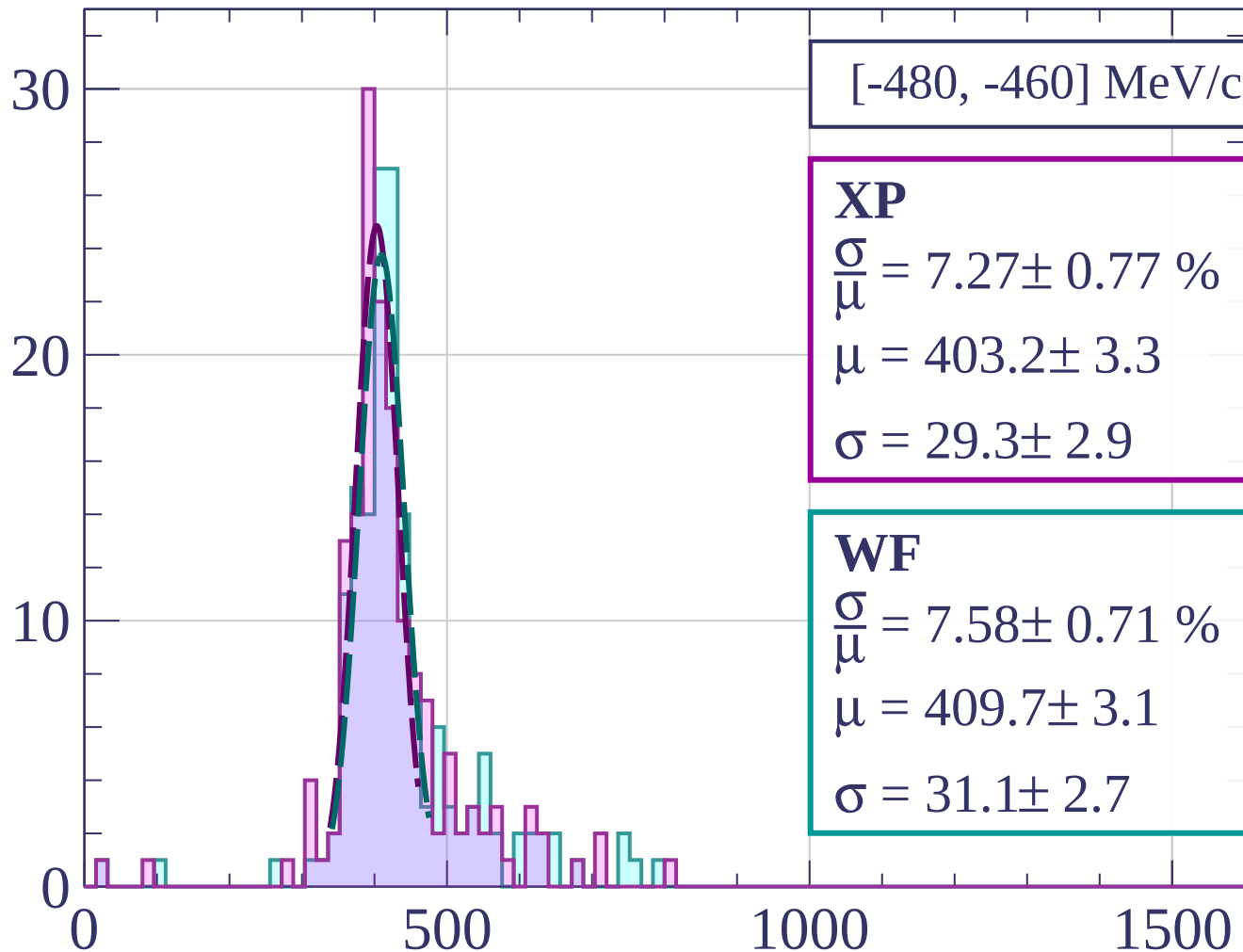
$$\frac{\sigma}{\mu} = 8.36 \pm 1.08 \%$$

$$\mu = 409.8 \pm 4.4$$

$$\sigma = 34.2 \pm 4.1$$

dE/dx [ADC counts/cm]

Count



[-480, -460] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.27 \pm 0.77 \%$$

$$\mu = 403.2 \pm 3.3$$

$$\sigma = 29.3 \pm 2.9$$

WF

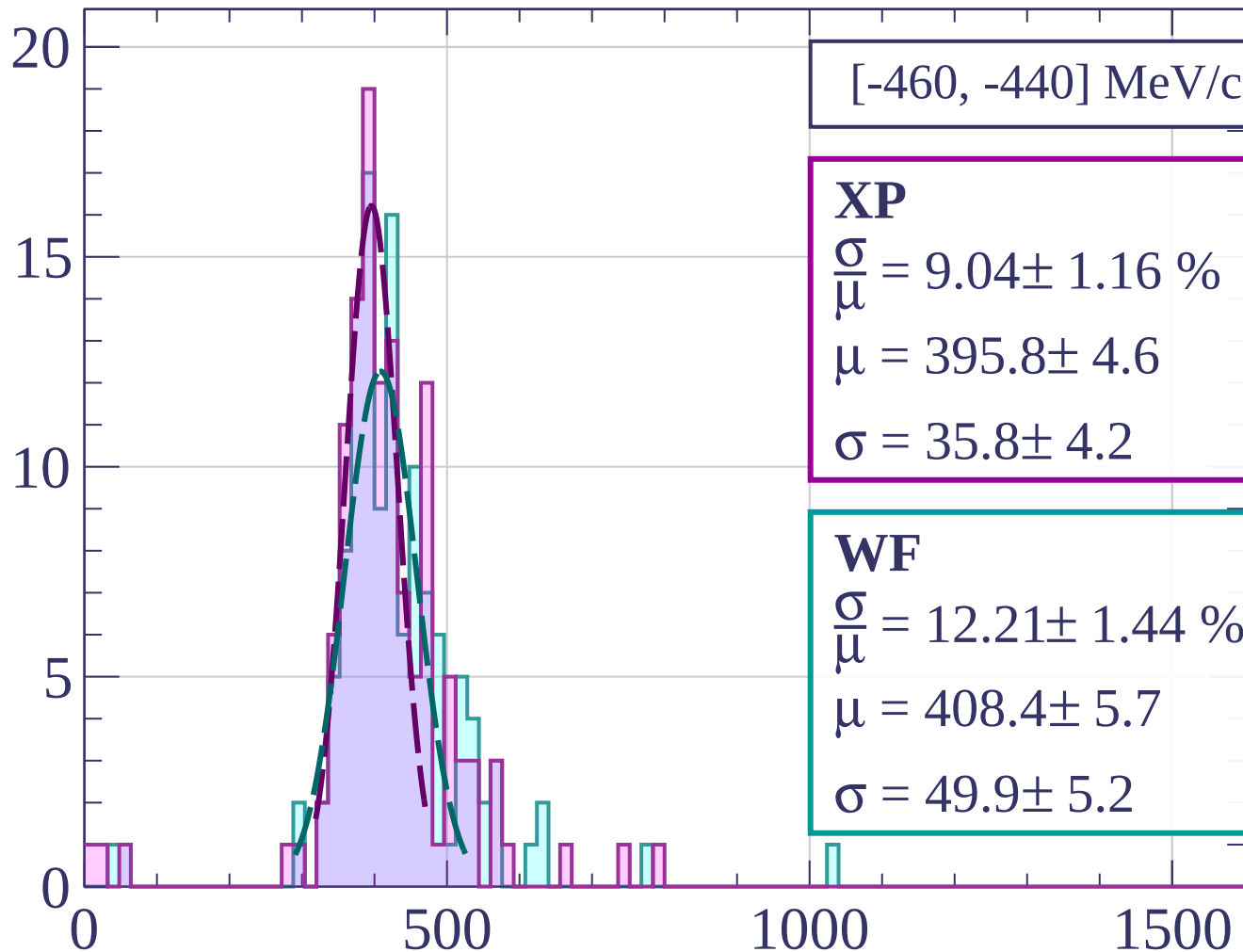
$$\frac{\sigma}{\mu} = 7.58 \pm 0.71 \%$$

$$\mu = 409.7 \pm 3.1$$

$$\sigma = 31.1 \pm 2.7$$

dE/dx [ADC counts/cm]

Count



[-460, -440] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.04 \pm 1.16 \%$$

$$\mu = 395.8 \pm 4.6$$

$$\sigma = 35.8 \pm 4.2$$

WF

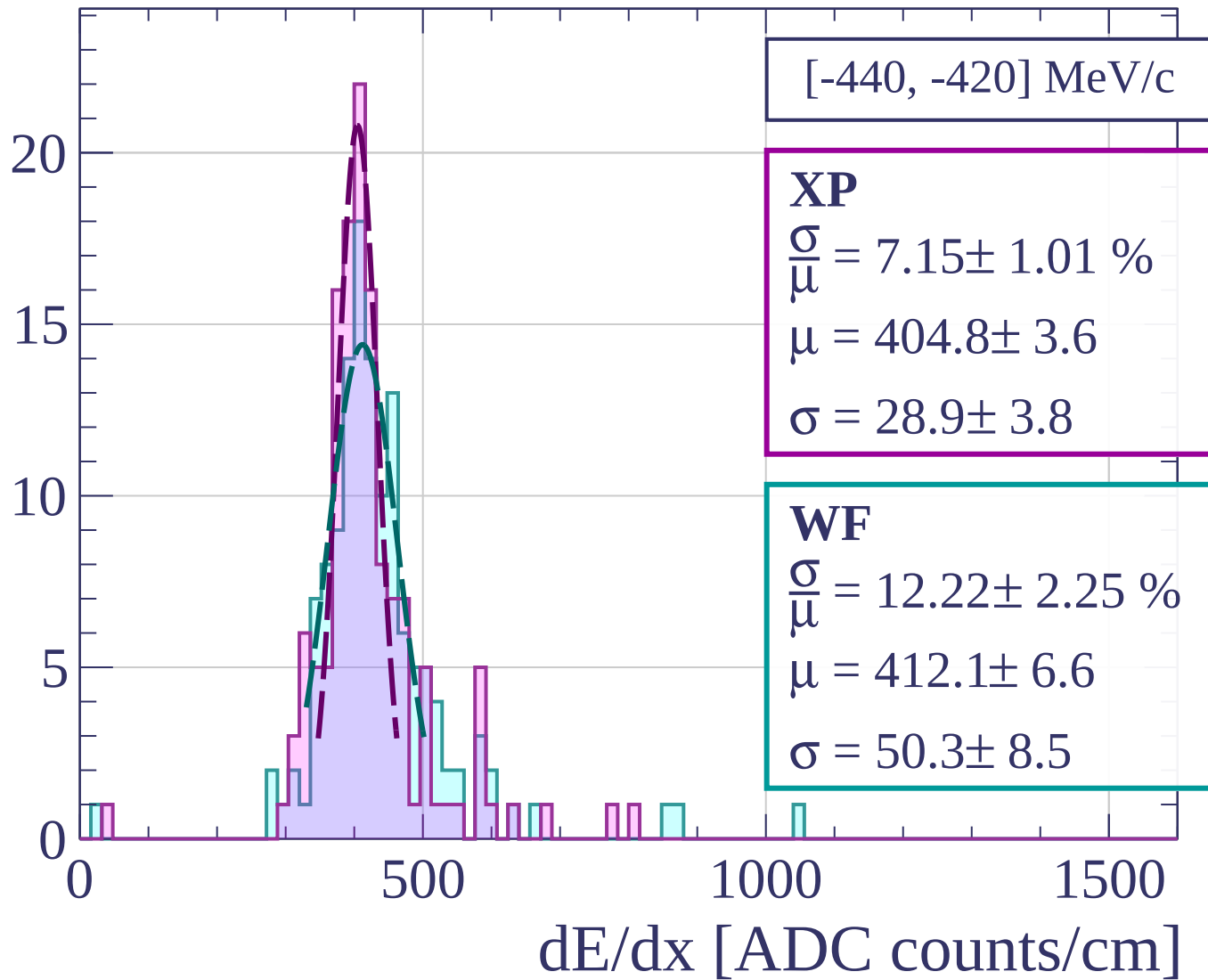
$$\frac{\sigma}{\mu} = 12.21 \pm 1.44 \%$$

$$\mu = 408.4 \pm 5.7$$

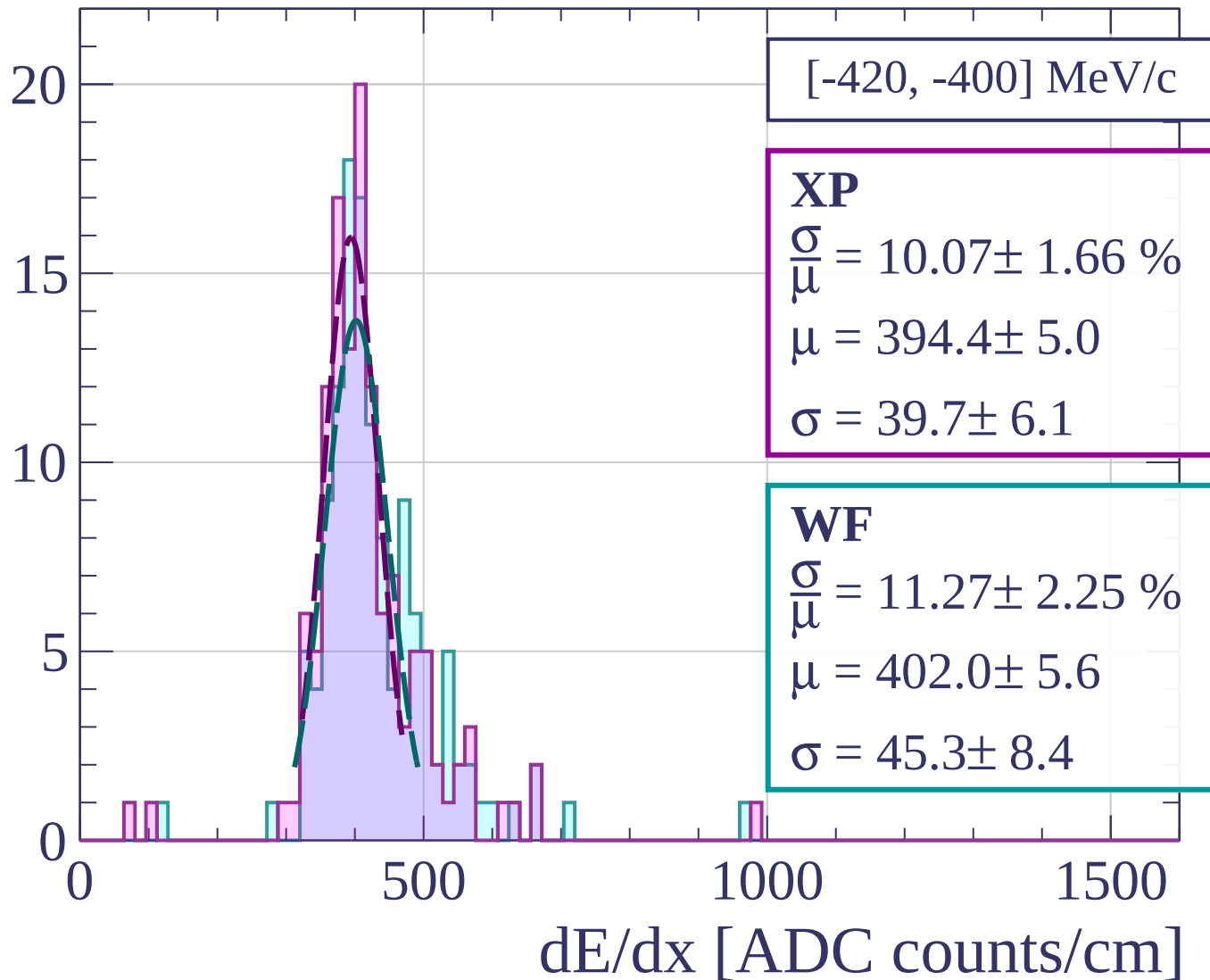
$$\sigma = 49.9 \pm 5.2$$

dE/dx [ADC counts/cm]

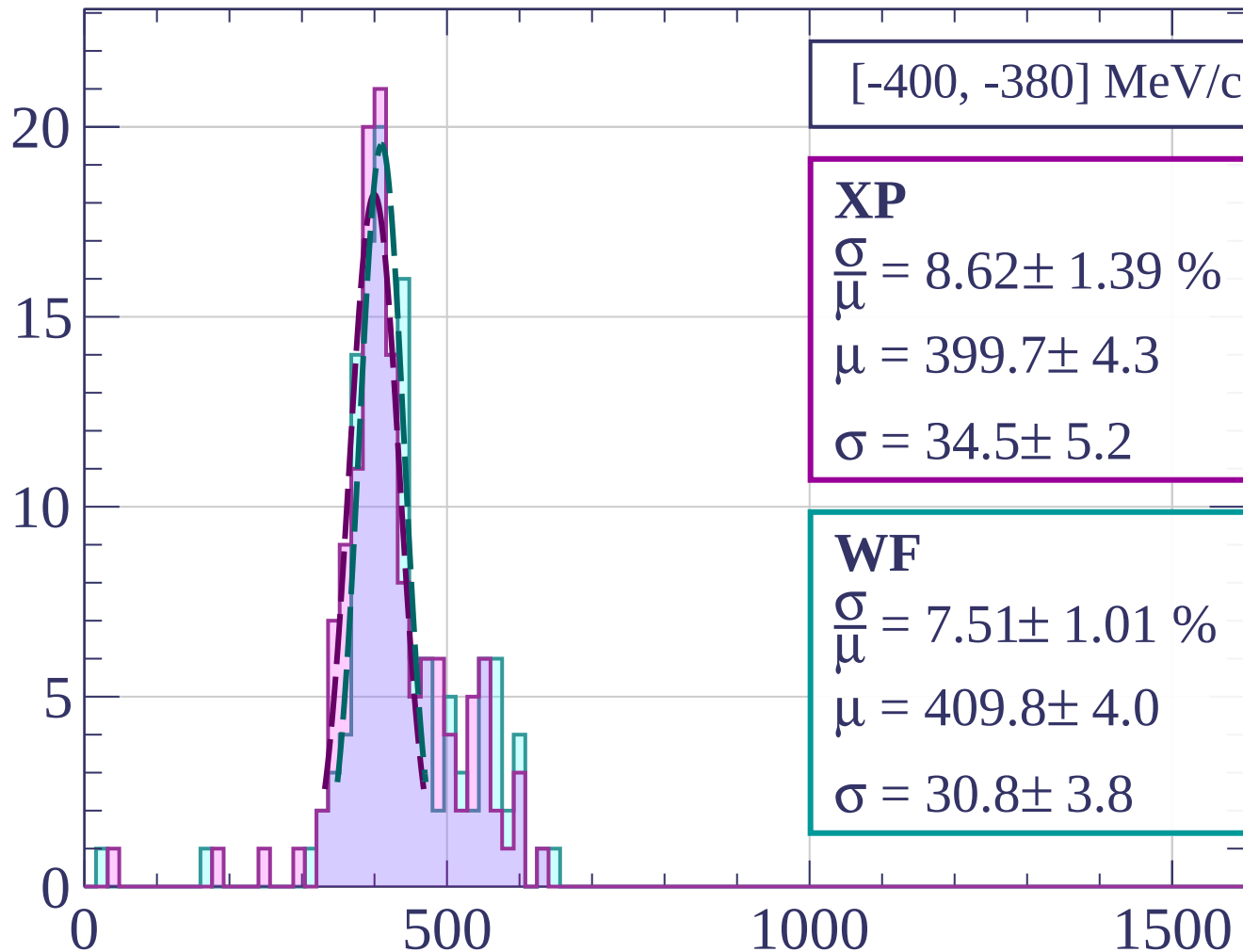
Count



Count



Count



[-400, -380] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.62 \pm 1.39 \%$$

$$\mu = 399.7 \pm 4.3$$

$$\sigma = 34.5 \pm 5.2$$

WF

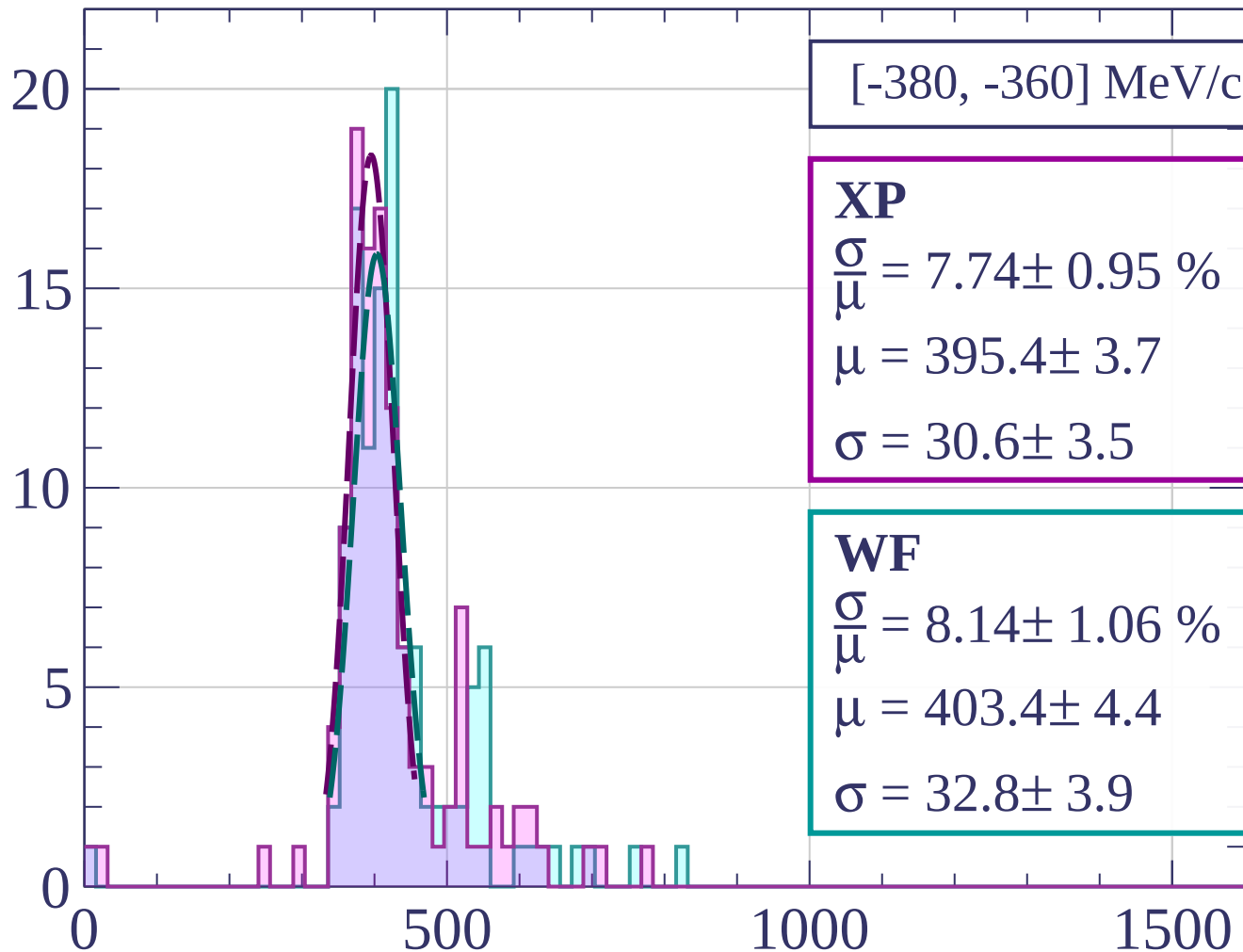
$$\frac{\sigma}{\mu} = 7.51 \pm 1.01 \%$$

$$\mu = 409.8 \pm 4.0$$

$$\sigma = 30.8 \pm 3.8$$

dE/dx [ADC counts/cm]

Count



[-380, -360] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.74 \pm 0.95 \%$$

$$\mu = 395.4 \pm 3.7$$

$$\sigma = 30.6 \pm 3.5$$

WF

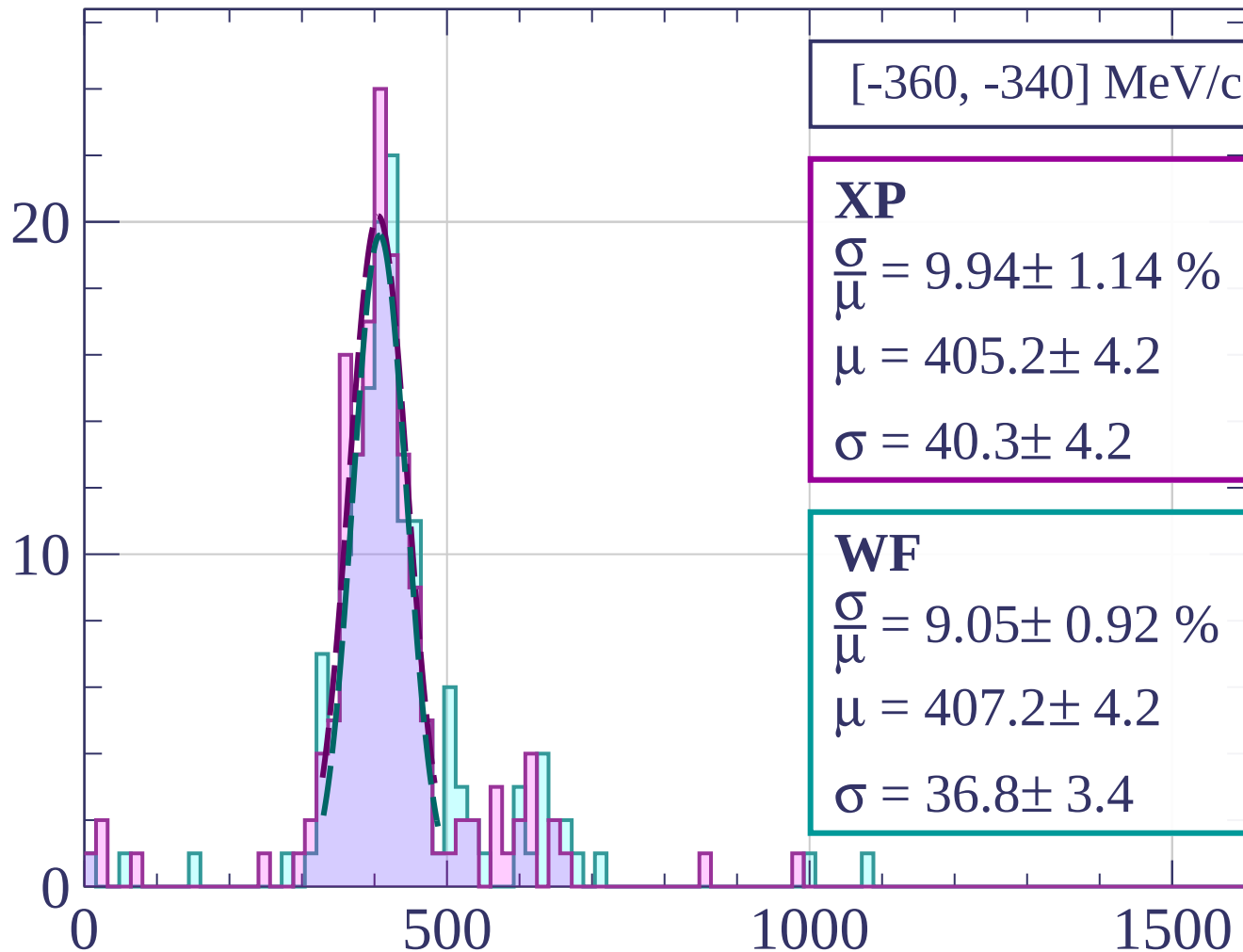
$$\frac{\sigma}{\mu} = 8.14 \pm 1.06 \%$$

$$\mu = 403.4 \pm 4.4$$

$$\sigma = 32.8 \pm 3.9$$

dE/dx [ADC counts/cm]

Count



[-360, -340] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.94 \pm 1.14 \%$$

$$\mu = 405.2 \pm 4.2$$

$$\sigma = 40.3 \pm 4.2$$

WF

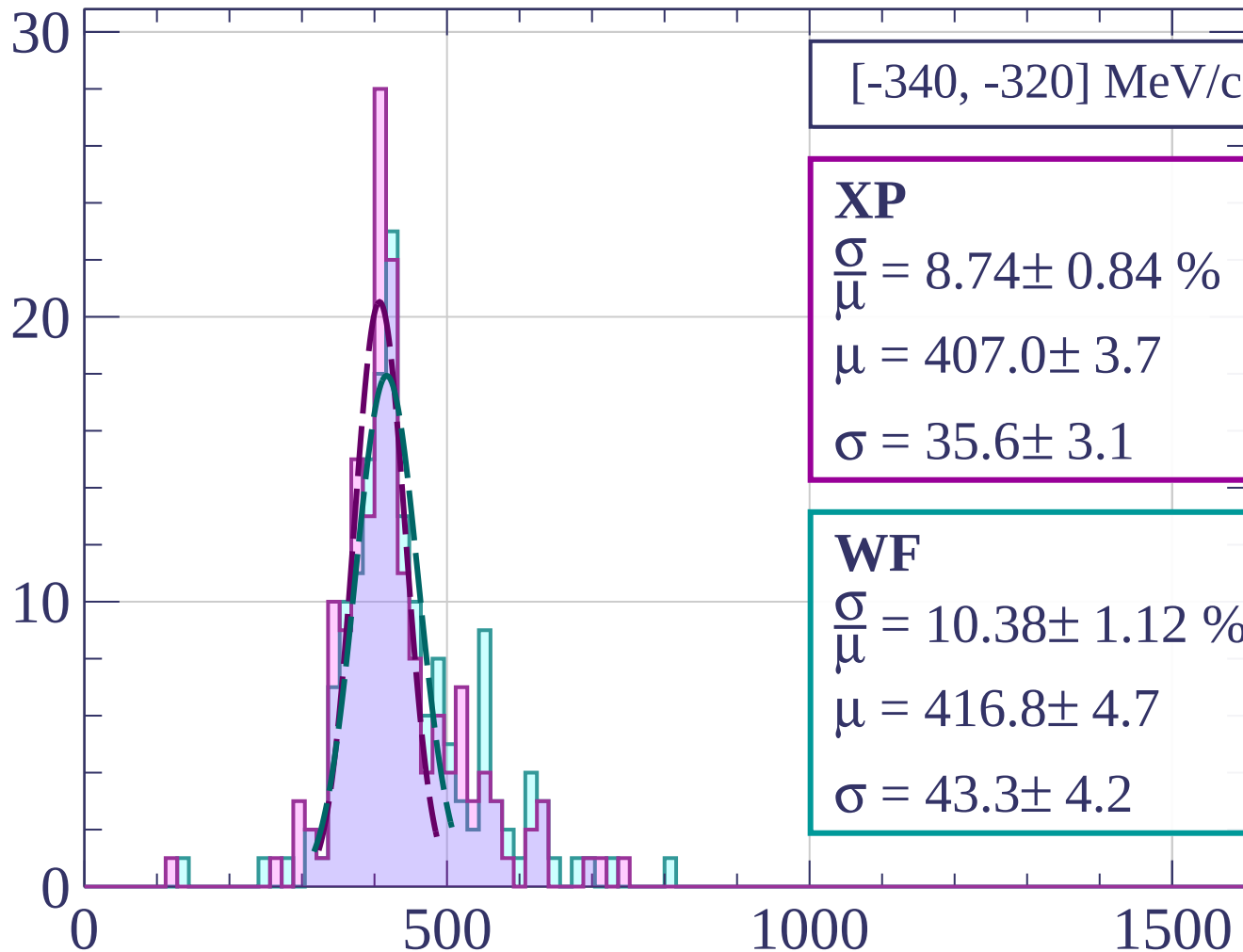
$$\frac{\sigma}{\mu} = 9.05 \pm 0.92 \%$$

$$\mu = 407.2 \pm 4.2$$

$$\sigma = 36.8 \pm 3.4$$

dE/dx [ADC counts/cm]

Count



[-340, -320] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.74 \pm 0.84 \%$$

$$\mu = 407.0 \pm 3.7$$

$$\sigma = 35.6 \pm 3.1$$

WF

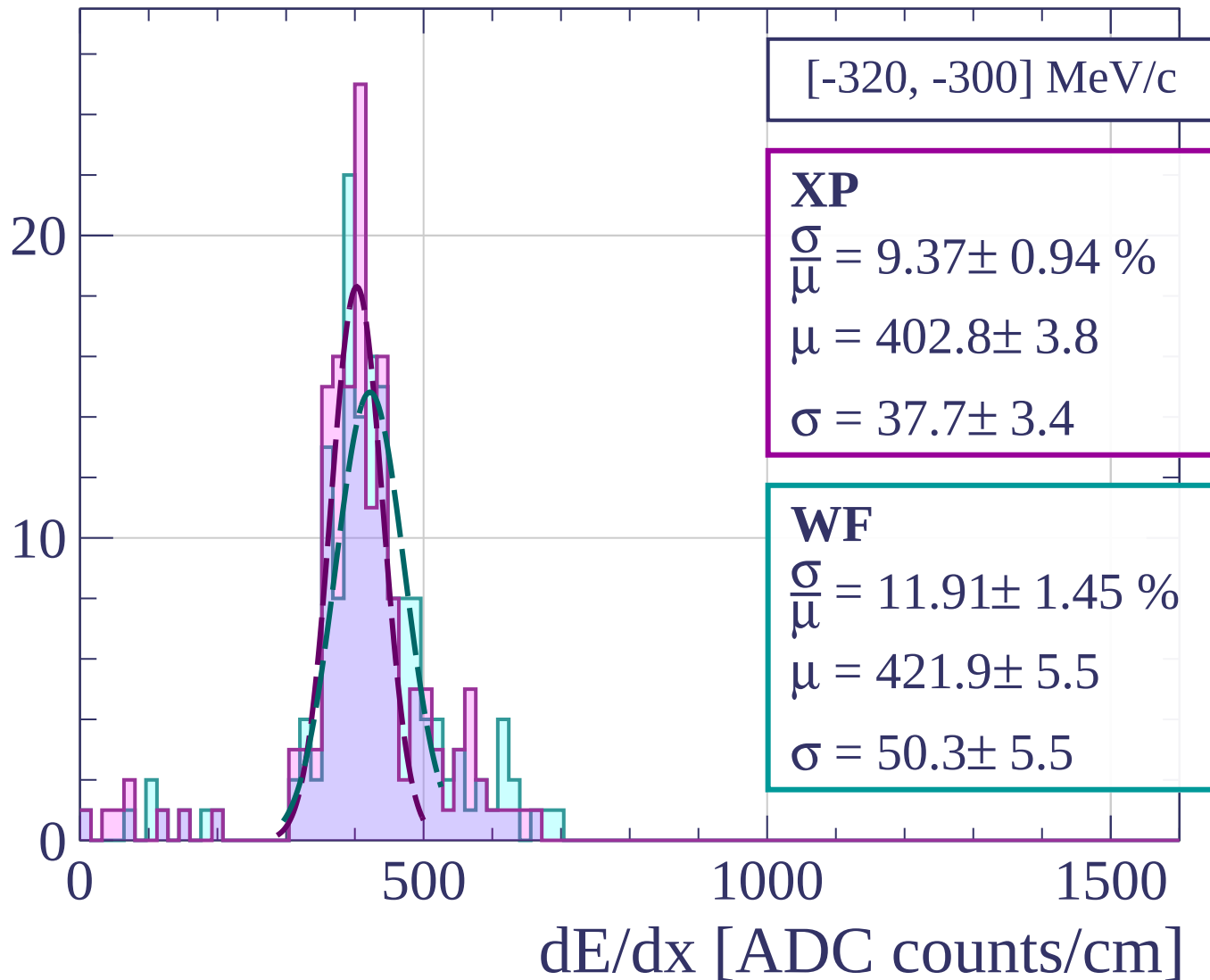
$$\frac{\sigma}{\mu} = 10.38 \pm 1.12 \%$$

$$\mu = 416.8 \pm 4.7$$

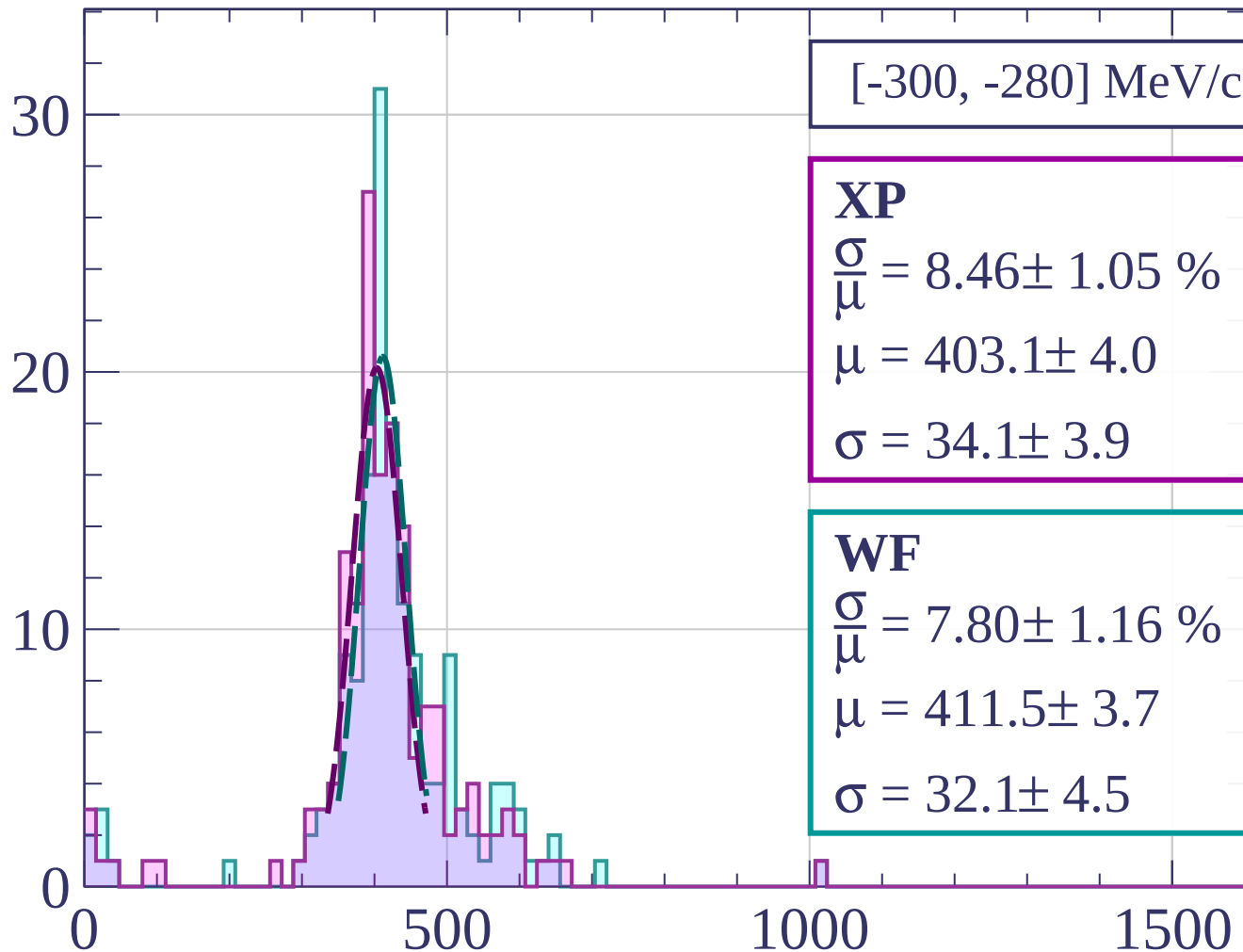
$$\sigma = 43.3 \pm 4.2$$

dE/dx [ADC counts/cm]

Count



Count



[-300, -280] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.46 \pm 1.05 \%$$

$$\mu = 403.1 \pm 4.0$$

$$\sigma = 34.1 \pm 3.9$$

WF

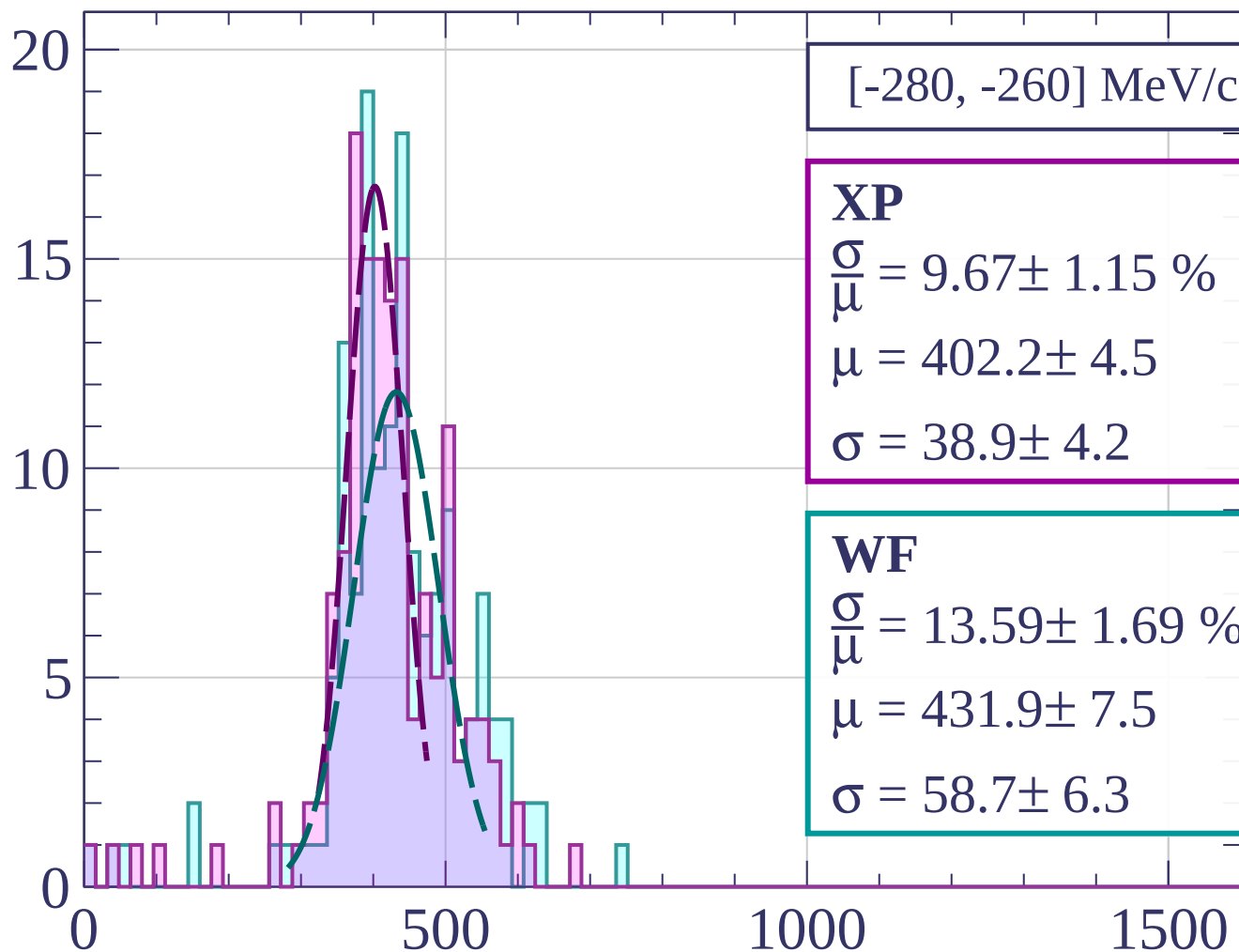
$$\frac{\sigma}{\mu} = 7.80 \pm 1.16 \%$$

$$\mu = 411.5 \pm 3.7$$

$$\sigma = 32.1 \pm 4.5$$

dE/dx [ADC counts/cm]

Count



[-280, -260] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.67 \pm 1.15 \%$$

$$\mu = 402.2 \pm 4.5$$

$$\sigma = 38.9 \pm 4.2$$

WF

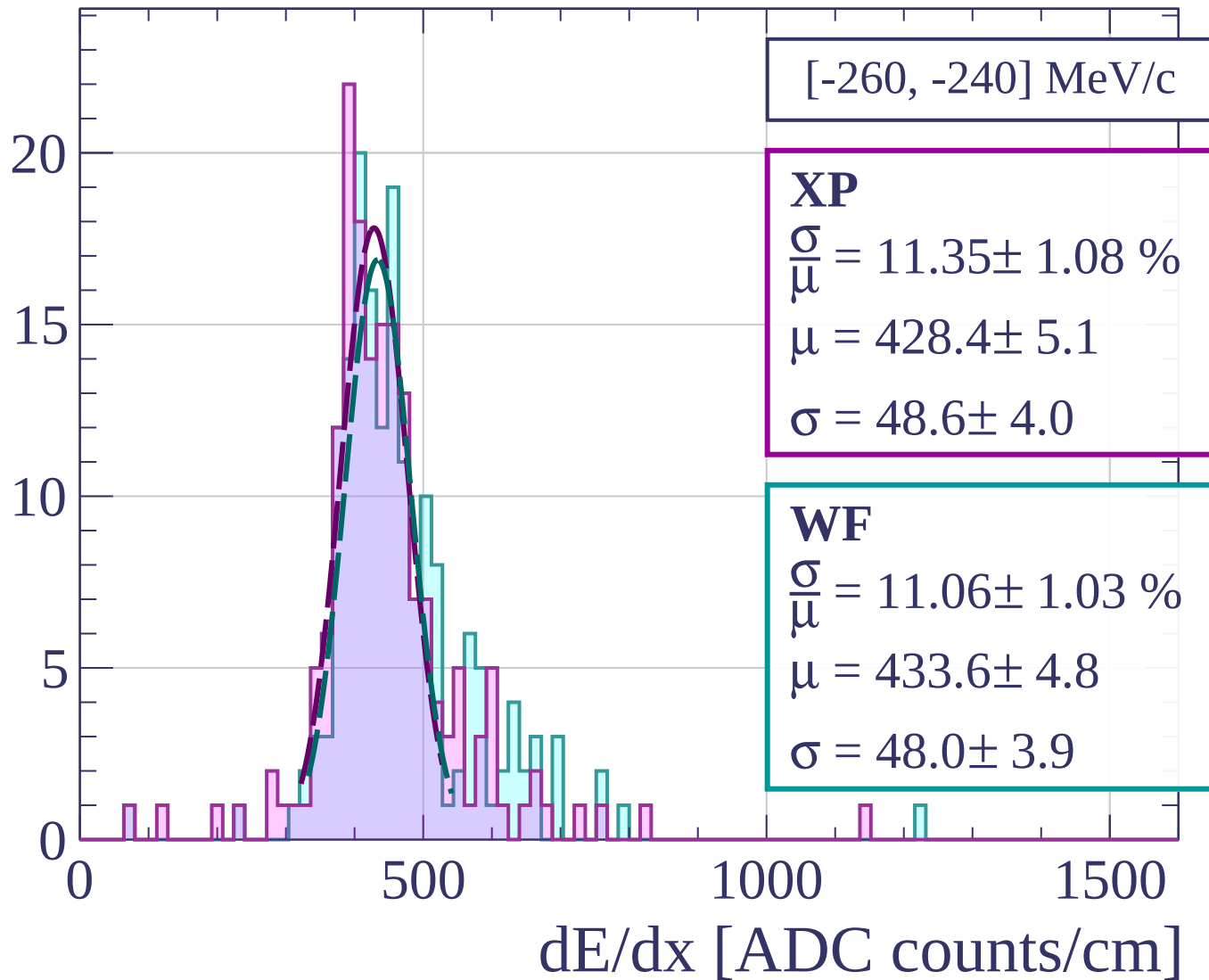
$$\frac{\sigma}{\mu} = 13.59 \pm 1.69 \%$$

$$\mu = 431.9 \pm 7.5$$

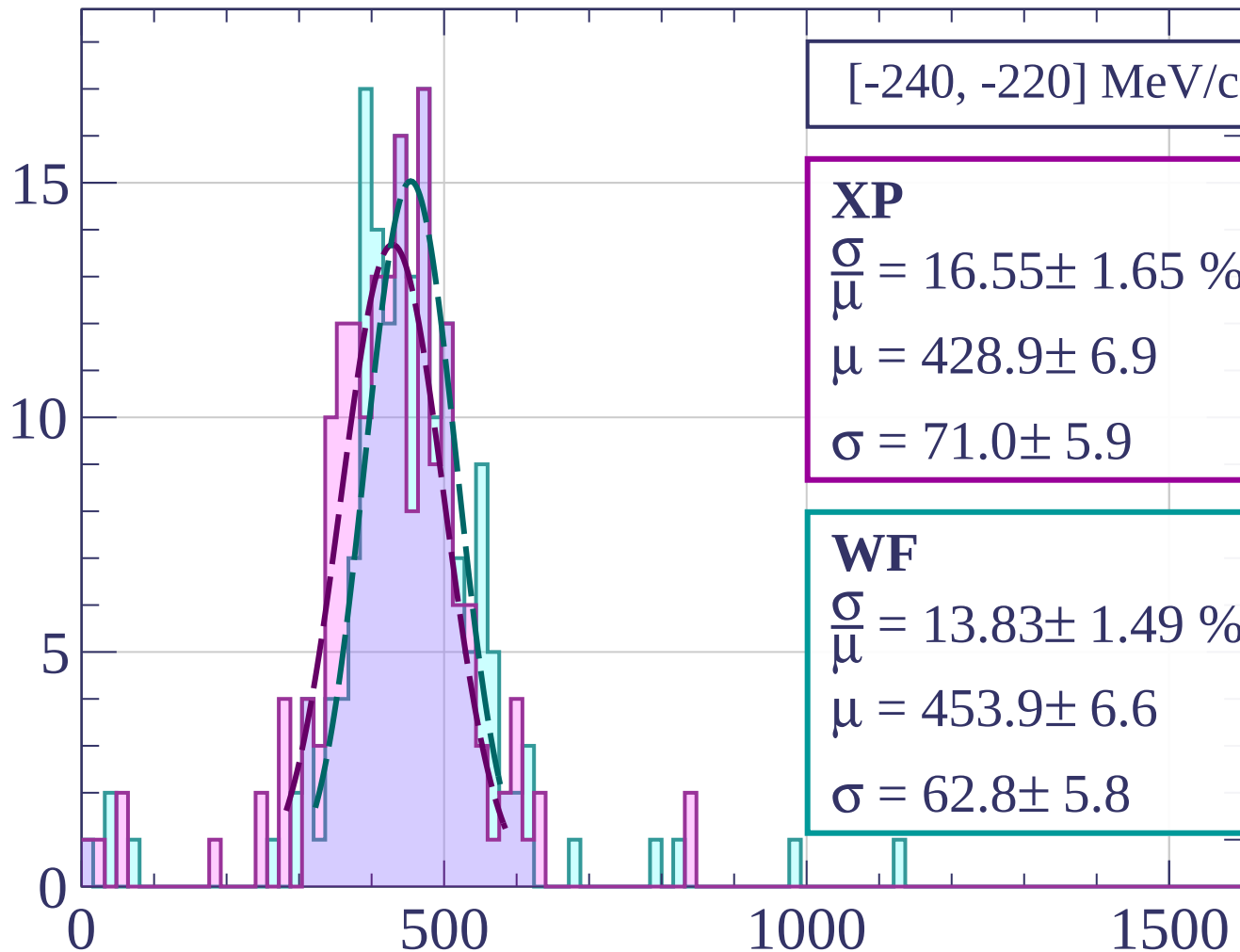
$$\sigma = 58.7 \pm 6.3$$

dE/dx [ADC counts/cm]

Count



Count



[-240, -220] MeV/c

XP

$$\frac{\sigma}{\mu} = 16.55 \pm 1.65 \%$$

$$\mu = 428.9 \pm 6.9$$

$$\sigma = 71.0 \pm 5.9$$

WF

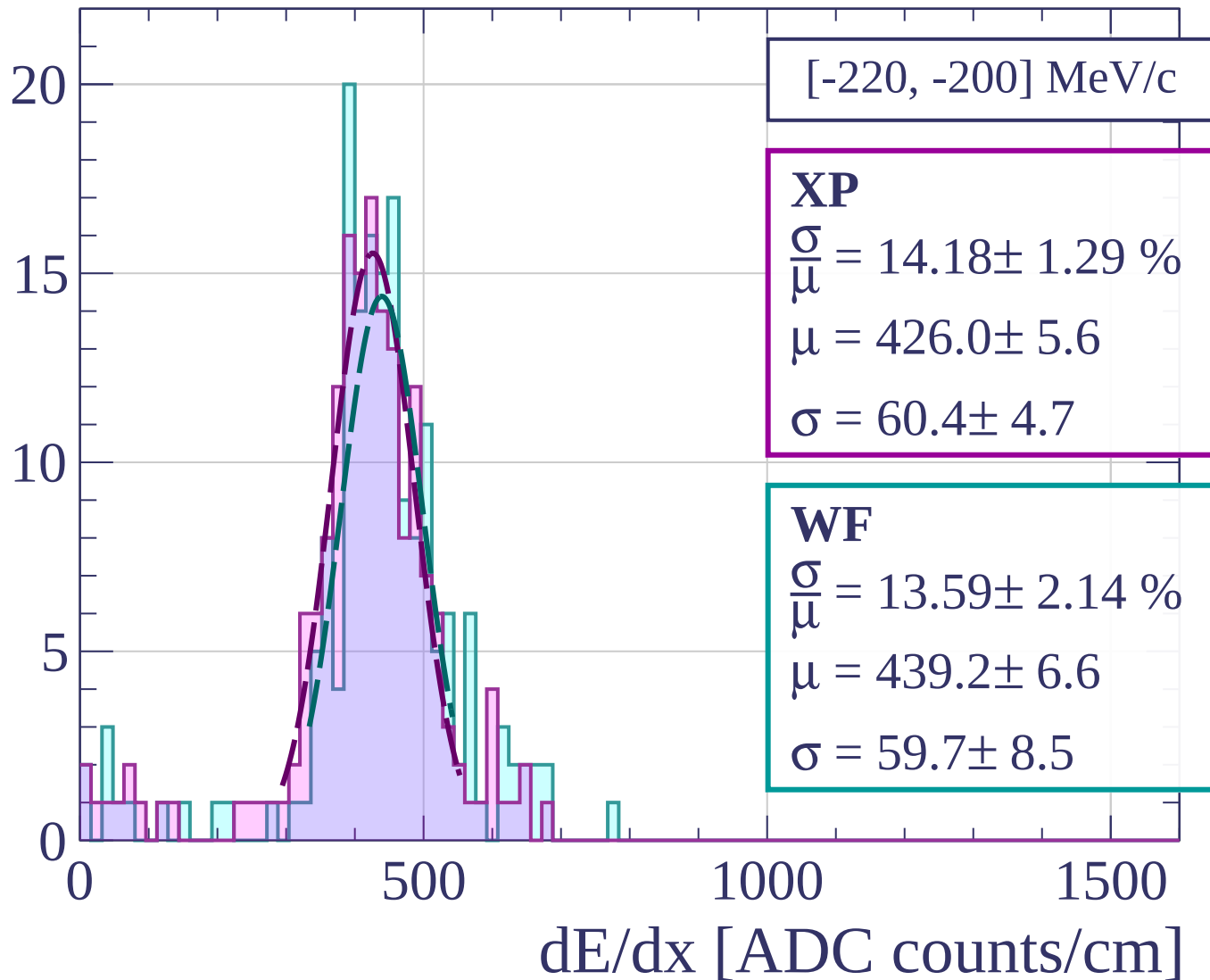
$$\frac{\sigma}{\mu} = 13.83 \pm 1.49 \%$$

$$\mu = 453.9 \pm 6.6$$

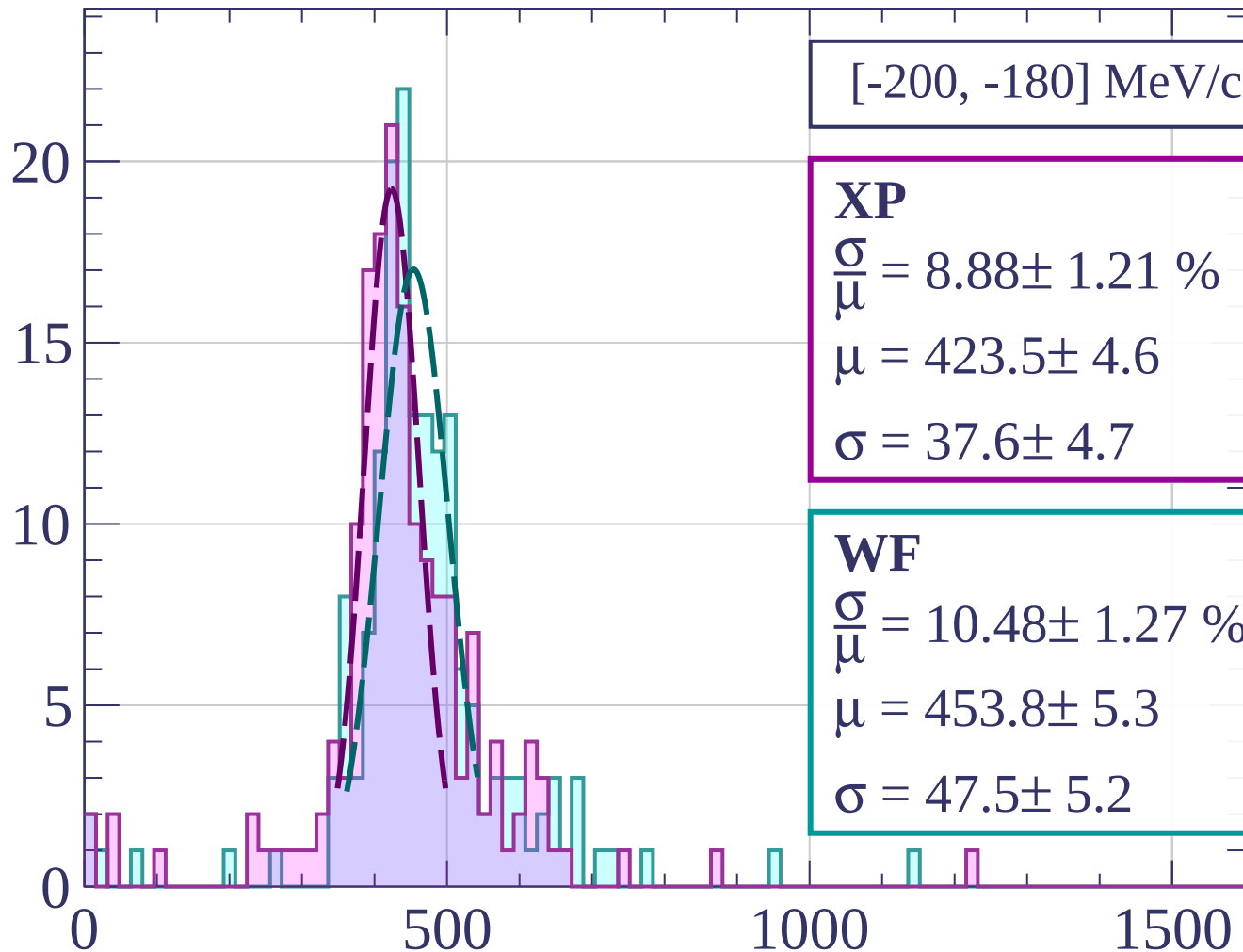
$$\sigma = 62.8 \pm 5.8$$

dE/dx [ADC counts/cm]

Count



Count



[-200, -180] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.88 \pm 1.21 \%$$

$$\mu = 423.5 \pm 4.6$$

$$\sigma = 37.6 \pm 4.7$$

WF

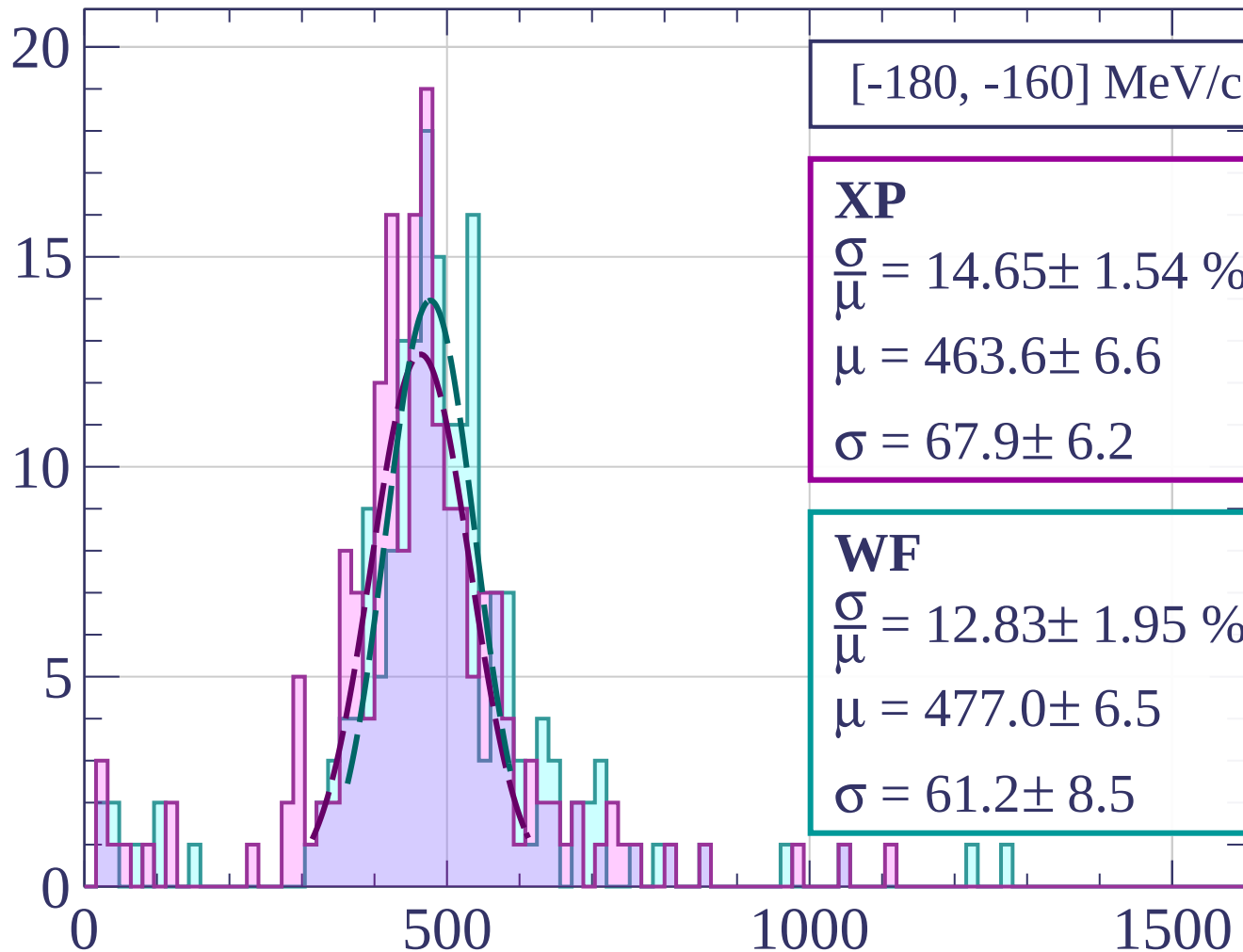
$$\frac{\sigma}{\mu} = 10.48 \pm 1.27 \%$$

$$\mu = 453.8 \pm 5.3$$

$$\sigma = 47.5 \pm 5.2$$

dE/dx [ADC counts/cm]

Count



[-180, -160] MeV/c

XP

$$\frac{\sigma}{\mu} = 14.65 \pm 1.54 \%$$

$$\mu = 463.6 \pm 6.6$$

$$\sigma = 67.9 \pm 6.2$$

WF

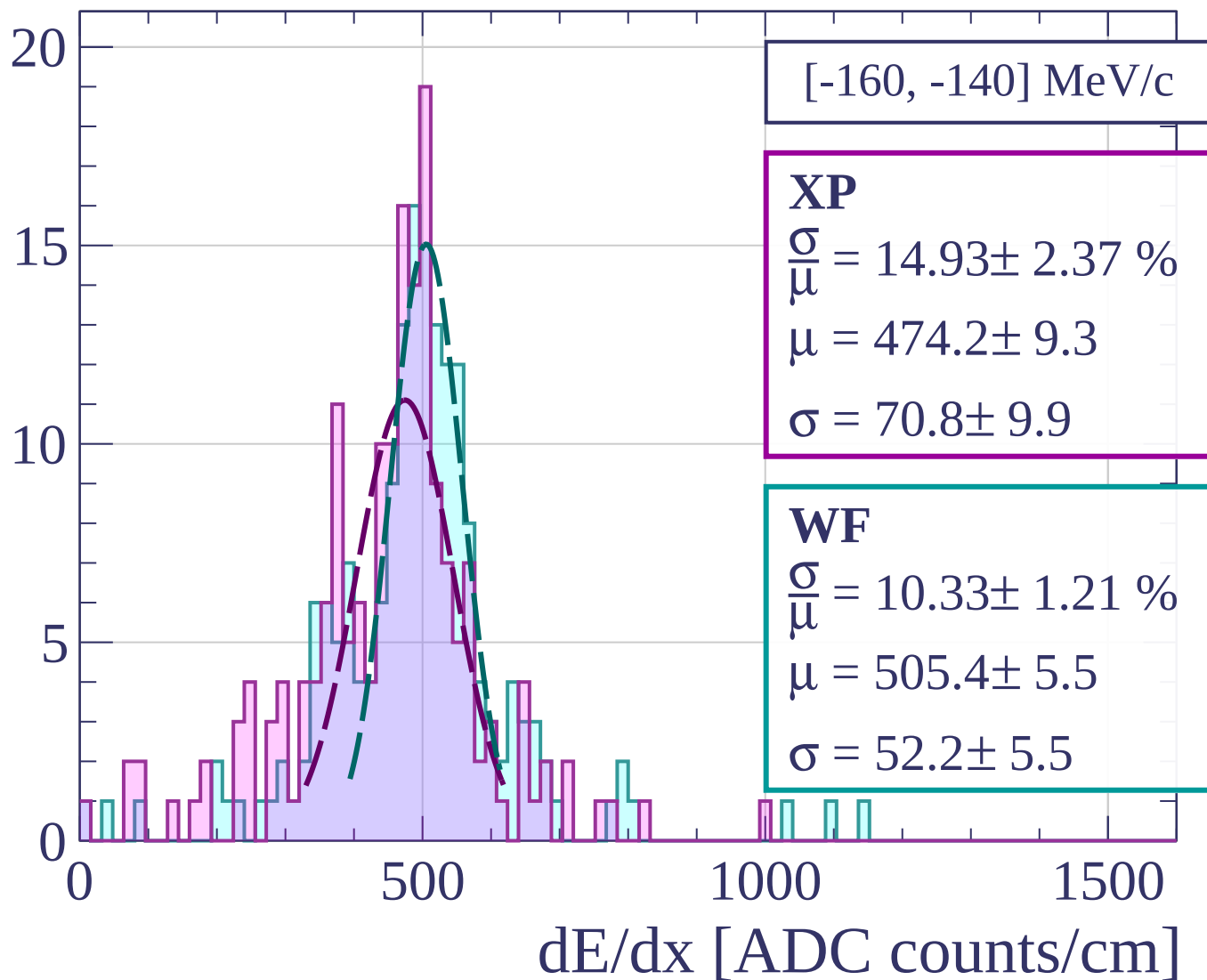
$$\frac{\sigma}{\mu} = 12.83 \pm 1.95 \%$$

$$\mu = 477.0 \pm 6.5$$

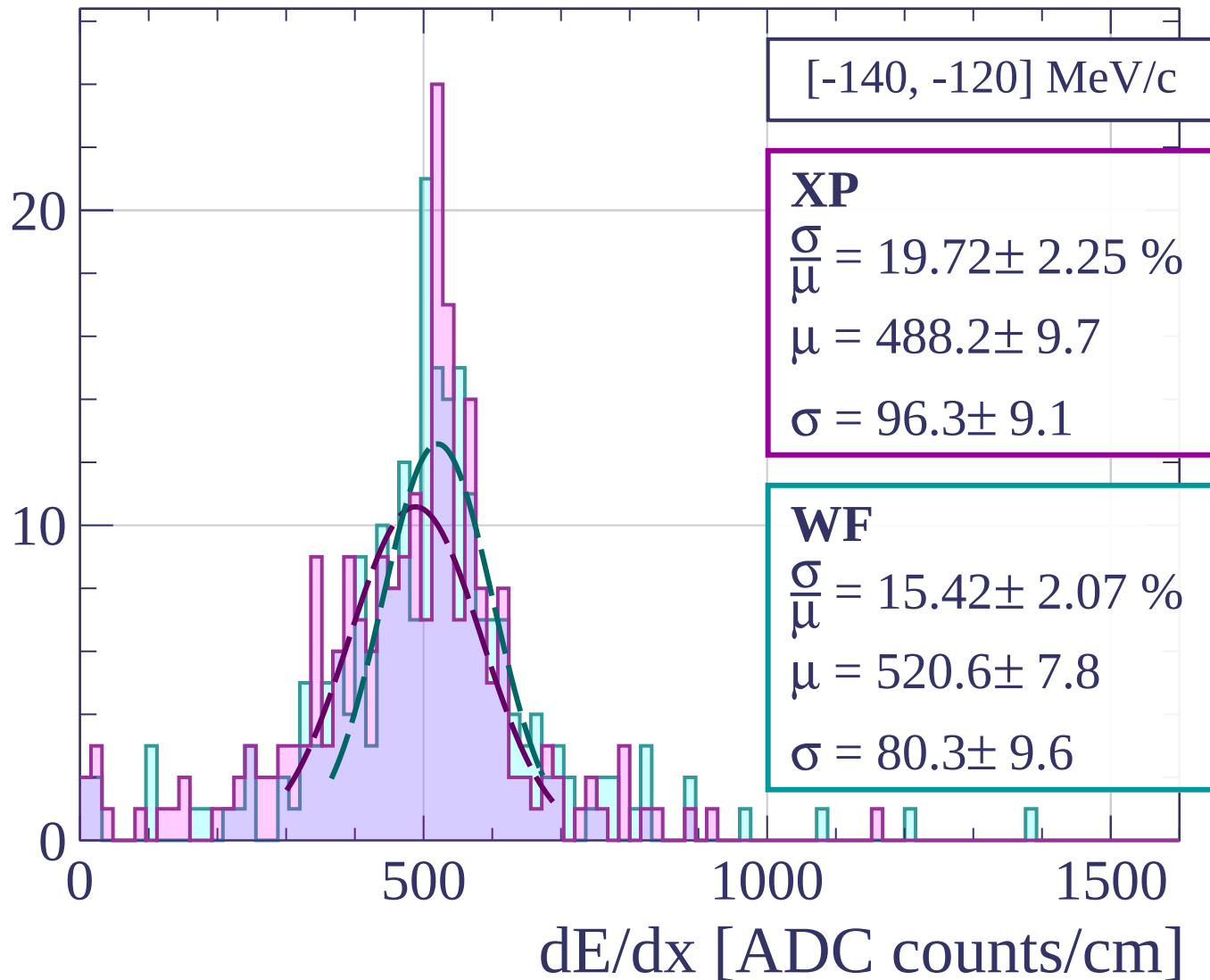
$$\sigma = 61.2 \pm 8.5$$

dE/dx [ADC counts/cm]

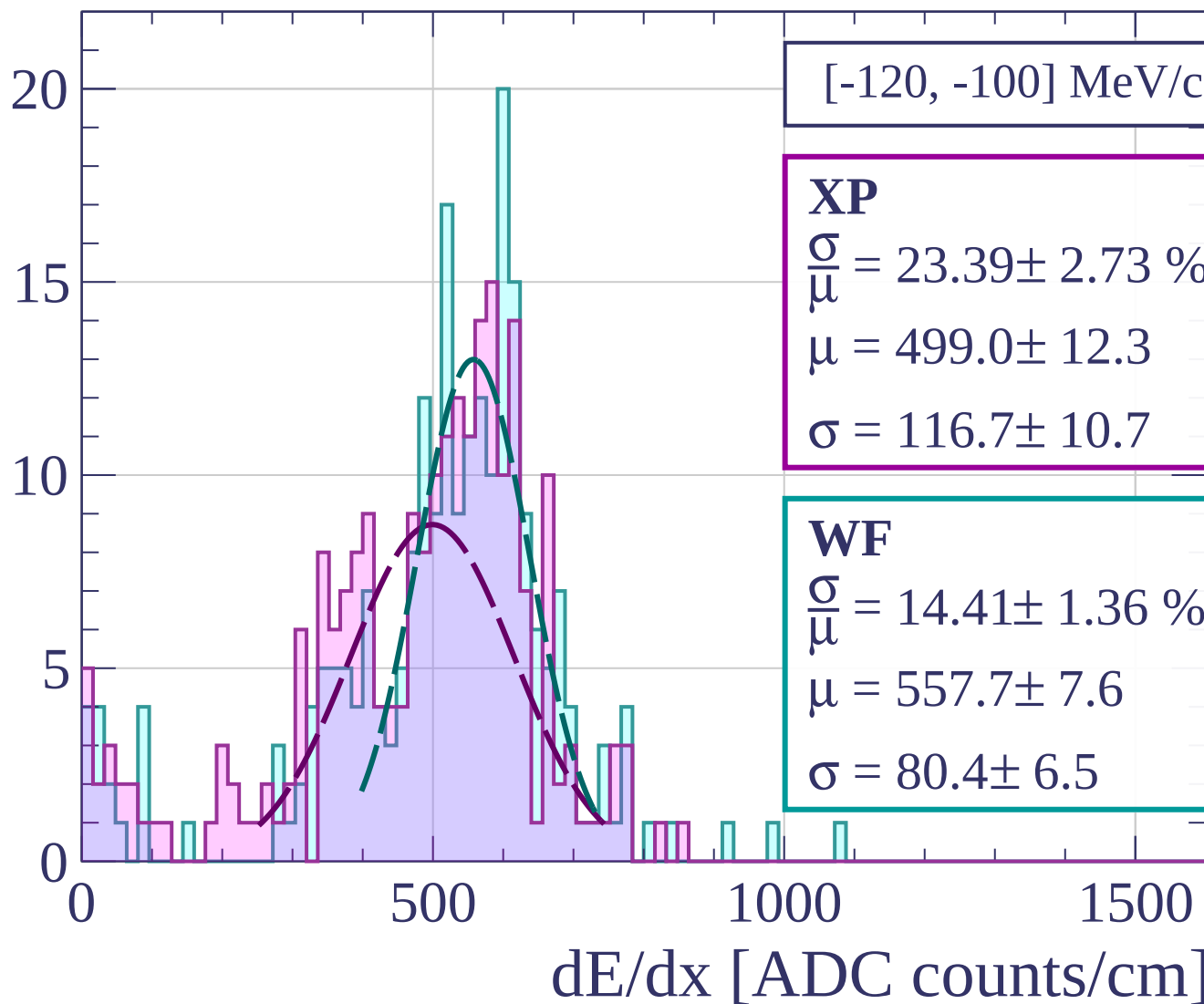
Count



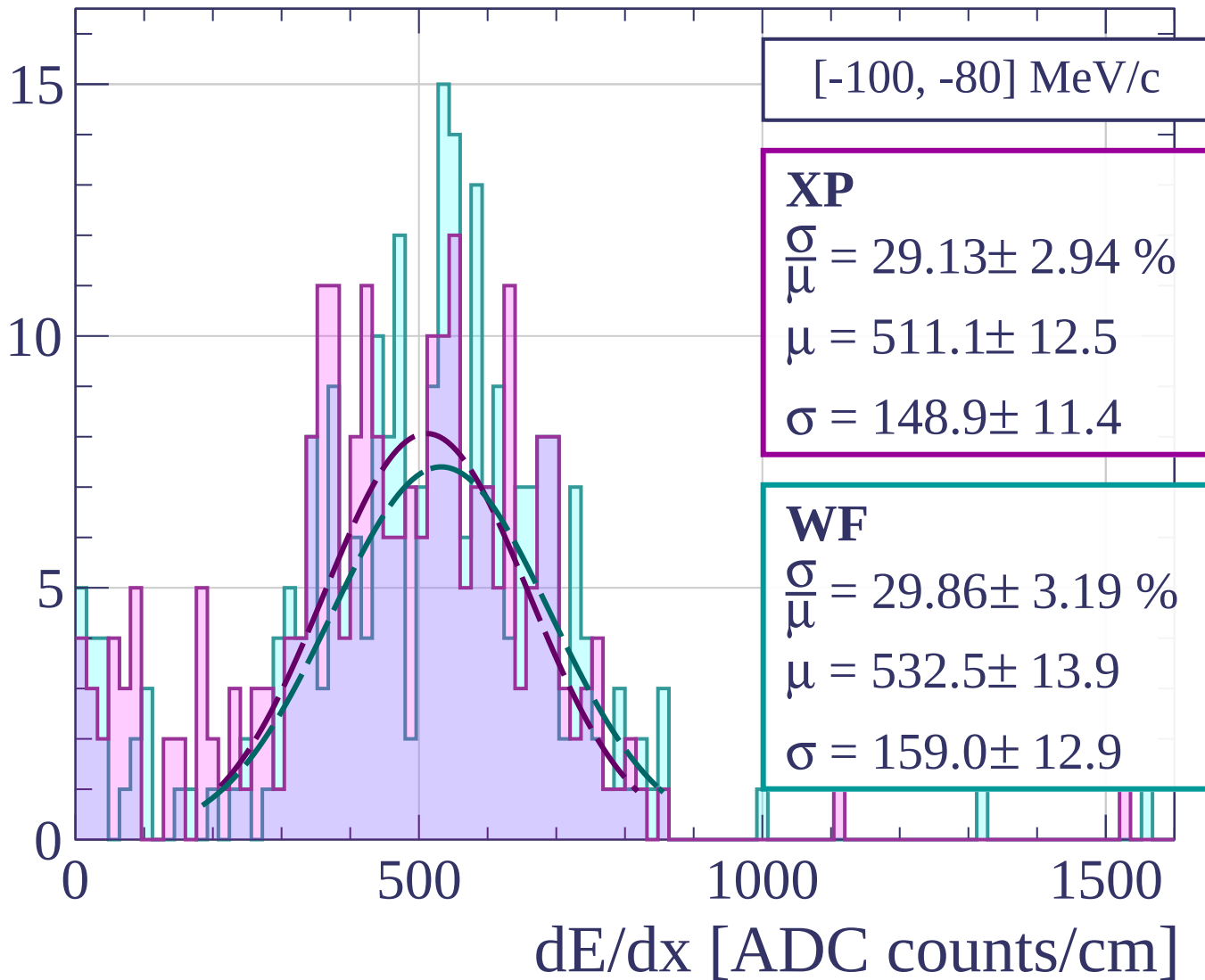
Count



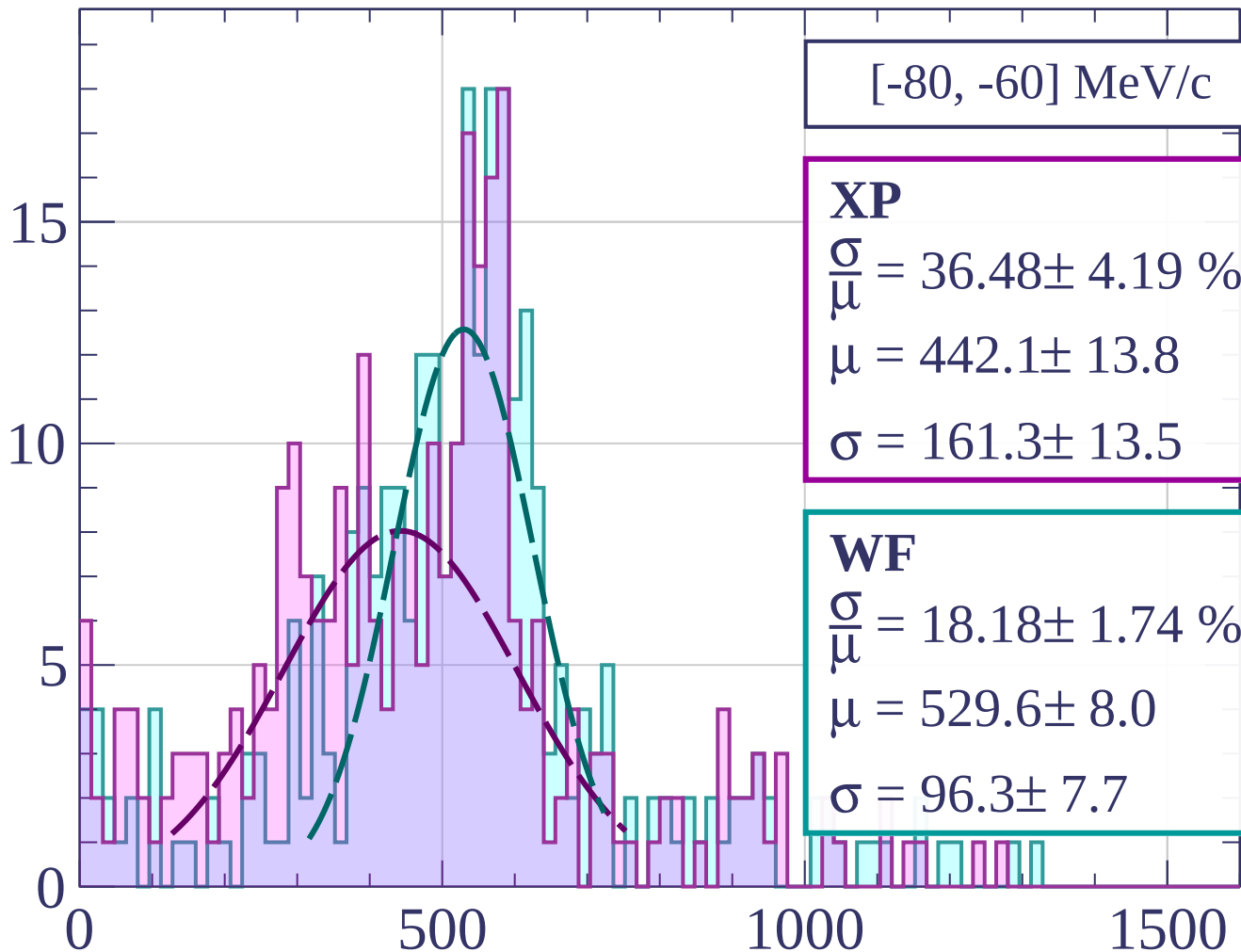
Count



Count



Count



[-80, -60] MeV/c

XP

$$\frac{\sigma}{\mu} = 36.48 \pm 4.19 \%$$

$$\mu = 442.1 \pm 13.8$$

$$\sigma = 161.3 \pm 13.5$$

WF

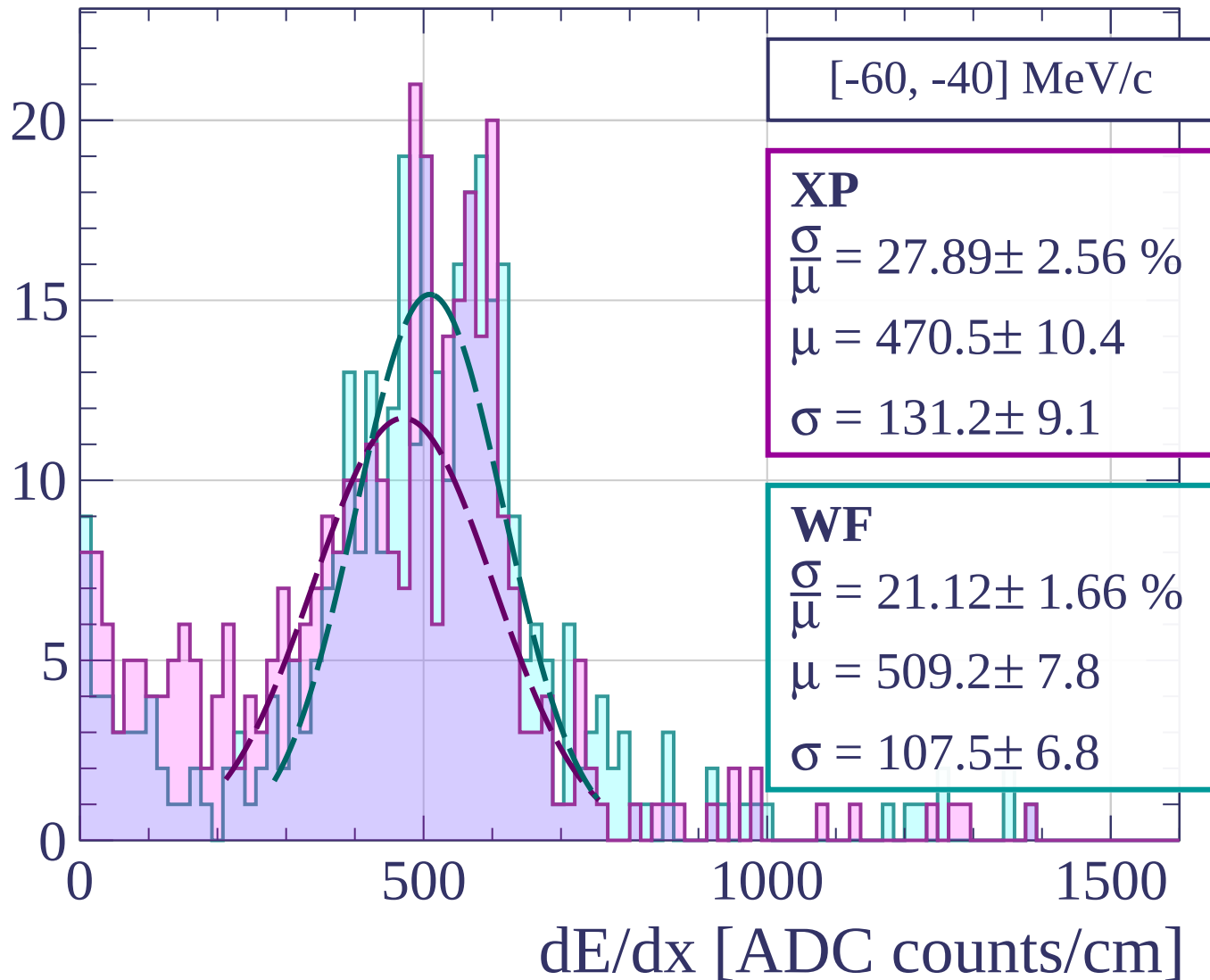
$$\frac{\sigma}{\mu} = 18.18 \pm 1.74 \%$$

$$\mu = 529.6 \pm 8.0$$

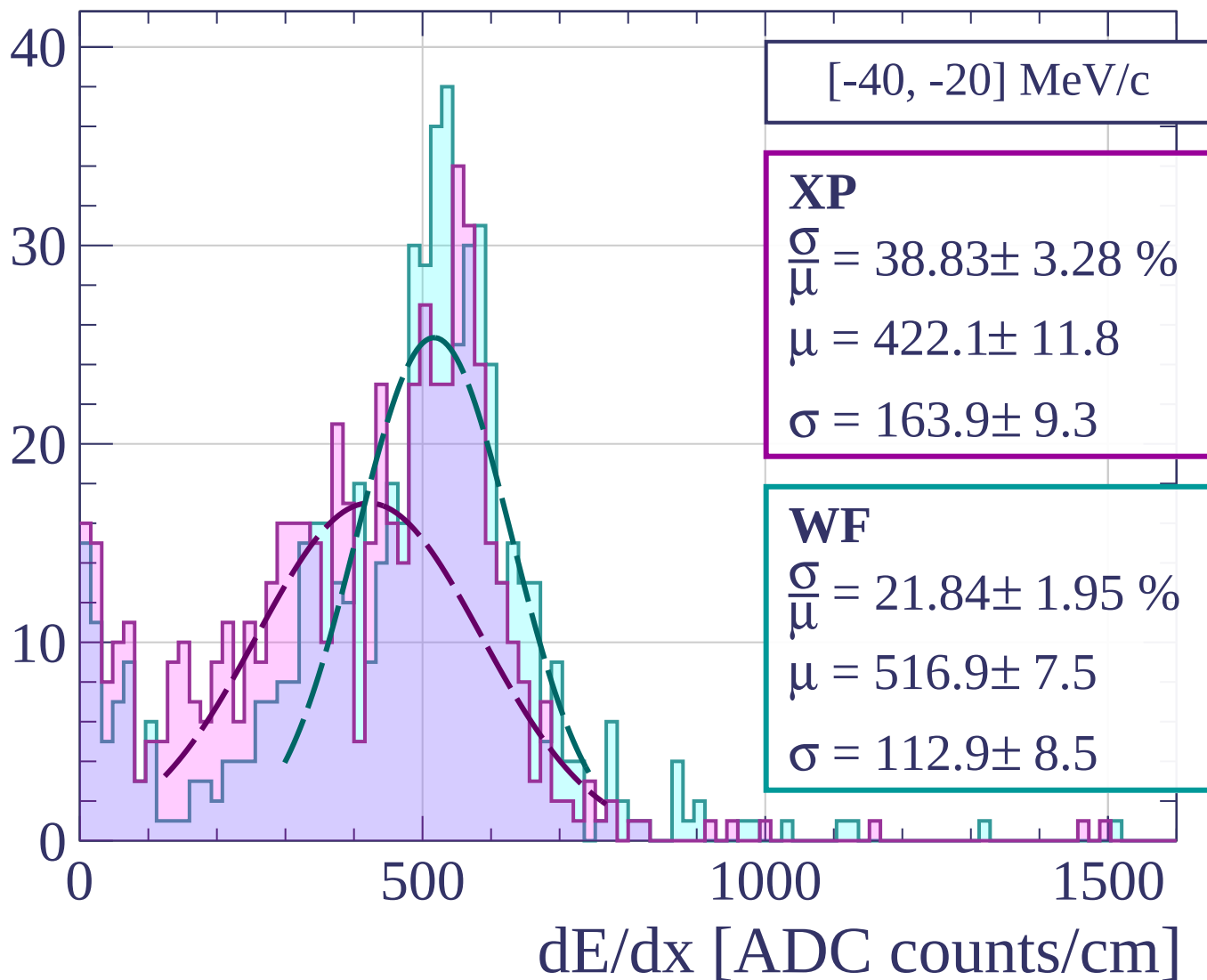
$$\sigma = 96.3 \pm 7.7$$

dE/dx [ADC counts/cm]

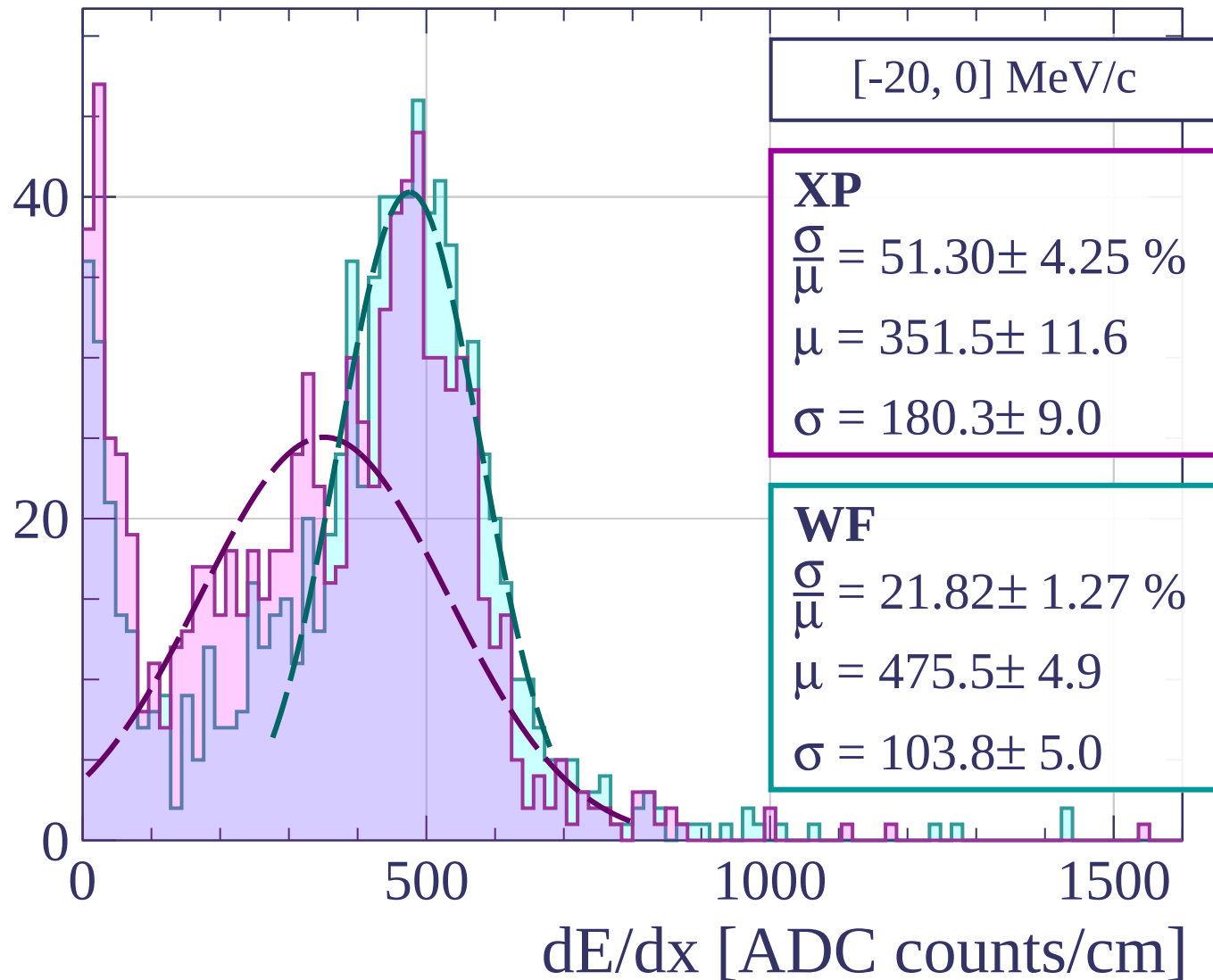
Count



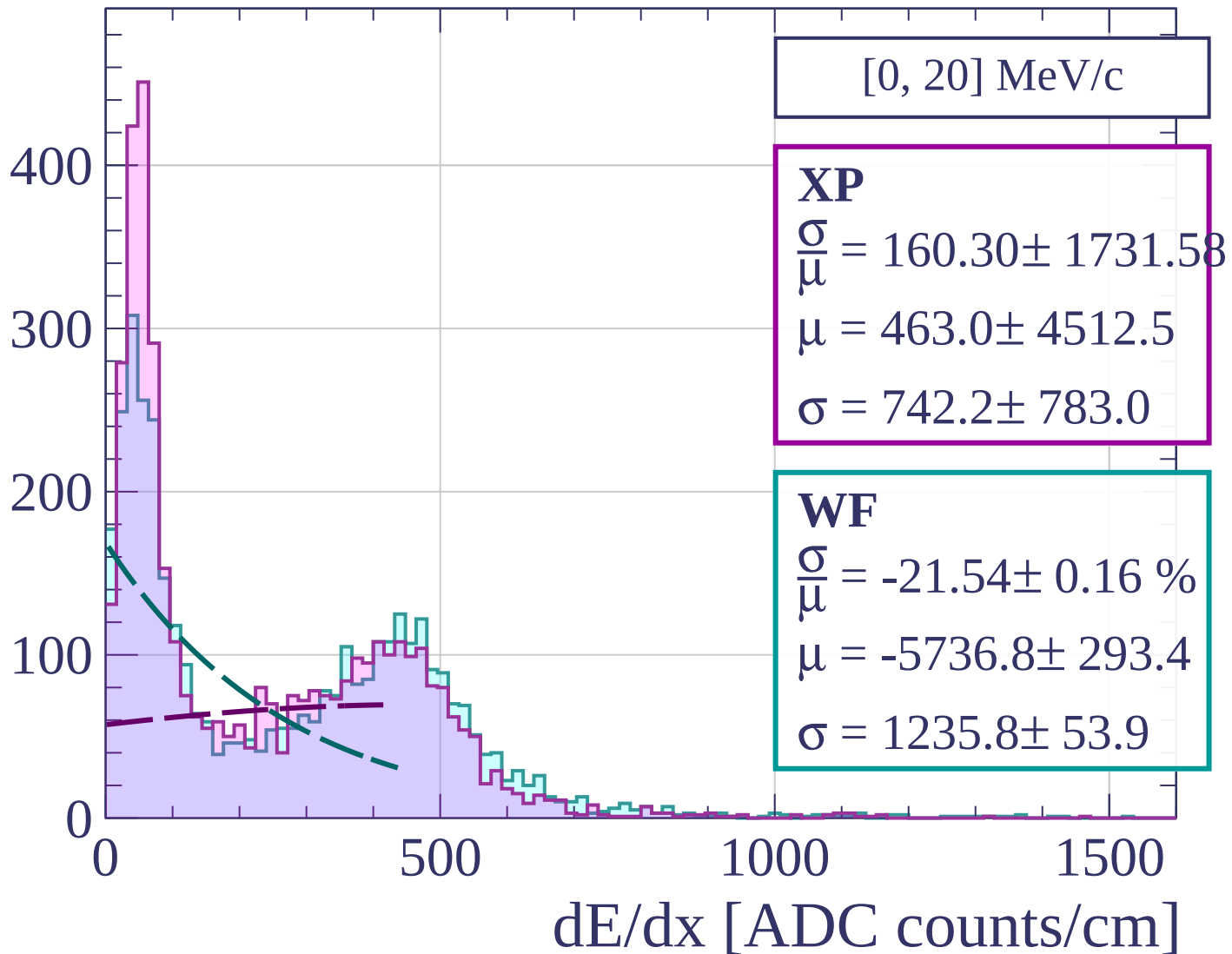
Count

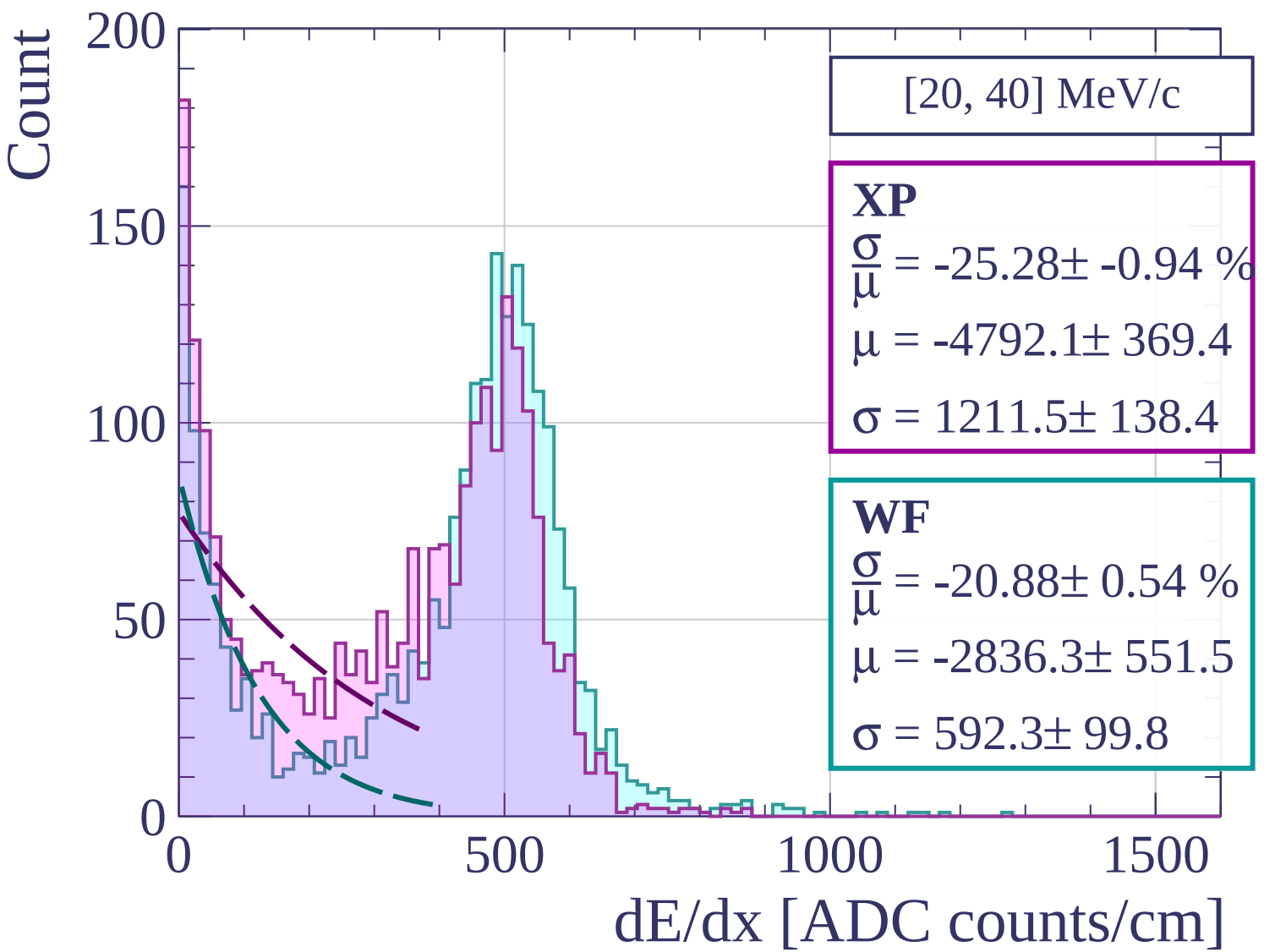


Count

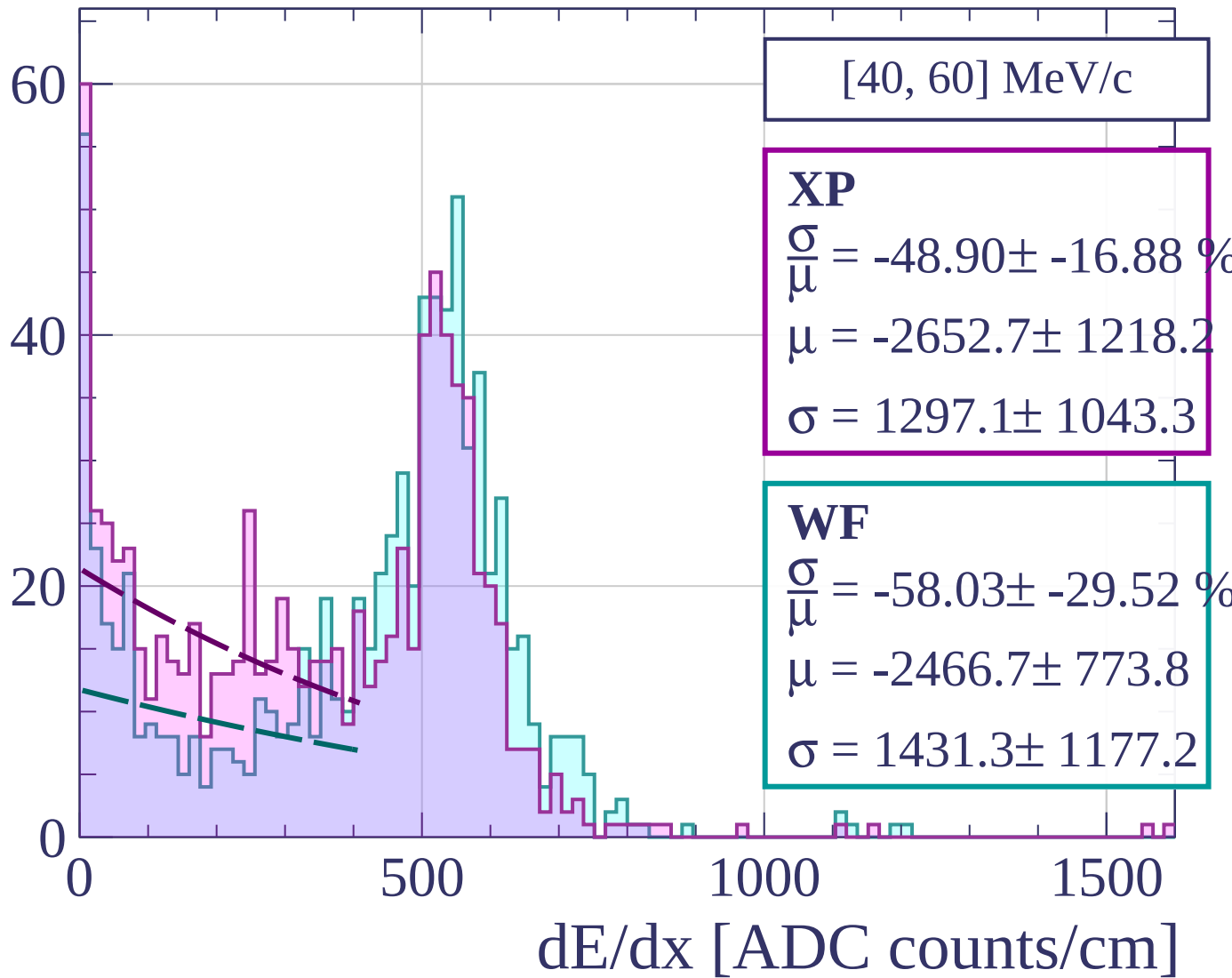


Count

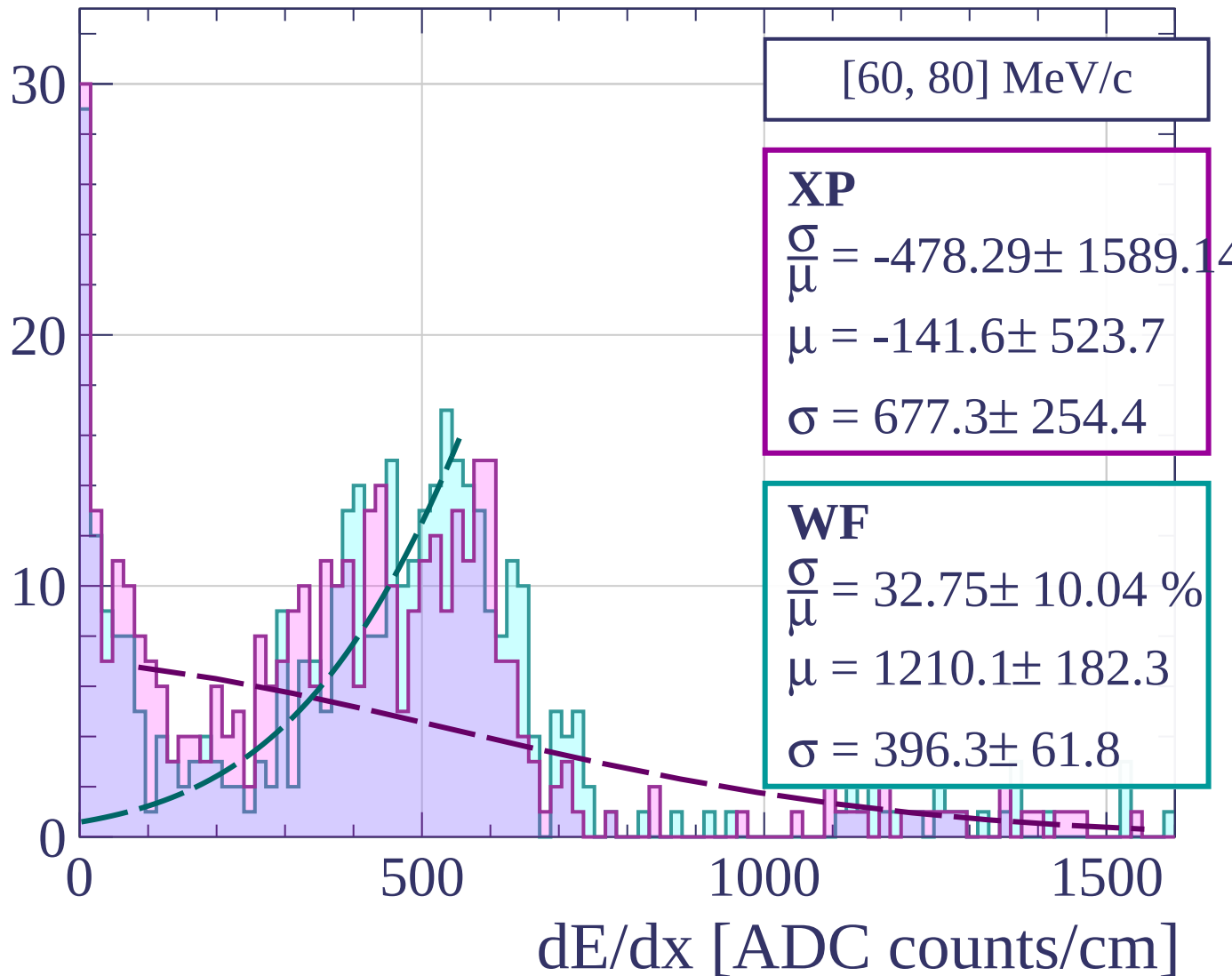




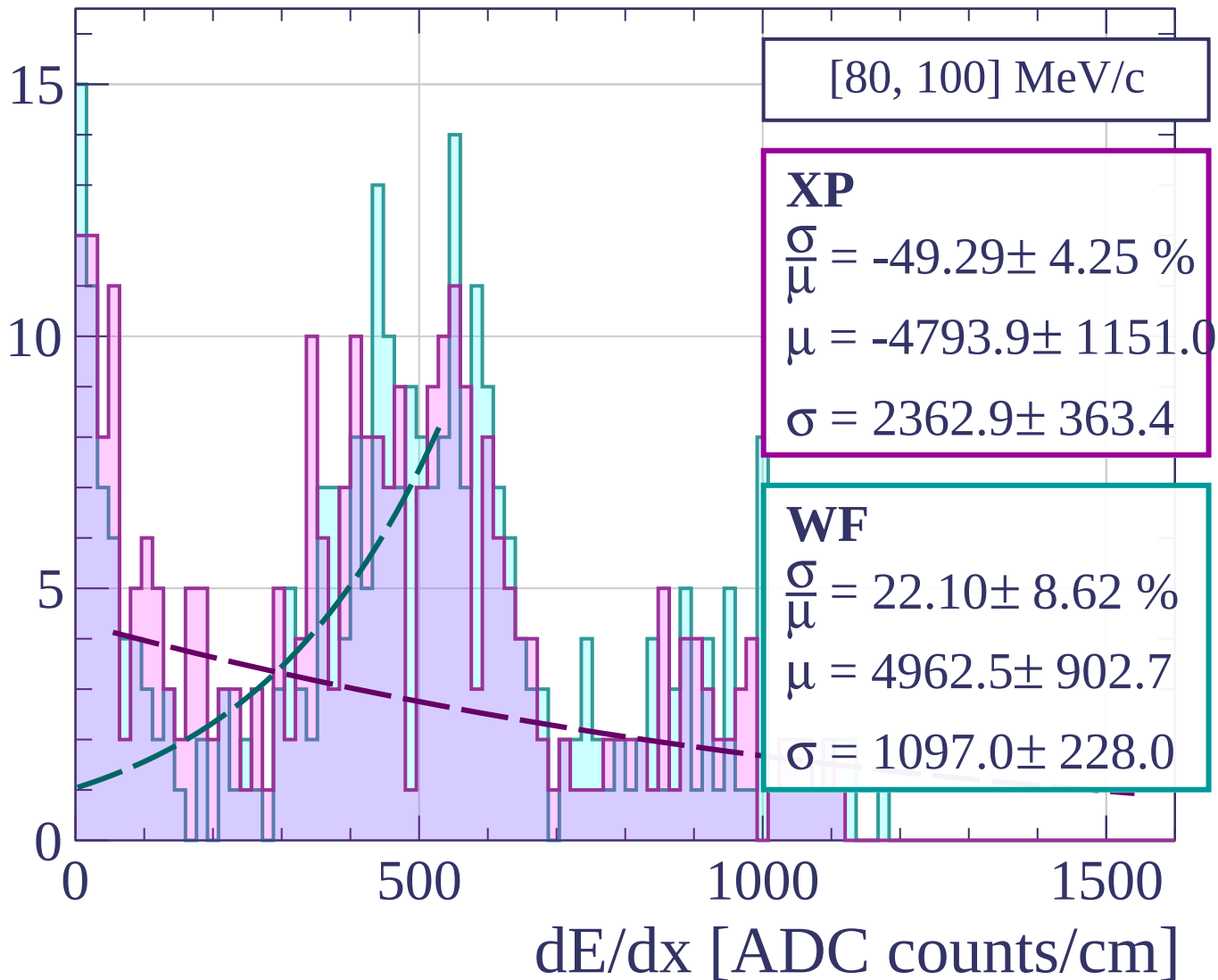
Count



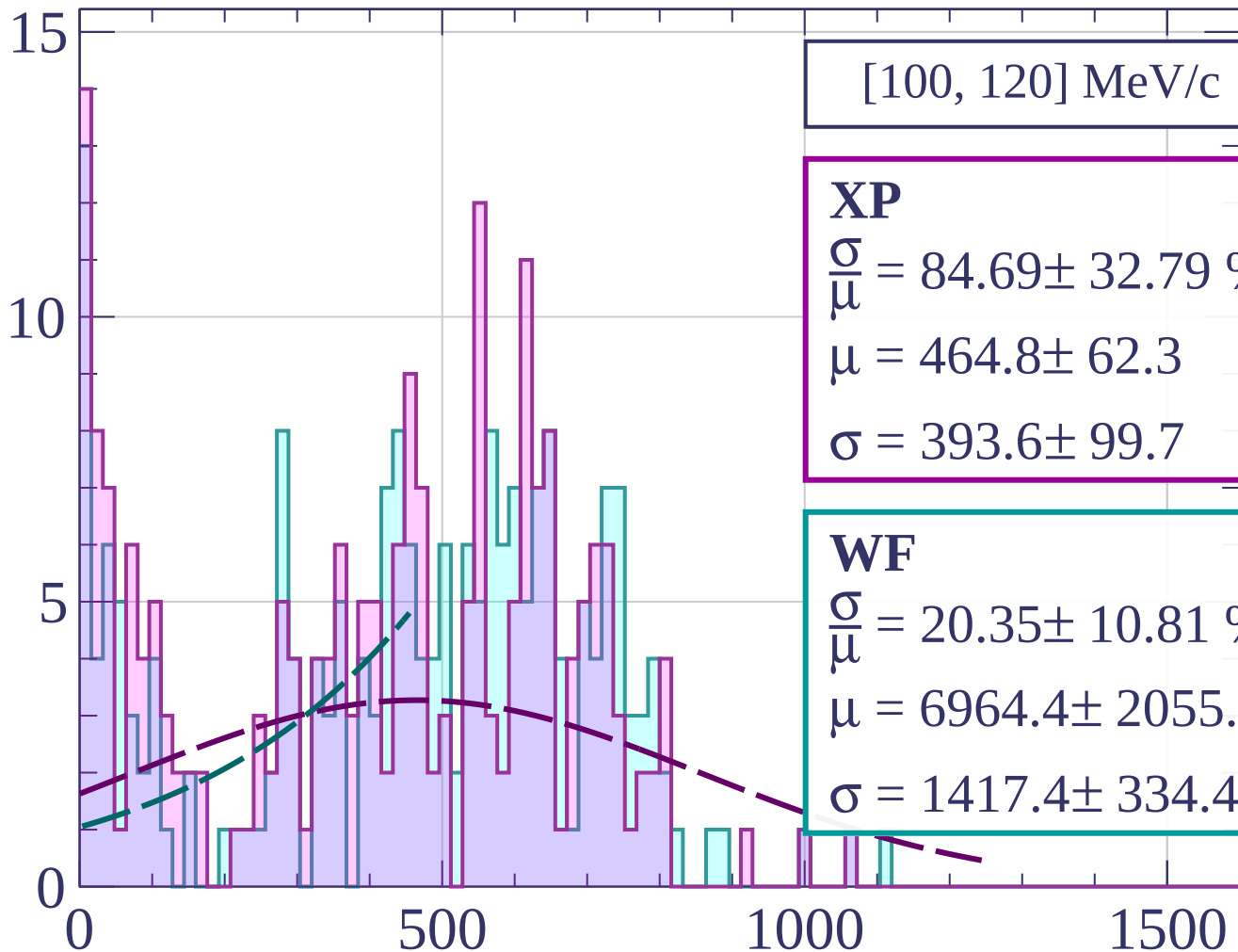
Count



Count



Count



[100, 120] MeV/c

XP

$$\frac{\sigma}{\mu} = 84.69 \pm 32.79 \%$$

$$\mu = 464.8 \pm 62.3$$

$$\sigma = 393.6 \pm 99.7$$

WF

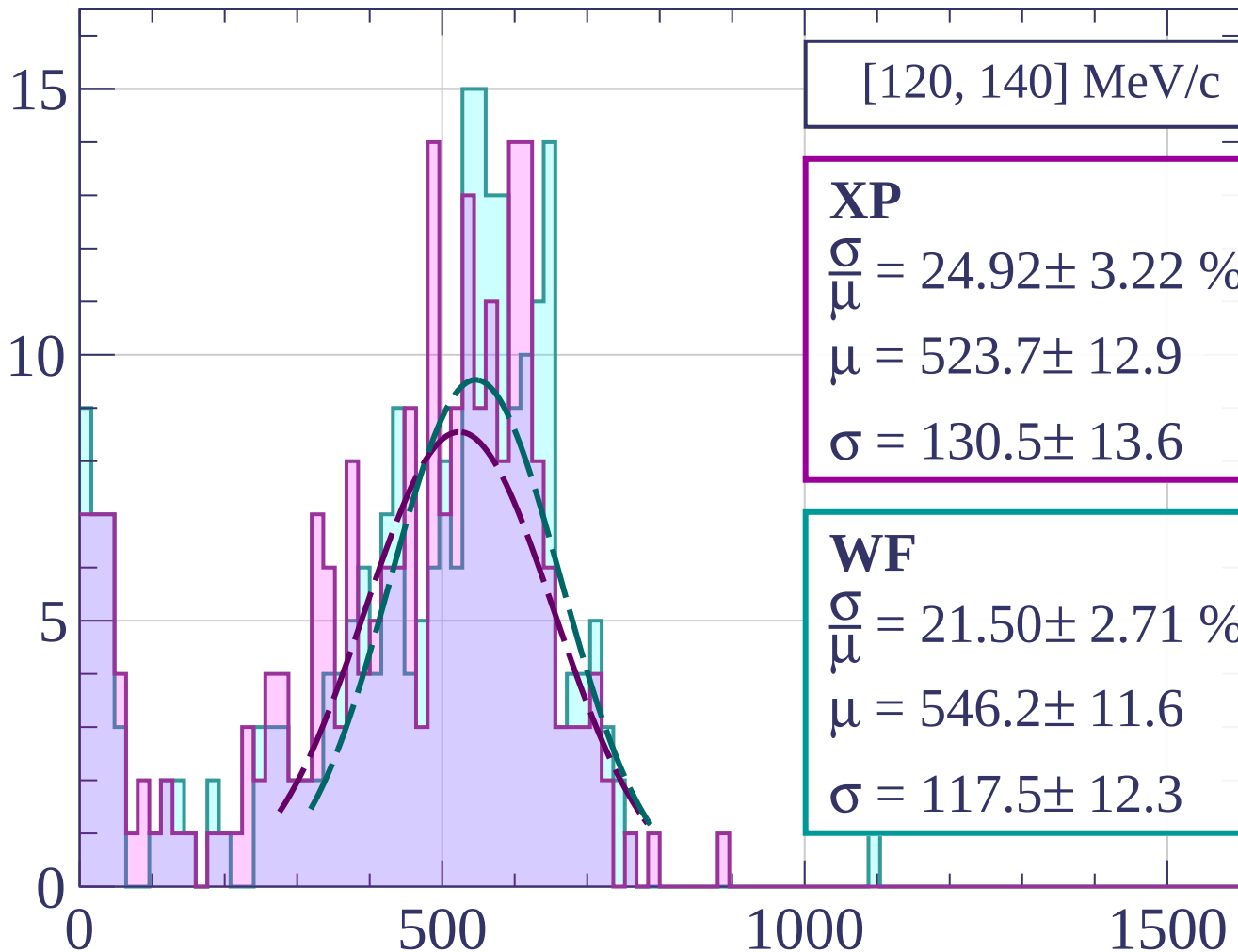
$$\frac{\sigma}{\mu} = 20.35 \pm 10.81 \%$$

$$\mu = 6964.4 \pm 2055.7$$

$$\sigma = 1417.4 \pm 334.4$$

dE/dx [ADC counts/cm]

Count



[120, 140] MeV/c

XP

$$\frac{\sigma}{\mu} = 24.92 \pm 3.22 \%$$

$$\mu = 523.7 \pm 12.9$$

$$\sigma = 130.5 \pm 13.6$$

WF

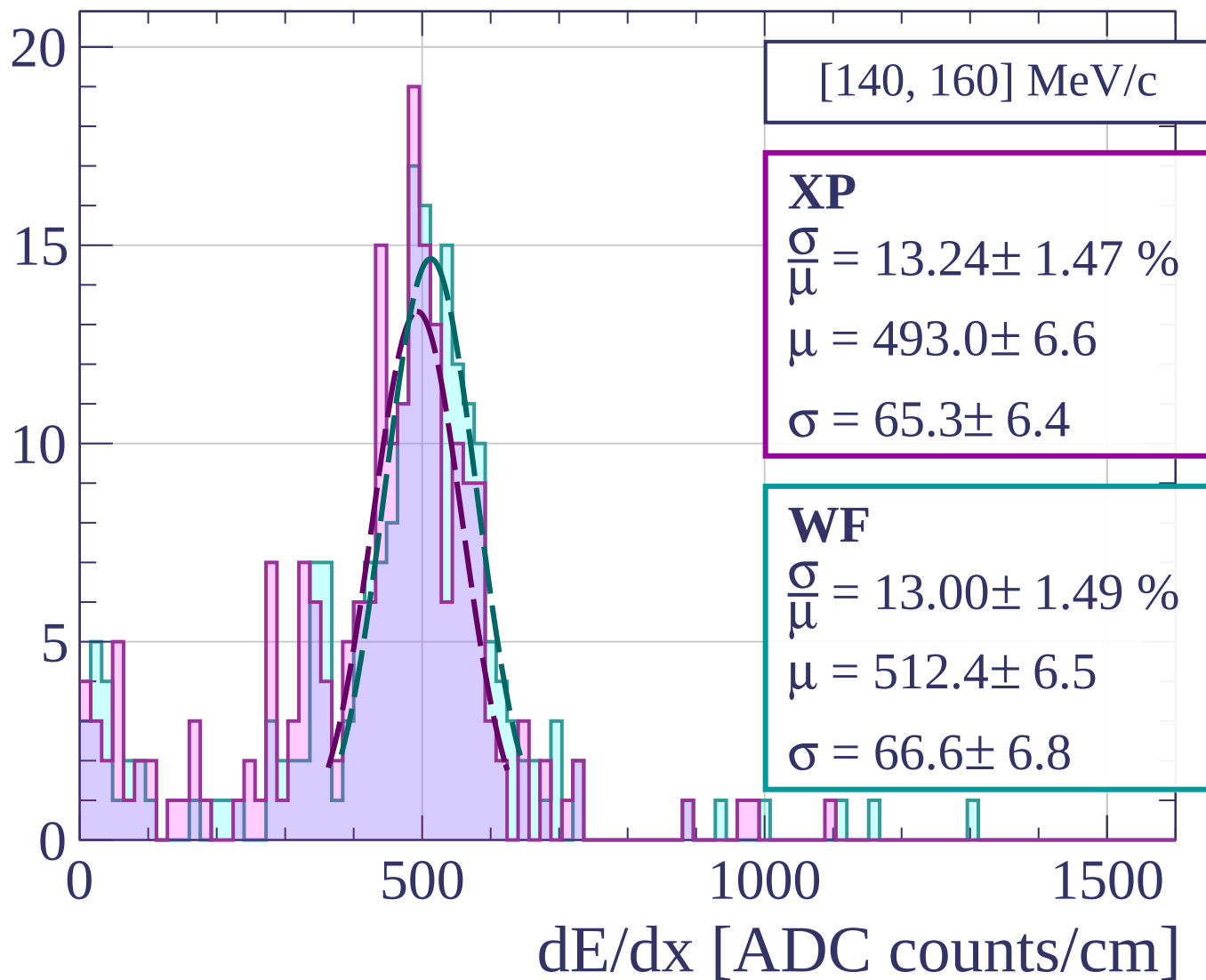
$$\frac{\sigma}{\mu} = 21.50 \pm 2.71 \%$$

$$\mu = 546.2 \pm 11.6$$

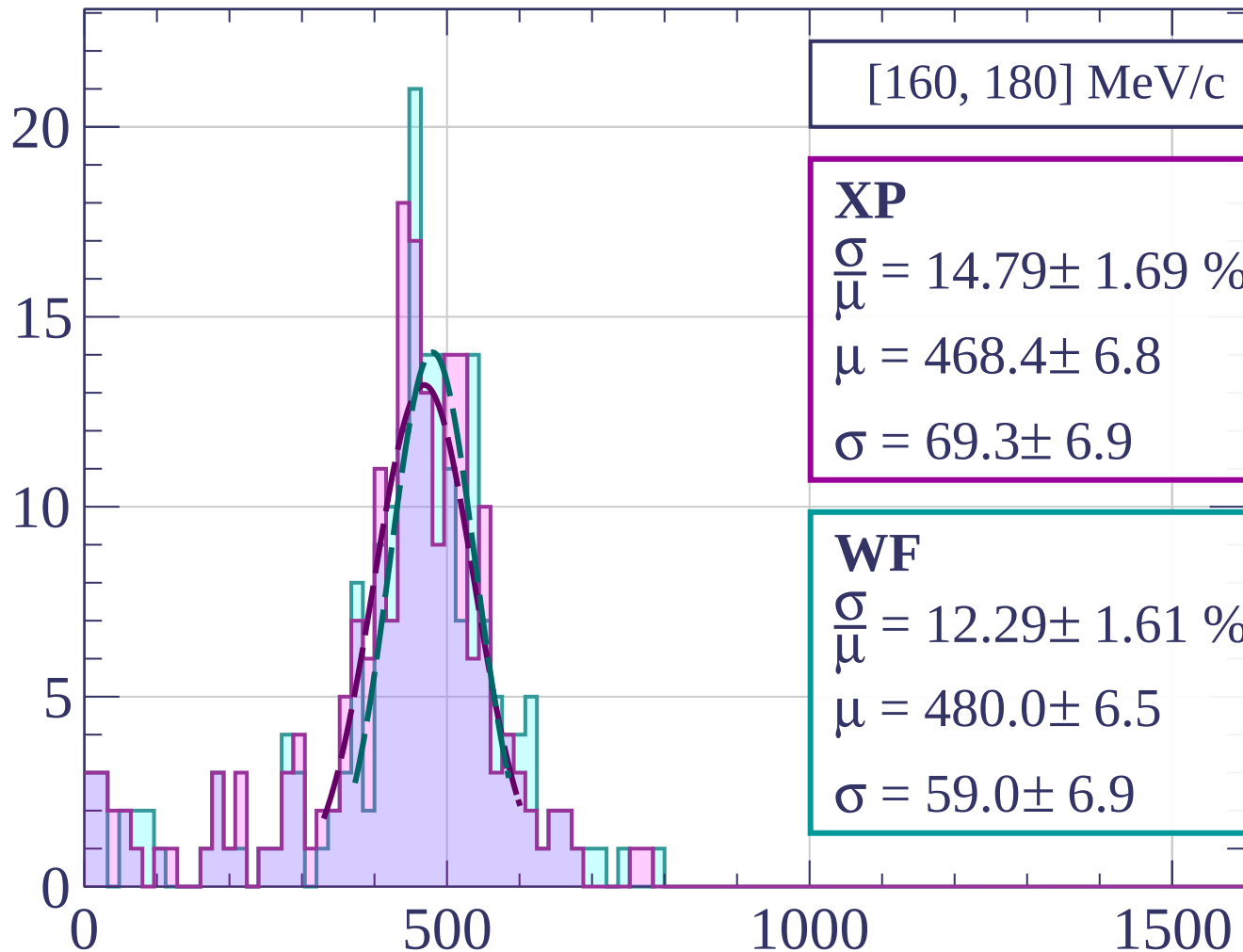
$$\sigma = 117.5 \pm 12.3$$

dE/dx [ADC counts/cm]

Count

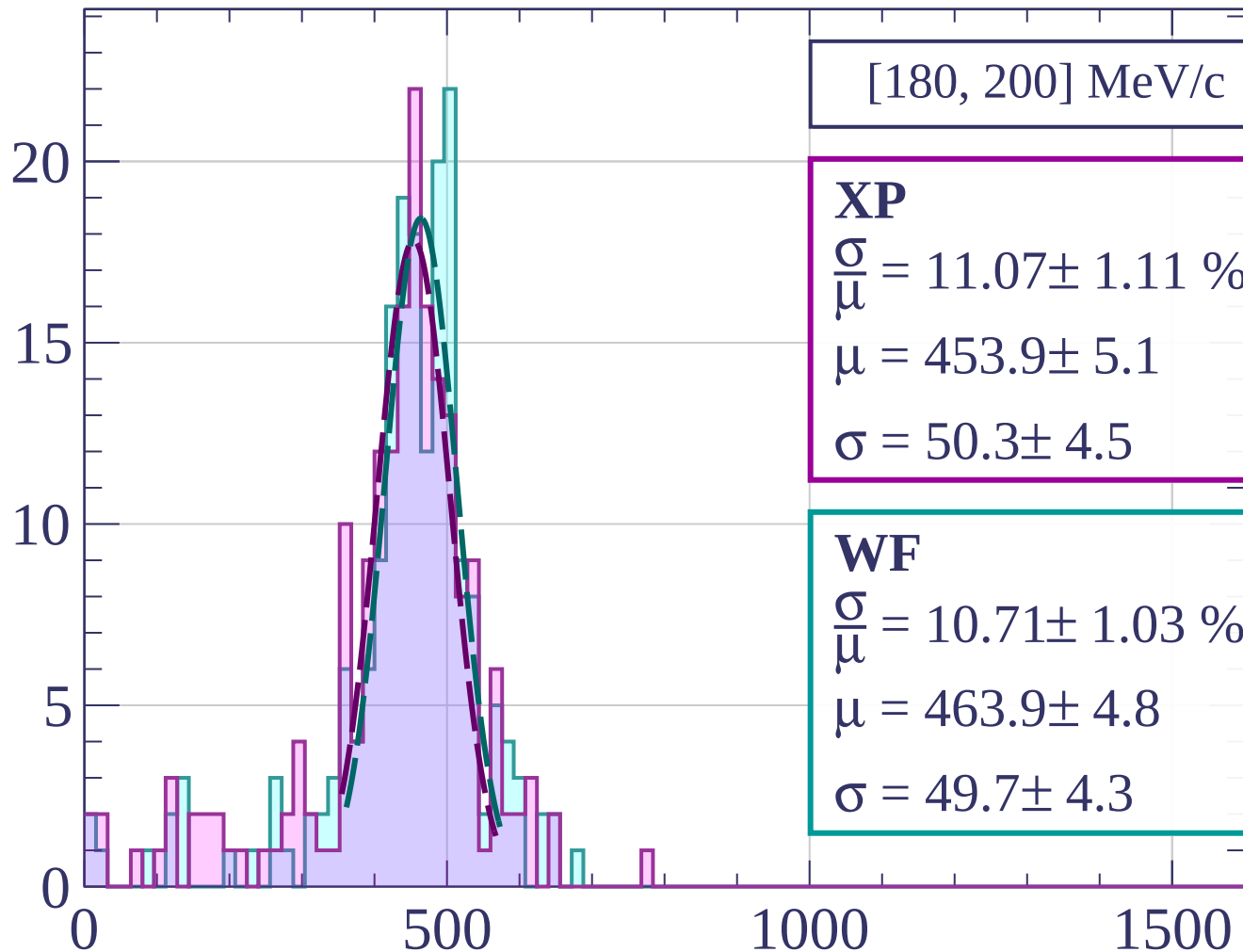


Count



dE/dx [ADC counts/cm]

Count



[180, 200] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.07 \pm 1.11 \%$$

$$\mu = 453.9 \pm 5.1$$

$$\sigma = 50.3 \pm 4.5$$

WF

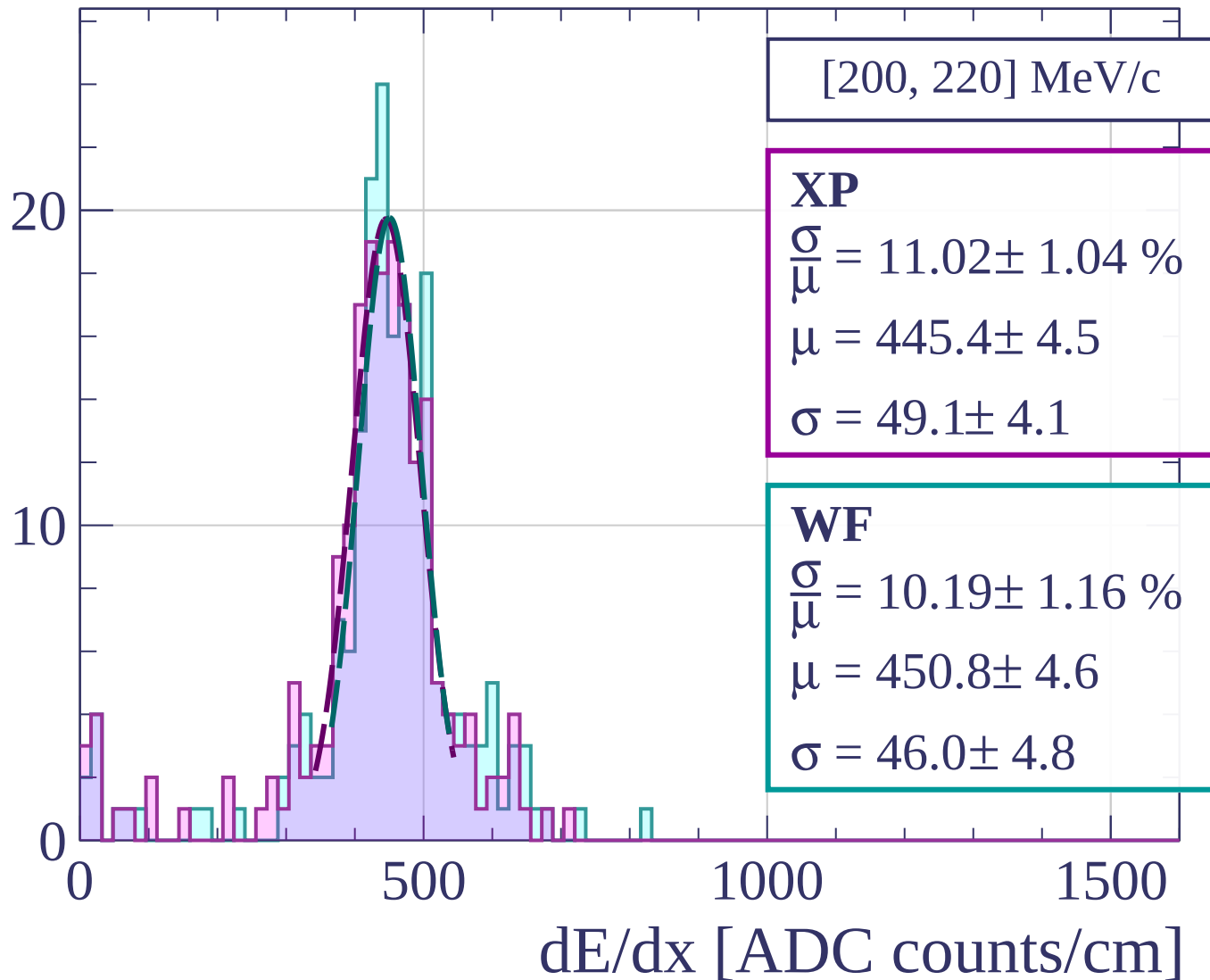
$$\frac{\sigma}{\mu} = 10.71 \pm 1.03 \%$$

$$\mu = 463.9 \pm 4.8$$

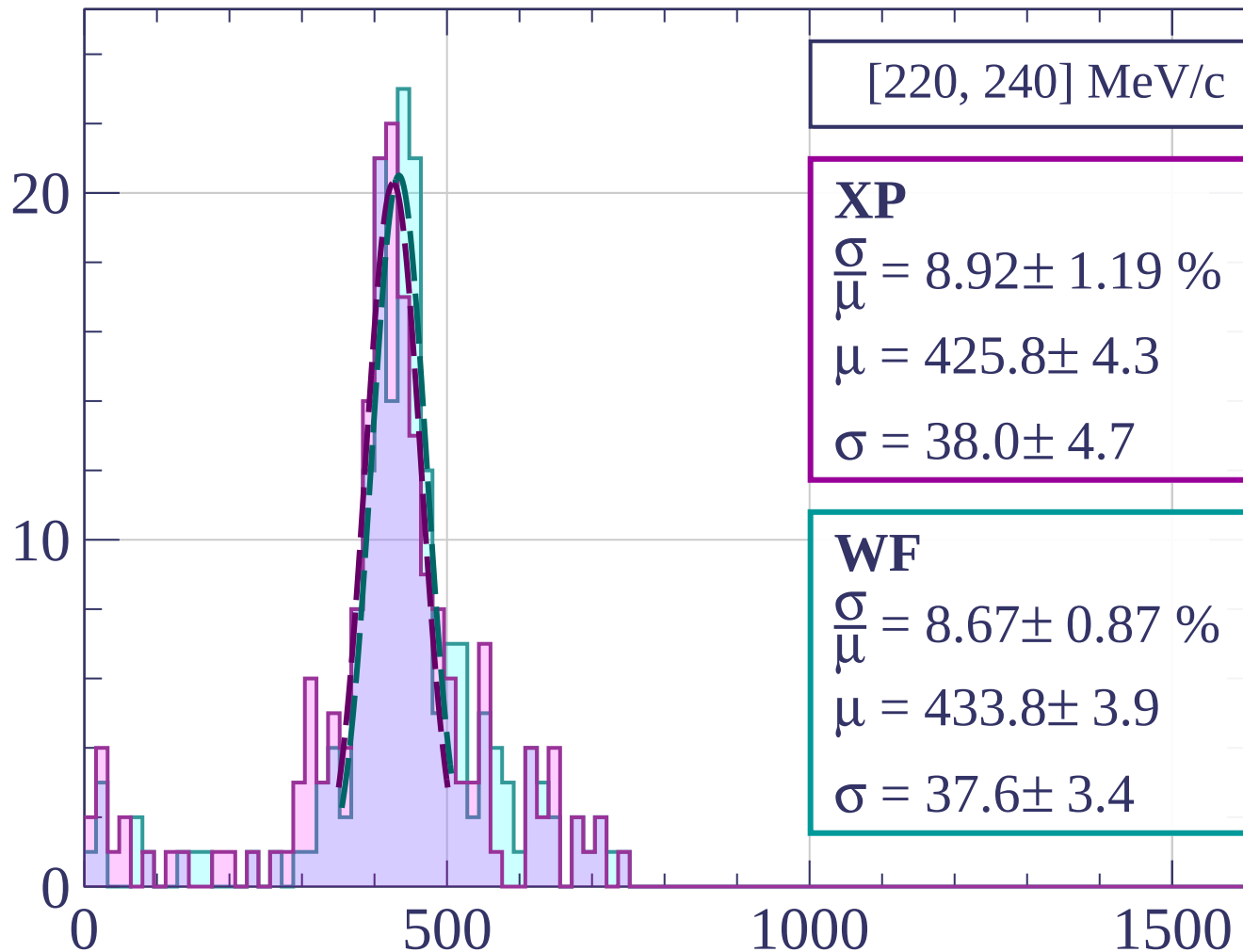
$$\sigma = 49.7 \pm 4.3$$

dE/dx [ADC counts/cm]

Count



Count



[220, 240] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.92 \pm 1.19 \%$$

$$\mu = 425.8 \pm 4.3$$

$$\sigma = 38.0 \pm 4.7$$

WF

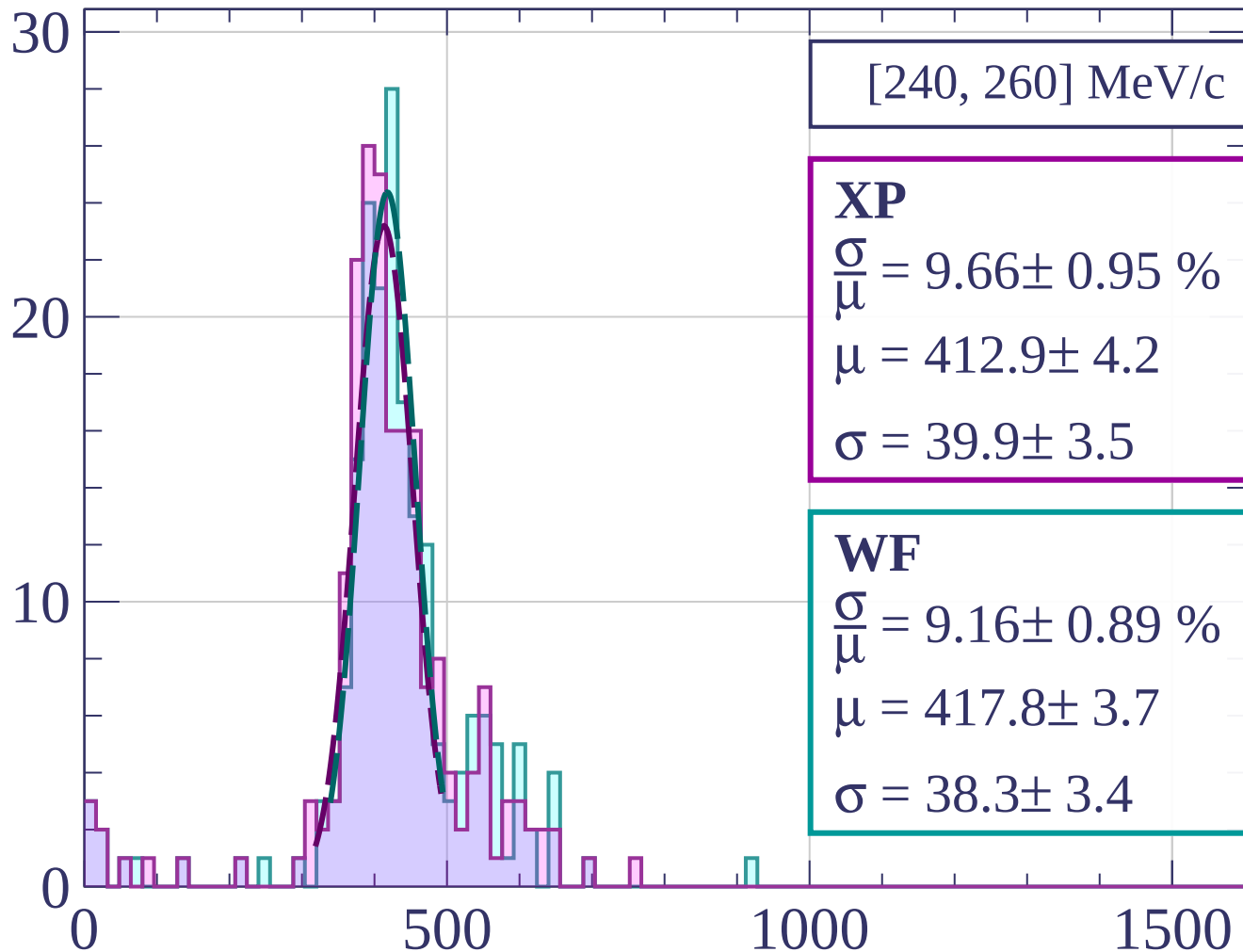
$$\frac{\sigma}{\mu} = 8.67 \pm 0.87 \%$$

$$\mu = 433.8 \pm 3.9$$

$$\sigma = 37.6 \pm 3.4$$

dE/dx [ADC counts/cm]

Count



[240, 260] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.66 \pm 0.95 \%$$

$$\mu = 412.9 \pm 4.2$$

$$\sigma = 39.9 \pm 3.5$$

WF

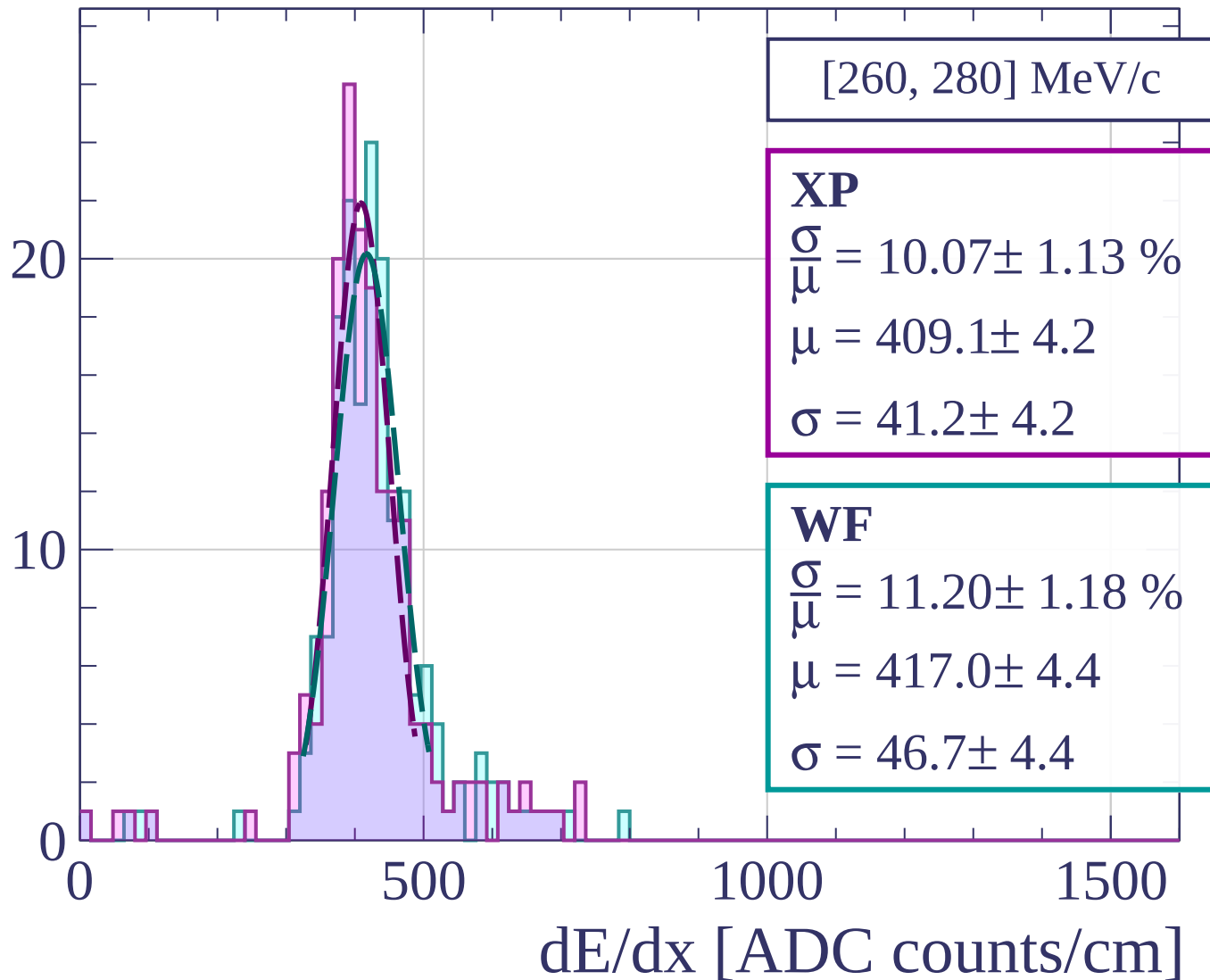
$$\frac{\sigma}{\mu} = 9.16 \pm 0.89 \%$$

$$\mu = 417.8 \pm 3.7$$

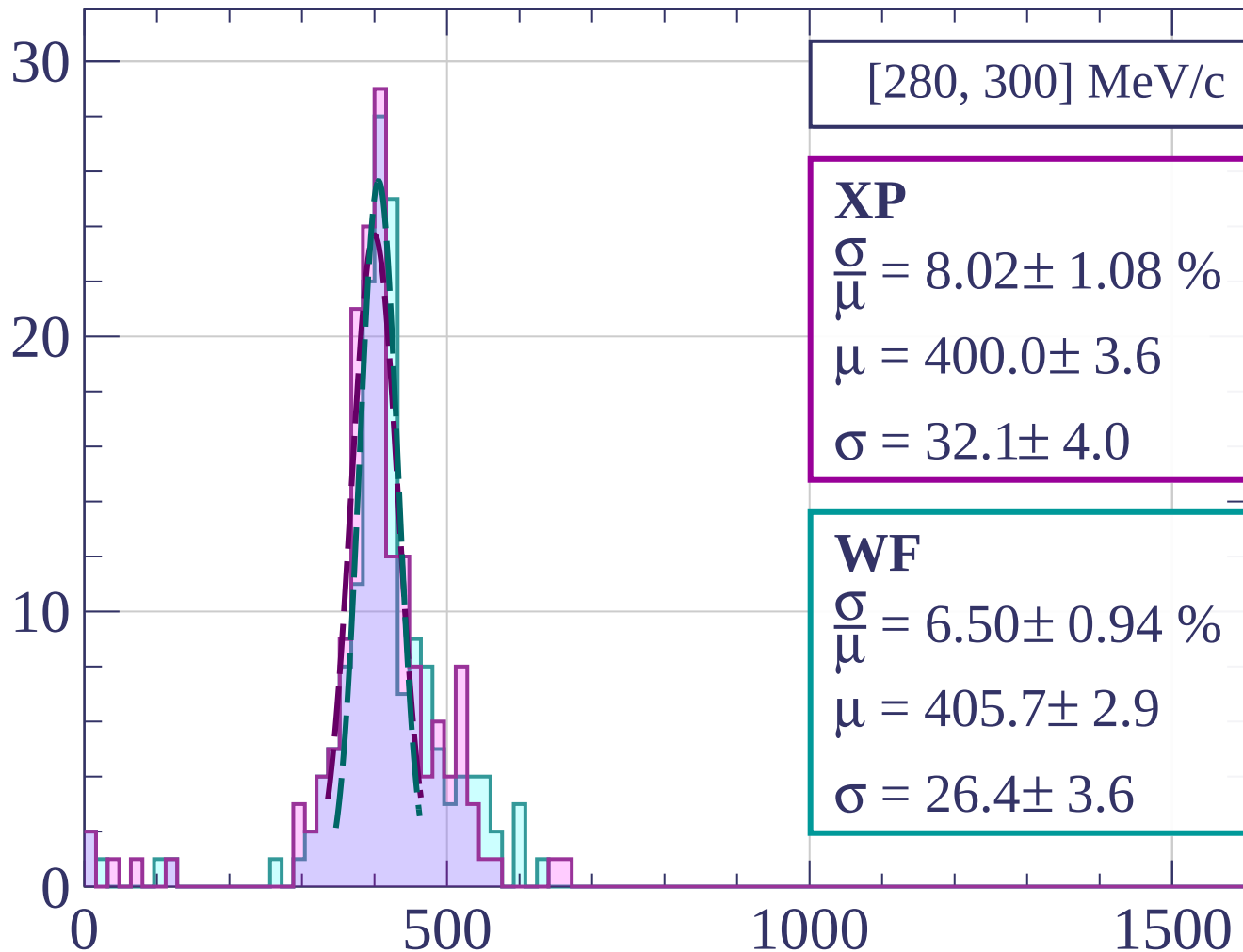
$$\sigma = 38.3 \pm 3.4$$

dE/dx [ADC counts/cm]

Count

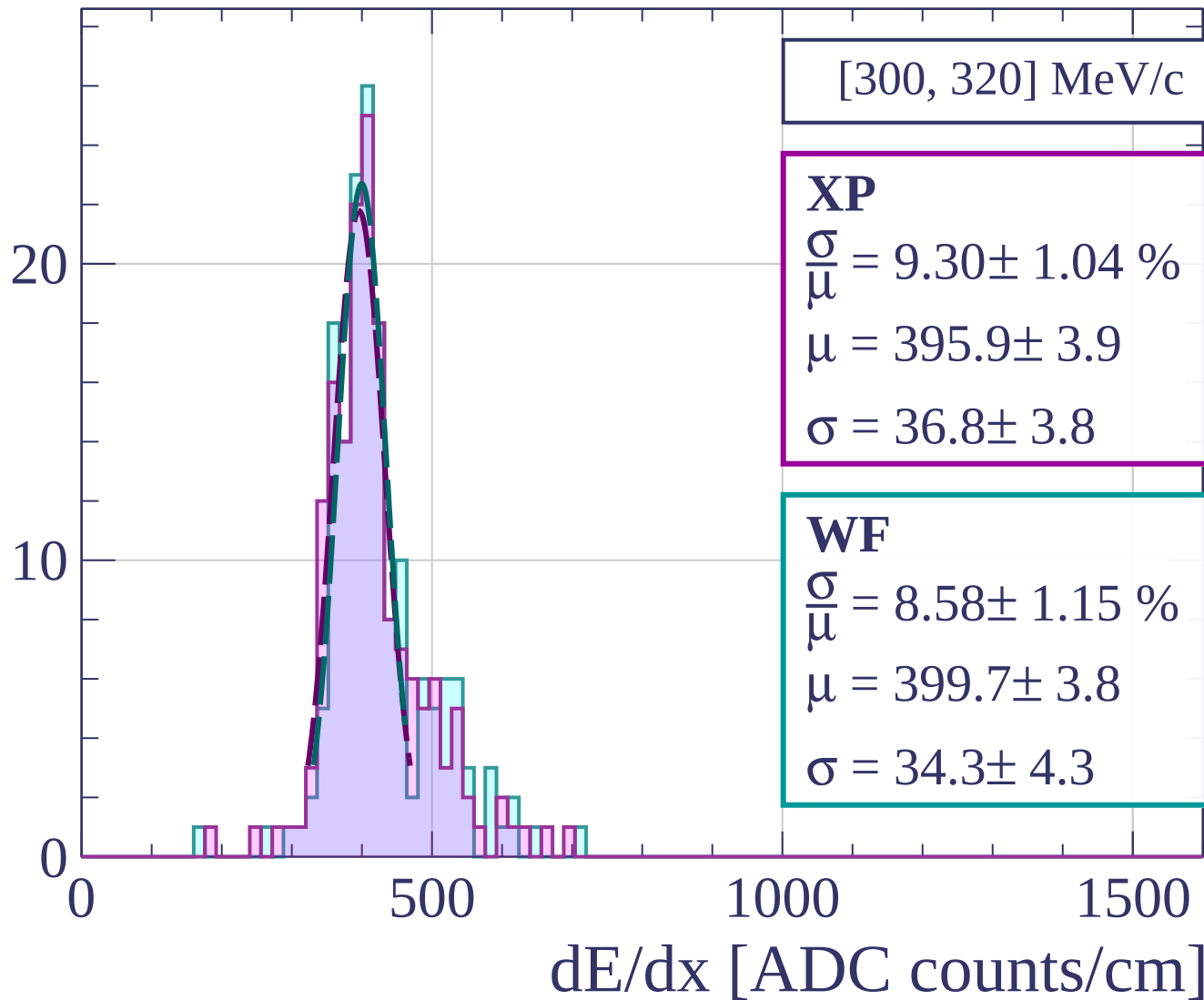


Count

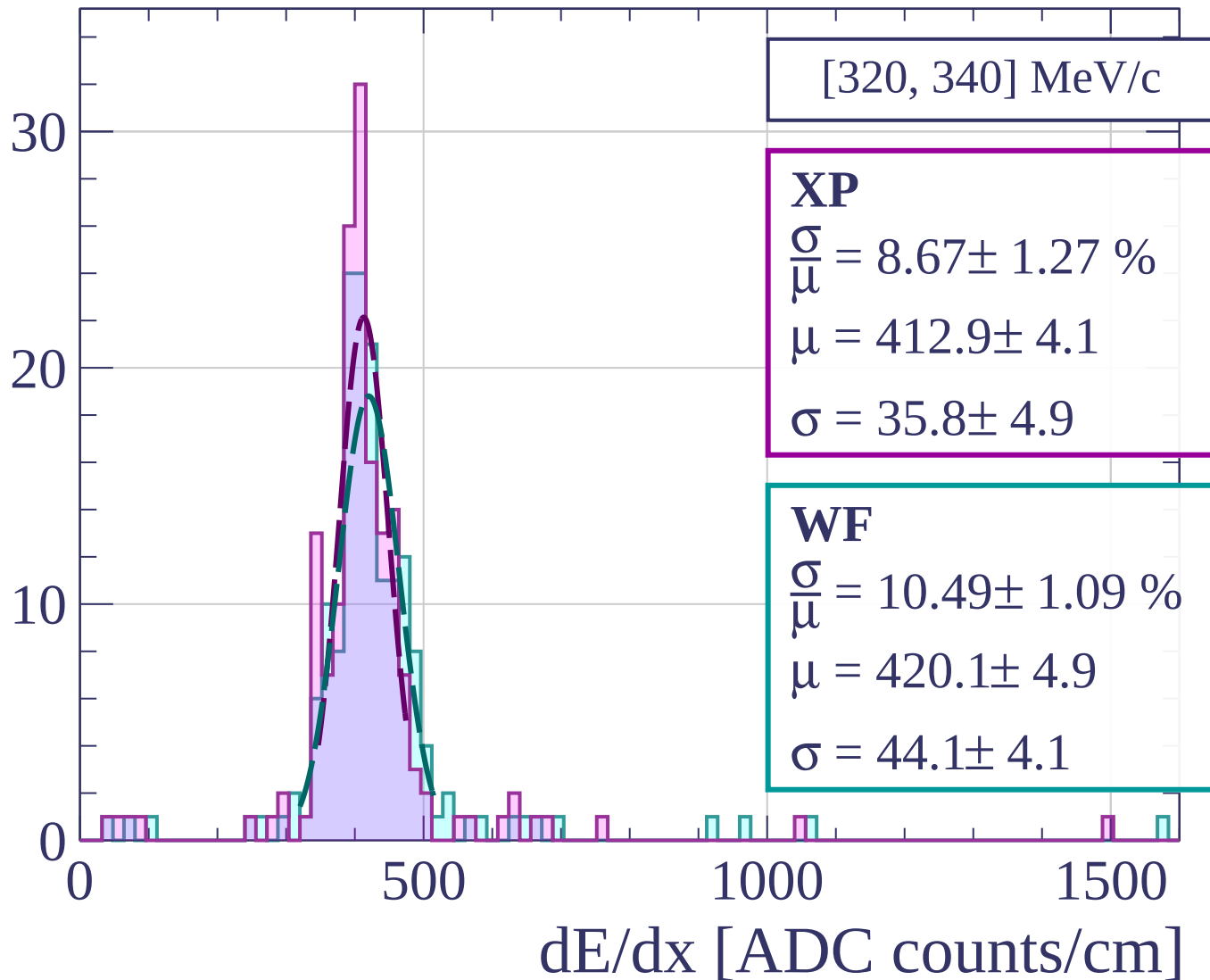


dE/dx [ADC counts/cm]

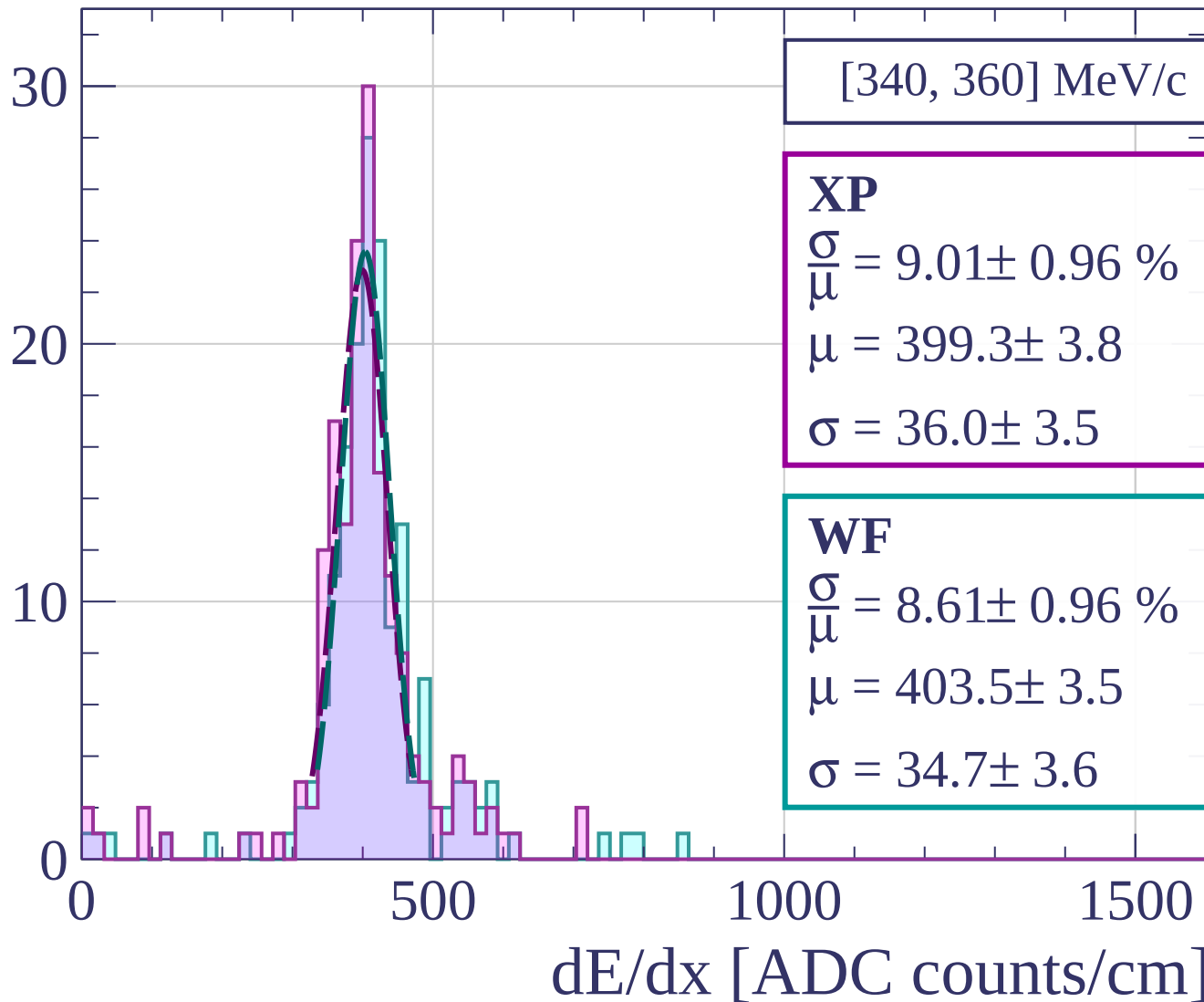
Count



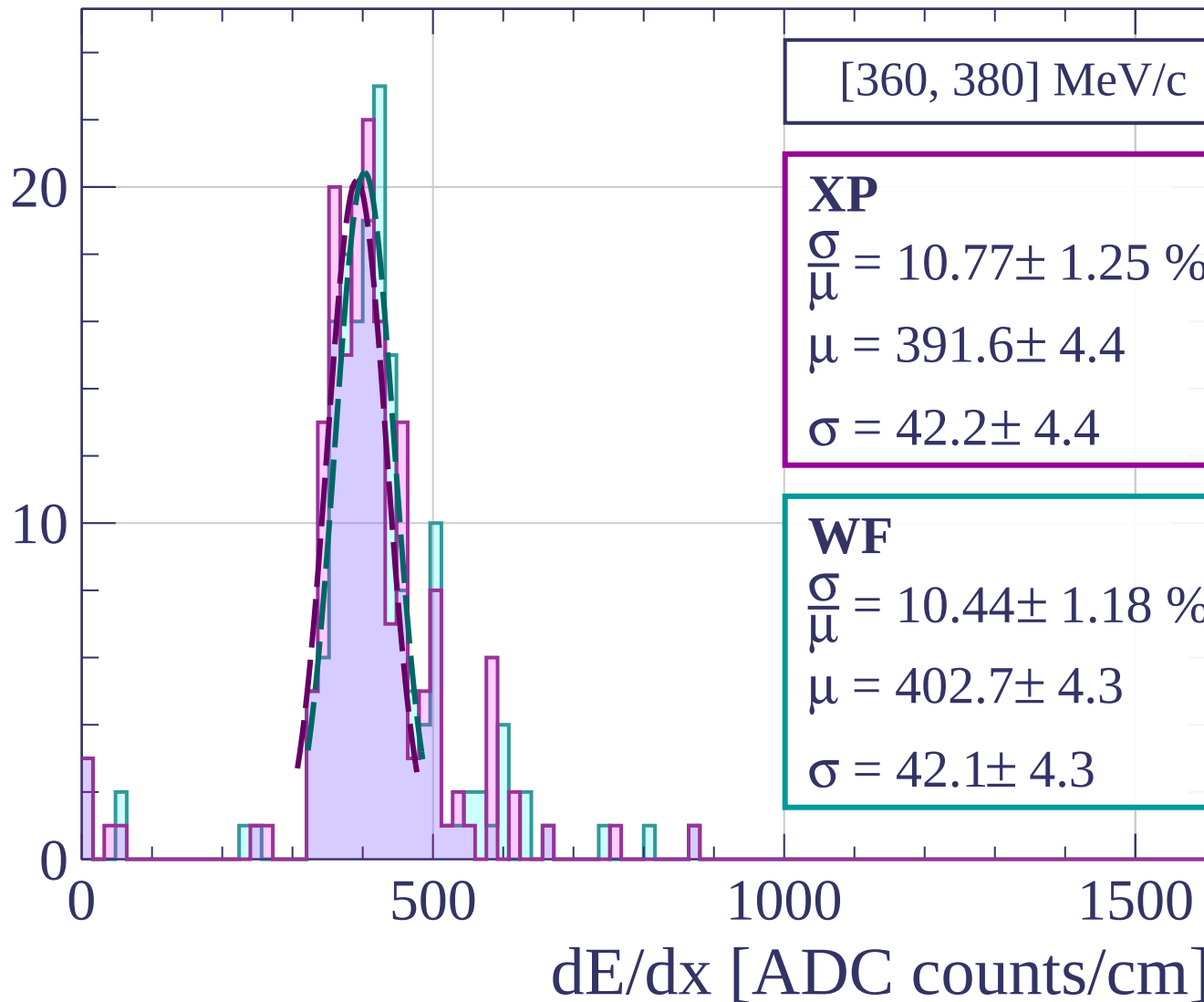
Count



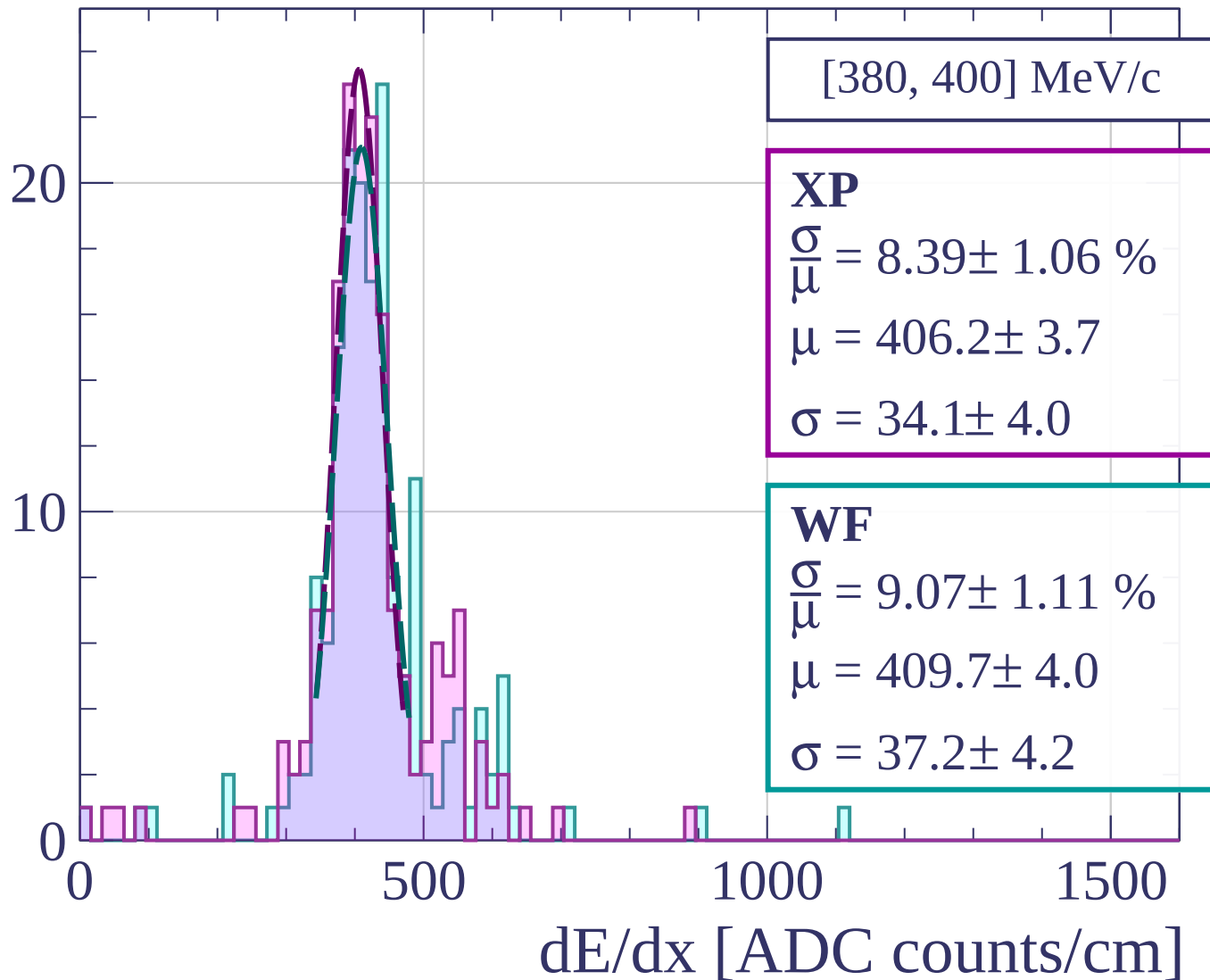
Count



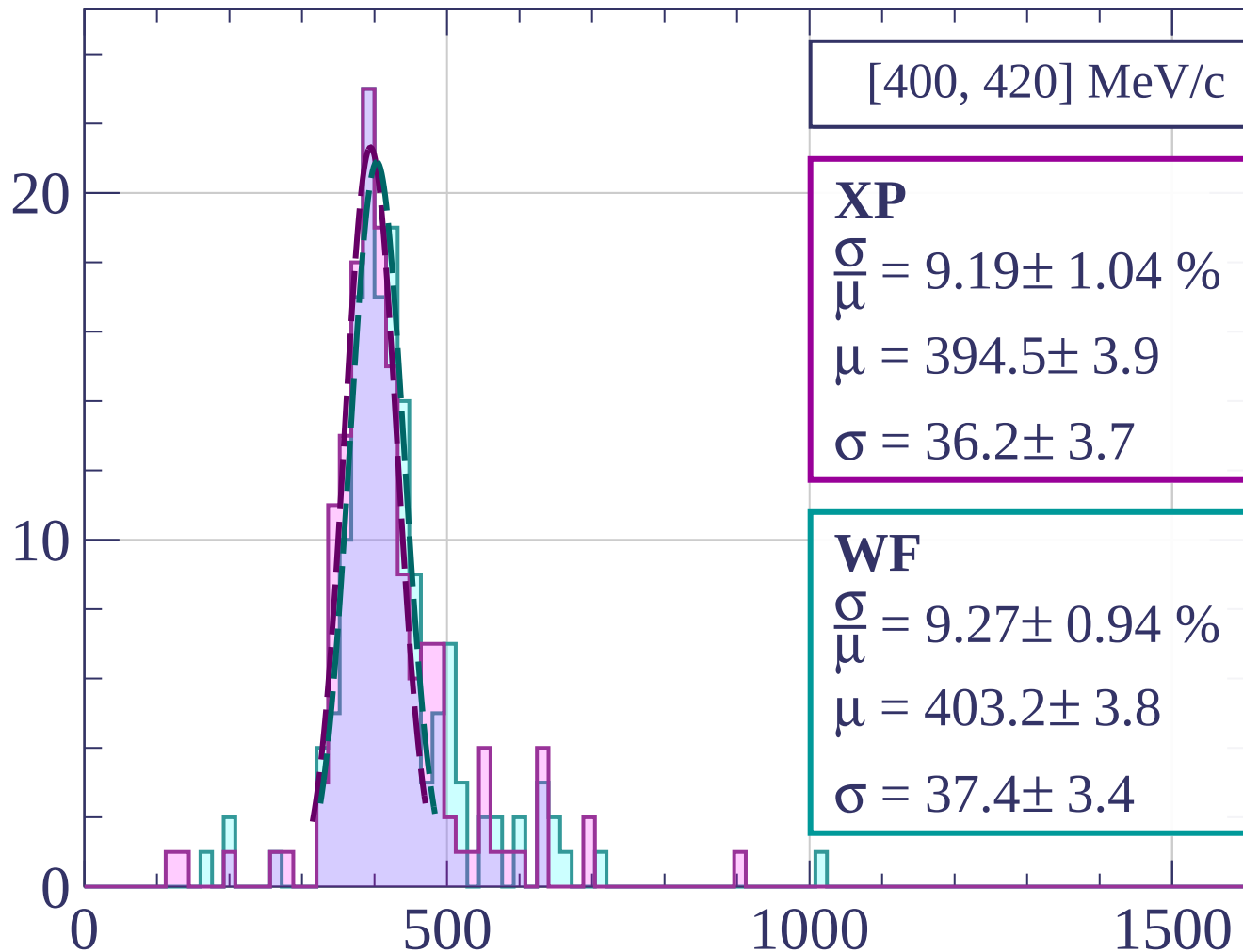
Count



Count



Count



[400, 420] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 9.19 \pm 1.04 \%$$

$$\mu = 394.5 \pm 3.9$$

$$\sigma = 36.2 \pm 3.7$$

WF

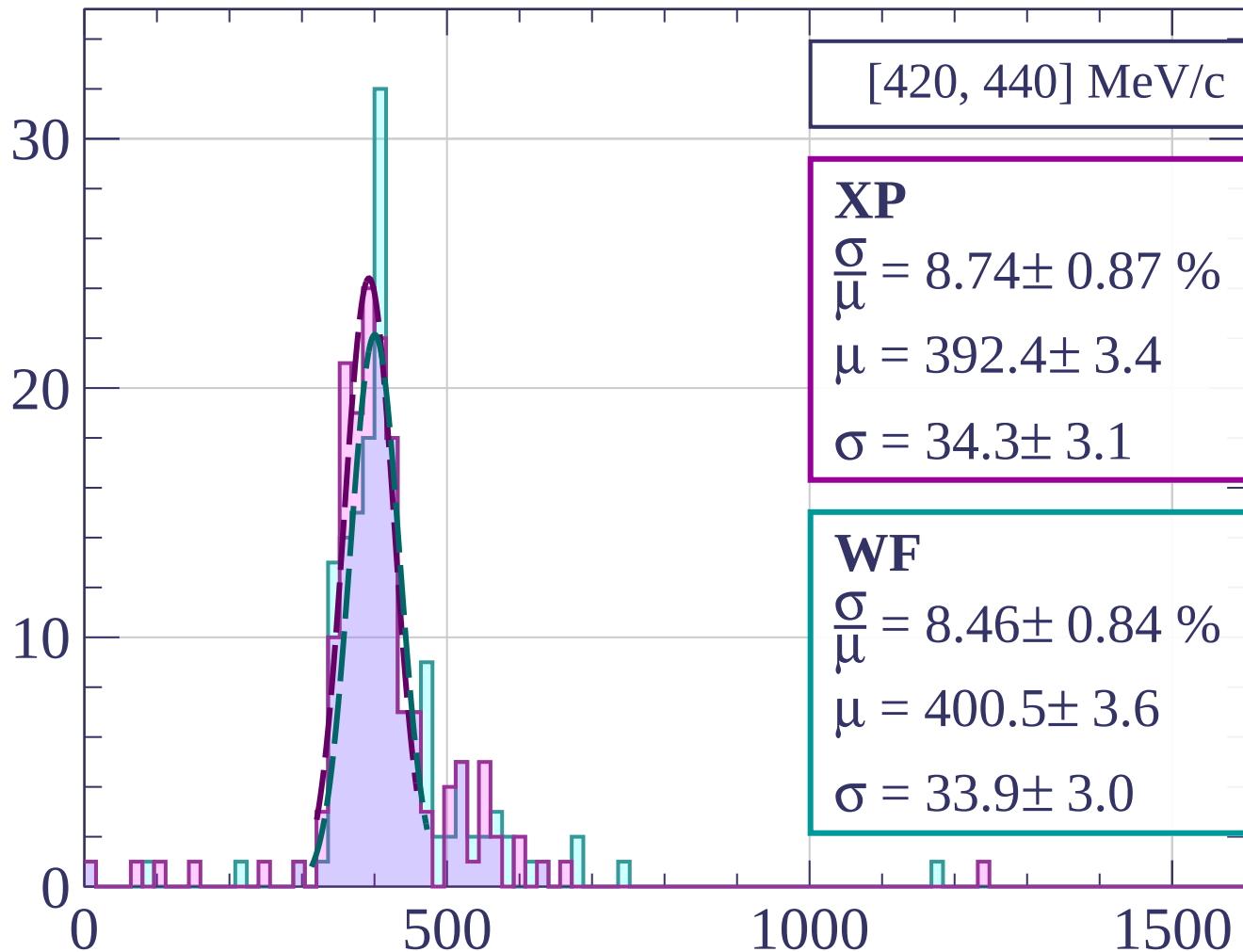
$$\frac{\sigma}{\bar{\mu}} = 9.27 \pm 0.94 \%$$

$$\mu = 403.2 \pm 3.8$$

$$\sigma = 37.4 \pm 3.4$$

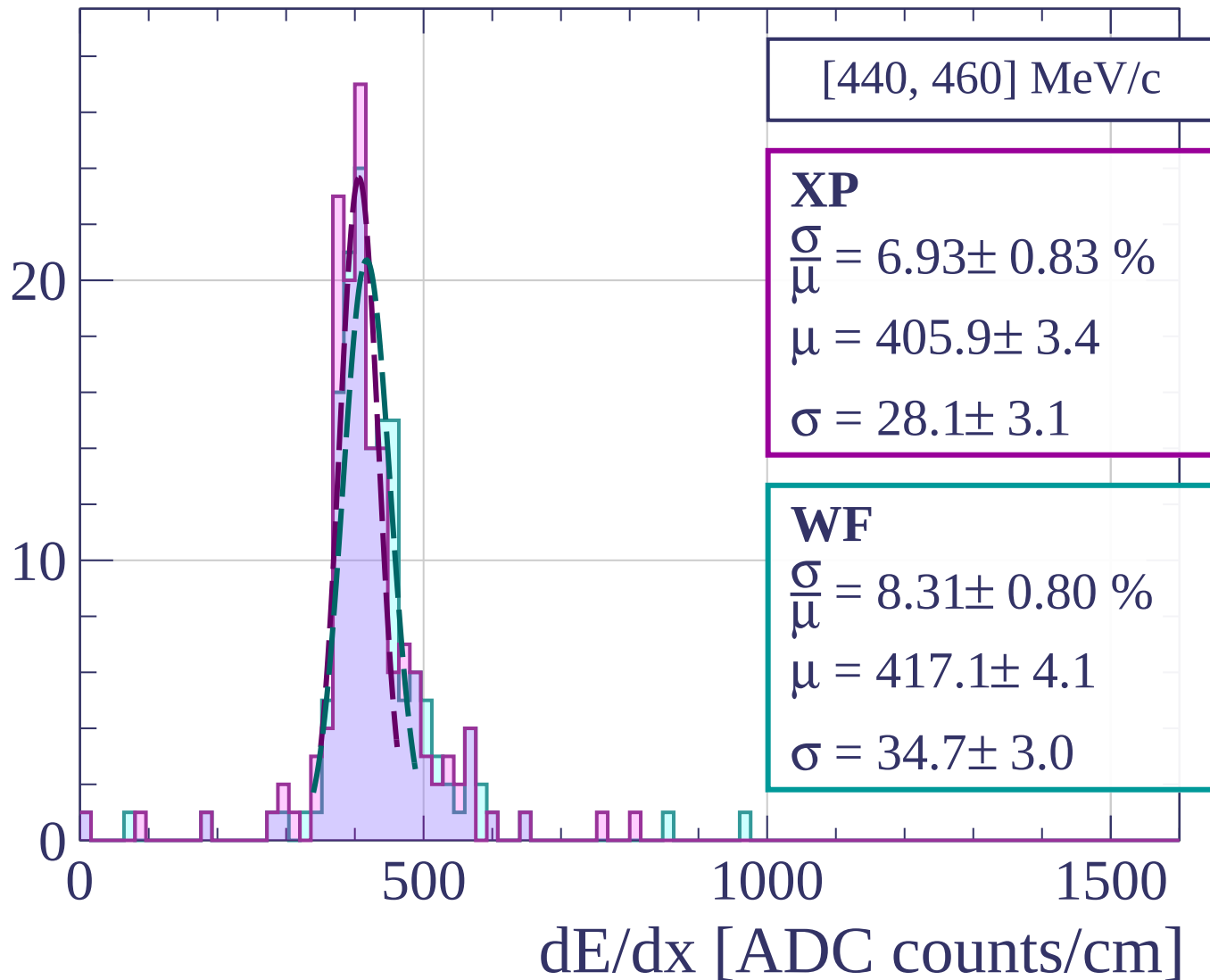
dE/dx [ADC counts/cm]

Count

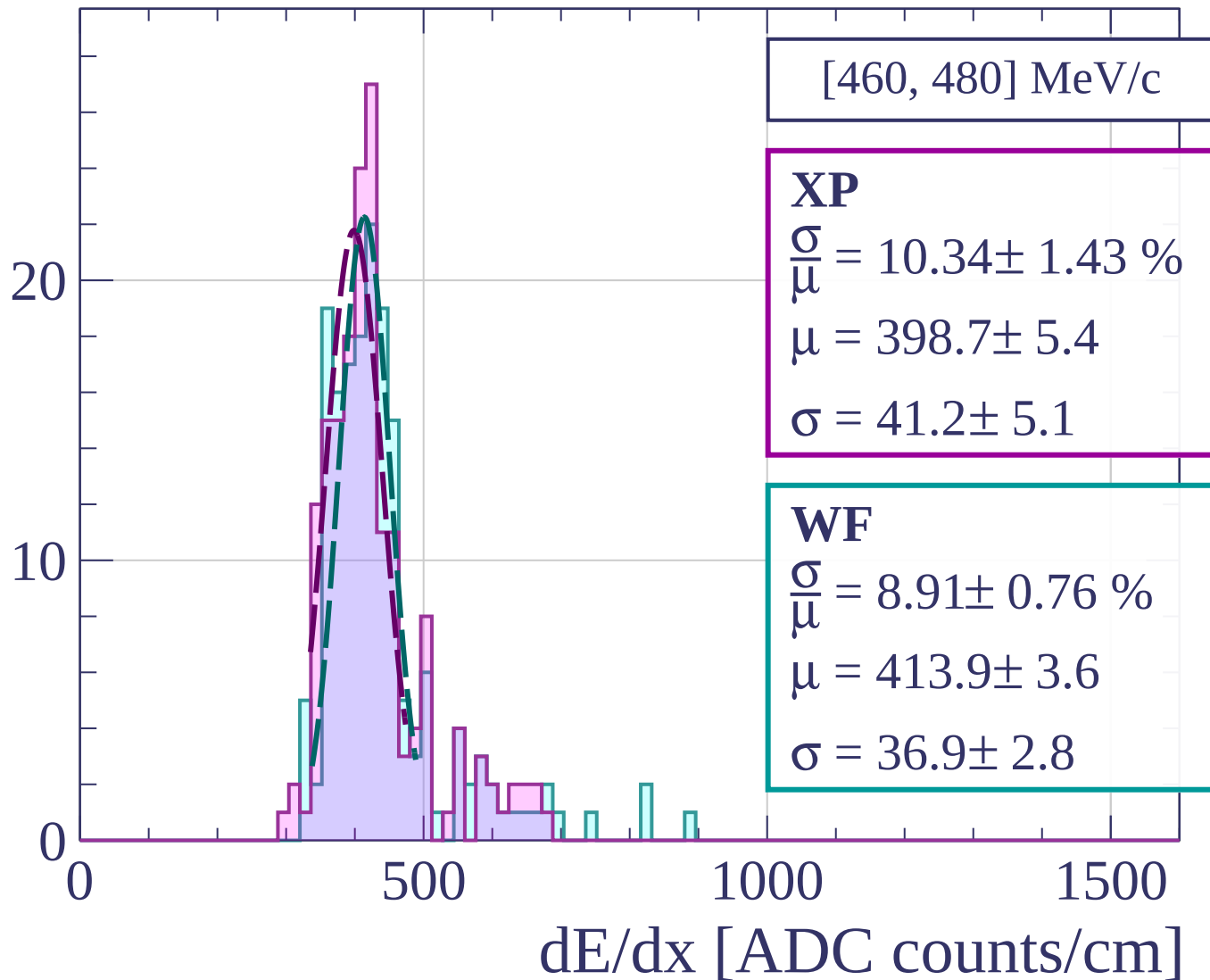


dE/dx [ADC counts/cm]

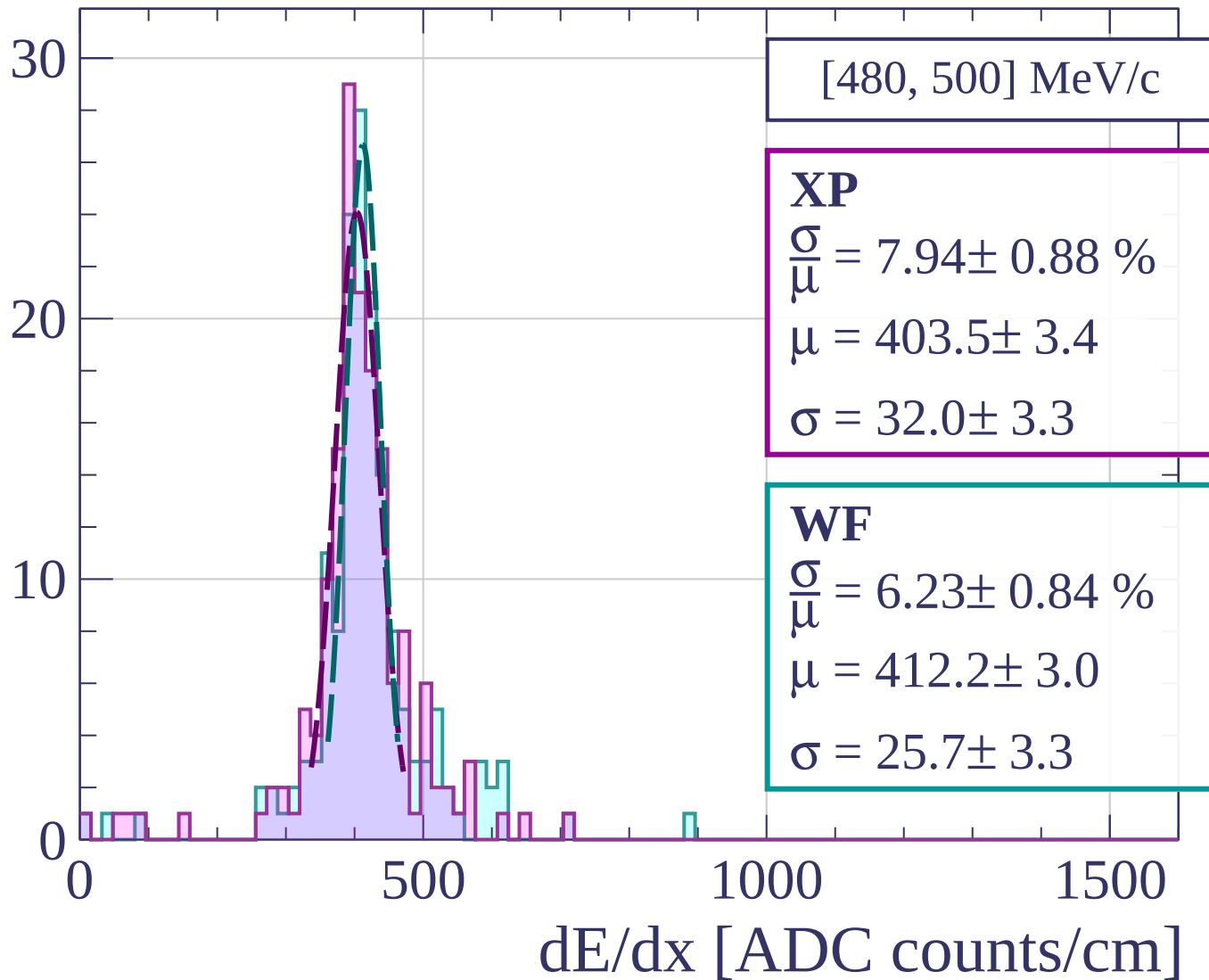
Count



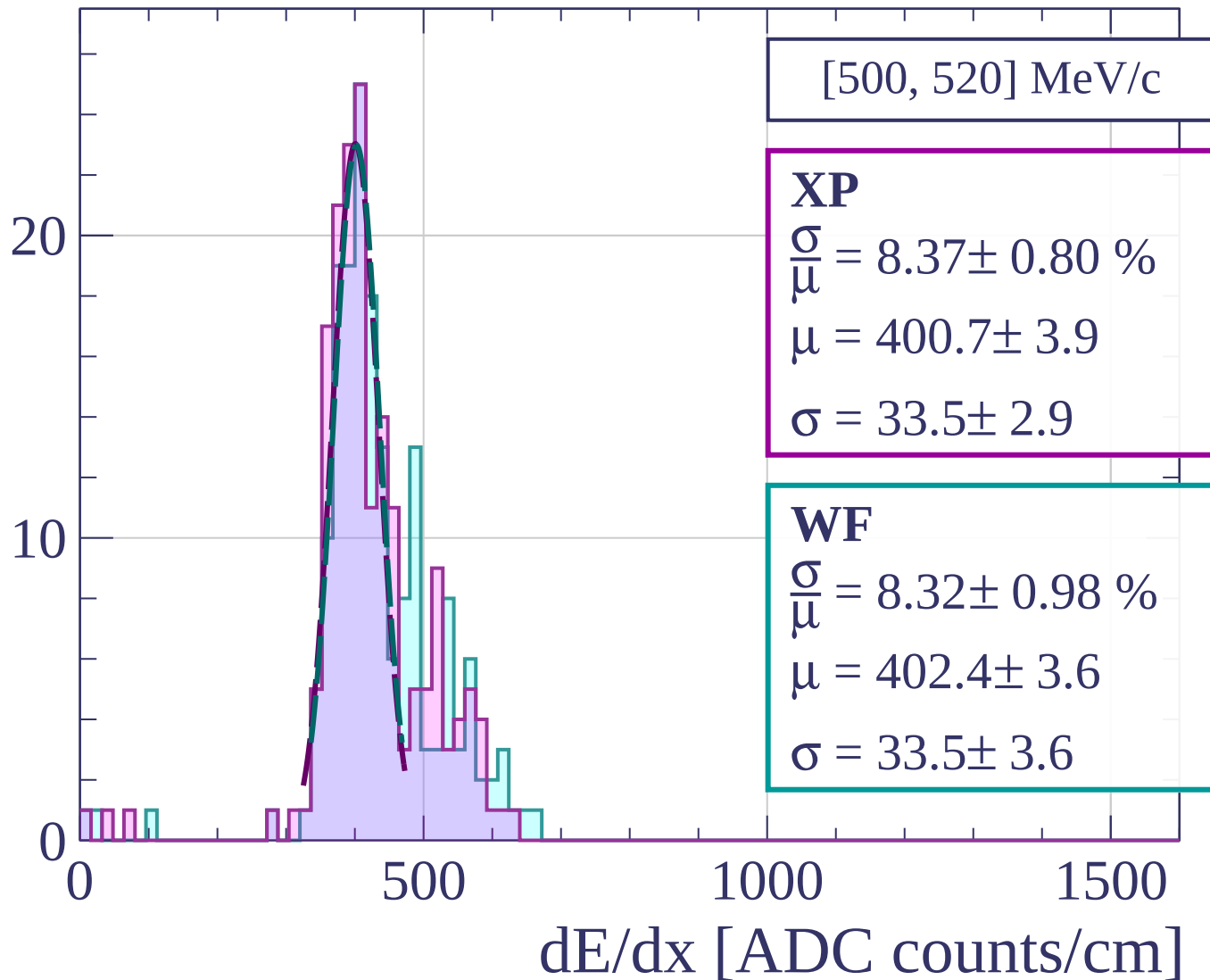
Count



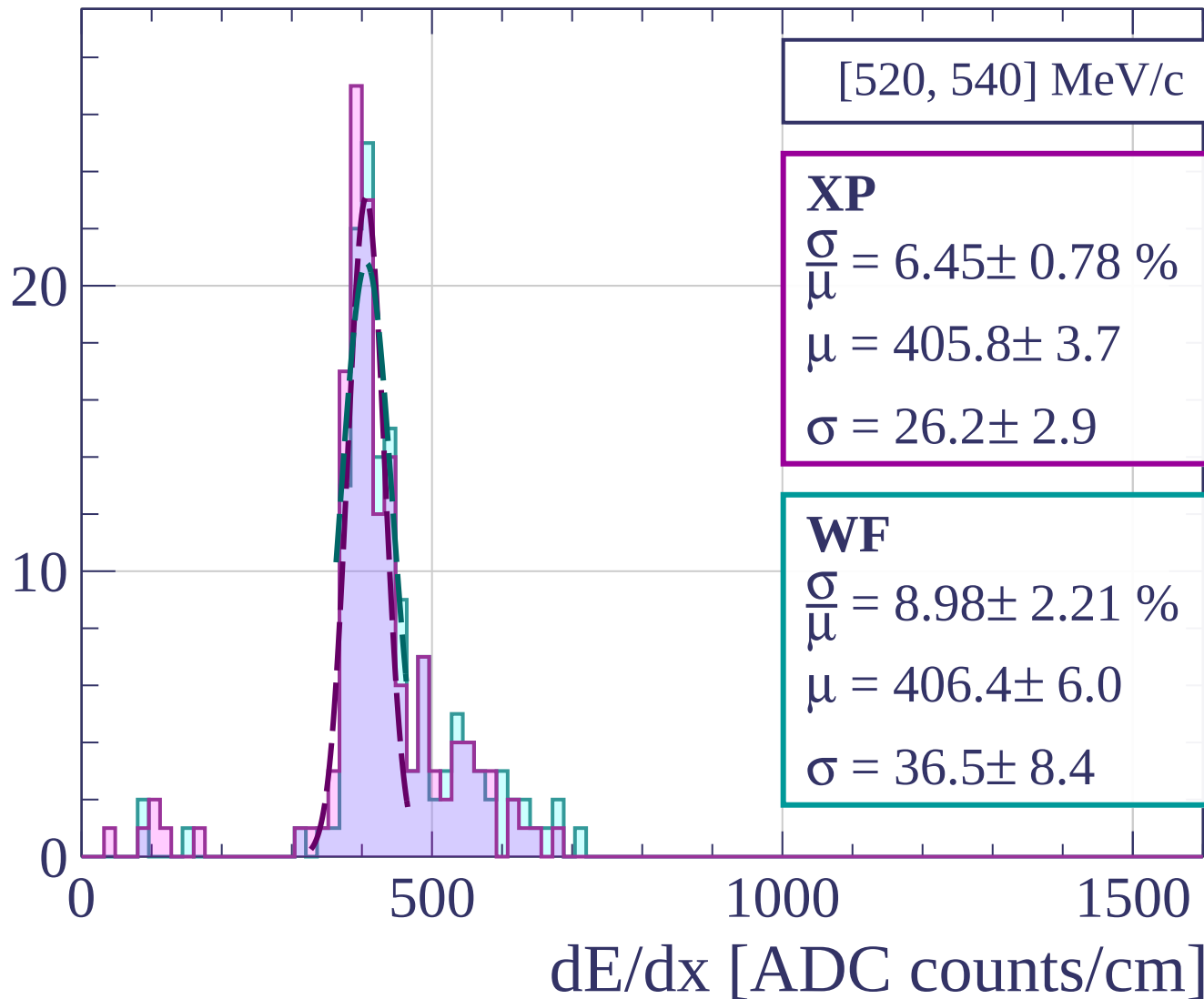
Count



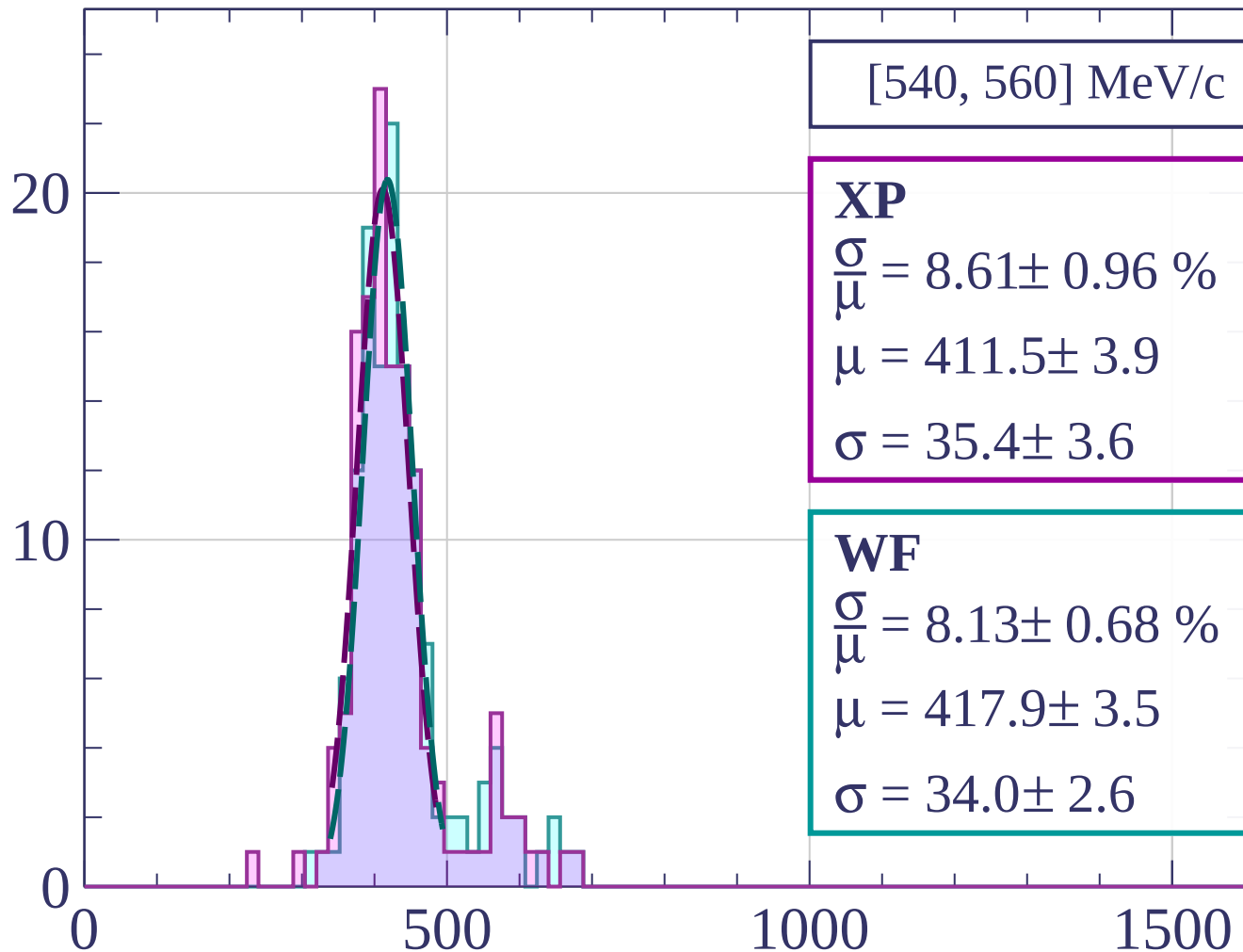
Count



Count



Count



[540, 560] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.61 \pm 0.96 \%$$

$$\mu = 411.5 \pm 3.9$$

$$\sigma = 35.4 \pm 3.6$$

WF

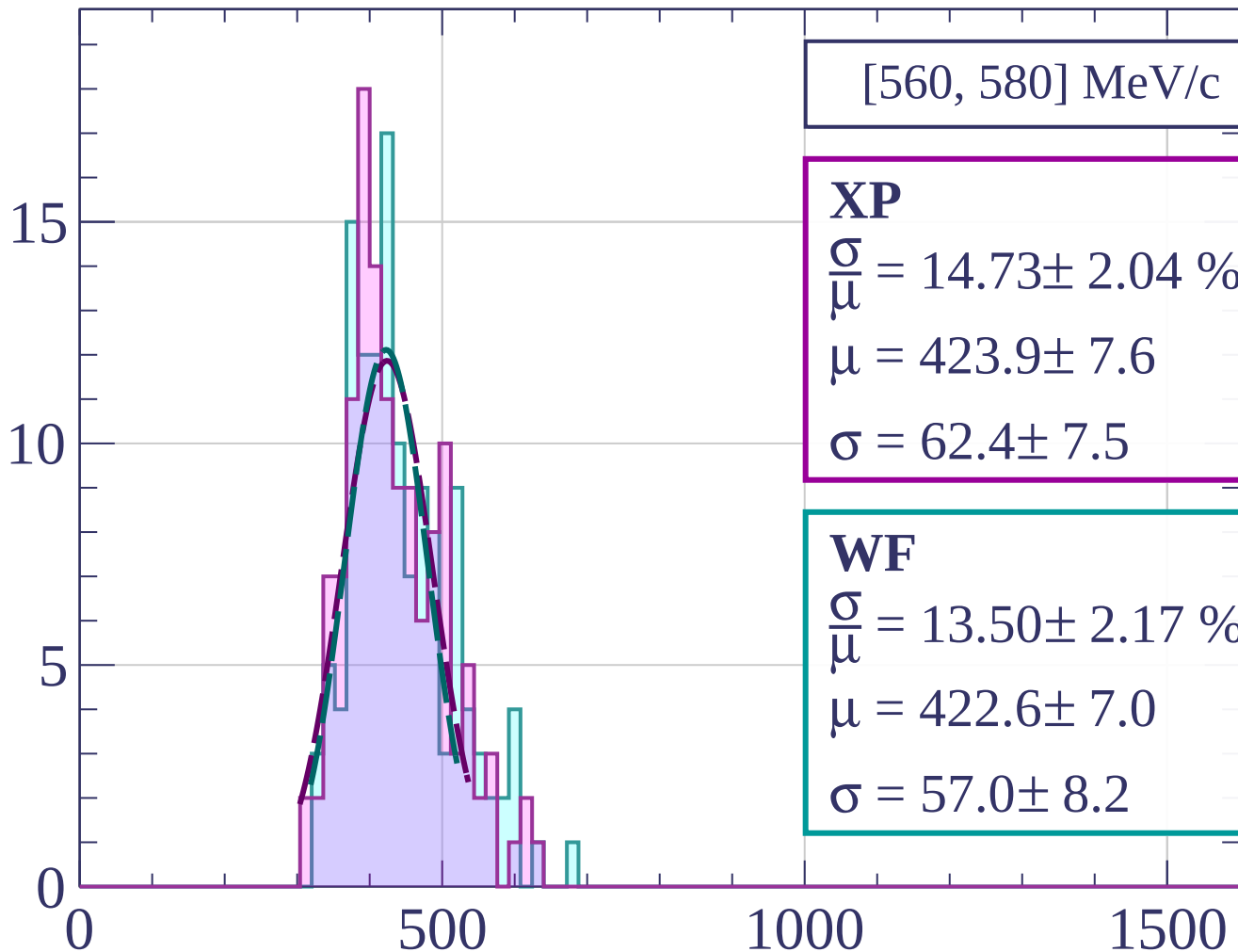
$$\frac{\sigma}{\mu} = 8.13 \pm 0.68 \%$$

$$\mu = 417.9 \pm 3.5$$

$$\sigma = 34.0 \pm 2.6$$

dE/dx [ADC counts/cm]

Count



[560, 580] MeV/c

XP

$$\frac{\sigma}{\mu} = 14.73 \pm 2.04 \%$$

$$\mu = 423.9 \pm 7.6$$

$$\sigma = 62.4 \pm 7.5$$

WF

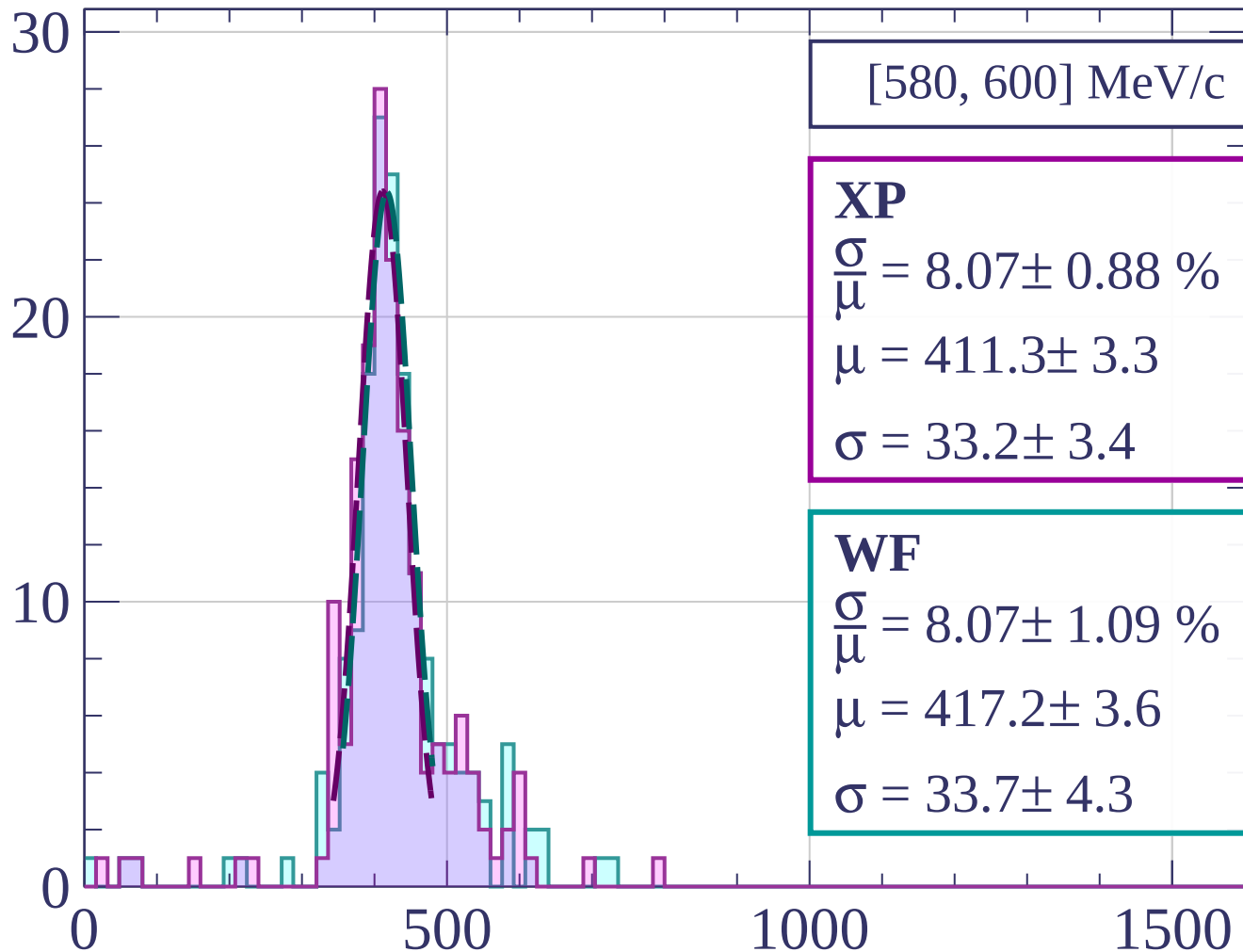
$$\frac{\sigma}{\mu} = 13.50 \pm 2.17 \%$$

$$\mu = 422.6 \pm 7.0$$

$$\sigma = 57.0 \pm 8.2$$

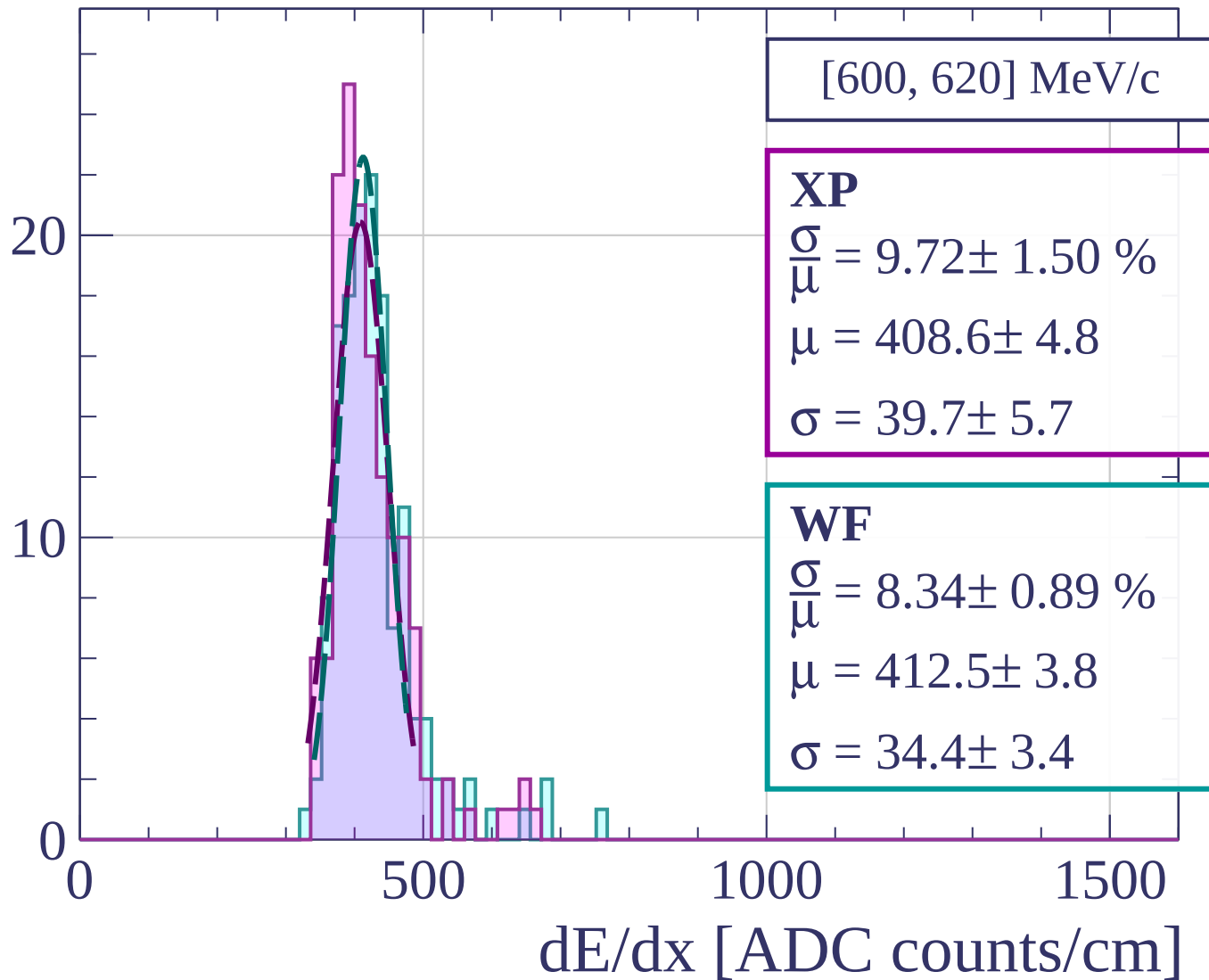
dE/dx [ADC counts/cm]

Count

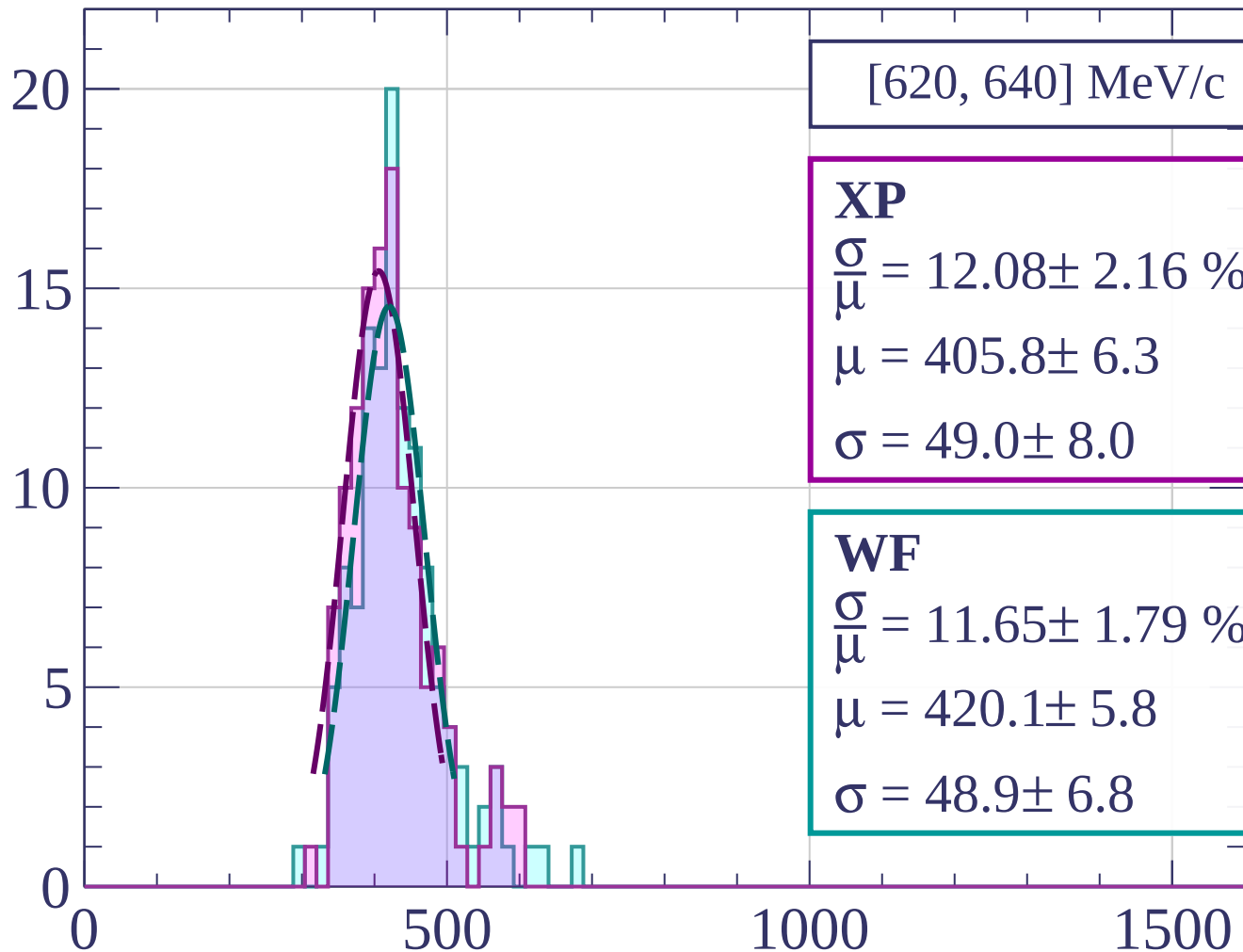


dE/dx [ADC counts/cm]

Count



Count



[620, 640] MeV/c

XP

$$\frac{\sigma}{\mu} = 12.08 \pm 2.16 \%$$

$$\mu = 405.8 \pm 6.3$$

$$\sigma = 49.0 \pm 8.0$$

WF

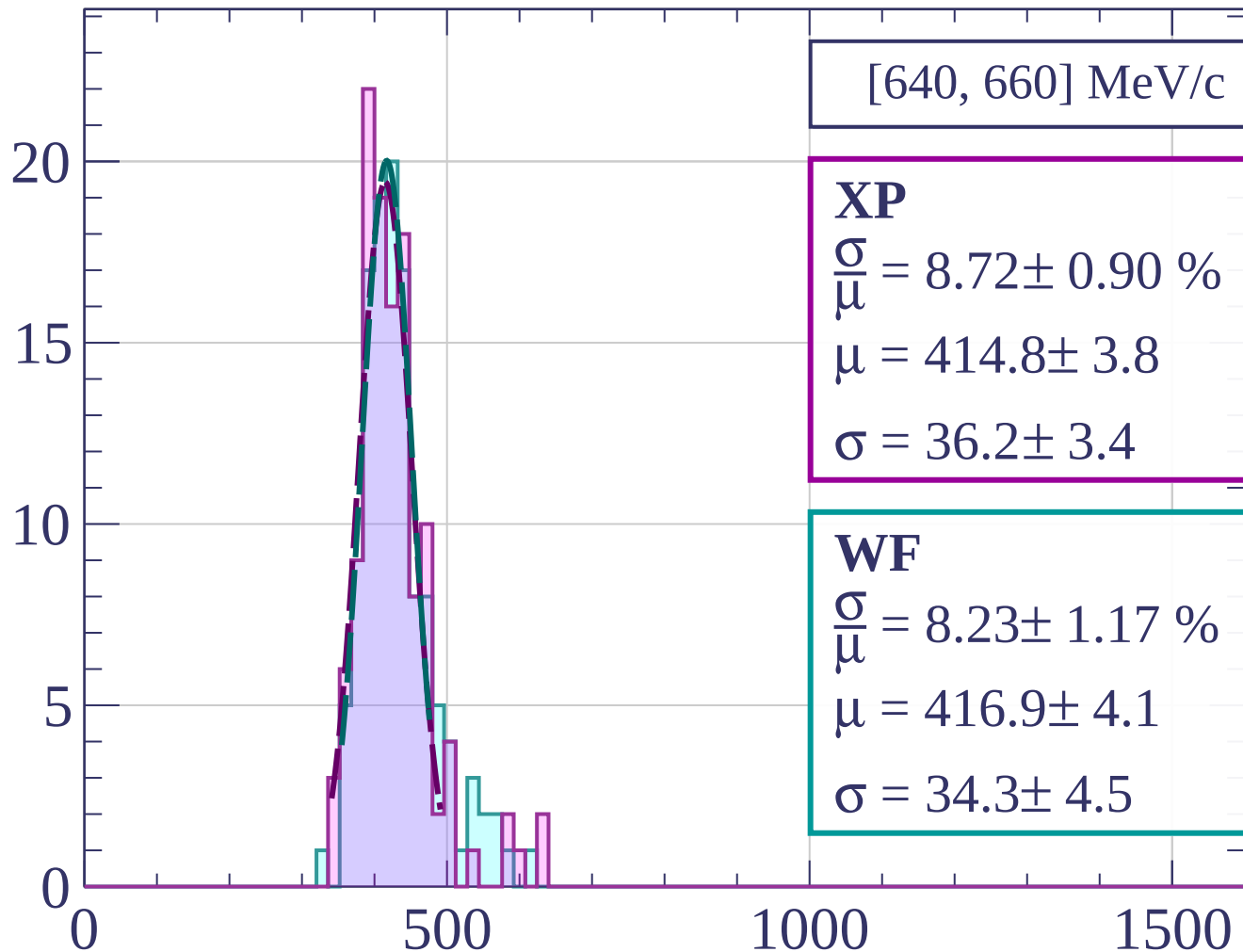
$$\frac{\sigma}{\mu} = 11.65 \pm 1.79 \%$$

$$\mu = 420.1 \pm 5.8$$

$$\sigma = 48.9 \pm 6.8$$

dE/dx [ADC counts/cm]

Count



[640, 660] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.72 \pm 0.90 \%$$

$$\mu = 414.8 \pm 3.8$$

$$\sigma = 36.2 \pm 3.4$$

WF

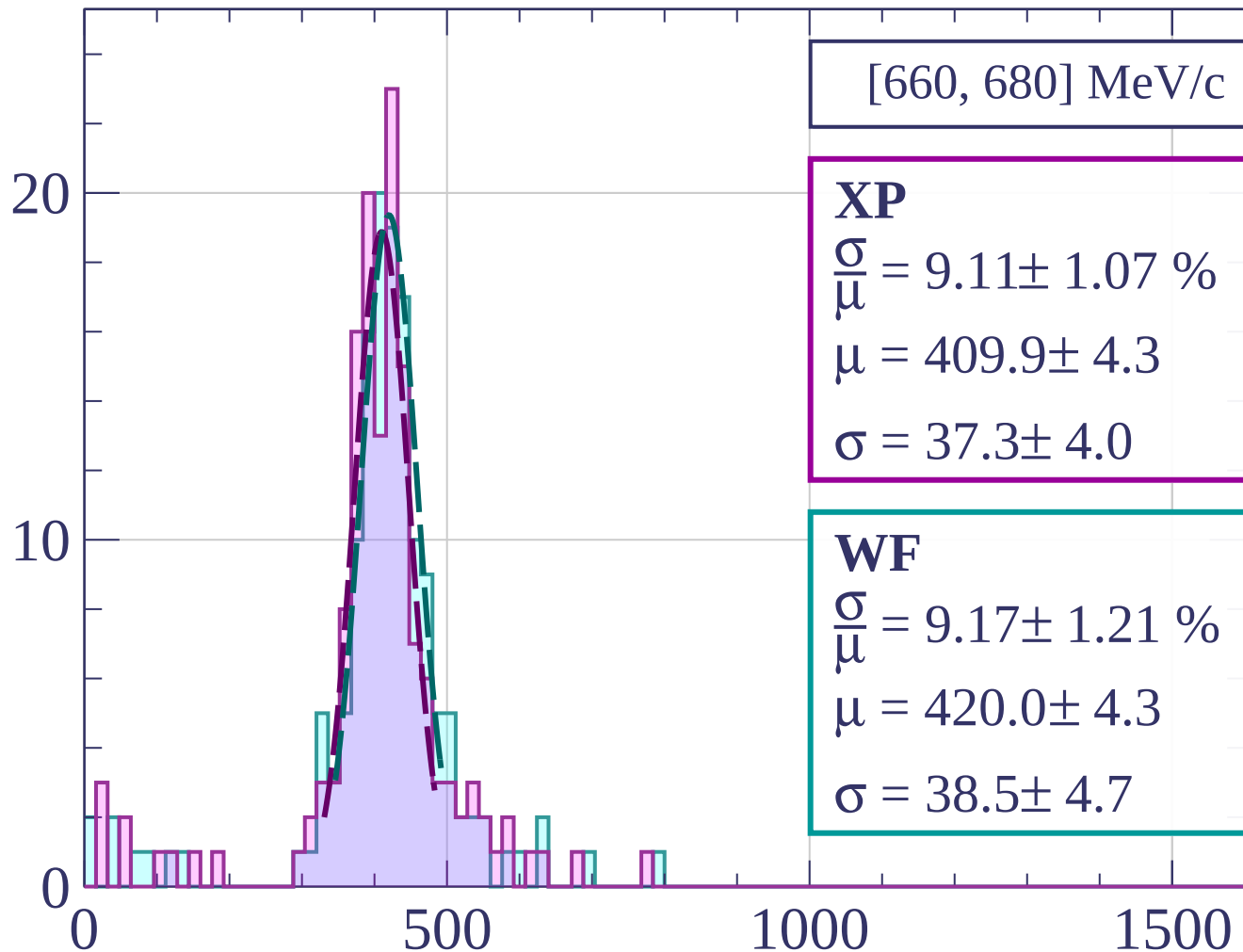
$$\frac{\sigma}{\mu} = 8.23 \pm 1.17 \%$$

$$\mu = 416.9 \pm 4.1$$

$$\sigma = 34.3 \pm 4.5$$

dE/dx [ADC counts/cm]

Count



[660, 680] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.11 \pm 1.07 \%$$

$$\mu = 409.9 \pm 4.3$$

$$\sigma = 37.3 \pm 4.0$$

WF

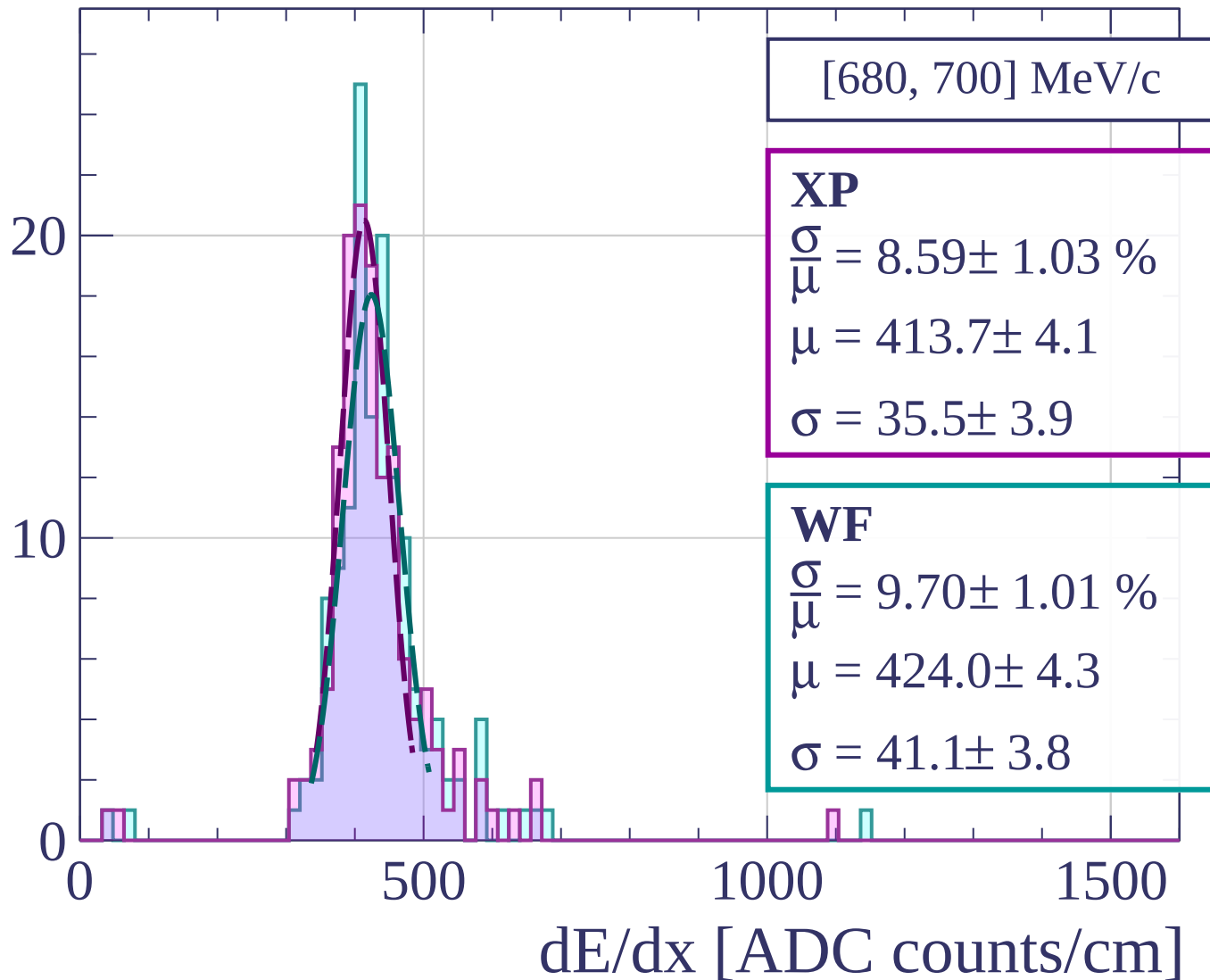
$$\frac{\sigma}{\mu} = 9.17 \pm 1.21 \%$$

$$\mu = 420.0 \pm 4.3$$

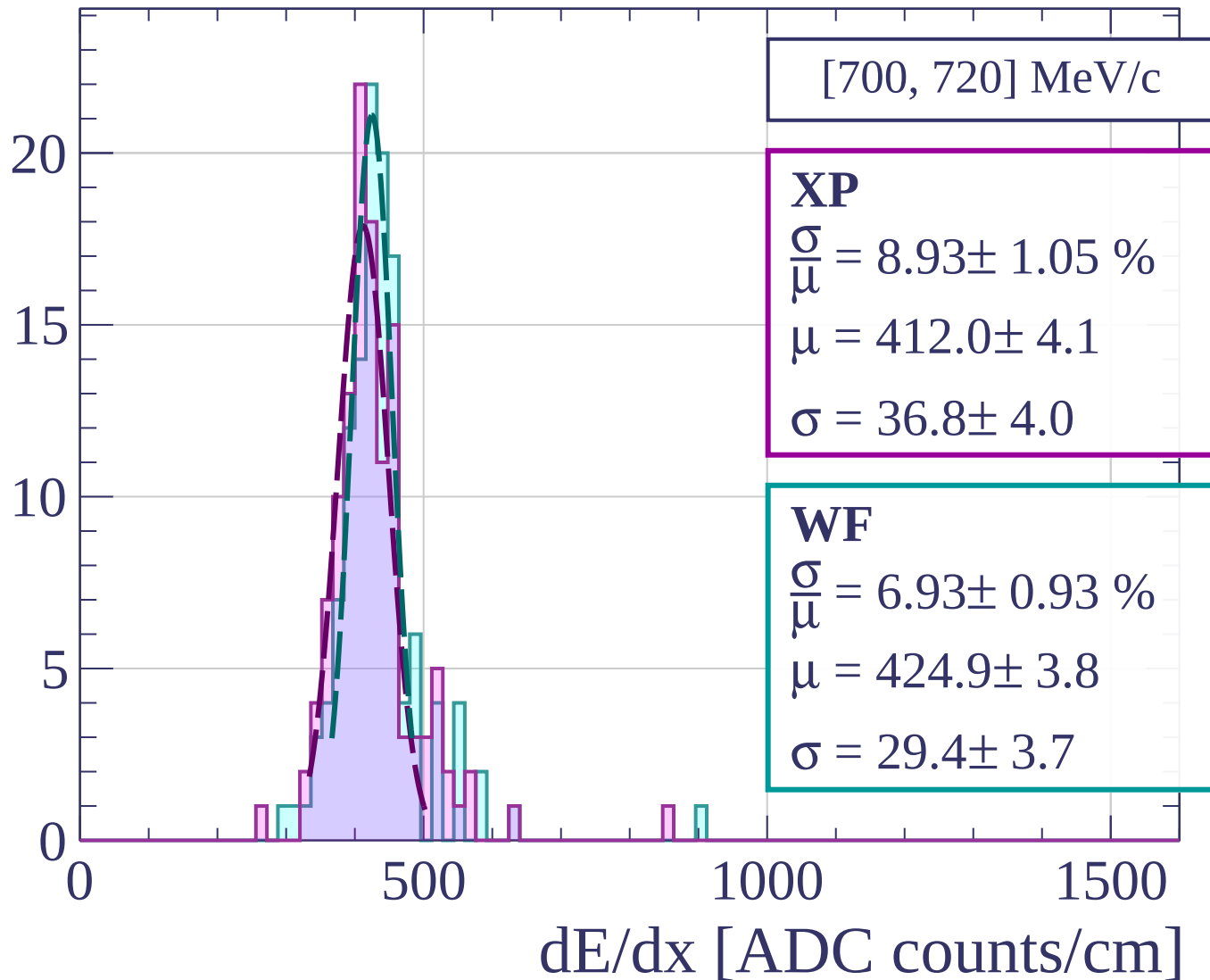
$$\sigma = 38.5 \pm 4.7$$

dE/dx [ADC counts/cm]

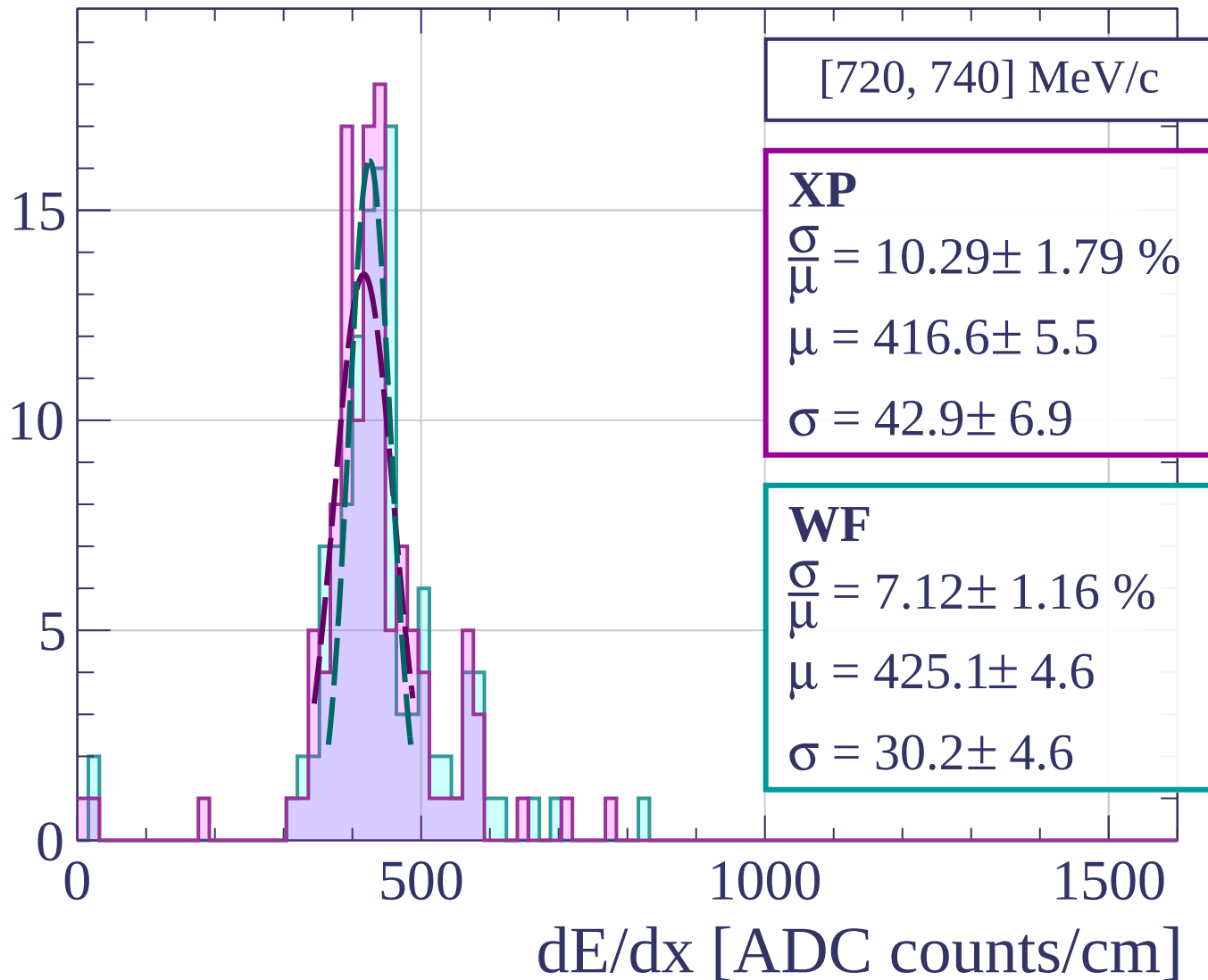
Count



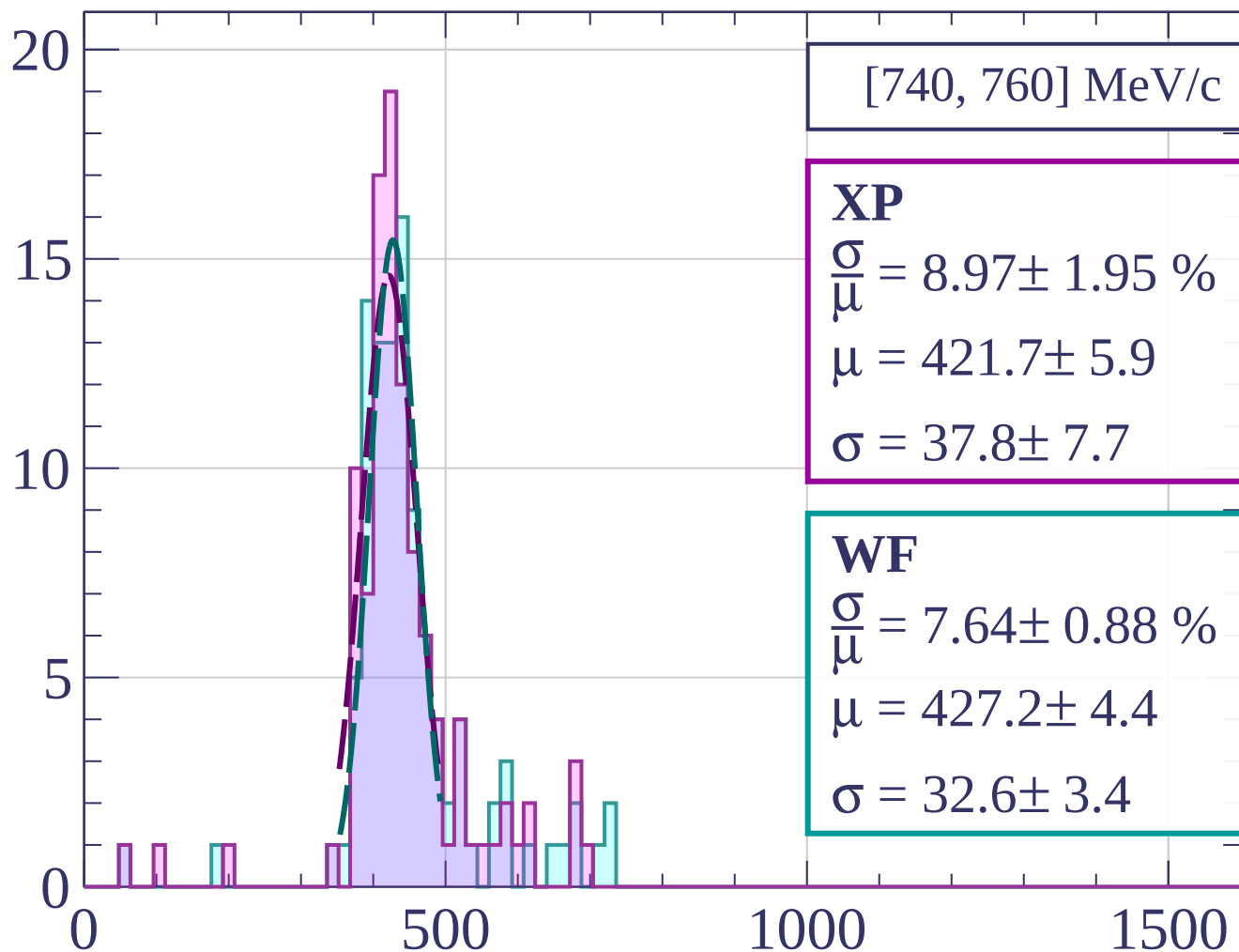
Count



Count



Count



[740, 760] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.97 \pm 1.95 \%$$

$$\mu = 421.7 \pm 5.9$$

$$\sigma = 37.8 \pm 7.7$$

WF

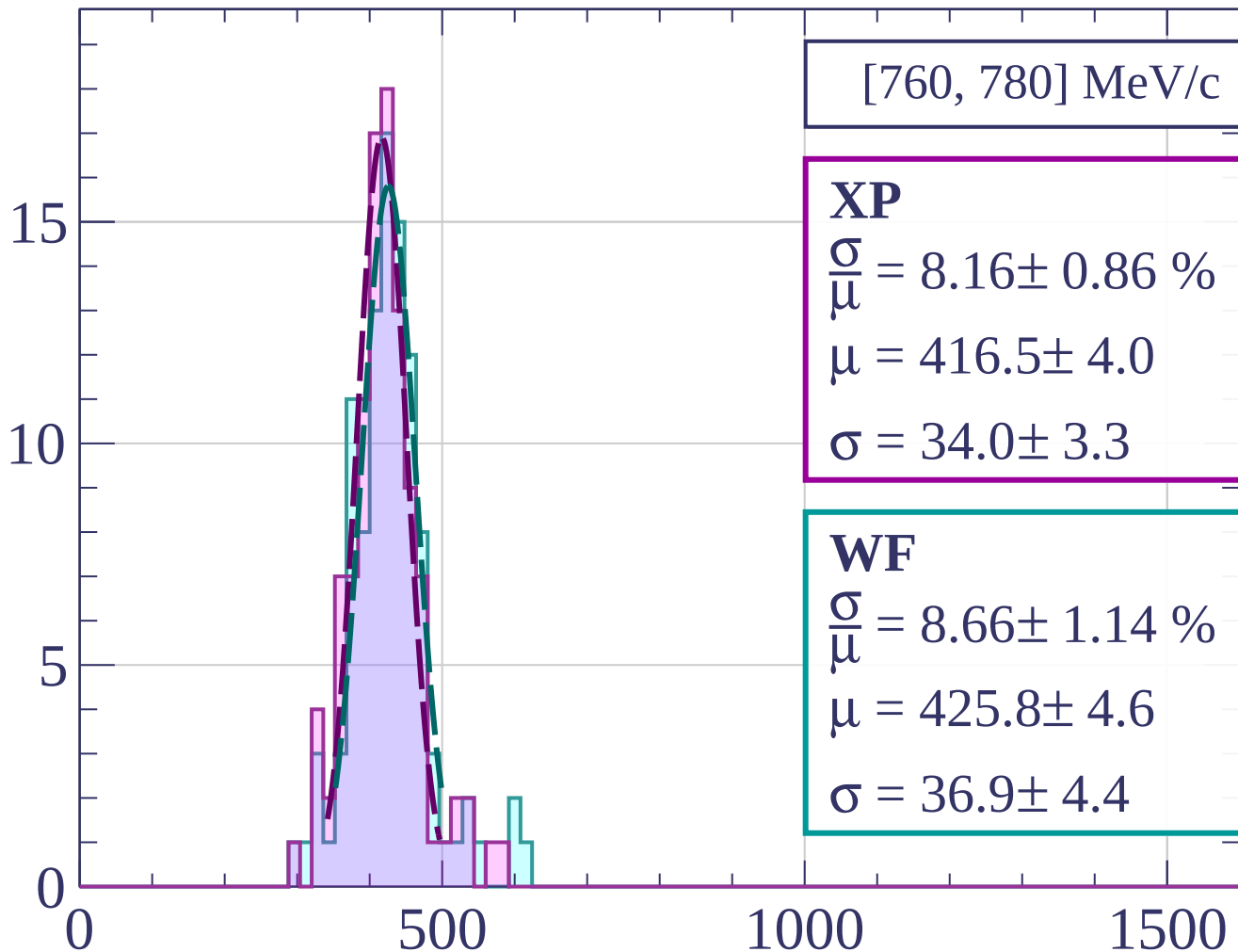
$$\frac{\sigma}{\mu} = 7.64 \pm 0.88 \%$$

$$\mu = 427.2 \pm 4.4$$

$$\sigma = 32.6 \pm 3.4$$

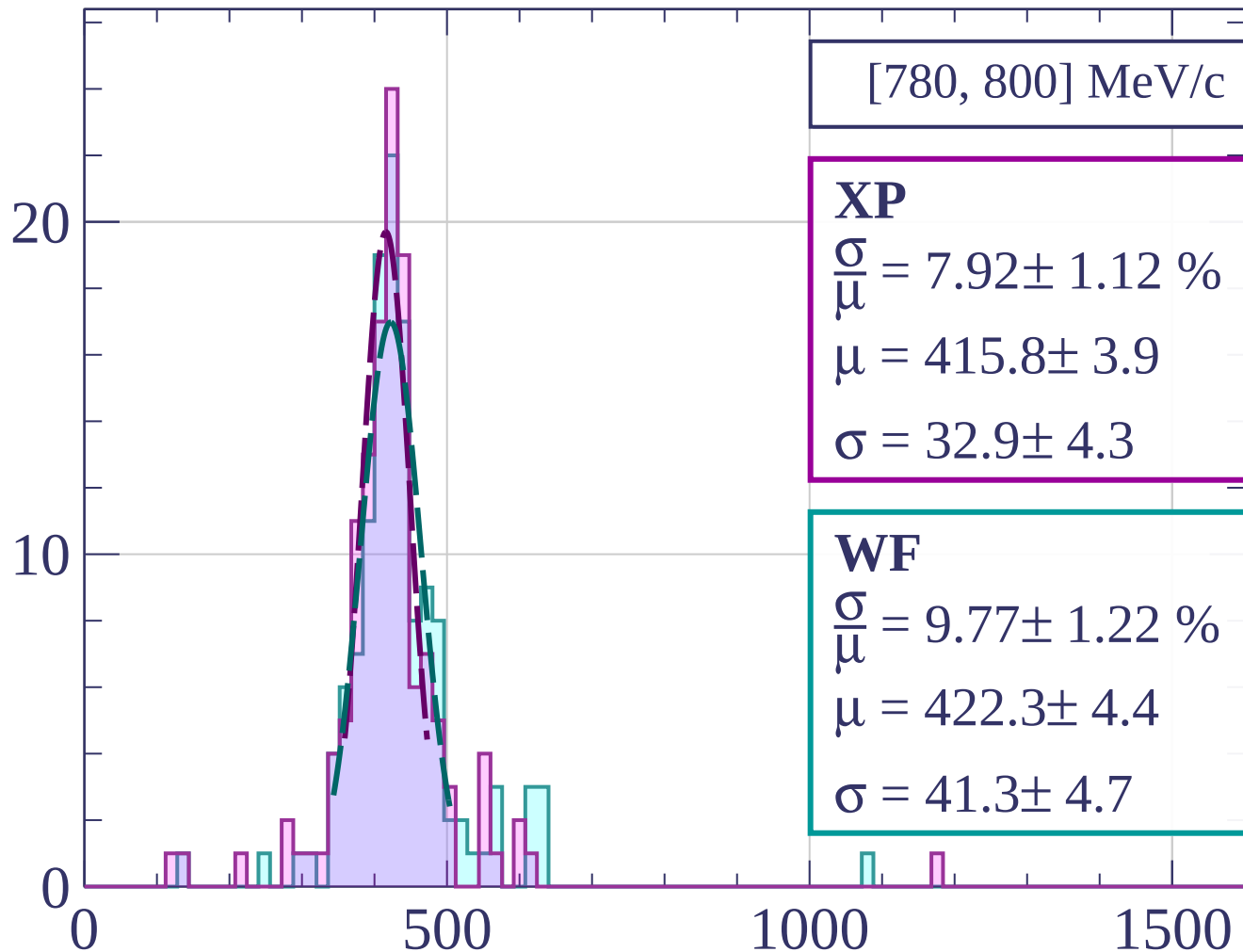
dE/dx [ADC counts/cm]

Count



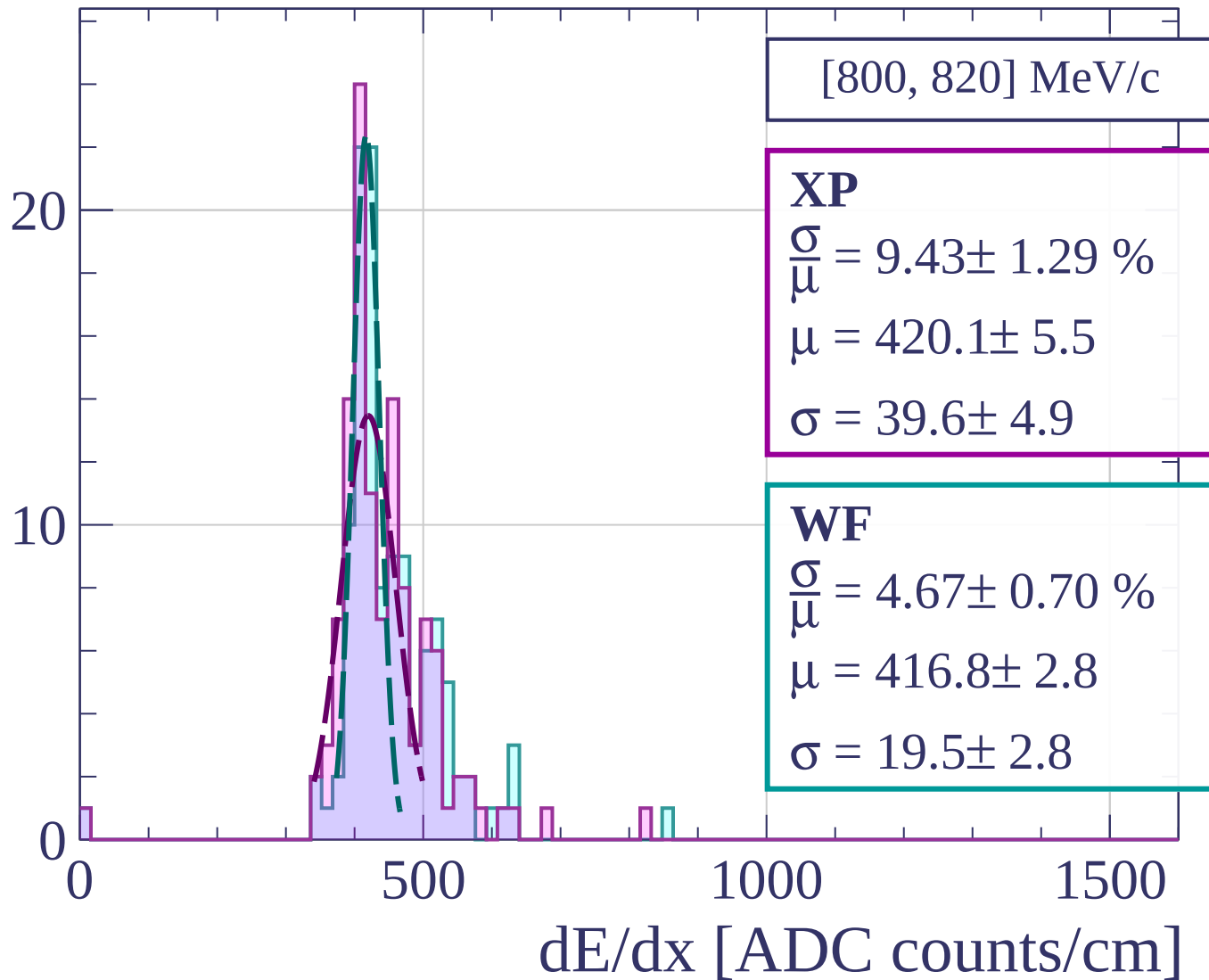
dE/dx [ADC counts/cm]

Count

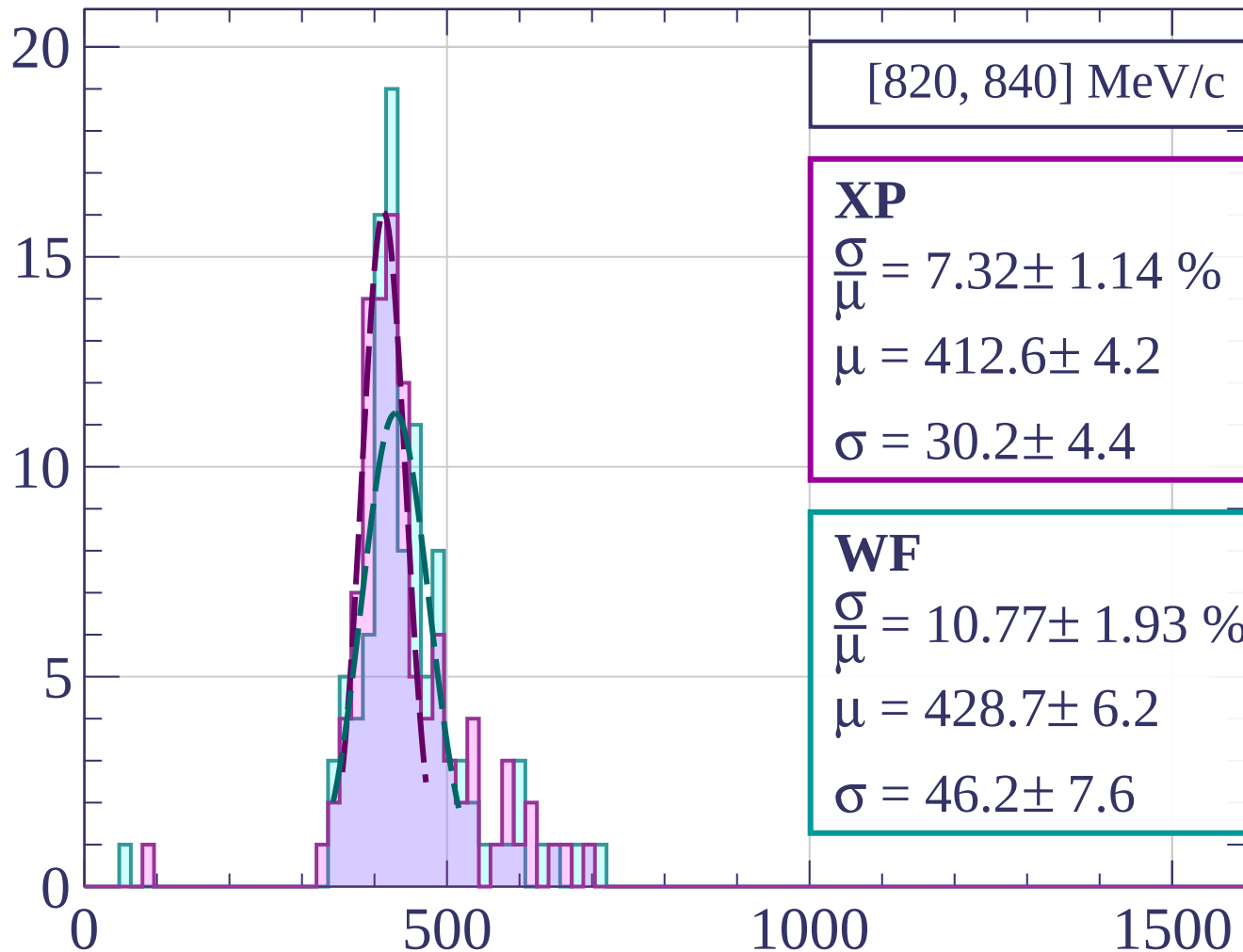


dE/dx [ADC counts/cm]

Count



Count



[820, 840] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.32 \pm 1.14 \%$$

$$\mu = 412.6 \pm 4.2$$

$$\sigma = 30.2 \pm 4.4$$

WF

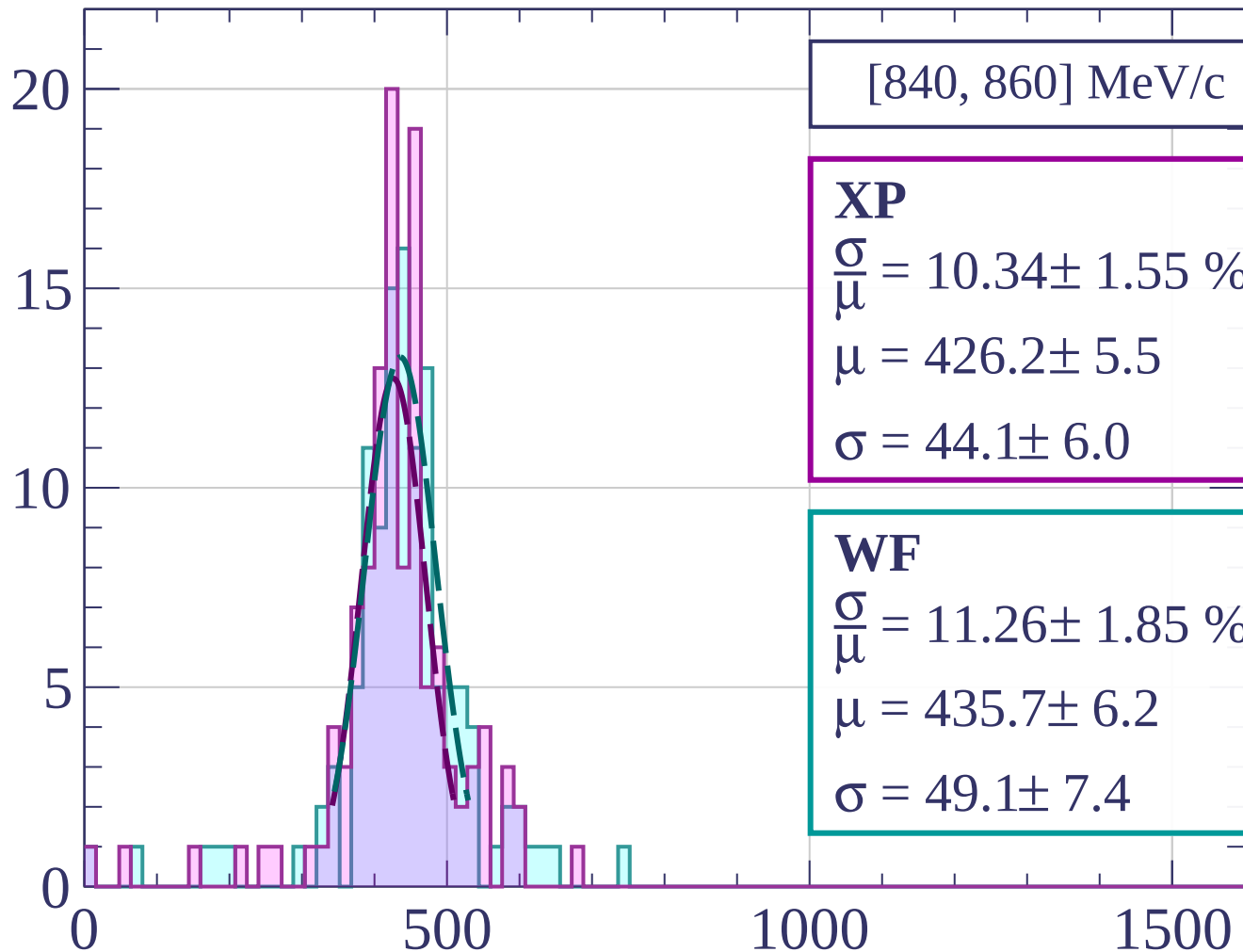
$$\frac{\sigma}{\mu} = 10.77 \pm 1.93 \%$$

$$\mu = 428.7 \pm 6.2$$

$$\sigma = 46.2 \pm 7.6$$

dE/dx [ADC counts/cm]

Count



[840, 860] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.34 \pm 1.55 \%$$

$$\mu = 426.2 \pm 5.5$$

$$\sigma = 44.1 \pm 6.0$$

WF

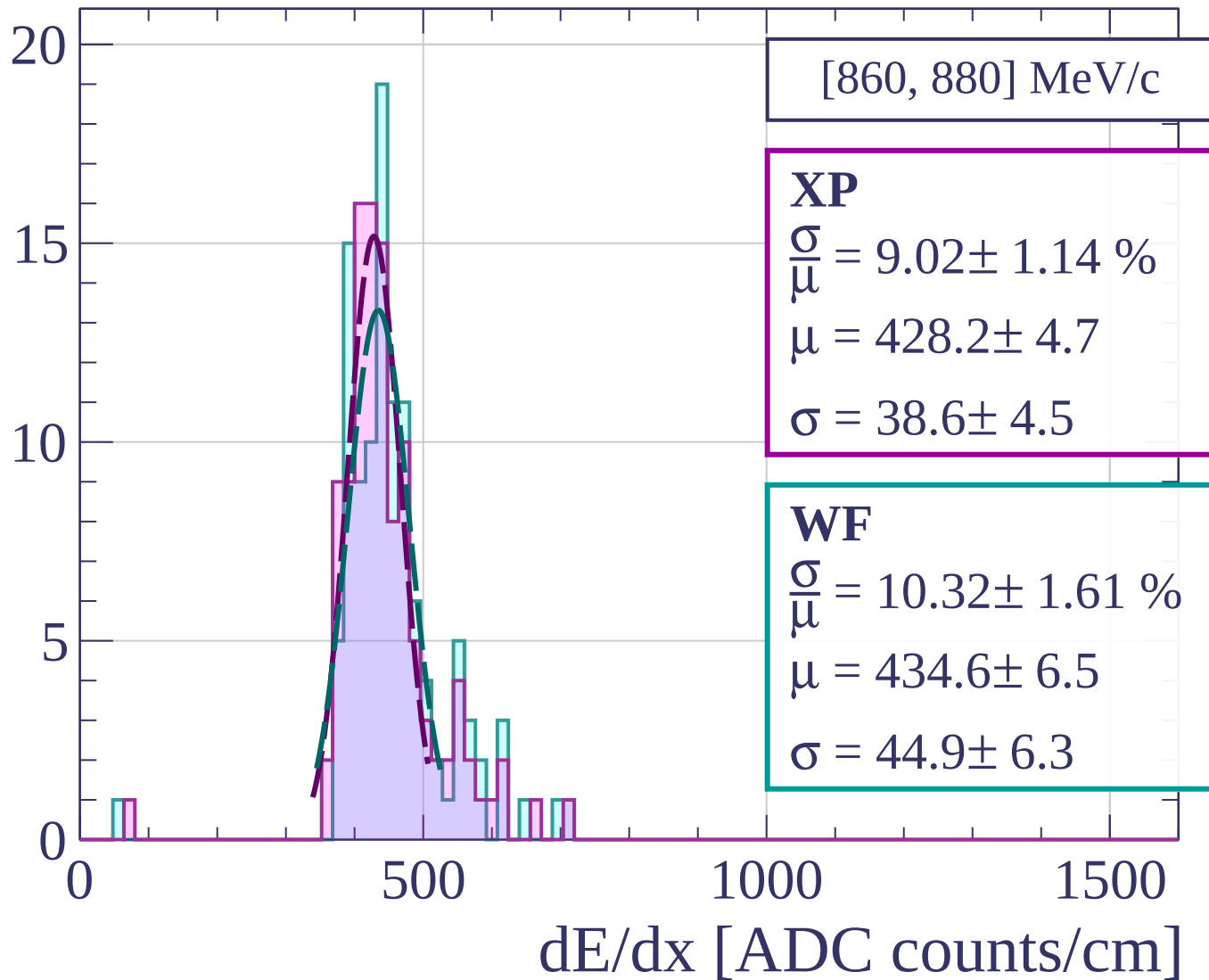
$$\frac{\sigma}{\mu} = 11.26 \pm 1.85 \%$$

$$\mu = 435.7 \pm 6.2$$

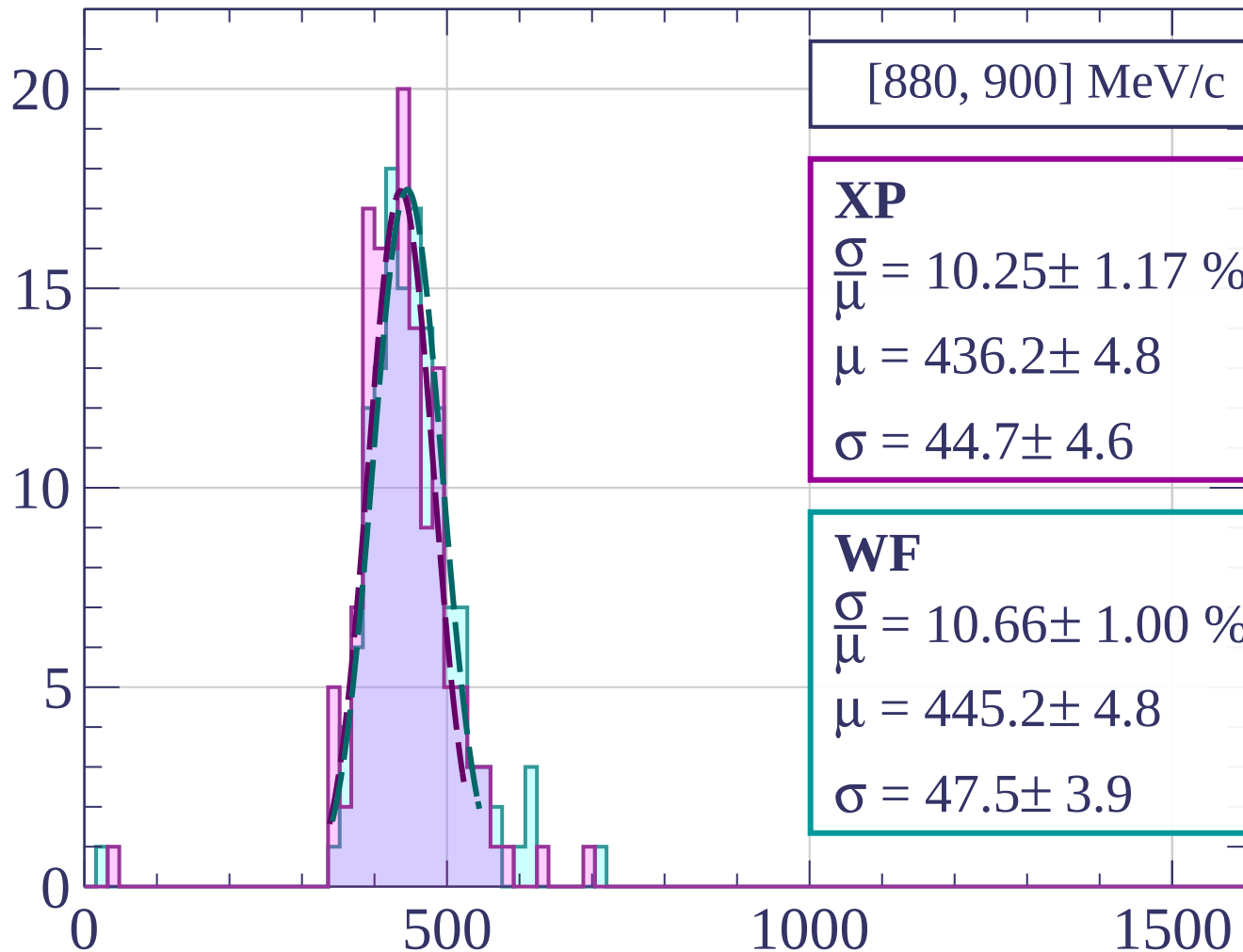
$$\sigma = 49.1 \pm 7.4$$

dE/dx [ADC counts/cm]

Count

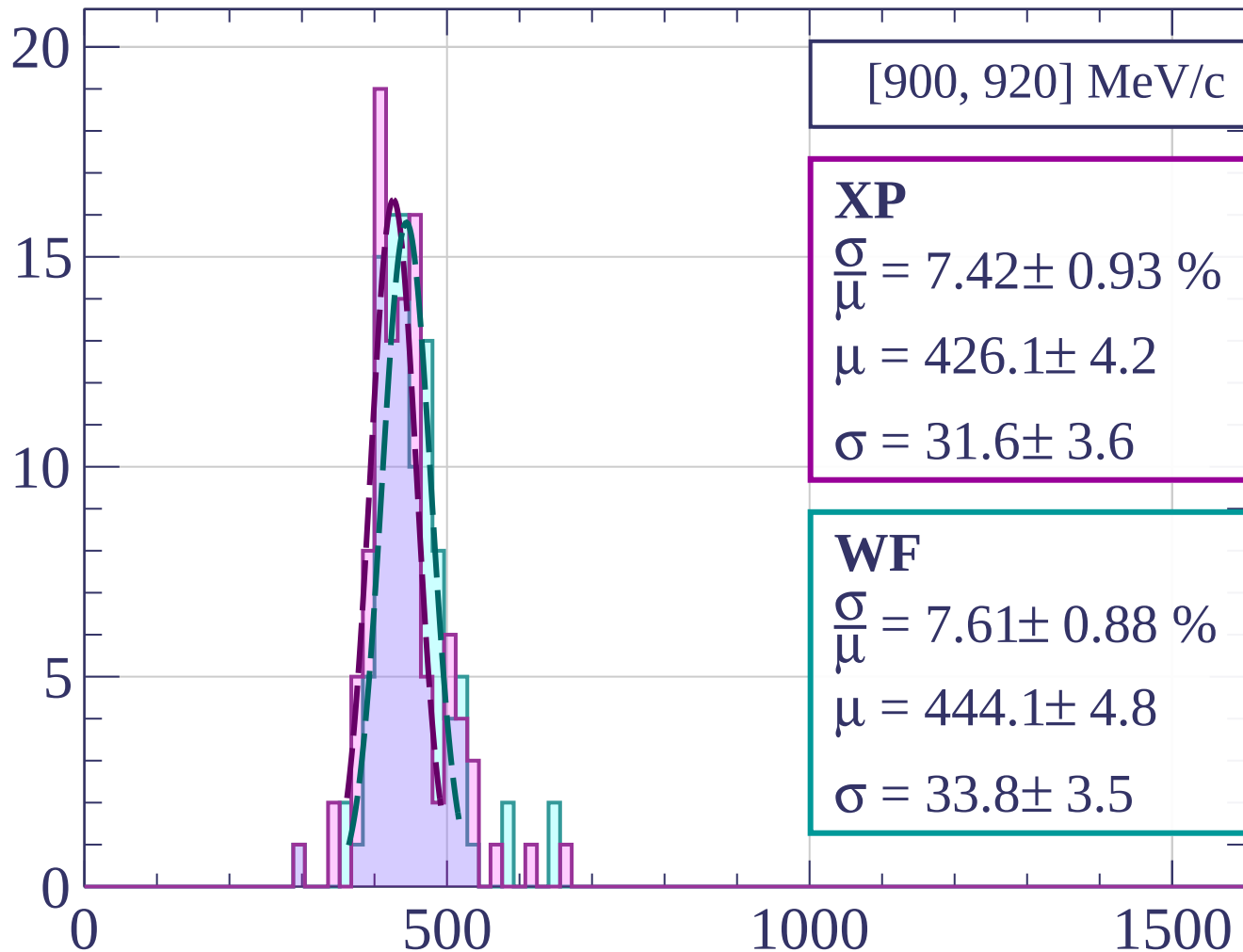


Count



dE/dx [ADC counts/cm]

Count



[900, 920] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.42 \pm 0.93 \%$$

$$\mu = 426.1 \pm 4.2$$

$$\sigma = 31.6 \pm 3.6$$

WF

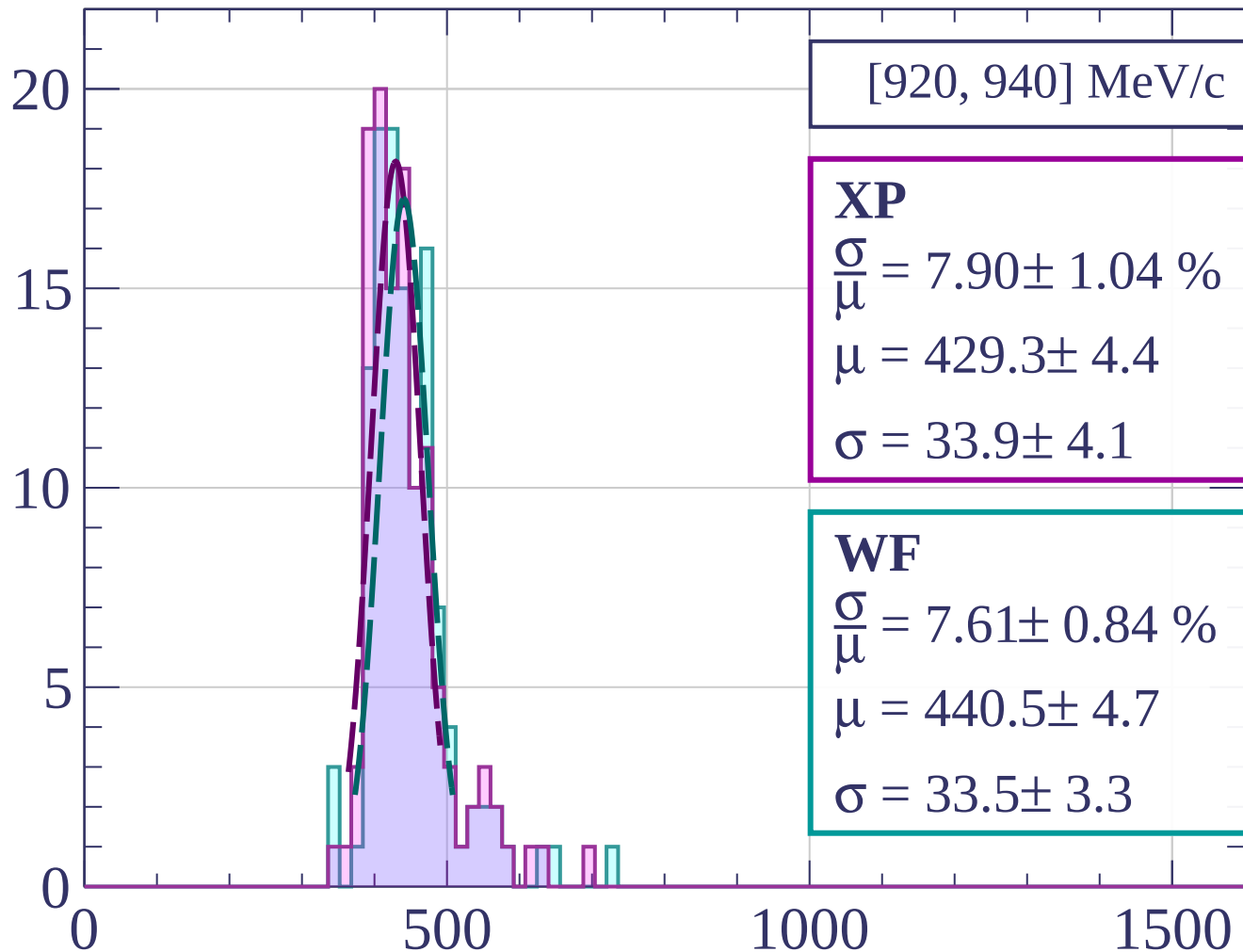
$$\frac{\sigma}{\mu} = 7.61 \pm 0.88 \%$$

$$\mu = 444.1 \pm 4.8$$

$$\sigma = 33.8 \pm 3.5$$

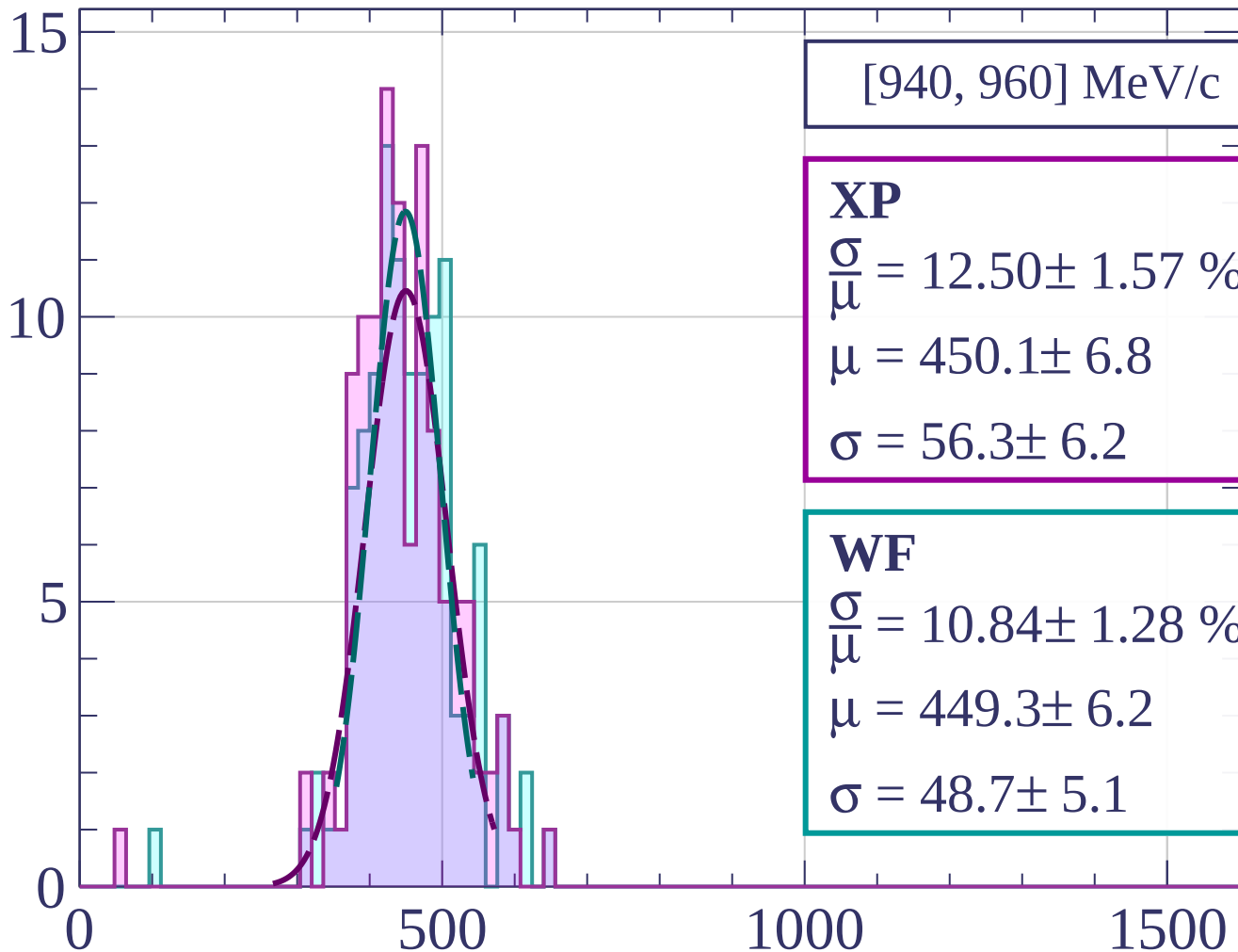
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[940, 960] MeV/c

XP

$$\frac{\sigma}{\mu} = 12.50 \pm 1.57 \%$$

$$\mu = 450.1 \pm 6.8$$

$$\sigma = 56.3 \pm 6.2$$

WF

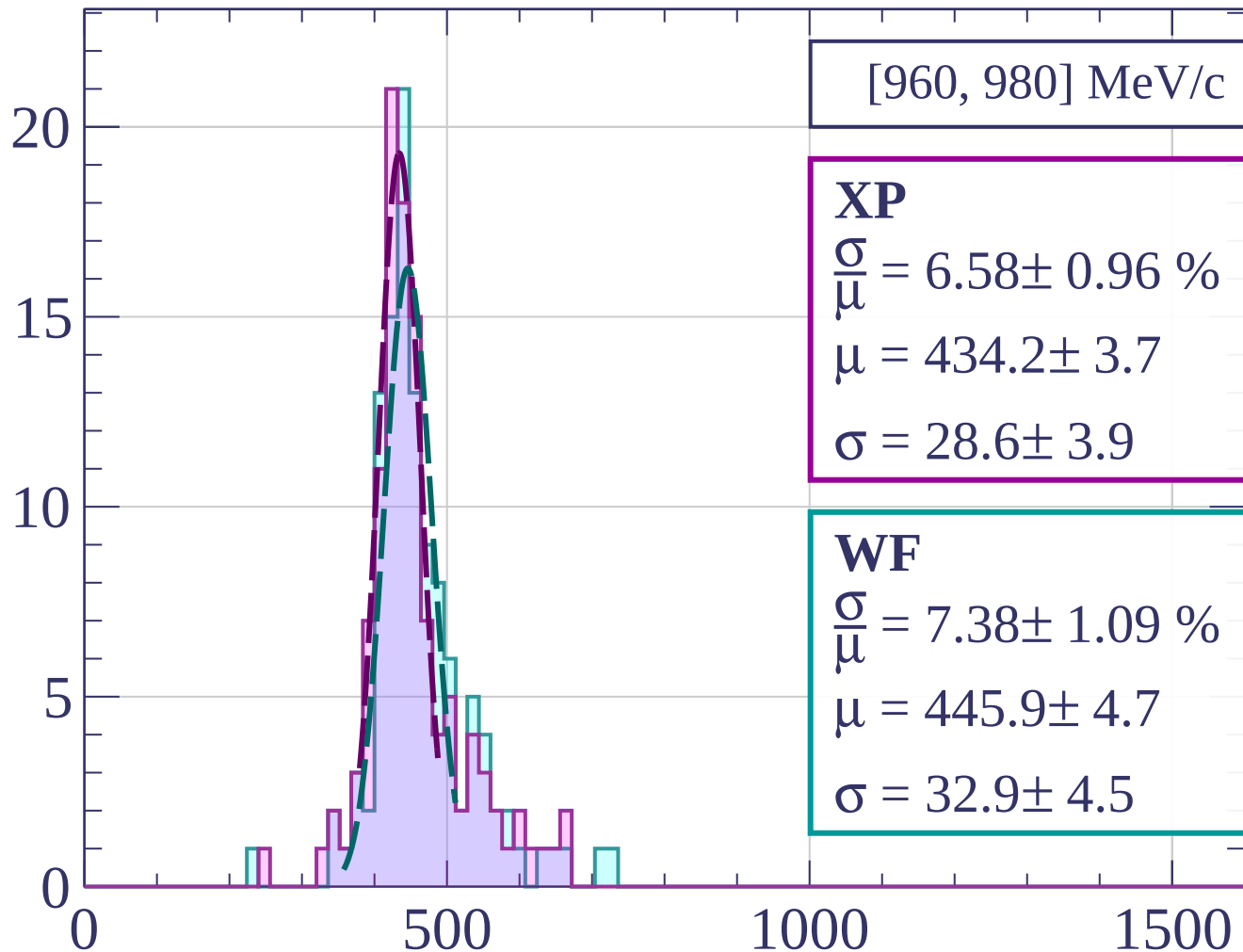
$$\frac{\sigma}{\mu} = 10.84 \pm 1.28 \%$$

$$\mu = 449.3 \pm 6.2$$

$$\sigma = 48.7 \pm 5.1$$

dE/dx [ADC counts/cm]

Count



[960, 980] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.58 \pm 0.96 \%$$

$$\mu = 434.2 \pm 3.7$$

$$\sigma = 28.6 \pm 3.9$$

WF

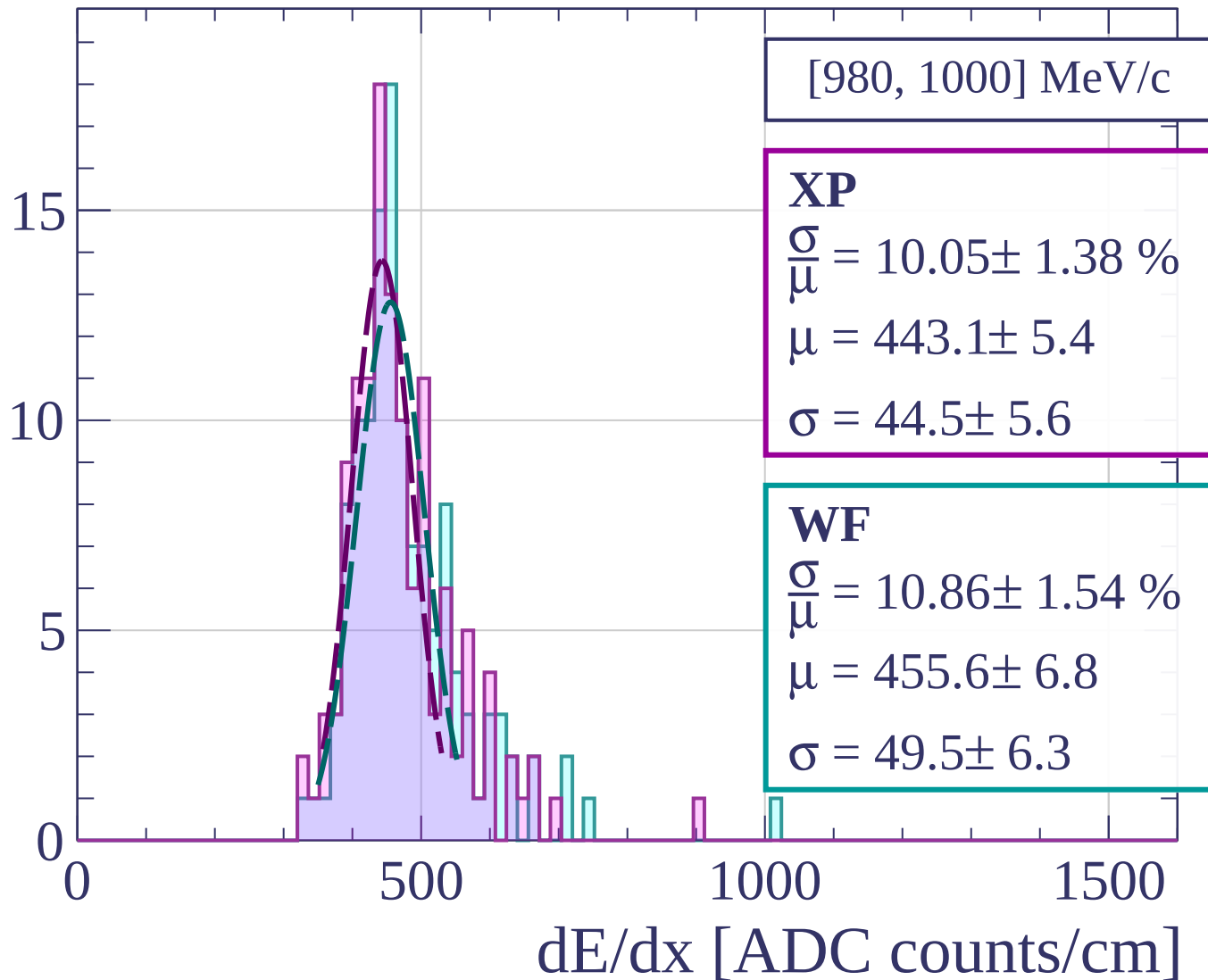
$$\frac{\sigma}{\mu} = 7.38 \pm 1.09 \%$$

$$\mu = 445.9 \pm 4.7$$

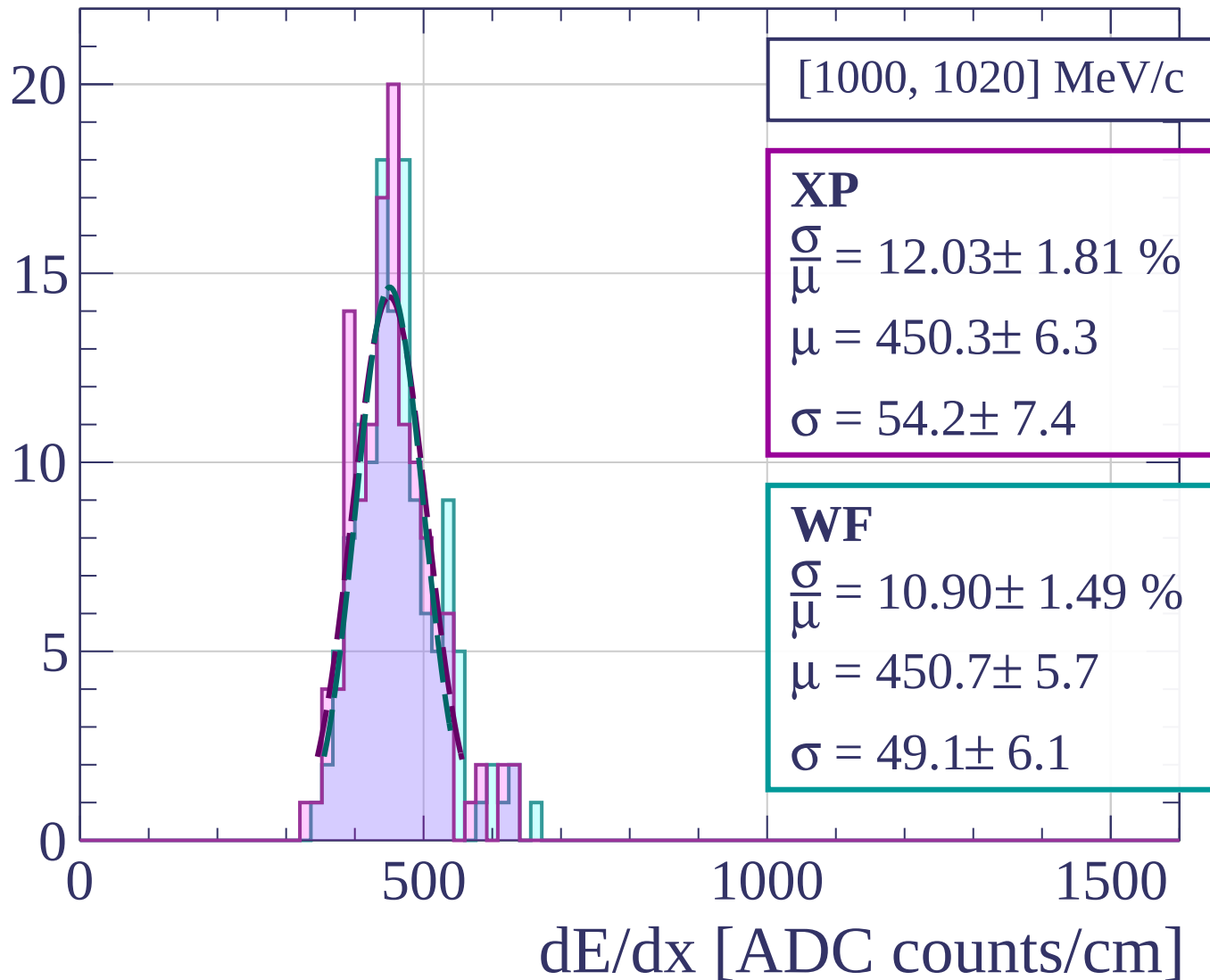
$$\sigma = 32.9 \pm 4.5$$

dE/dx [ADC counts/cm]

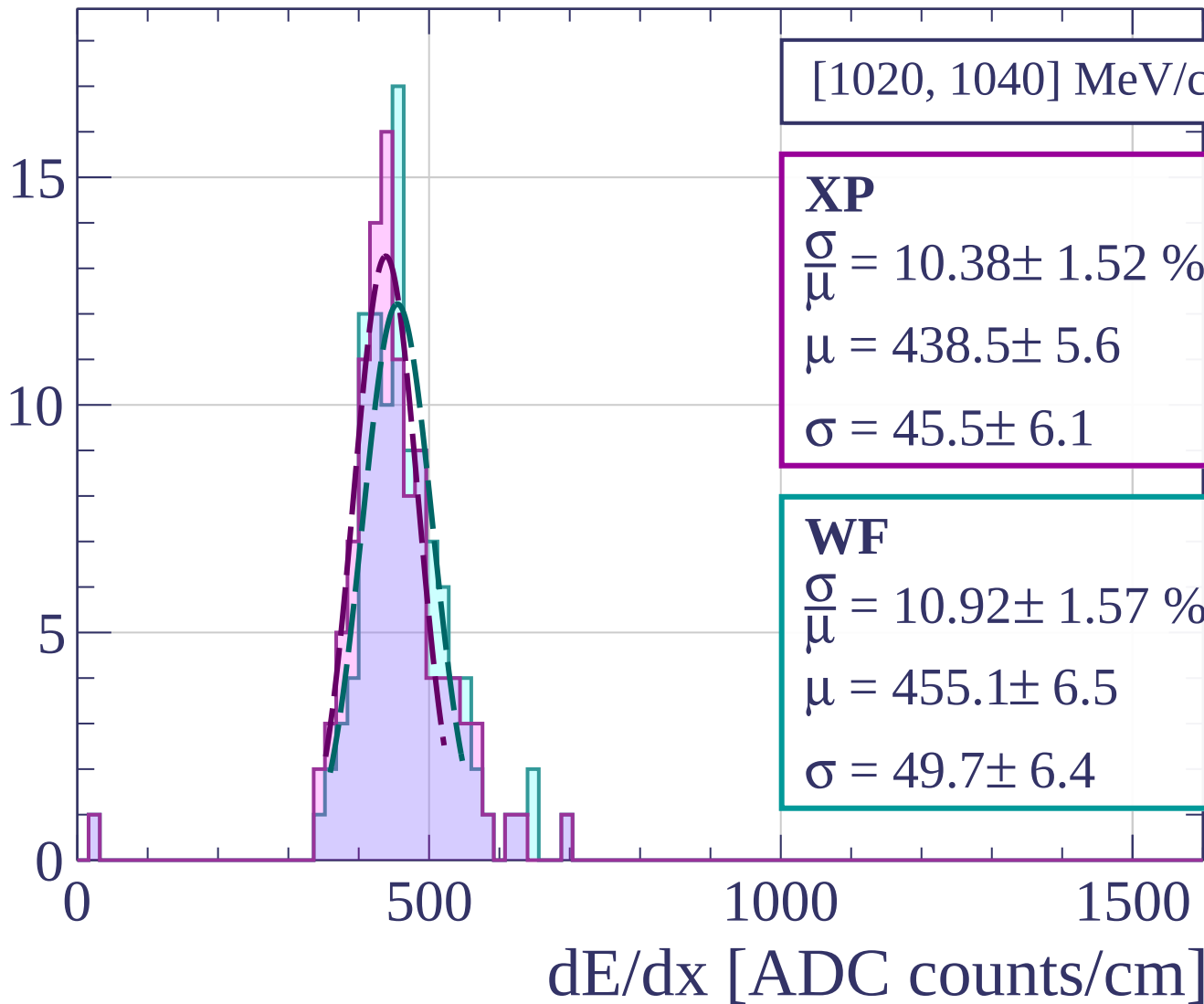
Count



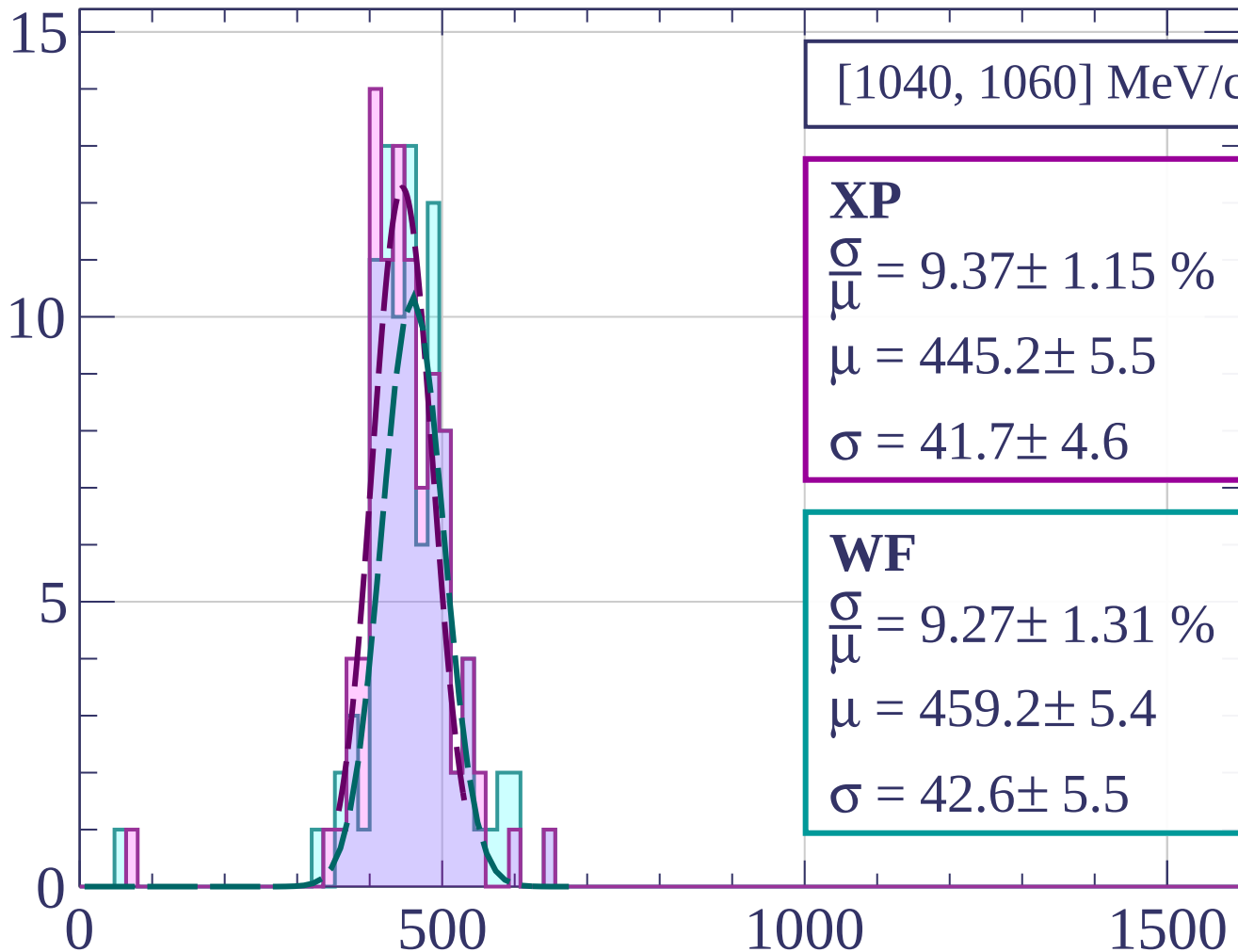
Count



Count



Count



[1040, 1060] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.37 \pm 1.15 \%$$

$$\mu = 445.2 \pm 5.5$$

$$\sigma = 41.7 \pm 4.6$$

WF

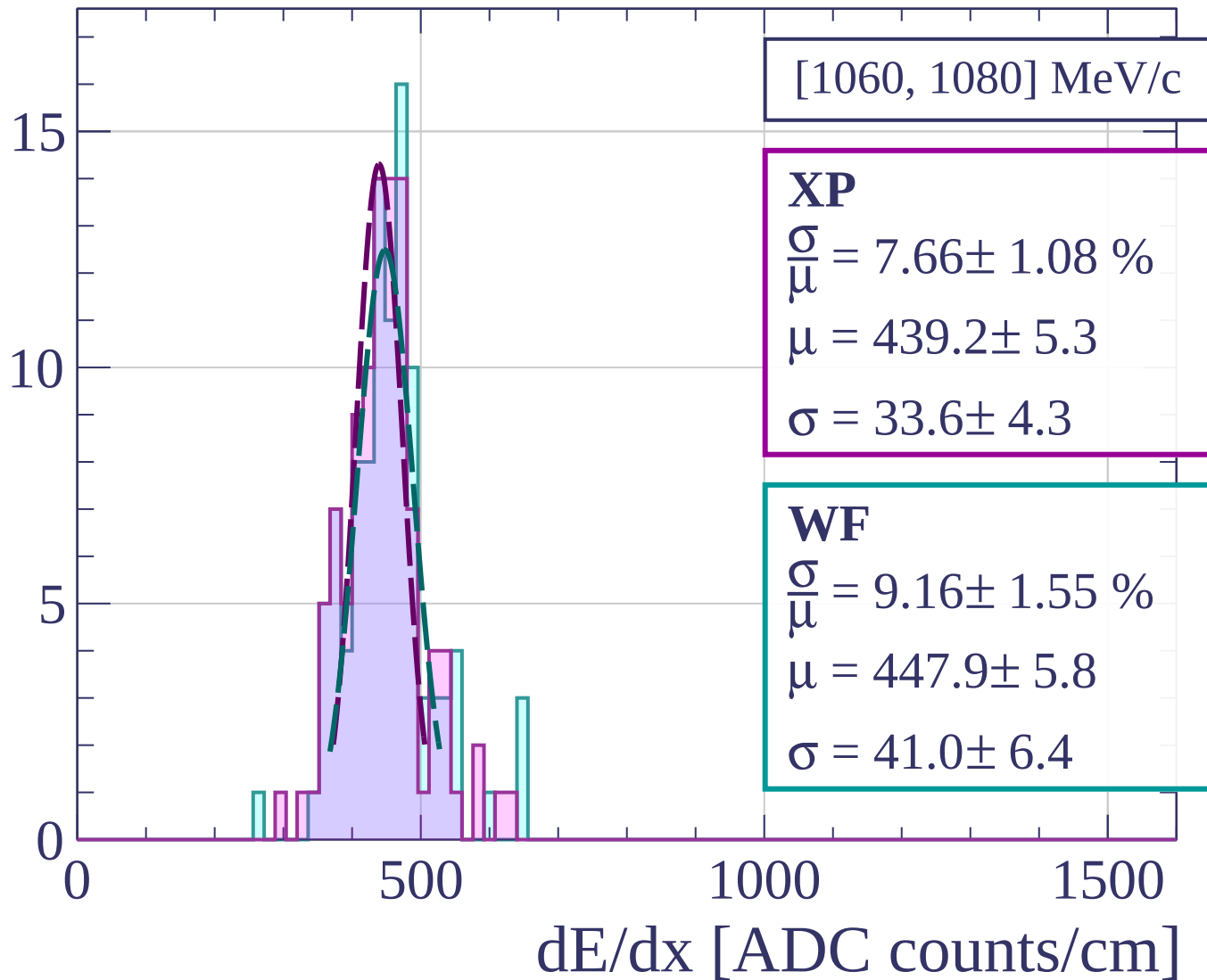
$$\frac{\sigma}{\mu} = 9.27 \pm 1.31 \%$$

$$\mu = 459.2 \pm 5.4$$

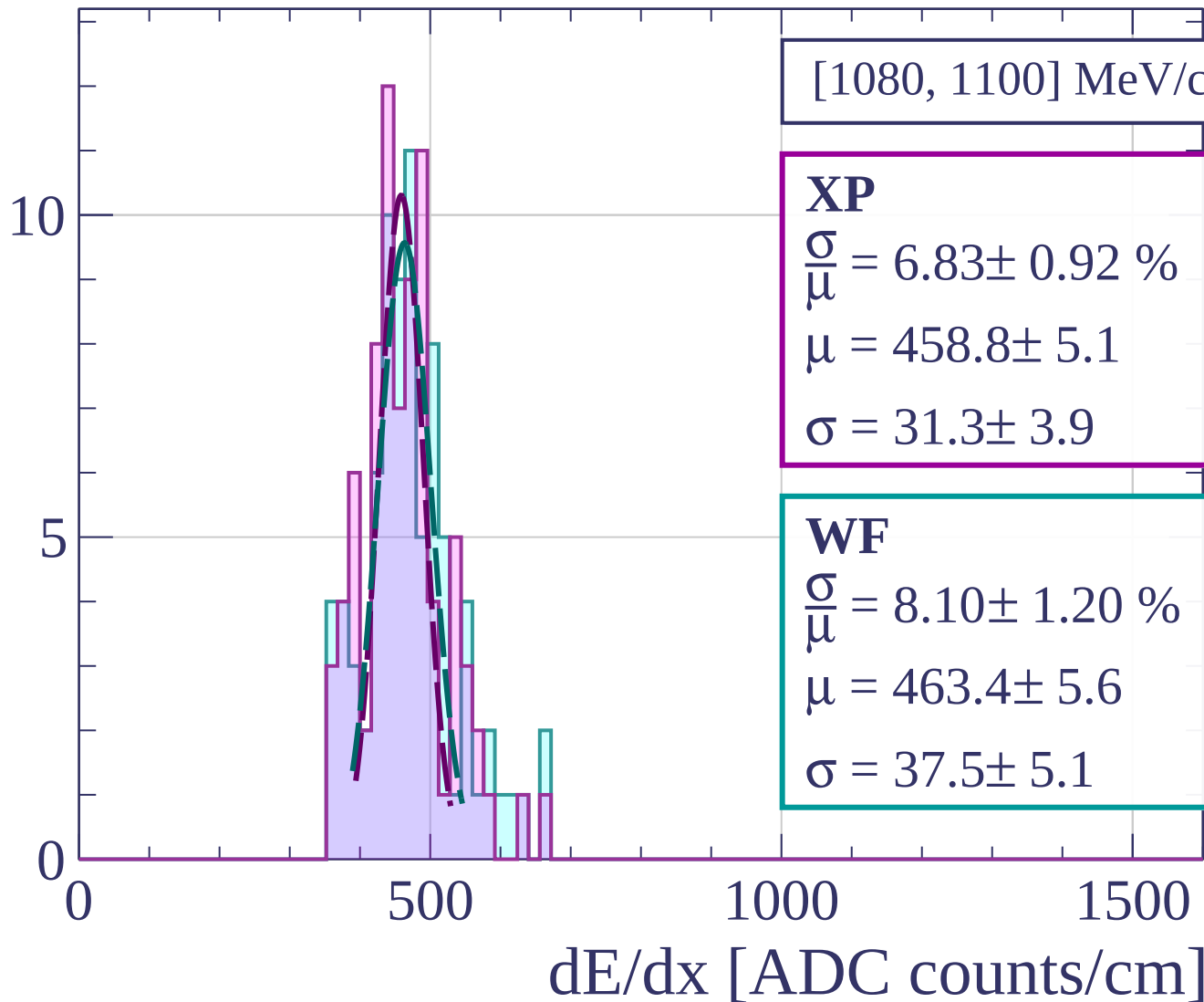
$$\sigma = 42.6 \pm 5.5$$

dE/dx [ADC counts/cm]

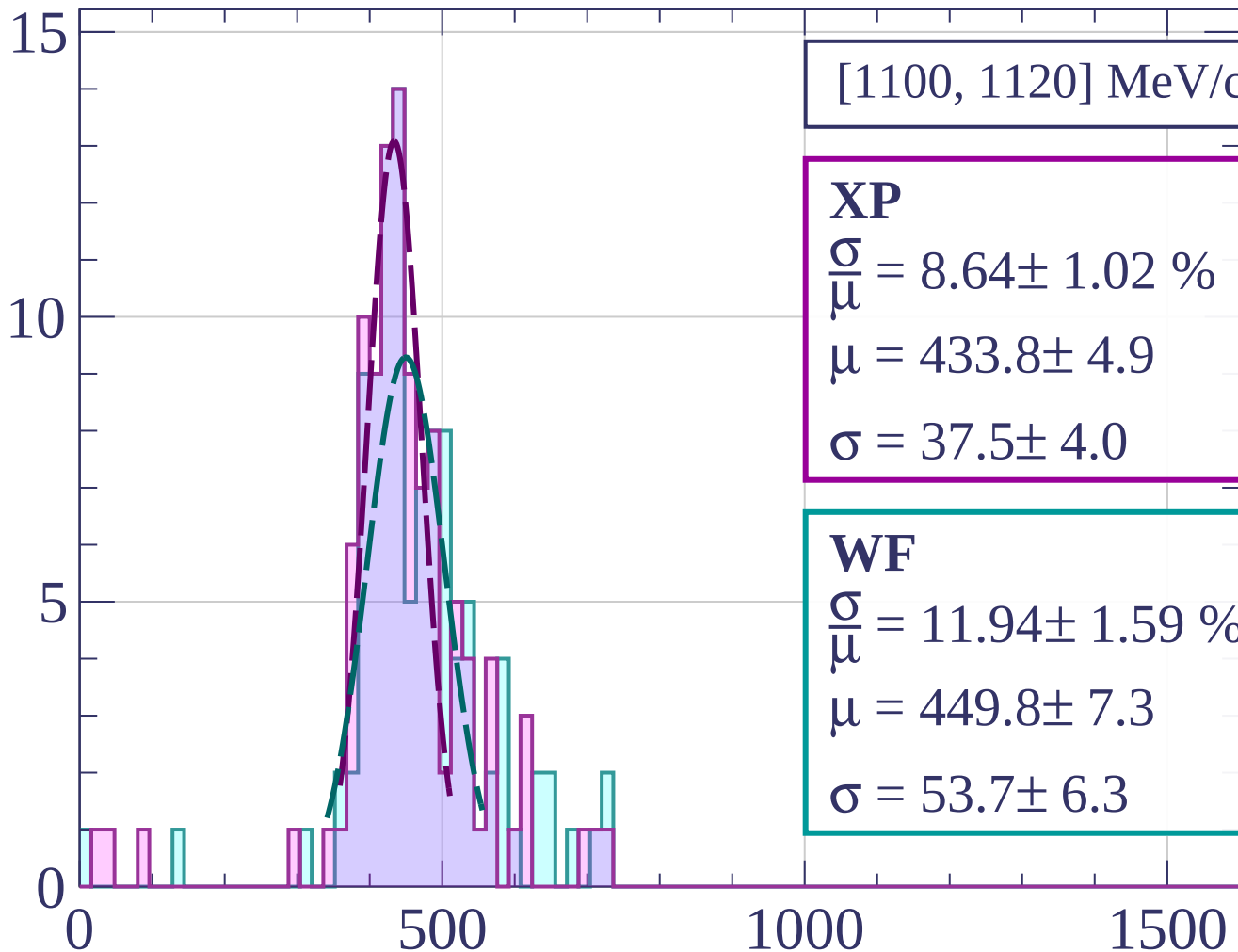
Count



Count



Count



[1100, 1120] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.64 \pm 1.02 \%$$

$$\mu = 433.8 \pm 4.9$$

$$\sigma = 37.5 \pm 4.0$$

WF

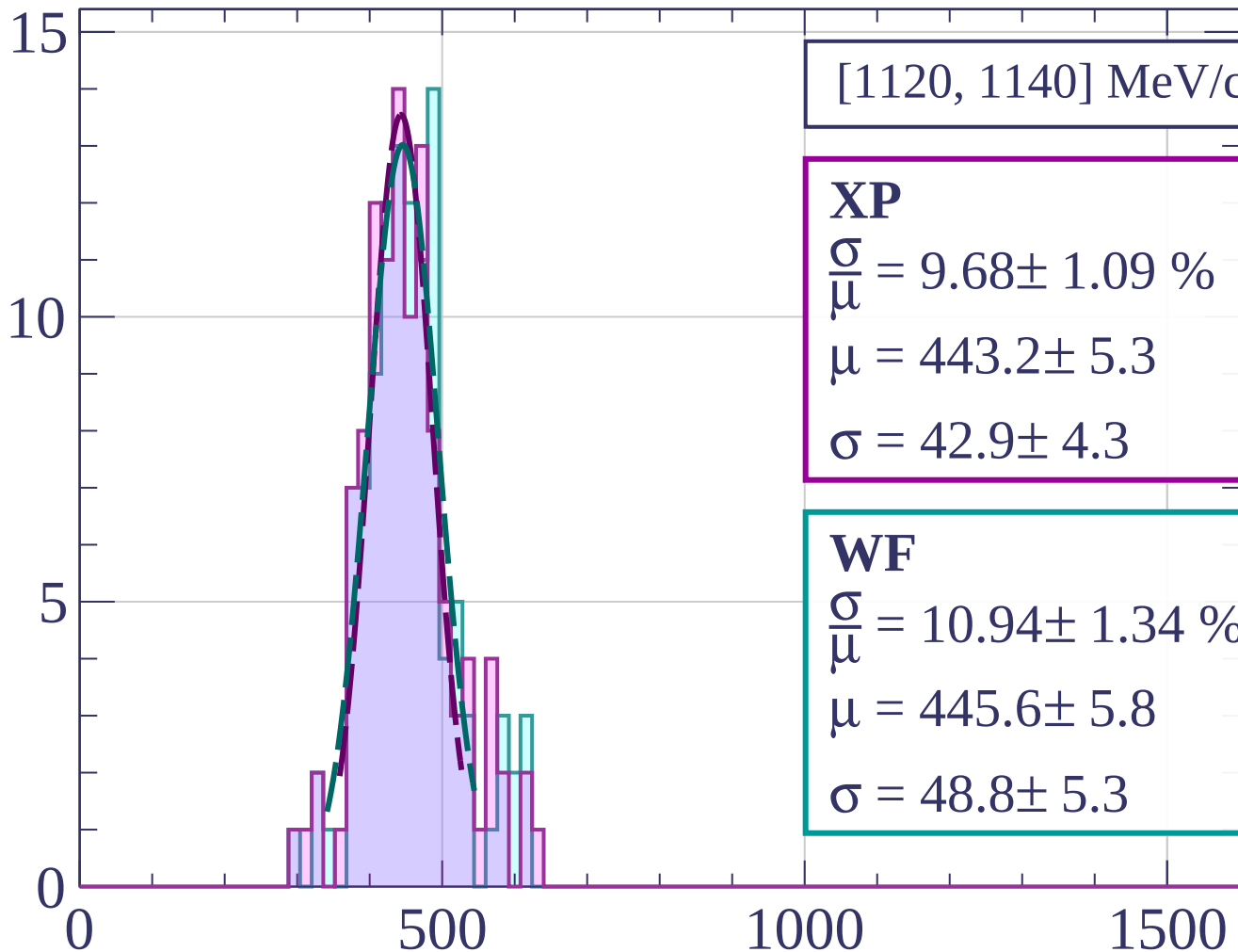
$$\frac{\sigma}{\mu} = 11.94 \pm 1.59 \%$$

$$\mu = 449.8 \pm 7.3$$

$$\sigma = 53.7 \pm 6.3$$

dE/dx [ADC counts/cm]

Count



[1120, 1140] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.68 \pm 1.09 \%$$

$$\mu = 443.2 \pm 5.3$$

$$\sigma = 42.9 \pm 4.3$$

WF

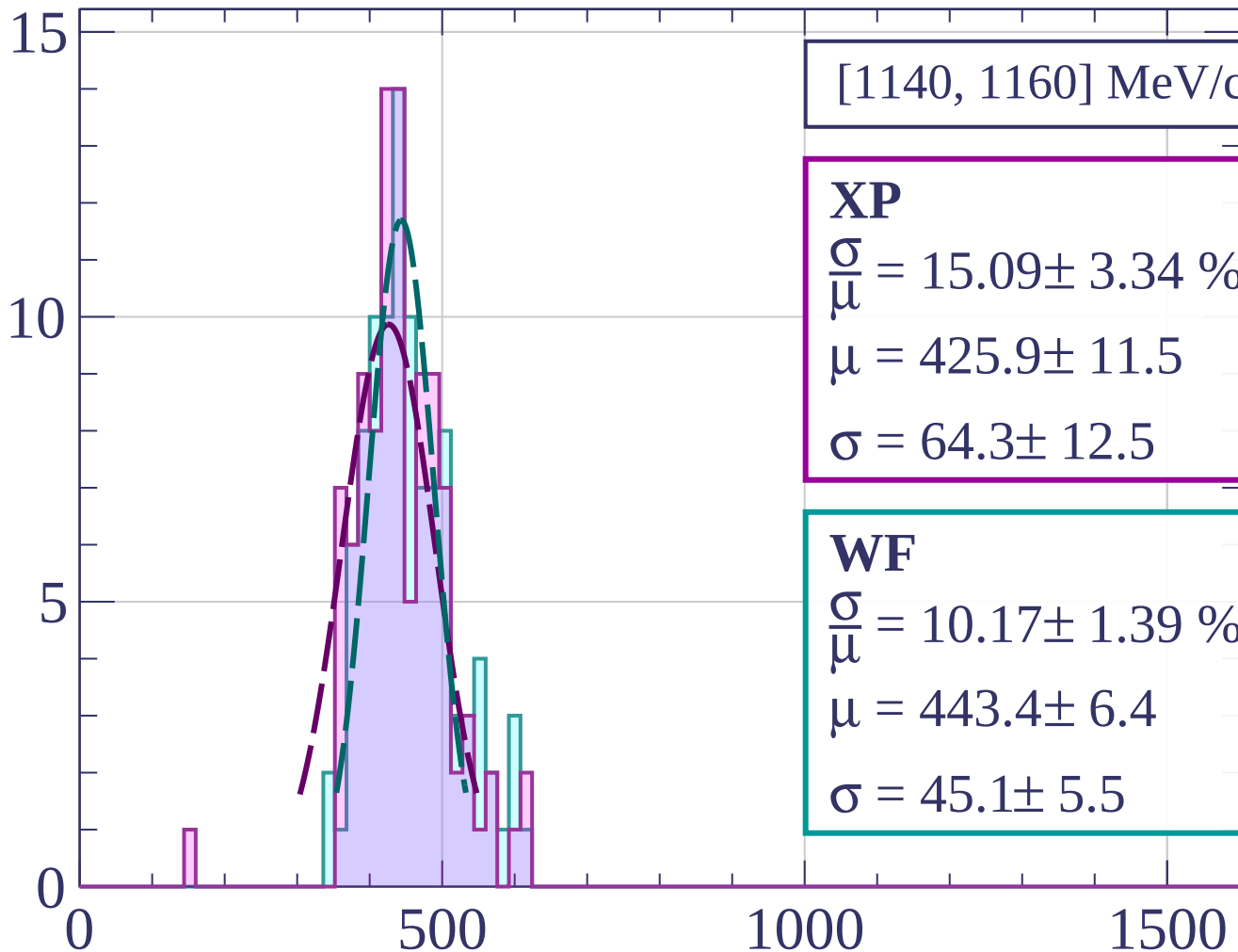
$$\frac{\sigma}{\mu} = 10.94 \pm 1.34 \%$$

$$\mu = 445.6 \pm 5.8$$

$$\sigma = 48.8 \pm 5.3$$

dE/dx [ADC counts/cm]

Count



[1140, 1160] MeV/c

XP

$$\frac{\sigma}{\mu} = 15.09 \pm 3.34 \%$$

$$\mu = 425.9 \pm 11.5$$

$$\sigma = 64.3 \pm 12.5$$

WF

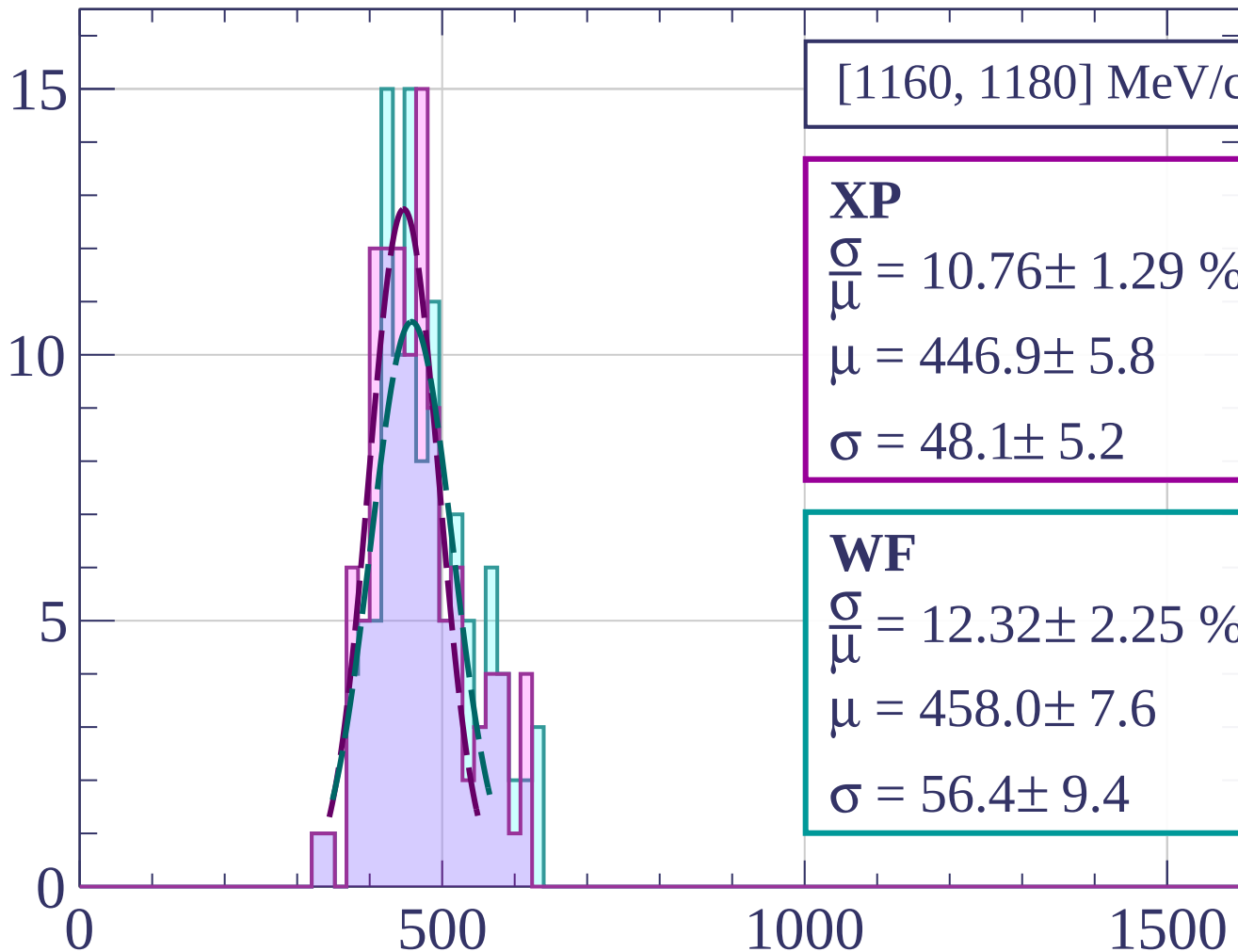
$$\frac{\sigma}{\mu} = 10.17 \pm 1.39 \%$$

$$\mu = 443.4 \pm 6.4$$

$$\sigma = 45.1 \pm 5.5$$

dE/dx [ADC counts/cm]

Count



[1160, 1180] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.76 \pm 1.29 \%$$

$$\mu = 446.9 \pm 5.8$$

$$\sigma = 48.1 \pm 5.2$$

WF

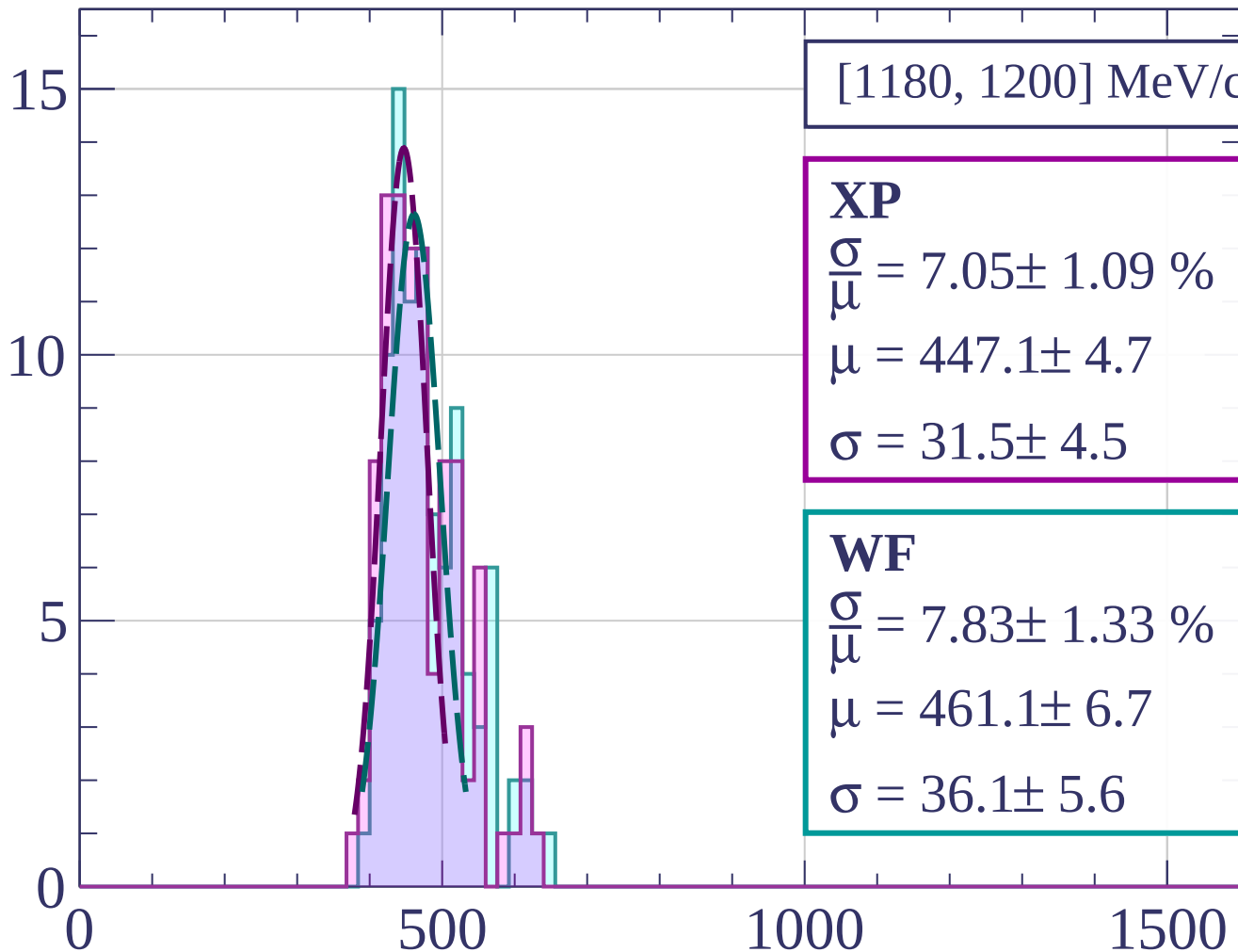
$$\frac{\sigma}{\mu} = 12.32 \pm 2.25 \%$$

$$\mu = 458.0 \pm 7.6$$

$$\sigma = 56.4 \pm 9.4$$

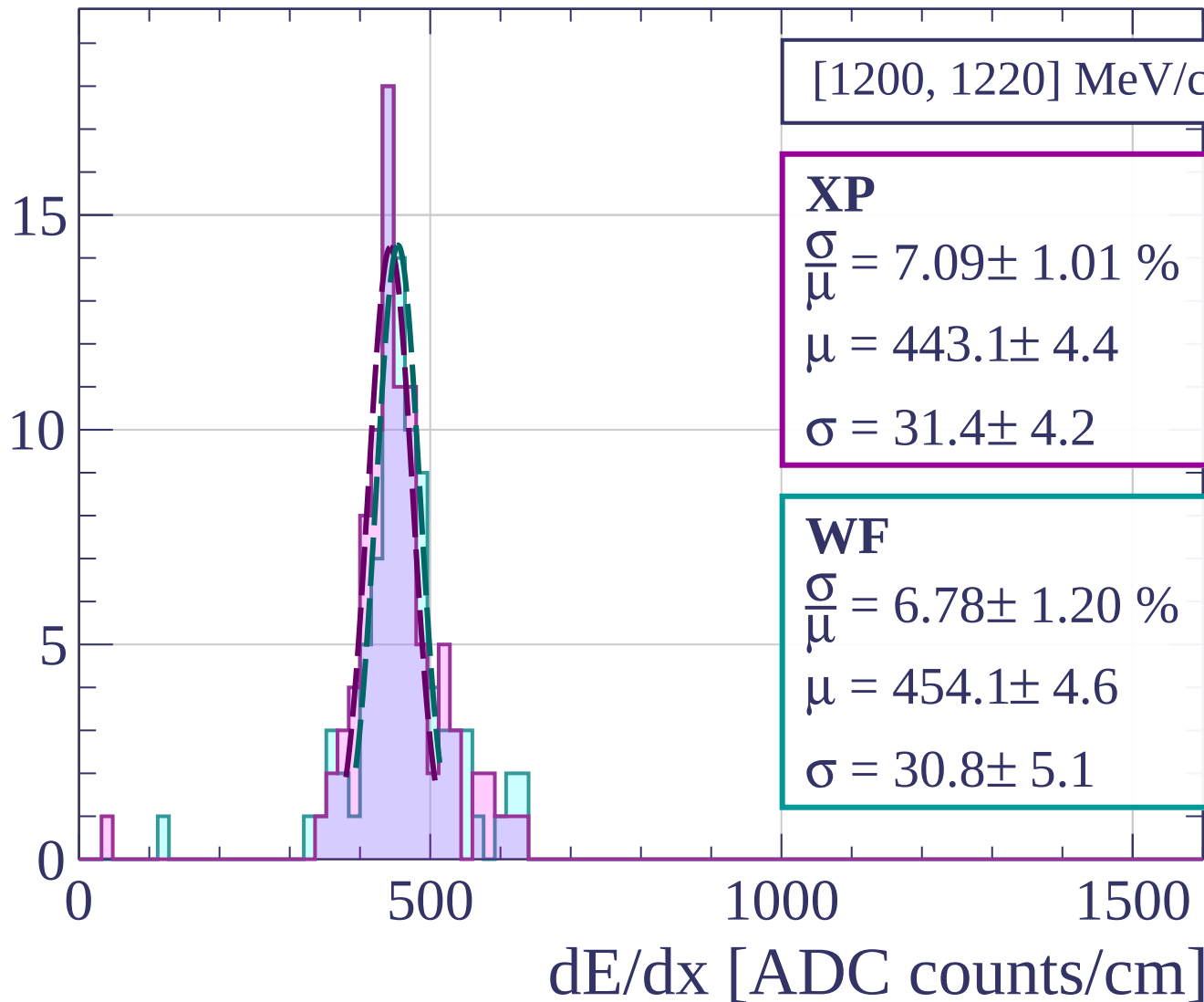
dE/dx [ADC counts/cm]

Count

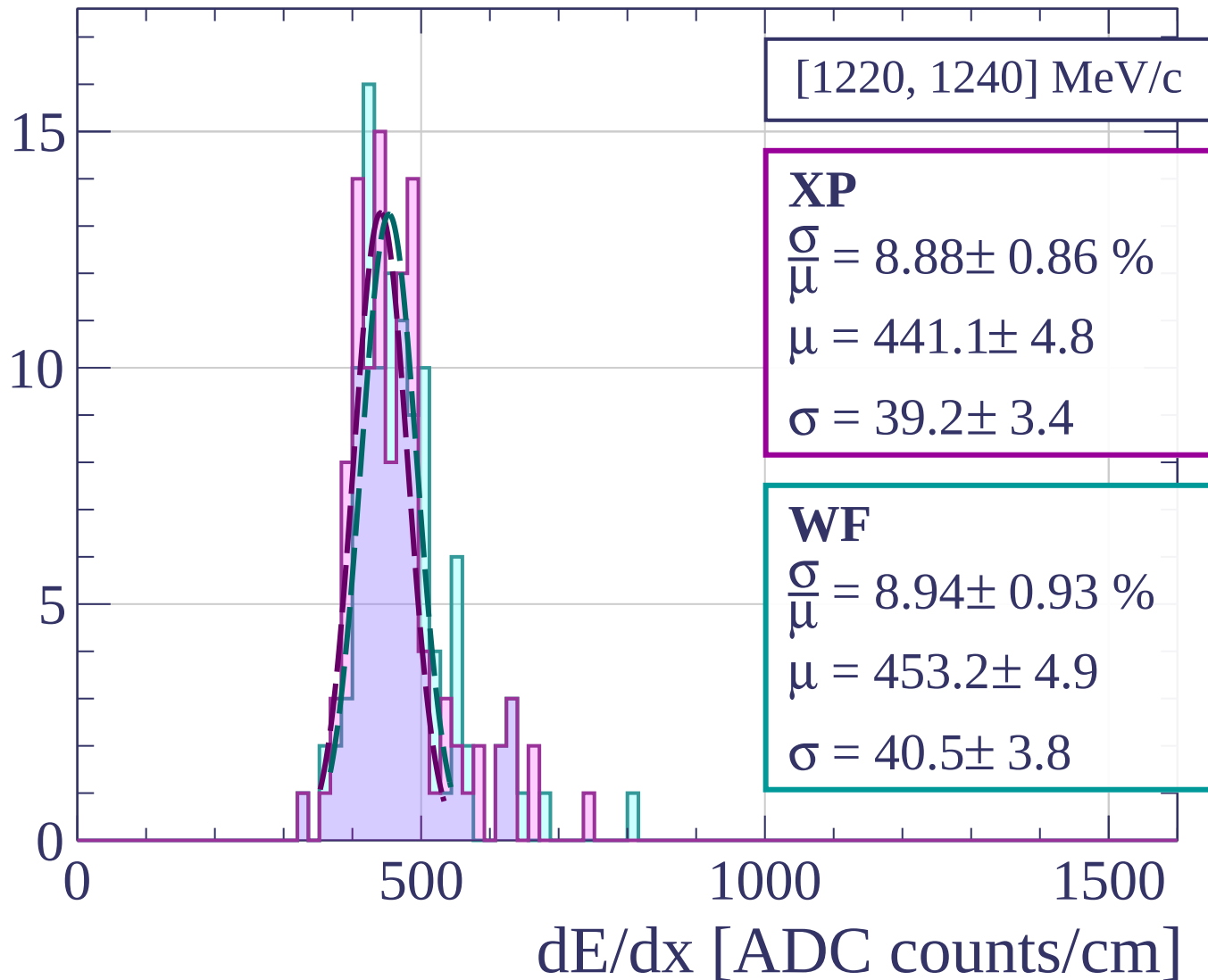


dE/dx [ADC counts/cm]

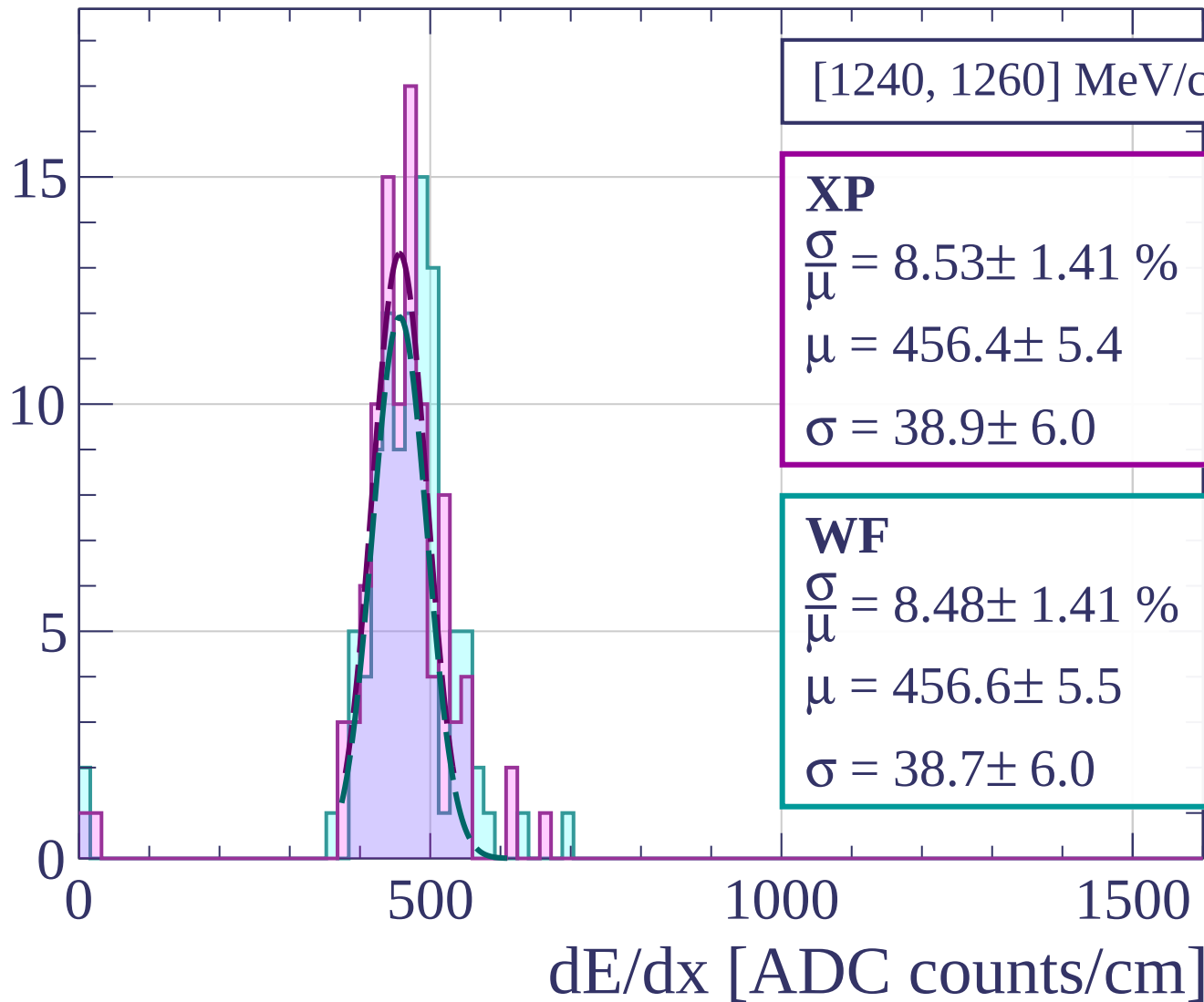
Count



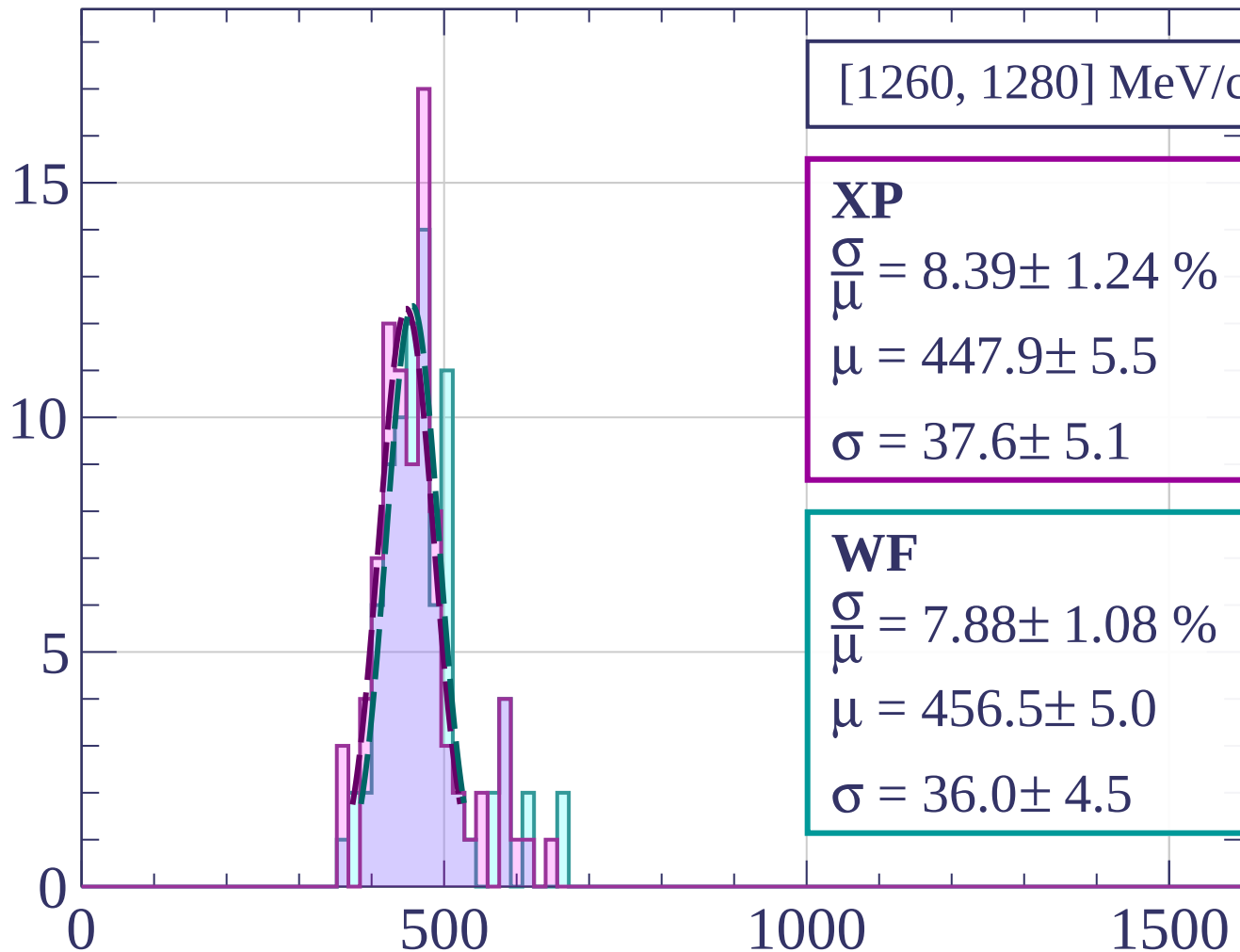
Count



Count



Count



[1260, 1280] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.39 \pm 1.24 \%$$

$$\mu = 447.9 \pm 5.5$$

$$\sigma = 37.6 \pm 5.1$$

WF

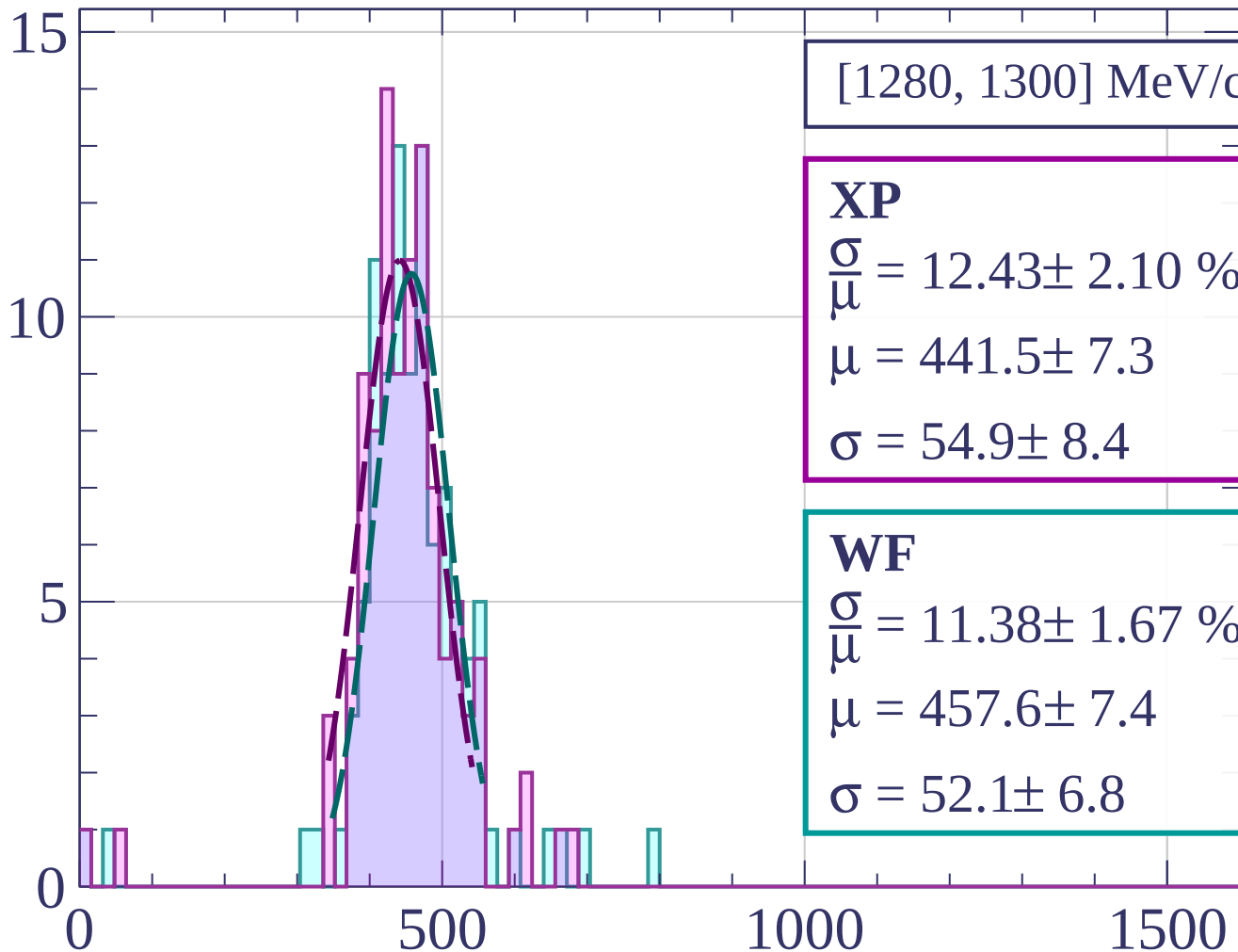
$$\frac{\sigma}{\mu} = 7.88 \pm 1.08 \%$$

$$\mu = 456.5 \pm 5.0$$

$$\sigma = 36.0 \pm 4.5$$

dE/dx [ADC counts/cm]

Count



[1280, 1300] MeV/c

XP

$$\frac{\sigma}{\mu} = 12.43 \pm 2.10 \%$$

$$\mu = 441.5 \pm 7.3$$

$$\sigma = 54.9 \pm 8.4$$

WF

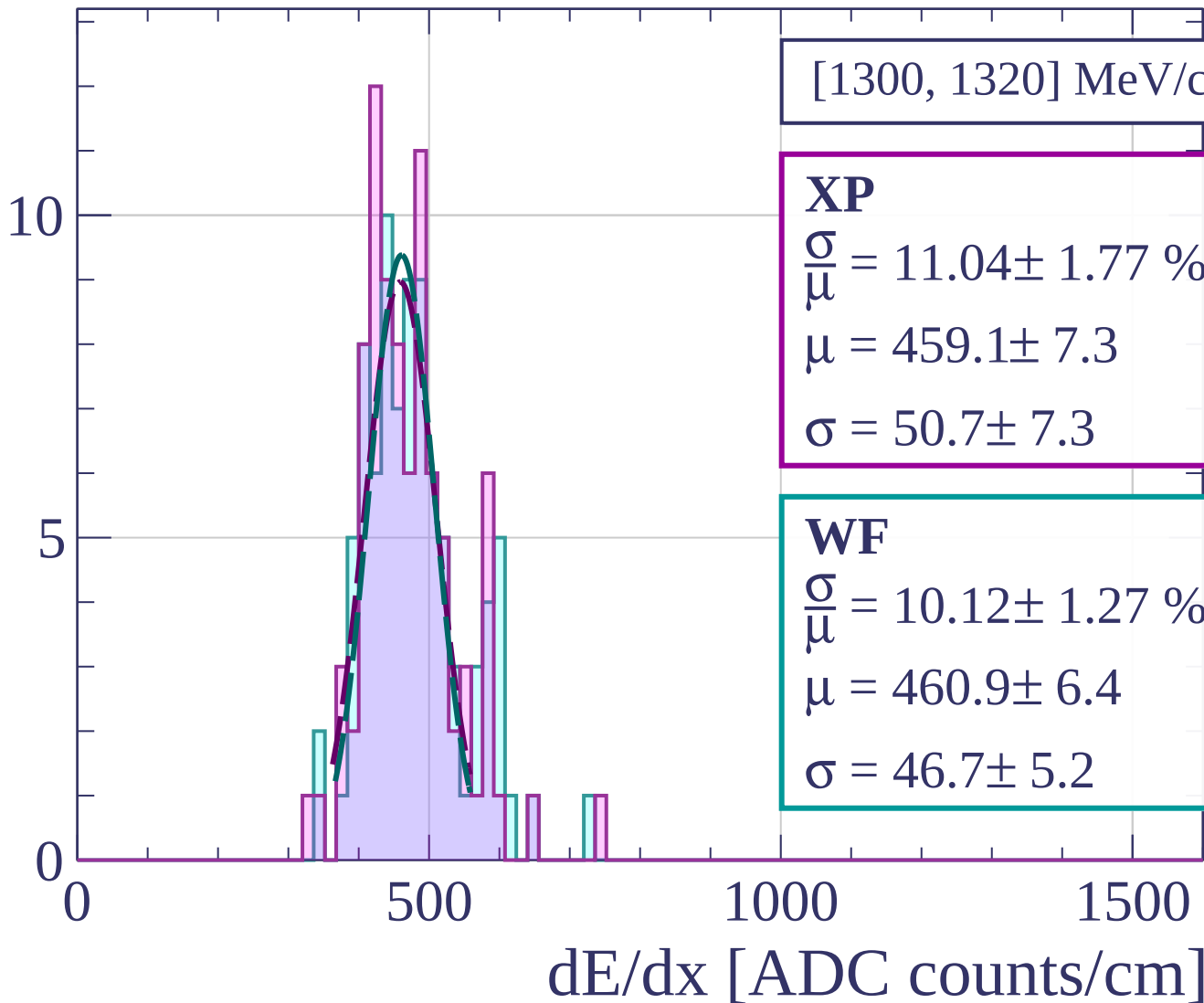
$$\frac{\sigma}{\mu} = 11.38 \pm 1.67 \%$$

$$\mu = 457.6 \pm 7.4$$

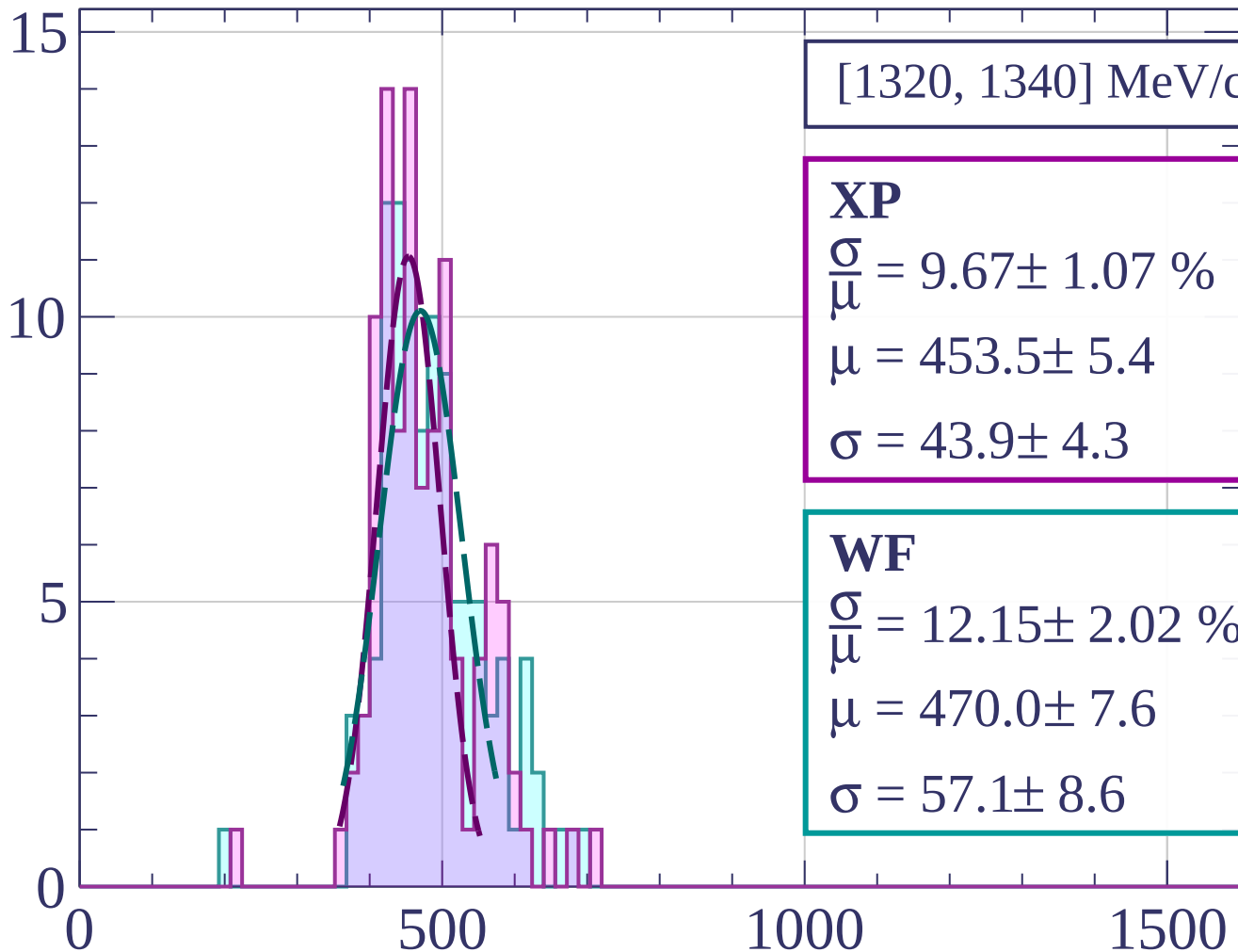
$$\sigma = 52.1 \pm 6.8$$

dE/dx [ADC counts/cm]

Count

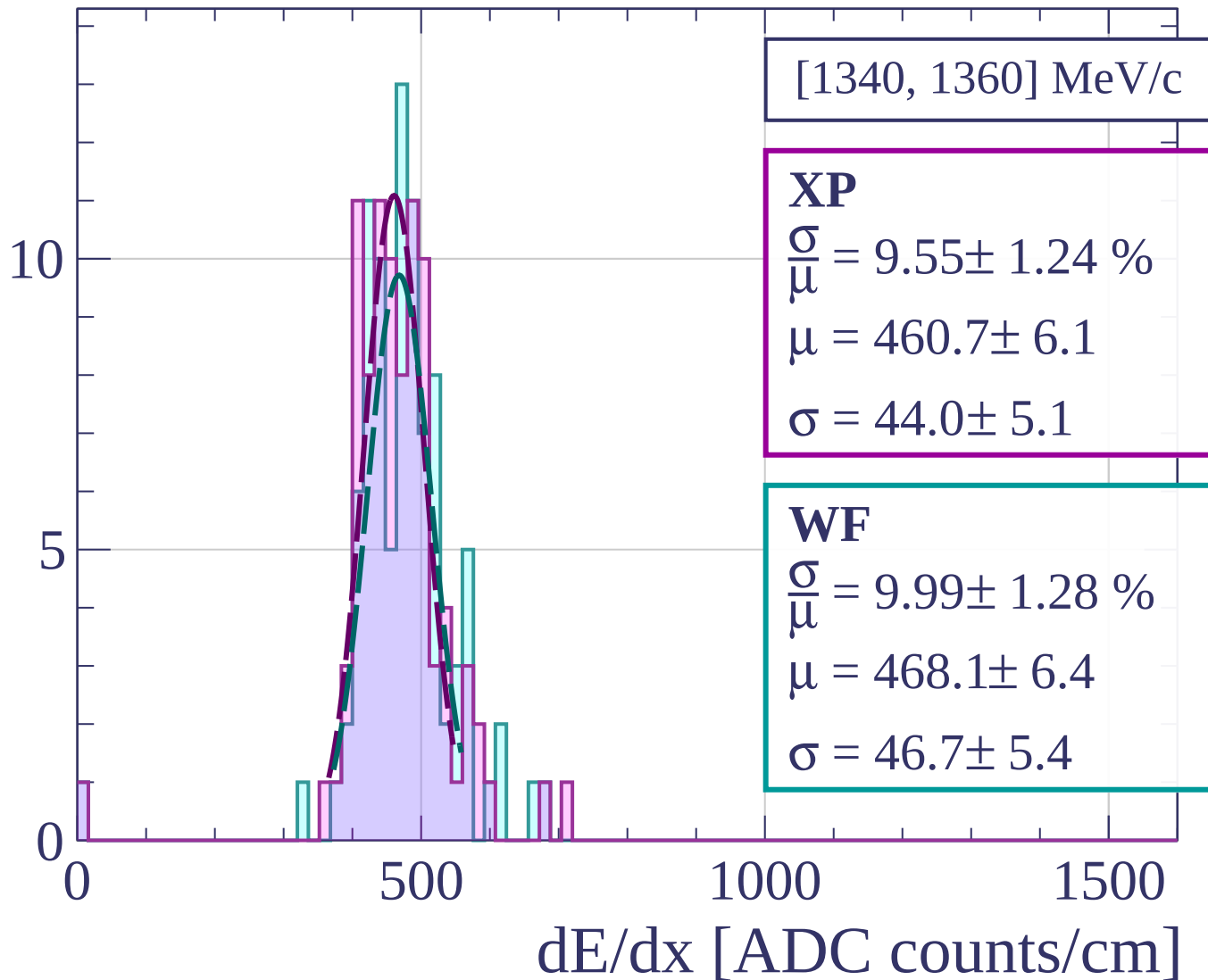


Count

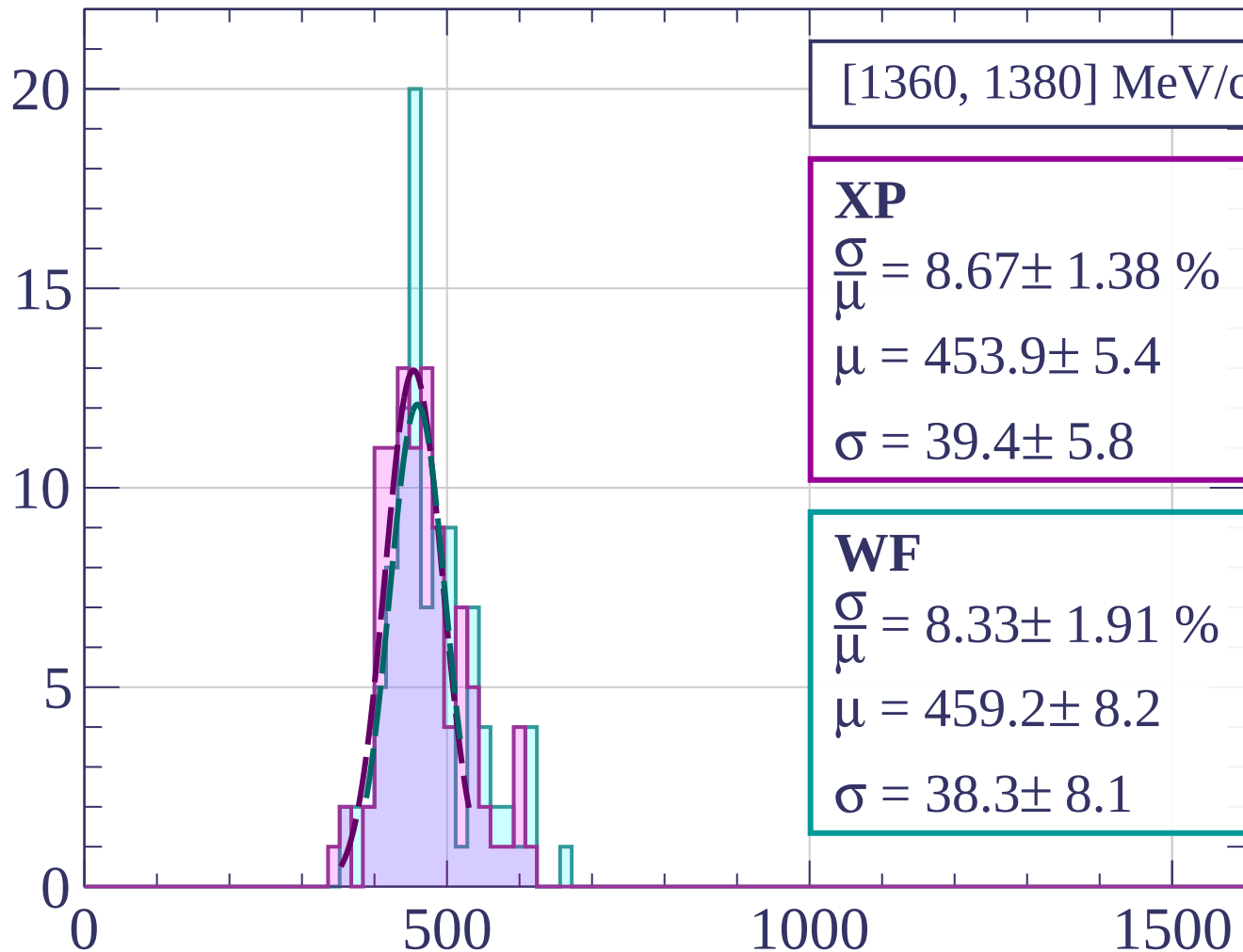


dE/dx [ADC counts/cm]

Count



Count



[1360, 1380] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.67 \pm 1.38 \%$$

$$\mu = 453.9 \pm 5.4$$

$$\sigma = 39.4 \pm 5.8$$

WF

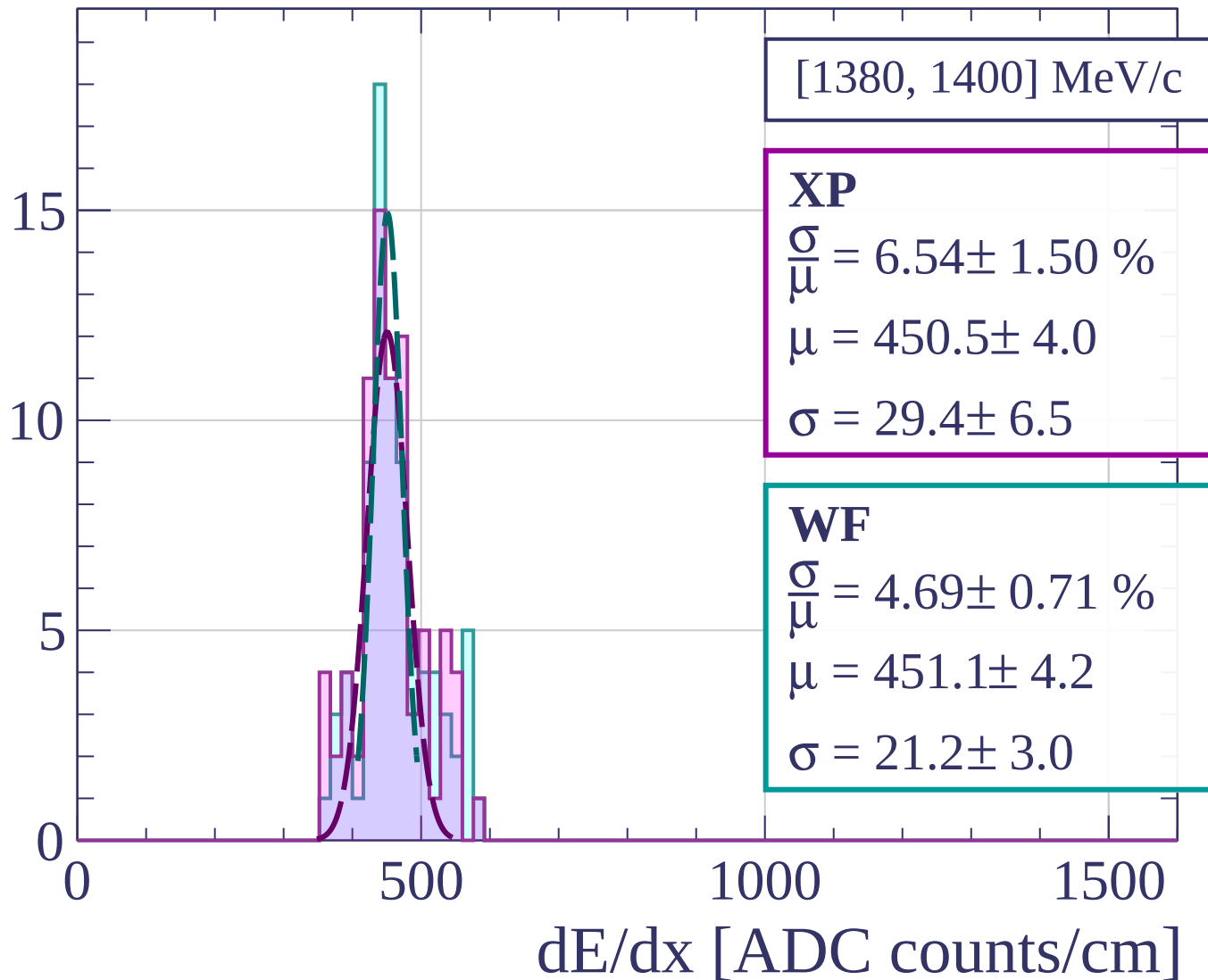
$$\frac{\sigma}{\mu} = 8.33 \pm 1.91 \%$$

$$\mu = 459.2 \pm 8.2$$

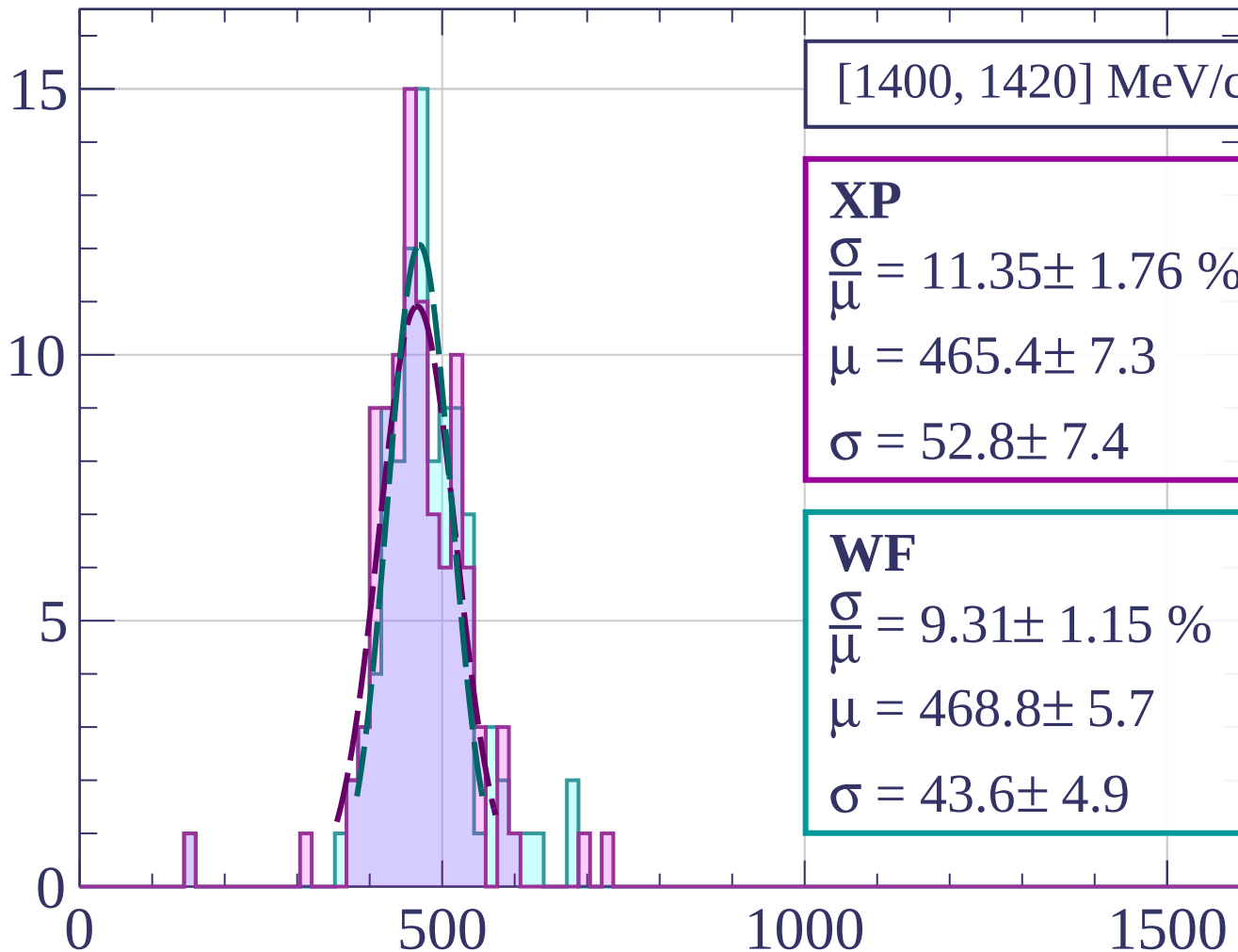
$$\sigma = 38.3 \pm 8.1$$

dE/dx [ADC counts/cm]

Count



Count



[1400, 1420] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.35 \pm 1.76 \%$$

$$\mu = 465.4 \pm 7.3$$

$$\sigma = 52.8 \pm 7.4$$

WF

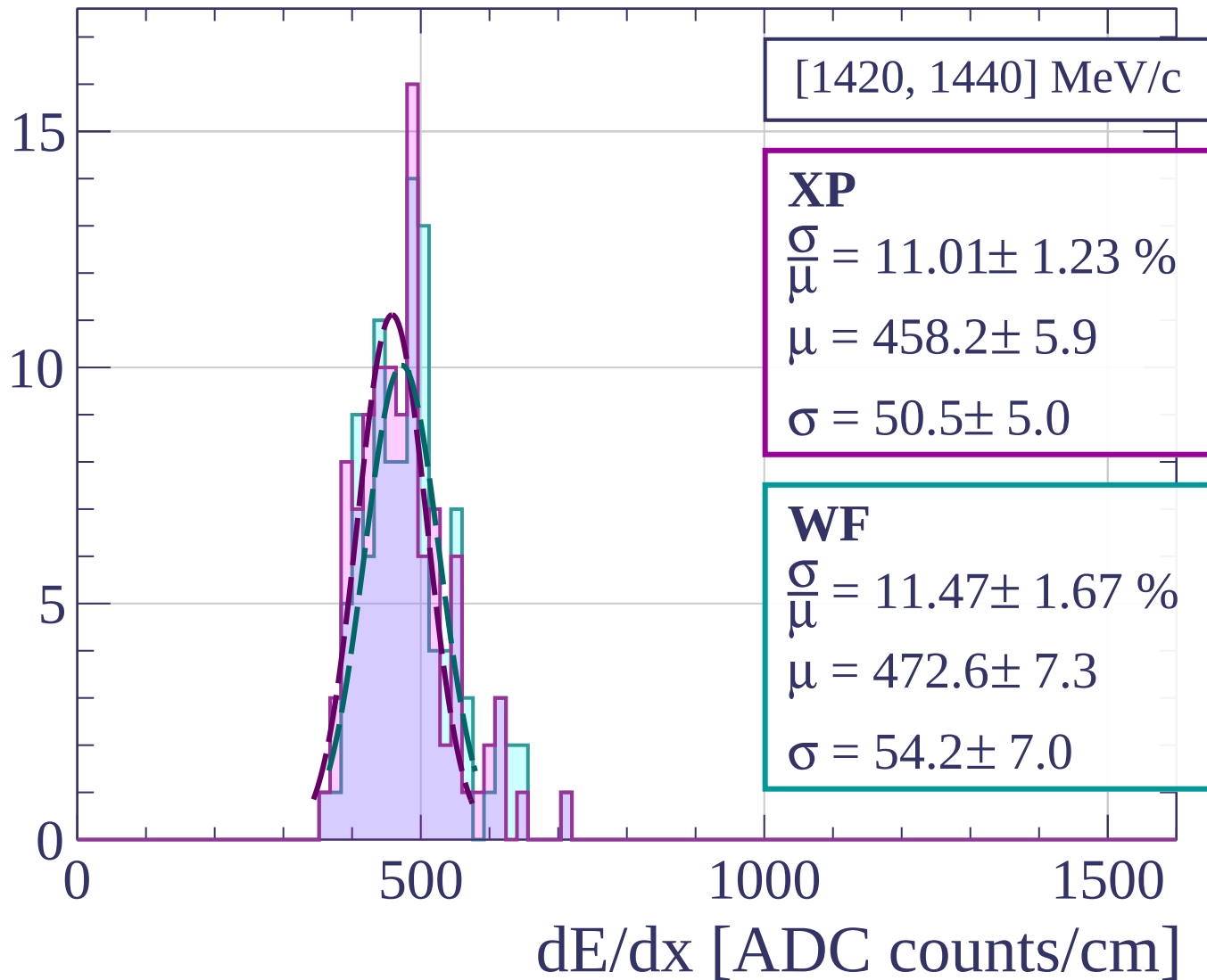
$$\frac{\sigma}{\mu} = 9.31 \pm 1.15 \%$$

$$\mu = 468.8 \pm 5.7$$

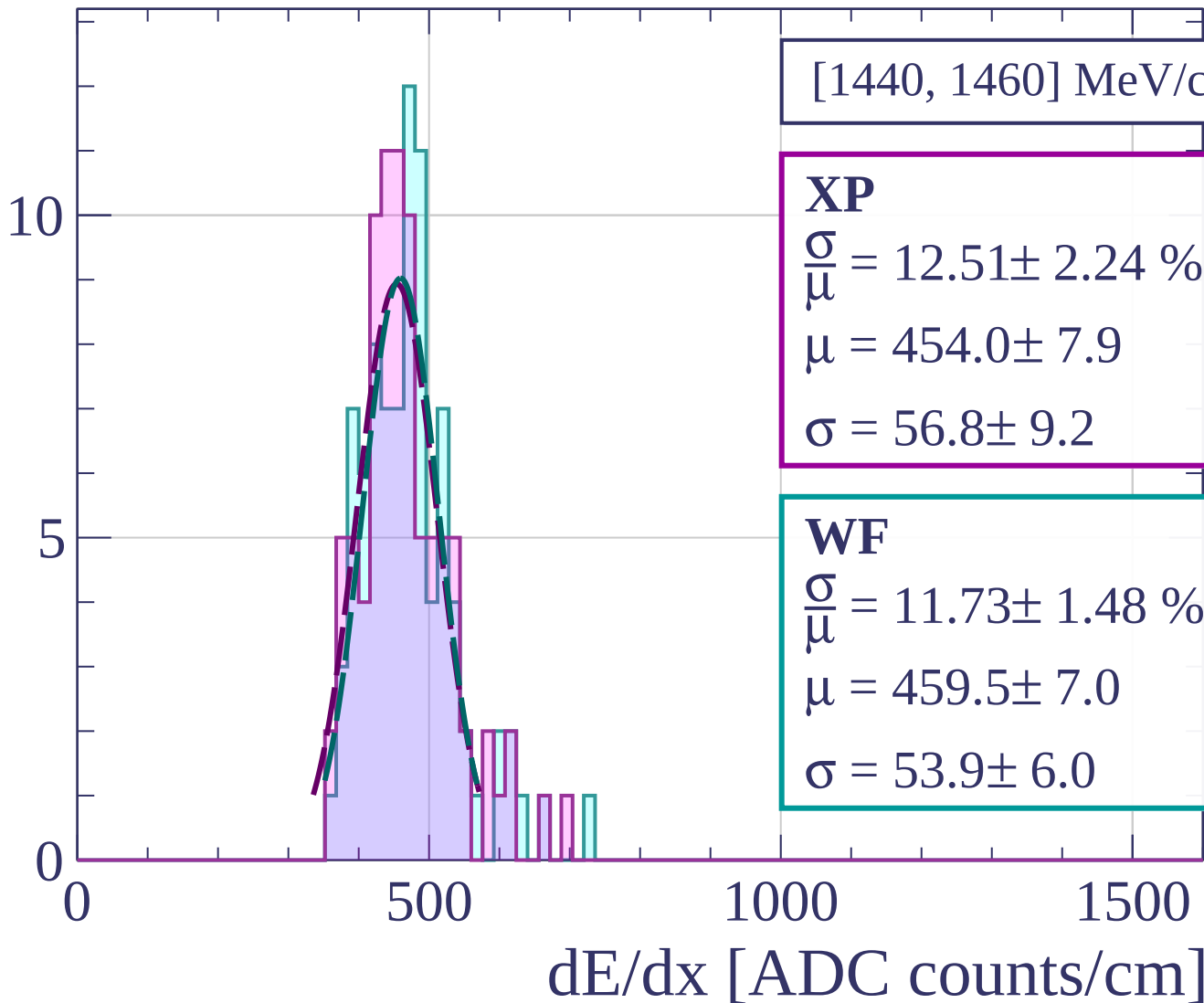
$$\sigma = 43.6 \pm 4.9$$

dE/dx [ADC counts/cm]

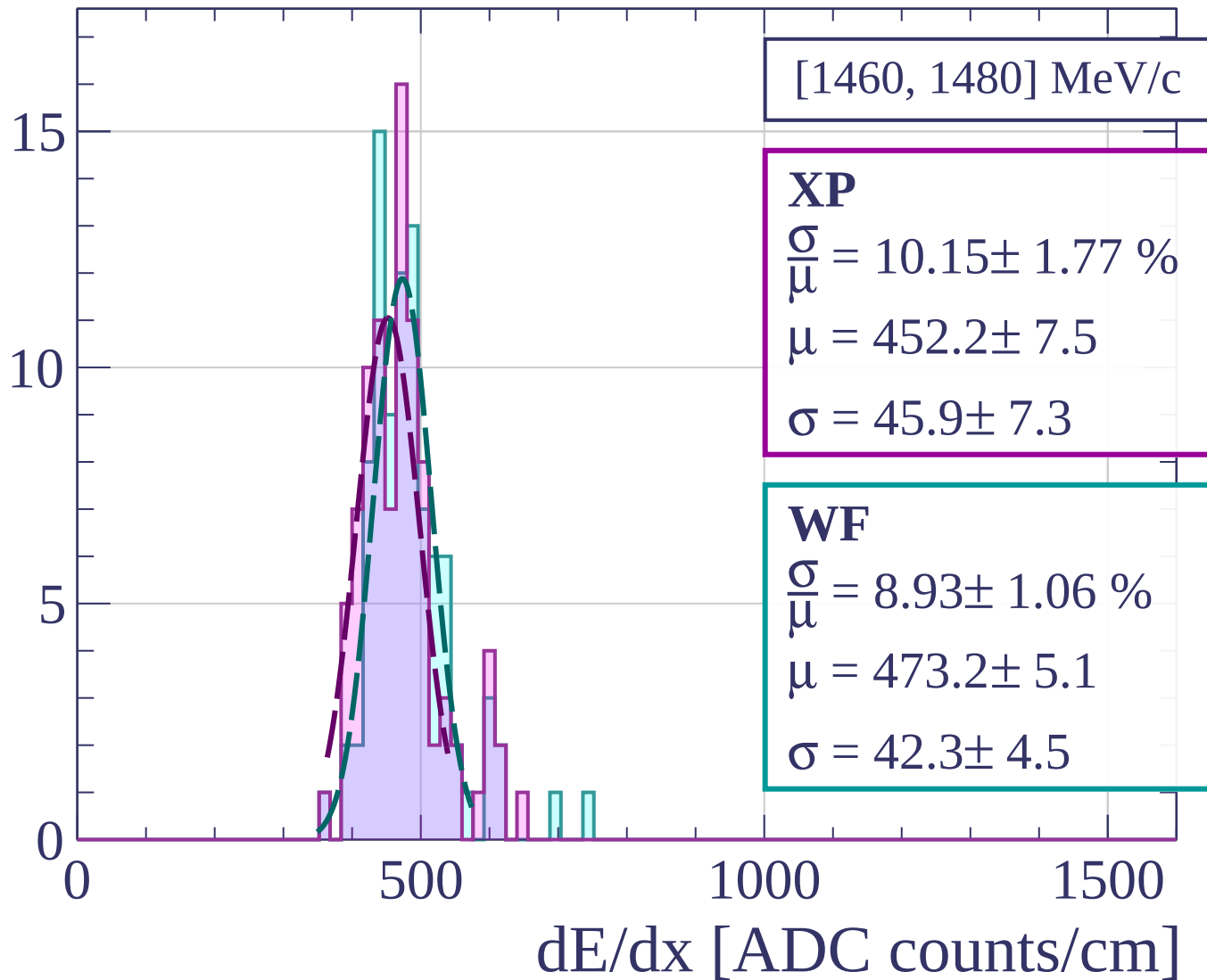
Count



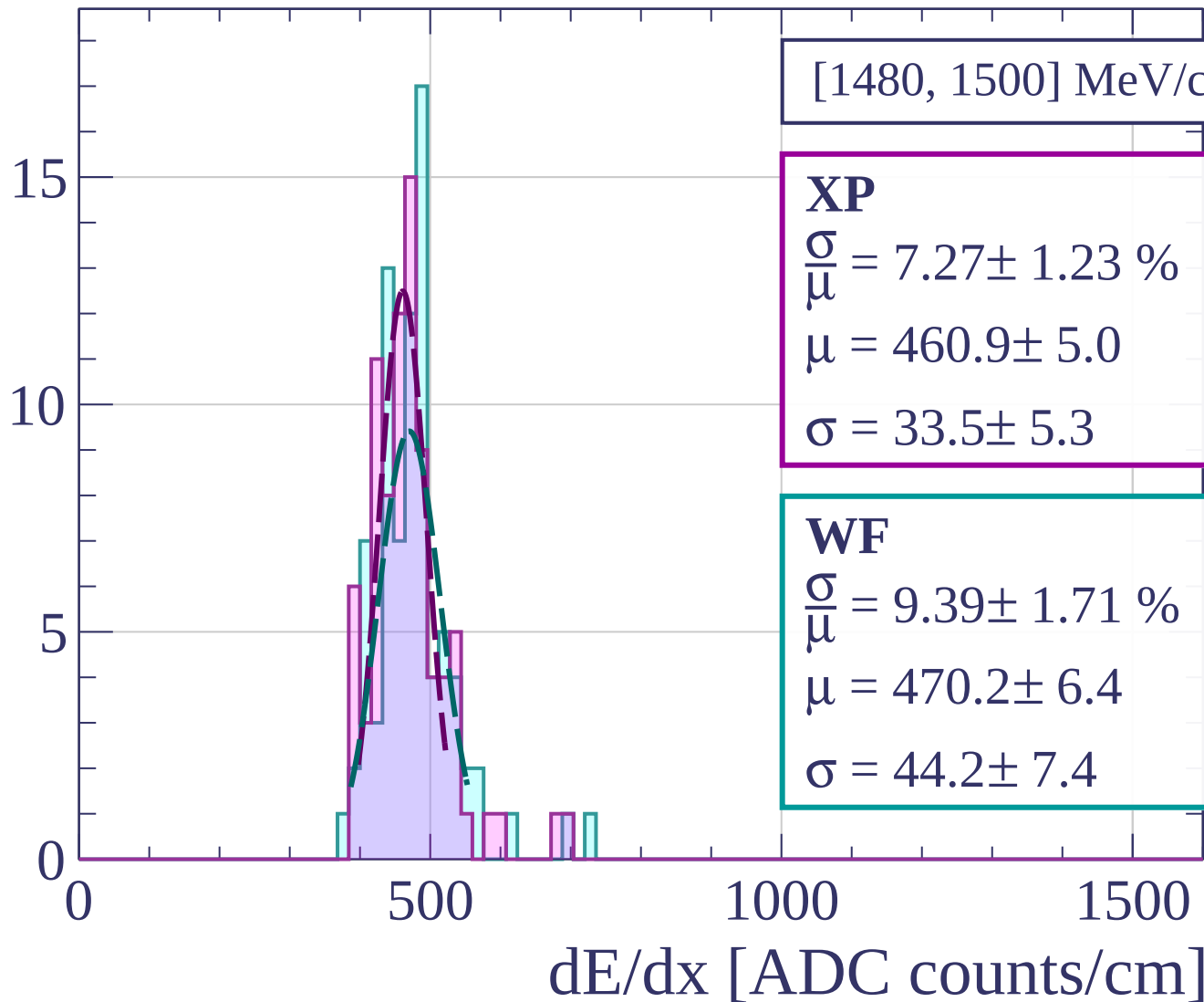
Count



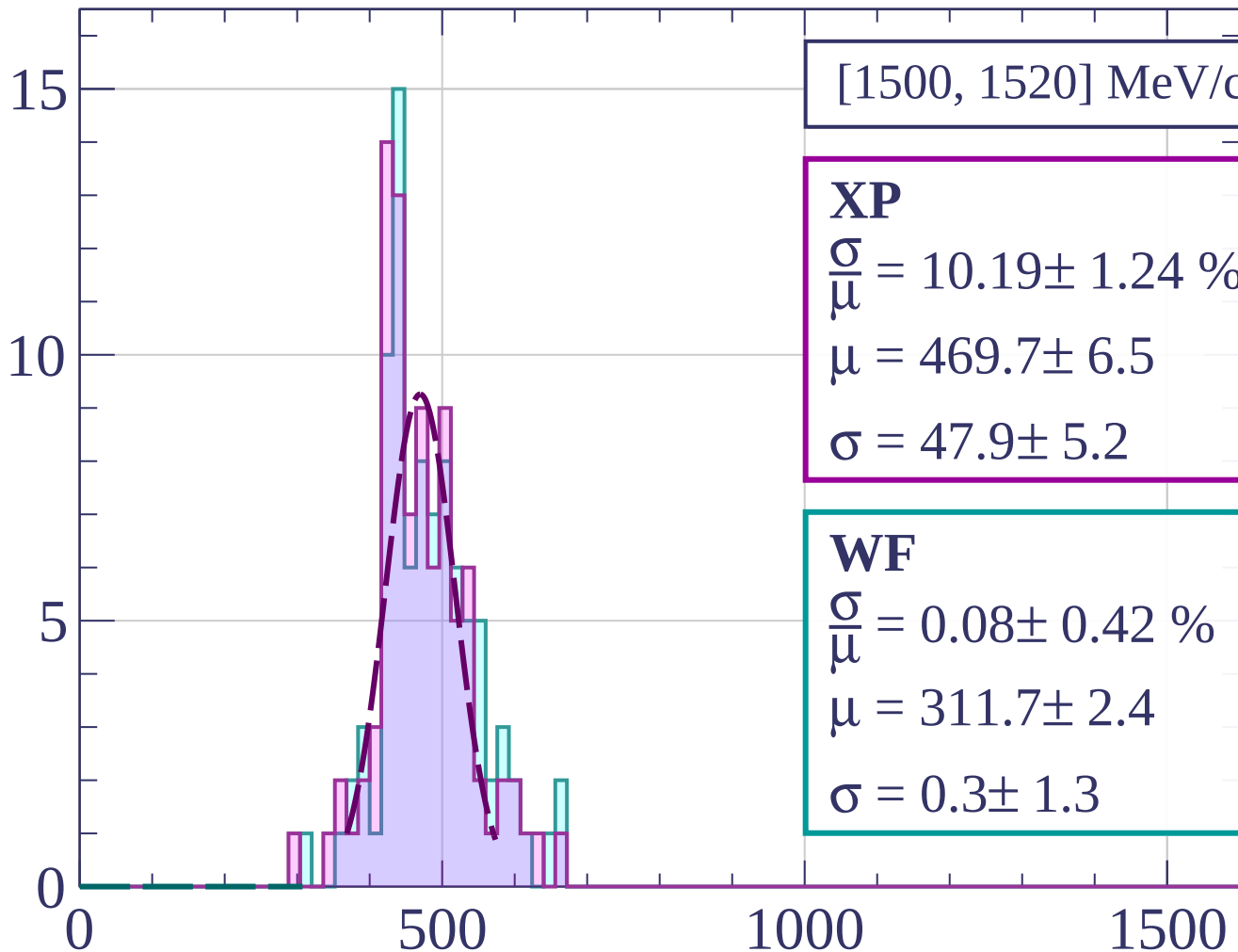
Count



Count

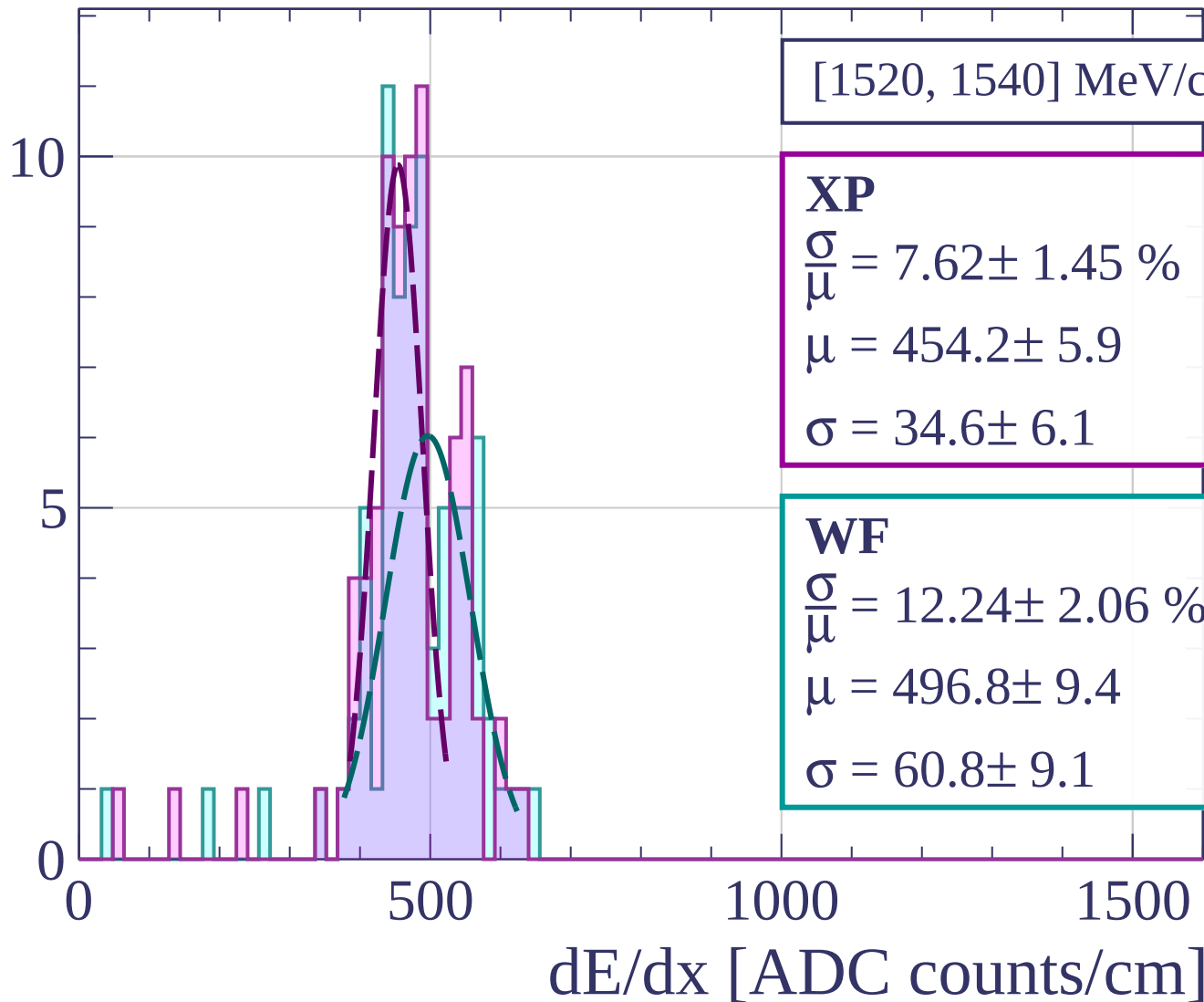


Count

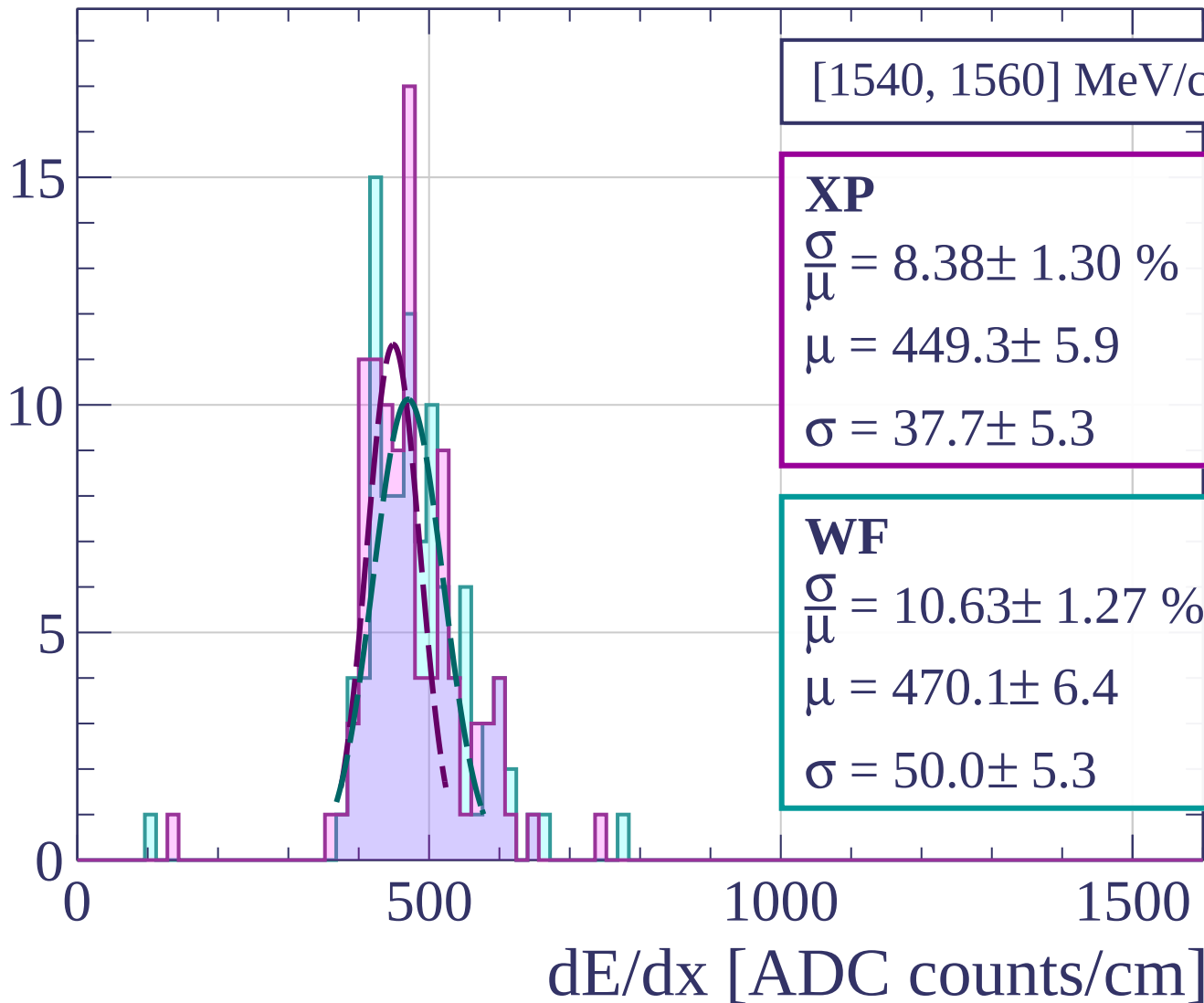


dE/dx [ADC counts/cm]

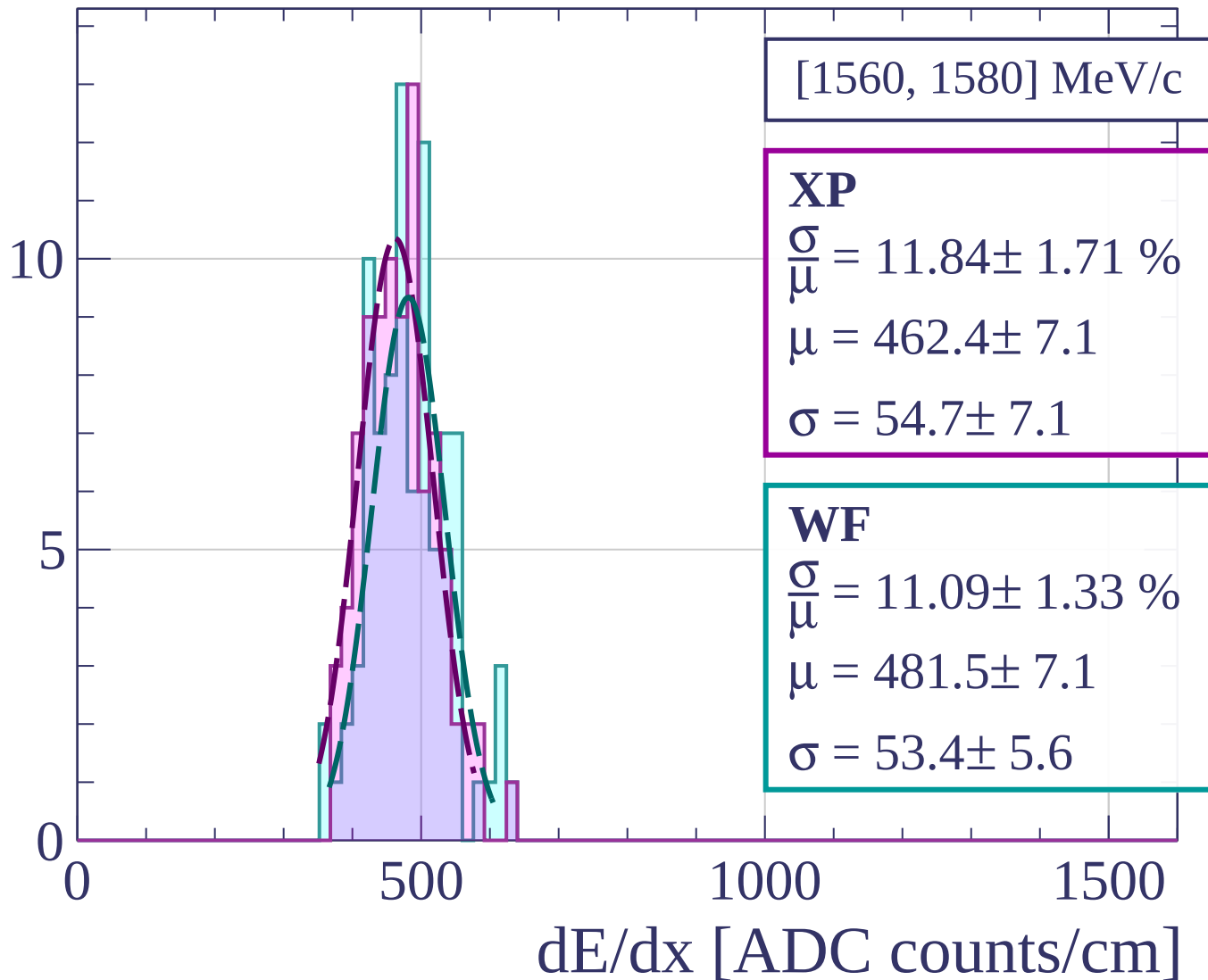
Count



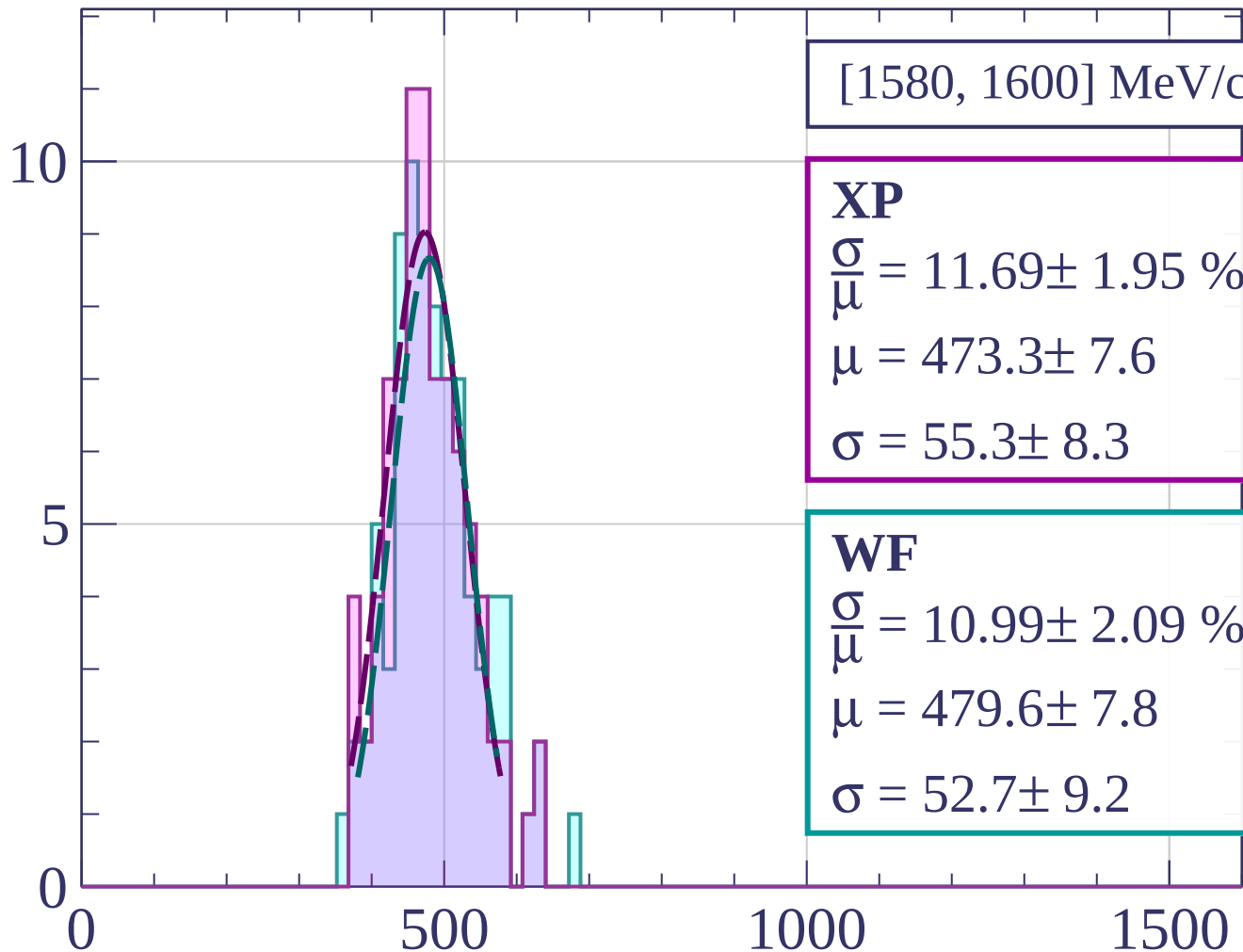
Count



Count

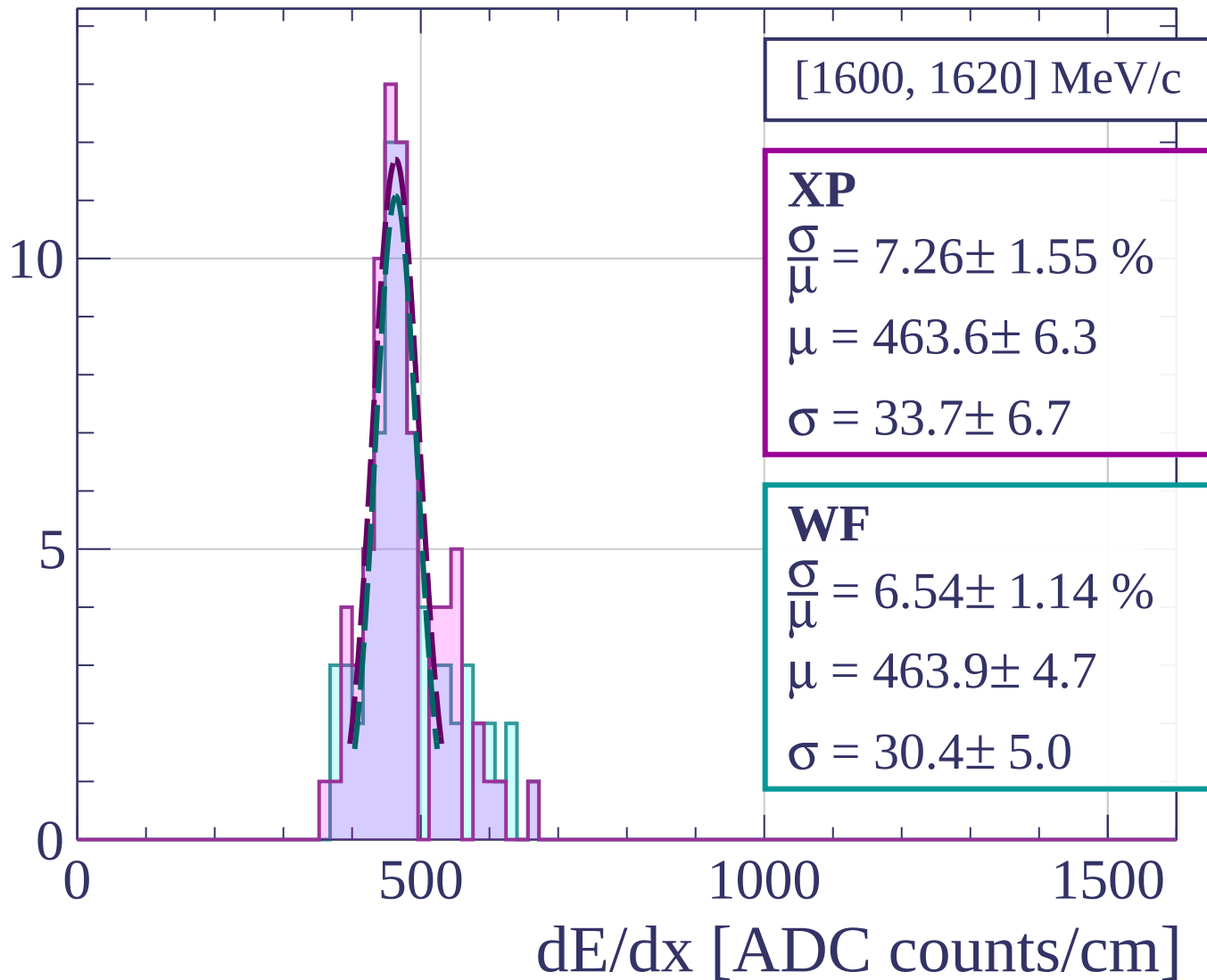


Count

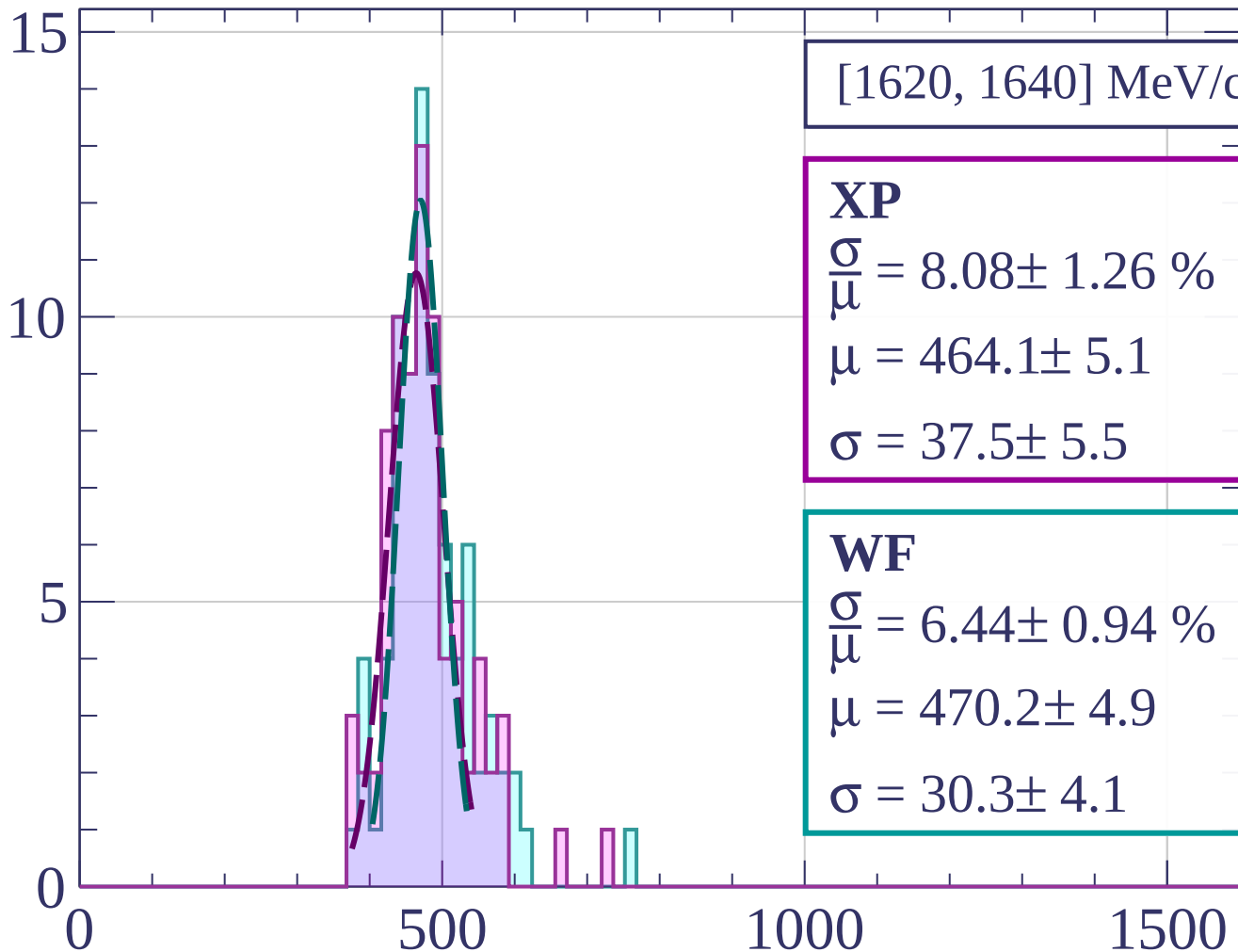


dE/dx [ADC counts/cm]

Count



Count



[1620, 1640] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.08 \pm 1.26 \%$$

$$\mu = 464.1 \pm 5.1$$

$$\sigma = 37.5 \pm 5.5$$

WF

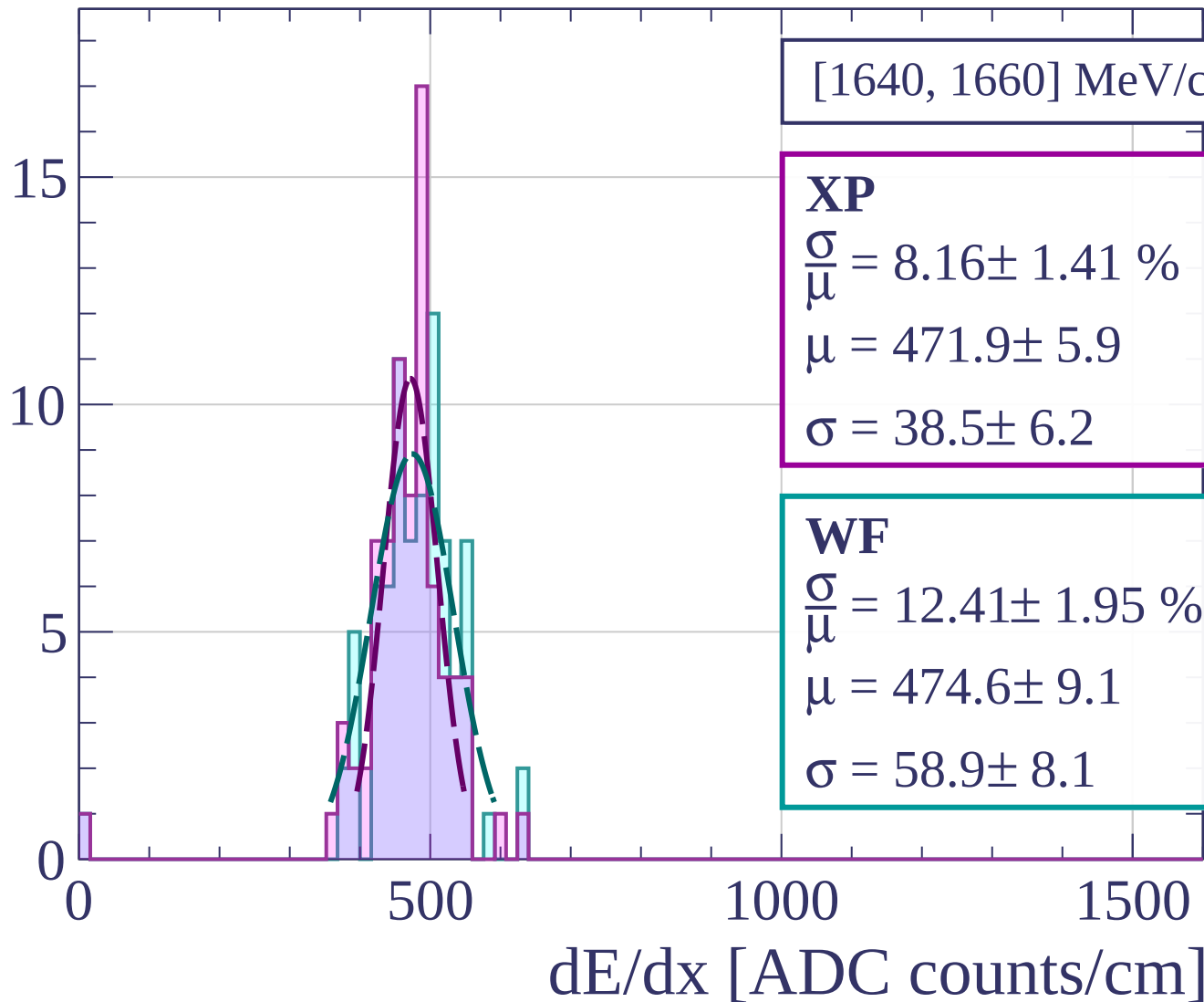
$$\frac{\sigma}{\mu} = 6.44 \pm 0.94 \%$$

$$\mu = 470.2 \pm 4.9$$

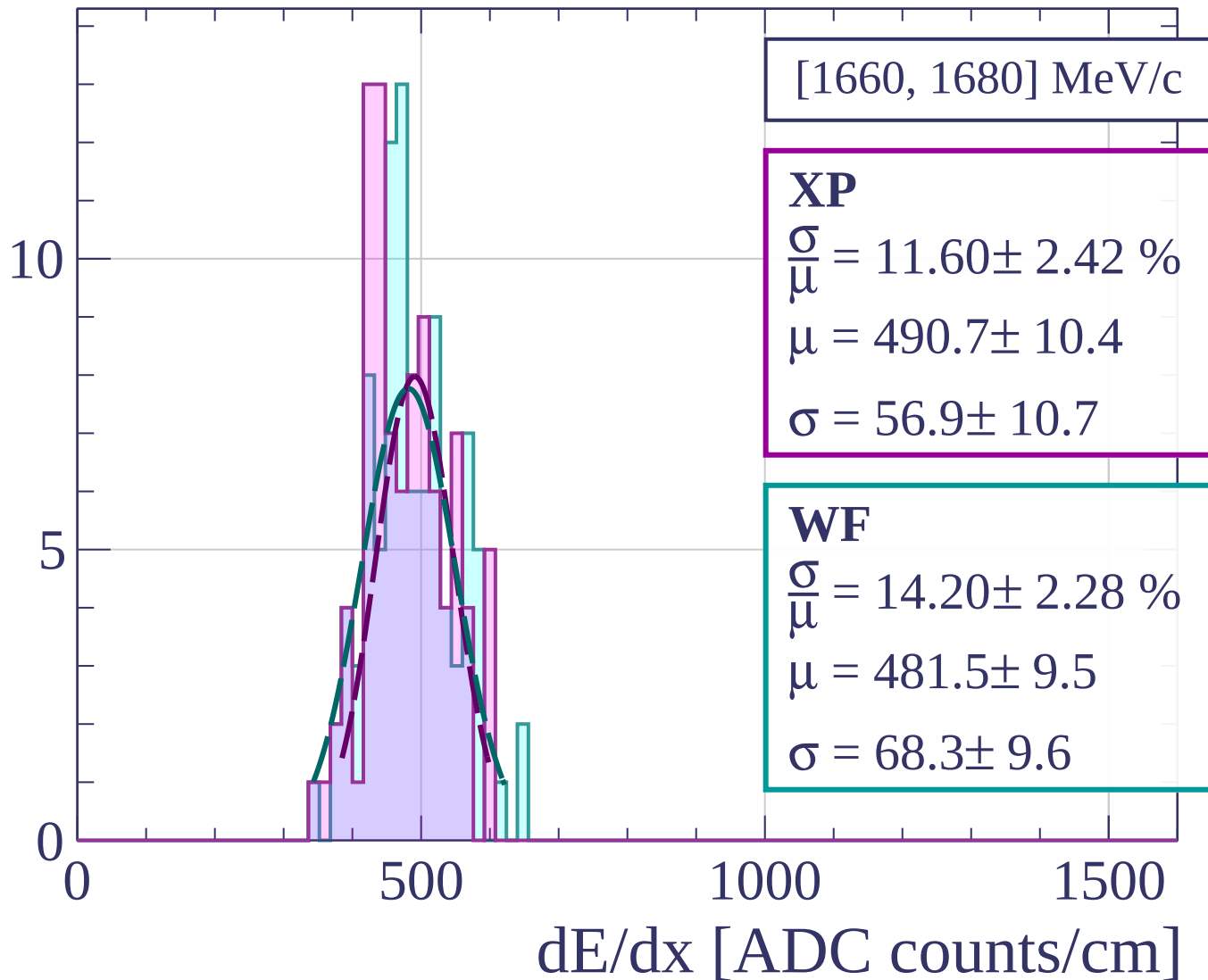
$$\sigma = 30.3 \pm 4.1$$

dE/dx [ADC counts/cm]

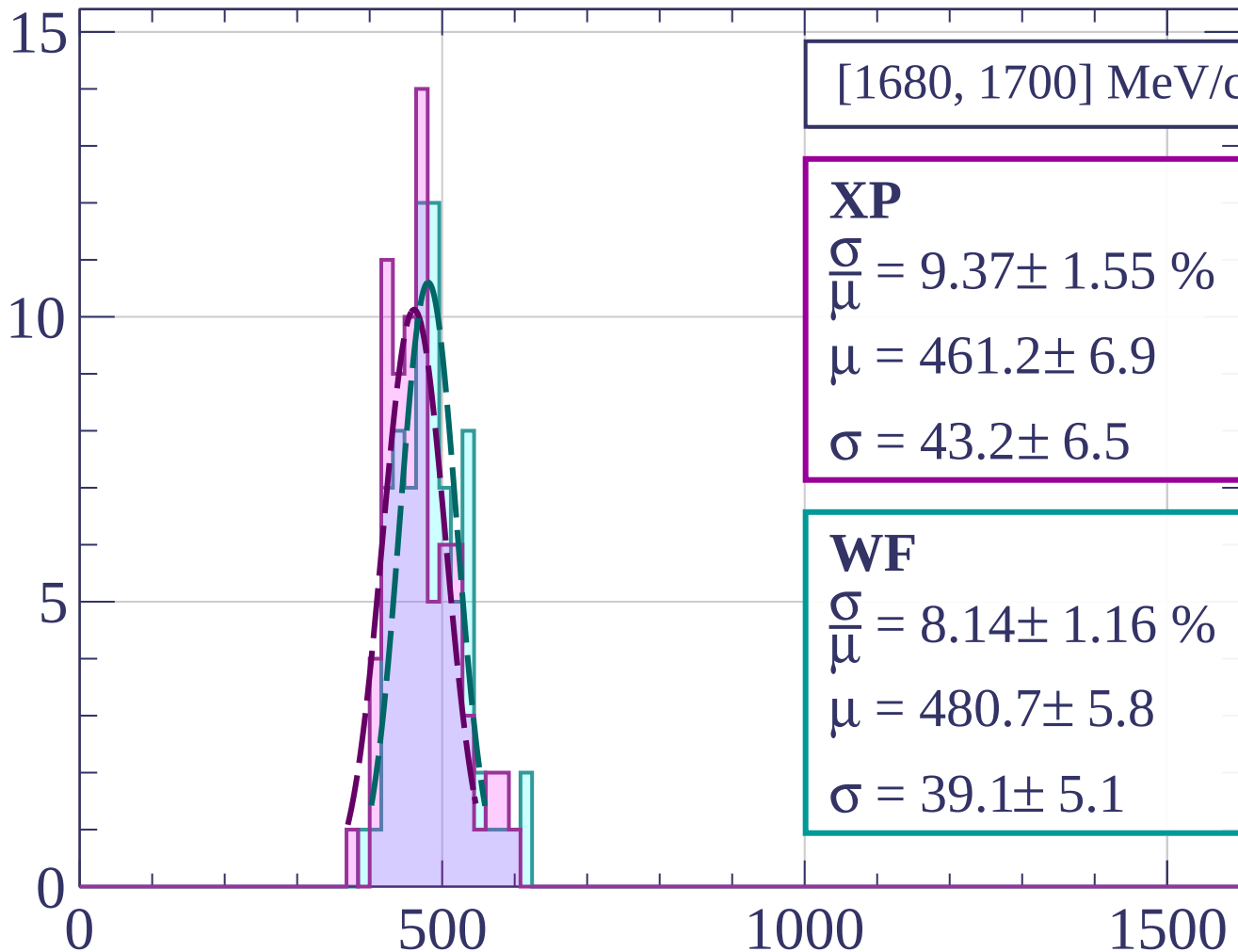
Count



Count

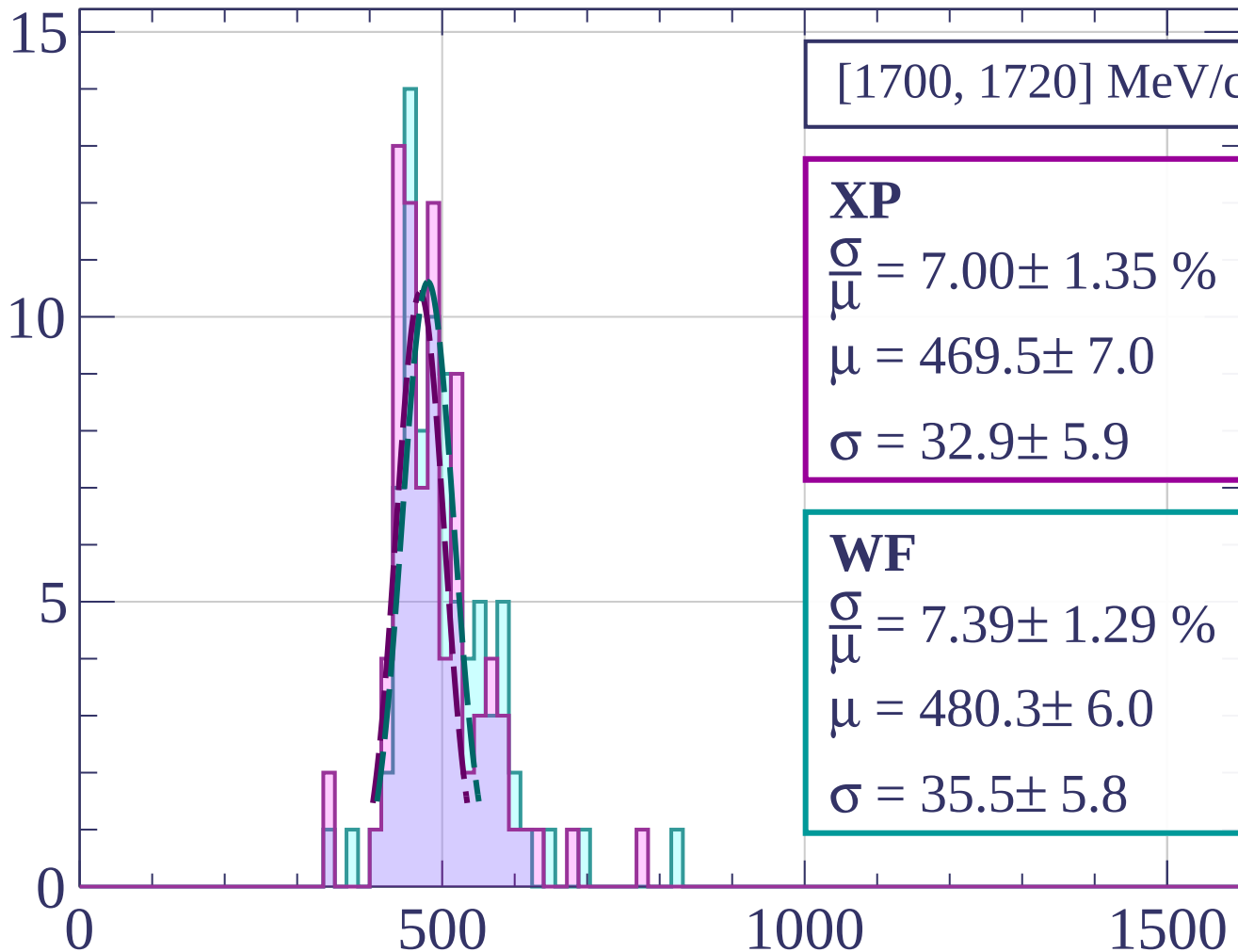


Count



dE/dx [ADC counts/cm]

Count



[1700, 1720] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.00 \pm 1.35 \%$$

$$\mu = 469.5 \pm 7.0$$

$$\sigma = 32.9 \pm 5.9$$

WF

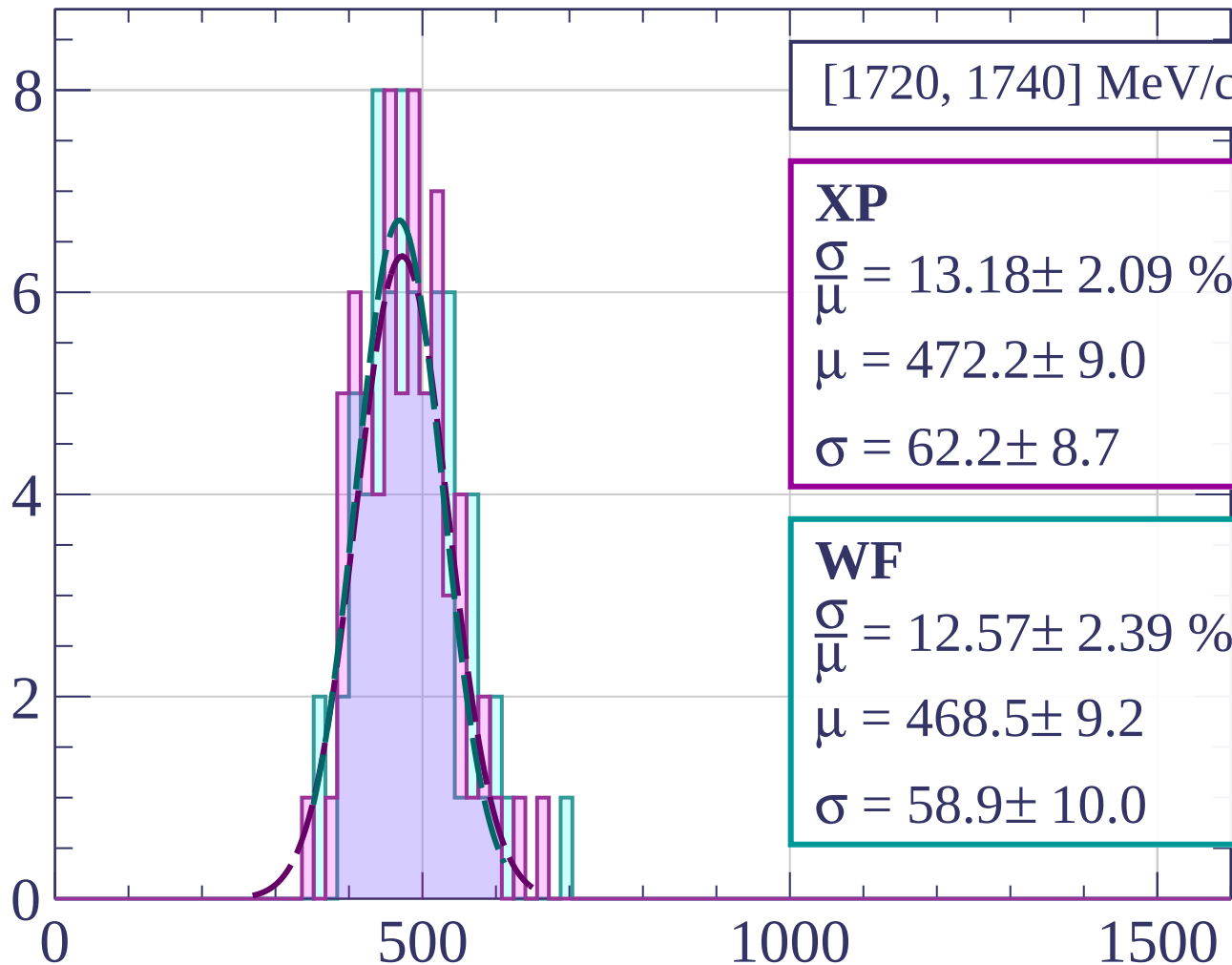
$$\frac{\sigma}{\mu} = 7.39 \pm 1.29 \%$$

$$\mu = 480.3 \pm 6.0$$

$$\sigma = 35.5 \pm 5.8$$

dE/dx [ADC counts/cm]

Count



[1720, 1740] MeV/c

XP

$$\frac{\sigma}{\mu} = 13.18 \pm 2.09 \%$$

$$\mu = 472.2 \pm 9.0$$

$$\sigma = 62.2 \pm 8.7$$

WF

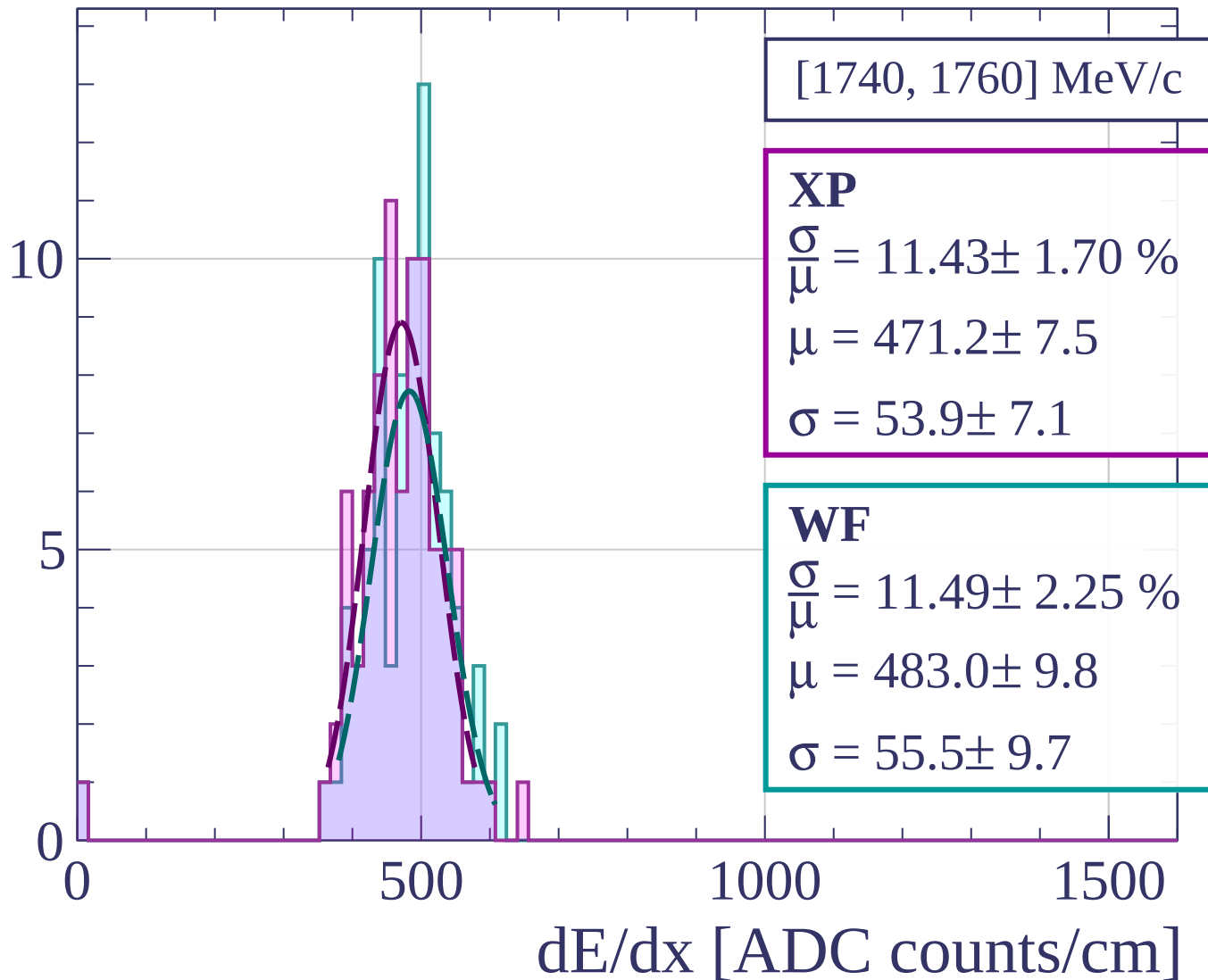
$$\frac{\sigma}{\mu} = 12.57 \pm 2.39 \%$$

$$\mu = 468.5 \pm 9.2$$

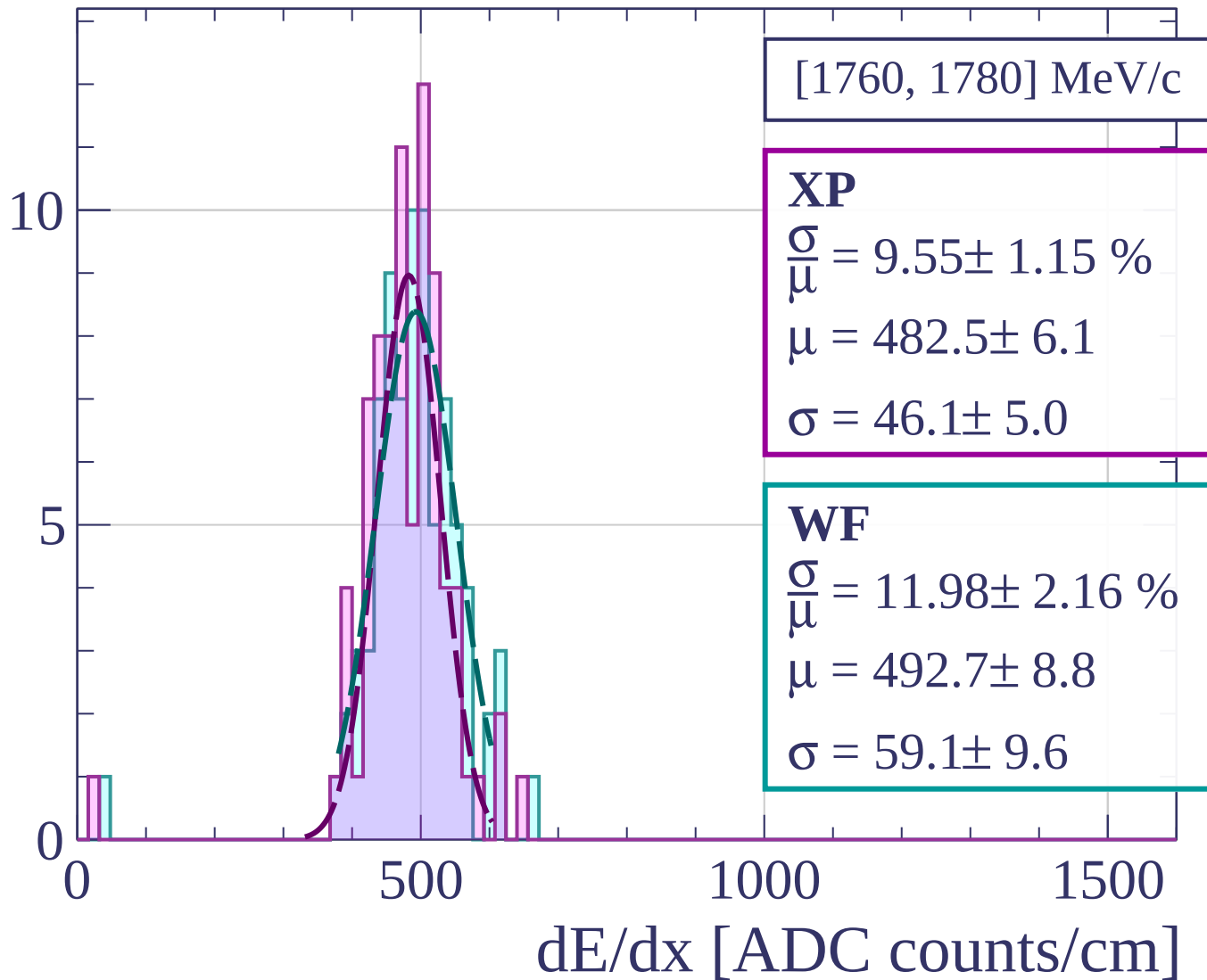
$$\sigma = 58.9 \pm 10.0$$

dE/dx [ADC counts/cm]

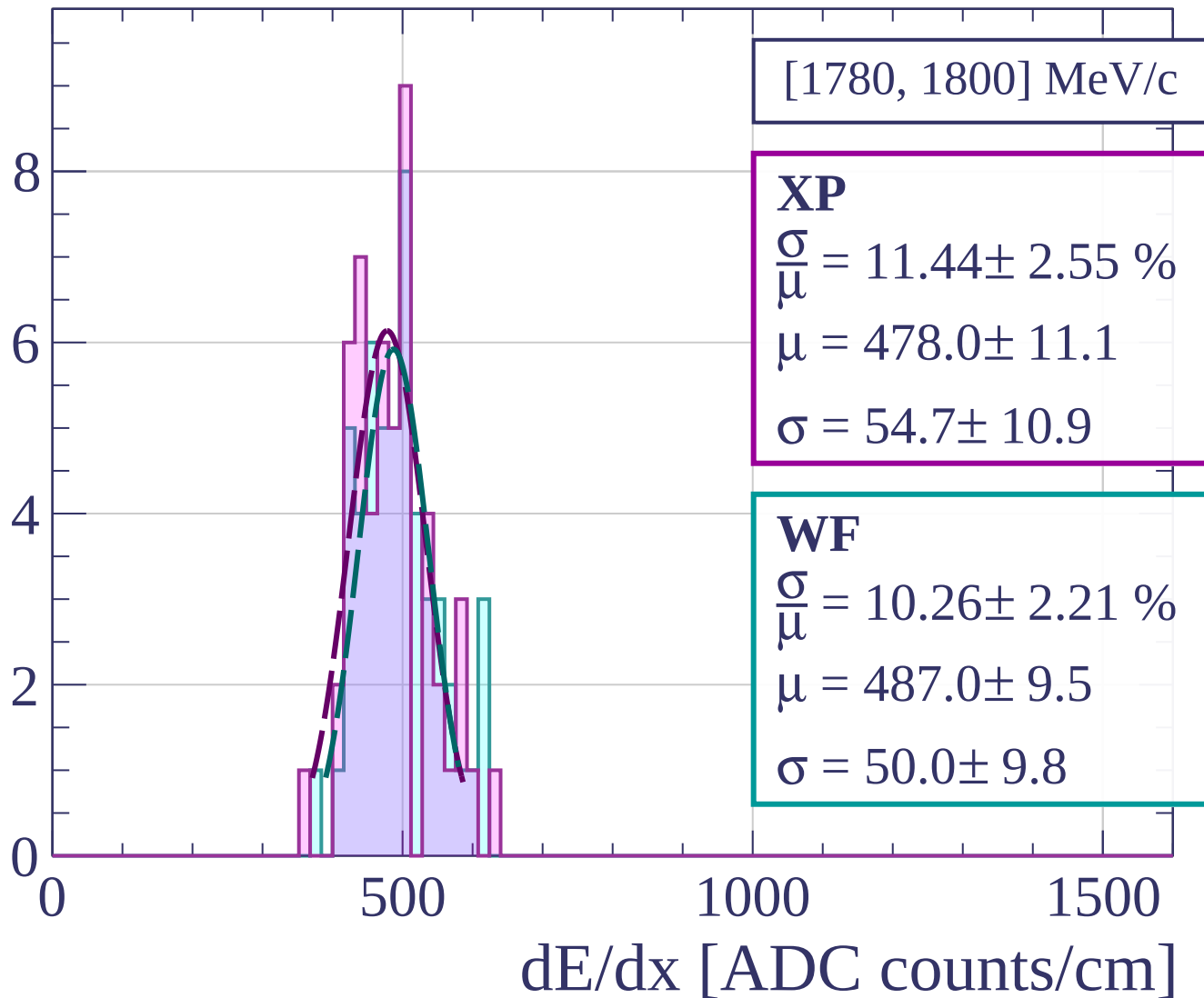
Count



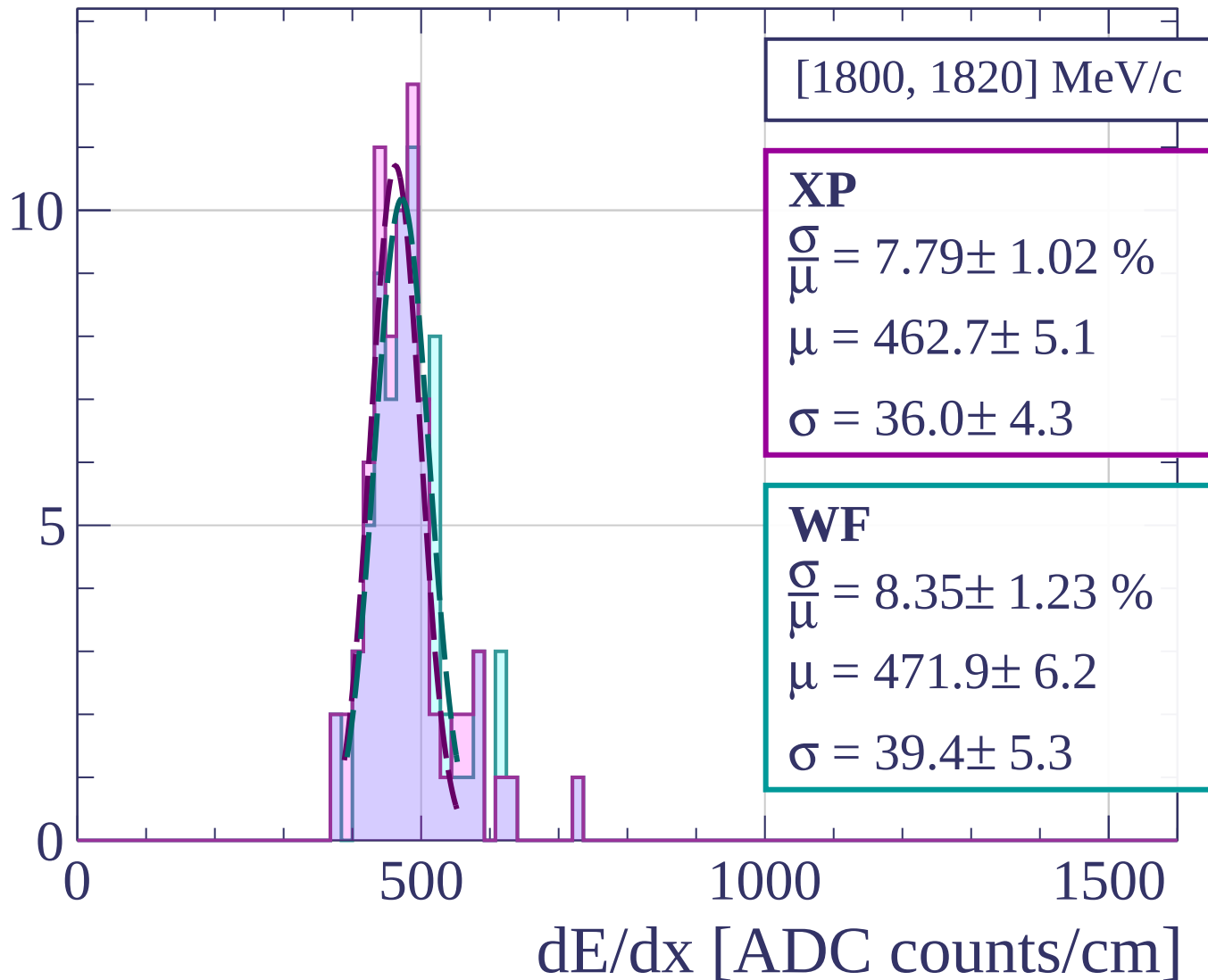
Count



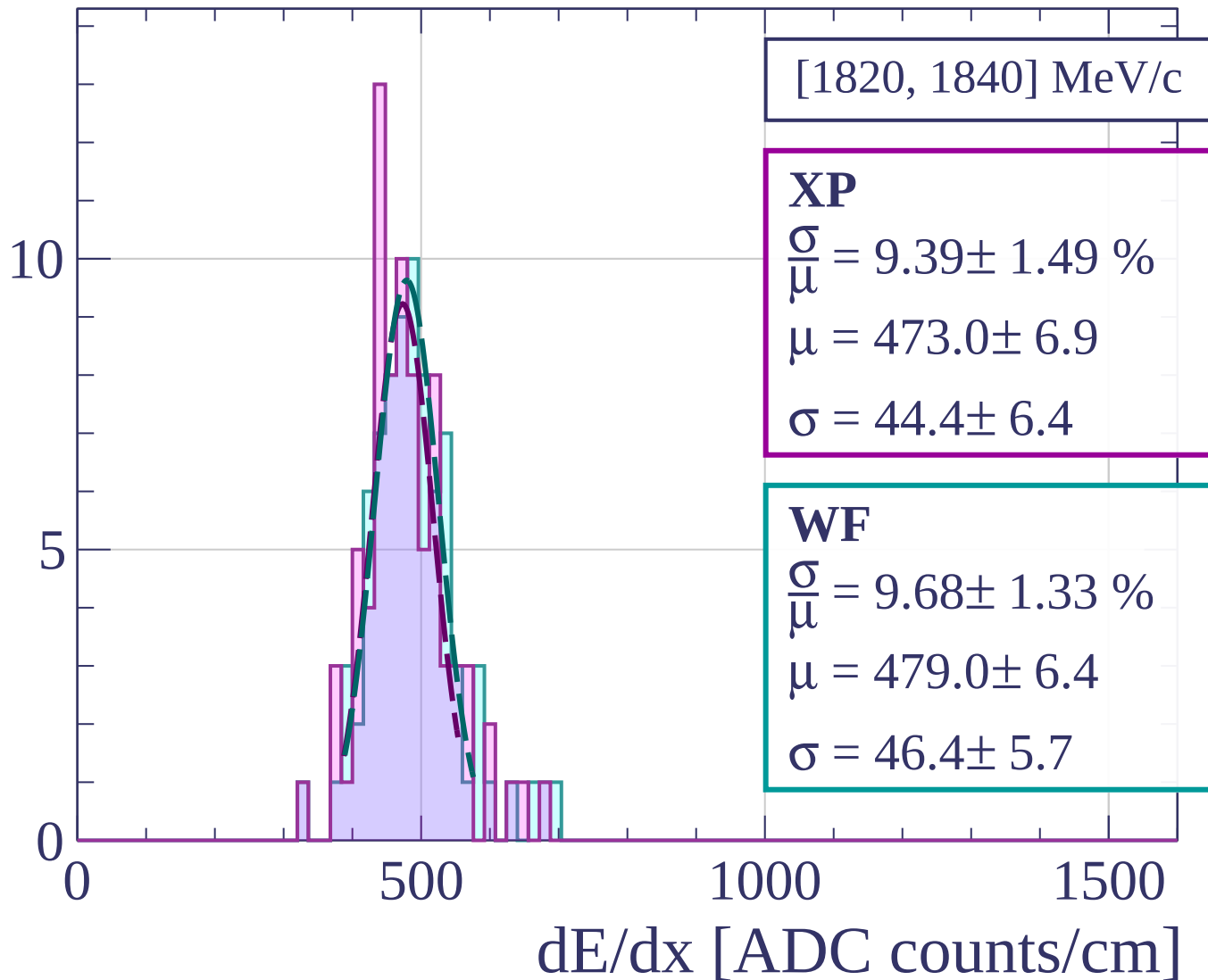
Count



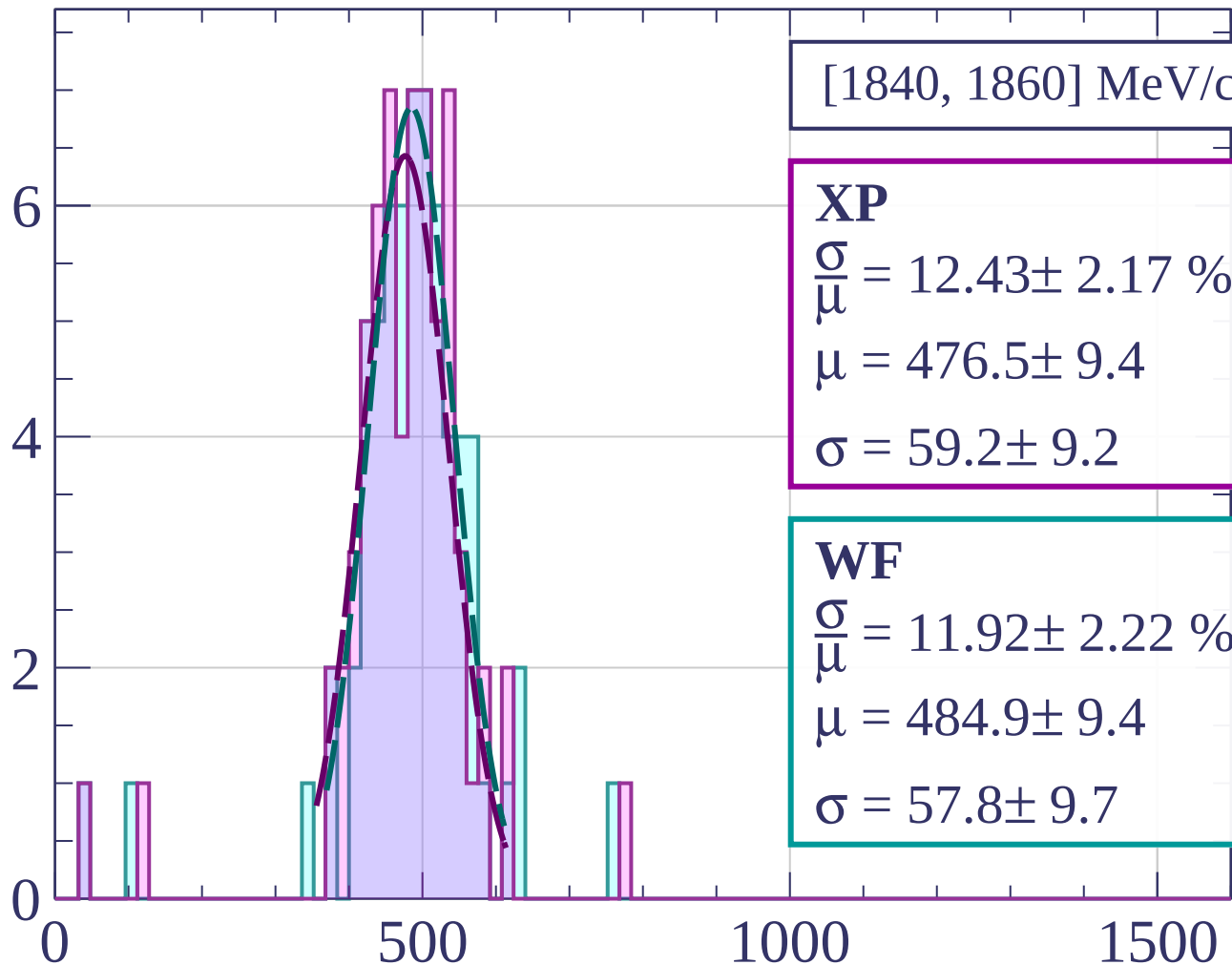
Count



Count



Count



[1840, 1860] MeV/c

XP

$$\frac{\sigma}{\mu} = 12.43 \pm 2.17 \%$$

$$\mu = 476.5 \pm 9.4$$

$$\sigma = 59.2 \pm 9.2$$

WF

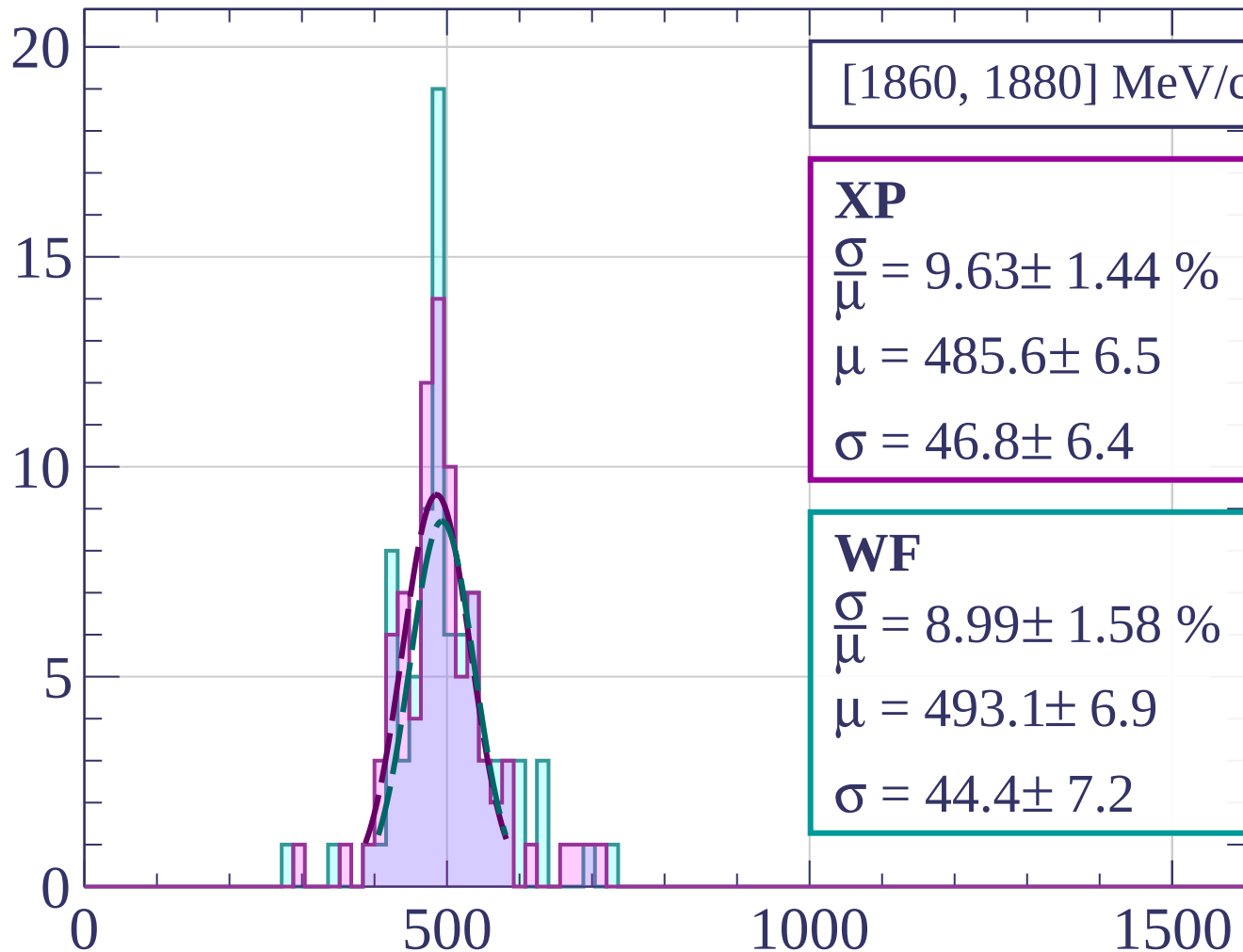
$$\frac{\sigma}{\mu} = 11.92 \pm 2.22 \%$$

$$\mu = 484.9 \pm 9.4$$

$$\sigma = 57.8 \pm 9.7$$

dE/dx [ADC counts/cm]

Count



[1860, 1880] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.63 \pm 1.44 \%$$

$$\mu = 485.6 \pm 6.5$$

$$\sigma = 46.8 \pm 6.4$$

WF

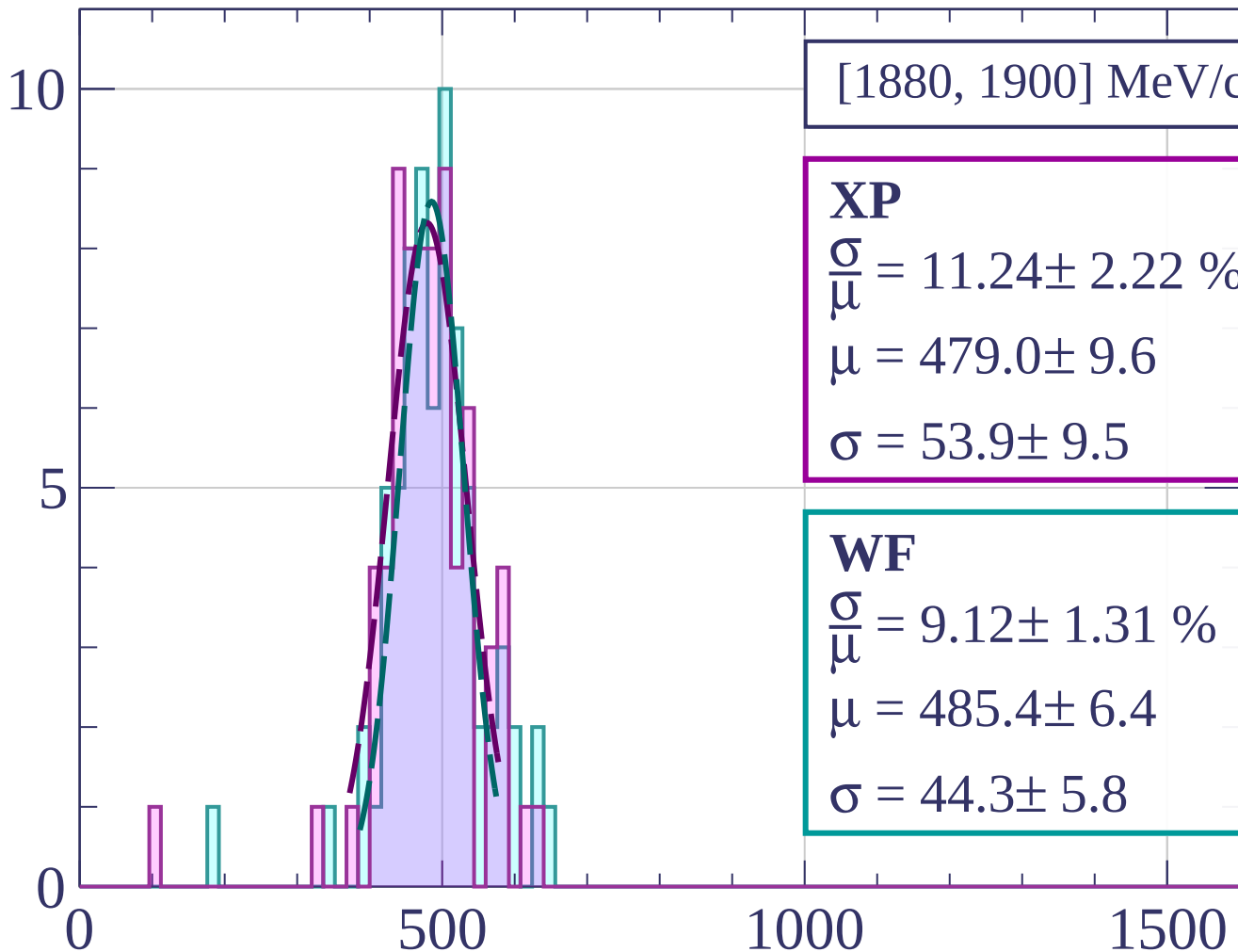
$$\frac{\sigma}{\mu} = 8.99 \pm 1.58 \%$$

$$\mu = 493.1 \pm 6.9$$

$$\sigma = 44.4 \pm 7.2$$

dE/dx [ADC counts/cm]

Count



[1880, 1900] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.24 \pm 2.22 \%$$

$$\mu = 479.0 \pm 9.6$$

$$\sigma = 53.9 \pm 9.5$$

WF

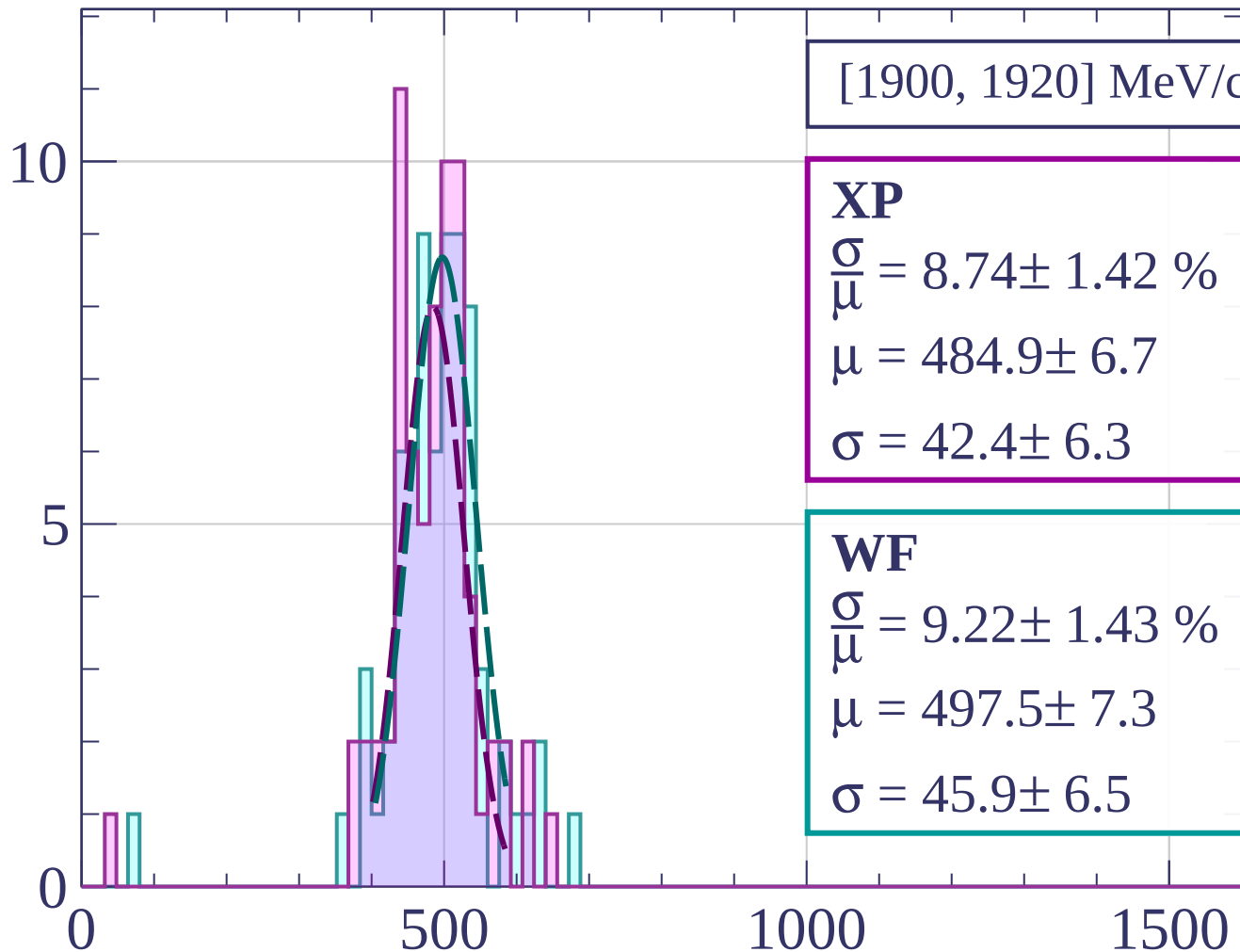
$$\frac{\sigma}{\mu} = 9.12 \pm 1.31 \%$$

$$\mu = 485.4 \pm 6.4$$

$$\sigma = 44.3 \pm 5.8$$

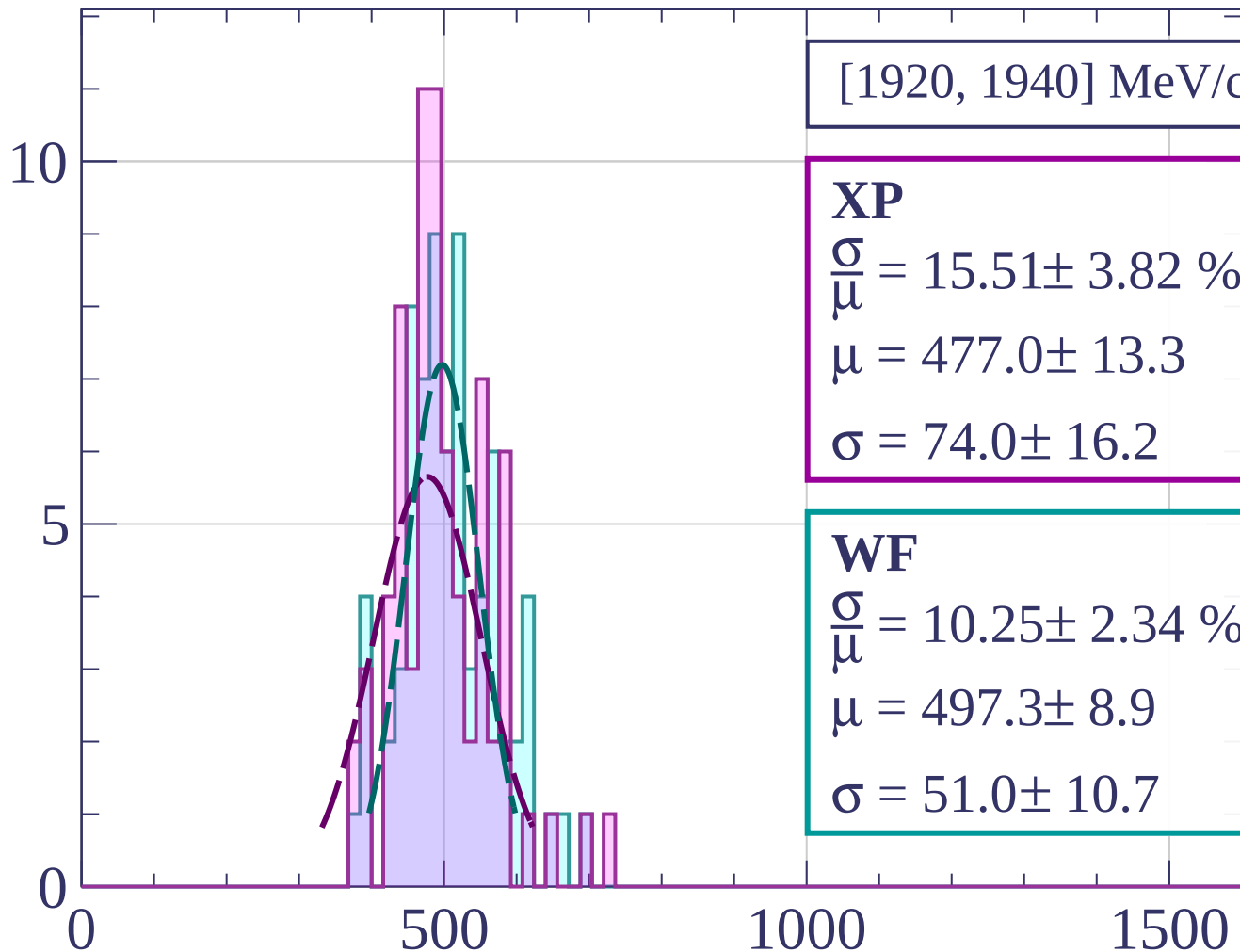
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[1920, 1940] MeV/c

XP

$$\frac{\sigma}{\mu} = 15.51 \pm 3.82 \%$$

$$\mu = 477.0 \pm 13.3$$

$$\sigma = 74.0 \pm 16.2$$

WF

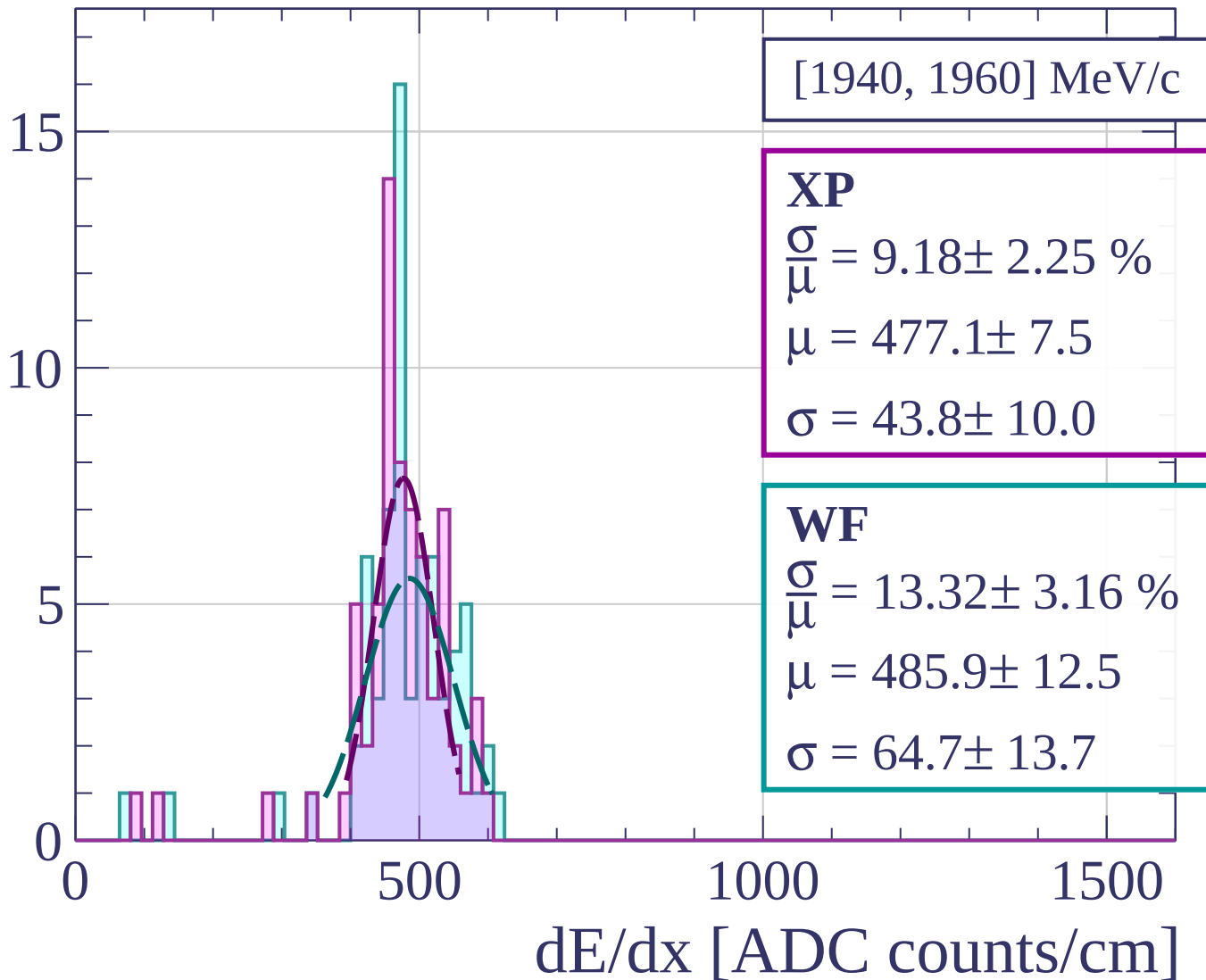
$$\frac{\sigma}{\mu} = 10.25 \pm 2.34 \%$$

$$\mu = 497.3 \pm 8.9$$

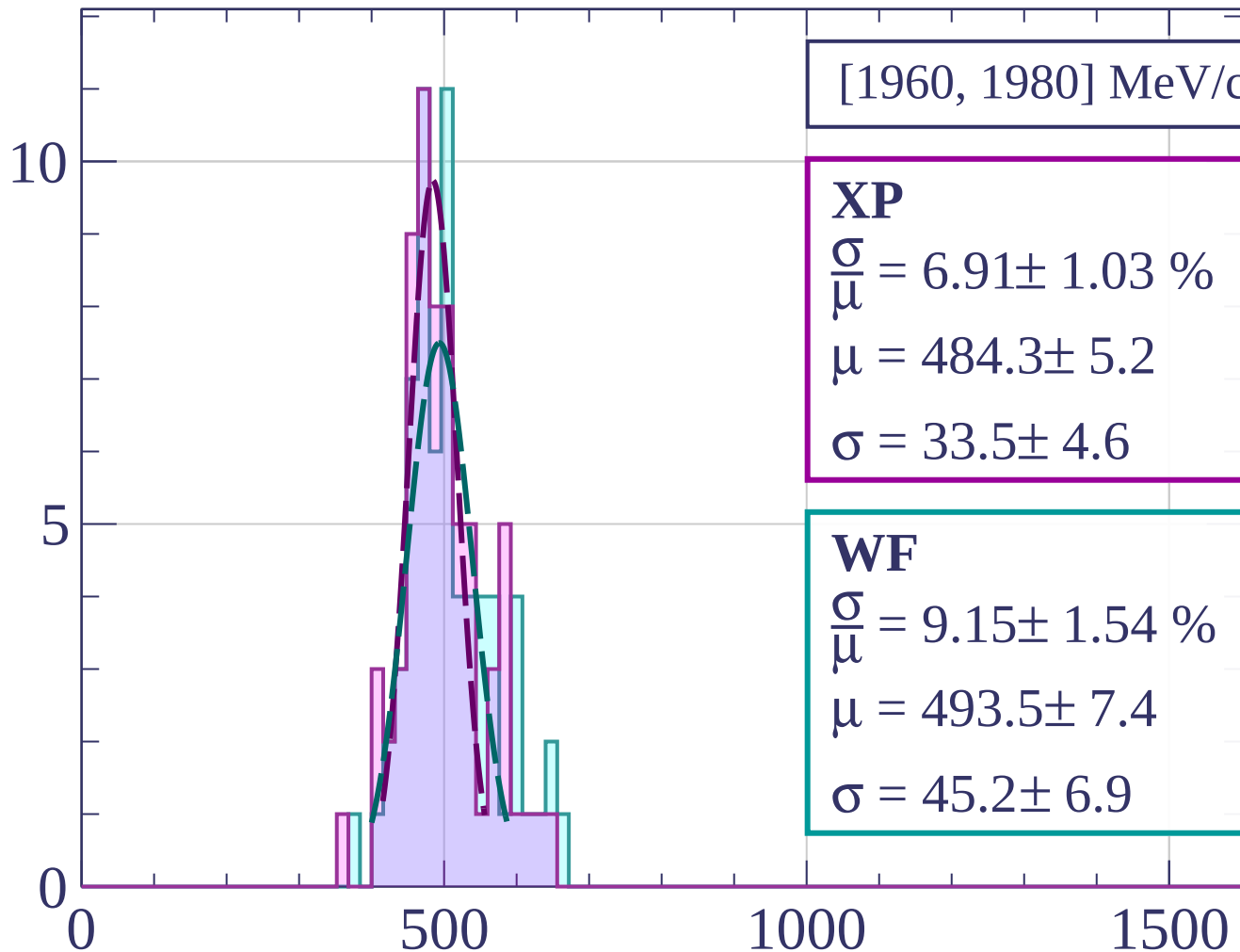
$$\sigma = 51.0 \pm 10.7$$

dE/dx [ADC counts/cm]

Count

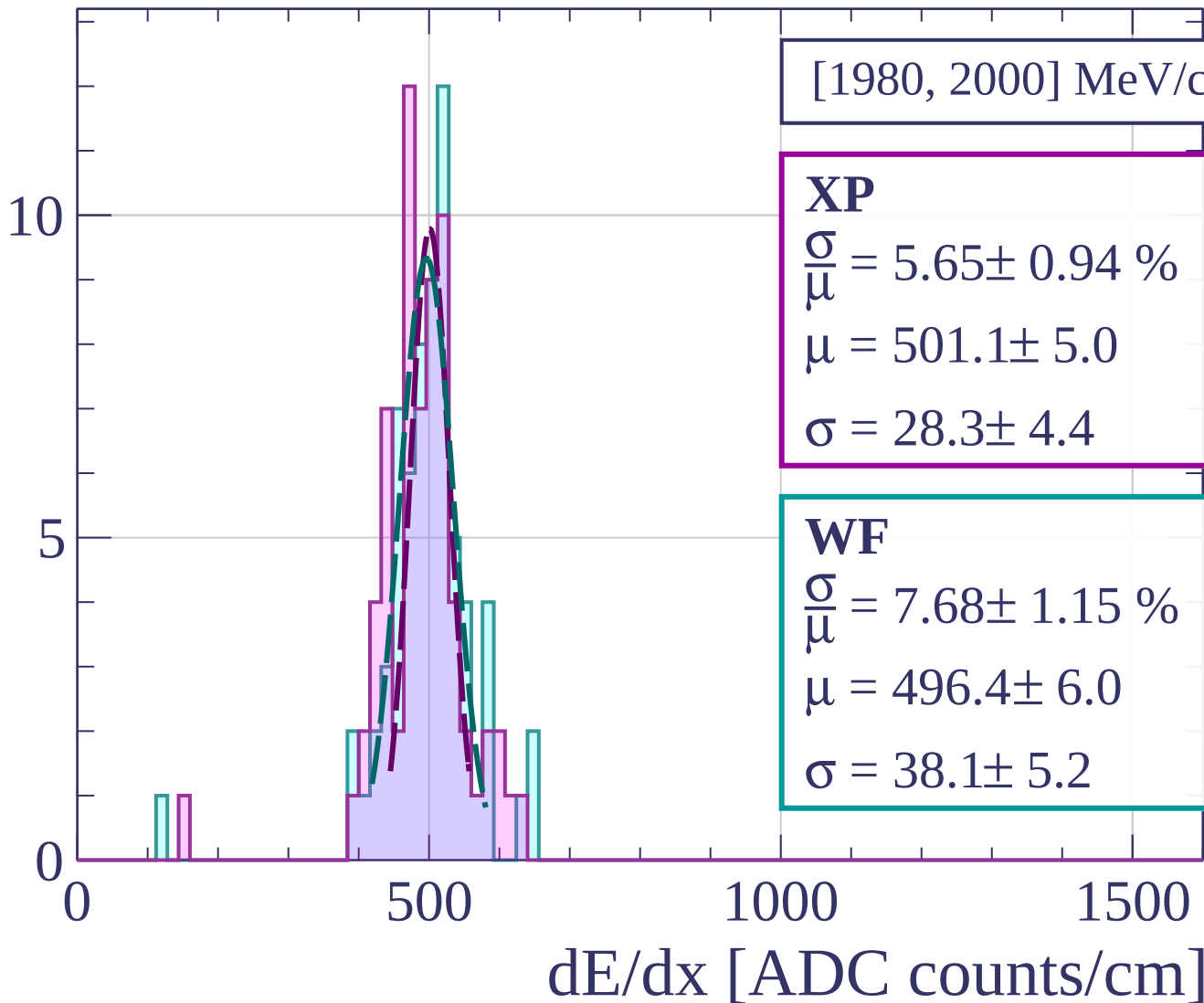


Count

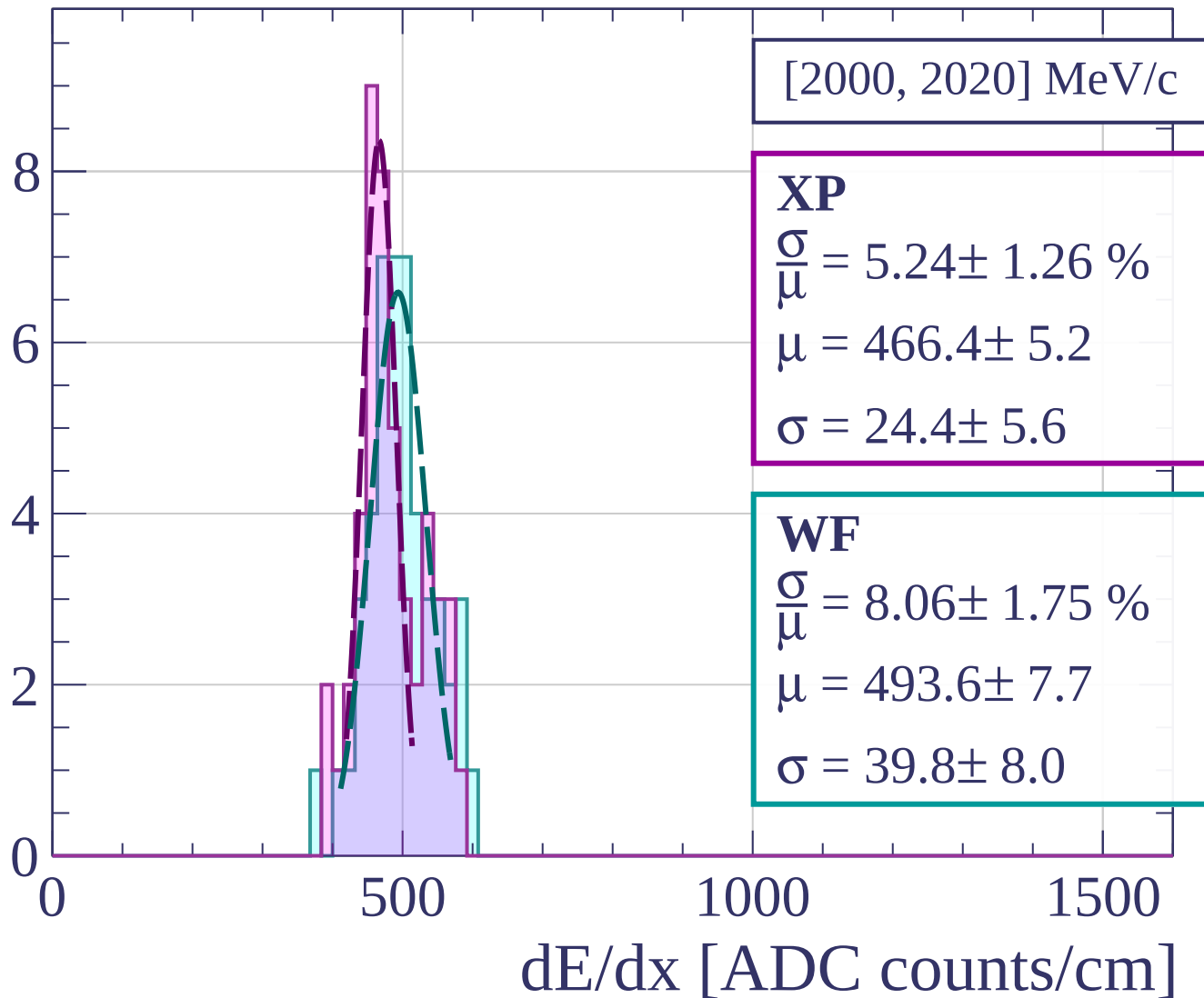


dE/dx [ADC counts/cm]

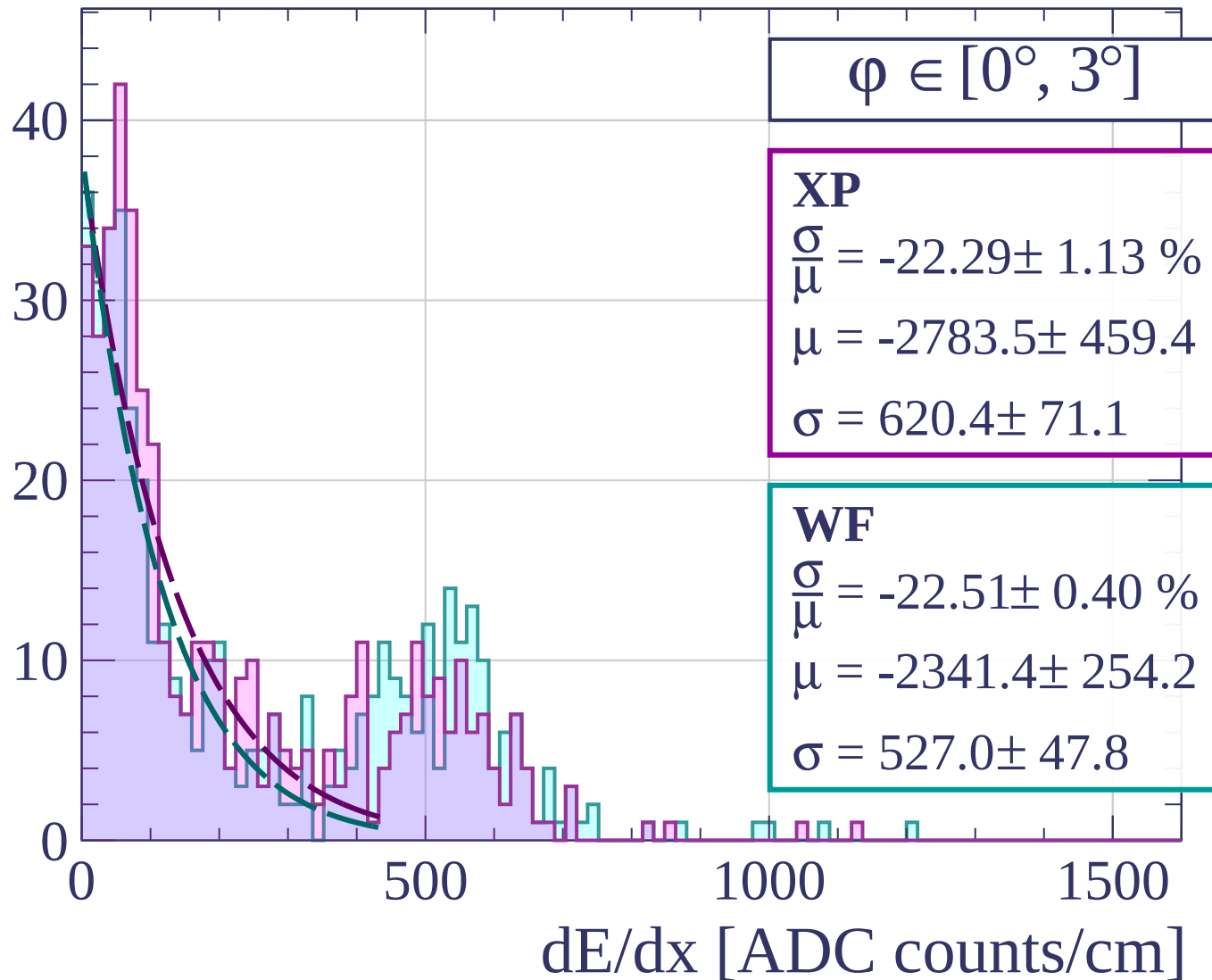
Count



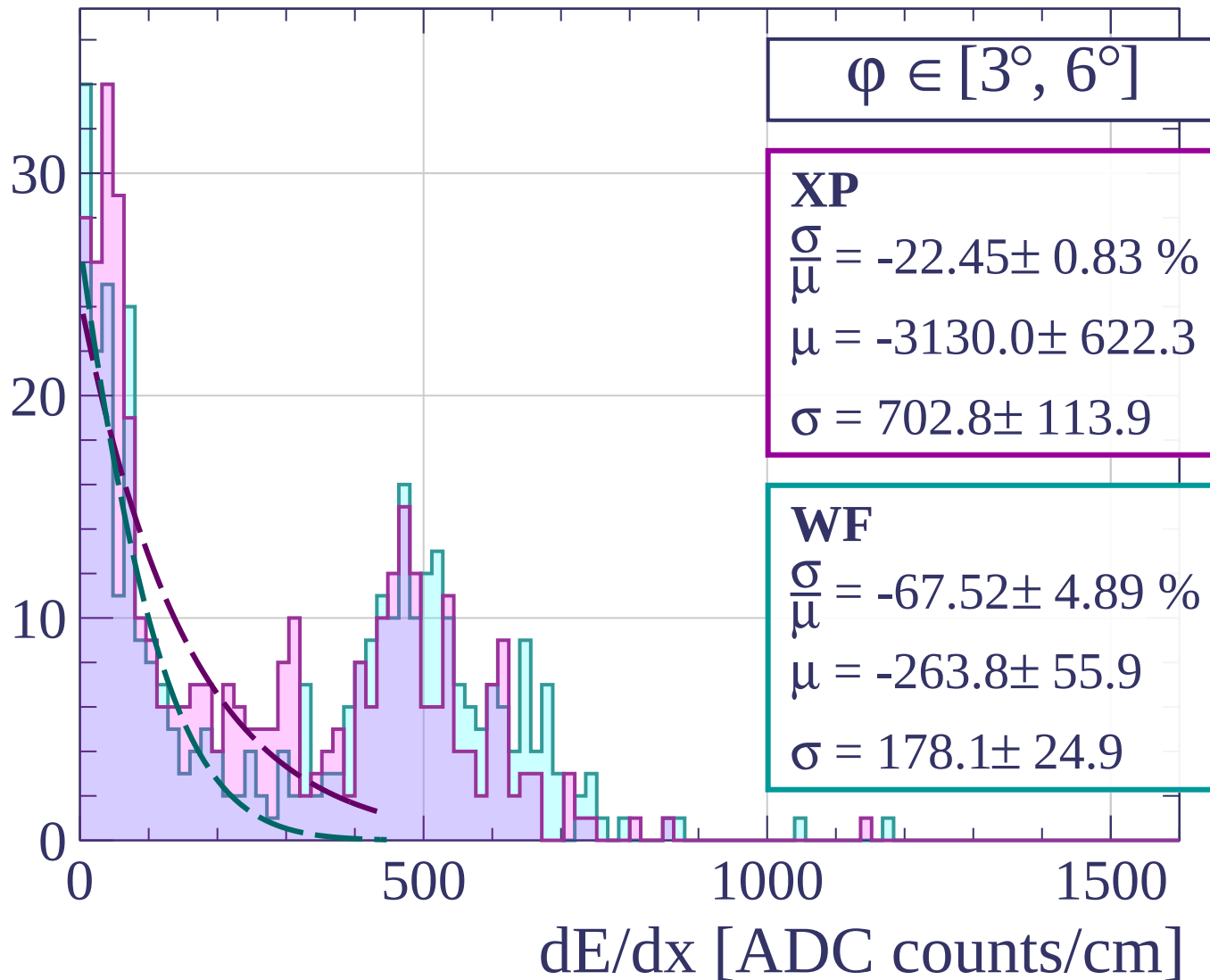
Count



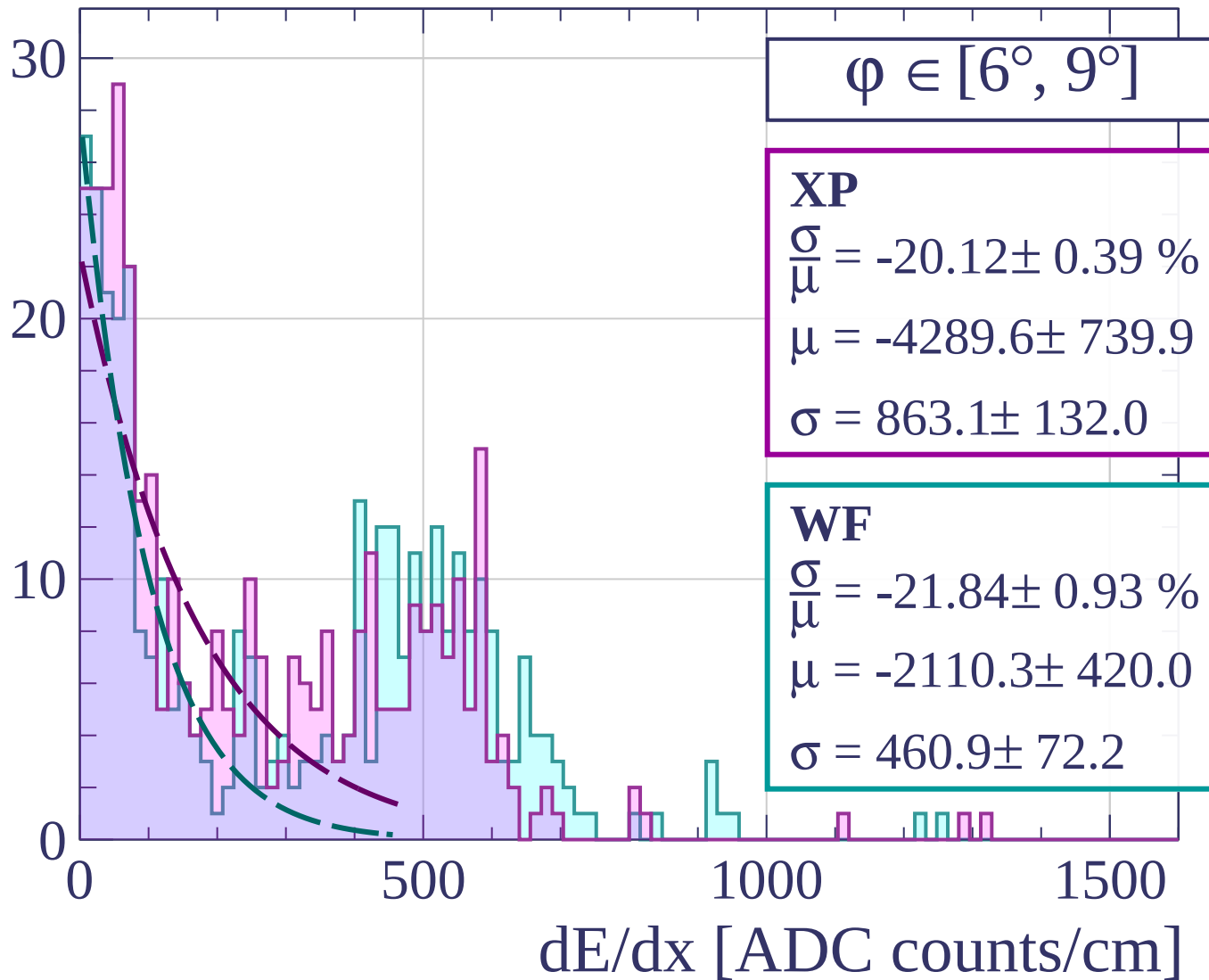
Count



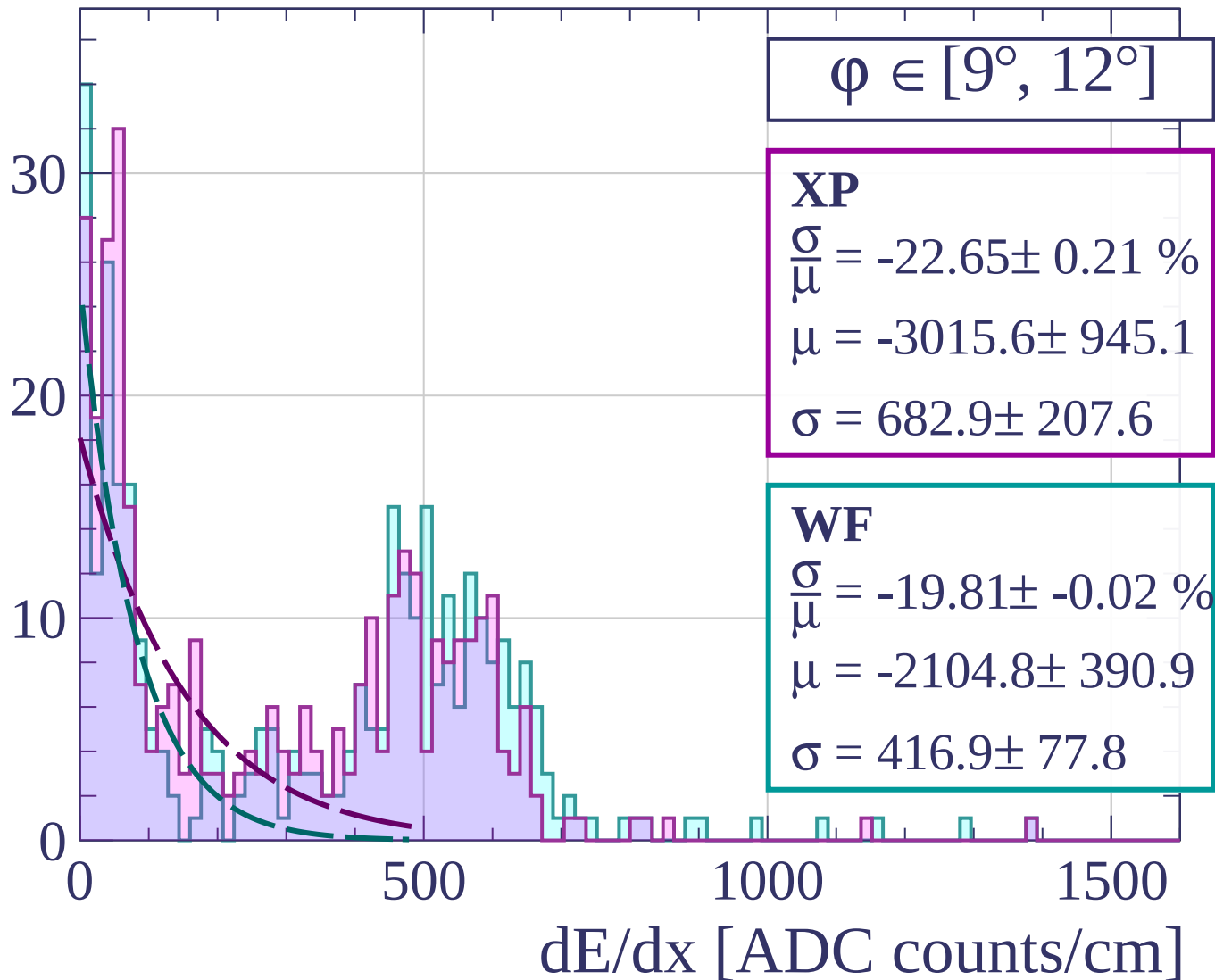
Count



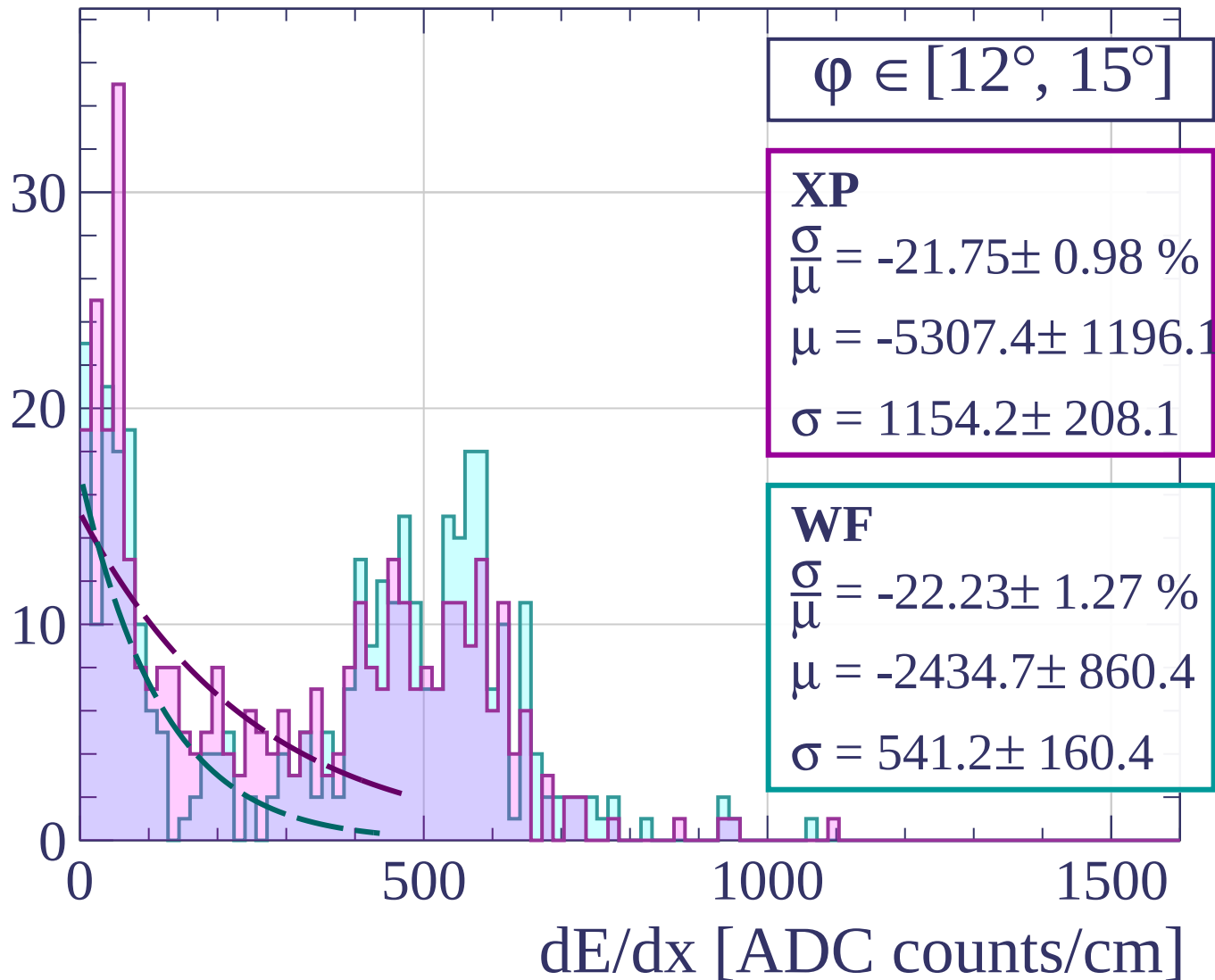
Count



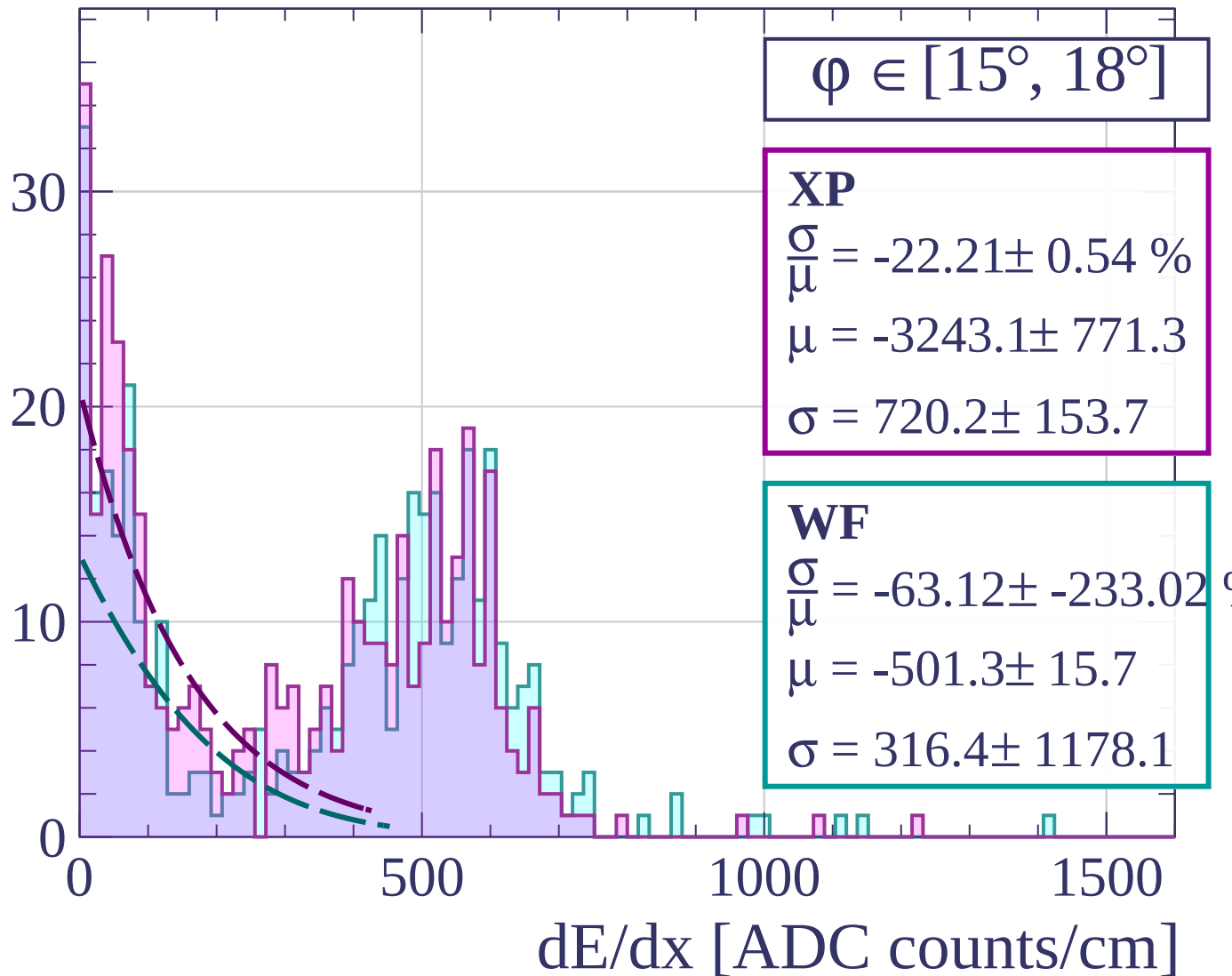
Count



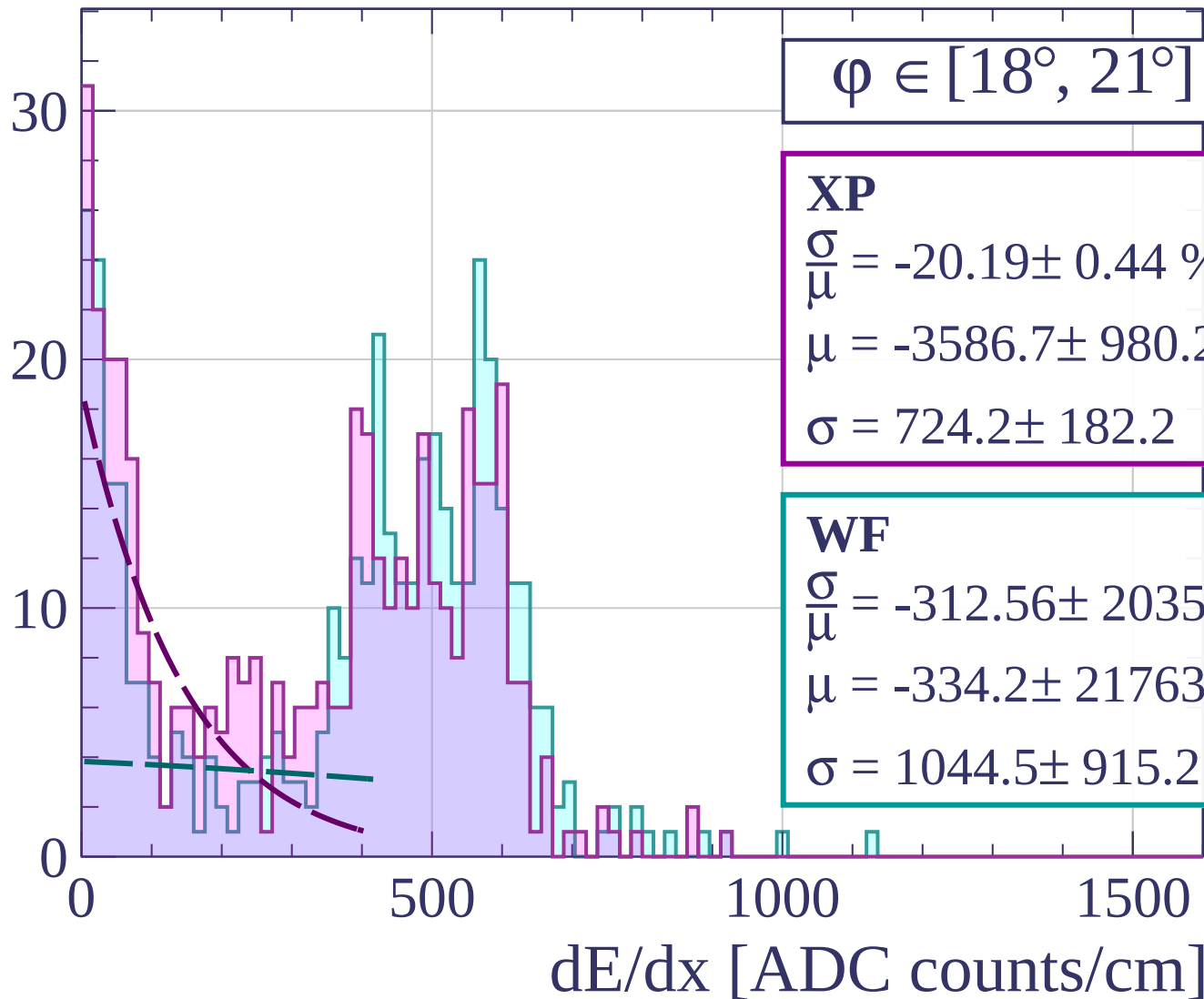
Count



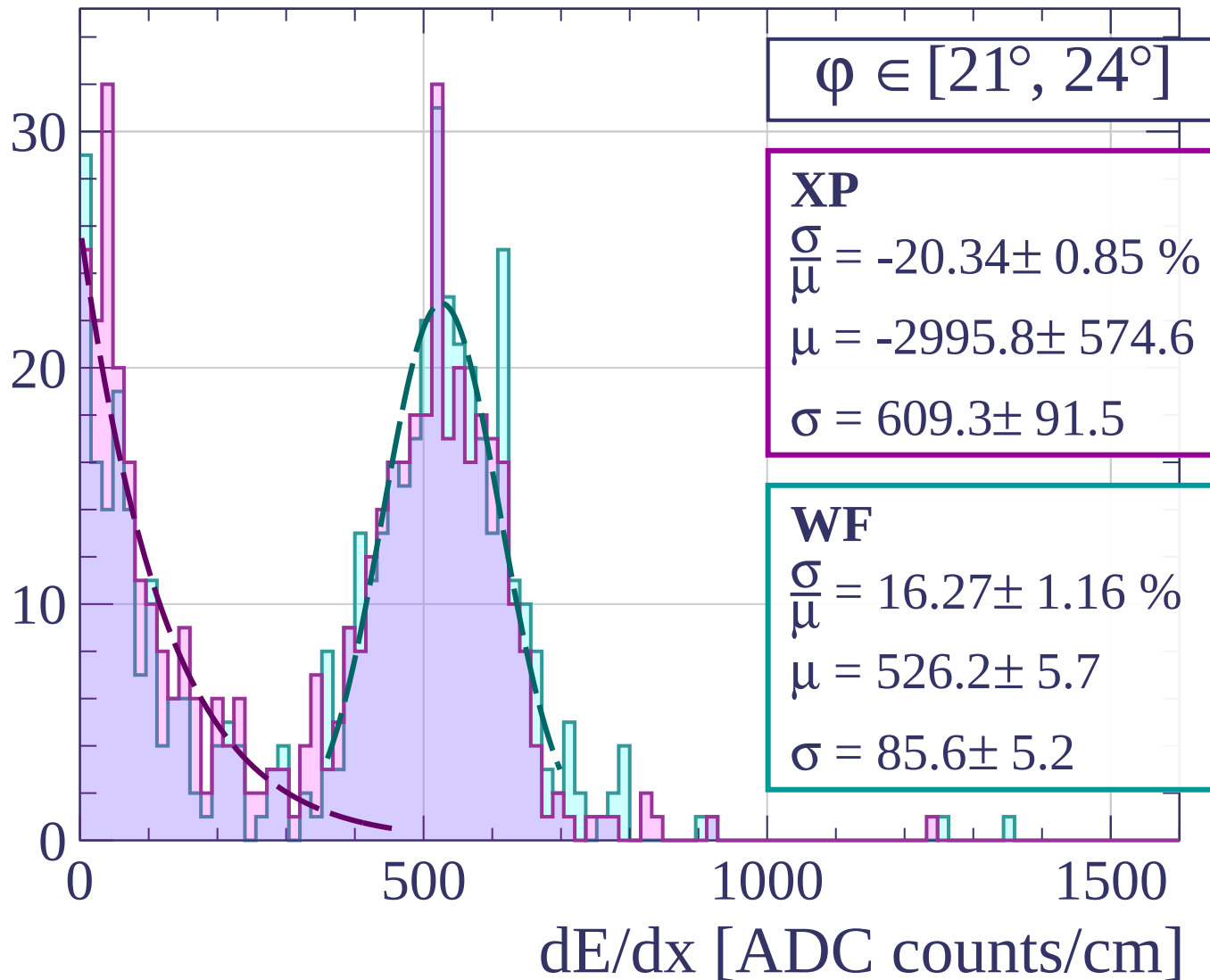
Count



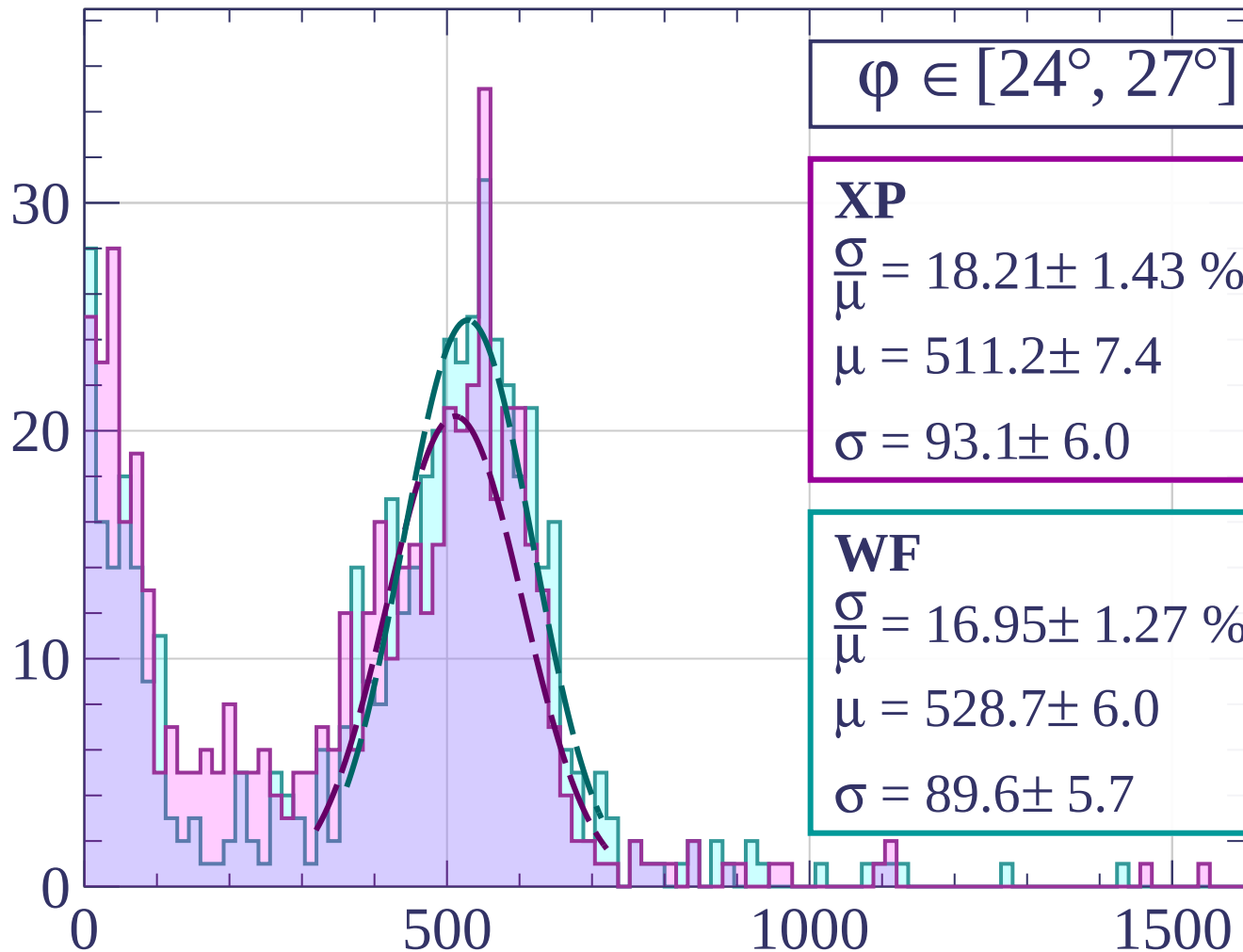
Count



Count

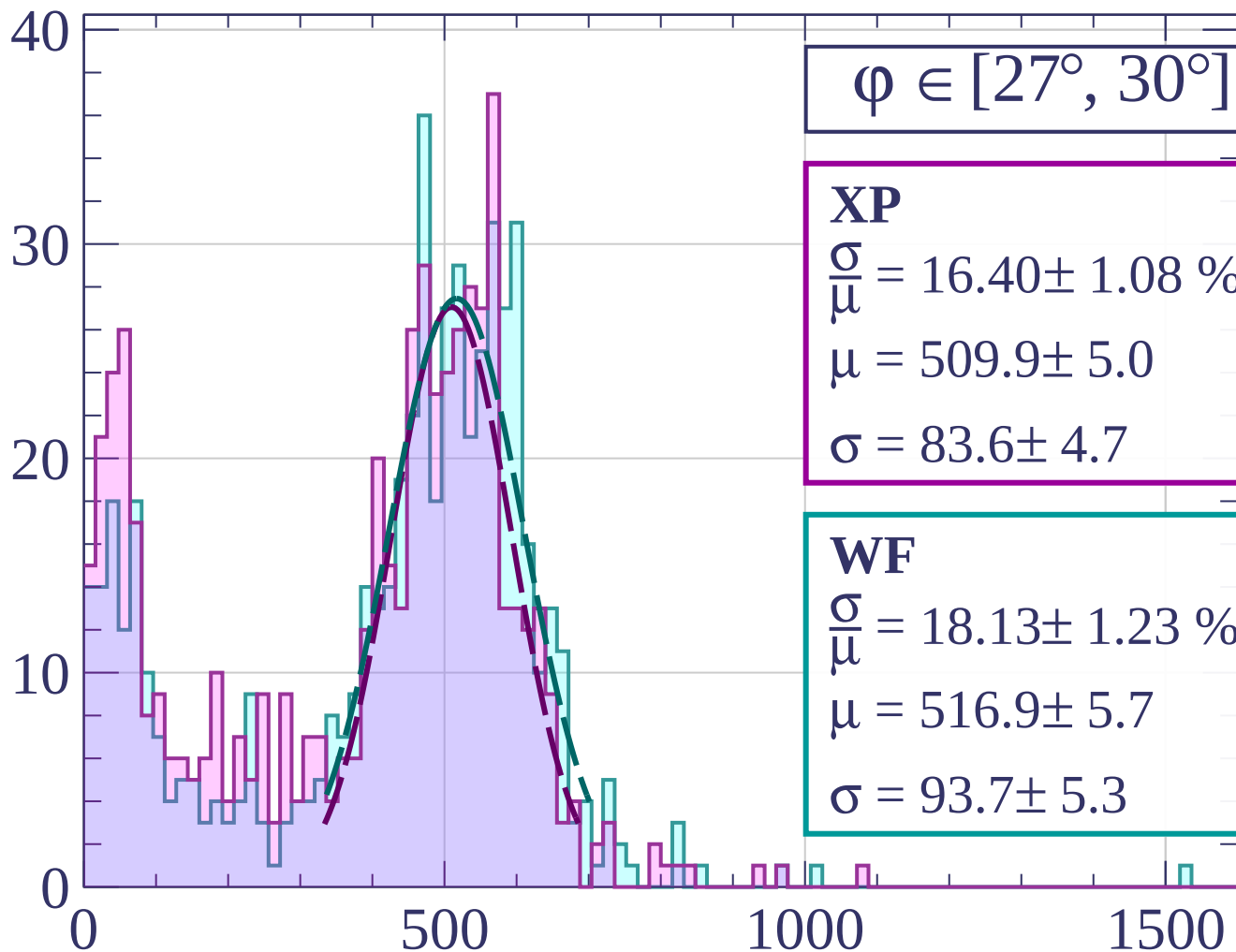


Count



dE/dx [ADC counts/cm]

Count



$\phi \in [27^\circ, 30^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.40 \pm 1.08 \%$$

$$\mu = 509.9 \pm 5.0$$

$$\sigma = 83.6 \pm 4.7$$

WF

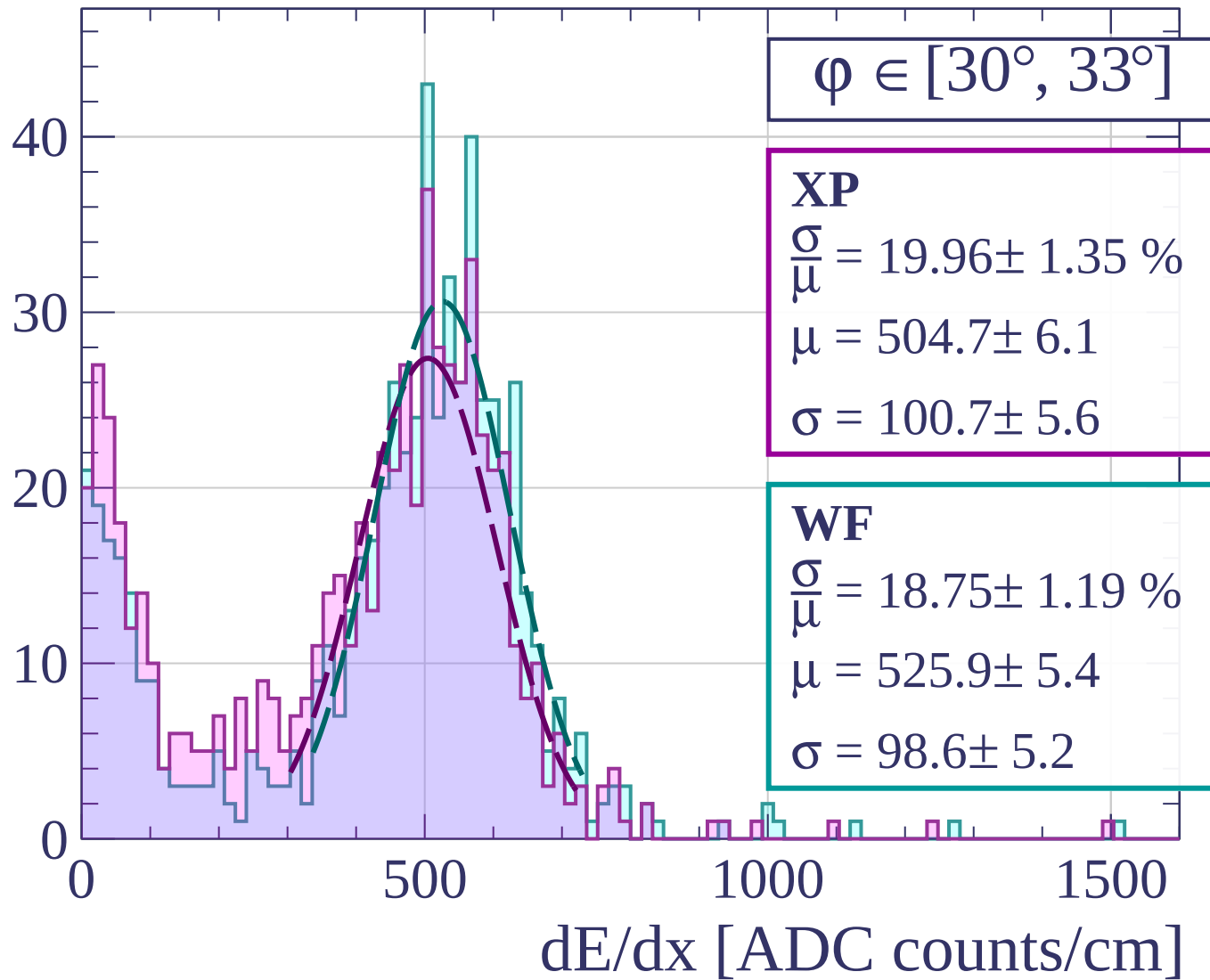
$$\frac{\sigma}{\mu} = 18.13 \pm 1.23 \%$$

$$\mu = 516.9 \pm 5.7$$

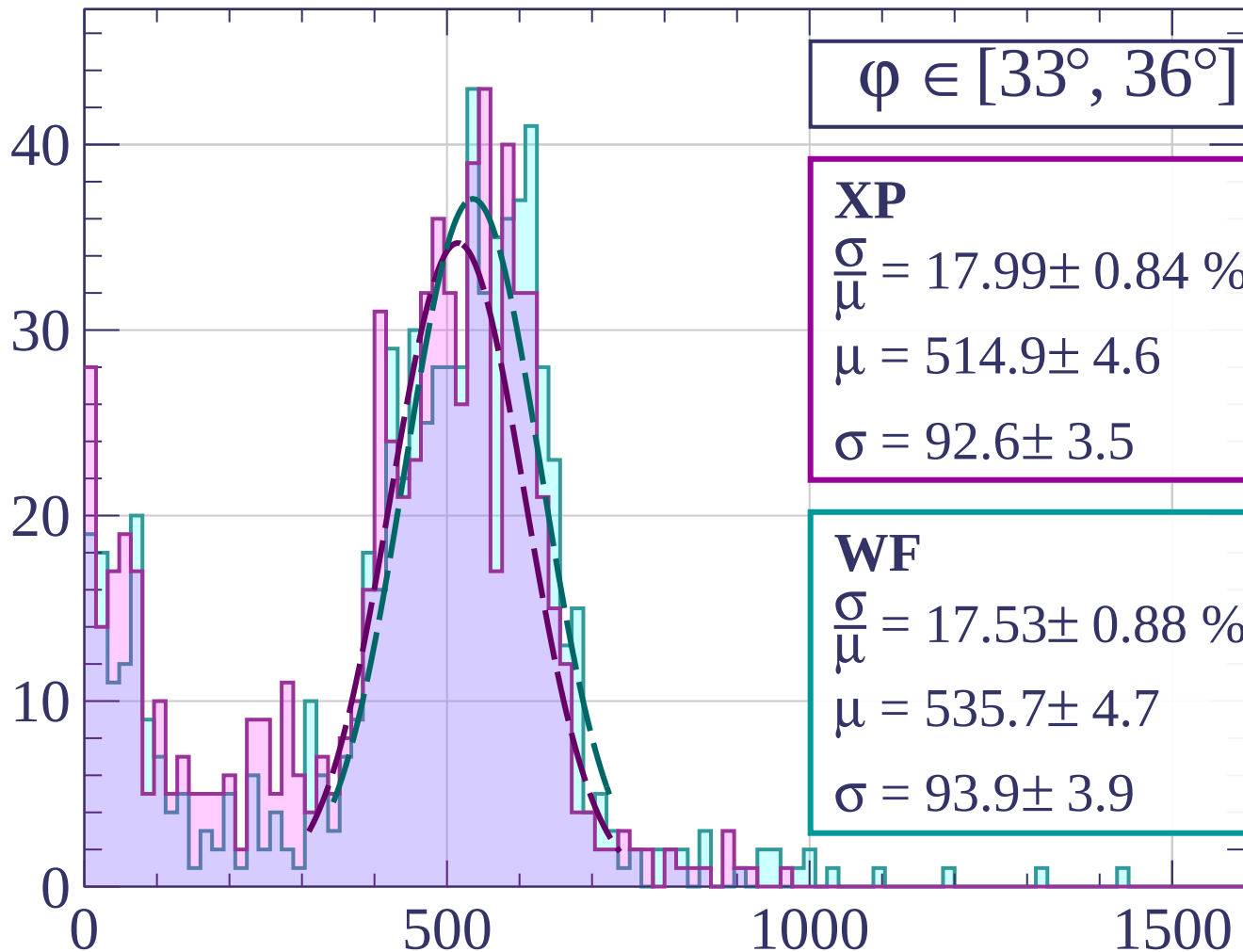
$$\sigma = 93.7 \pm 5.3$$

dE/dx [ADC counts/cm]

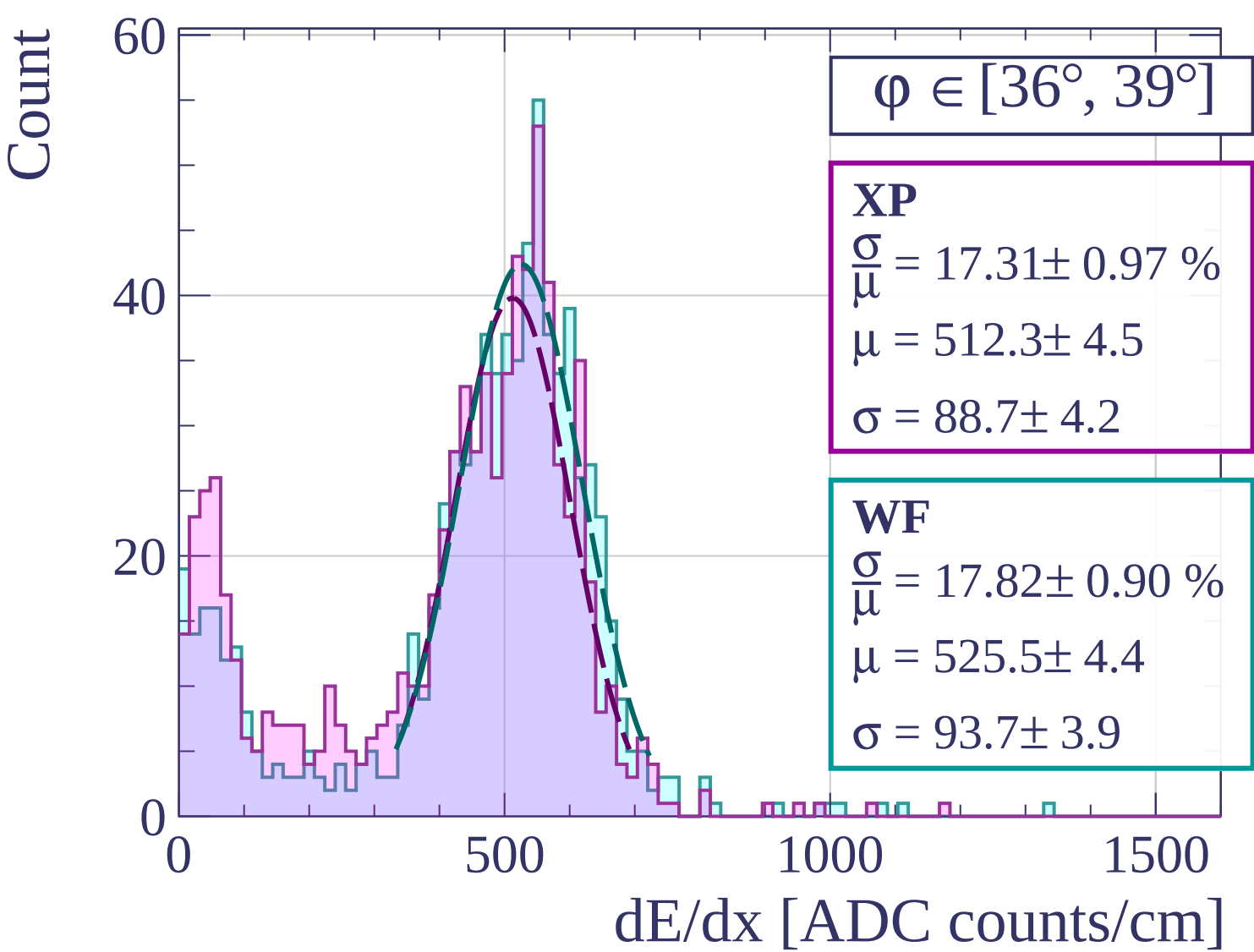
Count



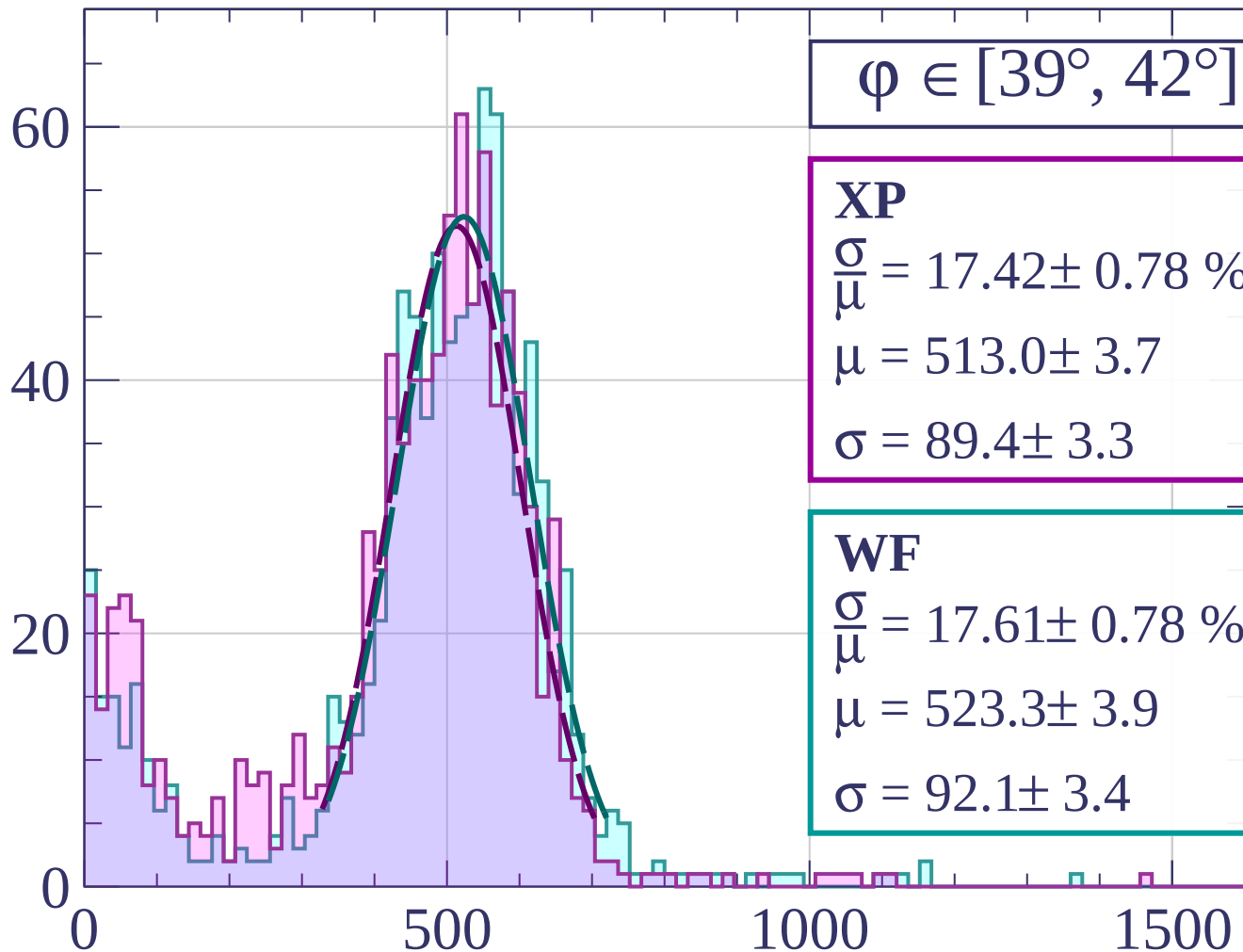
Count



dE/dx [ADC counts/cm]

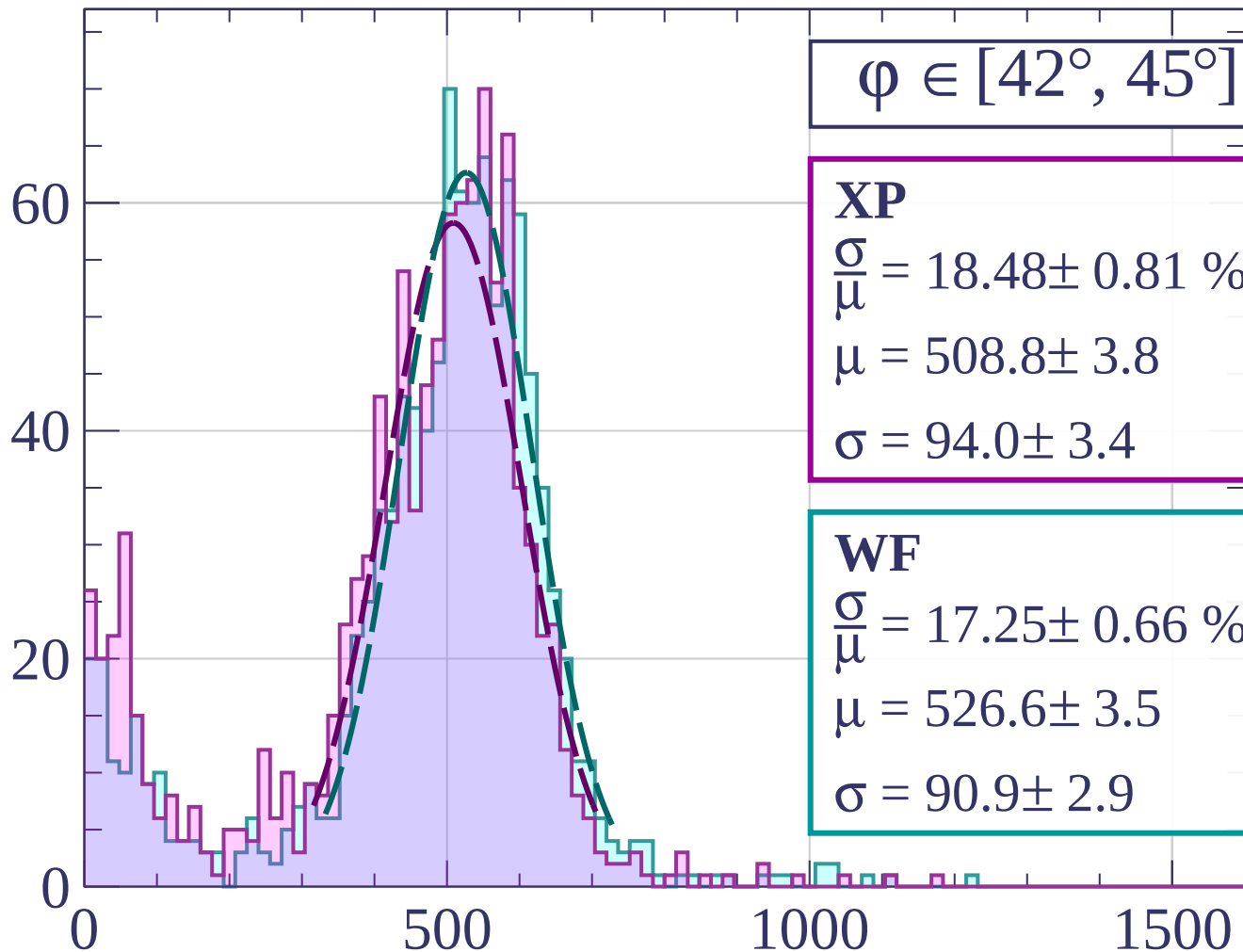


Count



dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

$\varphi \in [45^\circ, 48^\circ]$

XP

$$\frac{\sigma}{\mu} = 17.28 \pm 0.64 \%$$

$$\mu = 500.9 \pm 3.1$$

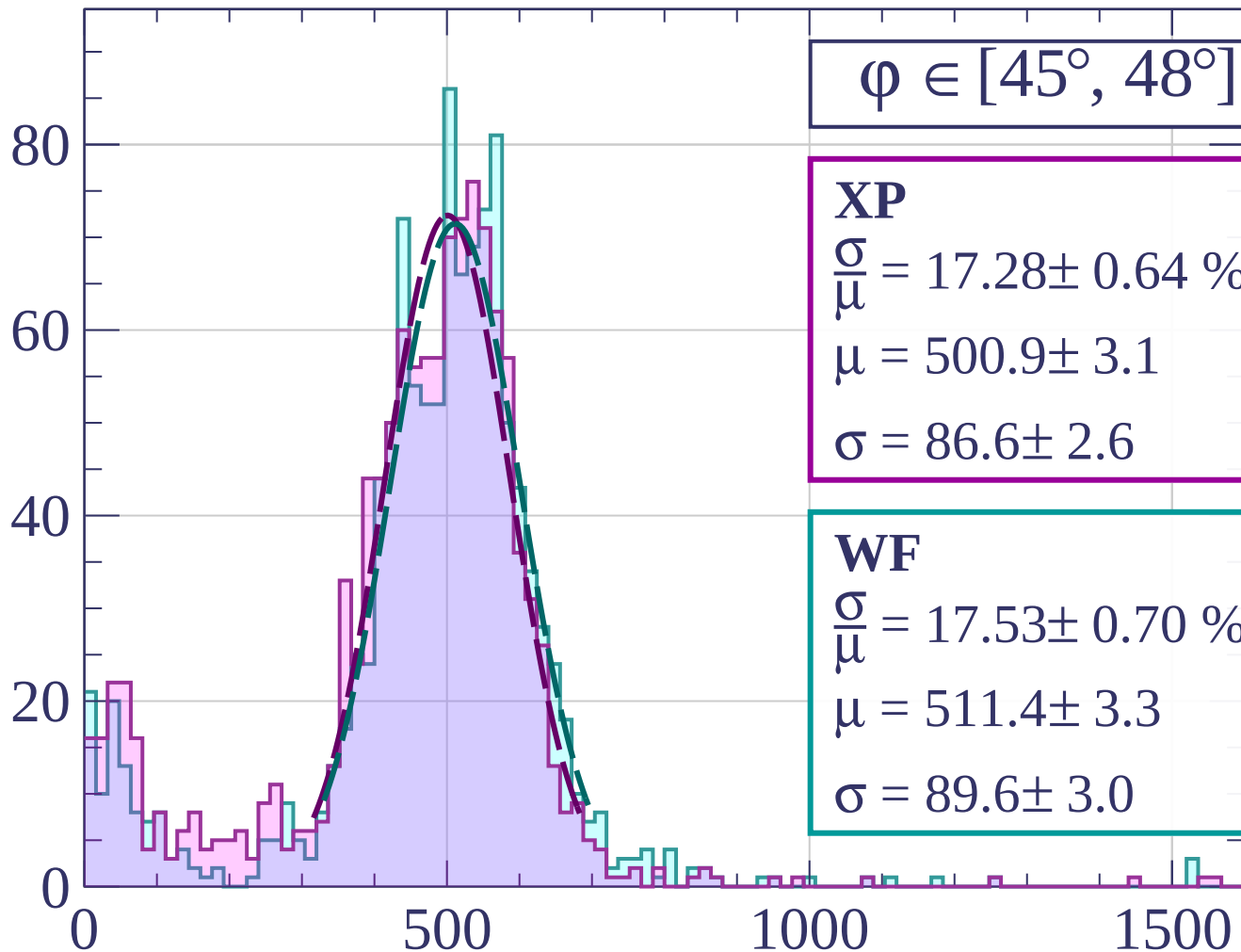
$$\sigma = 86.6 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 17.53 \pm 0.70 \%$$

$$\mu = 511.4 \pm 3.3$$

$$\sigma = 89.6 \pm 3.0$$



dE/dx [ADC counts/cm]

Count

$\varphi \in [48^\circ, 51^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.69 \pm 0.61 \%$$

$$\mu = 501.4 \pm 2.8$$

$$\sigma = 83.7 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 16.55 \pm 0.57 \%$$

$$\mu = 515.0 \pm 2.8$$

$$\sigma = 85.2 \pm 2.5$$

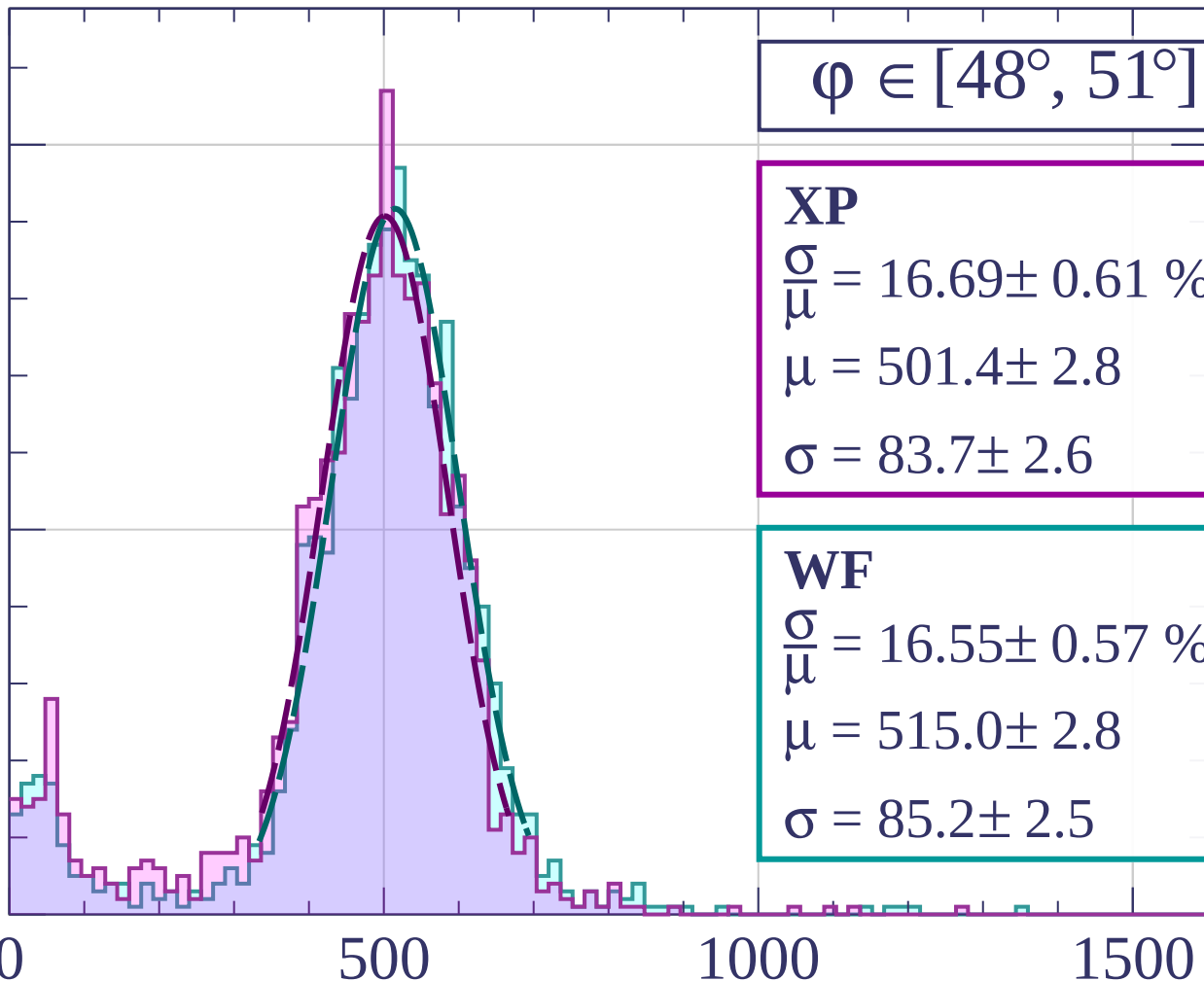
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\varphi \in [51^\circ, 54^\circ]$

XP

$$\frac{\sigma}{\mu} = 17.32 \pm 0.55 \%$$

$$\mu = 499.1 \pm 2.6$$

$$\sigma = 86.4 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 16.99 \pm 0.48 \%$$

$$\mu = 504.8 \pm 2.5$$

$$\sigma = 85.8 \pm 2.0$$

0

500

1000

1500

dE/dx [ADC counts/cm]

100

50

Count

$\phi \in [54^\circ, 57^\circ]$

XP

$$\frac{\sigma}{\mu} = 17.46 \pm 0.50 \%$$

$$\mu = 496.2 \pm 2.4$$

$$\sigma = 86.6 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 17.33 \pm 0.47 \%$$

$$\mu = 504.7 \pm 2.4$$

$$\sigma = 87.5 \pm 2.0$$

100

50

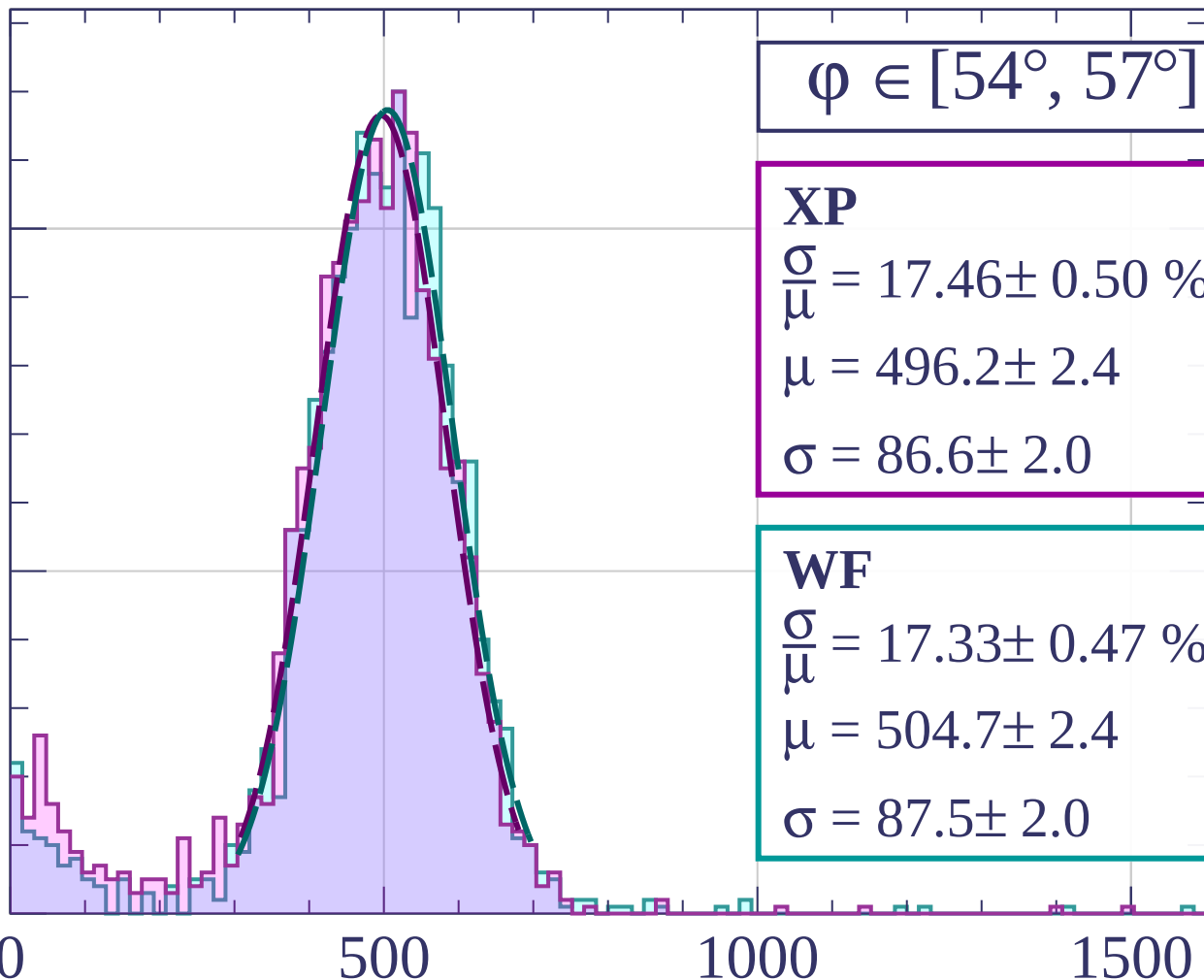
0

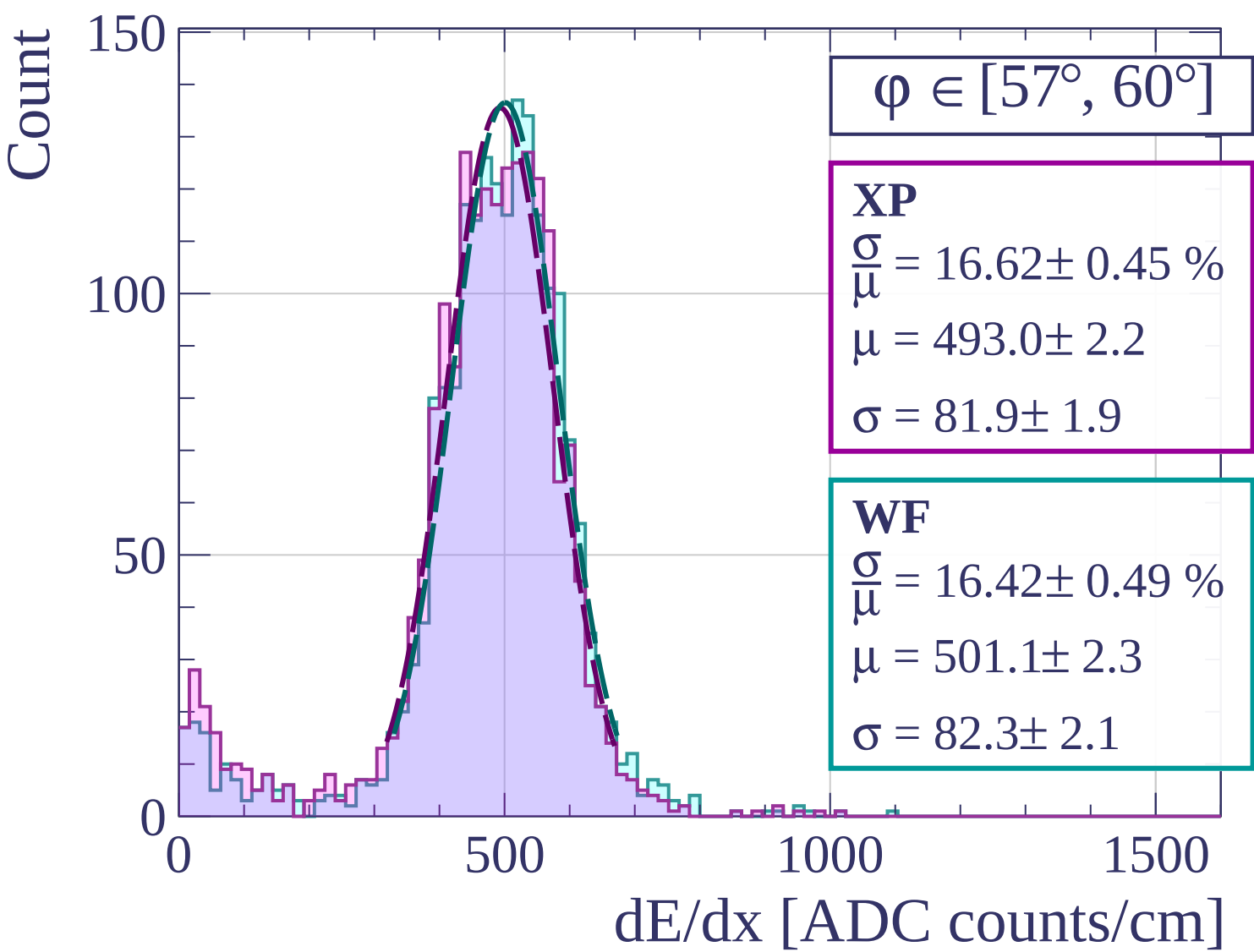
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\phi \in [60^\circ, 63^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.04 \pm 0.37 \%$$

$$\mu = 490.7 \pm 1.9$$

$$\sigma = 78.7 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 16.67 \pm 0.43 \%$$

$$\mu = 501.1 \pm 2.1$$

$$\sigma = 83.5 \pm 1.8$$

150

100

50

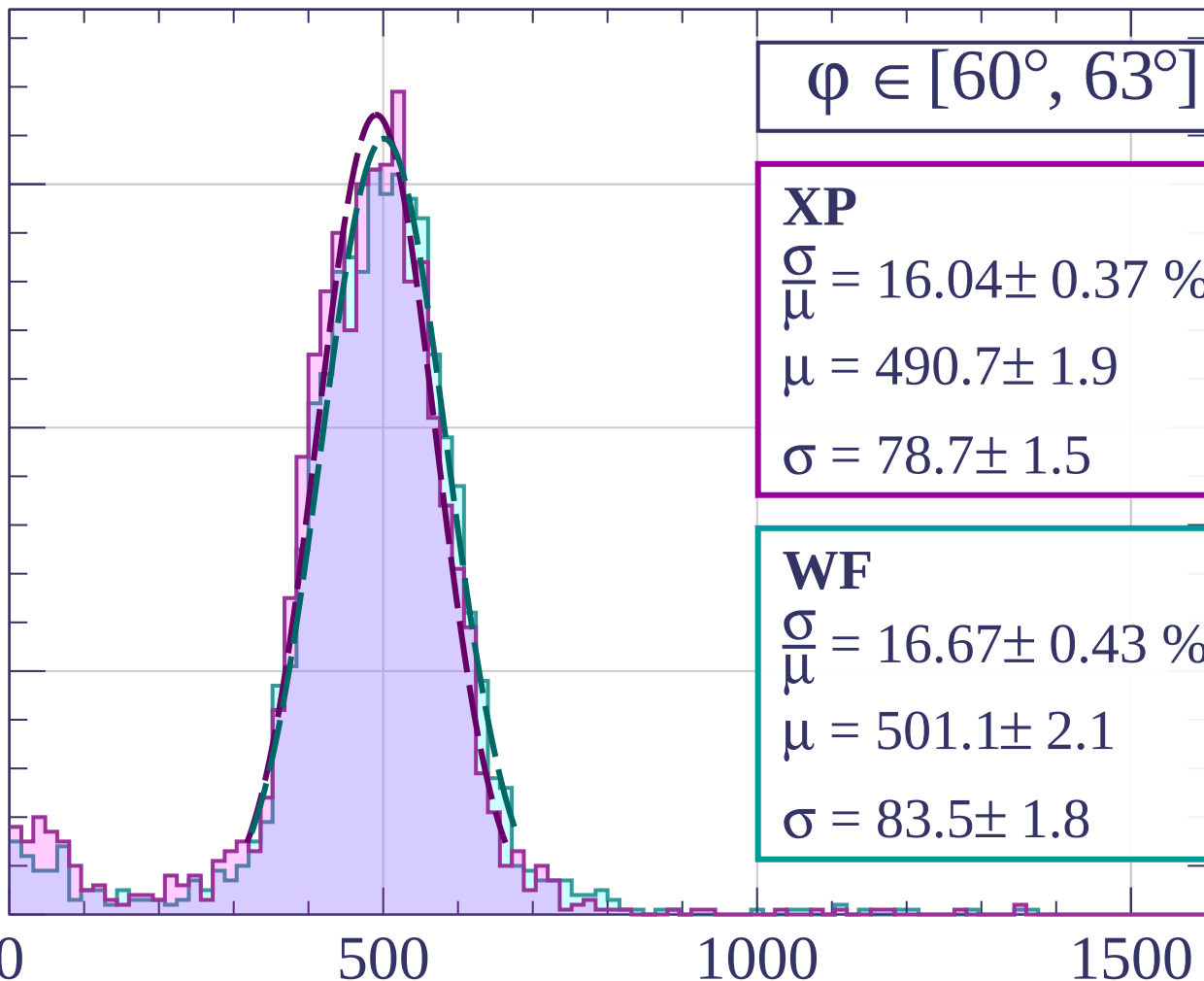
0

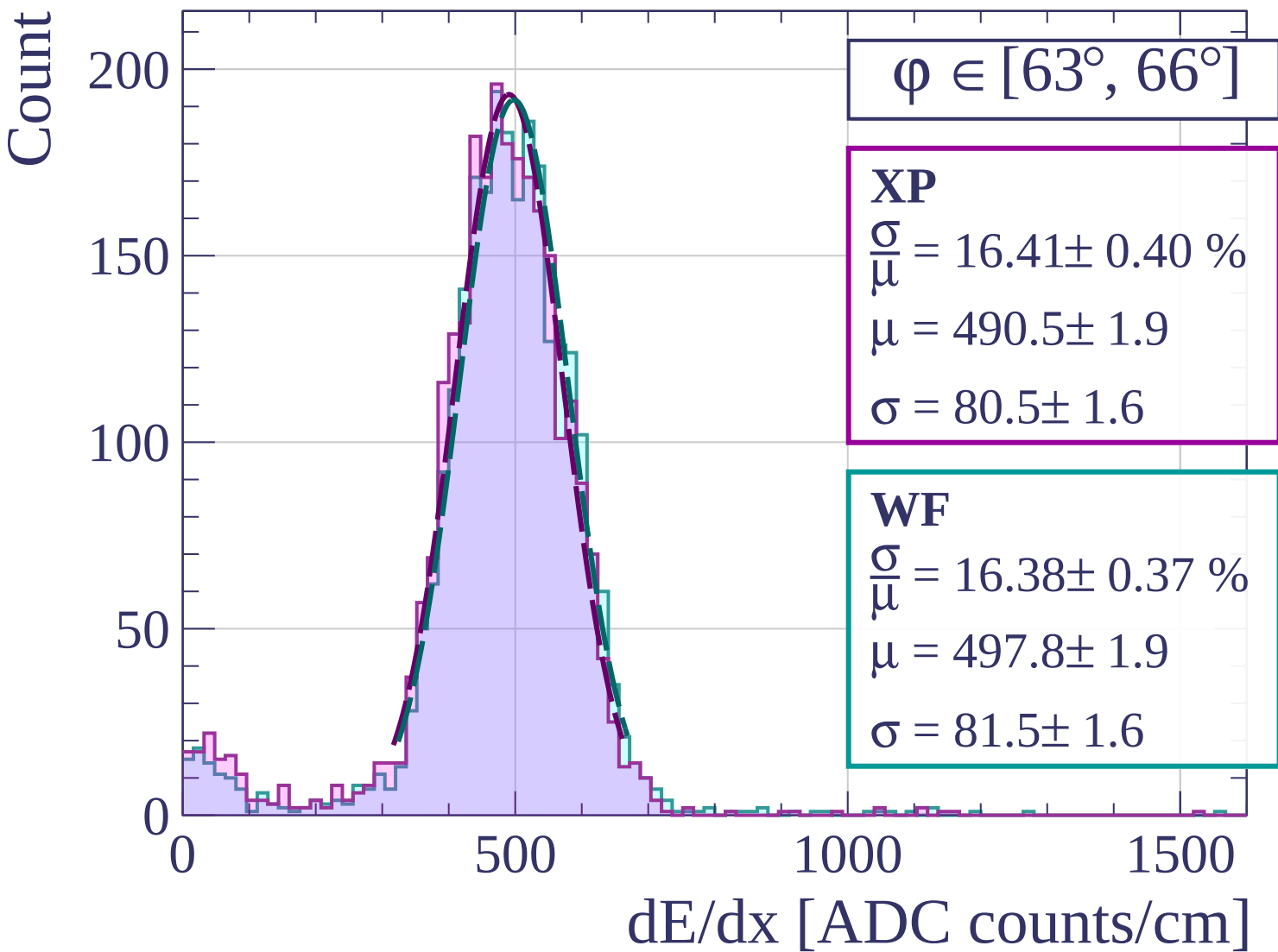
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\phi \in [66^\circ, 69^\circ]$

XP

$$\frac{\sigma}{\mu} = 15.87 \pm 0.34 \%$$

$$\mu = 484.0 \pm 1.7$$

$$\sigma = 76.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 15.98 \pm 0.35 \%$$

$$\mu = 492.4 \pm 1.8$$

$$\sigma = 78.7 \pm 1.4$$

200

150

100

50

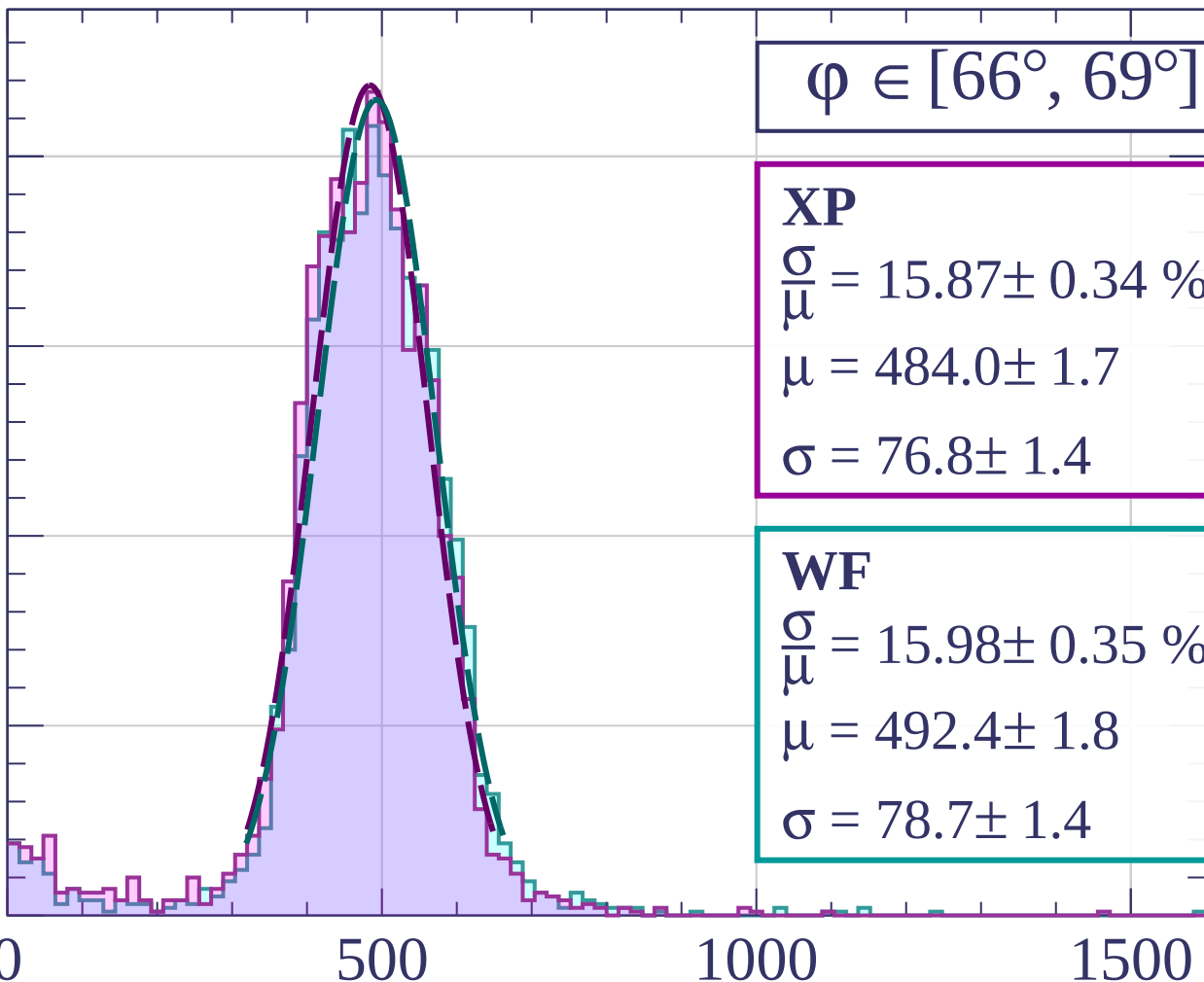
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [69^\circ, 72^\circ]$

XP

$$\frac{\sigma}{\mu} = 15.65 \pm 0.33 \%$$

$$\mu = 485.5 \pm 1.6$$

$$\sigma = 76.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 16.11 \pm 0.34 \%$$

$$\mu = 494.0 \pm 1.6$$

$$\sigma = 79.6 \pm 1.4$$

200

100

0

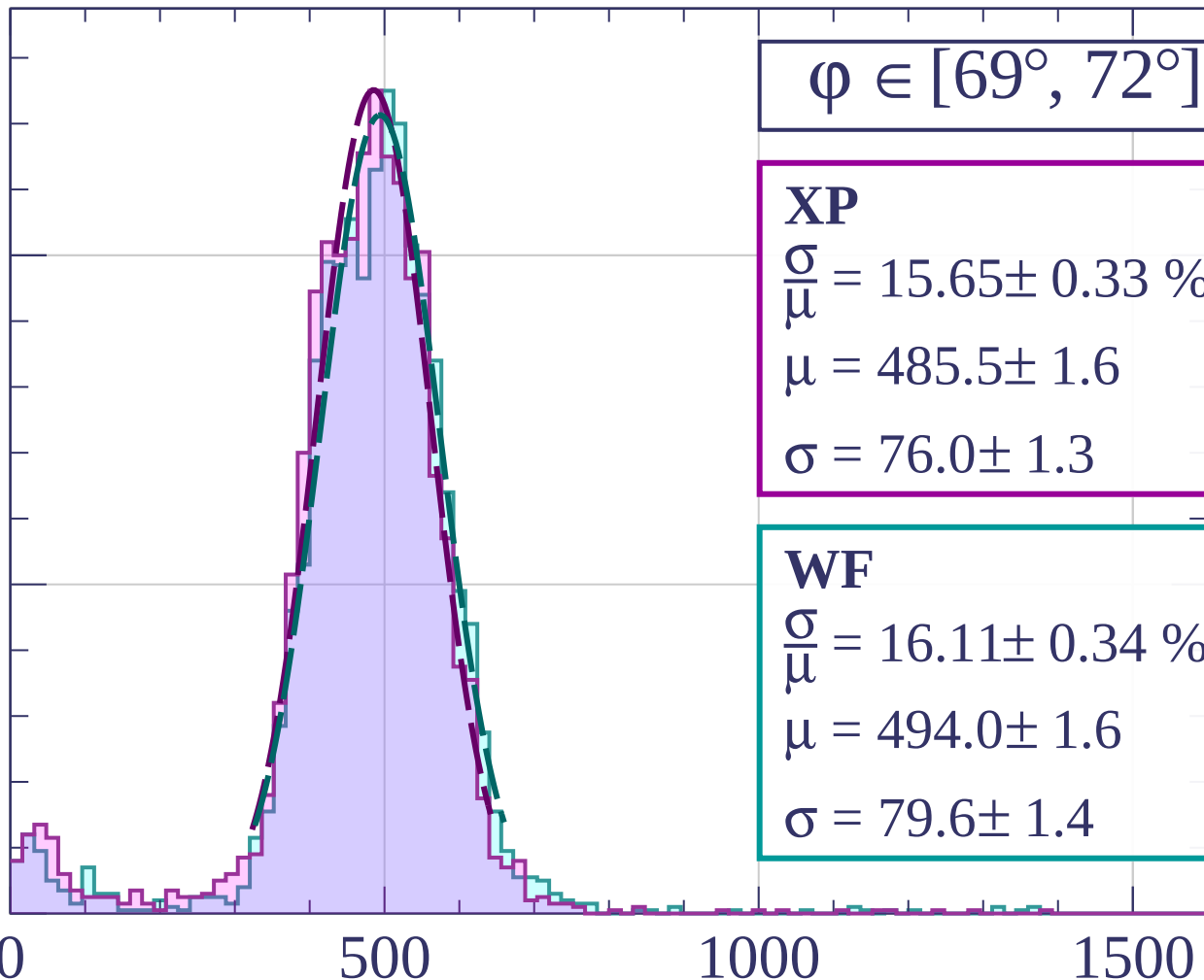
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [72^\circ, 75^\circ]$

XP

$$\frac{\sigma}{\mu} = 15.88 \pm 0.33 \%$$

$$\mu = 482.1 \pm 1.5$$

$$\sigma = 76.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 15.73 \pm 0.31 \%$$

$$\mu = 492.1 \pm 1.6$$

$$\sigma = 77.4 \pm 1.3$$

200

100

0

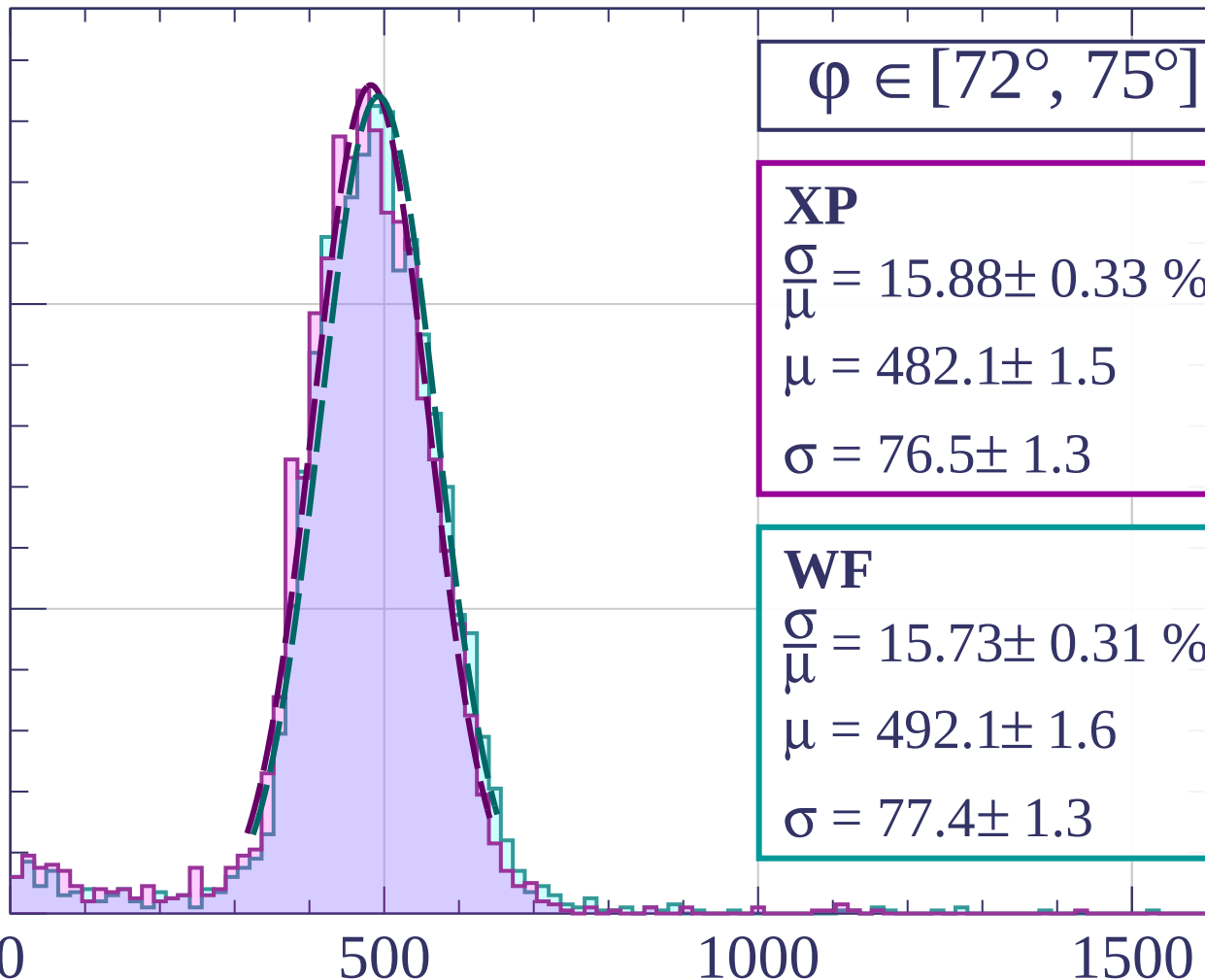
0

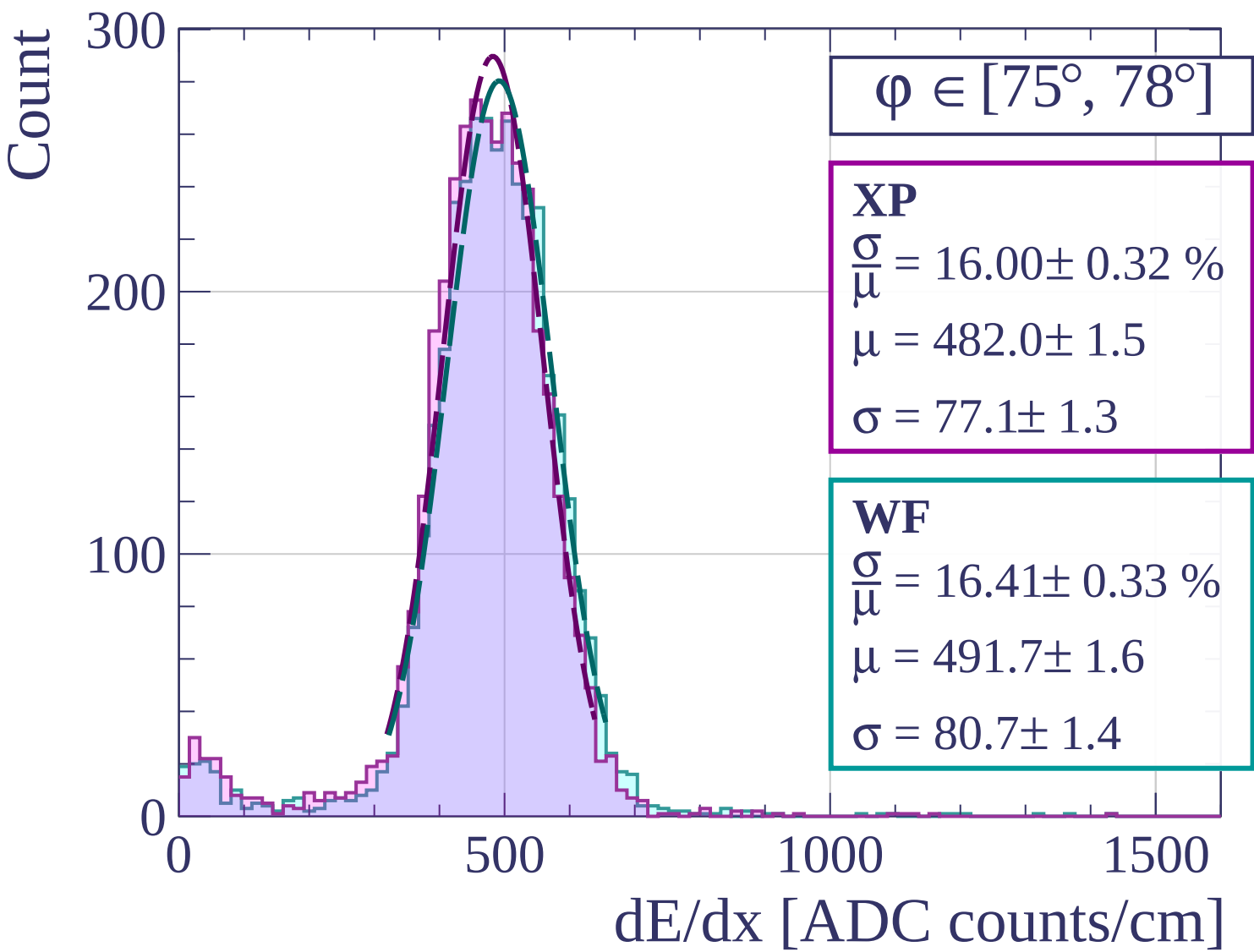
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\varphi \in [78^\circ, 81^\circ]$

XP

$$\frac{\sigma}{\mu} = 15.97 \pm 0.31 \%$$

$$\mu = 482.0 \pm 1.5$$

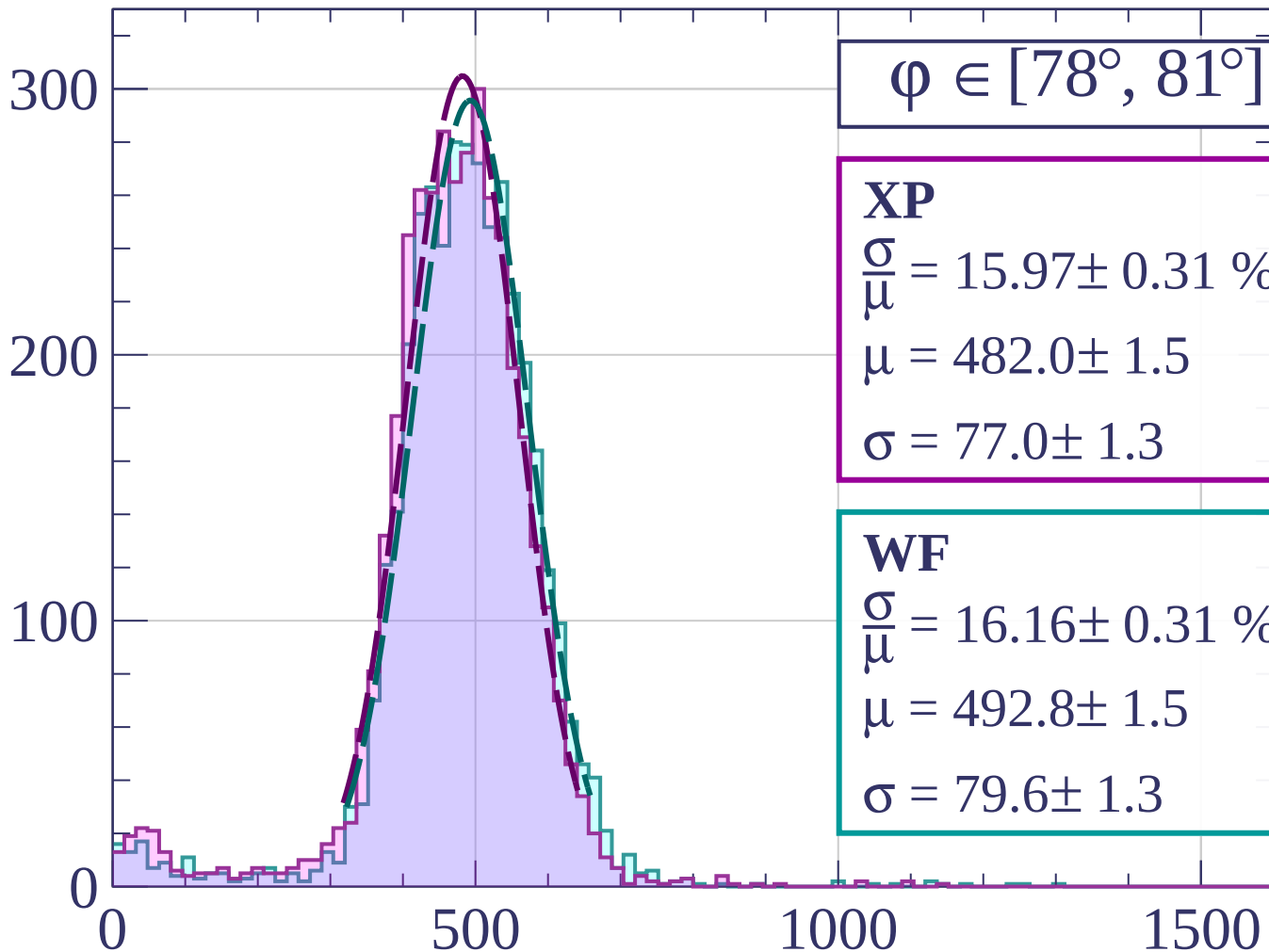
$$\sigma = 77.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 16.16 \pm 0.31 \%$$

$$\mu = 492.8 \pm 1.5$$

$$\sigma = 79.6 \pm 1.3$$



dE/dx [ADC counts/cm]

Count

$\varphi \in [81^\circ, 84^\circ]$

XP

$$\frac{\sigma}{\mu} = 15.69 \pm 0.35 \%$$

$$\mu = 478.1 \pm 1.5$$

$$\sigma = 75.0 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 15.45 \pm 0.33 \%$$

$$\mu = 490.0 \pm 1.4$$

$$\sigma = 75.7 \pm 1.4$$

300

200

100

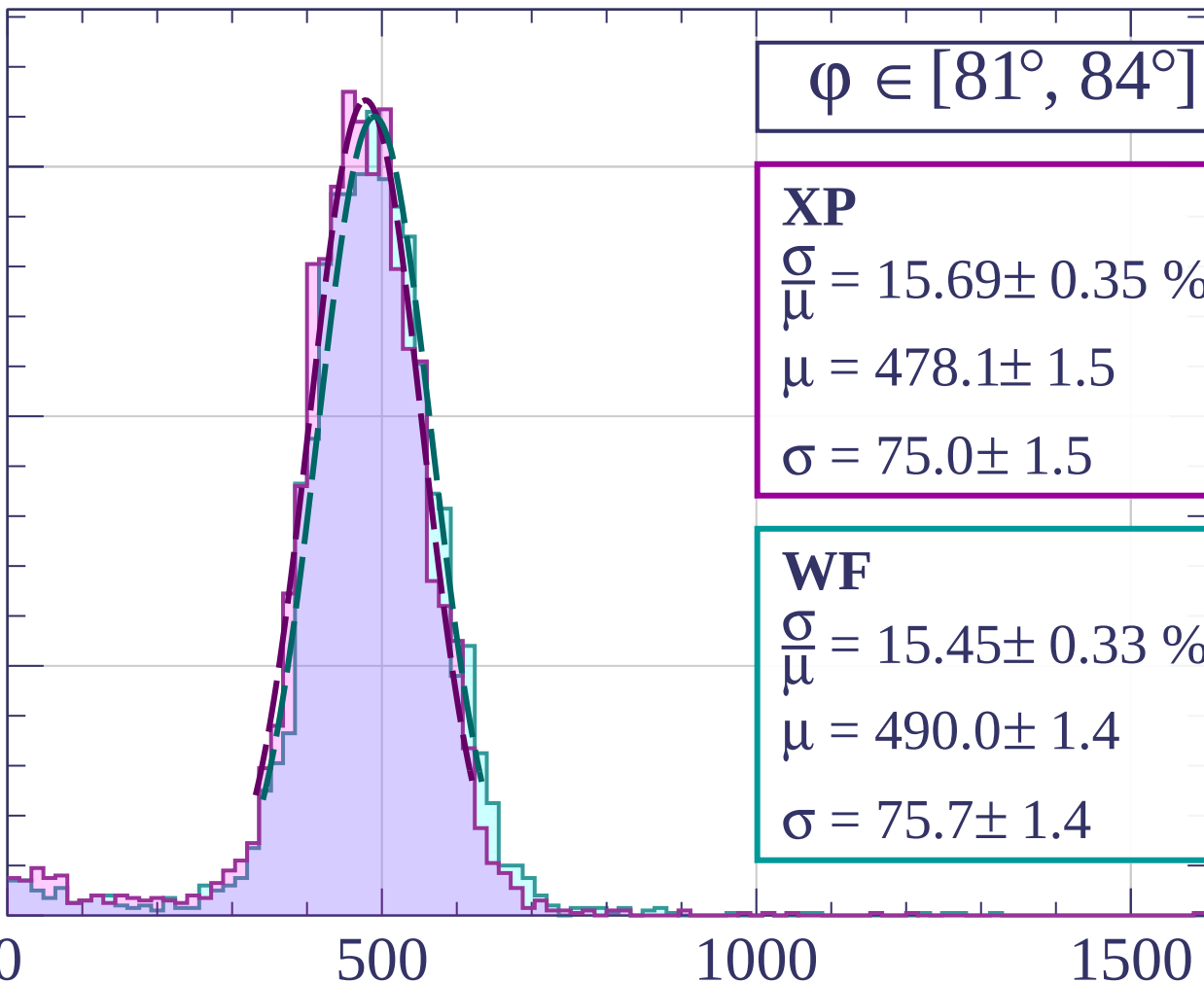
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [84^\circ, 87^\circ]$

XP

$$\frac{\sigma}{\mu} = 15.39 \pm 0.30 \%$$

$$\mu = 476.2 \pm 1.4$$

$$\sigma = 73.3 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 15.88 \pm 0.29 \%$$

$$\mu = 488.3 \pm 1.4$$

$$\sigma = 77.6 \pm 1.2$$

300

200

100

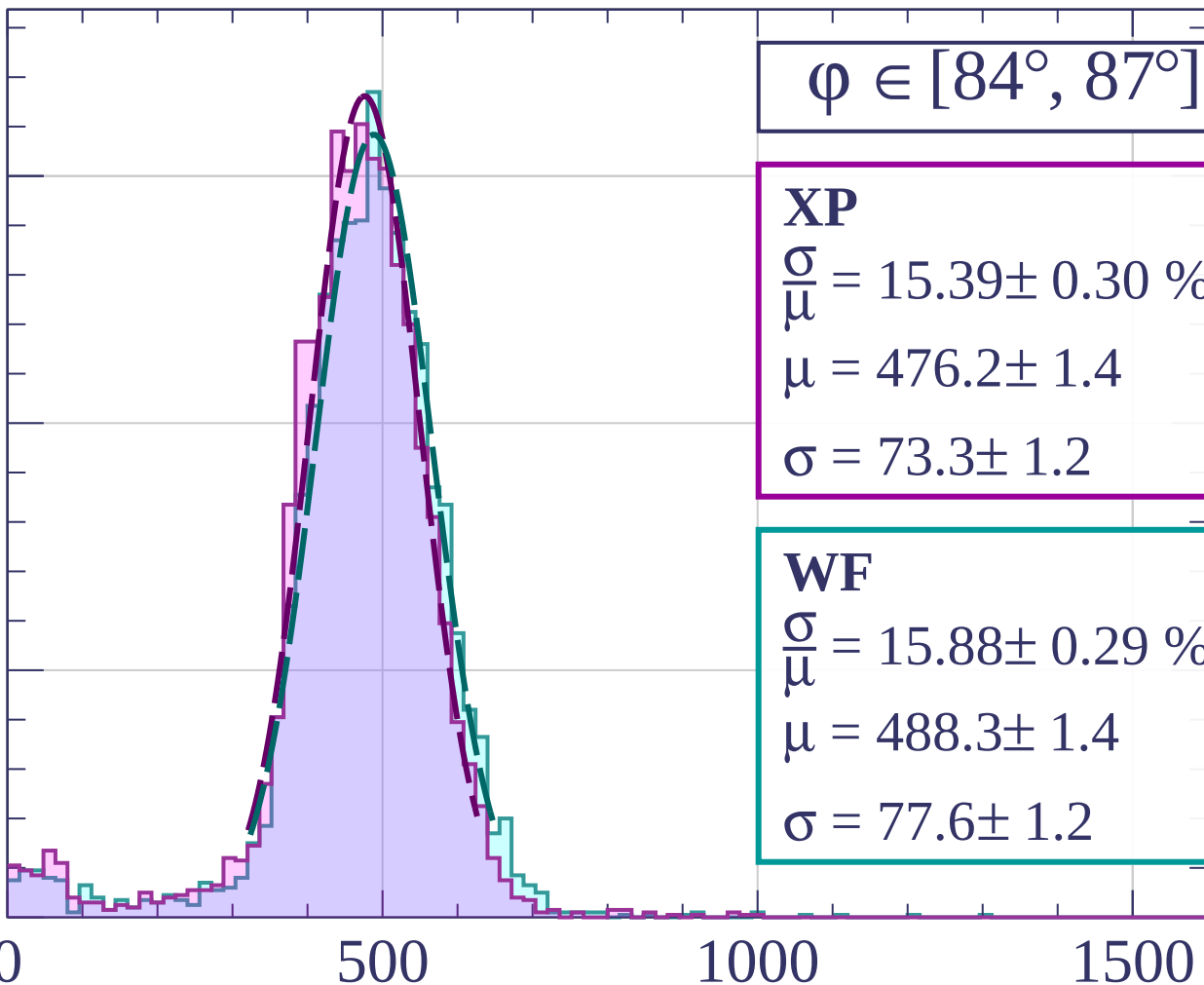
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [87^\circ, 90^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.18 \pm 0.28 \%$$

$$\mu = 474.7 \pm 1.4$$

$$\sigma = 76.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 16.61 \pm 0.32 \%$$

$$\mu = 490.6 \pm 1.5$$

$$\sigma = 81.5 \pm 1.3$$

300

200

100

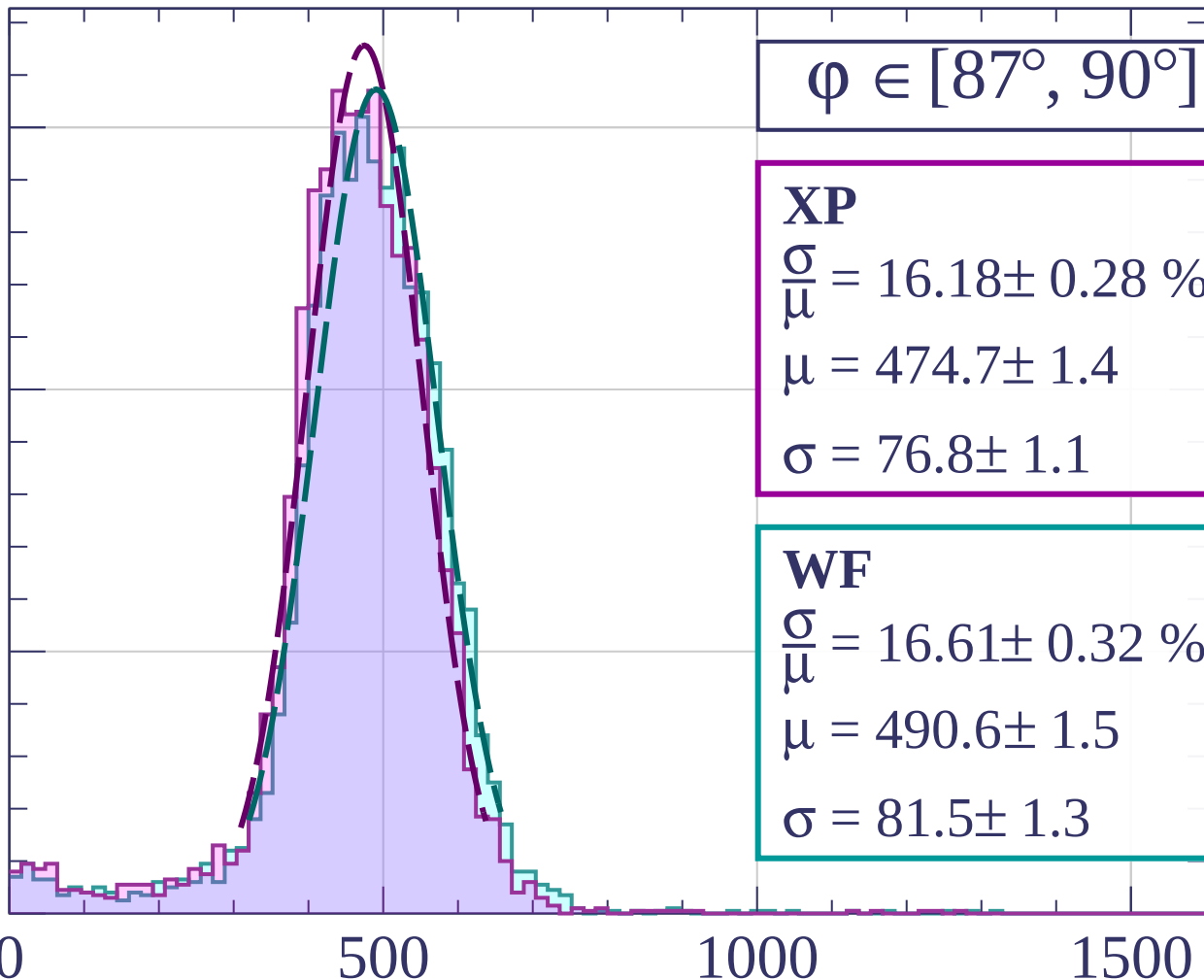
0

500

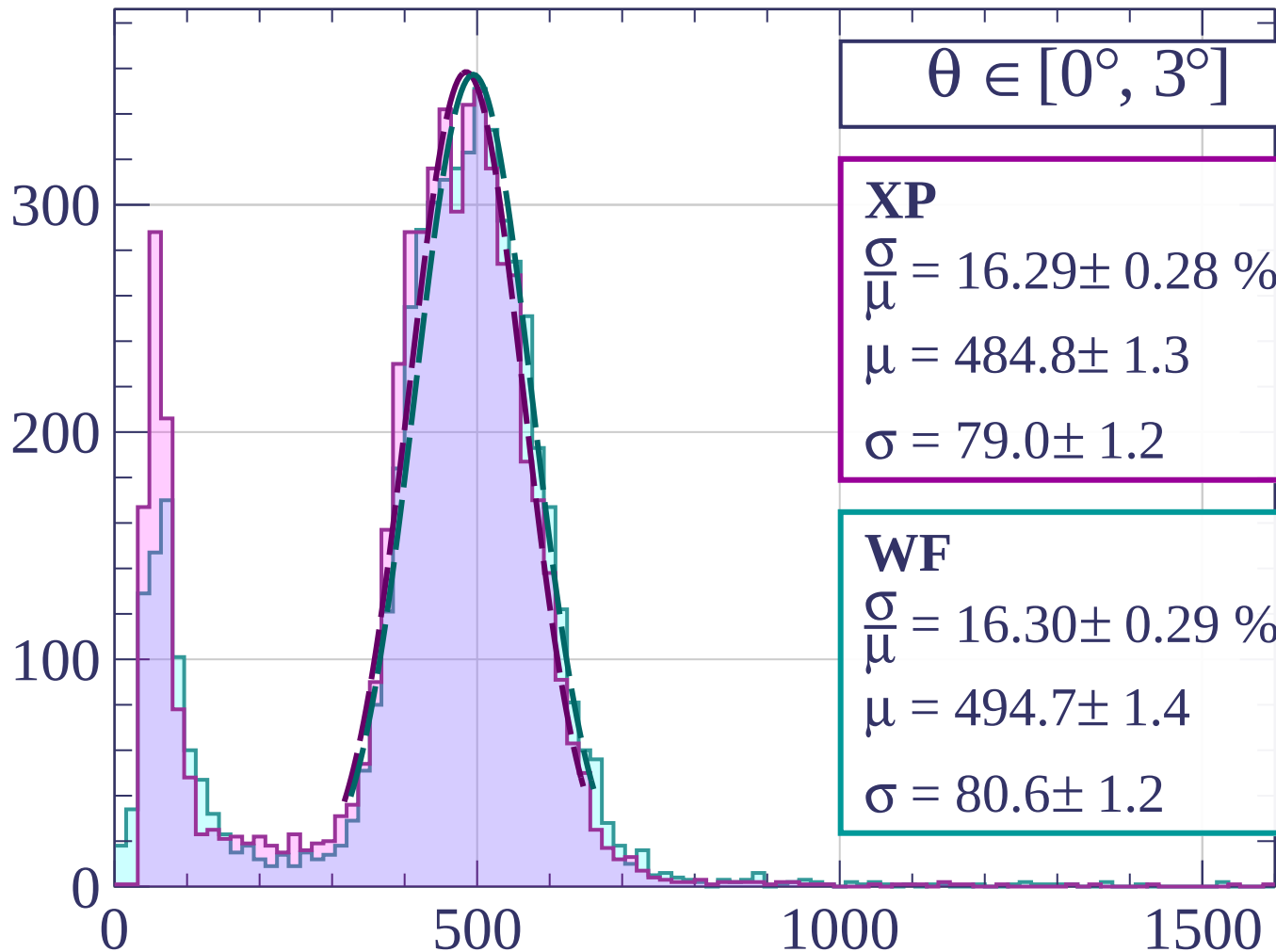
1000

1500

dE/dx [ADC counts/cm]



Count



$\theta \in [0^\circ, 3^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.29 \pm 0.28 \%$$

$$\mu = 484.8 \pm 1.3$$

$$\sigma = 79.0 \pm 1.2$$

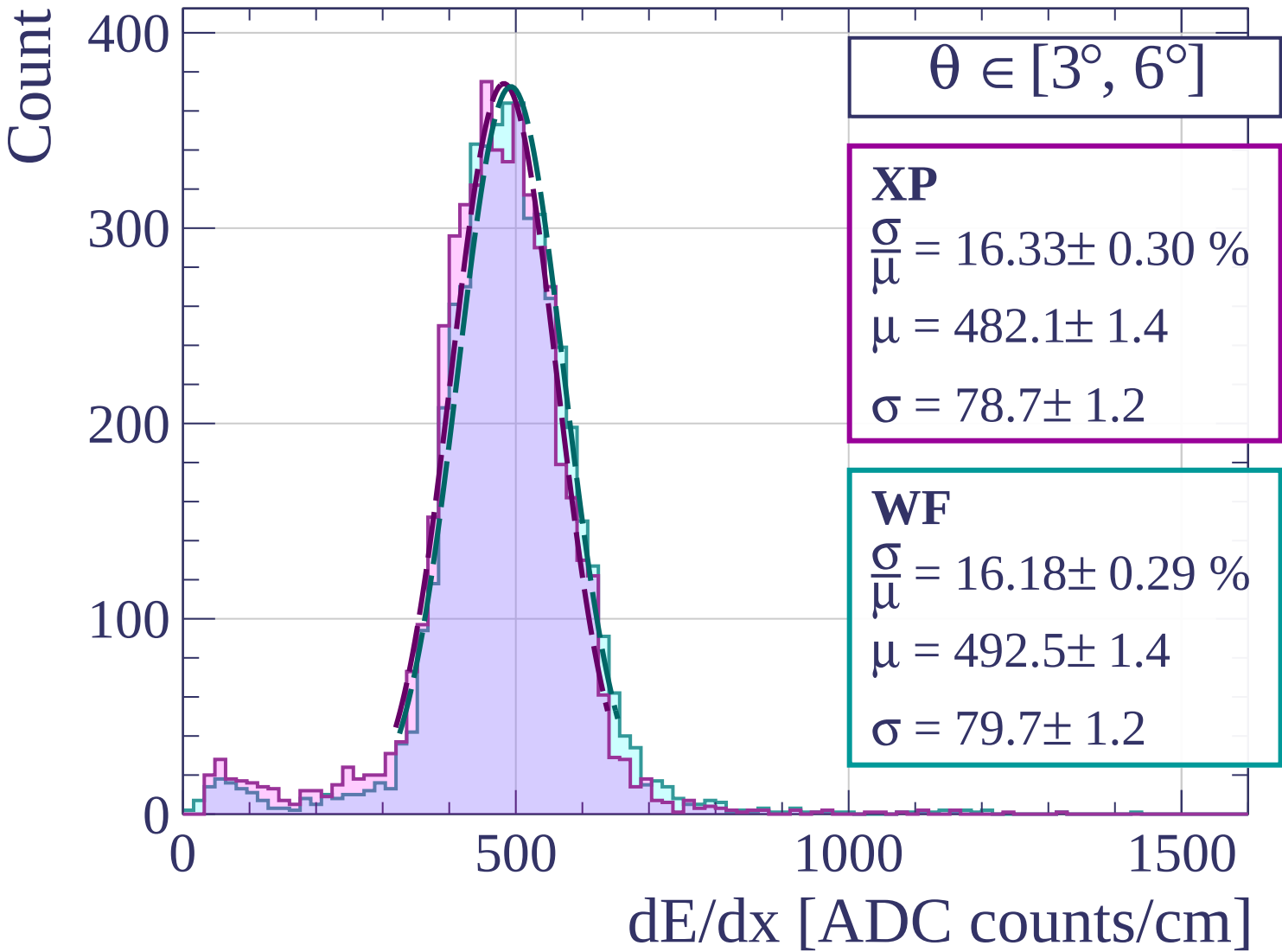
WF

$$\frac{\sigma}{\mu} = 16.30 \pm 0.29 \%$$

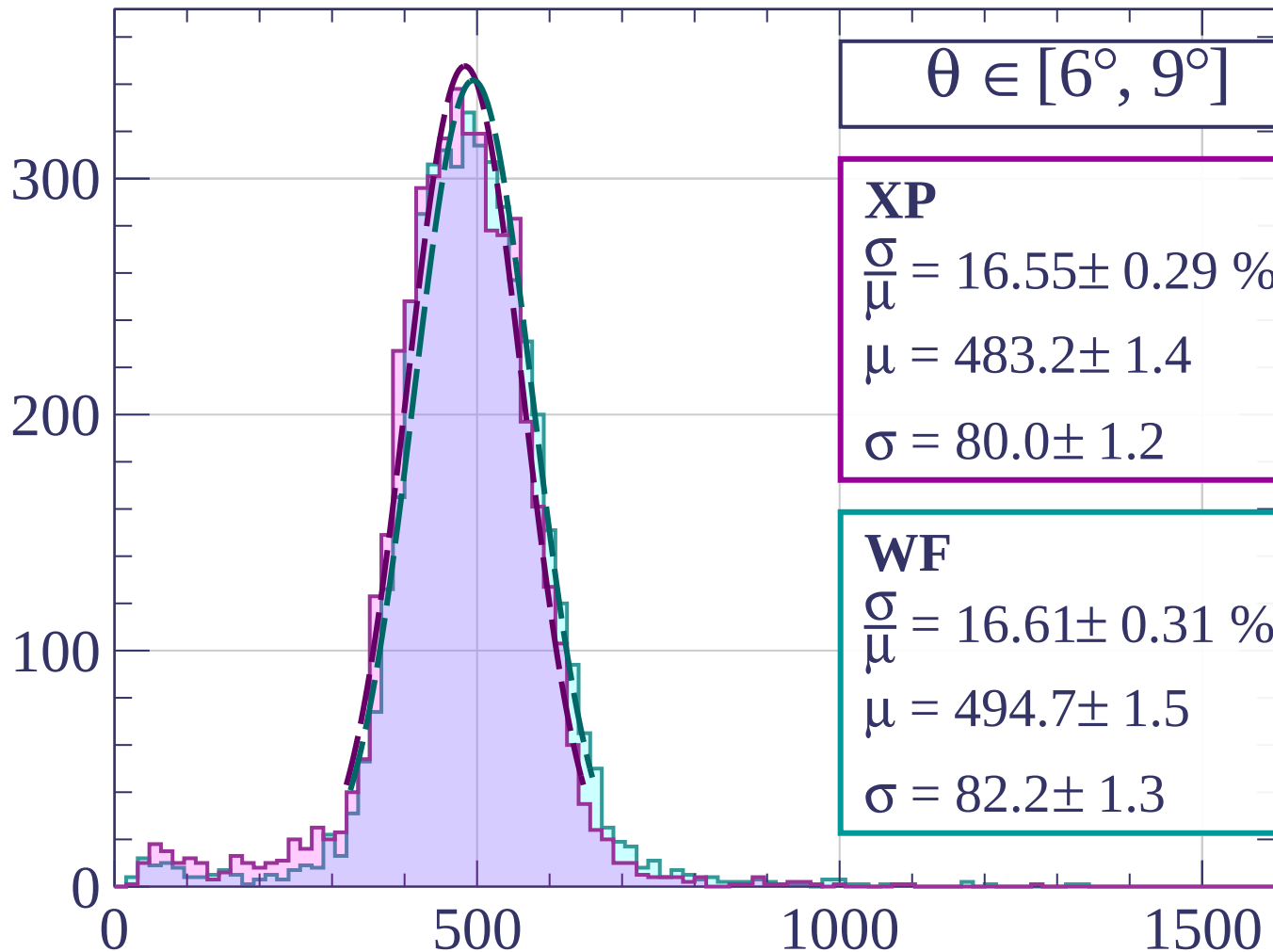
$$\mu = 494.7 \pm 1.4$$

$$\sigma = 80.6 \pm 1.2$$

dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]

Count

$\theta \in [9^\circ, 12^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.62 \pm 0.31 \%$$

$$\mu = 481.7 \pm 1.5$$

$$\sigma = 80.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 16.96 \pm 0.33 \%$$

$$\mu = 492.7 \pm 1.5$$

$$\sigma = 83.6 \pm 1.4$$

300

200

100

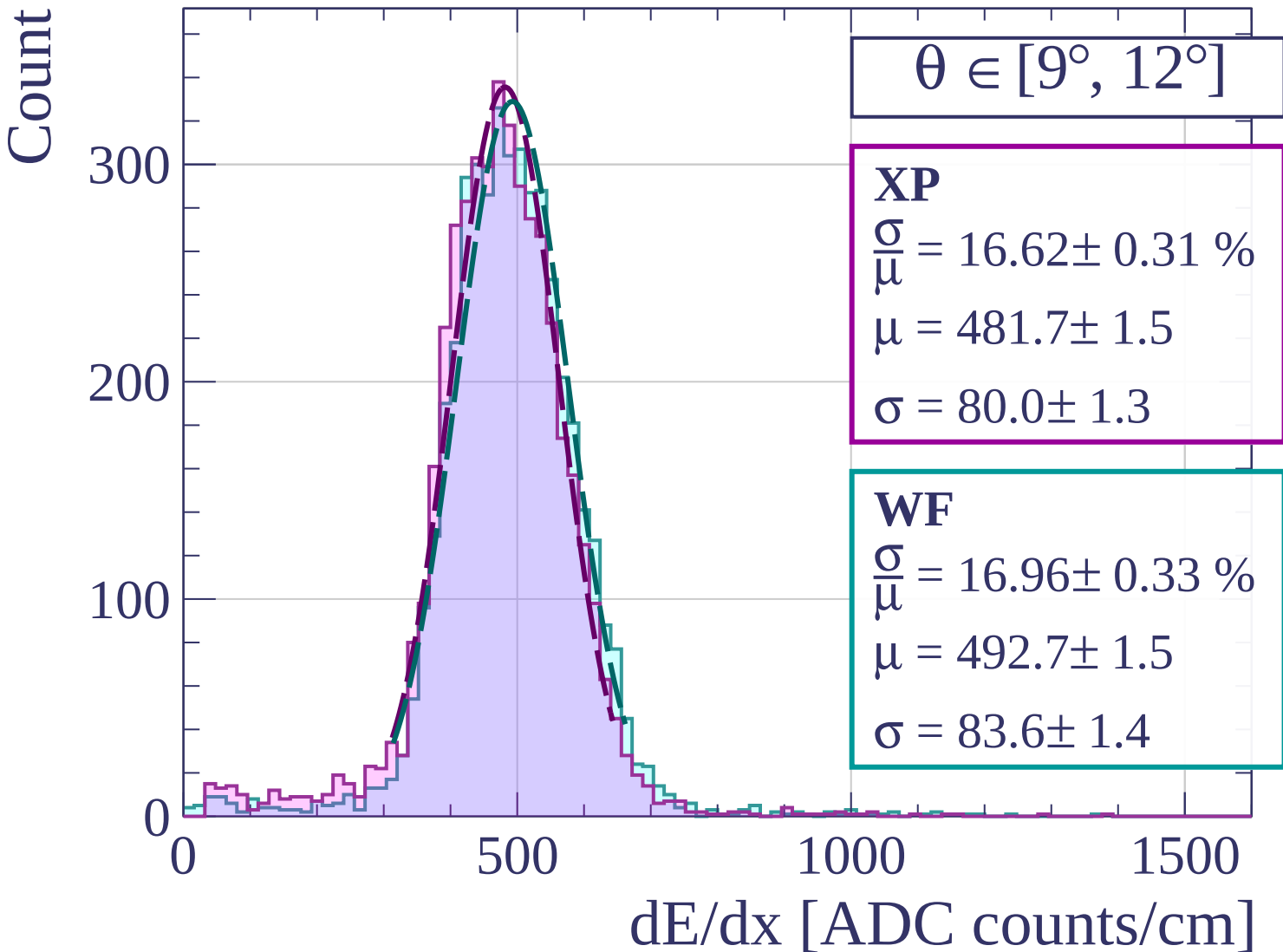
0

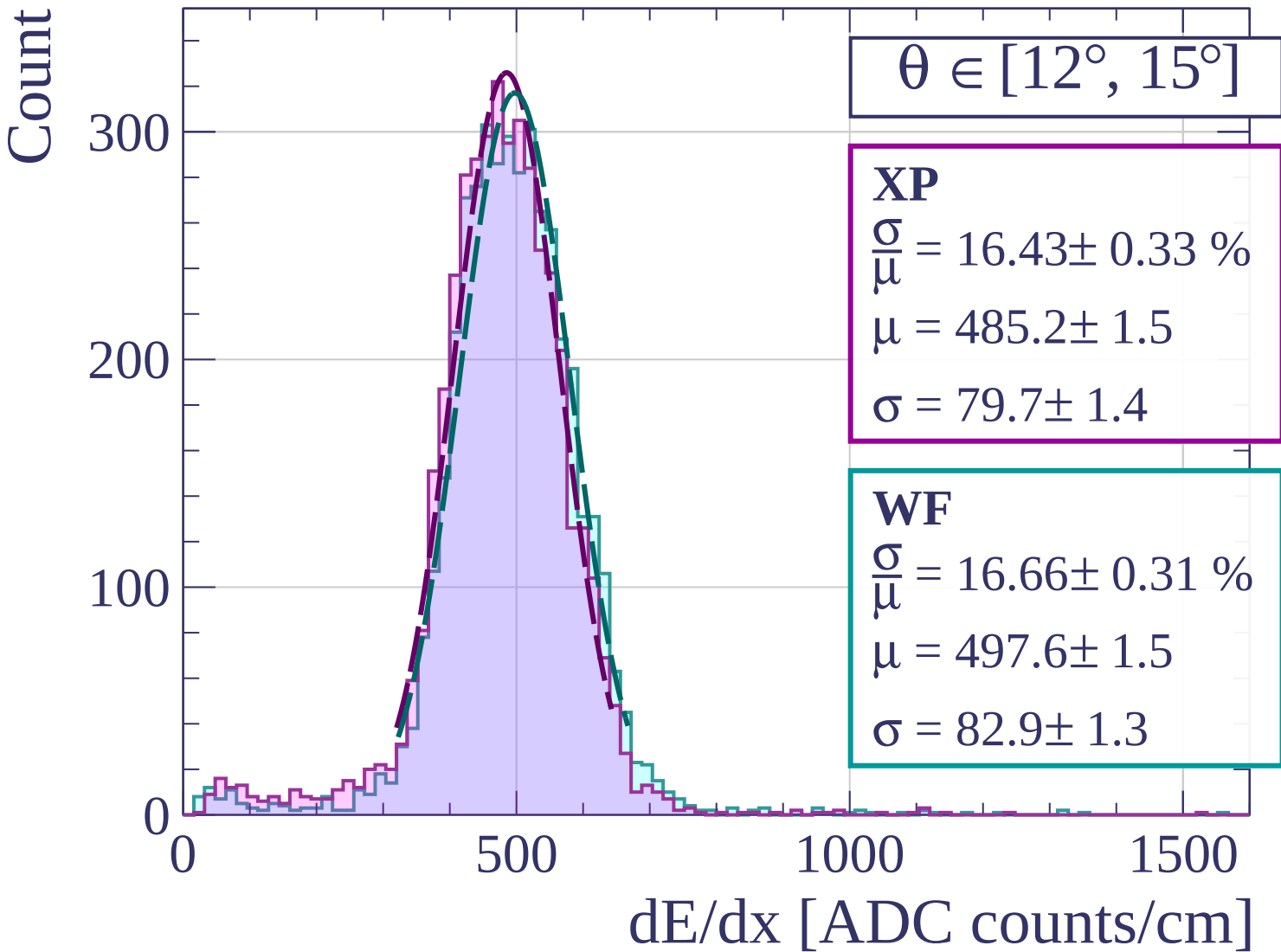
500

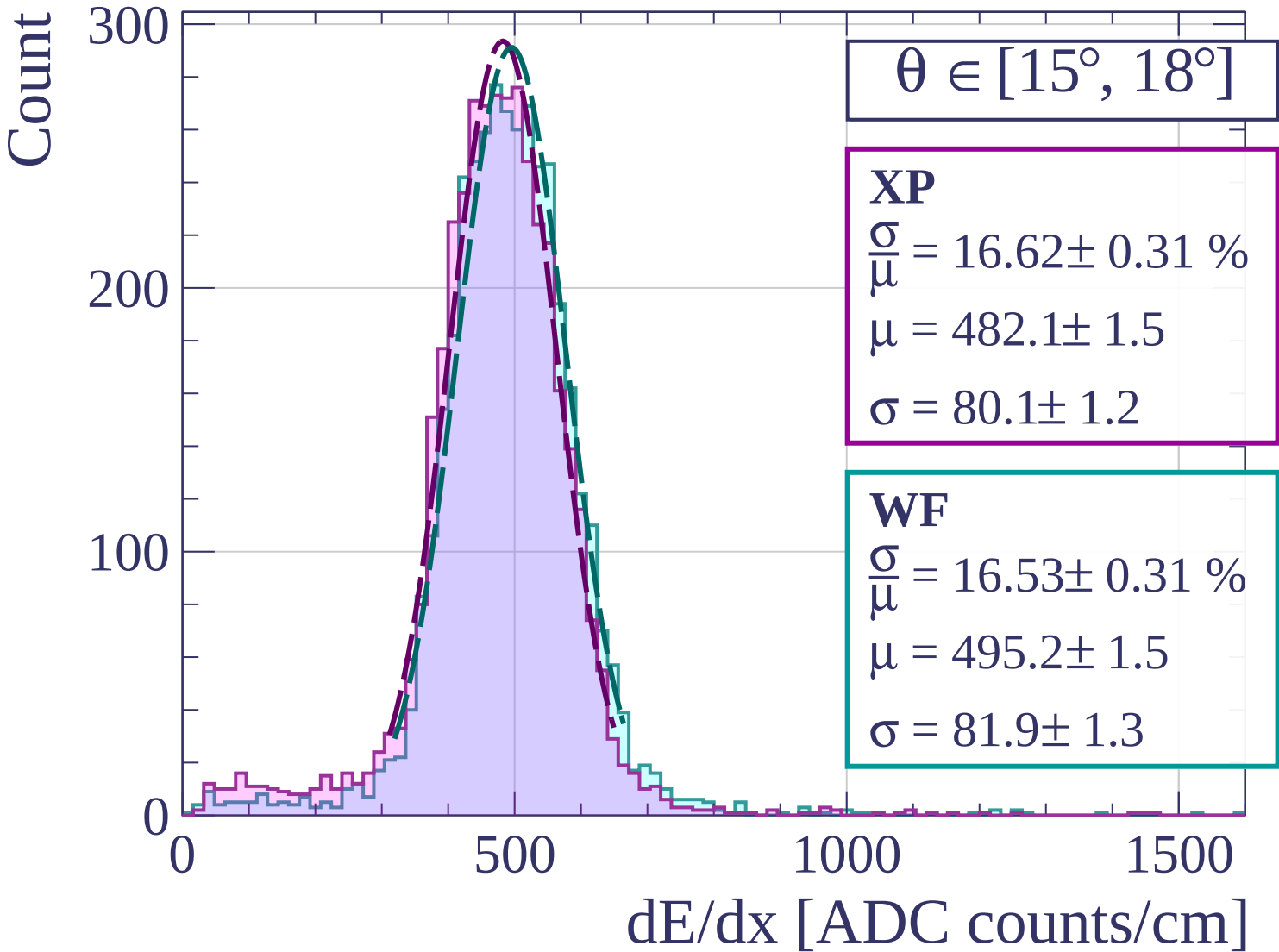
1000

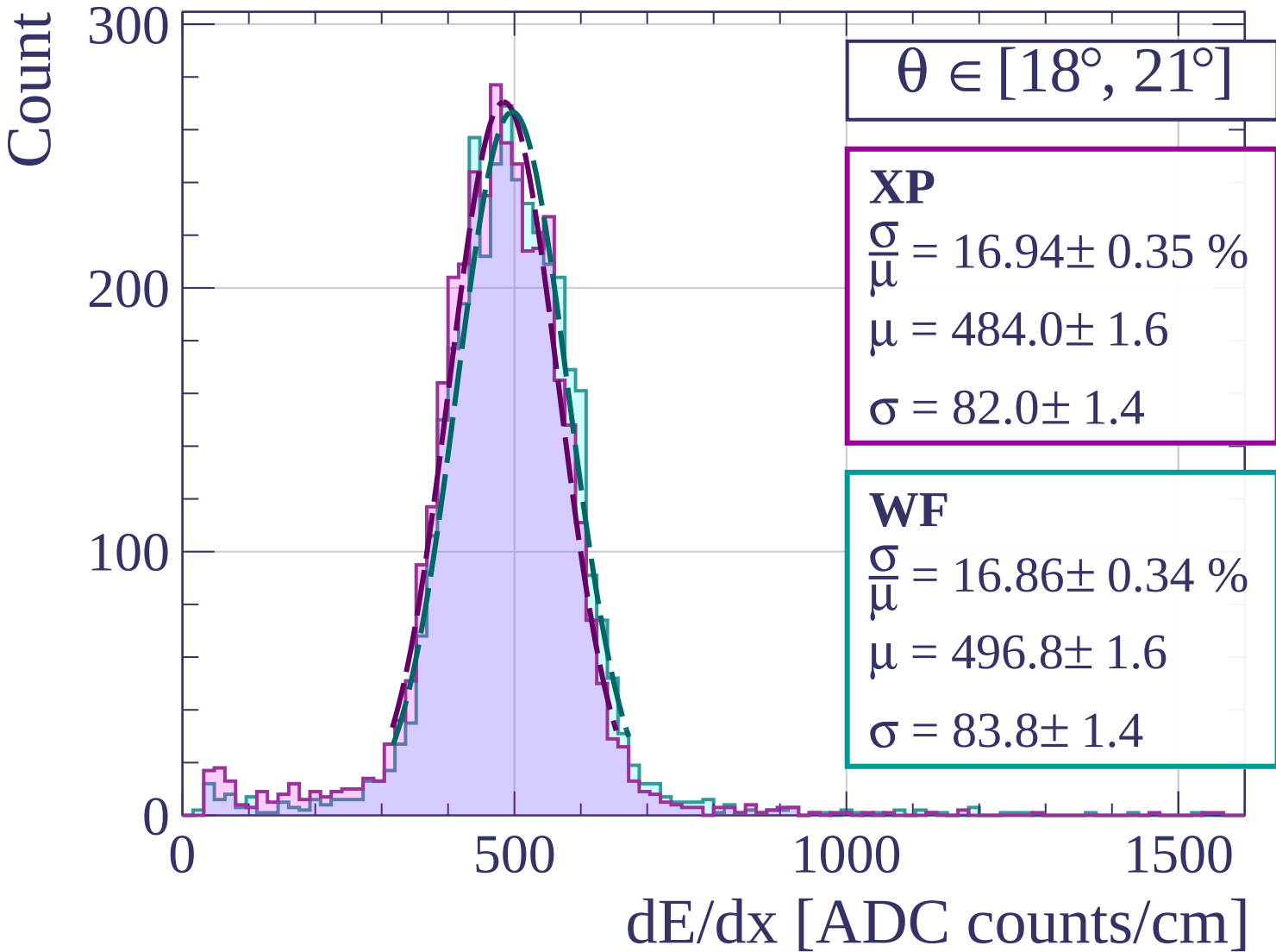
1500

dE/dx [ADC counts/cm]









Count

$\theta \in [21^\circ, 24^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.59 \pm 0.37 \%$$

$$\mu = 485.1 \pm 1.7$$

$$\sigma = 80.5 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 16.45 \pm 0.39 \%$$

$$\mu = 497.5 \pm 1.8$$

$$\sigma = 81.9 \pm 1.6$$

200

100

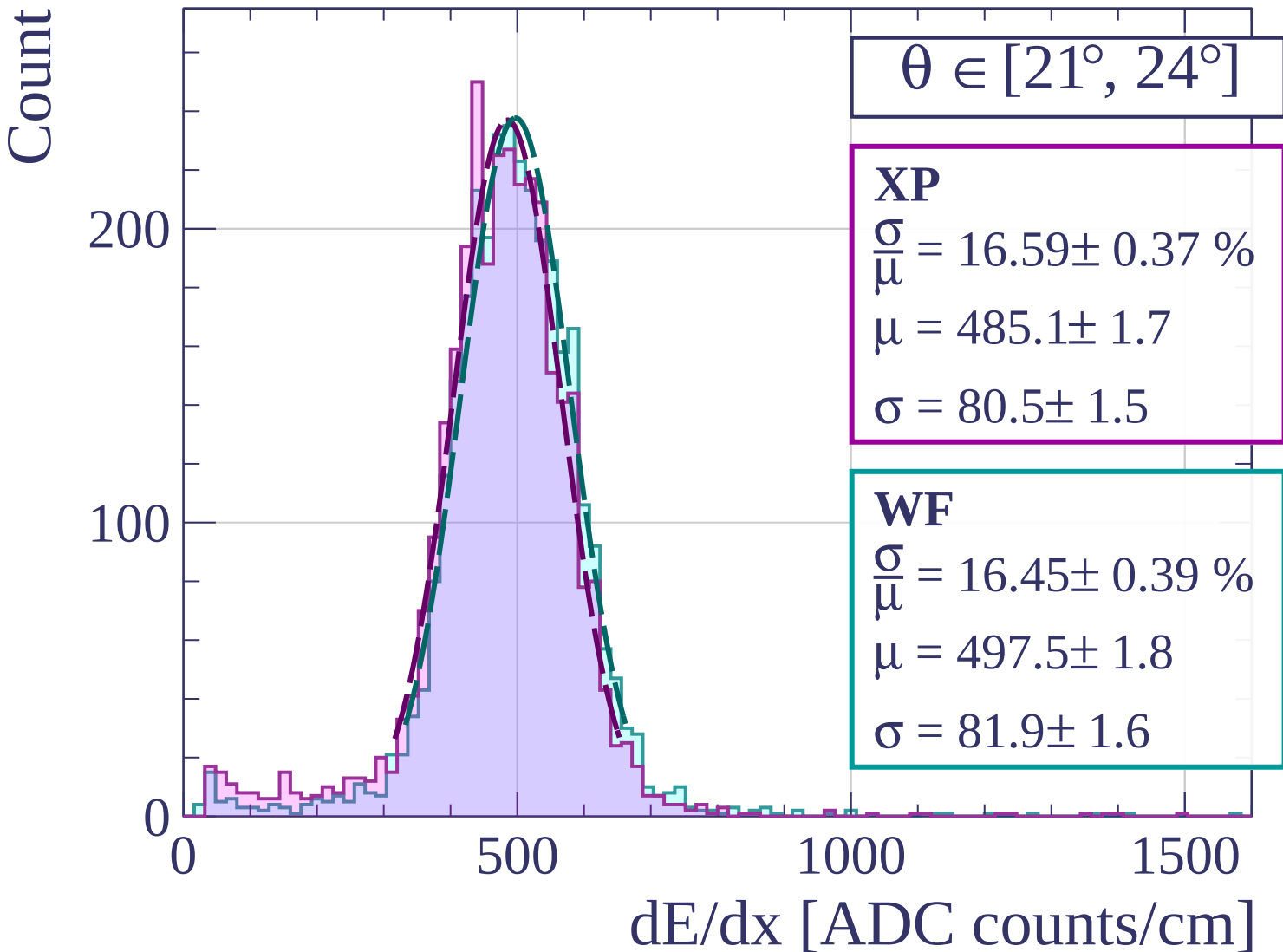
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [24^\circ, 27^\circ]$

XP

$$\frac{\sigma}{\mu} = 15.96 \pm 0.37 \%$$

$$\mu = 488.9 \pm 1.8$$

$$\sigma = 78.0 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 15.99 \pm 0.36 \%$$

$$\mu = 498.1 \pm 1.8$$

$$\sigma = 79.7 \pm 1.5$$

200

150

100

50

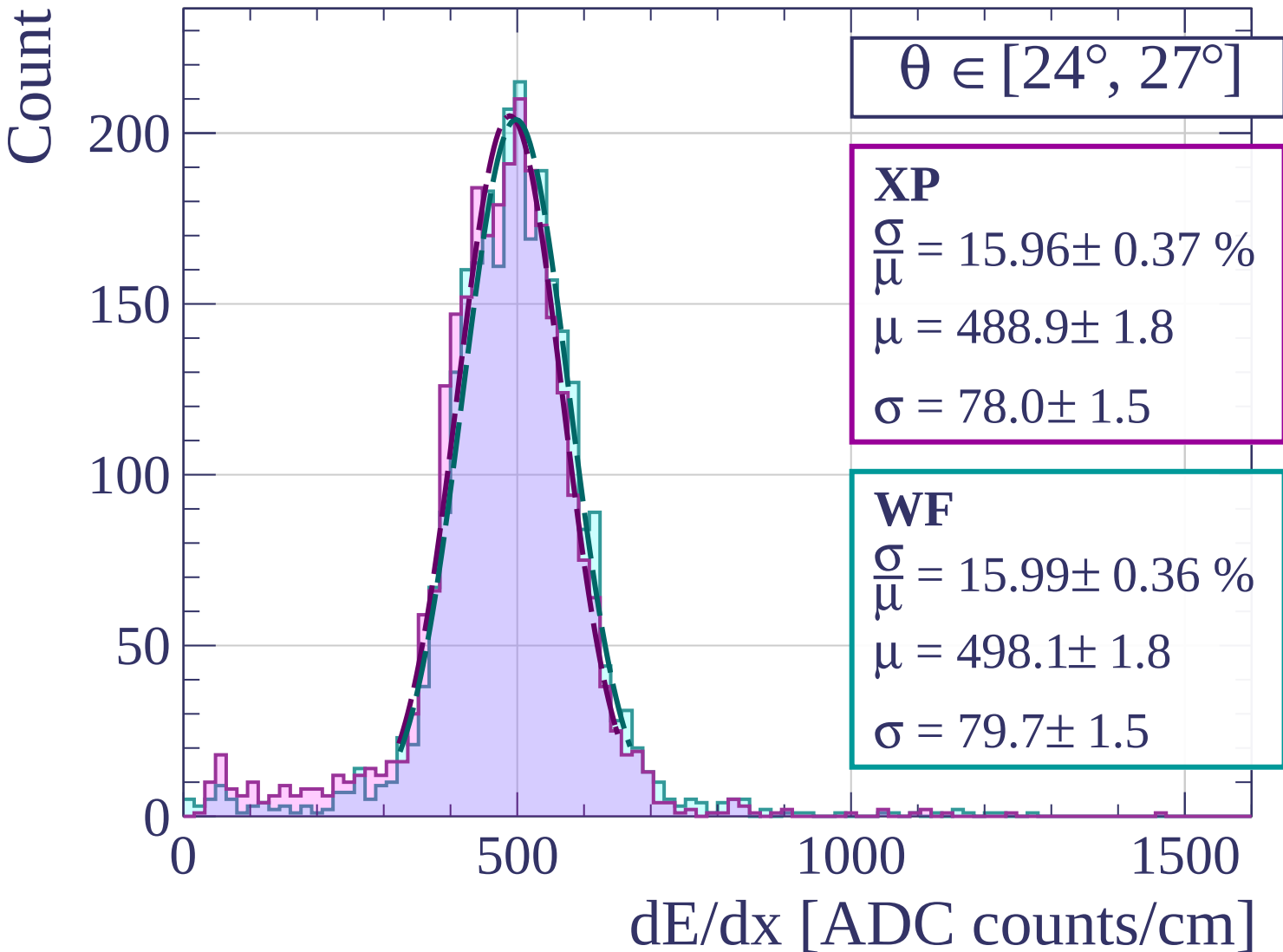
0

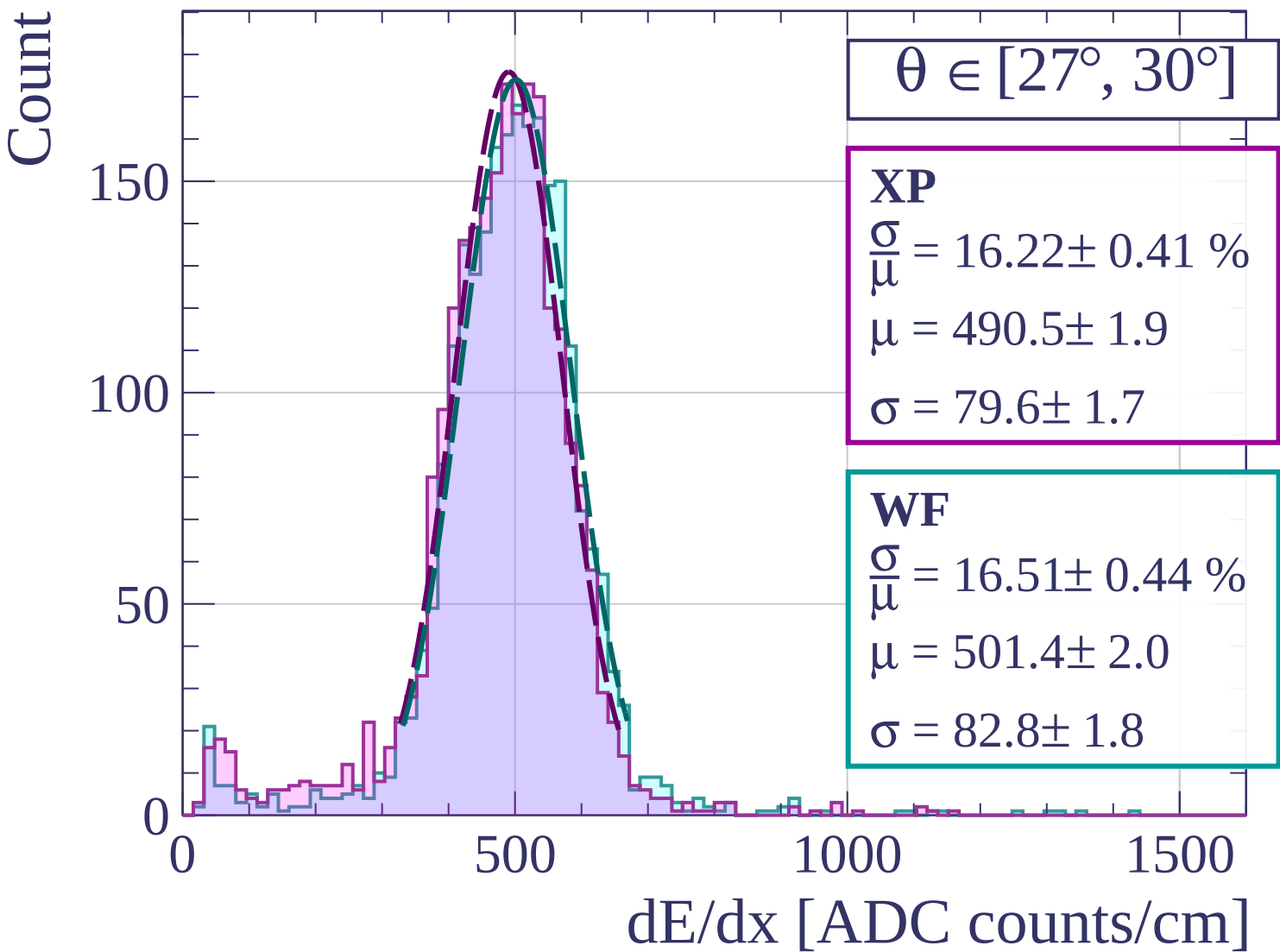
500

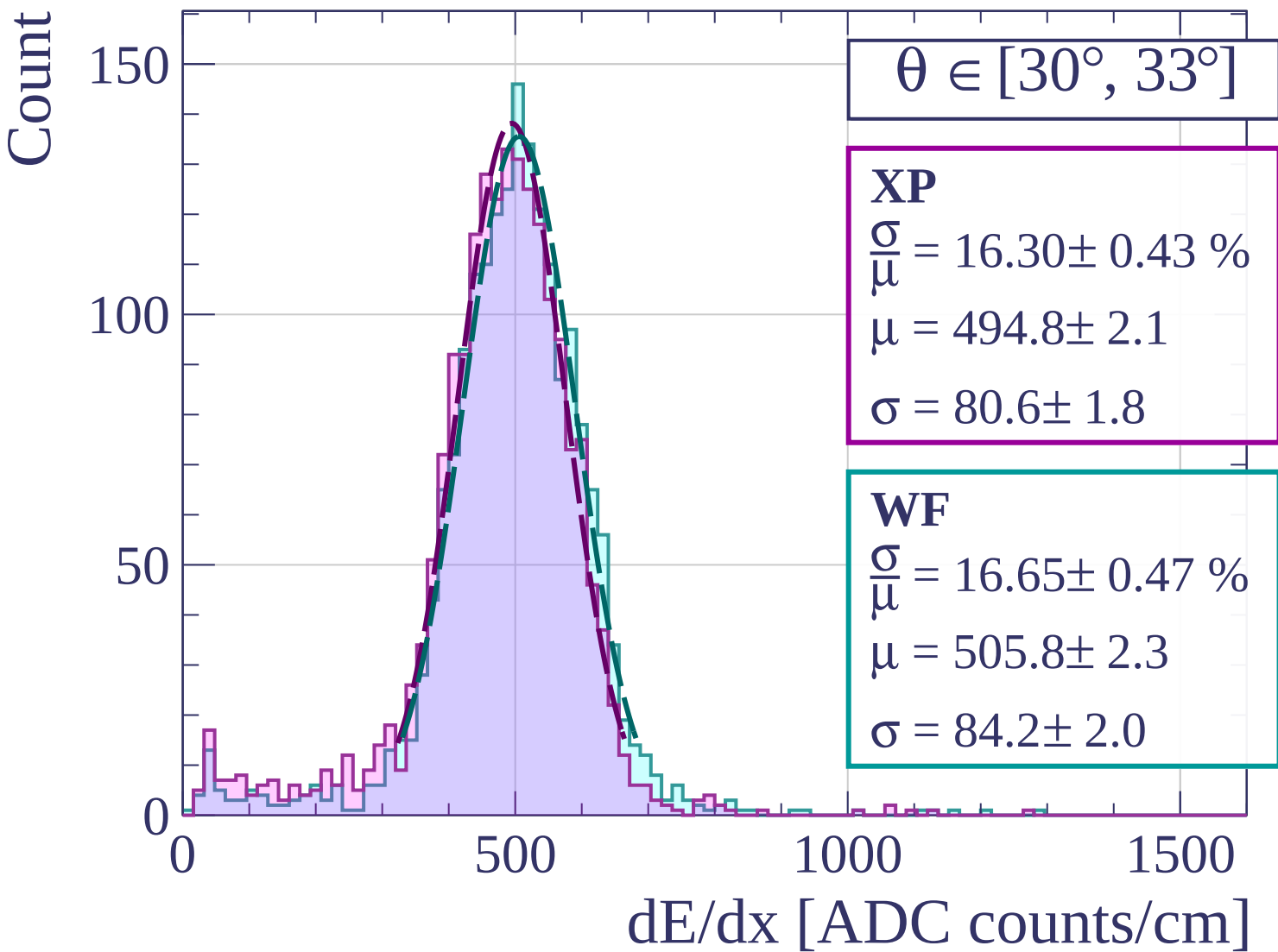
1000

1500

dE/dx [ADC counts/cm]







Count

$\theta \in [33^\circ, 36^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.74 \pm 0.49 \%$$

$$\mu = 498.3 \pm 2.5$$

$$\sigma = 83.4 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 17.59 \pm 0.58 \%$$

$$\mu = 511.9 \pm 2.8$$

$$\sigma = 90.0 \pm 2.5$$

100

50

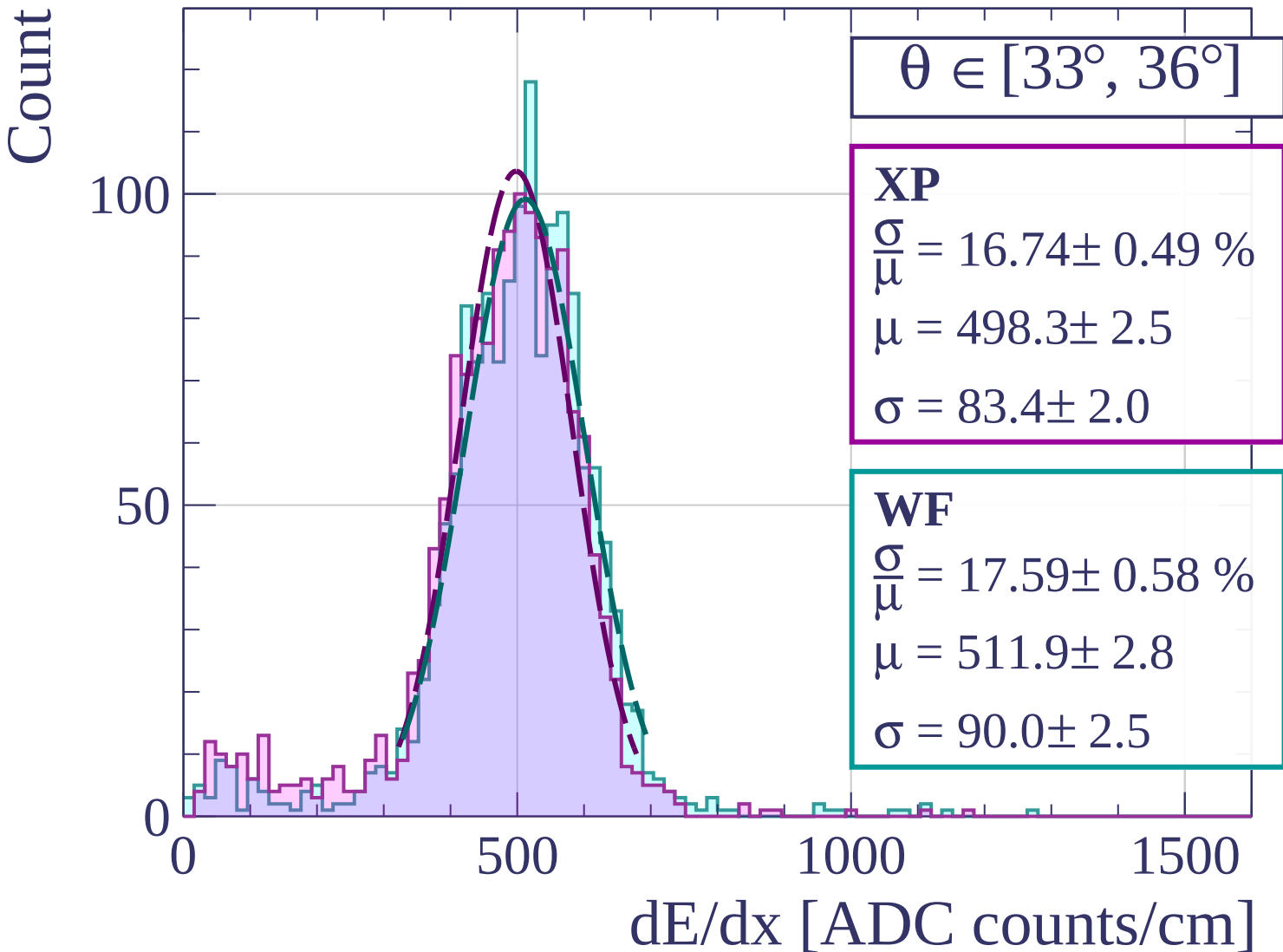
0

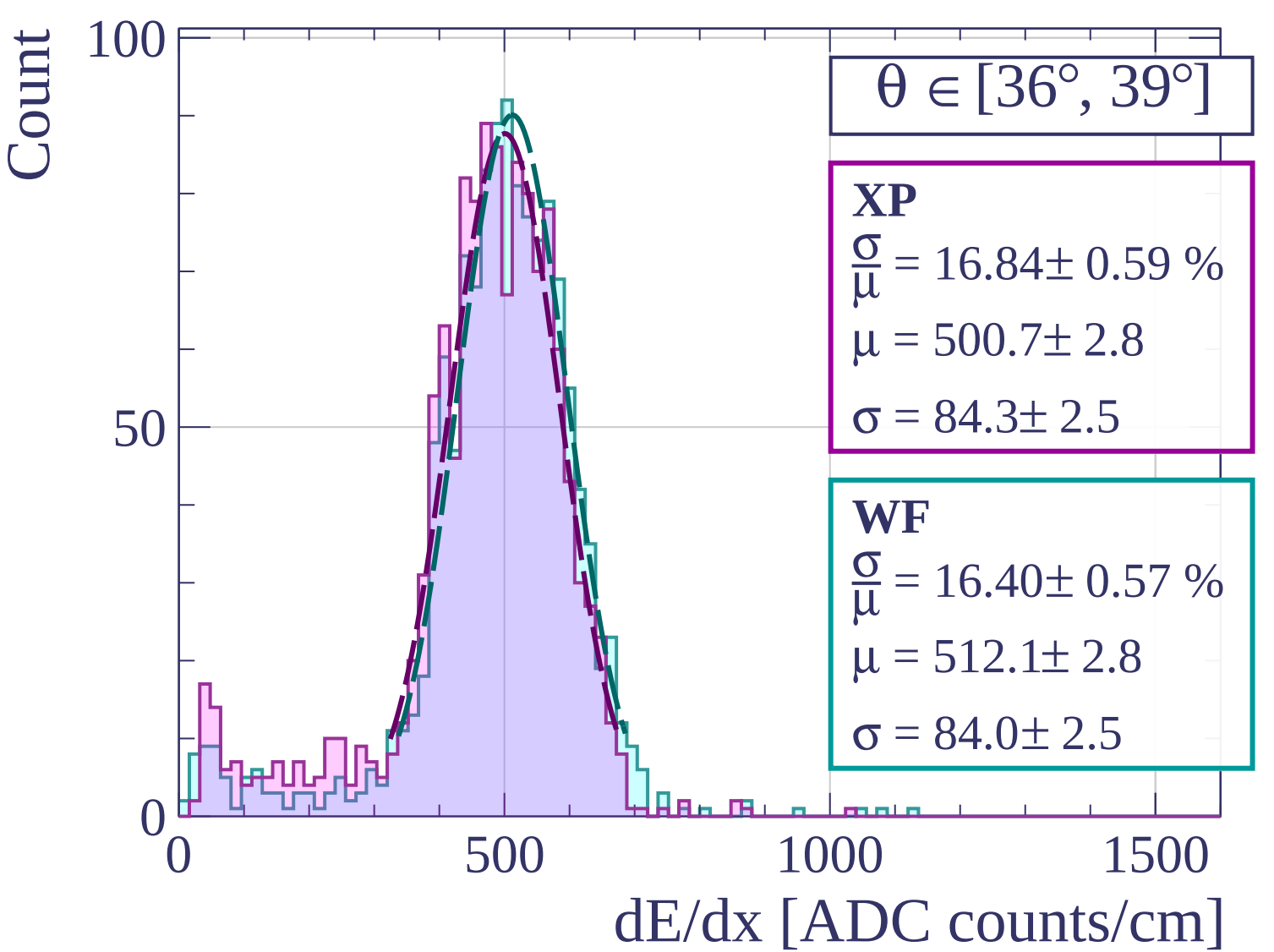
500

1000

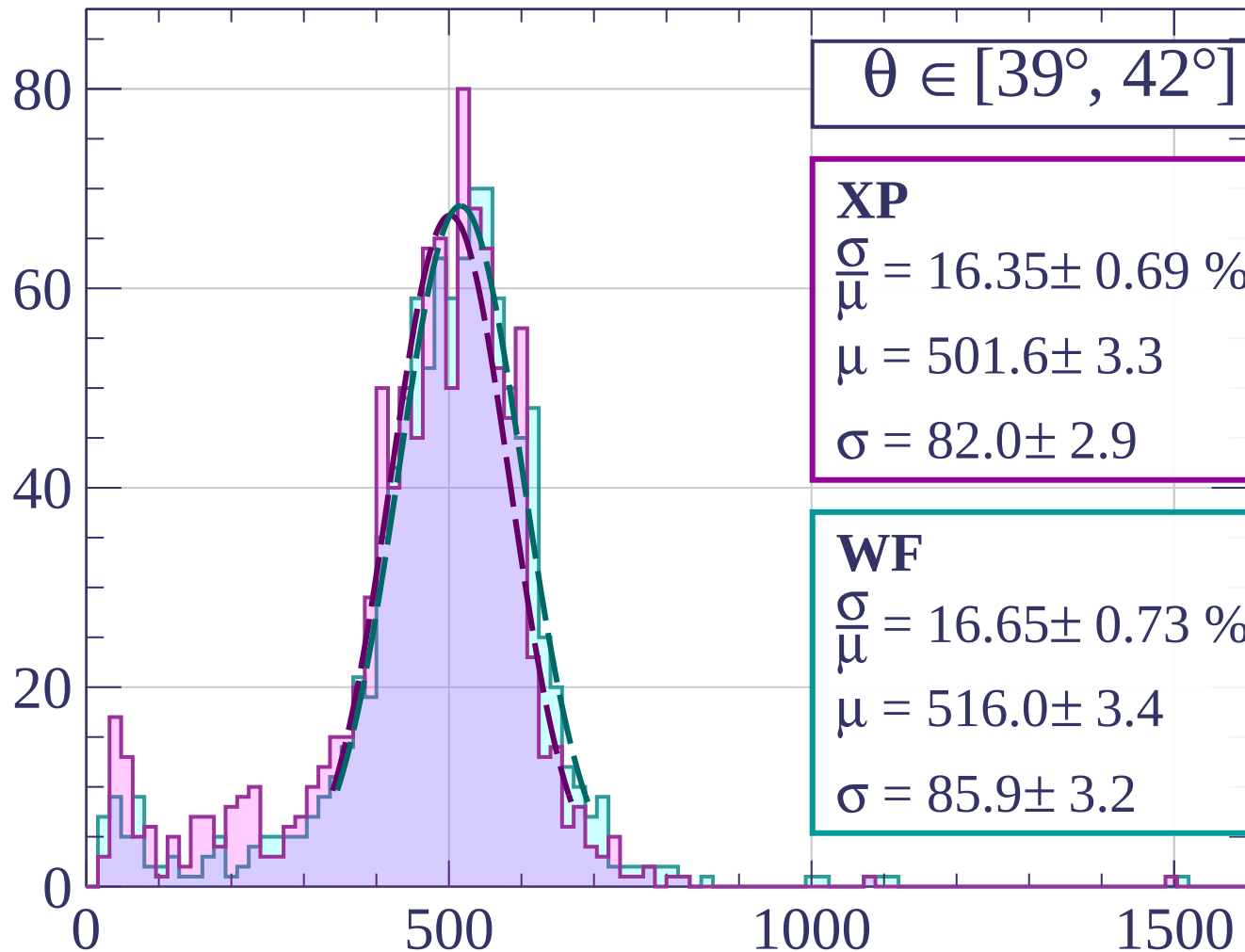
1500

dE/dx [ADC counts/cm]





Count



$\theta \in [39^\circ, 42^\circ]$

XP

$$\frac{\sigma}{\mu} = 16.35 \pm 0.69 \%$$

$$\mu = 501.6 \pm 3.3$$

$$\sigma = 82.0 \pm 2.9$$

WF

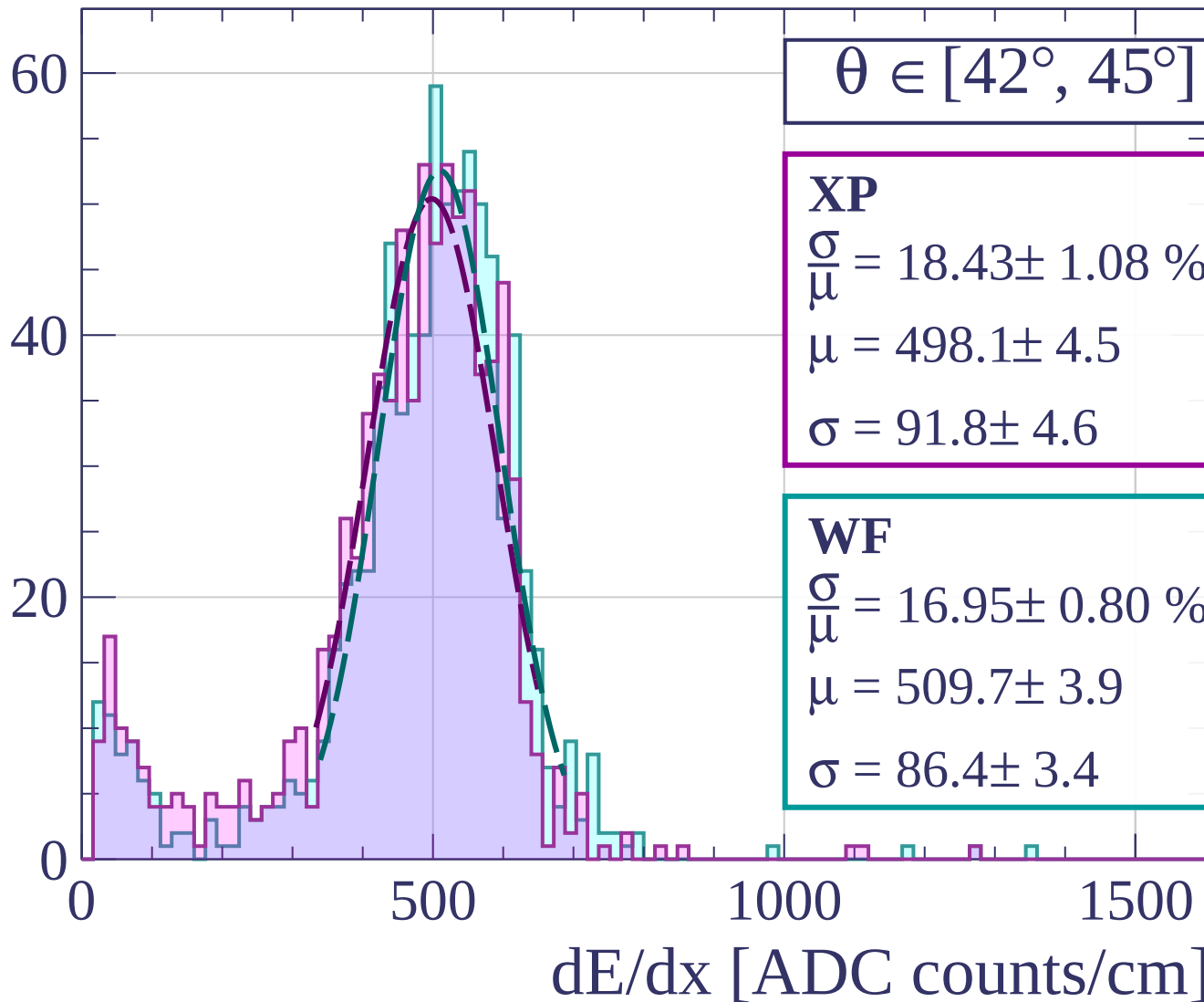
$$\frac{\sigma}{\mu} = 16.65 \pm 0.73 \%$$

$$\mu = 516.0 \pm 3.4$$

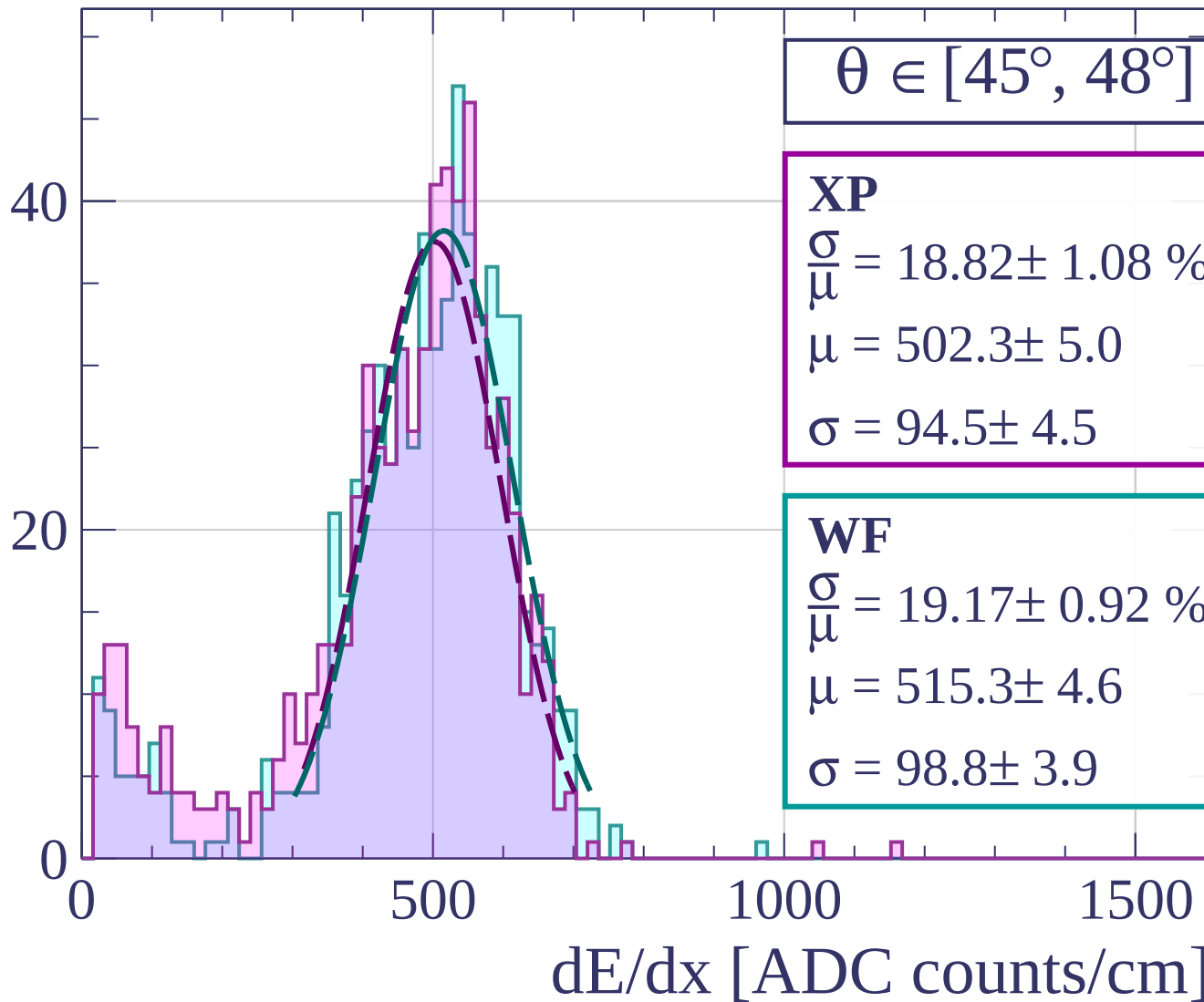
$$\sigma = 85.9 \pm 3.2$$

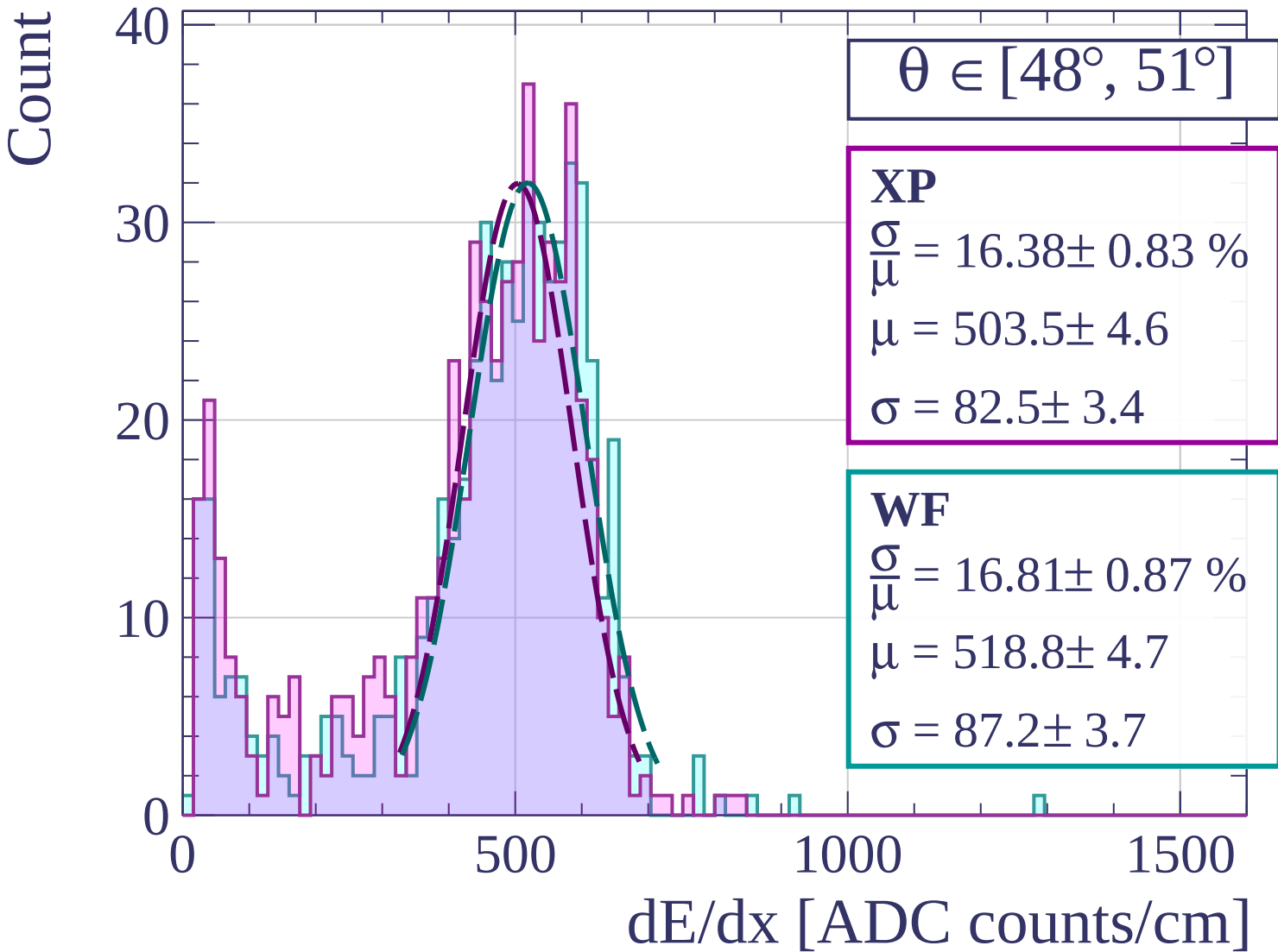
dE/dx [ADC counts/cm]

Count

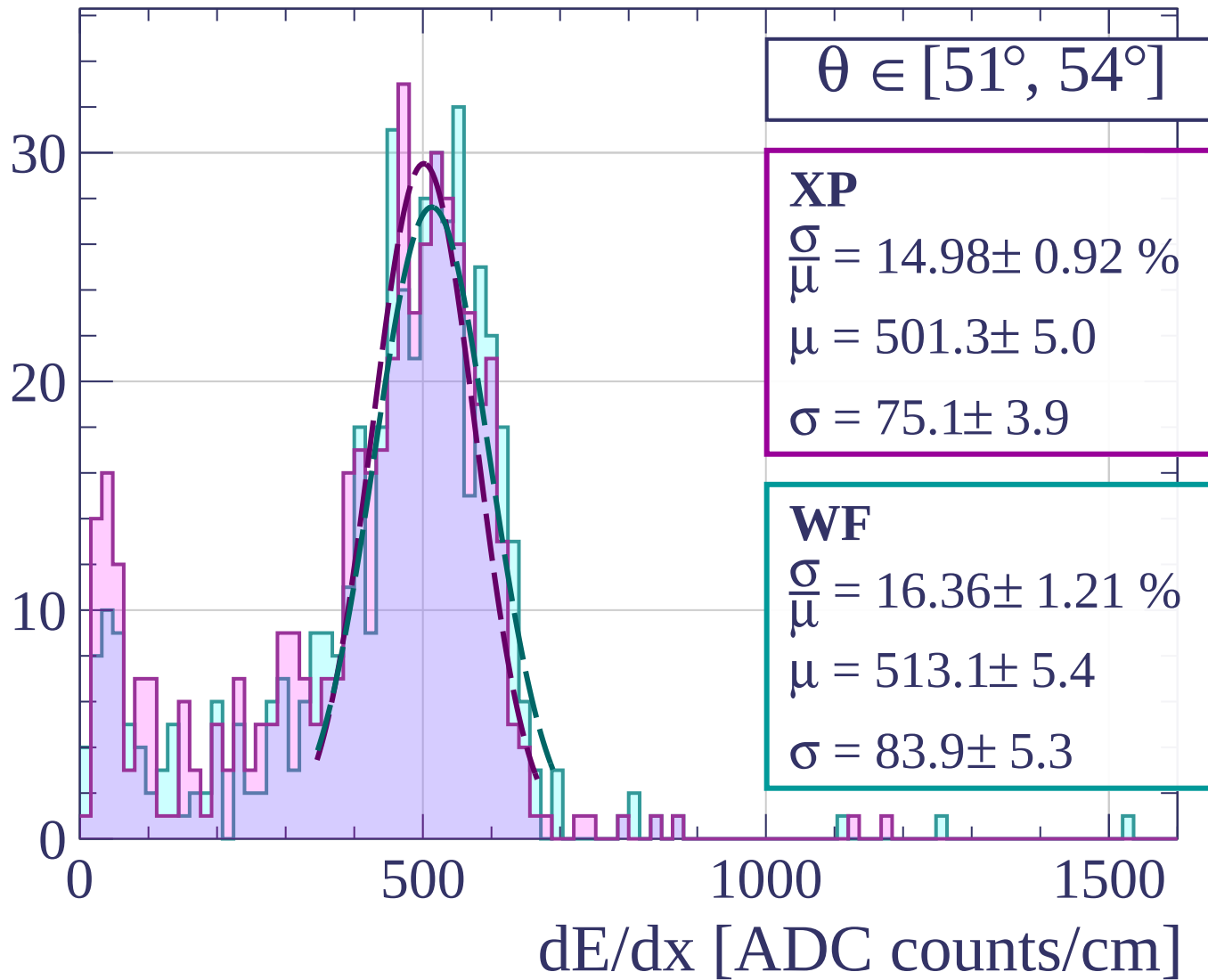


Count

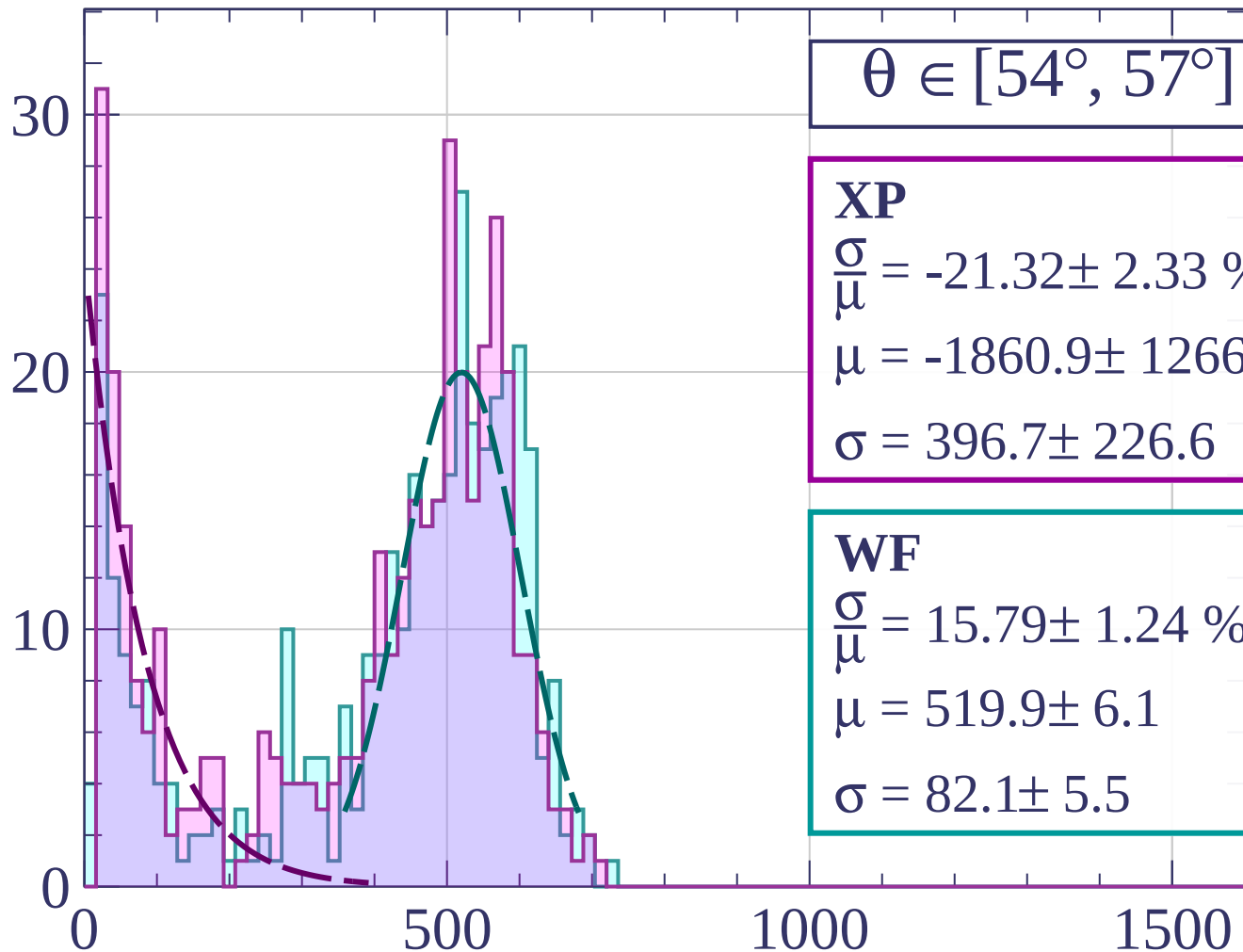




Count



Count



$\theta \in [54^\circ, 57^\circ]$

XP

$$\frac{\sigma}{\mu} = -21.32 \pm 2.33 \%$$

$$\mu = -1860.9 \pm 1266.2$$

$$\sigma = 396.7 \pm 226.6$$

WF

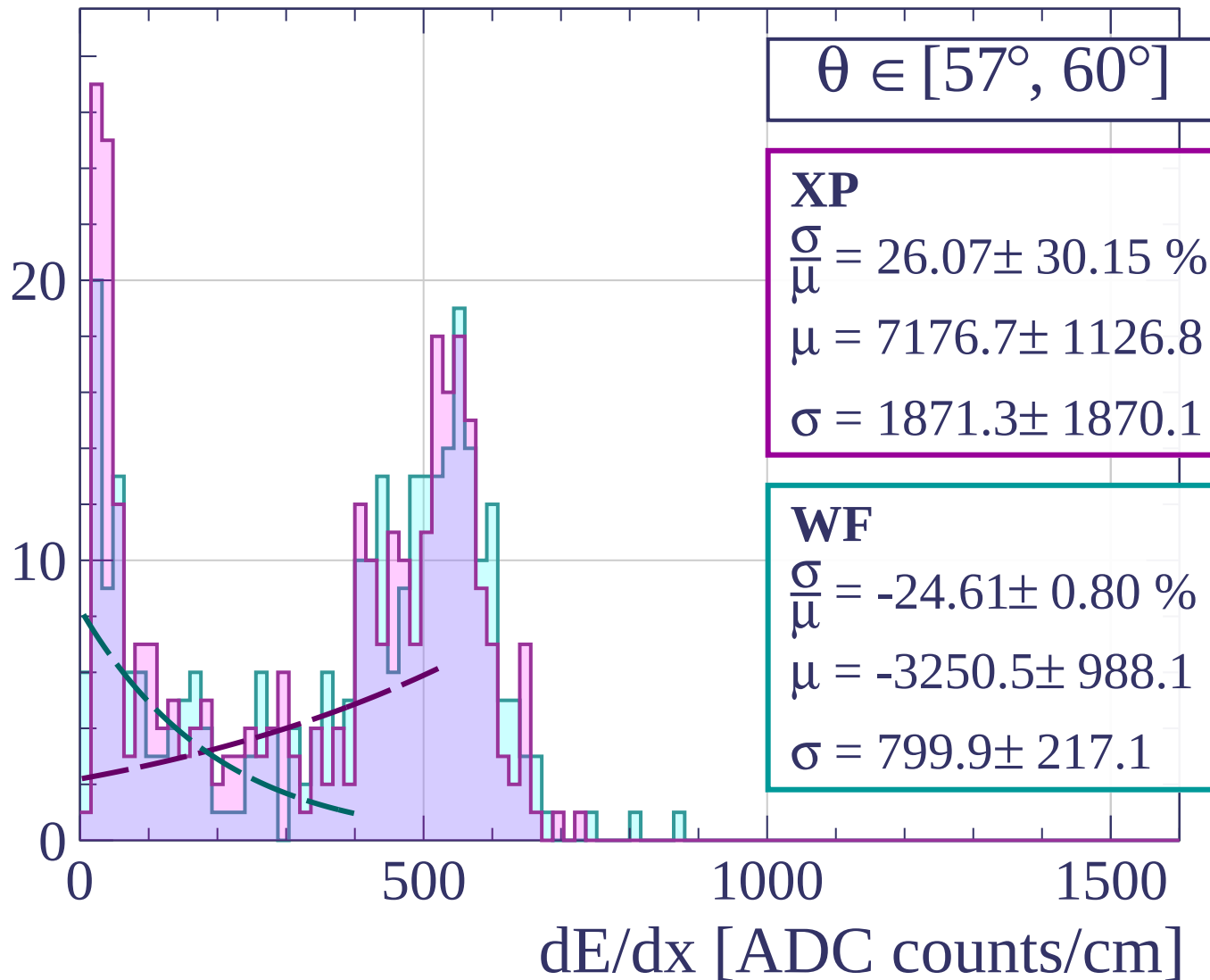
$$\frac{\sigma}{\mu} = 15.79 \pm 1.24 \%$$

$$\mu = 519.9 \pm 6.1$$

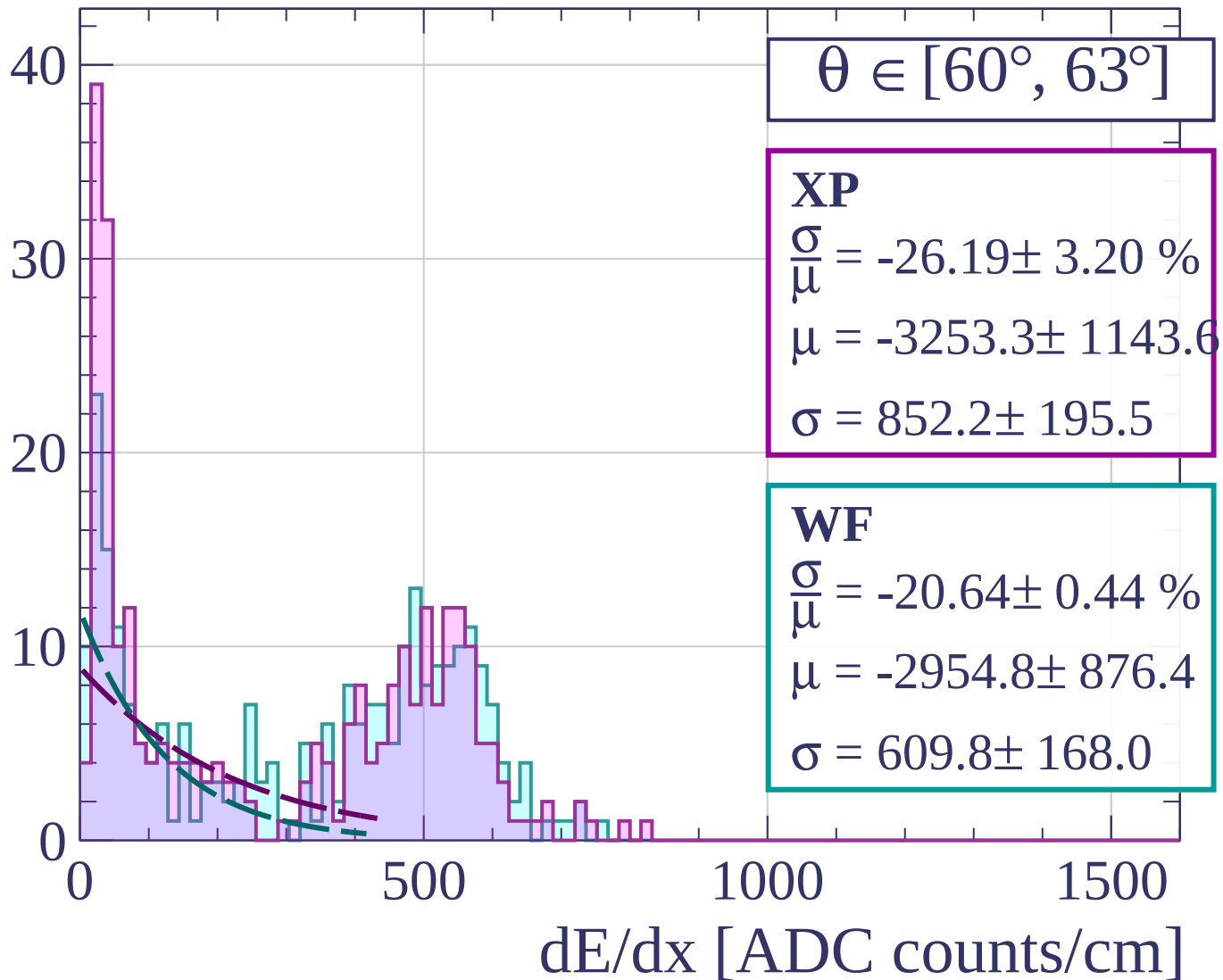
$$\sigma = 82.1 \pm 5.5$$

dE/dx [ADC counts/cm]

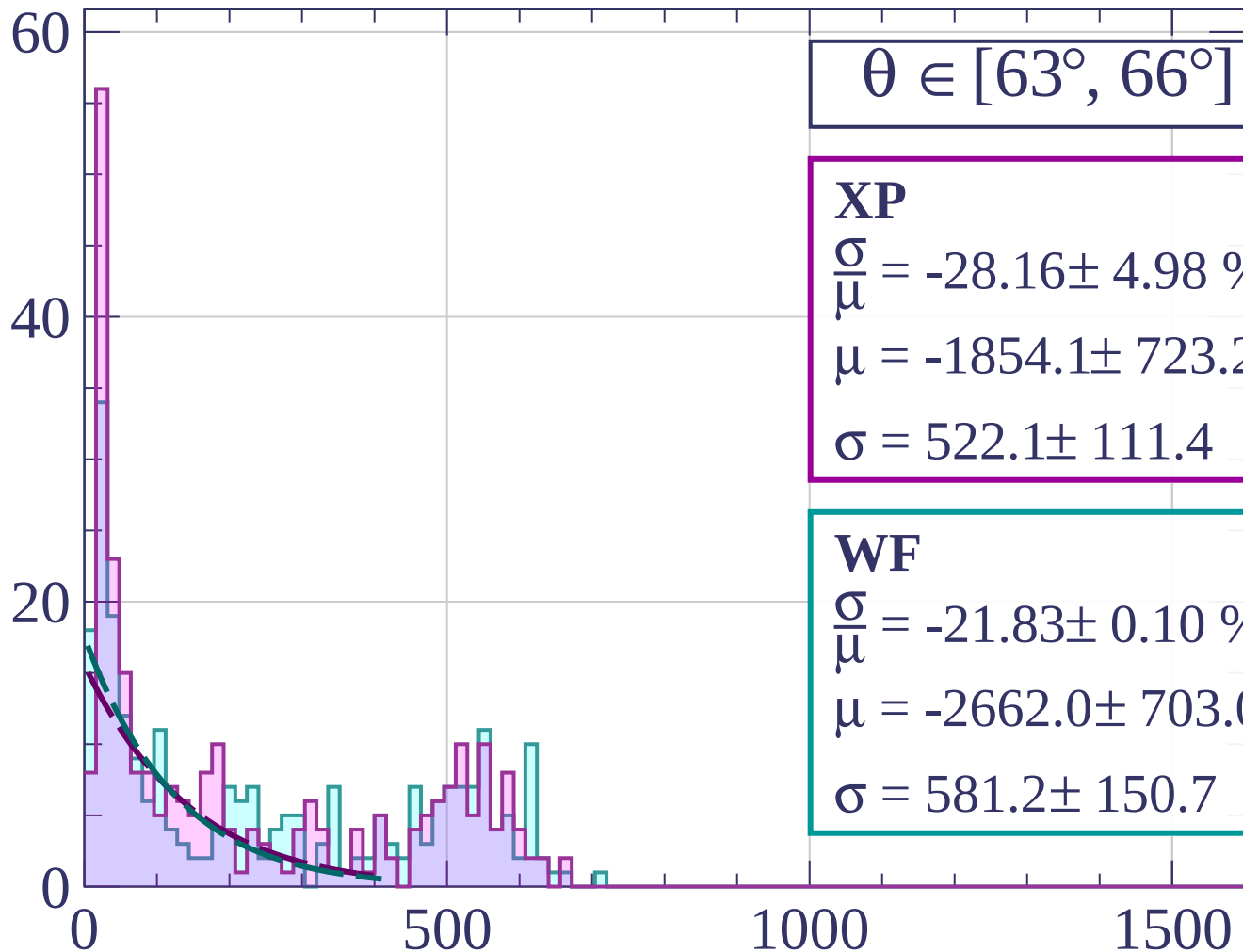
Count



Count

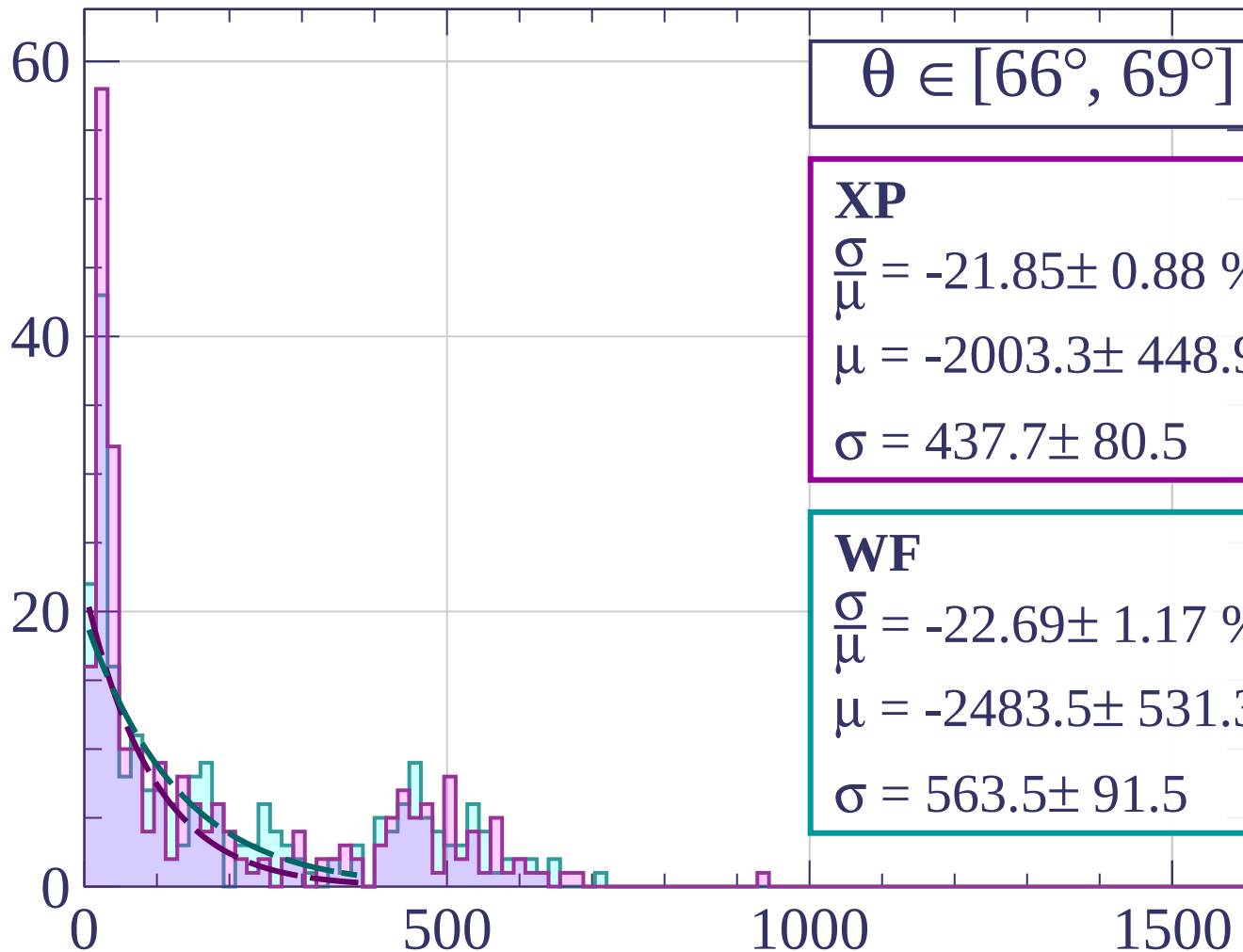


Count



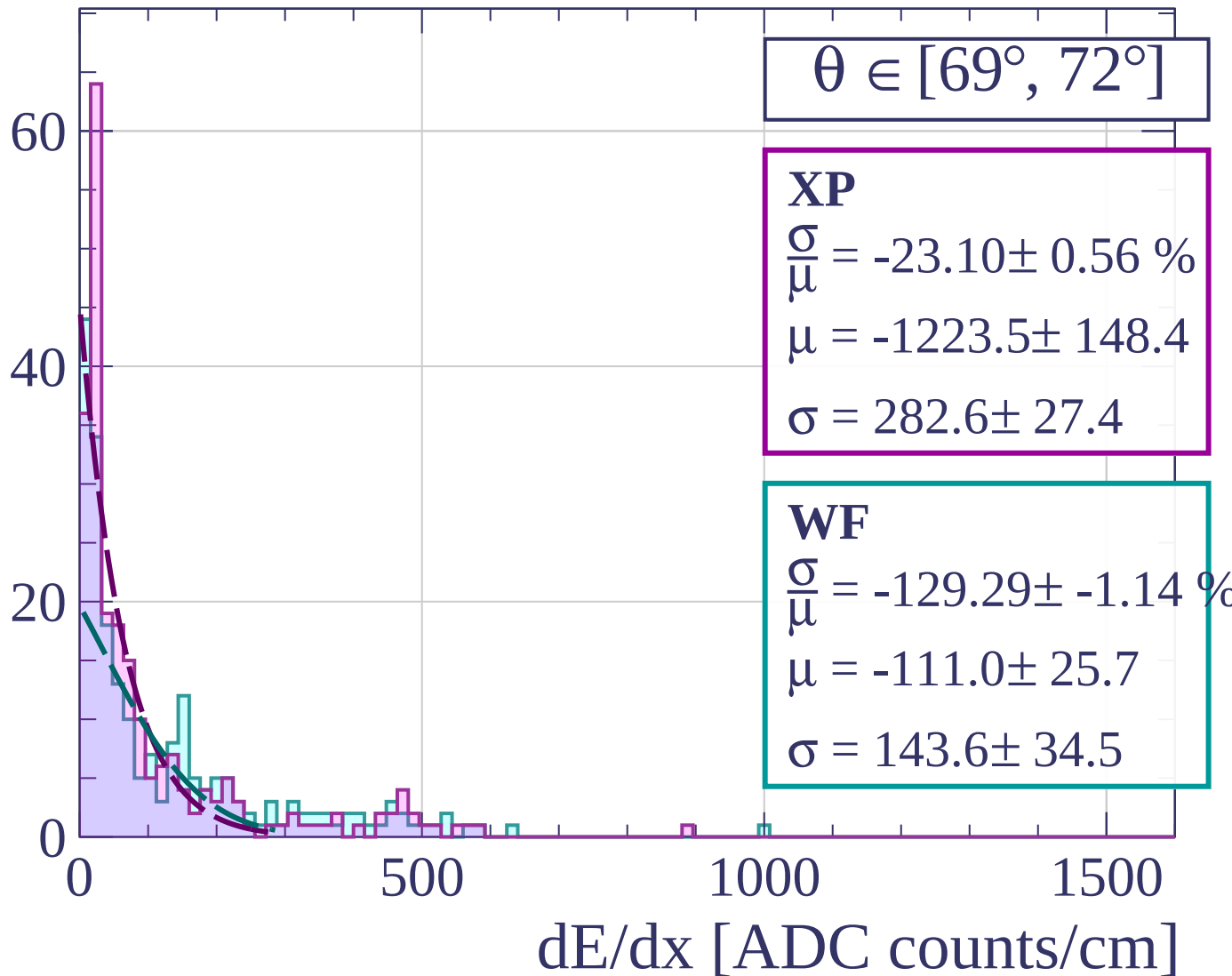
dE/dx [ADC counts/cm]

Count

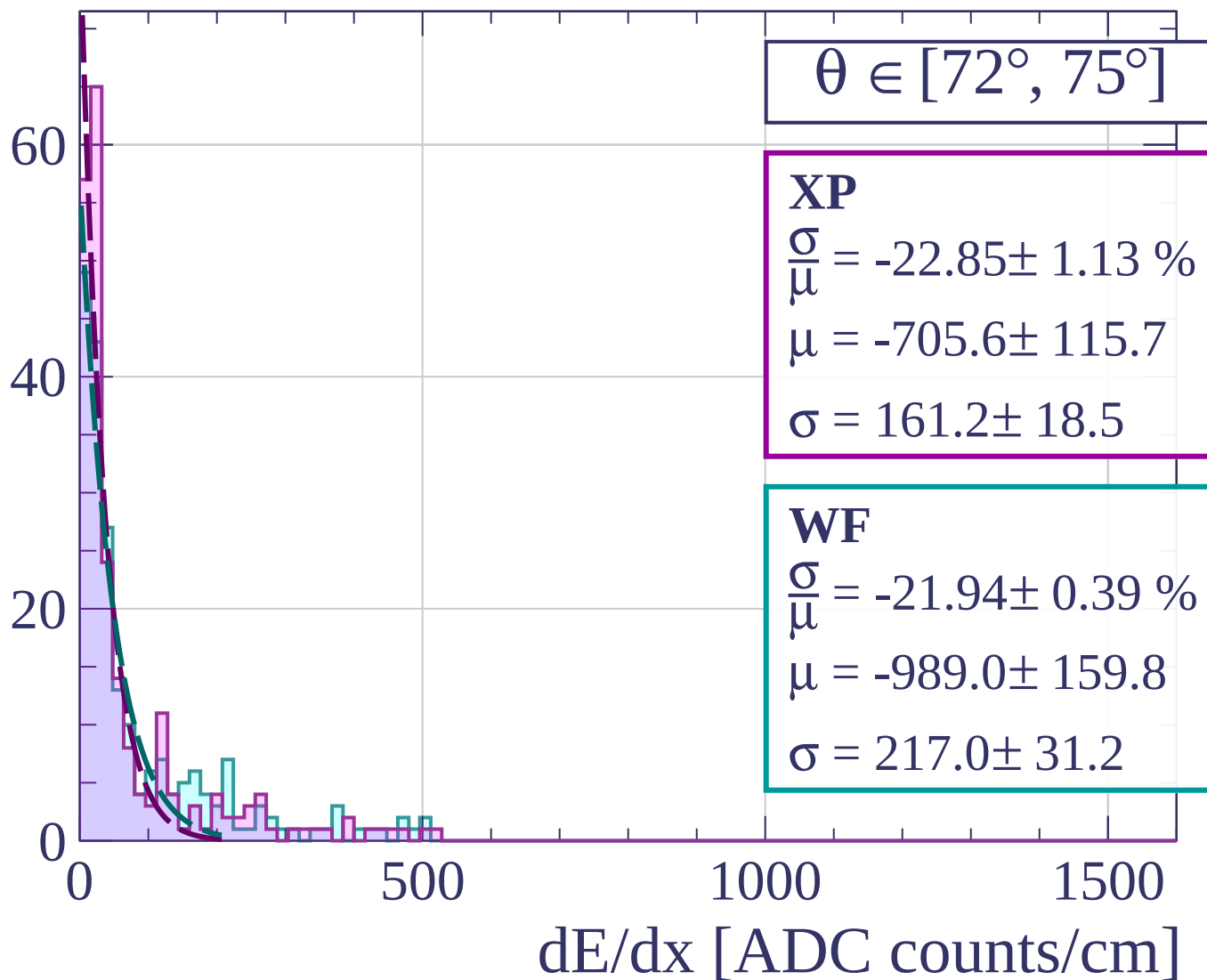


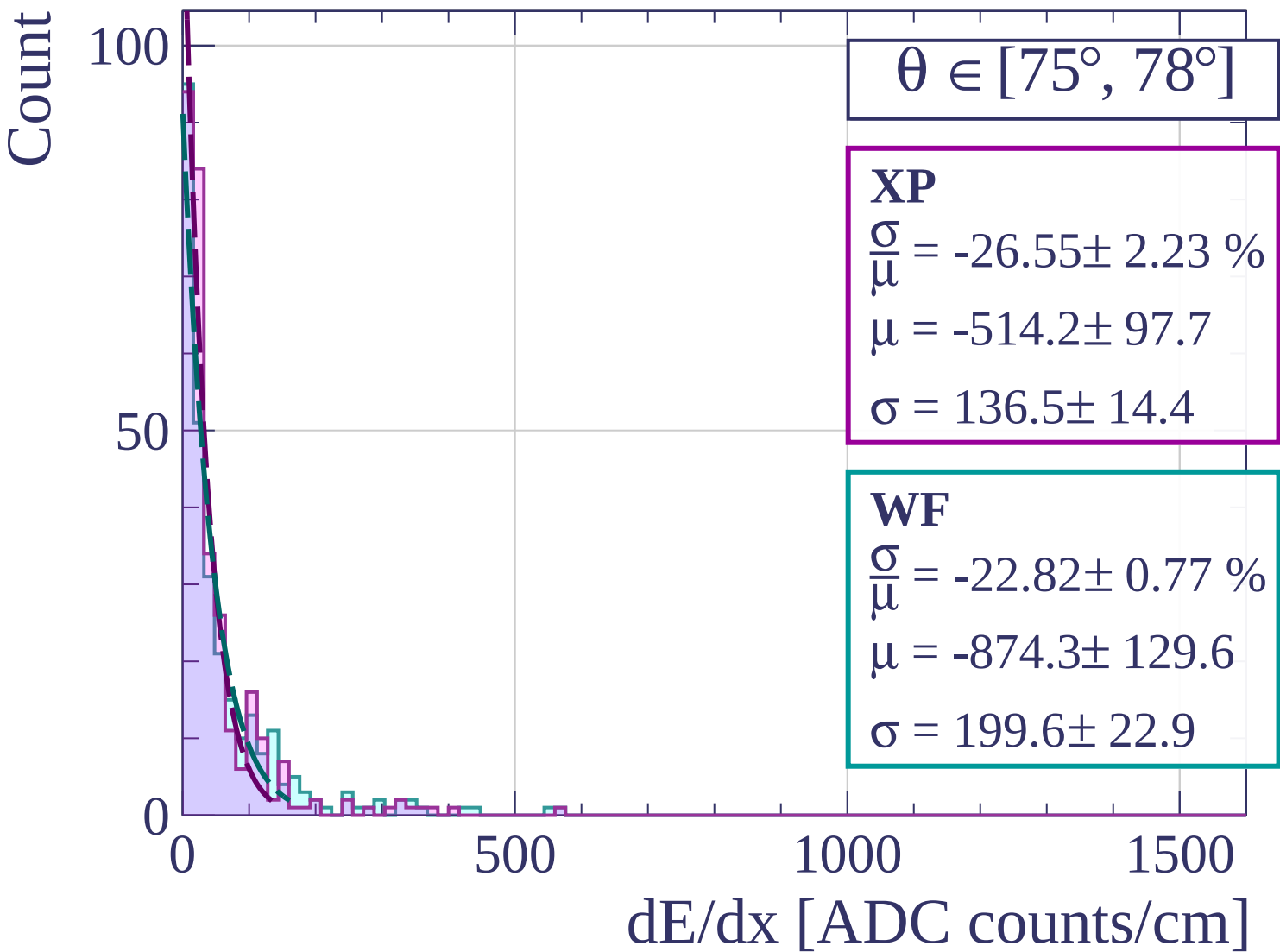
dE/dx [ADC counts/cm]

Count



Count





Count

$$\theta \in [78^\circ, 81^\circ]$$

XP

$$\frac{\sigma}{\mu} = 1.07 \pm 5.35 \%$$

$$\mu = 23.7 \pm 0.4$$

$$\sigma = 0.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = -21.44 \pm 0.58 \%$$

$$\mu = -529.8 \pm 83.0$$

$$\sigma = 113.6 \pm 14.7$$

100

50

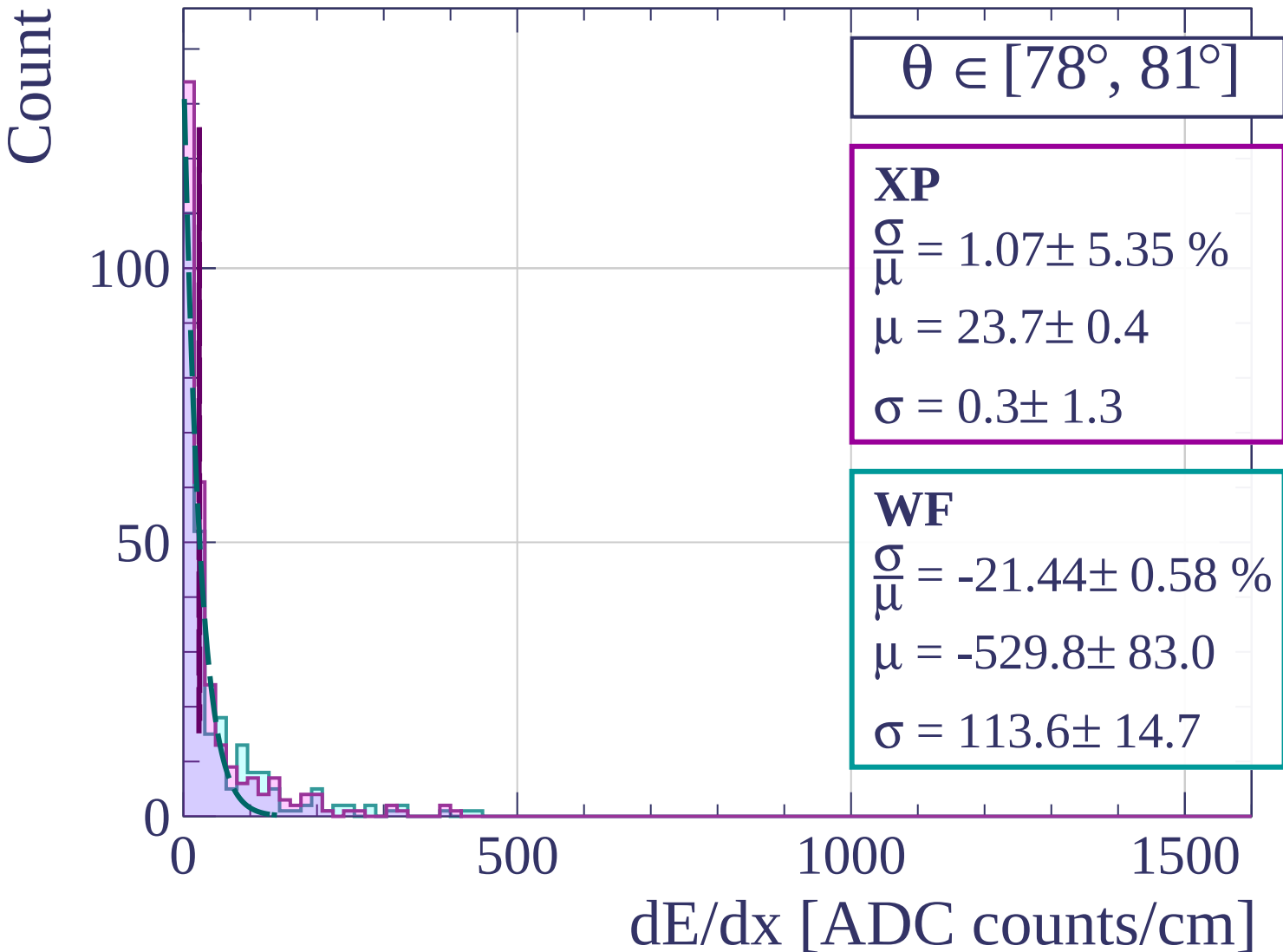
0

500

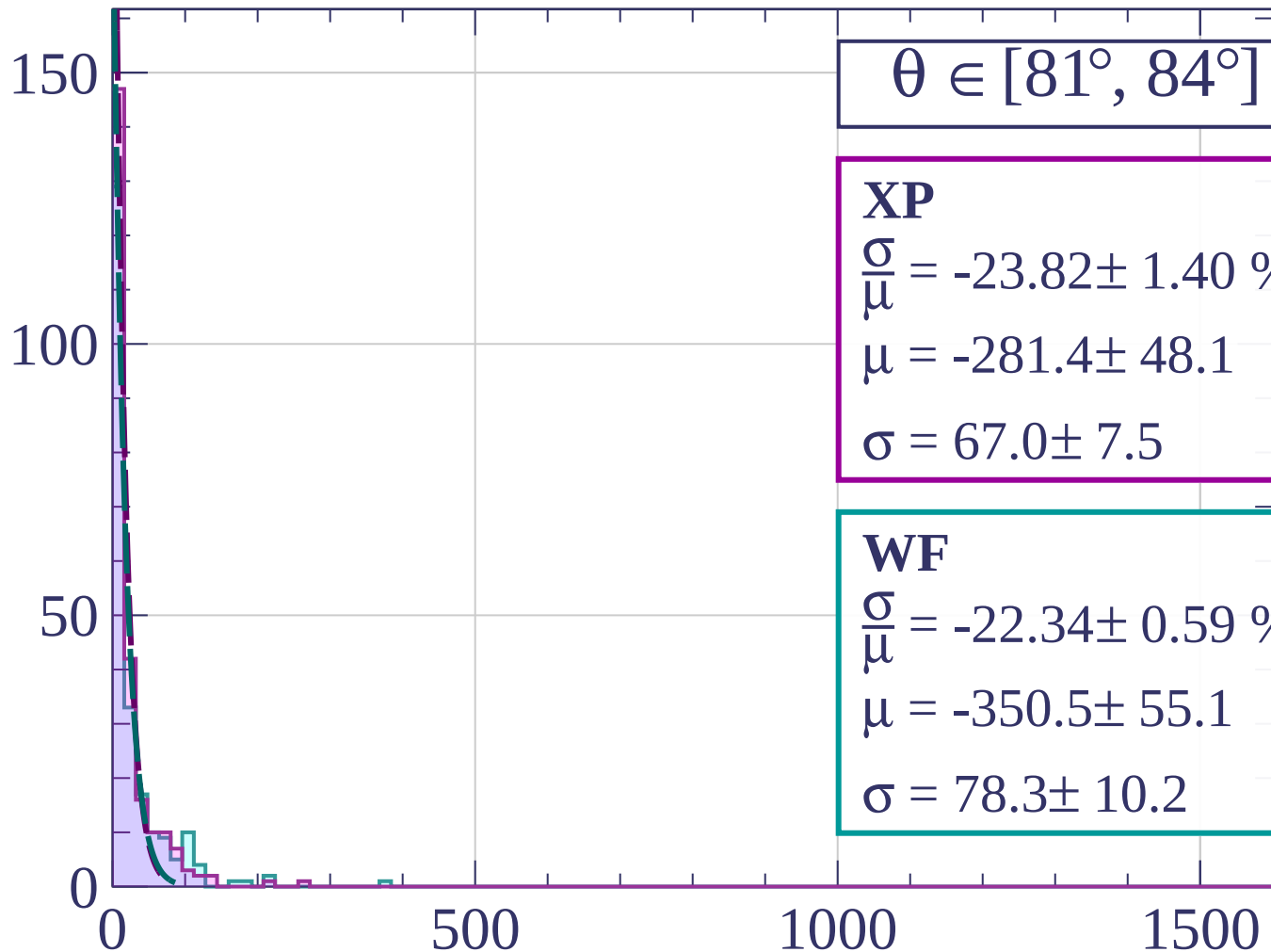
1000

1500

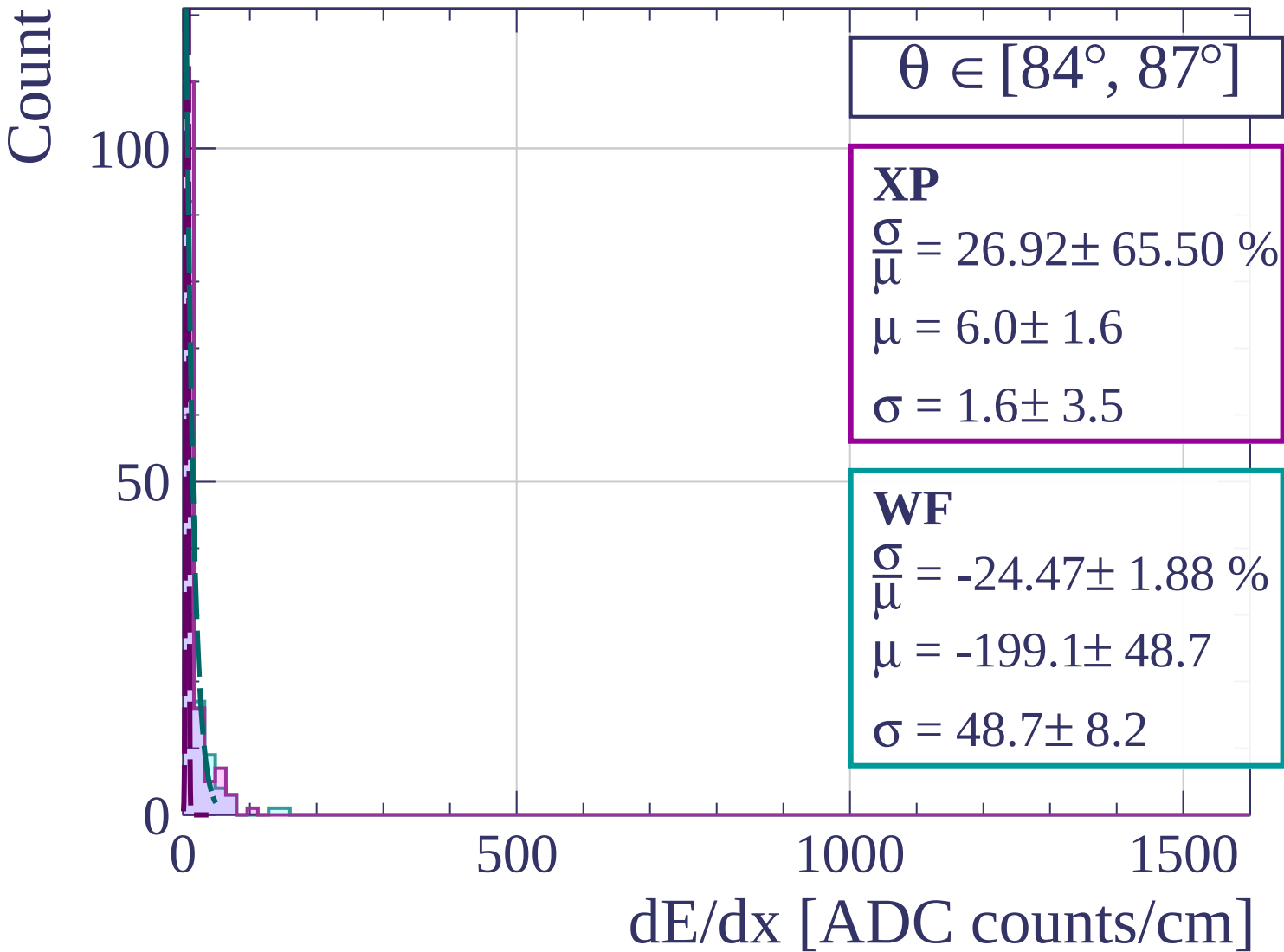
dE/dx [ADC counts/cm]



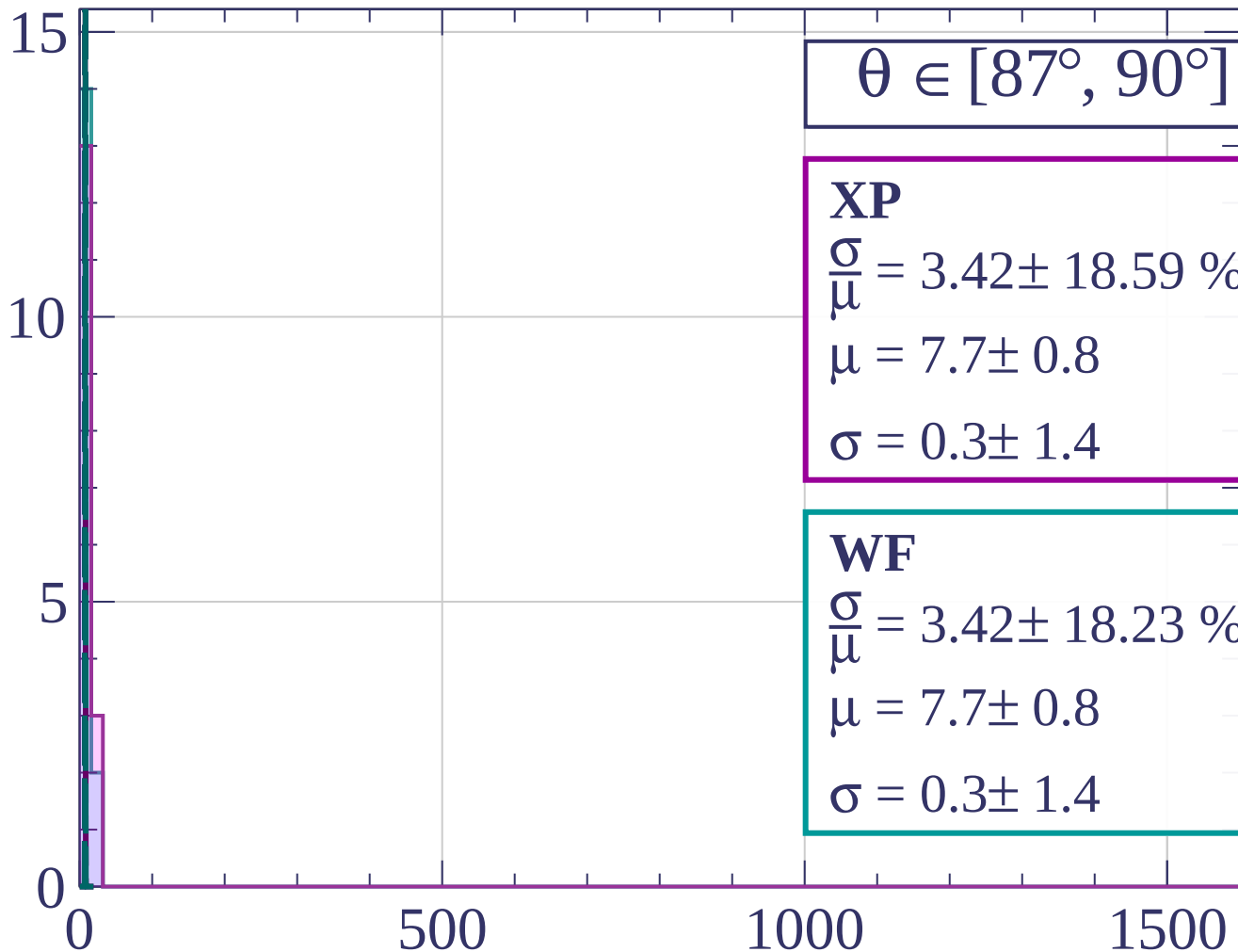
Count



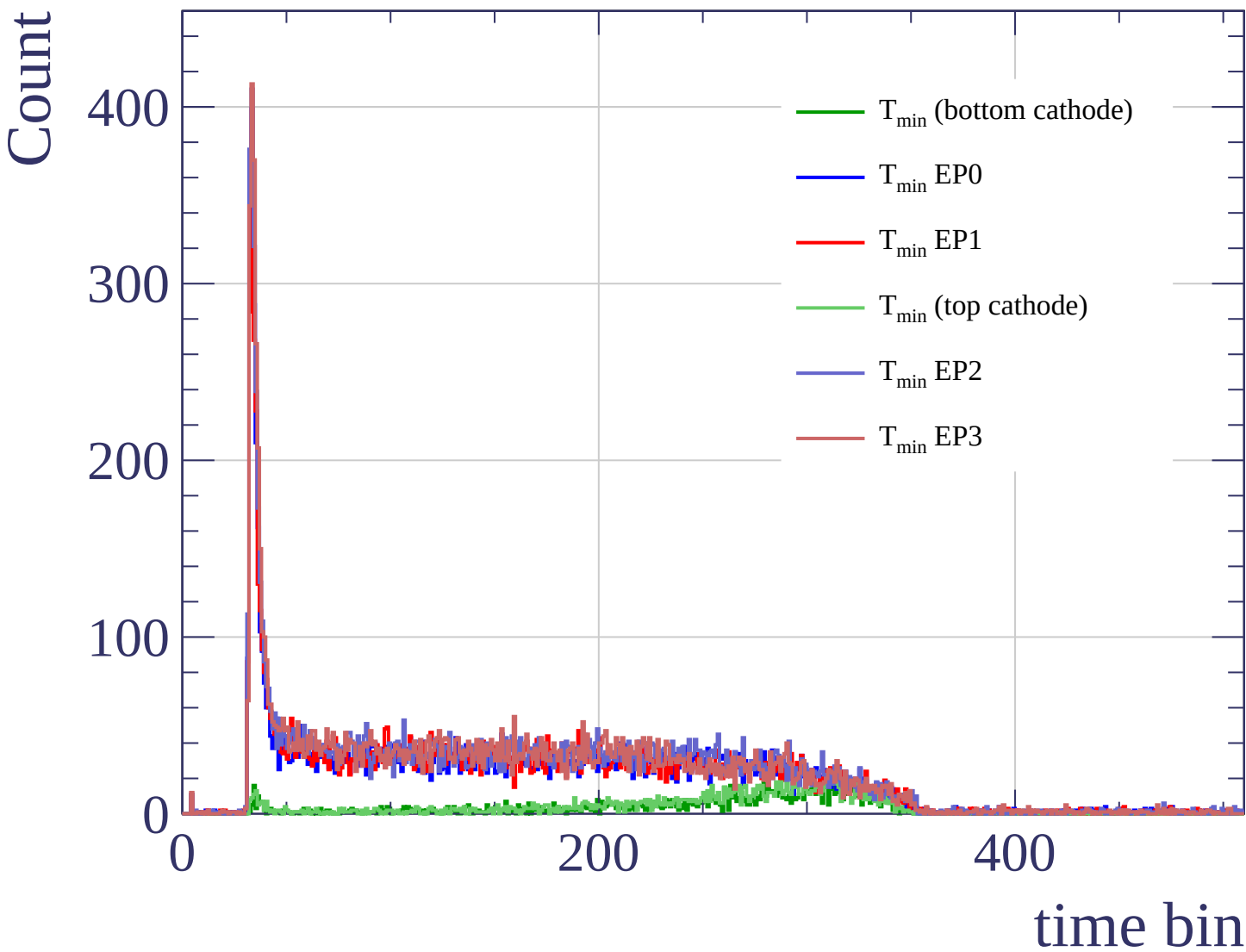
dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]



Count

600

400

200

0

$T_{\max}$ (bottom cathode)

$T_{\max}$ EP0

$T_{\max}$ EP1

$T_{\max}$ (top cathode)

$T_{\max}$ EP2

$T_{\max}$ EP3

200

400

time bin

